

Book I of Euclid's Elements

Formal Reconstruction

Technical Documentation

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Document Status

This volume is the technical documentation of the CGJteamLab formal reconstruction of Book I of Euclid's *Elements*. Its purpose is to record the mathematical architecture of the reconstruction, the Lean formalization, the dependency structure of individual propositions, the reusable Hilbert and scissors-congruence infrastructure, and the relation between the formal development and the classical sources.

Jurek Róžański is the author and editor of this documentation. His academic affiliation is the Military University of Technology, Warsaw, Poland. CGJteamLab is the name of the research project; it is not used here as an institutional affiliation or as the name of an organizational unit of the Military University of Technology.

This document is project documentation rather than a replacement for the Lean source code. The formal source remains authoritative for theorem statements, hypotheses, dependencies, and proof status.

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Introduction

From Hilbert’s Foundations to Synthetic Practice

This volume documents a formal reconstruction of the first part of Euclid’s *Elements*, Book I, over a Hilbert-style foundation in Lean 4.

The original objective was comparatively modest: to determine whether classical Euclidean proofs could be reconstructed faithfully in a modern proof assistant without replacing synthetic geometry by coordinates, real-number measurement, or algebraic computation.

The reconstruction of Propositions I.1–I.48 has revealed a more specific problem.

Hilbert’s axioms provide a foundation for geometry. They separate incidence, order, congruence, and continuity, and they make explicit many assumptions that remain implicit in Euclid. But there is still a substantial distance between such a foundational system and the language in which ordinary synthetic proofs are actually carried out.

A mathematician moves almost invisibly between statements such as:

- three points are collinear in a different order;
- one point lies between two others;
- two descriptions determine the same ray;
- an angle may be transported along congruent rays;
- a proper part of a segment is smaller than the whole;
- equal parts may be added to or removed from segment inequalities;
- a configuration suggested by a diagram really has the required order and separation properties.

None of these operations is mysterious. But neither are they merely syntactic details. In a formal proof they must be represented by definitions, lemmas, and explicit chains of inference.

This is the layer that the present reconstruction has progressively made visible.

Book Zero

Beeson, Narboux, and Wiedijk use the informal name *Book Zero* for fundamental preliminary theorems that are used in Euclid’s Book I but do not themselves depend on propositions of Book

I. Their Book Zero includes elementary results about congruence, betweenness, collinearity and noncollinearity, segment order, rays, laying off segments, angle comparison, and plane separation.¹

The CGJteamLab development adopts this terminology. In the present Hilbert-style reconstruction, Book Zero plays the same structural role, but now specifically as an intermediate theory between Hilbert’s foundational axioms and the level at which ordinary synthetic arguments are naturally conducted. The formalization of Book I suggests the following interpretation.

Book Zero functions as an intermediate language between Hilbert’s axioms and classical synthetic reasoning:

$$\boxed{\text{Hilbert axioms} \longrightarrow \text{Book Zero} \longrightarrow \text{classical synthetic geometry}}$$

Its purpose is not to replace Hilbert’s foundation and not to introduce a second axiomatic system. Rather, it packages the elementary consequences of that foundation at the level at which geometric arguments naturally operate.

This role becomes particularly clear in Proposition I.21. There the proof requires what looks informally like arithmetic with segments: parts are added, removed, compared, and inequalities are transported through congruence. No numerical length function is necessary. Betweenness, congruence, and strict segment comparison already carry the required arithmetic through geometric witnesses.

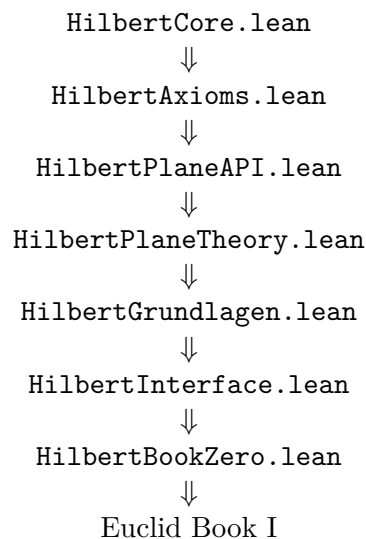
Thus Book Zero is not merely a library for shortening Lean proofs. It is an attempt to identify the missing operational layer between foundational geometry and working synthetic mathematics.

Current Formal Architecture

The production library now makes a sharper distinction between Hilbert’s source axioms and the derived planar theory used by Book I. In particular, `HilbertAxioms.lean` is now a source-faithful transcription of Hilbert’s five axiom groups. It records incidence, order, congruence, parallel uniqueness, and continuity in a form close to the *Grundlagen*; it is not the repository of the developed plane theory.

¹Michael Beeson, Julien Narboux, and Freek Wiedijk, “Proof-checking Euclid,” *Annals of Mathematics and Artificial Intelligence* 85 (2019), 213–257, doi:10.1007/s10472-018-9606-x.

The current Book I route is organized as follows:



The role of the intermediate files is important. The established planar theorem corpus remains in `HilbertPlaneTheory.lean`. `HilbertPlaneAPI.lean` is a compatibility and transport layer between the source-faithful axioms and the working planar vocabulary. The frozen legacy declarations retained in `HilbertAxiomsWork.lean` support that compatibility layer; they are not a second foundational axiom system.

Thus the dependency summaries in the proposition chapters should be read as *local mathematical dependency diagrams*, not as exhaustive Lean import graphs. A chapter may begin its diagram at `HilbertInterface` or Book Zero when that is the relevant working layer. Repeating the complete chain from `HilbertCore.lean` in every proposition would obscure rather than clarify the proposition-specific mathematics.

The higher-dimensional branch is separate from this Book I route. Spatial incidence is now entered through `HilbertSpaceIncidence.lean`; Book I continues to use the planar theory above.

Euclid and Hilbert Do Not Have the Same Dependency Graph

The reconstruction also makes clear that the order of Euclid’s propositions cannot simply be identified with the logical dependency order of a Hilbert-style theory.

Several different situations occur.

First, some Euclidean propositions correspond to principles that are already foundational in Hilbert geometry. Euclid’s Proposition I.4, for example, proves SAS by superposition, whereas SAS belongs to the Hilbert congruence structure. Similarly, the angle-copying construction of Proposition I.23 is already contained in Hilbert’s angle-construction principle.

Second, some Euclidean propositions genuinely require more than neutral Hilbert geometry. Propositions I.1 and I.22 expose intersection or continuity assumptions that cannot be obtained from incidence, order, and congruence alone. A different boundary appears at Proposition I.29. The criteria I.27 and I.28 derive parallelism from angle relations in neutral geometry, whereas I.29 reverses the direction and requires the Euclidean uniqueness of parallels.

The distinction is sharper than the numerical order of the *Elements* suggests. Proposition I.31 comes after I.29, but its existence statement is already neutral: an angle can be transported and

I.27 then proves that the constructed line is parallel. Only uniqueness of that parallel requires the Euclidean parallel principle. Thus the sequence of propositions and the logical layers of the theory cross one another.

A second structural boundary appears at Proposition I.35. Here the subject changes from congruence and parallelism to equality of rectilinear figures. The reconstruction does not introduce numerical area. Instead it follows Hilbert’s Chapter IV and represents the relevant equality by a synthetic theory of equidecomposition and equicomplementability. Proposition I.36 immediately confirms that this is not an isolated device: its content argument is obtained by applying I.35 twice and using transitivity, while the new work lies in constructing the intermediate parallelogram.

Third, and most importantly for the present project, many propositions require no stronger axiom but depend on a considerable intermediate theory of order, rays, segment comparison, angle comparison, noncollinearity, separation, and transport.

It is this third class that explains much of the apparent complexity of formal synthetic geometry.

What Formalization Reveals

A classical diagram compresses information.

It may show immediately that a point lies inside an angle, that two points occur on the same ray, that one point is between two others, or that an auxiliary construction has a particular orientation. A proof assistant cannot obtain these facts by looking at the figure.

Consequently formalization repeatedly exposes obligations that are easy to overlook in textbook prose:

$$\text{diagrammatic configuration} \not\equiv \text{formal order information.}$$

Proposition I.24 provides a particularly clear example. Euclid’s construction suggests one relative position of the auxiliary points, but that position is not forced merely by the stated construction. The formal proof must analyze the possible configurations explicitly.

The same phenomenon reappears in the theory of parallels. In I.27 the informal argument must exclude an intersection on either side of the transversal. In I.29 the decisive step is not another angle calculation but uniqueness: a neutrally constructed candidate parallel must be identified with the given parallel. Proposition I.31 then shows that existence and uniqueness, often compressed into one familiar statement, belong to different logical layers.

Such phenomena should not be regarded simply as inconveniences caused by Lean. They reveal logical structure already present in the mathematics.

The Purpose of This Documentation

The chapters that follow are therefore not intended merely as line-by-line explanations of Lean source files.

For each proposition we record several distinct objects:

- Euclid’s classical construction and proof;

- the corresponding reconstruction over the Hilbert foundation;
- the actual dependency architecture of the Lean theorem;
- the Hilbert and Book Zero components used in the proof;
- formal obligations hidden by the traditional diagram;
- the relationship of the proposition to earlier and later propositions of Book I.

The appendices compare this reconstruction with the Hilbert-style developments of Greenberg and Hartshorne and with Wilkins’s detailed study of the internal dependency structure of Euclid’s Book I. These comparisons are not included merely as historical commentary. They help distinguish genuine foundational dependencies from intermediate lemmas that a formal development must make explicit, and they expose alternative proof paths, converse relations, and configuration cases hidden by the linear order of the *Elements*.

Current Scope

The present volume covers the complete Euclid Book I, through Proposition I.48.

Forty-eight propositions are still too small a sample from which to claim a final architecture for synthetic geometry. They do, however, cross several genuine structural boundaries. Proposition I.29 marks the transition from neutral geometry to the Euclidean theory of parallels. Proposition I.35 marks another transition, from congruence and parallelism to a synthetic theory of equality of rectilinear figures. The later propositions then test whether these working layers remain usable through the closing constructions and the Pythagorean theorem together with its converse.

The treatment beginning at I.35 shows that the content transition can be handled without introducing numerical measurement. Hilbert’s equicomplementability theory and the formal scissors calculus provide a synthetic content layer that can be reused by subsequent propositions.

Nevertheless, the evidence is already substantial enough to identify a recurring structure.

Hilbert supplied the foundational theory. Euclid supplied the classical sequence of geometric problems and proofs. Between them lies a layer of elementary synthetic operations that is normally left implicit because mathematicians use it automatically.

Formal reconstruction forces that layer into view.

The principal question of the project is therefore no longer only

Can Euclid’s proofs be formalized over Hilbert geometry?

It has become the more structural question

What intermediate theory is required to turn Hilbert’s axioms into a practical language for classical synthetic geometry?

The reconstruction of Book I is one way of investigating that question.

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Chapter 1

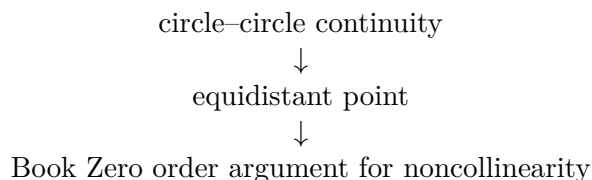
Proposition I.1

1.1 Introduction

Proposition I.1 is the first construction in Euclid’s *Elements*: given a nondegenerate segment AB , construct an equilateral triangle on that base. The familiar proof draws two circles of radius AB and uses a common point C as the third vertex.

The current formal reconstruction reveals more structure than this classical picture. The public theorem first obtains a point C equidistant from A and B , then proves separately that C cannot lie on the line AB . In the present production source, the equidistant-point theorem is no longer a provisional black box: it is proved from the project’s general circle–circle continuity principle.

Thus I.1 already exhibits three layers:



This makes I.1 an especially clear test of the boundary between Euclid’s visible construction and the lower Hilbert/Book Zero infrastructure.

1.2 Classical Configuration

Let AB be the given finite straight line, with $A \neq B$. Euclid draws the circle centered at A through B and the circle centered at B through A . If C is a common point of these circles, then

$$AC \cong AB, \quad BC \cong BA \cong AB.$$

Joining C to A and B therefore gives an equilateral triangle ABC .

Foundational caution. The familiar two-circle picture suppresses the main existence issue. The permission to draw the two circles does not by itself prove that they meet. A rigorous axiomatization needs a circle-intersection or continuity principle. This is not a defect specific to the formal reconstruction: it is exactly the gap hidden by the classical diagram.

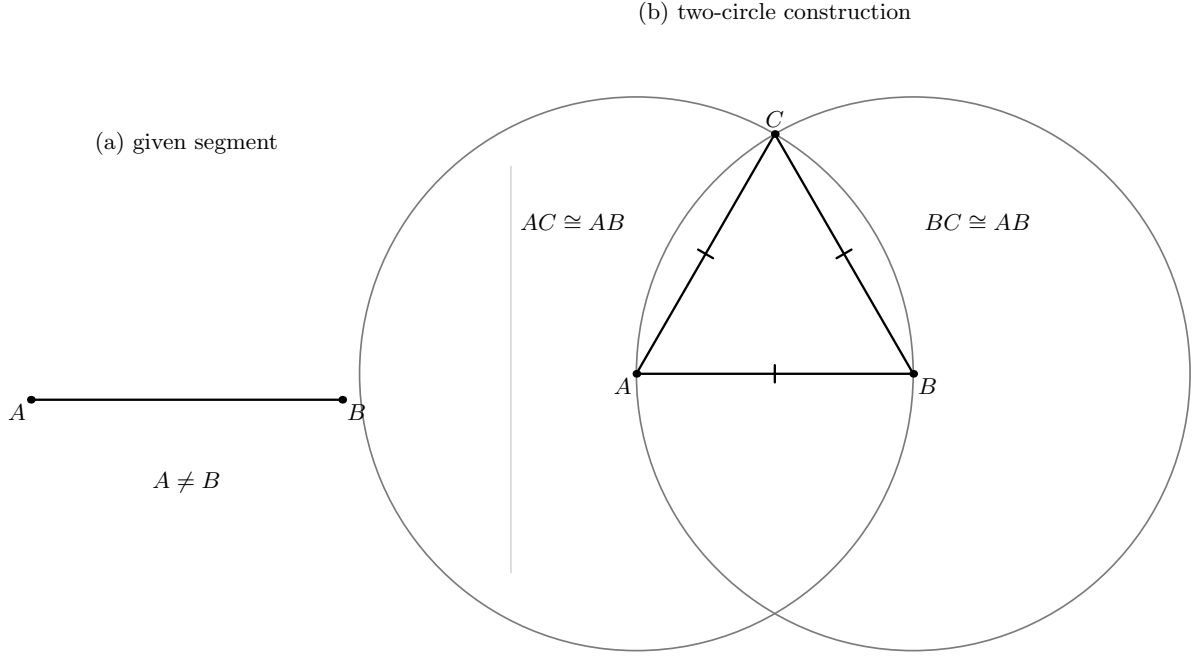


Figure 1.1: Euclid's construction for Proposition I.1. (a) The input is a nondegenerate segment AB . (b) Two circles with centers A, B and radius congruent to AB meet at C ; the three sides of ABC are then congruent.

1.3 Statement

Theorem 1 (Euclid I.1). *For every nondegenerate segment AB there exists a point C such that ABC is a genuine equilateral triangle:*

$$\neg \text{Collinear}(A, B, C), \quad AC \cong AB, \quad BC \cong AB.$$

The current production declaration is:

```

theorem euclid_proposition_1
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B : Geo.Point)
  (hAB : A ≠ B) :
  ∃ C : Geo.Point,
    ¬ PrimCollinear Geo A B C ∧
    Geo.Congruent A C A B ∧
    Geo.Congruent B C A B := by
  ...

```

The formal statement makes the nondegeneracy of the triangle explicit. Euclid's diagram visually places C off the base line; Lean requires this as a theorem.

Diagrammatic audit. The preserved Book1R chapter already reduced the classical construction from four incremental snapshots to two stable states. That reduction is sound and is retained. The formal part, however, was still represented mainly by a flowchart and described `hilbert_equidistant_point_exists` as a temporary existence principle. Both points are now outdated.

The audited chapter uses three geometric figures with seven panels:

2 classical panels + 2 continuity panels + 3 collinearity branches = 7 panels in 3 figures.

The three branch panels are required because the production proof performs a genuine betweenness trichotomy.

1.4 Classical Proof

Euclid's construction can be separated into the following steps.

1. By Postulate III, draw the circle centered at A through B .
2. Draw the circle centered at B through A .
3. Use a circle-circle intersection principle to obtain a common point C .
4. By Postulate I, join C to A and B .

Since C belongs to the first circle,

$$AC \cong AB.$$

Since C belongs to the second circle,

$$BC \cong BA.$$

Endpoint reversal does not change a segment, so $BA \cong AB$, and Common Notion 1 gives the required equality of the three sides.

The historical dependency structure is therefore

$$\text{Postulates I and III} + \text{circle-circle intersection} + \text{Common Notion 1} \longrightarrow \text{I.1.}$$

No earlier proposition of Book I is available or needed.

1.5 The Formal Existence Engine

The current source proves `hilbert_equidistant_point_exists`; it is not an axiom. The proof reduces the special equilateral configuration to the general continuity axiom `hilbert_circle_circle_intersection`.

Starting from $A \neq B$, the proof first extends AB beyond B . A transient point R supplies a ray from B . On that ray, Hilbert III.1 lays off a copy of BA , producing Y . Same-ray transport then yields

$$A - B - Y, \quad BY \cong BA.$$

The point R is used only to activate segment construction and disappears from the stable geometric state, so it is intentionally not drawn.

Now consider two circles represented synthetically by their centers and radius segments:

- circle 1 has center A and radius AB ;
- circle 2 has center B and radius BA .

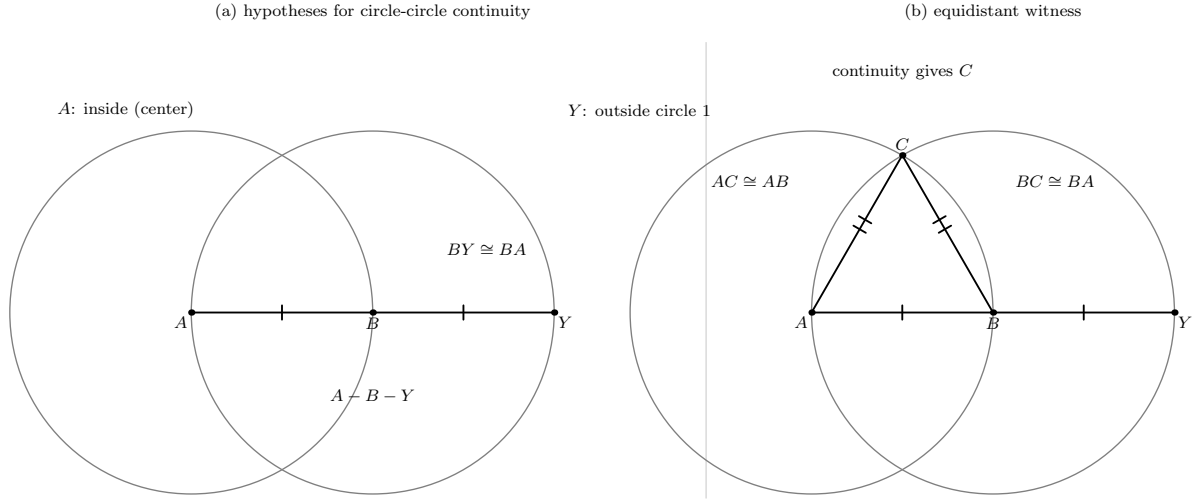


Figure 1.2: The formal continuity engine hidden below `hilbert_equidistant_point_exists`. (a) The construction gives $A - B - Y$ and $BY \cong BA$. Thus A is the center-case inside witness for circle 1, while Y lies on circle 2 and outside circle 1. The drawn circles are schematic: their visible crossing is not used as a proof. (b) The continuity axiom supplies a common point C , yielding $AC \cong AB$ and $BC \cong BA$.

The point A lies on circle 2 and is the center of circle 1. The point Y also lies on circle 2, because $BY \cong BA$. Since $A - B - Y$, the segment AB is a proper initial part of AY , so Y is outside circle 1 in the synthetic segment-order sense.

The decisive source call is structurally:

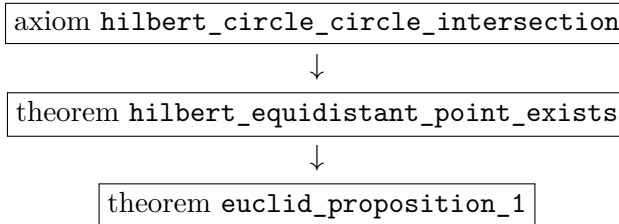
```
rcases
  hilbert_circle_circle_intersection
    Geo
    A B
    B A
    A Y
    ... with
  ⟨C, hAC_AB, hBC_BA⟩
```

After reversing the second radius orientation, the theorem returns exactly

$$AC \cong AB, \quad BC \cong AB.$$

1.6 Continuity Boundary

The source-level distinction is important. The chain is



The continuity axiom is stated synthetically. One circle contains two points X, Y on a second

circle; X is inside the first circle (or is its center), and Y is outside it. The conclusion is the existence of a common point of the two circles. No coordinates, real-valued radii, or distance function are introduced.

For I.1 the generic axiom is specialized in a particularly economical way: $X = A$ is the center case, and Y is the explicitly constructed outside point. Thus the equidistant-point theorem is a genuine derived theorem from the project's accepted circle-continuity layer.

1.7 The Synthetic Noncollinearity Stage

Once C is obtained, the production theorem still has to prove that A, B, C form a genuine triangle. First it excludes $C = A$ and $C = B$. Both degenerate cases would make a nonzero segment congruent to a null segment; `bookZero_nullSegment1` rules this out.

Assume next, for contradiction, that A, B, C are collinear. Since the three points are pairwise distinct, Hilbert betweenness trichotomy gives exactly three cases:

$$A - B - C, \quad B - A - C, \quad A - C - B.$$

Each case contradicts the fact that a proper part cannot be congruent to the whole.

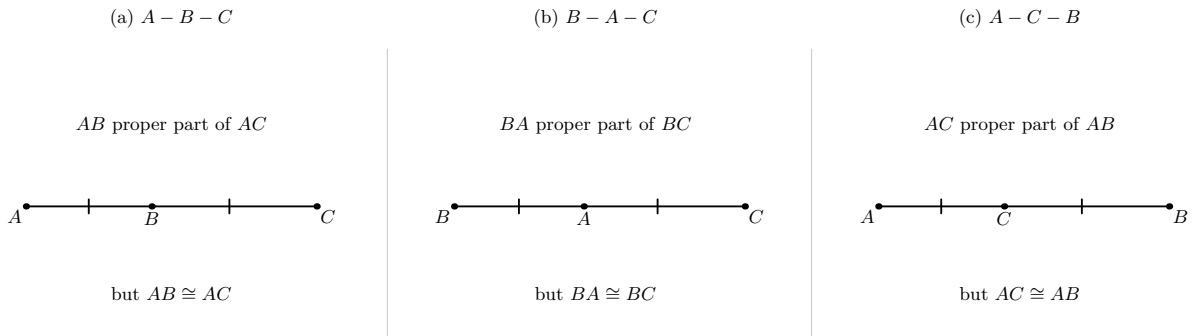


Figure 1.3: The three genuine branches of the formal collinearity reductio. In each order, one of the equidistant segments becomes a proper part of a segment to which it is congruent. Book Zero #45, `bookZero_45_partNotEqualWhole`, excludes all three cases.

More explicitly:

- If $A - B - C$, then AB is a proper part of AC , contradicting $AB \cong AC$.
- If $B - A - C$, then BA is a proper part of BC , while congruence transport gives $BA \cong BC$.
- If $A - C - B$, then AC is a proper part of AB , contradicting $AC \cong AB$.

Hence A, B, C are noncollinear.

1.8 Lean Formalization

At the Book I level the proof has three visible phases.

First, obtain the equidistant witness:

```
obtain ⟨C, hAC, hBC⟩ :=
  hilbert_equidistant_point_exists Geo A B hAB
```

Second, exclude the degenerate endpoint identifications using `bookZero_nullSegment1`. Third, assume collinearity and split the three possible orders:

```
rcases
  hilbert_between_trichotomy
    Geo A B C
    hAB hBCne hACne hCol with
  hABC | hBAC | hACB
```

Every branch is closed by `bookZero_45_partNotEqualWhole`, with only congruence orientation changes between the three cases.

This is a useful example of why theorem length alone is a poor guide to the geometric architecture. The public proof contains a substantial order argument, while its existence theorem contains a separate continuity construction beneath the interface.

1.9 Hilbert and Book Zero Components Used

1.9.1 Circle–circle continuity

The foundational existence input is `hilbert_circle_circle_intersection`. It is an explicit continuity axiom in the current interface. Proposition I.1 does not introduce a new local existence axiom.

1.9.2 Segment construction and order

Hilbert III.1 constructs the outside witness Y on a chosen ray. The order layer then turns same-ray information into the strict betweenness configuration $A - B - Y$, from which $AB < AY$ is represented synthetically.

1.9.3 Null-segment exclusion

The public theorem directly uses `bookZero_nullSegment1` to exclude $C = A$ and $C = B$. This declaration remains an active Book Zero/interface convenience axiom rather than one of Hilbert’s historical congruence axioms. It should therefore be distinguished from the deliberately accepted continuity principle above.

1.9.4 Proper part versus whole

`bookZero_45_partNotEqualWhole` is the common contradiction engine for all three collinear orders. It makes explicit the elementary geometric fact that Euclid’s final diagram leaves visually obvious.

1.10 Relationship with the Foundations

I.1 has no earlier Book I theorem to call, so the foundational organization is exposed immediately. The classical and formal proofs now meet at a surprisingly close point: both ultimately require a circle-intersection principle. The difference is that the formal development states that principle explicitly and supplies the precise inside/outside witnesses needed to invoke it.

The remaining noncollinearity burden is then discharged without circles. This separation is conceptually useful:

continuity controls existence, order + congruence control nondegeneracy.

It is this second stage that demonstrates the role of Book Zero as a proof language rather than merely a repository of constructions.

1.11 Dependency Summary

The classical architecture is

$$\begin{array}{c}
 AB \text{ given} \\
 \downarrow \\
 \text{two circles of radius } AB \\
 \downarrow \\
 \text{circle-circle intersection } C \\
 \downarrow \\
 AC \cong AB, \quad BC \cong AB \\
 \downarrow \\
 \text{I.1.}
 \end{array}$$

The current formal architecture is

$$\begin{array}{c}
 A \neq B \\
 \downarrow \\
 A - B - Y, \quad BY \cong BA \\
 \downarrow \\
 \text{hilbert_circle_circle_intersection} \\
 \downarrow \\
 \text{hilbert_equidistant_point_exists} \\
 \downarrow \\
 C \neq A, \quad C \neq B \\
 \downarrow \\
 \text{three-way betweenness trichotomy under collinearity} \\
 \downarrow \\
 \text{bookZero_45_partNotEqualWhole} \\
 \downarrow \\
 \neg \text{Collinear}(A, B, C) \\
 \downarrow \\
 \text{euclid_proposition_1.}
 \end{array}$$

At the axiom boundary relevant to this chapter, the two explicit interface primitives to record are the accepted circle-circle continuity axiom and the separate Book Zero null-segment convenience axiom used in the endpoint degeneracy checks. There is no proposition-local axiom in `Proposition01.lean`.

1.12 Conclusion

Proposition I.1 is not merely the first easy construction in Book I. Once its hidden assumptions are exposed, it already displays the architecture that recurs throughout the reconstruction.

Euclid presents the geometry as two circles and their intersection. The current Lean development makes the same existence issue explicit through a general circle–circle continuity axiom, derives the special equidistant point from it, and then proves noncollinearity by a separate Book Zero order argument. The resulting theorem is therefore both a reconstruction of Euclid’s construction and an early demonstration of how the project divides proof obligations between continuity, congruence, and order.

Chapter 2

Proposition I.2

2.1 Introduction

Proposition I.2 is Euclid’s first general segment-transport construction. Given a prescribed point A and a given segment BC , the problem is to construct a segment beginning at A and congruent to BC .

In the *Elements* this is a genuine derived construction. Euclid uses Proposition I.1, produces sides of an equilateral triangle, draws two circles, and subtracts equal parts from equal wholes. The current Lean reconstruction has a fundamentally different architecture. Its public statement already includes a chosen ray AE , and the result is obtained by one application of Book Zero #49, `bookZero_49_layoff`.

The audit reveals one further layer. Book Zero #49 is itself only a thin wrapper around Hilbert’s primitive segment-construction axiom III.1, `HilbertCongruence.segment_construction`. Thus the complete formal chain is

$$\text{Hilbert III.1} \longrightarrow \text{bookZero_49_layoff} \longrightarrow \text{euclid_proposition_2}.$$

No hidden geometric construction lies below this chain: the layoff operation is foundational in the current Hilbert congruence structure.

2.2 Classical Configuration

Let A be the prescribed endpoint and let BC be the given segment. Euclid joins A to B and constructs the equilateral triangle ABD on AB by Proposition I.1. The sides DB and DA are then produced beyond B and A .

A circle centered at B with radius BC meets the produced ray DB at G . Hence

$$BG \cong BC.$$

A second circle, centered at D with radius DG , meets the produced ray DA at L . Hence

$$DL \cong DG.$$

Since ABD is equilateral,

$$DB \cong DA.$$

Subtracting these equal parts from the equal wholes DG and DL gives

$$BG \cong AL,$$

and therefore

$$AL \cong BC.$$

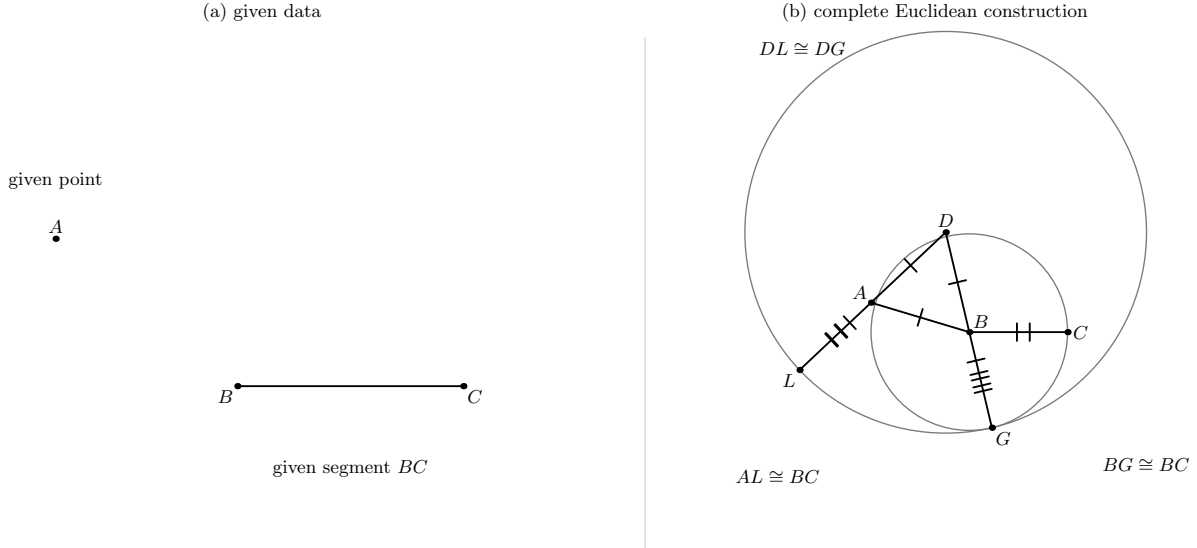


Figure 2.1: Euclid's construction for Proposition I.2. (a) The input is a prescribed point A and a given segment BC . (b) The equilateral triangle ABD and the two circles produce points G, L . Equal radii, the equilateral sides, and subtraction of equal parts yield $AL \cong BG \cong BC$.

Foundational caution. The completed diagram makes a classical suppression visible. The produced rays DB and DA must actually meet the two circles at the required points. A picture makes this look automatic, but a rigorous axiomatic reconstruction must supply the corresponding intersection/continuity facts. This issue is part of the feasibility of Euclid's classical construction; it does not enter the current Lean proof of I.2, because segment layoff is already primitive in the Hilbert congruence layer.

2.3 Statement

Theorem 2 (Euclid I.2). *Given a point A and a segment BC , construct from A a segment congruent to BC .*

The current production theorem is a directed strengthening of this classical statement: it asks for the copy on an already chosen nondegenerate ray AE . Its declaration is

```
theorem euclid_proposition_2
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A E B C : Geo.Point)
  (hAE : A ≠ E)
  (hBC : B ≠ C) :
  ∃ D : Geo.Point,
    HilbertSameRay Geo A E D ∧
```

```
Geo.Congruent A D B C := by
exact bookZero_49_layoff Geo A E B C hAE hBC
```

Thus the formal endpoint is stronger in positional information than Euclid’s text: not only is a congruent segment produced from A , but its second endpoint lies on a prescribed ray.

Diagrammatic audit. The older documentation expanded Euclid’s construction into a sequence of snapshots and represented the formal proof mainly by a logical flowchart. Neither choice is optimal under the current audit standard. The successive classical snapshots merely add the same auxiliary objects one at a time; they are not different cases. Conversely, the formal operation has a perfectly clear geometric state and does not need a flowchart in place of a figure.

The audited chapter therefore uses two geometric figures with three panels in total:

$$2 \text{ classical panels} + 1 \text{ formal panel} = 3 \text{ panels in 2 figures.}$$

2.4 Classical Proof

The direct classical construction can be organized as follows.

1. By Postulate 1, join A to the endpoint B of the given segment.
2. By Proposition I.1, construct the equilateral triangle ABD on AB .
3. By Postulate 2, produce DB beyond B and DA beyond A .
4. By Postulate 3, draw the circle with center B and radius BC ; let the produced ray DB meet it at G .
5. Draw the circle with center D and radius DG ; let the produced ray DA meet it at L .

The first circle gives $BG \cong BC$, and the second gives $DG \cong DL$. Because ABD is equilateral, $DB \cong DA$. The collinear orders on the two produced sides identify the wholes as

$$DG = DB + BG, \quad DL = DA + AL.$$

Subtracting equal parts from equal wholes gives $BG \cong AL$. Combining this with $BG \cong BC$ yields $AL \cong BC$.

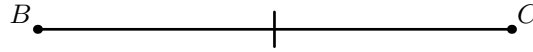
The historical dependency is therefore not merely $I.1 \rightarrow I.2$. At construction level it also uses Postulates 1–3, circle–line incidence/intersection, and the Common Notions for equality and subtraction.

2.5 Hilbert Reconstruction

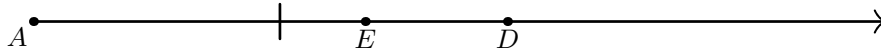
The formal theorem begins from a different interface. Besides the segment BC , a ray AE is already supplied. The goal is to lay the segment off on that ray.

At the Book I level there is no auxiliary point construction at all:

$$AD \cong BC$$



D lies on the same ray from A as E



chosen ray AE

Figure 2.2: The stable formal configuration for Proposition I.2. The chosen ray AE and the source segment BC are inputs; Hilbert III.1, exposed through Book Zero #49, produces D on the same ray with $AD \cong BC$. The drawn relative order of E and D is only a representative realization; the theorem asserts same-ray position, not a specific betweenness relation between them.

```
exact bookZero_49_layoff Geo A E B C hAE hBC
```

The substantive question is therefore what Book Zero #49 contains.

2.6 Book Zero #49 and the Hidden Layer

The current source defines the layoff theorem as follows:

```
theorem bookZero_49_layoff
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D : Geo.Point)
  (hAB : A ≠ B)
  (_hCD : C ≠ D) :
  ∃ X : Geo.Point,
    HilbertSameRay Geo A B X ∧
    Geo.Congruent A X C D := by

  rcases HilbertCongruence.segment_construction
    (Geo := Geo)
    C D
    A B
    hAB with
    ⟨X, hRayX, hAXCD⟩

  exact ⟨X, hRayX, hAXCD⟩
```

Thus Book Zero #49 is itself only a packaging layer. The actual existence principle is the field

HilbertCongruence.segment_construction, Hilbert III.1:

```
segment_construction :
  ∀ A B O R : Geo.Point,
    O ≠ R →
    ∃ X : Geo.Point,
      HilbertSameRay Geo O R X ∧
      Geo.Congruent O X A B
```

After substituting

$$O = A, \quad R = E, \quad (A, B)_{\text{source}} = (B, C),$$

this is exactly the conclusion of Proposition I.2.

This is a sharper architectural statement than “I.2 uses Book Zero”. The complete formal geometry is already present in Hilbert III.1; Book Zero only gives it the project-level name “layoff”, and the Book I theorem only re-exports that operation under Euclid’s proposition number.

2.7 API Audit: the Source-Segment Nondegeneracy Hypothesis

The source inspection also exposes a small but real interface redundancy. The public theorem assumes

$$hBC : B \neq C.$$

Book Zero #49 receives this as its second hypothesis, but names it `_hCD`; it is not used in the proof. The primitive `segment_construction` currently requires only that the prescribed ray be nondegenerate, $O \neq R$; it does not require the segment being copied to have distinct endpoints.

Consequently, in the current implementation the formal dependency is

$$hAE : A \neq E \text{ is consumed, whereas } hBC : B \neq C \text{ is API-level only.}$$

This audit does not change the production theorem. The extra hypothesis may be retained deliberately to match Hilbert’s traditional convention that a “segment” is proper, but it should not be described as logically necessary for the present proof term. If the API is simplified later, both `bookZero_49_layoff` and `euclid_proposition_2` are natural places to reconsider this assumption together.

2.8 Formal Proof Architecture

The exact formal chain is

$$\begin{array}{c}
 hAE : A \neq E \quad \text{chosen ray } AE \quad \text{source endpoints } B, C \\
 \downarrow \\
 \text{HilbertCongruence.segment_construction} \quad (\text{Hilbert III.1}) \\
 \downarrow \\
 \exists D, \quad \text{HilbertSameRay Geo A E D}, \quad AD \cong BC \\
 \downarrow \\
 \text{bookZero_49_layoff} \\
 \downarrow \\
 \text{euclid_proposition_2.}
 \end{array}$$

There is no branch, no order trichotomy, and no proposition-local helper. There is also no hidden derived theorem below III.1: the final existence step is axiomatic in the Hilbert congruence class.

2.9 Relationship with Earlier Propositions

2.9.1 Proposition I.1

Classically, I.1 is the decisive earlier theorem. The equilateral triangle ABD supplies the equality $DB \cong DA$, which is later subtracted from the equal radii $DG \cong DL$. Thus I.2 is the first immediate demonstration that the equilateral construction is not isolated: its symmetry becomes a tool for transporting a length.

Formally, there is no dependency on `euclid_proposition_1`. Hilbert III.1 already provides arbitrary segment layoff, so the classical auxiliary triangle disappears entirely from the Lean proof.

2.9.2 Transition to Proposition I.3

I.2 provides free transport of a segment from a prescribed endpoint. I.3 adds an order condition: the copied segment must be cut from inside a given longer segment. This explains the classical progression

I.2 : copy a length $\longrightarrow$ I.3 : cut that length from a greater segment.

In the current formal language, however, I.3 is compressed even further: its hypothesis `HilbertSegmentLess` already stores the required interior witness.

2.10 Hilbert and Book Zero Components Used

2.10.1 Hilbert III.1: Segment Construction

This is the actual mathematical engine of the formal proposition. It lays off any source endpoint pair on a prescribed nondegenerate ray and returns both the same-ray condition and the segment congruence.

2.10.2 Book Zero #49: Layoff

The Book Zero theorem introduces no new geometry beyond III.1. Its role is terminological and architectural: it exposes the primitive construction in a form convenient for later synthetic proofs.

2.10.3 Same-Ray Structure

The positional conclusion

`HilbertSameRay Geo A E D`

is stronger than collinearity. It fixes the direction from A in which the copy is laid off. This is precisely why the formal statement is a directed strengthening of Euclid's original endpoint-only formulation.

2.10.4 Segment Congruence

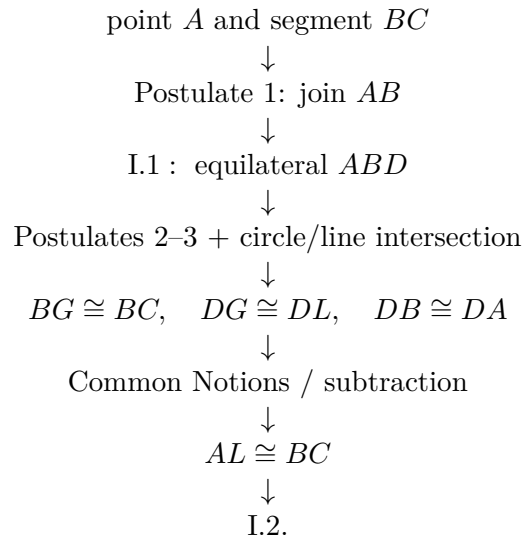
The metric conclusion is purely synthetic:

$$AD \cong BC.$$

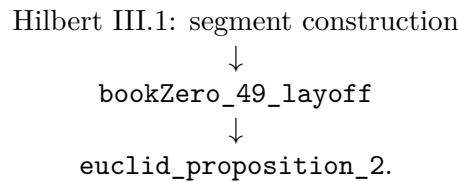
No numerical length function, coordinates, or metric arithmetic is used.

2.11 Dependency Summary

The classical dependency can be summarized as



The formal dependency is much shorter:



No new geometric axiom is introduced locally by `Proposition02.lean`. The Book I theorem is a thin interface over a foundational congruence operation.

2.12 Conclusion

Proposition I.2 is one of the clearest early examples of the difference between Euclid’s historical construction order and the Hilbert foundation. For Euclid, transporting a length to a prescribed endpoint requires the full auxiliary mechanism of I.1, produced straight lines, two circles, and subtraction of equal segments. In the current formal development the same operation has already become Hilbert III.1.

The audit therefore identifies three distinct layers:

Euclid’s derived construction, Book Zero layoff interface, Hilbert III.1 primitive.

The public Book I theorem is short not because the geometry has been ignored, but because the relevant construction has been moved to the axiomatic foundation. The unused source-segment nondegeneracy hypothesis also provides a concrete API-cleanliness point for possible later refactoring.

Chapter 3

Proposition I.3

3.1 Introduction

Proposition I.3 asks for a segment-copying construction with an order constraint. Given a greater segment AB and a shorter segment CD , one must find an interior point E of AB such that

$$AE \cong CD.$$

Classically, Euclid solves the problem by combining Proposition I.2 with a circle centered at A . The present formal reconstruction has a very different architecture. The hypothesis is not a primitive numerical or order-theoretic inequality. It is the witness-based predicate `HilbertSegmentLess`. Its definition already asserts the existence of a point inside the greater segment cutting off a congruent initial part.

Thus I.3 is a particularly clean example of a change of mathematical language. Euclid proves the existence of the desired point by a geometric construction; Book Zero defines strict segment comparison by the existence of exactly such a point. At the public Book I level, the Lean theorem only unpacks the witness and reverses the orientation of segment congruence.

3.2 Classical Configuration

Let AB be the greater segment and CD the shorter one. Euclid first uses Proposition I.2 to construct from A an auxiliary segment AF satisfying

$$AF \cong CD.$$

A circle centered at A through F then meets the segment AB at E . Since AE and AF are radii of the same circle,

$$AE \cong AF,$$

and hence

$$AE \cong CD.$$

The inequality $CD < AB$ is what places the circle intersection on the open segment from A to B , rather than beyond B . The final positional statement is therefore

$$A - E - B.$$

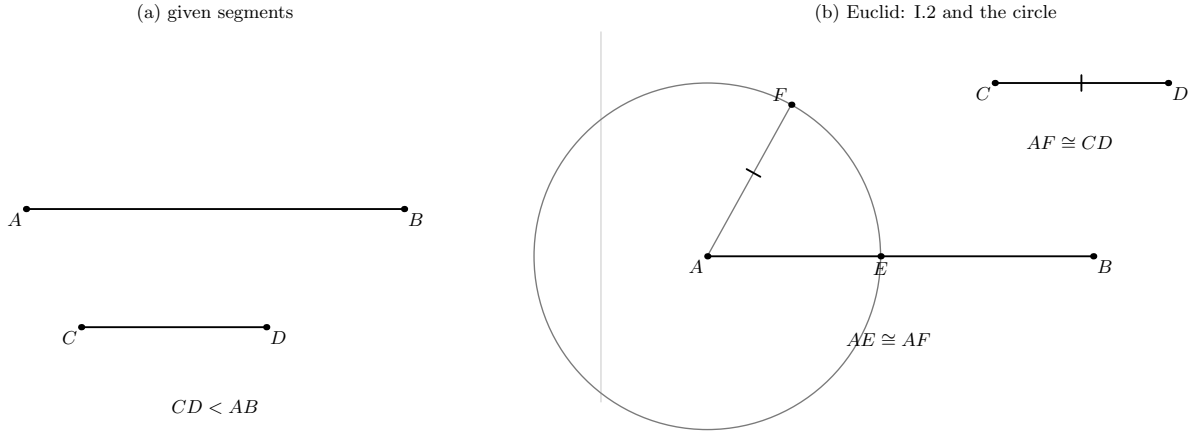


Figure 3.1: Euclid's construction for Proposition I.3. (a) The input consists of the greater segment AB and the shorter segment CD . (b) Proposition I.2 supplies $AF \cong CD$; the circle centered at A through F cuts AB at E , so $AE \cong AF \cong CD$.

3.3 Statement

Theorem 3 (Euclid I.3). *Let AB and CD be segments, and suppose CD is strictly shorter than AB . Then there exists a point E such that*

$$A - E - B$$

and

$$AE \cong CD.$$

The current production declaration is:

```

theorem euclid_proposition_3
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D : Geo.Point)
  (hCDAB : HilbertSegmentLess Geo C D A B) :
  ∃ E : Geo.Point,
    Geo.Between A E B ∧
    Geo.Congruent A E C D := by

rcases hCDAB with ⟨E, hAEB, hCDAE⟩

exact ⟨E, hAEB,
  hilbert_congruent_symmetry Geo C D A E hCDAE⟩

```

Diagrammatic audit. The preserved chapter displayed the final segment configuration and a logical flowchart. The flowchart is unnecessary once the definition of `HilbertSegmentLess` is inspected, while the final configuration alone does not show Euclid's actual construction.

The revised chapter therefore uses two geometric figures with three panels in total:

$$2 \text{ classical panels} + 1 \text{ formal panel} = 3 \text{ panels in 2 figures.}$$

The formal panel is intentionally simple: there is no hidden construction below the predicate at this point, because the predicate itself is defined by the required geometric witness.

3.4 Classical Proof

Start with the two unequal segments, with AB the greater. By Proposition I.2, construct from the point A a segment AF congruent to the given shorter segment CD :

$$AF \cong CD.$$

By Postulate 3, draw the circle with center A and radius AF . Let this circle meet the greater segment AB at E . Because E and F lie on the same circle centered at A ,

$$AE \cong AF.$$

Together with $AF \cong CD$, Common Notion 1 gives

$$AE \cong CD.$$

Since the radius represented by CD is shorter than AB , the intersection point lies between A and B . Hence

$$A - E - B.$$

This is the required construction.

The direct classical dependency is therefore

$$\text{I.2} \quad + \quad \text{Postulate 3 (circle)} \quad + \quad \text{Common Notion 1.}$$

Proposition I.1 is not a direct dependency of I.3.

3.5 Formal Meaning of Segment Inequality

The essential formal fact is the definition

```
def HilbertSegmentLess
  (Geo : Geometry.Geo)
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D : Geo.Point) : Prop :=
  ∃ P : Geo.Point,
    Geo.Between C P D ∧
    Geo.Congruent A B C P
```

Thus

$$\text{HilbertSegmentLess Geo C D A B}$$

expands to

$$\exists E, \quad A - E - B, \quad CD \cong AE.$$

This is already the full existential content of Proposition I.3, except that the congruence is oriented as $CD \cong AE$ rather than $AE \cong CD$.

This point is foundationally important. In the formal language used here, “ $CD < AB$ ” is not merely an abstract comparison from which the cut-off point must later be constructed. The comparison is defined synthetically by the existence of an initial subsegment of AB congruent to CD .

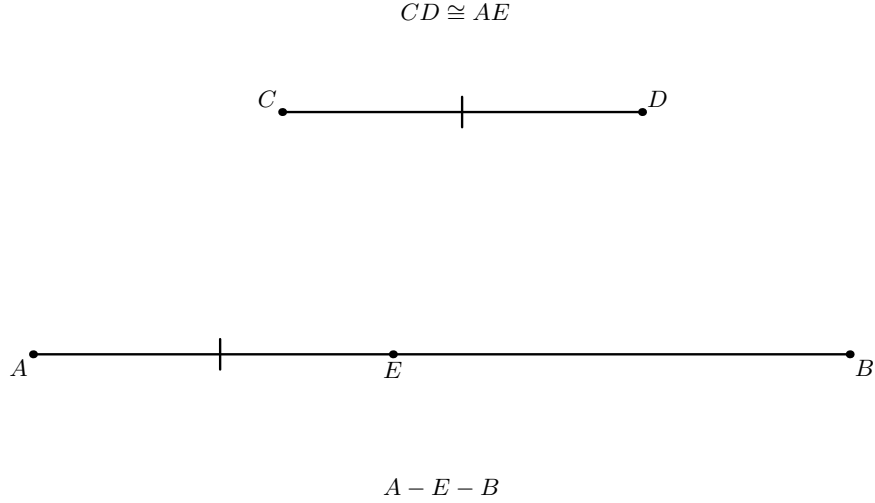


Figure 3.2: The geometric witness contained in `HilbertSegmentLess Geo C D A B`. The predicate already supplies $A - E - B$ and $CD \cong AE$; the public theorem only reverses the congruence orientation.

3.6 Formal Proof Architecture

The complete formal chain is

```

hCDAB : HilbertSegmentLess Geo C D A B
      ↓
  unfold/extract the witness
      ↓
  E,  A - E - B,  CD ≅ AE
      ↓
hilbert_congruent_symmetry
      ↓
  A - E - B,  AE ≅ CD
      ↓
  euclid_proposition_3.

```

There is no theorem call corresponding to Euclid's use of I.2, and there is no formal circle construction in the proof. Nor is there a deeper hidden proof engine beneath `HilbertSegmentLess`: it is a definition, not a theorem whose proof could conceal further geometry.

The only nontrivial operation after witness extraction is symmetry of segment congruence:

```
hilbert_congruent_symmetry Geo C D A E hCDAE
```

which changes

$$CD \cong AE$$

to

$$AE \cong CD.$$

3.7 Relationship with Earlier Propositions

3.7.1 Proposition I.2

Proposition I.2 is the decisive earlier proposition in Euclid’s proof. It allows the length of the shorter segment CD to be reproduced from the specified point A , producing the auxiliary radius AF .

The formal proof of I.3 does not call `euclid_proposition_2`. The constructional content that I.2 supplies classically has already been absorbed into the witness-based language of strict segment comparison.

3.7.2 Proposition I.1

I.1 is not a direct dependency of Euclid’s I.3. Any appearance of the equilateral-triangle construction occurs inside Euclid’s proof of I.2, not as a new step of I.3 itself. The classical dependency should therefore be recorded as

$$\text{I.2} \longrightarrow \text{I.3},$$

not as a direct edge from I.1 to I.3.

3.7.3 Transition to Proposition I.4

I.3 closes the initial sequence of explicit segment constructions. I.4 then changes character completely: it is a rigidity theorem for triangles rather than a point-construction theorem. In the formal development the contrast is even sharper, because I.3 is witness extraction from a definition, while I.4 is a wrapper around the developed Hilbert SAS theorem.

3.8 Hilbert and Book Zero Components Used

3.8.1 HilbertSegmentLess

This definition is the central object of the formal proposition:

$$AB < CD \iff \exists P (C - P - D \wedge AB \cong CP).$$

It provides a synthetic notion of strict segment comparison without introducing numerical lengths.

For I.3 the arguments are instantiated in the reversed role

$$CD < AB,$$

so the witness lies inside AB and cuts off a segment congruent to CD .

3.8.2 Segment Congruence Symmetry

The definition returns

$$CD \cong AE,$$

while the proposition states

$$AE \cong CD.$$

The theorem `hilbert_congruent_symmetry` is therefore the only reusable theorem called after the witness is unpacked.

3.8.3 Betweenness

The open-segment condition is represented directly by `Geo.Between`. Consequently the phrase “cut off from the greater straight line” is not inferred from a picture: the output explicitly contains

`Geo.Between A E B.`

3.9 Formalization Notes

Proposition I.3 is one of the clearest examples in Book I where theorem size can be misleading. The Lean proof is extremely short, but not because a circle construction has been compressed into automation. Rather, the input predicate has been chosen so that its witness is exactly the output point required by the proposition.

This is a legitimate and useful synthetic design choice. The relation `HilbertSegmentLess` is not numerical: it defines strict comparison by internal geometric configuration. It therefore supports later proofs by turning inequalities immediately into betweenness and congruence data.

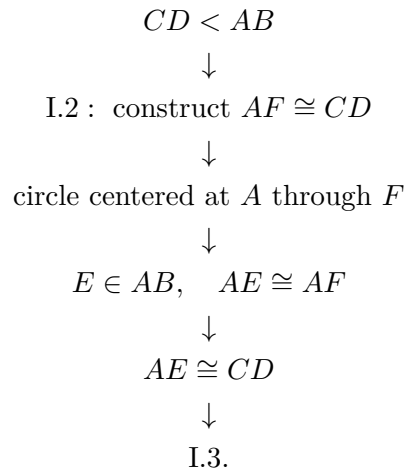
At the same time, documentation must keep the two layers distinct:

- Euclid constructs the witness using I.2 and a circle;
- Book Zero uses the existence of such a witness as the definition of strict segment comparison;
- I.3 then exposes that witness as a Book I theorem.

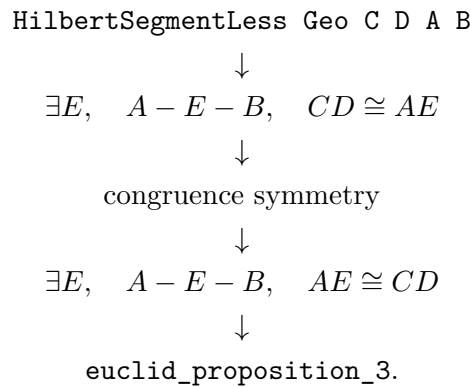
The diagrammatic audit reflects this distinction. The classical figure shows the construction, while the formal figure shows only the geometric state encoded by the comparison predicate. A logical flowchart would add no new geometric information.

3.10 Dependency Summary

The classical architecture is



The formal architecture is much shorter:



No local axiom is introduced by `euclid_proposition_3`; the theorem uses the Book Zero definition of strict segment comparison and the existing congruence-symmetry theorem.

3.11 Conclusion

Proposition I.3 completes Euclid’s first sequence of segment constructions. Classically, it reuses I.2 and adds one circle to cut a prescribed length from a greater segment. Formally, the same geometric endpoint has already been built into the definition of strict segment comparison.

The proposition therefore illustrates a recurring theme of the reconstruction: a classical construction may reappear not as a long theorem proof but as the witness structure of a carefully chosen synthetic predicate. The short Lean proof is mathematically informative precisely because the underlying notion `HilbertSegmentLess` has made the relevant geometry explicit in its definition.

Chapter 4

Proposition I.4

4.1 Introduction

Proposition I.4 is Euclid’s first triangle-rigidity theorem. Given two nondegenerate triangles with two corresponding sides and the included angle congruent, it proves the remaining side and the remaining corresponding angles congruent. In modern notation the input is

$$AB \cong DE, \quad AC \cong DF, \quad \angle BAC \cong \angle EDF,$$

and the new conclusions are

$$BC \cong EF, \quad \angle ABC \cong \angle DEF, \quad \angle ACB \cong \angle DFE.$$

Historically, Euclid proves this by superposition. The present Hilbert reconstruction cannot use the physical-language operation of moving one triangle onto the other. At the public Book I level the Lean theorem is only a wrapper around `TriangleCongruentFromSAS`. The audit therefore has to descend below that name.

That descent reveals the main correction to the preserved documentation. The compatibility theorem is itself a wrapper for `SAS`, and the project’s full Hilbert Theorem 12 is not merely one primitive SAS axiom call. Hilbert III.5 directly supplies the two remaining angle congruences, but the third side is derived by a separate construction-and-uniqueness argument. The latter constructs a segment witness on a ray, uses SAS once more, invokes the uniqueness clause of angle construction III.4, and finally uses incidence uniqueness to identify the witness with the target vertex.

4.2 Classical Configuration and Superposition

The classical theorem has no auxiliary construction in the sense of Propositions I.1–I.3. Its complete input consists of two triangles with the SAS data displayed in panel (a) of Figure 4.1. Euclid then argues by imagining one triangle applied to the other.

This is the only genuinely classical geometric state needed for the theorem, plus the conceptual coincidence state created by superposition. There is no Euclidean point construction to expand into further snapshots.

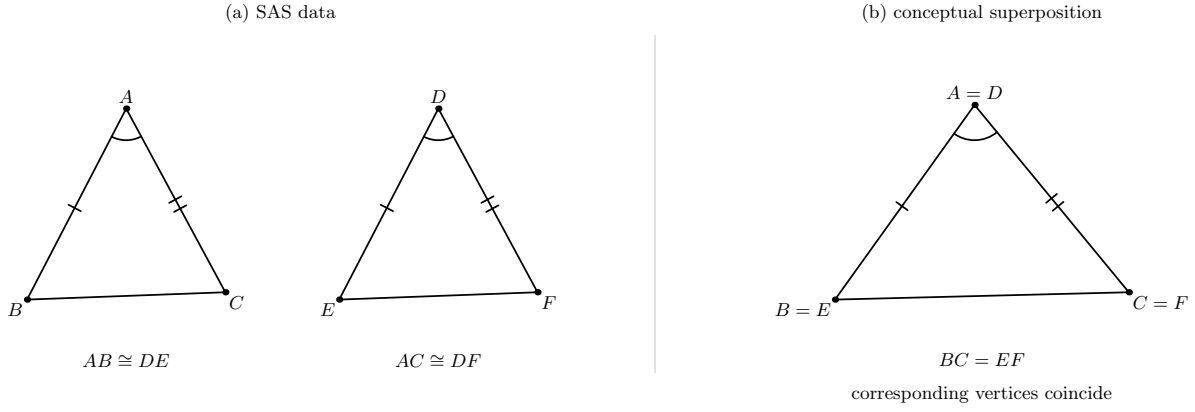


Figure 4.1: Euclid’s geometric data and the conceptual superposition step. (a) The two triangles satisfy the two side congruences and the included angle congruence. (b) Under Euclid’s superposition, A is placed on D and AB on DE ; the hypotheses force $B = E$ and then $C = F$. The second panel records the conclusion of the conceptual placement, not a Hilbert construction.

4.3 Statement

Theorem 4 (Euclid I.4). *Let ABC and DEF be nondegenerate triangles. Assume*

$$AB \cong DE, \quad AC \cong DF, \quad \angle BAC \cong \angle EDF.$$

Then

$$BC \cong EF,$$

$$\angle ABC \cong \angle DEF, \quad \angle ACB \cong \angle DFE.$$

Equivalently, all corresponding side and angle data are collected in the project structure $\text{TriangleCongruenceResult}$

The current production declaration is:

```

theorem euclid_proposition_4
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D E F : Geo.Point)
  (hABC : not Collinear Geo A B C)
  (hDEF : not Collinear Geo D E F)
  (hAB : Geo.Congruent A B D E)
  (hAngle : Geo.AngleCongruent B A C E D F)
  (hAC : Geo.Congruent A C D F) :
  TriangleCongruenceResult Geo A B C D E F := by
exact
  TriangleCongruentFromSAS
    Geo A B C D E F
    hABC hDEF hAB hAngle hAC

```

Diagrammatic audit. The preserved chapter used one ordinary triangle picture and one logical flowchart. The triangle picture did not visualize Euclid’s distinctive superposition step, while the flowchart stopped at the compatibility theorem and therefore hid the actual geometric engine below it.

The revised audit replaces those two items by two geometric figures with four panels in total:

$$2 \text{ classical panels} + 2 \text{ formal panels} = 4 \text{ panels in 2 figures.}$$

No panel is introduced for endpoint reversals, noncollinearity permutations, or structure packaging, because those operations do not create a new geometric state.

4.4 Classical Proof

Euclid begins with the two triangles ABC and DEF . Conceive triangle ABC as placed on triangle DEF so that A falls on D and the side AB lies along DE . Since

$$AB \cong DE,$$

the endpoint B falls on E .

The included-angle hypothesis

$$\angle BAC \cong \angle EDF$$

then makes the ray AC fall along the ray DF . Since

$$AC \cong DF,$$

the endpoint C falls on F .

Thus the third side BC coincides with EF , so

$$BC \cong EF.$$

The remaining angles likewise coincide pairwise, yielding

$$\angle ABC \cong \angle DEF, \quad \angle ACB \cong \angle DFE.$$

The classical proof therefore does not depend directly on Propositions I.1–I.3. Its distinctive proof principle is superposition together with the common notions governing coincidence and equality. Later propositions, starting immediately with I.5, reuse I.4 as a theorem.

4.5 Formal Proof Architecture

At the public level the chain is short:

```

euclid_proposition_4
  ↓
TriangleCongruentFromSAS
  ↓
SAS.

```

The second arrow matters. The compatibility theorem contains no mathematical argument of its own:

```

theorem TriangleCongruentFromSAS ... := by
  exact SAS Geo A B C D E F

```

The theorem `SAS`, described in the source as Hilbert’s Theorem 12, splits the work into two components:

```

have hAngles :=
  hilbert_sas_remaining_angles
    Geo A B C D E F hABC hDEF hAB hAC hAngleA

have hNeeded :=
  hilbert_sas_third_side_and_angle
    Geo A B C D E F hABC hDEF hAB hAC hAngleA

```

The first component supplies the remaining angles. The second supplies the third side and one of those angles. The final `TriangleCongruenceResult` is only the packaging of these facts.

Thus the deeper dependency is

$$\begin{array}{c}
 AB \cong DE, AC \cong DF, \angle BAC \cong \angle EDF \\
 \downarrow \\
 \text{hilbert_sas_remaining_angles} \\
 \downarrow \\
 \angle ABC \cong \angle DEF, \quad \angle ACB \cong \angle DFE
 \end{array}$$

together with the separate branch

$$\begin{array}{c}
 \text{hilbert_sas_third_side_and_angle} \\
 \downarrow \\
 \text{construct } X \in \text{ray}(EF), EX \cong BC \\
 \downarrow \\
 \text{SAS gives the required angle at } D \\
 \downarrow \\
 \text{III.4 angle uniqueness forces } DX \text{ and } DF \text{ to be one ray} \\
 \downarrow \\
 X, F \in EF \text{ and } X, F \in \text{ray}(D\cdot) \\
 \downarrow \\
 X = F \\
 \downarrow \\
 BC \cong EF.
 \end{array}$$

4.6 The Hidden Hilbert SAS Engine

4.6.1 The angle consequences are primitive III.5 consequences

The theorem `hilbert_sas_remaining_angles` applies the primitive congruence axiom `HilbertCongruence`. `sas` twice. The first application uses the triangles in their stated order and returns the angle at B/E . The second swaps B and C in both triangles and returns the angle at C/F . No new points are constructed in this stage.

This distinction is important: in the present architecture the primitive III.5 mechanism gives the remaining *angles*, not the complete `TriangleCongruenceResult` by itself.

4.6.2 The third side requires an auxiliary segment witness

To derive

$$BC \cong EF,$$

`hilbert_sas_third_side_and_angle` invokes Hilbert segment construction. Specialized to I.4, it chooses a point X on the ray from E through F such that

$$EX \cong BC.$$

This is the state shown in panel (a) of Figure 4.2. The original triangle DEF is shown in grey; X is allowed to be a priori different from F . The dashed segment DX is the candidate second ray whose direction will be identified by the subsequent angle argument.

The angle at E between ED and EX is the same angle as the angle between ED and EF , because X and F lie on the same ray from E . The already obtained angle congruence at B/E therefore permits another SAS application, now to the triangles BAC and EDX . It yields

$$\angle BAC \cong \angle EDX.$$

Together with the original hypothesis

$$\angle BAC \cong \angle EDF,$$

the two candidate rays DX and DF make the same prescribed angle with DE on the selected side.

4.6.3 Angle uniqueness and incidence identify the endpoint

The proof now invokes Hilbert's angle-construction axiom III.4. Internally it constructs a witness Z for the unique ray from D making the prescribed angle with DE . The uniqueness field is applied both to F and to X , showing that

$$\text{ray}(DZ) = \text{ray}(DF) = \text{ray}(DX).$$

But X and F already lie on the same line through E . If $X \neq F$, the line through X, F would be simultaneously the line EF and the line through D and the common ray, forcing D onto EF . This contradicts the nondegeneracy of triangle DEF . Hence

$$X = F.$$

Substituting into $EX \cong BC$ gives the required third-side congruence

$$BC \cong EF.$$

The witness Z has no role in Euclid's classical presentation. It is shown only in the formal panel because it is the carrier of the unique ray returned by the III.4 API. By contrast, no separate figure is created for the two primitive III.5 angle calls: they enrich the original pair of triangles but do not create a new geometric configuration.

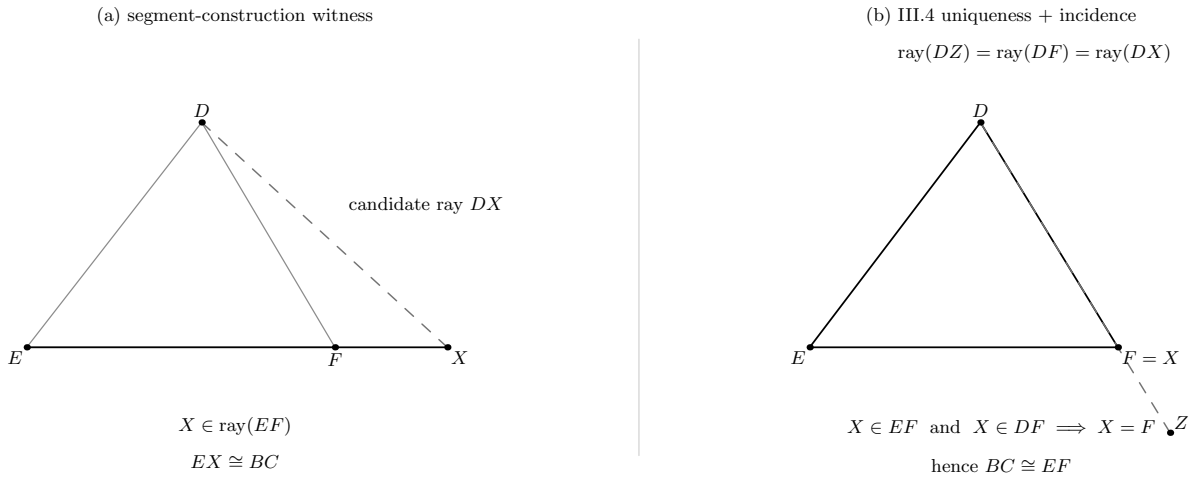


Figure 4.2: The derived third-side mechanism inside Hilbert Theorem 12. (a) Segment construction places X on the ray EF with $EX \cong BC$. The angle equality later derived for DX is deliberately not drawn while X is still displayed as distinct from F : drawing equal rays already would encode the conclusion. (b) Hilbert III.4 supplies the unique ray represented by DZ ; both F and X lie on it. Together with their collinearity on EF , incidence uniqueness forces $X = F$, hence $BC \cong EF$.

4.7 Lean Formalization

The production theorem itself is a genuine thin wrapper:

```
exact
  TriangleCongruentFromSAS
    Geo
    A B C
    D E F
    hABC
    hDEF
    hAB
    hAngle
    hAC
```

One level lower, the compatibility declaration is also a wrapper:

```
exact SAS Geo A B C D E F
```

The first substantial theorem is therefore **SAS**. Its source comment explicitly records the architecture: only the third side and the angle at C require the derived construction argument; the remaining fields are hypotheses or direct consequences of III.5.

Inside `hilbert_sas_third_side_and_angle`, the decisive witness is created by

```
rcases HilbertCongruence.segment_construction
  (Geo := Geo) B C E F hBC hEF with
  <X, hRay, hBCX>
```

where the displayed variable names are schematic for the specialization $A' = D$, $B' = E$, $C' = F$. The actual source then uses line incidence, same-side transport, SAS, and `HilbertCongruence`.

`angle_construction` to prove $X = F$.

There is no proposition-local helper theorem, no case split, and no new axiom inside `Proposition04.lean`. The complexity is entirely below the Book I wrapper.

4.8 Relationship with Earlier Propositions

4.8.1 No direct dependency on I.1–I.3

The classical proof of I.4 does not invoke Propositions I.1, I.2, or I.3. Those propositions establish early construction machinery, whereas I.4 is a rigidity theorem proved by superposition. They are adjacent in Book I but not direct proposition-level predecessors in the I.4 proof DAG.

The formal theorem likewise does not call the Lean theorems for I.1–I.3. Segment construction used inside Hilbert Theorem 12 belongs to the Hilbert congruence layer, not to Euclid I.2 or I.3 as a Book I dependency.

4.8.2 Transition to I.5

Proposition I.5 is the first immediate consumer of this rigidity principle. Its classical proof builds auxiliary triangles and applies I.4 twice. In the formal library the same SAS infrastructure is already available below the Book I layer and is reused by the Hilbert theorems that I.5 calls.

Thus I.4 is a structural turning point:

$$\text{I.1–I.3: constructions} \quad \longrightarrow \quad \text{I.4: triangle rigidity.}$$

4.9 Hilbert Components Used

4.9.1 Hilbert III.5: primitive SAS angle rigidity

The primitive method `HilbertCongruence.sas` is applied to obtain corresponding angle congruence under the two-side-and-included-angle hypotheses. A second application after swapping the two non-apex vertices supplies the other remaining angle.

4.9.2 Hilbert III.1: segment construction

The third side is not assumed to follow automatically from the primitive SAS angle statement. A point X is laid off on the ray EF with $EX \cong BC$. This creates the auxiliary triangle used to force the endpoint.

4.9.3 Hilbert III.4: uniqueness of angle construction

After the second SAS comparison, DX and DF satisfy the same prescribed angle condition on the same side of DE . The uniqueness clause of III.4 identifies their rays.

4.9.4 Incidence uniqueness

Once X, F lie both on the line EF and on the common ray from D , the uniqueness of the line through two distinct points rules out $X \neq F$. This incidence step is what converts ray uniqueness into equality of the constructed endpoint.

No Euclidean parallel axiom, numerical length, coordinates, or numerical angle measure enters this chain.

4.10 Formalization Notes

The principal audit correction is foundational rather than cosmetic. The old chapter correctly observed that the public theorem is short, but it collapsed three distinct levels into one phrase, “Hilbert SAS”:

primitive III.5 $\neq$ derived Hilbert Theorem 12 $\neq$ Book I wrapper.

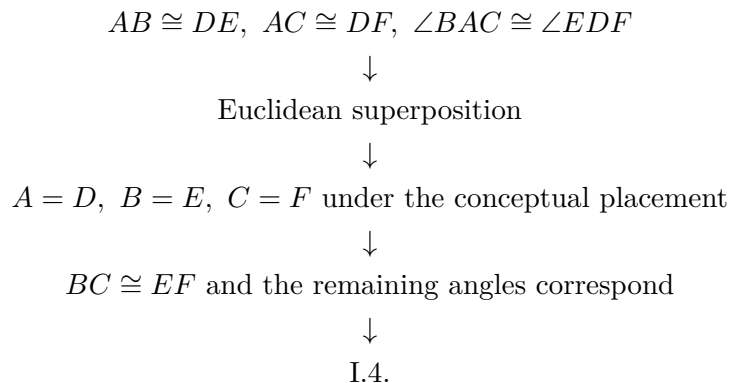
The current code separates them cleanly:

1. `HilbertCongruence.sas` supplies primitive angle rigidity;
2. `hilbert_sas_third_side_and_angle` derives the third side by construction and uniqueness;
3. `SAS` packages the full Hilbert Theorem 12 result;
4. `TriangleCongruentFromSAS` is a compatibility name;
5. `euclid_proposition_4` exposes this infrastructure as Euclid I.4.

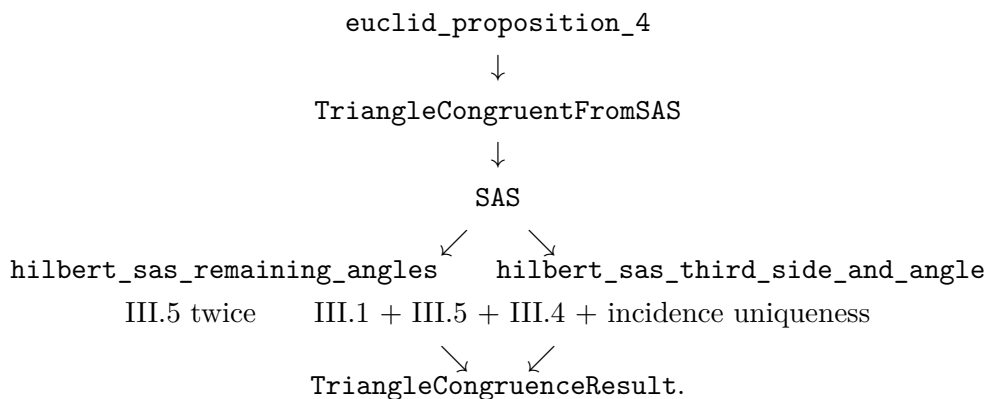
This explains why the Book I proof can consist of a single theorem call without making the underlying geometry trivial. The proof burden has moved below the proposition boundary, but it has not disappeared.

4.11 Dependency Summary

The classical path is



The formal path is



The audited diagram set mirrors exactly this division: the first figure records Euclid’s superposition logic, while the second records the hidden construction required for the third side in the formal Hilbert theorem.

4.12 Conclusion

Proposition I.4 is the first major point at which Euclid’s order of exposition and the Hilbert/Lean foundational order diverge. For Euclid, SAS is a theorem proved by superposition. In the reconstruction, primitive SAS angle rigidity belongs to the congruence axioms, while the complete theorem needed by Book I is Hilbert Theorem 12, derived above those axioms.

The current production theorem is therefore short for a precise reason. It is a wrapper around a reusable theorem whose nontrivial third-side proof has already been carried out by segment construction, angle uniqueness, and incidence. The revised figures make both sides visible: Euclid’s conceptual superposition and the actual synthetic construction hidden below the Lean compatibility name.

Chapter 5

Proposition I.5

5.1 Introduction

Proposition I.5 is Euclid’s first systematic theorem about isosceles triangles. From

$$AB \cong AC$$

it proves both the equality of the interior base angles

$$\angle ABC \cong \angle ACB$$

and, after the equal sides are produced beyond B and C , the equality of the angles below the base,

$$\angle CBD \cong \angle BCE.$$

The classical proof is the celebrated *pons asinorum*. It is not a one-line application of SAS. Euclid extends the two equal sides, introduces two further points F, G , joins them crosswise to the opposite base vertices, and applies Proposition I.4 twice. Common Notion 3 is used first on segments and then on angles.

The current formal reconstruction is organized differently. At the public Book I level, Proposition I.5 merely composes two established theorems: `hilbert_isosceles_base_angles` and `hilbert_adjacent_angles_congruent`. The present audit descends one layer below both calls. The first is a short Pappus-style self-comparison by SAS; the second is Hilbert’s Theorem 14 and contains a genuine auxiliary construction with three copied segments.

5.2 The Classical Configuration

Let ABC be a nondegenerate isosceles triangle with

$$AB \cong AC.$$

The initial data contain no auxiliary points.

5.2.1 Euclid’s auxiliary construction

Produce AB beyond B and AC beyond C . The outer points D, E in the formal statement record these produced rays:

$$A - B - D, \quad A - C - E.$$

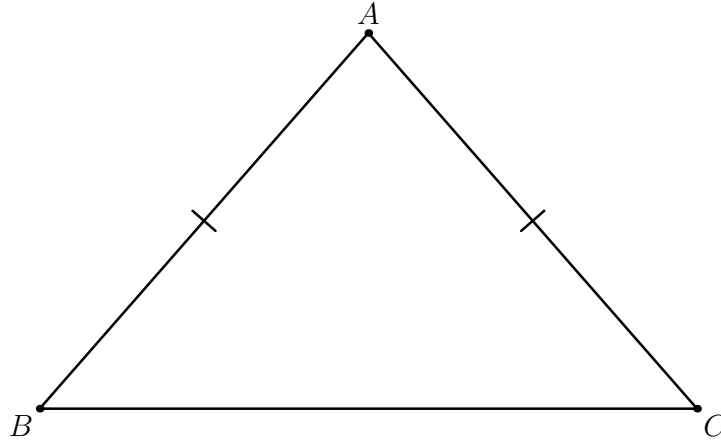


Figure 5.1: The initial isosceles triangle of Proposition I.5, with $AB \cong AC$.

Euclid then chooses an arbitrary point F on the first produced side. On the other produced side Proposition I.3 cuts off AG so that

$$AF \cong AG.$$

Because $AB \cong AC$, Common Notion 3 gives

$$BF \cong CG.$$

Finally join F to C and G to B .

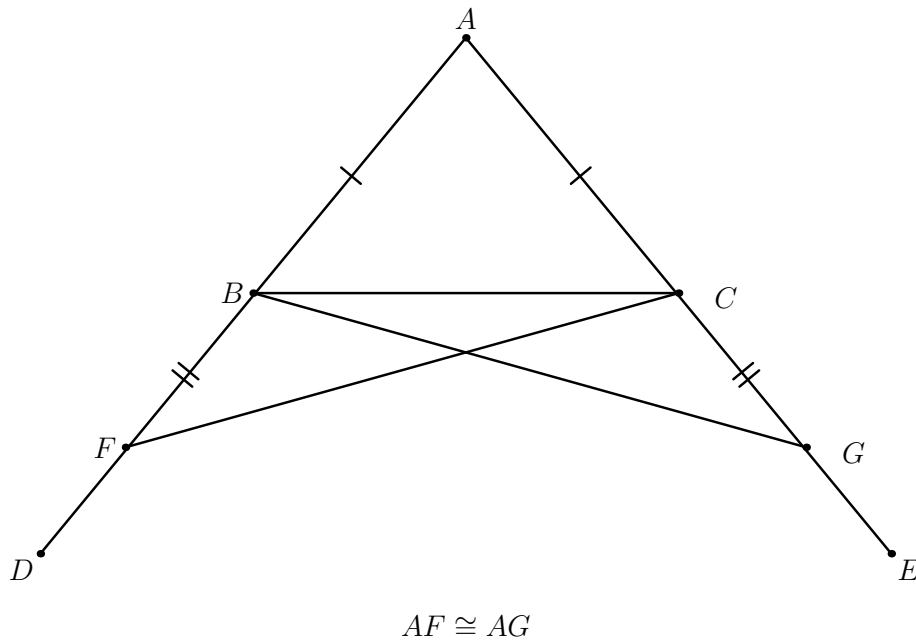


Figure 5.2: The complete classical configuration. The direct I.3 construction is $AF \cong AG$; after subtracting the given $AB \cong AC$, Common Notion 3 gives $BF \cong CG$. The cross-joins FC and GB prepare the two SAS comparisons.

This corrects an ambiguity in the preserved chapter: G is not introduced by directly prescribing $CG \cong BF$. Euclid prescribes $AG \cong AF$; the shorter equality follows by subtraction.

5.3 Statement

Theorem 5 (Euclid I.5). *Let ABC be a nondegenerate triangle satisfying*

$$AB \cong AC.$$

Let the equal sides be produced beyond the base vertices so that

$$A - B - D, \quad A - C - E.$$

Then

$$\angle ABC \cong \angle ACB$$

and

$$\angle CBD \cong \angle BCE.$$

The production Lean declaration has the corresponding pair-valued conclusion:

```
theorem euclid_proposition_5
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D E : Geo.Point)
  (hABC : not Collinear Geo A B C)
  (hABAC : Geo.Congruent A B A C)
  (hABD : Geo.Between A B D)
  (hACE : Geo.Between A C E) :
  Geo.AngleCongruent A B C A C B /\
  Geo.AngleCongruent C B D B C E := by
  ...
```

Diagrammatic audit. The preserved Book1R chapter had already made one good compression: the successive classical snapshots were merged into one complete auxiliary configuration. That decision is retained. The input triangle is also kept, because it separates the theorem data from Euclid’s larger reductio-free construction.

The old formal architecture flowchart is replaced by geometry. The formal figure has two panels. Panel (a) displays the Pappus-style SAS comparison hidden in `hilbert_isosceles_base_angles`: the same geometric triangle is read as ABC and ACB . Panel (b) records the genuinely new construction inside Hilbert Theorem 14: three segments from the second linear pair are copied onto the three rays of the first angle, producing witnesses X, Y, Z .

Thus the audited set contains three figures and four panels.

5.4 Classical Proof

The first SAS comparison is between the triangles

$$\triangle FAC \quad \text{and} \quad \triangle GAB.$$

The two side pairs are

$$FA \cong GA, \quad AC \cong AB,$$

and the included angles at A are the same angle formed by the two produced sides, with the rays read in the opposite order. Proposition I.4 therefore yields, among other consequences,

$$FC \cong GB$$

and the corresponding angle congruences needed for the second comparison.

Now compare

$$\triangle FBC \quad \text{and} \quad \triangle GCB.$$

The side equalities are

$$FB \cong GC, \quad FC \cong GB,$$

and the included angles at F, G are congruent by the first SAS result and same-ray identification along the produced sides. A second application of I.4 gives the equality of the angles under the base. Since F and D lie on the same produced ray from B , and G and E lie on the same produced ray from C , this is precisely

$$\angle CBD \cong \angle BCE.$$

The first SAS comparison also gives equality of two larger angles containing the original base angles. The second SAS comparison gives equality of the parts cut off from them. Common Notion 3 therefore subtracts equal angles from equal angles and yields

$$\angle ABC \cong \angle ACB.$$

The direct classical dependency chain is therefore

$$\text{I.3} + 2 \times \text{I.4} + \text{Common Notion 3} \longrightarrow \text{I.5}.$$

Proposition I.2 is an indirect dependency through I.3, not an additional proposition-level step in Euclid's I.5 proof.

5.5 Proof Architecture of the Reconstruction

At the public Book I level the proof is exceptionally compact:

$$\begin{array}{c}
 AB \cong AC \\
 \downarrow \\
 \text{hilbert_isosceles_base_angles} \\
 \downarrow \\
 \angle ABC \cong \angle ACB \\
 \downarrow \quad A - B - D, \quad A - C - E \\
 \text{hilbert_adjacent_angles_congruent} \\
 \downarrow \\
 \angle CBD \cong \angle BCE.
 \end{array}$$

The source contains exactly these two substantive calls:

```

have hBase :
  Geo.AngleCongruent A B C A C B :=
  hilbert_isosceles_base_angles
  Geo A B C hABC hABAC

```

```

have hExterior :
  Geo.AngleCongruent C B D B C E :=
  hilbert_adjacent_angles_congruent
  Geo
  A B C D
  A C B E
  hABD
  hACE
  hABC
  hACB
  hBase

```

The only additional local step is the rotated noncollinearity hypothesis A, C, B required by the second theorem.

5.6 The Two Hidden Formal Engines

5.6.1 The isosceles theorem: Pappus-style SAS

The theorem `hilbert_isosceles_base_angles` contains no auxiliary construction. It compares the same triangle in the two vertex orders

$$\triangle ABC \quad \text{and} \quad \triangle ACB.$$

The side data are $AB \cong AC$ and its symmetric form $AC \cong AB$. The included angle at A is congruent to itself after reversing its two rays. A single call to SAS therefore returns the base-angle congruence.

This is the classical Pappus shortcut in formal form. It is mathematically substantive but creates no new point, line, ray, or incidence relation.

5.6.2 Hilbert Theorem 14: adjacent congruent angles

The second theorem is deeper. In its general form it begins with two linear pairs

$$A - O - C, \quad A' - O' - C',$$

and a congruence

$$\angle AOB \cong \angle A'O'B'.$$

It constructs three points on the rays of the first angle:

$$X \in \text{ray}(OA), \quad Y \in \text{ray}(OB), \quad Z \in \text{ray}(OC),$$

with

$$OX \cong O'A', \quad OY \cong O'B', \quad OZ \cong O'C'.$$

The proof then uses the previously derived SAS consequences, transports the betweenness $A - O - C$ to $X - O - Z$, applies segment additivity, performs a second SAS comparison, and finishes with a final SAS in the adjacent orientation. This is Hilbert's neutral Theorem 14.

When Theorem 14 is specialized to Proposition I.5, the first linear pair is $A - B - D$ and the second is $A - C - E$. The public theorem therefore inherits all of this neutral construction even though none of the witnesses appears in the statement of `euclid_proposition_5`.

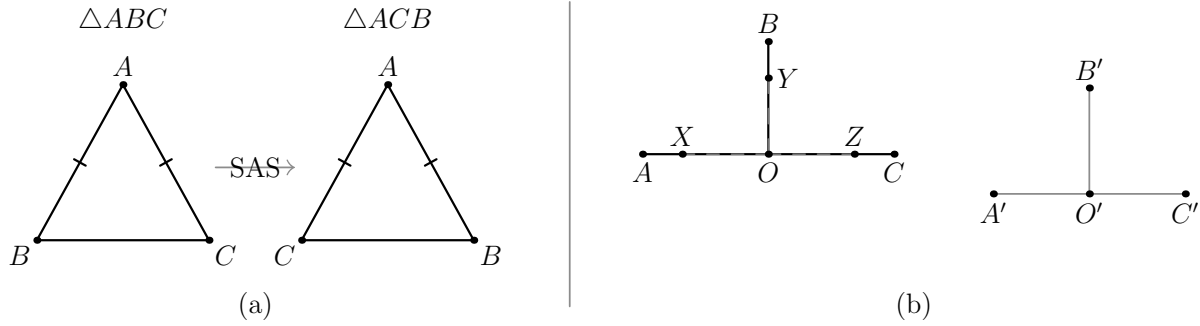


Figure 5.3: Formal engines below Proposition I.5. (a) The base-angle theorem compares the same isosceles triangle as ABC and ACB and closes by SAS. (b) In the general adjacent-angle theorem, $A - O - C$ and $A' - O' - C'$ are linear pairs; the proof copies $O'A', O'B', O'C'$ onto the rays OA, OB, OC , producing X, Y, Z before the SAS/additivity chain.

5.7 Hilbert Reconstruction

The formal reconstruction does not replay Euclid's points F, G or his two applications of Proposition I.4. Instead the two classical outputs have already been separated into reusable Hilbert theorems.

For the interior base angles, triangle symmetry plus SAS is enough. For the angles below the base, the two betweenness hypotheses provide the linear pairs required by Hilbert Theorem 14. Thus the formal architecture is modular in a precise sense: the Book I theorem packages two independently proved neutral results rather than hiding a new proposition-local construction.

No Euclidean parallel axiom occurs in either formal engine. The proof belongs to the incidence/order/congruence layer.

5.8 Lean Formalization

The production theorem first obtains

```
have hBase :
  Geo.AngleCongruent A B C A C B :=
  hilbert_isosceles_base_angles
  Geo A B C hABC hABAC
```

and rotates noncollinearity:

```
have hACB : not Collinear Geo A C B := by
  intro h
  exact hABC (PrimCollinearRotate Geo A C B h)
```

It then invokes Hilbert Theorem 14 through `hilbert_adjacent_angles_congruent` and returns

```
exact And.intro hBase hExterior
```

The current `Proposition05.lean` contains no proposition-local helper theorem, no local axiom declaration, and no geometric case split. This documentation audit read the production source on the repository default branch; it did not run a fresh `lake build` or `#print axioms` check, so

no stronger compilation claim is made here.

5.9 Relationship with Earlier Propositions

5.9.1 Proposition I.3

I.3 is the direct constructional dependency in Euclid's proof. After F is chosen on one produced side, I.3 cuts from the other produced side the segment AG satisfying $AG \cong AF$.

The earlier chapter wording that described G as if $CG \cong BF$ were the prescribed construction conflated the construction with its first derived consequence. The equality $BF \cong CG$ follows only after subtracting the given $AB \cong AC$ from $AF \cong AG$.

5.9.2 Proposition I.4

I.4 is used twice classically. The first SAS comparison creates the cross side equality $FC \cong GB$ and the auxiliary angle congruences. The second uses those results together with $BF \cong CG$ to obtain the angles below the base and the equal angle-parts needed for the final subtraction.

Formally the SAS principle appears lower in the architecture. It is the entire proof engine of `hilbert_isosceles_base_angles`, and it also appears inside the independent proof of Hilbert Theorem 14.

5.9.3 Transition to Proposition I.6

I.5 proves the forward implication

$$AB \cong AC \implies \angle ABC \cong \angle ACB.$$

Proposition I.6 proves the converse. In the present Hilbert architecture the two Book I propositions are therefore short endpoints of different general congruence results rather than mutually dependent formal proofs.

5.10 Hilbert Components Used

5.10.1 `hilbert_isosceles_base_angles`

This is Hilbert Theorem 11 in the current interface. Its proof is the Pappus-style SAS comparison of ABC with ACB .

5.10.2 `hilbert_adjacent_angles_congruent`

This is Hilbert Theorem 14. Its source comment records the mathematical engine explicitly: three points are constructed on the rays of the first angle; Theorem 12 is used twice, segment additivity III.3 once, and the final rigidity step is SAS III.5.

5.10.3 Order and same-ray transport

The relations $A - B - D$ and $A - C - E$ are not decorative. They provide the two linear pairs required by Theorem 14 and allow angle representatives to be replaced along the corresponding rays without changing the angles.

5.11 Dependency Summary

The classical and formal DAGs are genuinely different.

Classical Euclid

```

-----
AB ~= AC
  |
produce AB and AC
  |
choose F; I.3 constructs G with AF ~= AG
  |
CN3: BF ~= CG
  |
I.4 on FAC and GAB
  |
FC ~= GB + angle data
  |
I.4 on FBC and GCB
  |
angles below base equal
+ CN3 subtraction of equal angles
  |
base angles equal

```

Formal reconstruction

```

-----
AB ~= AC
  |
hilbert_isosceles_base_angles
(Pappus-style SAS)
  |
angle ABC ~= angle ACB
  |
A-B-D, A-C-E
  |
hilbert_adjacent_angles_congruent
(Hilbert Theorem 14)
  |
angle CBD ~= angle BCE

```

No coordinate representation, numerical length, numerical angle measure, or Euclidean parallel principle is introduced.

5.12 Conclusion

Proposition I.5 is an instructive boundary between the historical and formal architectures of Book I. Euclid builds a sizeable auxiliary configuration so that Proposition I.4 can be used twice. The current formal theorem is short because both outputs have already been abstracted into neutral Hilbert infrastructure.

The audit exposes both sides of that fact. The classical figure retains the full F, G construction and corrects the distinction between the prescribed $AF \cong AG$ and the derived $BF \cong CG$. The formal figure removes the old flowchart and shows the actual mathematics below the two interface calls: a Pappus-style SAS proof for the base angles and Hilbert's three-witness construction for adjacent angles.

Chapter 6

Proposition I.6

6.1 Introduction

Proposition I.6 is the converse of the isosceles-triangle theorem. Where Proposition I.5 shows that equal sides imply equal opposite angles, Proposition I.6 starts from

$$\angle ABC \cong \angle ACB$$

and concludes

$$AB \cong AC.$$

Euclid proves the result by reductio. Assuming that the two sides are unequal, he cuts from the longer one a segment equal to the shorter and uses SAS to force an angle to equal a proper part of itself. The current formal reconstruction is organized differently. The public theorem applies the already derived ASA side theorem to the same triangle with its two base vertices interchanged.

The present audit also descends one layer below the public call. The theorem `hilbert_asa_sides` is not a content-free wrapper: its proof contains an auxiliary segment construction, SAS, Hilbert III.4 angle-construction uniqueness, and an incidence-uniqueness argument. Those geometric steps are recorded explicitly below.

6.2 The Classical Configuration

Let ABC be a nondegenerate triangle with equal base angles:

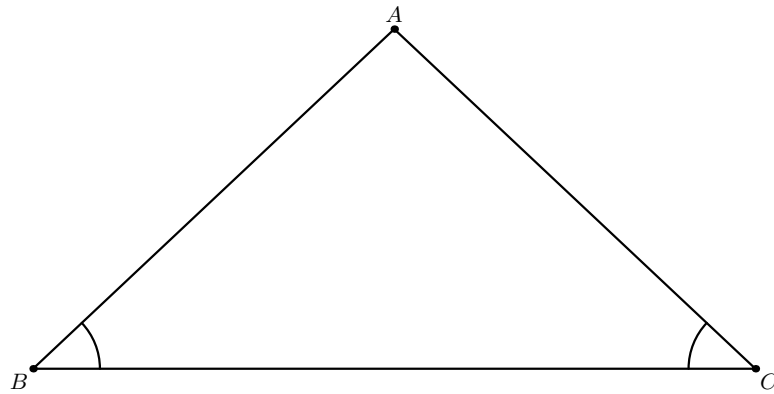
$$\angle ABC \cong \angle ACB.$$

The desired conclusion is $AB \cong AC$.

6.2.1 Euclid's reductio construction

Assume that the two sides are unequal. Interchanging B and C if necessary, suppose that AB is the longer side. Proposition I.3 cuts from AB a segment DB congruent to AC , with

$$A - D - B, \quad DB \cong AC.$$



$$\angle ABC \cong \angle ACB$$

Figure 6.1: The data of Proposition I.6: a nondegenerate triangle with $\angle ABC \cong \angle ACB$.

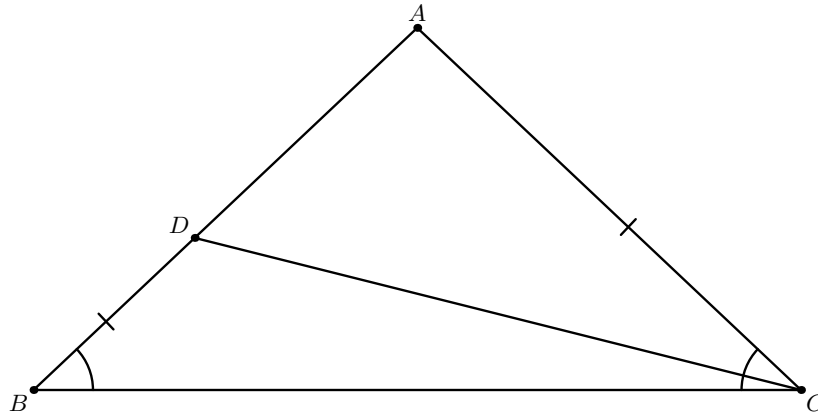
Join D to C .

Because D lies on AB , the angle DBC is the same angle as ABC . Hence the hypothesis gives

$$\angle DBC \cong \angle ACB.$$

Together with $DB \cong AC$ and the common side BC , Proposition I.4 applies to $\triangle DBC$ and $\triangle ACB$.

SAS on $\triangle DBC$ and $\triangle ACB$



$$A - D - B, \quad DB \cong AC$$

Figure 6.2: Euclid's contradiction configuration. Proposition I.3 gives $A - D - B$ and $DB \cong AC$; after joining D to C , SAS compares $\triangle DBC$ with $\triangle ACB$.

SAS gives, in particular,

$$\angle BCD \cong \angle ACB.$$

By the hypothesis the latter angle is congruent to $\angle ACB$. But the ray CD lies inside $\angle ACB$, so $\angle BCD$ is a proper part of $\angle ACB$. Thus the whole angle would be congruent to a proper part of itself, contradicting Common Notion 5. Therefore the unequal-side assumption is impossible.

The classical dependency chain is therefore

$$\text{I.3} + \text{I.4} + \text{Common Notion 5} \longrightarrow \text{I.6.}$$

No use of Proposition I.5 is required in the proof of its converse.

6.3 Statement

Theorem 6 (Euclid I.6). *Let ABC be a nondegenerate triangle. If*

$$\angle ABC \cong \angle ACB,$$

then

$$AB \cong AC.$$

The current Lean declaration has exactly this shape:

```
theorem euclid_proposition_6
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C : Geo.Point)
  (hABC : not Collinear Geo A B C)
  (hAngles : Geo.AngleCongruent A B C A C B) :
  Geo.Congruent A B A C := by
  ...
```

Diagrammatic audit. The two classical figures are retained: the input triangle and the single stable auxiliary configuration after the point D has been cut from the longer side. No further classical snapshot is needed, because SAS and the proper-angle contradiction add no new point or incidence state.

The formal layer requires one additional two-panel figure. Panel (a) specializes the auxiliary construction hidden inside `hilbert_asa_sides`: when the theorem is applied to the two vertex orders BCA and CBA , it constructs a point X on the ray BA satisfying $BX \cong CA$. Panel (b) records the subsequent rigidity step: Hilbert III.4 forces the ray CX to coincide with the ray CA ; since X also lies on the line BA , incidence uniqueness identifies X with A . The construction then yields $AB \cong AC$.

Thus the audited set is

$$2 \text{ classical panels} + 2 \text{ formal panels} = 4 \text{ panels in 3 figures.}$$

6.4 Proof Architecture of the Reconstruction

At the public Book I level the proof is short:

$$\begin{array}{c} \angle ABC \cong \angle ACB \\ \downarrow \\ \text{reorient the two equal angles and the common side } BC \\ \downarrow \\ \text{hilbert_asa_sides on } BCA \text{ and } CBA \\ \downarrow \\ AB \cong AC. \end{array}$$

The current source prepares two cyclic noncollinearity variants, reverses the common segment BC to CB , reorients the given angle congruence, and then calls

```

have hASA :=
  hilbert_asa_sides
    Geo
    B C A
    C B A
    hBCA
    hCBA
    hBC
    hAngleB
    hAngleC

```

The final endpoint reversals are representational only.

The deeper mathematical engine is:

$$\begin{aligned}
 & \text{ASA data for } ABC \text{ and } A'B'C' \\
 & \quad \downarrow \\
 & X \in \text{ray}(B'C'), \quad B'X \cong BC \\
 & \quad \downarrow \text{SAS} \\
 & \quad \angle B'A'X \cong \angle BAC \\
 & \quad \downarrow \text{given angle at } A \\
 & \quad \angle B'A'X \cong \angle B'A'C' \\
 & \quad \downarrow \text{Hilbert III.4 uniqueness} \\
 & \quad \text{ray}(A'X) = \text{ray}(A'C') \\
 & \quad \downarrow \text{incidence uniqueness} \\
 & \quad X = C' \\
 & \quad \downarrow \\
 & AC \cong A'C', \quad BC \cong B'C'.
 \end{aligned}$$

For I.6 this specializes to $X \in \text{ray}(BA)$ with $BX \cong CA$, followed by $\text{ray}(CX) = \text{ray}(CA)$, and therefore $X = A$.

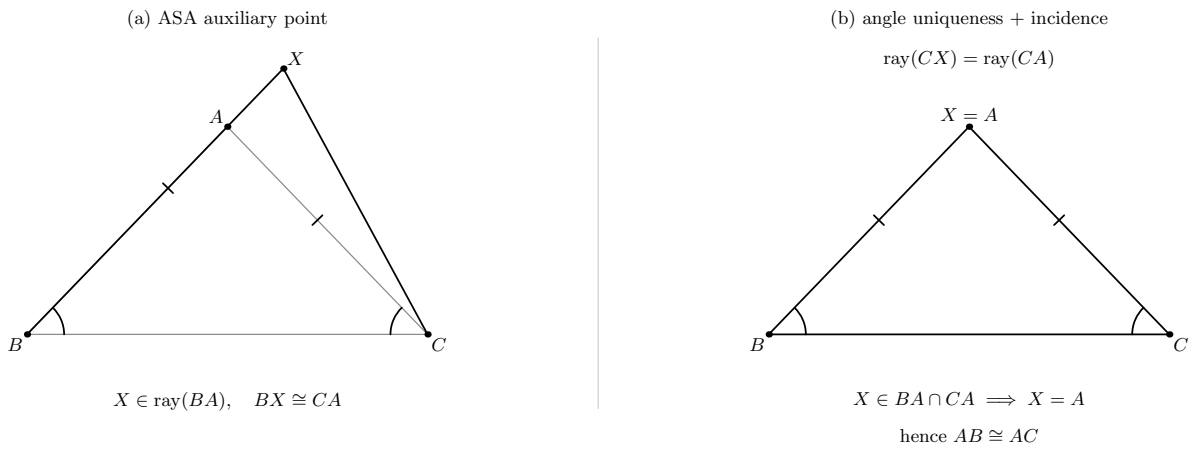


Figure 6.3: The formal ASA engine specialized to Proposition I.6. Left: the segment-construction witness X lies on the ray BA and satisfies $BX \cong CA$. Right: angle-construction uniqueness forces CX onto the ray CA ; the two carrier lines meet at A , hence $X = A$ and $AB \cong AC$.

6.5 Hilbert Reconstruction

6.5.1 The public theorem

The formal Proposition I.6 contains no reductio and introduces no local geometric witness. It compares the triangle with itself after interchanging B and C . The side between the two equal angles is the same segment BC , so ASA immediately supplies the two remaining side congruences.

The decisive public dependency is therefore

$$\text{hilbert_asa_sides} \longrightarrow \text{euclid_proposition_6}.$$

6.5.2 The hidden ASA witness

The general theorem `hilbert_asa_sides` performs the geometric work that the public I.6 theorem does not show. It constructs X on the ray $B'C'$ so that $B'X \cong BC$. The construction deliberately puts X on the same side of the reference line as C' . SAS then transports the first given angle to the auxiliary triangle.

The second prescribed angle is now a uniqueness condition. Hilbert's angle-construction axiom III.4 says that on a fixed side of a fixed base ray there is only one ray making the prescribed congruent angle. Consequently $A'X$ and $A'C'$ are the same ray. Since X and C' already lie on the same carrier through B' , the two distinct carriers would otherwise have two common points. Incidence uniqueness therefore gives $X = C'$.

This is the actual mathematical content hidden below the public ASA call. It uses only neutral incidence, order, and congruence structure; no Euclidean parallel axiom enters Proposition I.6.

6.6 Lean Formalization

The current production theorem first derives the cyclic noncollinearity variants

```
have hBCA : not Collinear Geo B C A := ...
have hCBA : not Collinear Geo C B A := ...
```

and prepares the common side in the required orientation:

```
have hBC : Geo.Congruent B C C B := by
  exact
    (Geo.congruent_reverse_second
      B C B C).mp
    (hilbert_congruent_reflexive Geo B C)
```

The hypothesis is then reversed to provide the two ASA angle inputs. After `hilbert_asa_sides` returns the side conclusion, the proof closes by endpoint reversal:

```
exact
  (Geo.congruent_reverse_second
    A B C A).mp
  ((Geo.congruent_reverse_first
    B A C A).mp hASA.1)
```

There is no proposition-local helper theorem, no case split, no segment trichotomy, and no segment layoff in `Proposition06.lean` itself. Those items appeared in an older documentation layer and are removed from the current formal dependency description.

6.7 Relationship with Earlier Propositions

6.7.1 Proposition I.3

I.3 is a genuine classical dependency. It creates the point D on the assumed longer side with $DB \cong AC$. The current Lean theorem does not call `euclid_proposition_3`; the corresponding construction has been absorbed into the lower ASA theorem in a different form.

6.7.2 Proposition I.4

I.4 is the decisive classical congruence step. SAS compares $\triangle DBC$ and $\triangle ACB$ and forces the angle equality that contradicts the proper-part relation. The formal ASA engine also uses a reusable SAS theorem internally, but it is not a proposition-level call to `euclid_proposition_4`.

6.7.3 Proposition I.5

I.6 is the converse of I.5, but I.5 is not a dependency of Euclid’s proof of I.6. Together the two propositions establish

$$AB \cong AC \iff \angle ABC \cong \angle ACB$$

for a nondegenerate triangle.

6.7.4 Transition to Proposition I.7

The classical proof of I.7 uses I.5, not I.6. Thus I.6 is conceptually part of the early isosceles theory, but it should not be inserted as a spurious edge into the classical dependency DAG of I.7. The formal development, in turn, reorganizes both results around stronger reusable congruence and uniqueness theorems.

6.8 Hilbert Components Used

The actual formal dependency layer relevant to I.6 is:

- segment construction on a prescribed ray inside `hilbert_asa_sides`;
- same-ray and same-side transport;
- SAS angle and side consequences;
- Hilbert III.4 angle-construction uniqueness;
- line uniqueness from plane incidence;
- endpoint and angle orientation symmetries;

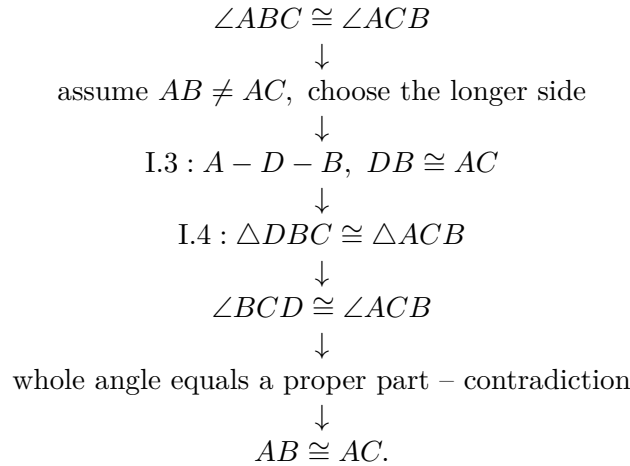
- the final ASA side theorem exposed to Proposition I.6.

Segment trichotomy, a two-branch comparison of AB and AC , and a Book-Zero layoff theorem are *not* dependencies of the current production proof.

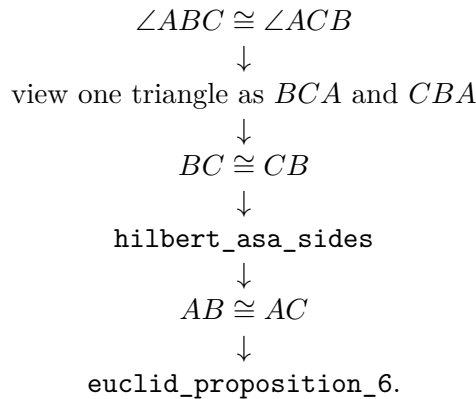
6.9 Dependency Summary

The two architectures should be kept separate.

Classical.



Formal.



Inside the reusable ASA theorem, the essential hidden geometric chain is

segment construction $\rightarrow$ SAS $\rightarrow$ III.4 uniqueness $\rightarrow$ incidence uniqueness.

6.10 Conclusion

Proposition I.6 is a particularly clean example of proof reorganization. Euclid's argument uses an explicit cut on the longer side and SAS inside a reductio. The formal Book I theorem instead recognizes the statement as the side conclusion of an already established ASA theorem.

The short public proof should not be mistaken for absence of geometry. The ASA theorem below it contains a genuine witness X , a segment construction, SAS, angle-construction uniqueness,

and an incidence-rigidity step. Once that lower theorem is available, however, Proposition I.6 itself reduces to vertex reorientation and a single interface call.

Chapter 7

Proposition I.7

7.1 Introduction

Proposition I.7 is a uniqueness theorem. A fixed base AB and two prescribed distances from its endpoints determine at most one vertex on a chosen side of the base. In modern synthetic language, if C and D lie on the same side of the line AB and

$$AC \cong AD, \quad BC \cong BD,$$

then $C = D$.

The proposition is historically important because Euclid uses it immediately in Proposition I.8: after superposition, I.7 rules out a second position of the remaining vertex. The current Lean development reverses that logical order. The general theorem `HilbertSSS` already exists, so I.7 is proved by SSS followed by two uniqueness principles: uniqueness of the ray constructed by a prescribed angle, and uniqueness of a point at a prescribed distance on a ray.

The audit therefore separates three layers:

1. Euclid's representative classical contradiction;
2. the actual proposition-level Lean proof;
3. reusable lower theorems, especially `HilbertSSS`, whose internal SSS engine has already been exposed in the audit of I.8.

7.2 The Classical Configuration

Let A, B be distinct points on a fixed line and let C, D be points on the same side of that line. Assume

$$AC \cong AD, \quad BC \cong BD.$$

The reductio hypothesis is $C \neq D$.

Euclid joins C to D . This creates two isosceles triangles, ACD and BCD , sharing the side CD . In the representative positional case the angle-order relation visible after this join conflicts with the two isosceles base-angle equalities.

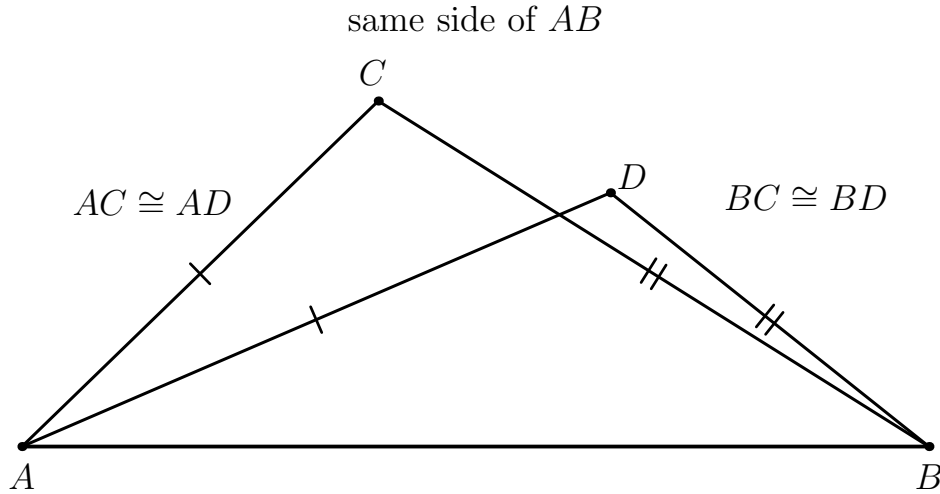


Figure 7.1: The forbidden hypothesis of Proposition I.7. The congruence marks encode the assumptions $AC \cong AD$ and $BC \cong BD$ while C, D are drawn as distinct points on the same side of AB . Since the theorem says that such a metric configuration cannot exist, the drawing is a reductio schematic rather than a metrically realizable Euclidean model.

7.3 Statement

Theorem 7 (Euclid I.7). *Let A, B be distinct points on a line base, and let C, D lie on the same side of base. If*

$$AC \cong AD, \quad BC \cong BD,$$

then

$$C = D.$$

The current production theorem records the carrier line explicitly. Besides $A \neq B$, it assumes that A and B lie on **base**, and packages the positional hypothesis as **HilbertSameSide**, applied to Geo, C, D and **base**. The target is the point equality $C = D$.

Diagrammatic audit. The preserved Book1R chapter already improved the old diagram set by merging two duplicate introductory pictures. The two resulting classical figures are retained: the forbidden hypothesis and Euclid’s joined configuration.

The old formal material, however, was only a theorem-call flowchart. The current audit replaces it by Figure 7.3, which records the actual proposition-level geometry in two panels. Panel (a) is the stable state after SSS: the two candidate rays from B make congruent angles with the common base ray and lie on the prescribed side. Panel (b) shows what the two uniqueness theorems accomplish: Hilbert III.4 identifies a single ray carrier through a witness X , and segment-construction uniqueness then collapses the two equal-distance candidates to $C = D$.

The internal construction used to prove **HilbertSSS** is not redrawn here, because it is a reusable lower theorem and was audited explicitly in Proposition I.8. Thus the final set is

$$2 \text{ classical panels} + 2 \text{ formal panels} = 4 \text{ panels in 3 figures.}$$

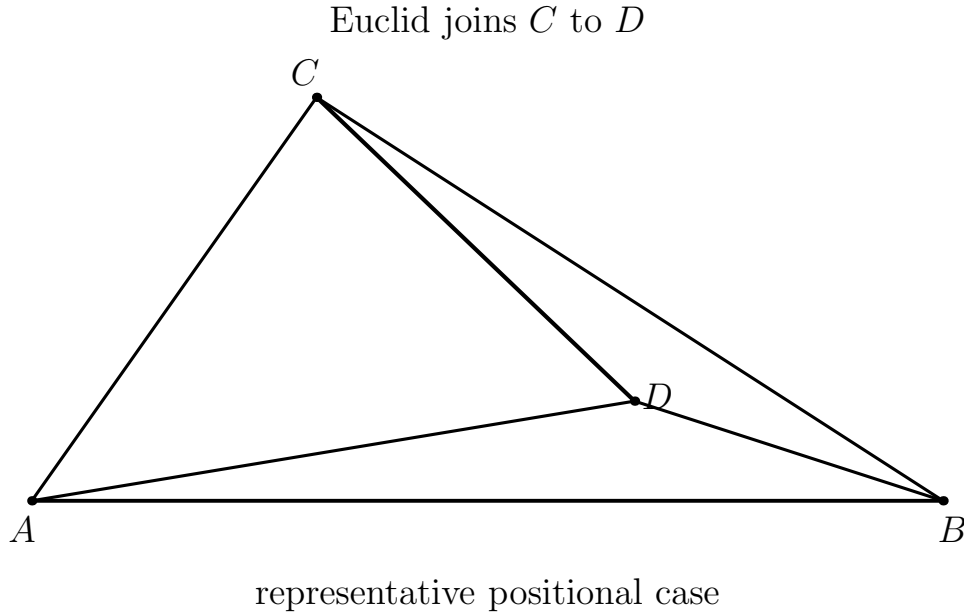


Figure 7.2: Euclid’s representative configuration after joining C to D . The picture records one positional case of the classical contradiction; it is not intended to encode every possible same-side position of two distinct candidate vertices.

7.4 Classical Proof

Assume $C \neq D$ and join C to D . From

$$AC \cong AD$$

triangle ACD is isosceles, so Proposition I.5 gives

$$\angle ACD \cong \angle CDA.$$

Similarly,

$$BC \cong BD$$

makes BCD isosceles, hence again by I.5,

$$\angle BCD \cong \angle CDB.$$

In Euclid’s displayed positional case, one of the latter angles is a proper part of the corresponding former angle. The two equalities above therefore force an angle to be both equal to and strictly greater than the corresponding angle, a contradiction. Euclid concludes that the two candidate vertices cannot be distinct.

A diagrammatic qualification is essential. The single classical drawing does not exhaust all possible same-side positions of C and D ; a fully expanded classical treatment requires additional positional cases. The formal theorem below does not inherit this limitation: its `HilbertSameSide` hypothesis covers the full same-side statement.

The classical proposition dependency relevant to Euclid’s argument is

$$\boxed{\text{I.5} \longrightarrow \text{I.7}}.$$

The previous documentation listed Proposition I.6 as part of the classical route. That edge is removed in this audit: the displayed I.7 contradiction uses the isosceles base-angle theorem I.5, not the converse theorem I.6.

7.5 Proof Architecture of the Reconstruction

The public source begins with a trivial equality split:

$$C = D \quad \text{or} \quad C \neq D.$$

The first branch is immediate and carries no geometric content. In the second branch, the actual proposition-level DAG is

$$\begin{array}{c}
 C, D \text{ on the same side of } AB, \quad AC \cong AD, \quad BC \cong BD \\
 \downarrow \\
 C, D \text{ off the base line} \\
 \downarrow \\
 ABC, ABD \text{ noncollinear} \\
 \downarrow \\
 AB \cong AB \text{ and the two given side congruences} \\
 \downarrow \text{HilbertSSS} \\
 \angle ABC \cong \angle ABD \\
 \downarrow \text{hilbert_angle_unique_common_ray} \\
 \exists X : \text{SameRay}(B; X, C) \wedge \text{SameRay}(B; X, D) \\
 \downarrow \text{hilbert_segment_construction_unique} \\
 C = D.
 \end{array}$$

This is not Euclid's angle-order contradiction. The formal proof turns the problem into a rigidity chain:

$$\text{SSS} \implies \text{unique ray} \implies \text{unique point on that ray}.$$

7.6 Hilbert Reconstruction

7.6.1 Off-line information and nondegeneracy

The same-side hypothesis already contains the information that both C and D are off the base line. The proof extracts those two facts and calls `hilbert_not_collinear_of_off_line` twice to obtain

$$\neg \text{Collinear}(A, B, C), \quad \neg \text{Collinear}(A, B, D).$$

These are the nondegeneracy obligations needed for the congruence step and for the subsequent angle-uniqueness theorem.

7.6.2 SSS creates equal angles at the common base vertex

The triangles ABC and ABD share AB . Together with

$$AC \cong AD, \quad BC \cong BD, \quad AB \cong AB,$$

the reusable theorem `HilbertSSS` gives complete triangle congruence data. I.7 consumes only the angle at B :

$$\angle ABC \cong \angle ABD.$$

At this point the formal proof has already left Euclid's historical order: SSS is available as lower infrastructure, while Euclid proves I.7 precisely in order to use it later inside his proof of I.8.

7.6.3 Hilbert III.4 forces a common ray

The current helper `hilbert_angle_unique_common_ray` is not merely a name for the conclusion. Its proof invokes `HilbertCongruence.angle_construction`. Applied here with common vertex B , reference ray BA , and the prescribed side of the base, it constructs a witness X for the unique ray making the required angle. The uniqueness field returned by Hilbert III.4 is then applied once to C and once to D . The result is

$$\text{SameRay}(B; X, C), \quad \text{SameRay}(B; X, D).$$

Thus the two initially distinct candidate directions have been identified at the ray level.

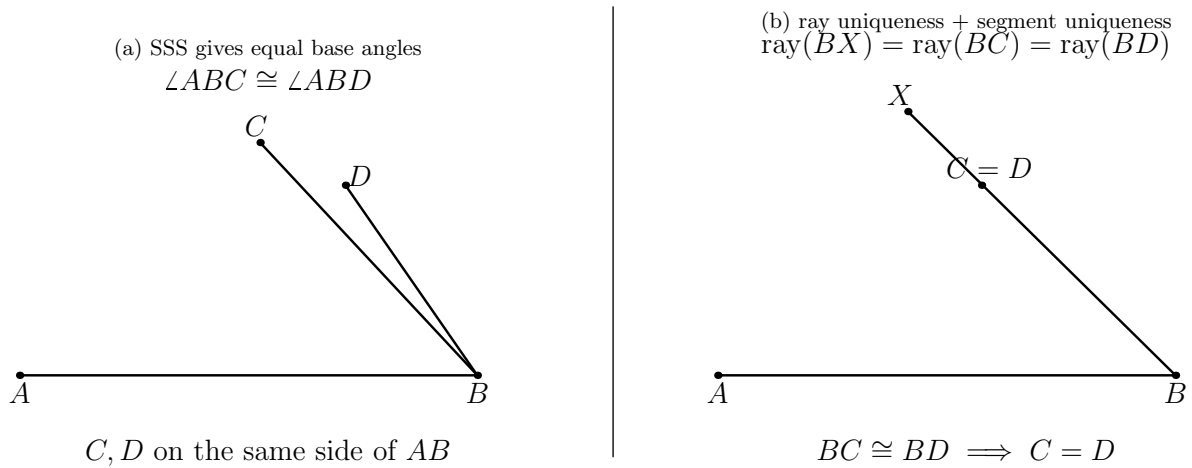


Figure 7.3: The proposition-level Hilbert mechanism. (a) SSS yields $\angle ABC \cong \angle ABD$ while C, D are on the prescribed side of the base. Hilbert III.4 uniqueness therefore identifies both candidate rays with one ray represented by BX . (b) Once the ray is common, the original congruence $BC \cong BD$ and uniqueness of segment construction on that ray leave only one endpoint, so $C = D$.

7.6.4 Segment uniqueness identifies the endpoint

The final step uses the original equality of distances from B . The proof first reverses the congruence orientation to obtain

$$BD \cong BC.$$

Both C and D lie on the ray represented by BX . The theorem `hilbert_segment_construction_unique` therefore says that two points laid off on that same ray at the same prescribed distance from B must be the same point. Hence $C = D$.

The witness X has no independent classical significance. It is a carrier witness for the unique ray supplied by the formal angle-construction API. This is why it is shown only in the formal figure and not introduced into the classical diagrams.

7.7 Lean Formalization

The current production proof applies SSS as follows:

```
have hSSS :=
```

```
HilbertSSS
  Geo
  A B C
  A B D
  hABC
  hABAB
  hBCBD
  hACAD
```

It extracts the angle at B and feeds it to the current same-ray uniqueness helper:

```
rcases
  hilbert_angle_unique_common_ray
    Geo
    A B C D
    base
    hAB
    hAbase
    hBbase
    hCbase
    hSame
    hAngleB with
  <X, hRayXC, hRayXD>
```

Finally, the two points are identified by segment-construction uniqueness:

```
exact
  hilbert_segment_construction_unique
    Geo
    B C
    B X
    C D
    hRayXC
    hRayXD
    (hilbert_congruent_reflexive Geo B C)
    hBD_BC
```

The angle helper itself currently has the proposition-level specification

$$\begin{aligned} \angle OLH \cong \angle OLK, \quad H, K \text{ on the same side of } OL \\ \implies \exists X \text{ (SameRay}(L; X, H) \wedge \text{SameRay}(L; X, K)). \end{aligned}$$

Its implementation uses the uniqueness component of Hilbert III.4 rather than an angle measure or coordinate calculation.

7.8 Relationship with Earlier and Later Propositions

7.8.1 Proposition I.5

I.5 is the decisive proposition in Euclid's displayed contradiction. The equalities $AC \cong AD$ and $BC \cong BD$ produce two isosceles triangles, and I.5 converts each side equality into a base-angle equality.

7.8.2 Proposition I.6 is not a dependency

The older chapter described I.5 and I.6 as a two-way bridge used in I.7. That overstates the classical proof. I.6 is not invoked in the displayed argument of Proposition I.7, and the current Lean theorem does not call `euclid_proposition_6`. It is therefore removed from both the historical and formal DAGs of this proposition.

7.8.3 Transition to Proposition I.8

The historical direction is

$$\text{I.7} \longrightarrow \text{I.8}.$$

After superposition in I.8, the remaining vertex is characterized by its two distances from the endpoints of the common base; I.7 excludes a second same-side realization.

The current formal library has the reverse dependency pattern at the reusable-theorem level:

$$\text{HilbertSSS} \longrightarrow \text{euclid_proposition_7}.$$

This is not circularity. The Hilbert theorem is established independently of Book I's proposition order, and the Book I endpoint is reconstructed from that stronger infrastructure.

7.9 Hilbert Components Used

The proposition-level proof uses the following established components:

- `HilbertSameSide` for the prescribed half-plane and off-line information;
- `hilbert_not_collinear_of_off_line` for triangle nondegeneracy;
- `HilbertSSS` for the angle congruence at B ;
- `hilbert_angle_unique_common_ray`, whose current proof uses Hilbert III.4 angle construction and its uniqueness field;
- `hilbert_segment_construction_unique` for the final endpoint uniqueness on the common ray;
- reflexivity, symmetry, and endpoint transport for segment congruence.

No numerical distance, angle measure, coordinate system, or Euclidean parallel axiom occurs in the proposition-level argument.

7.10 Formalization Notes

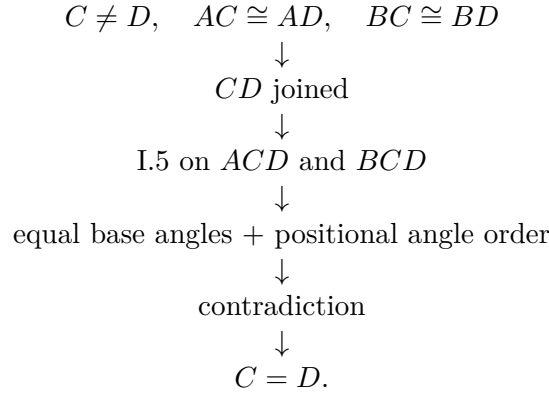
The proof has a useful modular structure. Its only local case split is whether $C = D$. The equality branch closes immediately; the distinct branch contains all of the geometry. There is no positional subdivision analogous to the classical region analysis.

The SSS call should be read as a reusable lower theorem, not as an unexplained primitive. Its internal Hilbert construction has already been audited in I.8, so recursively reproducing that construction here would obscure rather than clarify I.7's own contribution.

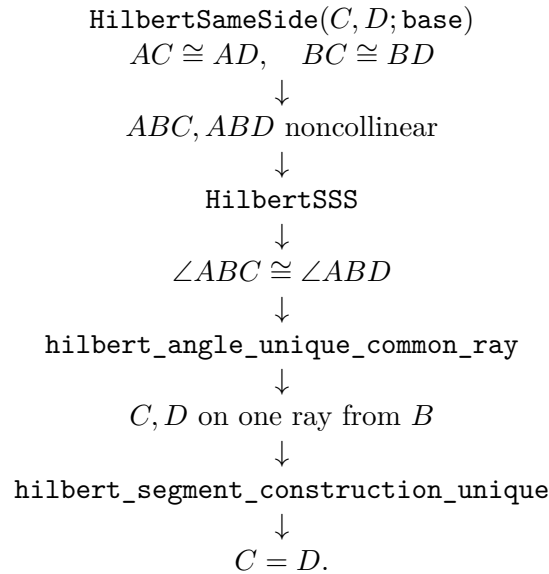
Conversely, the old theorem-call flowchart was too compressed because it hid the geometric meaning of the angle-uniqueness step. The new formal figure therefore exposes the one new stable state that matters locally: two same-side equal-angle candidates are forced onto one ray, after which segment uniqueness finishes the proof.

7.11 Dependency Summary

Classical representative route:



Current formal route:



7.12 Conclusion

Proposition I.7 is a rigidity theorem whose classical and formal proofs have opposite architectural roles. For Euclid, I.7 is preparation for SSS: the isosceles theorem and angle order exclude a second same-side vertex. In the current reconstruction, SSS is already available and I.7 becomes a concise corollary of stronger reusable uniqueness infrastructure.

The audit corrects the classical dependency by removing I.6, replaces the old formal flowchart by the actual common-ray geometry, and keeps the lower SSS engine modular rather than duplicating the I.8 audit. The result is a three-level account of the same theorem: Euclid's representative

contradiction, the proposition-level Lean proof, and the reusable Hilbert machinery that makes the formal proof short.

Chapter 8

Proposition I.8

8.1 Introduction

Proposition I.8 is Euclid’s Side–Side–Side rigidity theorem. Given two nondegenerate triangles with their three corresponding sides congruent, the proposition concludes that all three corresponding angles are congruent.

Historically, Euclid proves this by superposition and uses Proposition I.7 to exclude a second possible position of the remaining vertex. The current production theorem `euclid_proposition_8` is much shorter: it calls the reusable theorem `HilbertSSS` and exposes the three angle fields of the returned triangle-congruence result.

The diagrammatic audit cannot stop at that public call. Current source inspection shows that `HilbertSSS` itself contains substantial geometry. In Hilbert’s Theorem 18 construction, an auxiliary point is placed on the side of the base opposite the original third vertex, SAS is used once to transfer a side, Hilbert’s Theorem 17 supplies an opposite-side rigidity angle, and a second SAS application recovers the complete congruence data. The audited chapter therefore records both the public wrapper and this hidden mathematical engine.

8.2 The Classical Configuration

Let ABC and DEF be nondegenerate triangles with

$$AB \cong DE, \quad BC \cong EF, \quad AC \cong DF.$$

The task is to prove

$$\angle BAC \cong \angle EDF, \quad \angle ABC \cong \angle DEF, \quad \angle ACB \cong \angle DFE.$$

8.2.1 Euclid’s superposition

Euclid places one triangle on the other so that B coincides with E and C with F . The transported position of A will be denoted by G . If $G \neq D$, then G and D are two distinct points on the same side of the common base EF satisfying

$$EG \cong ED, \quad FG \cong FD.$$

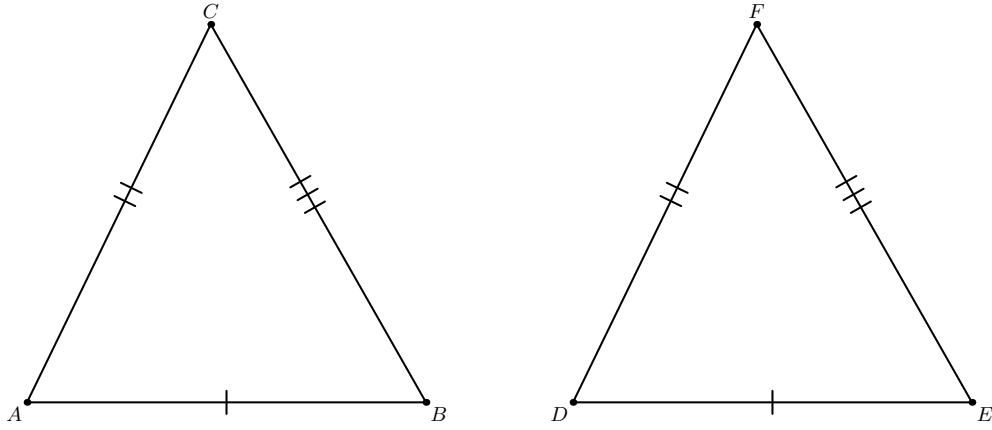


Figure 8.1: The data of Proposition I.8. The three pairs of corresponding sides of ABC and DEF are congruent.

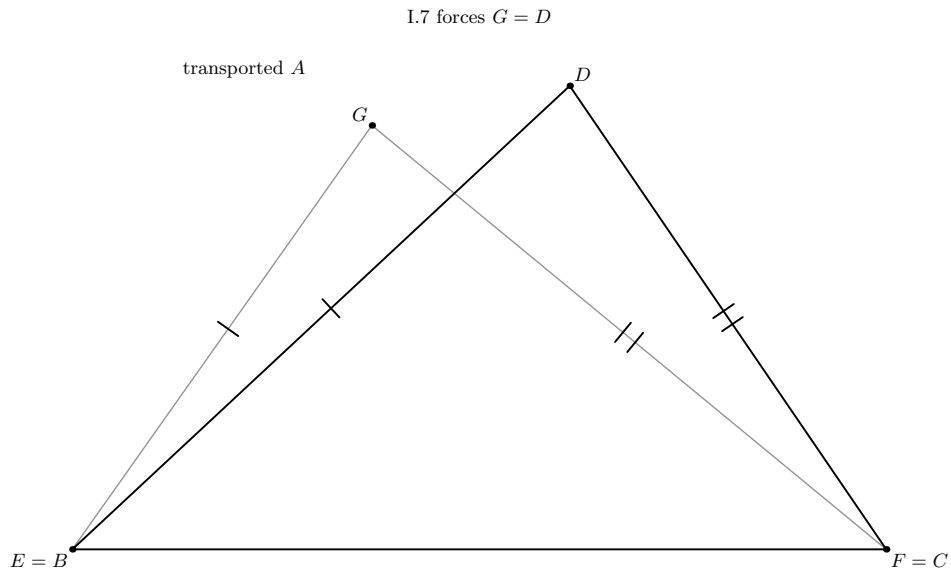


Figure 8.2: Euclid's superposition step. After the base has been identified, the transported vertex G and the target vertex D lie on the same side of EF with $EG \cong ED$ and $FG \cong FD$. Proposition I.7 forces $G = D$.

Proposition I.7 excludes exactly this configuration, so $G = D$.

Once the three vertices coincide under superposition, the corresponding angles coincide as well. This is the classical SSS argument.

8.3 Statement

Theorem 8 (Euclid I.8). *Let A, B, C, D, E, F be points such that A, B, C are noncollinear and*

$$AB \cong DE, \quad BC \cong EF, \quad AC \cong DF.$$

Then

$$\angle BAC \cong \angle EDF, \quad \angle ABC \cong \angle DEF, \quad \angle ACB \cong \angle DFE.$$

The current Lean statement assumes noncollinearity only for ABC :

```
(hABC : not Collinear Geo A B C)
```

Noncollinearity of DEF is derived inside `HilbertSSS` from the three side congruences. The public target is exactly the conjunction of the three angle-congruence statements.

Diagrammatic audit. The preserved Book1R chapter already contained two useful classical figures: the input pair of triangles and the superposition configuration. Both are retained. What was missing was the actual geometry hidden under the public `HilbertSSS` call. Figure 8.3 adds that material as one two-panel formal figure.

Panel (a) records the auxiliary point G constructed on the side of AB opposite C and matched by angle and segment data to the triangle DEF . Panel (b) records the stable two-center configuration after the first SAS step: C and G lie on opposite sides of AB with $AC \cong AG$ and $BC \cong BG$. This is precisely the input to Hilbert’s Theorem 17.

The internal proof of Theorem 17 itself contains further incidence cases. They are not expanded here: Theorem 17 is a reusable lower theorem called by SSS, just as the midpoint theorem was treated as a reusable component in the I.9 audit. The present figure records the proposition-relevant geometry at the level of the SSS engine, not the internal proof tree of every dependency. Thus the audited set is

$$2 \text{ classical panels} + 2 \text{ formal panels} = 4 \text{ panels in 3 figures.}$$

8.4 Classical Proof

The historical dependency is simple:

$$\text{I.7} \longrightarrow \text{I.8.}$$

Superposition identifies one side of the first triangle with the corresponding side of the second. The remaining vertex is then determined by its two distances from the endpoints of the common base. Proposition I.7 supplies the uniqueness on the chosen side of that base.

This is a historical dependency. The current Lean theorem does not call `euclid_proposition_7`.

8.5 Proof Architecture of the Reconstruction

At the public Book I level the architecture is only

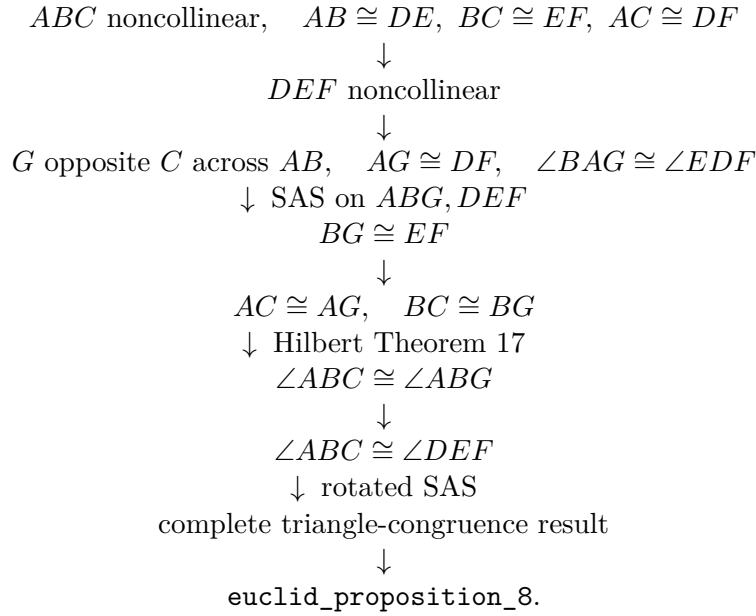
```

three side congruences
  ↓
HilbertSSS
  ↓
three angle congruences
  ↓
euclid_proposition_8.

```

For audit purposes this is too compressed. The current Hilbert SSS engine has the following

mathematical structure:



This formal proof is not Euclid’s superposition argument in disguise. It is Hilbert’s synthetic SSS construction organized through reusable angle, segment, separation, SAS, and rigidity theorems.

8.6 Hilbert Reconstruction

8.6.1 Derive noncollinearity of the second triangle

The API of `HilbertSSS` deliberately asks only for noncollinearity of the first triple. Its first geometric obligation is to show that the side data cannot describe a degenerate second triangle. The helper

```
hilbert_noncollinear_of_three_congruences
```

transports collinearity backwards through the three congruences and derives a contradiction with the hypothesis on ABC .

This matters for the documentation: the formal theorem is slightly stronger at the interface than a statement requiring both triangles to be declared nondegenerate by the caller.

8.6.2 Construct the auxiliary point on the opposite side

Let AB be the base line. The SSS proof calls the auxiliary construction

```
hilbert_sss_auxiliary_point
```

which produces G on the side of AB opposite C such that

$$AG \cong DF, \quad \angle BAG \cong \angle EDF.$$

The helper obtains the correct side by extending through A , copying the angle by Hilbert III.4, and laying off the required segment on the resulting ray.

Using the hypothesis $AB \cong DE$, SAS for ABG and DEF then gives

$$BG \cong EF.$$

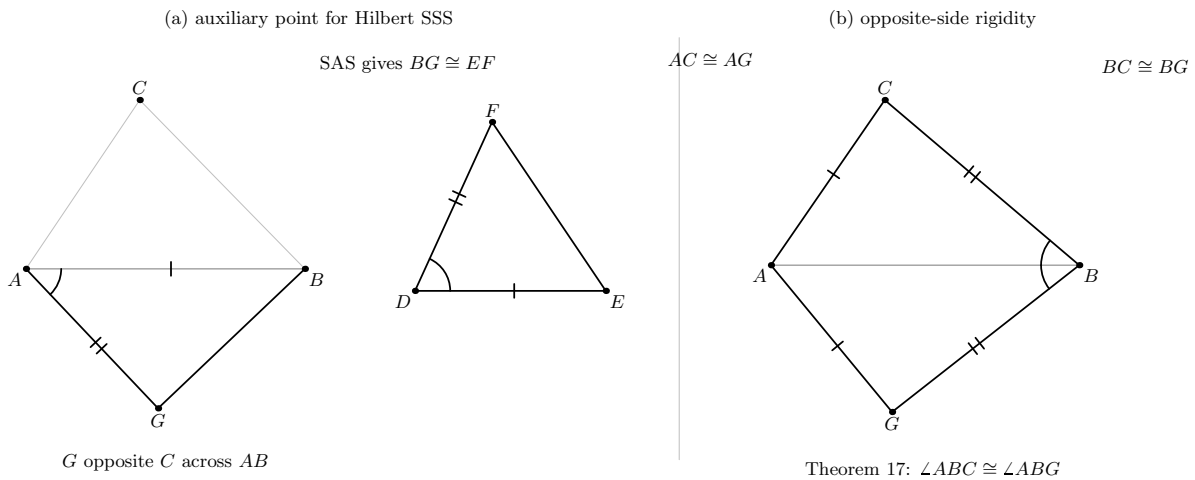


Figure 8.3: The hidden Hilbert SSS engine used by the current I.8 wrapper. (a) The auxiliary point G is constructed opposite C across AB and matched to DEF by $AB \cong DE$, $AG \cong DF$, and the included-angle congruence; SAS gives $BG \cong EF$. (b) Combining those results with the original SSS hypotheses gives $AC \cong AG$ and $BC \cong BG$. Hilbert's Theorem 17 converts this opposite-side two-center symmetry into the required angle rigidity at B .

8.6.3 Opposite-side rigidity

The original hypotheses and the first SAS result now give, by congruence transitivity,

$$AC \cong AG, \quad BC \cong BG.$$

The points C and G lie on opposite sides of AB . Hilbert's Theorem 17 therefore applies to the two centers A, B and the two points C, G , yielding

$$\angle ABC \cong \angle ABG.$$

The first SAS comparison already gives the corresponding angle relation between ABG and DEF , so transitivity yields

$$\angle ABC \cong \angle DEF.$$

8.6.4 Recover the remaining angles by a rotated SAS

With the angle at B now known, the proof reverses the first side representation and applies SAS to the rotated triples

$$BAC \quad \text{and} \quad EDF.$$

This second SAS call returns complete congruence data. The angle at C is put into the orientation required by the public theorem using the standard angle-reversal theorem. The resulting record contains exactly the three angle fields exported by I.8.

8.7 Lean Formalization

The current production endpoint is intentionally short:

```

theorem euclid_proposition_8
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D E F : Geo.Point)
  (hABC : not Collinear Geo A B C)
  (hAB : Geo.Congruent A B D E)
  (hBC : Geo.Congruent B C E F)
  (hAC : Geo.Congruent A C D F) :
  Geo.AngleCongruent B A C E D F /\
  Geo.AngleCongruent A B C D E F /\
  Geo.AngleCongruent A C B D F E := by
have hSSS :=
  HilbertSSS Geo A B C D E F hABC hAB hBC hAC
exact
  <hSSS.2.angleA,
    hSSS.2.angleB,
    hSSS.2.angleC>

```

The shortness of this code is an API fact, not a statement that SSS is primitive. The current `HilbertSSS` theorem packages Hilbert’s Theorem 18 and returns both the derived noncollinearity of DEF and a `TriangleCongruenceResult`.

There are no proposition-local helper theorems and no proposition-local case splits in `Proposition08.lean`. Source inspection of the current file finds no `sorry` and no proposition-local `axiom`. This documentation audit did not run a fresh `lake build` or `#print axioms`, so it makes no stronger compilation or axiomatic status claim.

8.8 Relationship with Earlier Propositions

8.8.1 Proposition I.7

I.7 is the decisive historical dependency. Under superposition it says that two distinct points on the same side of a fixed base cannot have the same two prescribed distances from the endpoints. The current Lean theorem I.8 does not call `euclid_proposition_7`; the Hilbert SSS theorem supplies a different rigidity proof.

8.8.2 Proposition I.4

I.4 and I.8 are the two early triangle-congruence landmarks of Book I. I.4 exposes SAS, while I.8 exposes SSS. In the present architecture the lower Hilbert SSS proof itself uses the reusable SAS theorem as an ingredient. This is a dependency of the Hilbert engine, not a call from I.8 to the Book I wrapper `euclid_proposition_4`.

8.8.3 Transition to Proposition I.9

Euclid uses I.8 immediately in I.9 to compare the two triangles surrounding the candidate angle bisector. The current I.9 implementation instead calls `HilbertSSS` directly. Thus I.8 and I.9 share the same lower triangle-rigidity interface even though the Book I call graph does not contain an edge from the formal I.9 theorem to the formal I.8 theorem.

8.9 Hilbert and Book Zero Components Used

8.9.1 Hilbert SSS

`HilbertSSS` is the direct dependency of the public theorem. It is Hilbert Theorem 18 in the project interface and is the mathematical engine whose structure is exposed in Figure 8.3.

8.9.2 Opposite-side geometry

The auxiliary point is deliberately constructed on the side of AB opposite the original third vertex. This is essential input to Hilbert Theorem 17 and is not visible in the public SSS statement.

8.9.3 Angle and segment construction

The helper `hilbert_sss_auxiliary_point` uses the congruence layer to copy the target angle and lay off the target side on the selected ray. The proof therefore remains synthetic: no coordinates, numerical lengths, or numerical angle measures are introduced.

8.9.4 SAS and Hilbert Theorem 17

The lower proof uses SAS twice. The first application determines BG in the auxiliary triangle; Theorem 17 then supplies an angle comparison across the base, and the final rotated SAS recovers the complete triangle-congruence record.

Theorem 17 is treated as a reusable lower theorem in this chapter. Its own internal incidence case split is not reproduced as proposition-specific geometry.

8.10 Formalization Notes

The main correction to the preserved Book1R chapter is architectural. The sentence “Proposition I.8 becomes a direct application of `HilbertSSS`” is correct only at the public endpoint. It is incomplete as a description of the mathematical proof because the current SSS theorem is a substantial derivation rather than a primitive axiom or an opaque declaration.

The audit therefore distinguishes three levels:

1. Euclid’s historical proof: superposition plus I.7;
2. the Book I API endpoint: one call to `HilbertSSS`;

3. the hidden Hilbert engine: auxiliary opposite-side construction, SAS, Theorem 17, and rotated SAS.

The formal figure is intentionally limited to two stable states. It does not turn every theorem call into a panel, and it does not recursively expand the proof of Theorem 17. This preserves modularity while still showing the geometry that the old theorem-call diagram concealed.

8.11 Dependency Summary

Classical:

$$\begin{array}{c}
 AB \cong DE, AC \cong DF, BC \cong EF \\
 \downarrow \\
 \text{superposition} \\
 \downarrow \\
 \text{I.7 fixes the remaining vertex} \\
 \downarrow \\
 \text{corresponding angles congruent} \\
 \downarrow \\
 \text{I.8.}
 \end{array}$$

Public formal endpoint:

$$\text{HilbertSSS} \longrightarrow \text{euclid_proposition_8.}$$

Hidden mathematical engine:

$$\begin{array}{c}
 \text{three side congruences and } ABC \text{ noncollinear} \\
 \downarrow \\
 DEF \text{ noncollinear} \\
 \downarrow \\
 G \text{ opposite } C \text{ across } AB \\
 \downarrow \\
 \text{angle copy} + \text{segment layoff} \\
 \downarrow \\
 \text{SAS}(ABG, DEF) \\
 \downarrow \\
 AC \cong AG, BC \cong BG \\
 \downarrow \\
 \text{Hilbert Theorem 17} \\
 \downarrow \\
 \angle ABC \cong \angle DEF \\
 \downarrow \\
 \text{rotated SAS} \\
 \downarrow \\
 \text{TriangleCongruenceResult} \\
 \downarrow \\
 \text{euclid_proposition_8.}
 \end{array}$$

8.12 Conclusion

Proposition I.8 is a particularly clear example of the difference between a short public formal theorem and the geometry supporting its interface. Euclid proves SSS by superposition and I.7. The Book I Lean endpoint merely calls `HilbertSSS`. One level lower, however, the Hilbert proof is a substantial synthetic argument built from an opposite-side auxiliary point, SAS, two-center rigidity, and a second SAS comparison.

The audited documentation now displays all three viewpoints without confusing them. The classical proof remains historically visible, the public Lean theorem remains accurately short, and the hidden SSS engine is no longer represented as if it were a primitive black box.

Chapter 9

Proposition I.9

9.1 Introduction

Proposition I.9 constructs a bisector of a given nondegenerate rectilinear angle. Euclid’s classical proof and the current Lean reconstruction are both synthetic, but they use different auxiliary configurations.

Euclid chooses equal segments on the two arms of the angle, constructs an equilateral triangle on the segment joining their endpoints, and invokes Proposition I.8. The current production theorem instead lays off a segment on one original ray, takes the midpoint of the segment joining that point to the other original endpoint, and applies the general Hilbert SSS theorem.

This distinction is mathematically significant. The formal proof does not call the Book I theorems I.1, I.3, or I.8. It uses the reusable Hilbert congruence and midpoint infrastructure directly.

9.2 The Classical Configuration

Let $\angle BAC$ be the given rectilinear angle. The hypothesis is nondegeneracy: the points A, B, C are not collinear. The objective is to construct a ray from A that divides the angle into two congruent angles.

9.2.1 Euclid’s construction

Choose a point D on the side AB . By Proposition I.3 cut off on the other side a segment AE satisfying

$$AD \cong AE.$$

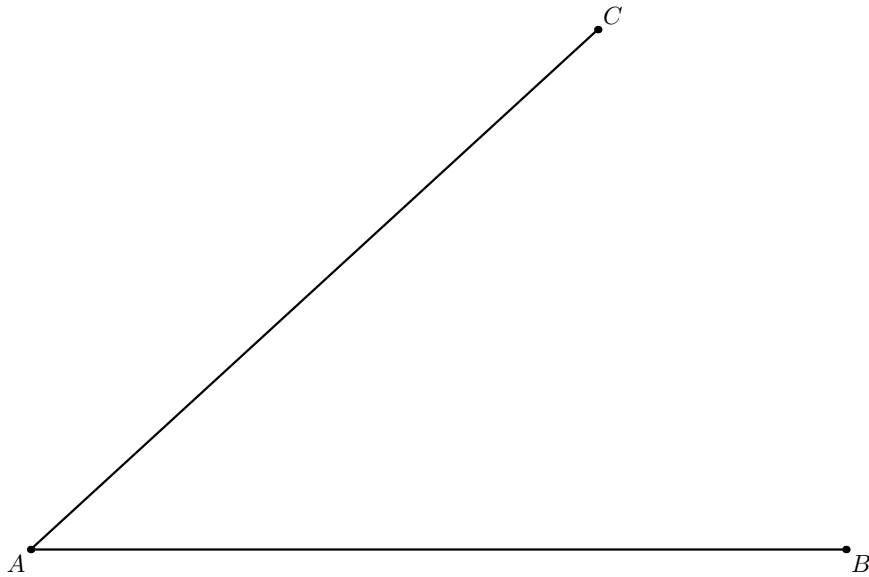
Join D to E , construct the equilateral triangle DEF on DE by Proposition I.1, and join A to F . Since

$$AD \cong AE, \quad DF \cong EF, \quad AF \cong AF,$$

Proposition I.8 gives

$$\angle DAF \cong \angle FAE.$$

Because D and E lie on the original rays AB and AC , the ray AF bisects the given angle.

Figure 9.1: The given nondegenerate angle $\angle BAC$.

9.3 Statement

Theorem 9 (Euclid I.9). *Let A, B, C be noncollinear. Then there exists a point M such that*

$$\angle BAM \cong \angle CAM.$$

Thus the ray determined by A and M bisects the angle whose arms are the rays AB and AC .

The order of the last two letters in the second displayed angle follows the current Lean endpoint literally:

```
Geo.AngleCongruent B A M C A M
```

The project's angle object is symmetric in its two arms, so this is the same geometric bisection normally written $\angle BAM \cong \angle MAC$.

Diagrammatic audit. The audited chapter uses three geometric figures. Figures 9.1 and 9.2 retain the two classical states already present in Book1R. The old non-geometric flowchart for the Hilbert reconstruction is replaced by Figure 9.3, which records the actual stable formal configuration. No additional panel is needed: the proof has no geometric case split, and the final midpoint configuration contains the earlier segment-layoff state without ambiguity.

9.4 Classical Proof

The classical dependency chain is

$$\text{I.3} \longrightarrow \text{I.1} \longrightarrow \text{I.8} \longrightarrow \text{I.9}.$$

Proposition I.3 supplies equal auxiliary segments on the two arms, Proposition I.1 supplies the equilateral triangle, and Proposition I.8 converts three side congruences into the required equality of angles.

This chain is historical, not the direct call graph of the current Lean implementation.

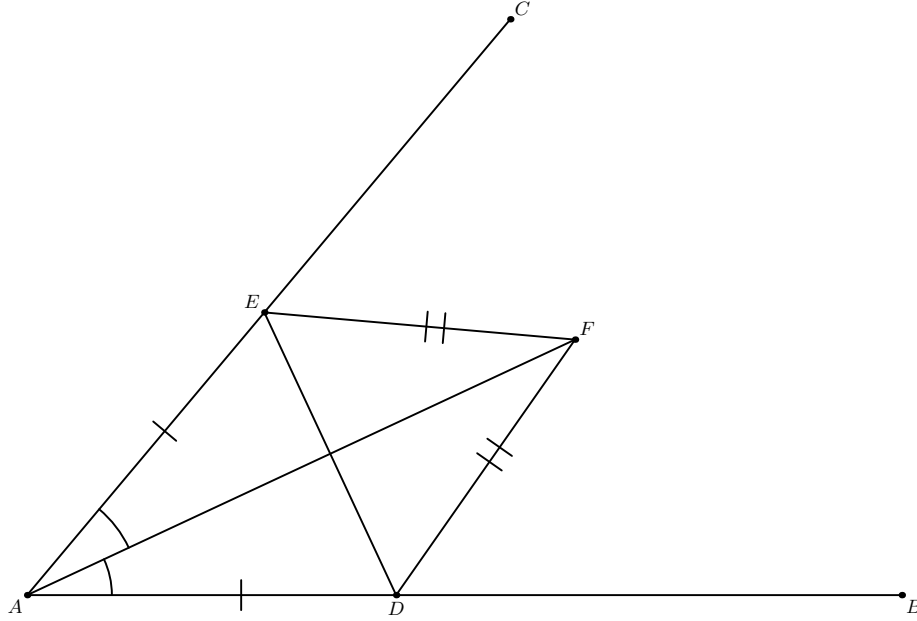


Figure 9.2: Euclid's construction for Proposition I.9. The points D, E satisfy $AD \cong AE$, and DEF is equilateral. With AF common, SSS for $\triangle ADF$ and $\triangle AEF$ gives equality of the two angles at A .

9.5 Proof Architecture of the Reconstruction

The current production theorem is not a thin wrapper. It performs the following proposition-specific construction:

$$\begin{aligned}
 &A, B, C \text{ noncollinear} \\
 &\quad \downarrow \\
 &D \text{ on ray } AB, \quad AD \cong AC \\
 &\quad \downarrow \\
 &M \text{ midpoint of } DC \\
 &\quad \downarrow \\
 &\triangle ADM \text{ and } \triangle ACM \text{ satisfy SSS} \\
 &\quad \downarrow \\
 &\angle DAM \cong \angle CAM \\
 &\downarrow \text{ same-ray transport } AD \rightsquigarrow AB \\
 &\angle BAM \cong \angle CAM.
 \end{aligned}$$

The midpoint call is deliberately treated here as a reusable theorem-level component. Unlike Proposition I.10, I.9 already has a substantial local geometric construction. Recursively expanding the internal proof of `HilbertMidpointExists` inside this chapter would duplicate a general midpoint theorem rather than clarify the proposition-specific geometry.

9.6 Hilbert Reconstruction

9.6.1 Lay off a congruent segment on the original ray

From noncollinearity the proof derives $A \neq B$. Hilbert III.1 is then used in the form

```
HilbertCongruence.segment_construction
```

```
(Geo := Geo) A C A B hAB
```

to obtain a point D satisfying

$$\text{SameRay}(A; B, D), \quad AD \cong AC.$$

The same-ray information is essential: it is what later allows the formal angle $\angle DAM$ to be transported back to the original arm AB .

9.6.2 Construct the midpoint of the cross-segment

The proof establishes that A, D, C are noncollinear and therefore $D \neq C$. It then invokes

```
HilbertMidpointExists Geo D C hDC
```

and obtains M with

$$D - M - C, \quad DM \cong MC.$$

The strict betweenness is not decorative bookkeeping. It supplies both the position of M on DC and the distinctness facts needed to prove that $\triangle ADM$ is nondegenerate.

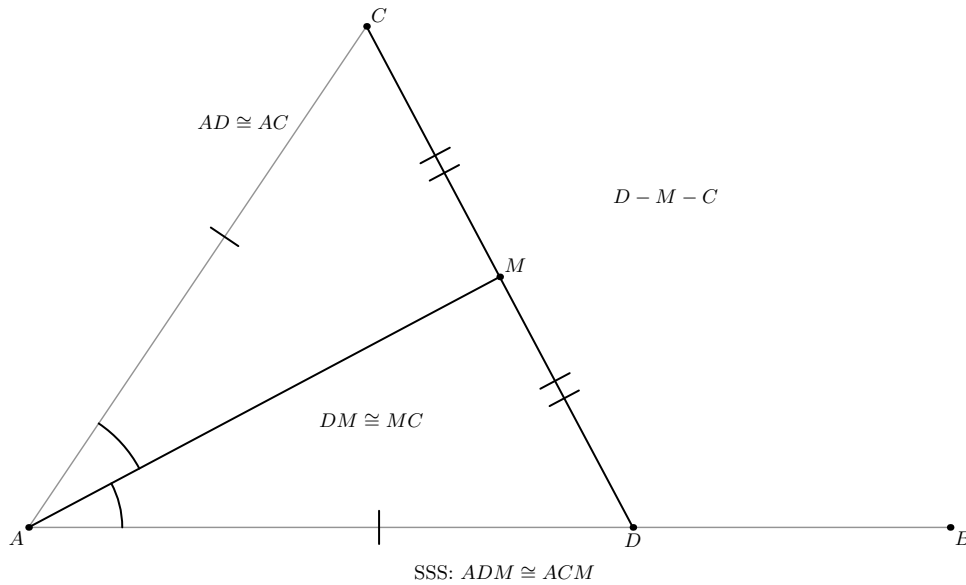


Figure 9.3: The formal Hilbert configuration used by the current `euclid_proposition_9`. Segment construction gives D on the original ray AB with $AD \cong AC$. The reusable midpoint theorem gives $D - M - C$ and $DM \cong MC$. Together with the common side AM , these are the three side congruences for SSS on $\triangle ADM$ and $\triangle ACM$.

9.6.3 SSS and the final ray transport

The midpoint congruence is reversed on its second segment so that the SSS arguments have the exact orientation expected by the theorem. Reflexivity supplies

$$AM \cong AM.$$

The current call is

```

have hSSS :=
  HilbertSSS
    Geo
    A D M
    A C M
    hADM
    hAD_AC
    hDM_CM
    hAM

```

The current version of `HilbertSSS` requires noncollinearity only for the first triangle; noncollinearity of the second is derived internally from the three side congruences. This is a small but concrete correction to older documentation that displayed an extra noncollinearity argument in the call.

From SSS the proof extracts

$$\angle DAM \cong \angle CAM.$$

Finally, symmetry of the same-ray relation gives D and B on the same ray from A , and `hilbert_angle_eq_of_sameRay_first` identifies $\angle DAM$ with $\angle BAM$. This yields the public conclusion.

9.7 Lean Formalization

The production source imports only

```

import CGJteamLab.HilbertInterface

```

and the public theorem has the form

```

theorem euclid_proposition_9
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C : Geo.Point)
  (hABC : not Collinear Geo A B C) :
  exists M : Geo.Point,
    Geo.AngleCongruent B A M C A M := by
  ...

```

There are no proposition-local helper theorems and no geometric case splits. The proof is nevertheless substantive: it contains segment construction, noncollinearity propagation, midpoint construction, SSS, and final same-ray angle transport.

Source inspection of the current `Proposition09.lean` finds no `sorry` and no proposition-local `axiom`. This documentation audit did not run a fresh `lake build` or `#print axioms`, so it makes no stronger compilation or axiomatic-status claim.

9.8 Relationship with Earlier Propositions

9.8.1 Proposition I.3

I.3 is part of Euclid’s historical construction: it cuts off on one arm a segment equal to the chosen segment on the other arm. The current Lean proof does not call `euclid_proposition_3`; it uses Hilbert III.1 segment construction directly.

9.8.2 Proposition I.1

Euclid constructs the equilateral triangle DEF . No equilateral triangle is constructed in the formal proof, and `euclid_proposition_1` is not a direct dependency of `euclid_proposition_9`.

9.8.3 Proposition I.8

I.8 is the classical SSS step. The formal proof uses the reusable theorem `HilbertSSS` directly instead of calling `euclid_proposition_8`. This is an API distinction, not a change from synthetic to numerical geometry.

9.8.4 Transition to Proposition I.10

Classically, I.10 immediately reuses I.9 to bisect the apex angle of an equilateral triangle. In the current Lean development, however, I.10 is exposed through the independent midpoint theorem rather than by calling `euclid_proposition_9`. The historical dependency and the formal call graph therefore diverge at exactly this point.

9.9 Hilbert and Book Zero Components Used

9.9.1 Hilbert III.1: segment construction

The construction of D is a direct use of the congruence axiom for laying off a segment on a prescribed ray.

9.9.2 Strict midpoint infrastructure

`HilbertMidpointExists` returns the strict midpoint predicate, hence both

$$D - M - C \quad \text{and} \quad DM \cong MC.$$

The theorem is reused as an interface component; its internal neutral proof is not duplicated inside the proposition-specific diagram.

9.9.3 Hilbert SSS

The theorem `HilbertSSS` supplies triangle congruence from the three side congruences. Its current API derives noncollinearity of the second triangle internally.

9.9.4 Same-ray angle transport

The final result from SSS is expressed using the constructed arm AD . `hilbert_sameRay_symm` and `hilbert_angle_eq_of_sameRay_first` replace it by the original arm AB without changing the geometric angle.

No Book Zero theorem is called directly by `Proposition09.lean`. The file works through the Hilbert interface imported above.

9.10 Formalization Notes

The principal documentary issue in I.9 is not a missing classical step but a misleading representation of the formal proof. A flowchart saying “segment construction – midpoint – SSS” records theorem calls, not the geometry that makes those calls meaningful. The audited Figure 9.3 instead shows the actual symmetric configuration A, D, M, C .

A second correction concerns the endpoint. The current theorem returns only an angle-congruence witness M ; it does not package an explicit “interior-of-angle” or side-of-line predicate in its conclusion. The proof contains strict betweenness $D - M - C$ and same-ray information for B, D , which are exactly the positional facts used by the argument. Documentation should not strengthen this to a formal side-of-line conclusion that the theorem does not state.

Finally, the midpoint theorem is a reusable lower-layer result. Expanding its entire internal construction here would obscure modularity and duplicate the midpoint audit. I.9 should document how the midpoint is *used*: it creates the symmetric cross-segment that makes the SSS comparison possible.

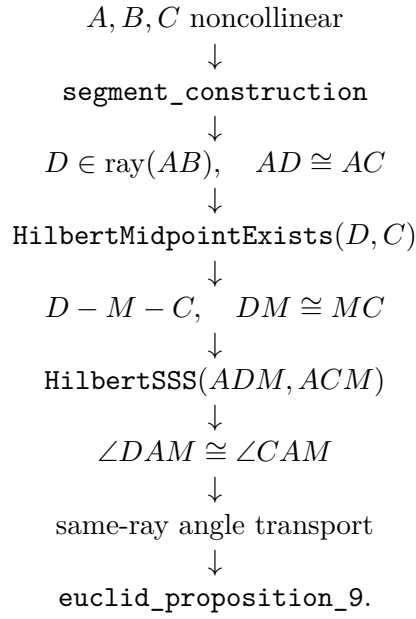
9.11 Dependency Summary

The classical and current formal dependency structures are:

Classical:

$$\text{I.3} \longrightarrow \text{I.1} \longrightarrow \text{I.8} \longrightarrow \text{I.9}.$$

Current formal reconstruction:



No numerical angle measure, coordinate system, or Euclidean parallel axiom is introduced by this proposition-level reconstruction.

9.12 Conclusion

Proposition I.9 is the first Book I construction in this sequence for which the current Hilbert proof has a visibly different auxiliary figure from Euclid’s proof. Euclid obtains symmetry from an equilateral triangle and I.8; Lean obtains it from segment layoff, a strict midpoint, and the reusable Hilbert SSS theorem.

The audited documentation now shows both pictures rather than replacing the formal geometry by a theorem-call flowchart. The formal endpoint is also recorded exactly: the constructed ray gives

$$\angle BAM \cong \angle CAM,$$

with the original arm AB recovered by same-ray transport.

Chapter 10

Proposition I.10

10.1 Introduction

Proposition I.10 constructs the midpoint of a given finite straight line. In Euclid’s development the construction is built from three earlier propositions: I.1 supplies an equilateral triangle, I.9 bisects its apex angle, and I.4 applies SAS to show that the two pieces of the original segment are congruent.

The current Lean endpoint has a very different surface shape. The theorem `euclid_proposition_10` is a thin wrapper around `HilbertMidpointExists`. A diagrammatic audit, however, must not stop at that wrapper. The interface theorem itself is only a packaging layer for Hilbert’s neutral midpoint theorem `hilbert_midpoint_exists` in `HilbertPlaneTheory`. That derived theorem contains a substantial synthetic construction involving angle construction, segment layoff, a neutral alternate-angle criterion, plane separation, Pasch, vertical angles, and AAS.

Thus Proposition I.10 is a useful example of three distinct levels that should not be conflated:

1. Euclid’s historical construction;
2. the short public Book I wrapper;
3. the hidden mathematical engine already proved in the Hilbert layer.

10.2 The Classical Configuration

Let AB be the given finite straight line. The objective is to construct a point on the open segment AB dividing it into two congruent parts.

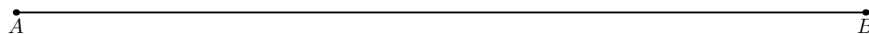


Figure 10.1: The given finite straight line AB .

10.2.1 Construction

Construct the equilateral triangle ABC on the segment AB by Proposition I.1. Bisect the angle $\angle ACB$ by Proposition I.9, and let the bisecting ray meet AB at D .

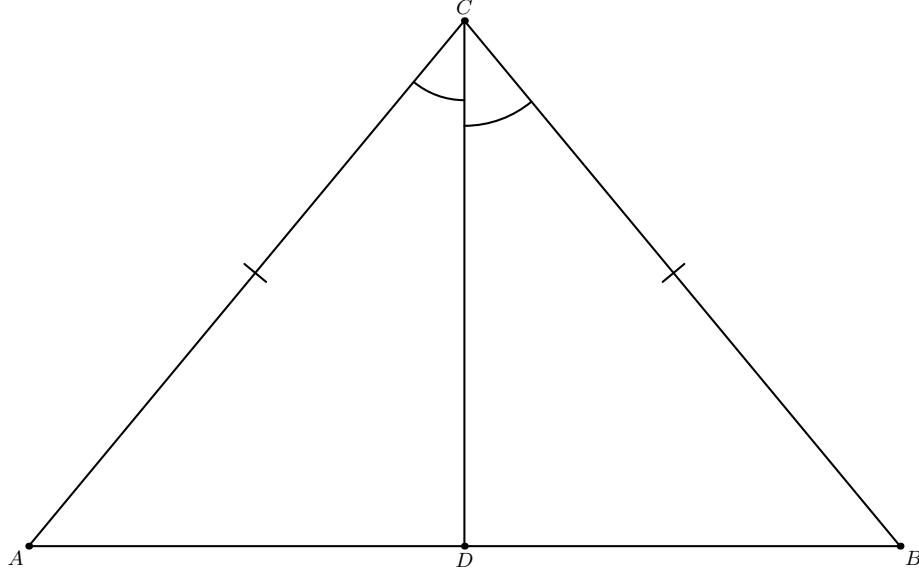


Figure 10.2: Euclid's construction for Proposition I.10. The equilateral triangle gives $AC \cong BC$; the ray CD bisects the angle at C . SAS for $\triangle ACD$ and $\triangle BCD$ yields $AD \cong DB$.

The segment CD is common to the two triangles. Hence

$$AC \cong BC, \quad CD \cong CD, \quad \angle ACD \cong \angle DCB.$$

By Proposition I.4,

$$AD \cong DB.$$

Since D lies between A and B , it is the required midpoint.

10.3 Statement

Theorem 10 (Euclid I.10). *Let AB be a nondegenerate segment, so that $A \neq B$. Then there exists a point M such that*

$$A - M - B \quad \text{and} \quad AM \cong MB.$$

Equivalently, every nondegenerate segment admits a strict Hilbert midpoint.

In the project this conclusion is represented by `HilbertIsMidpoint`. Thus the formal statement records strict betweenness as part of the midpoint data; equality of two segment magnitudes alone would not be sufficient.

Diagrammatic audit. Three geometric figures are used in the audited chapter. Figure 10.1 records the input segment. Figure 10.2 contains the complete Euclidean construction. Figure 10.3 records the lower Hilbert theorem hidden by the public wrapper. This additional formal figure is necessary because the actual mathematical proof of midpoint existence is not the one-line wrapper: it is already stored one layer below in `hilbert_midpoint_exists`.

10.4 Classical Proof

Euclid's proof is a compact synthesis of Propositions I.1, I.9, and I.4. Construct the equilateral triangle ABC on AB . Bisect $\angle ACB$ and let the bisector meet AB at D . Then

$$AC \cong BC, \quad CD \cong CD, \quad \angle ACD \cong \angle DCB.$$

SAS gives congruence of $\triangle ACD$ and $\triangle BCD$, hence $AD \cong DB$. The construction of D on the segment supplies the required order relation, so D is the midpoint of AB .

The classical dependency chain is therefore

$$\text{I.1} \longrightarrow \text{I.9} \longrightarrow \text{I.4} \longrightarrow \text{I.10}.$$

10.5 Proof Architecture of the Reconstruction

The current production theorem is

```
theorem euclid_proposition_10
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B : Geo.Point)
  (hAB : A != B) :
  exists M : Geo.Point,
    HilbertIsMidpoint Geo M A B := by
exact
  HilbertMidpointExists
    Geo A B hAB
```

At this level there is no proposition-local helper and no case split. It is important, however, not to mistake this short endpoint for the complete proof architecture. The current `HilbertInterface` explicitly documents `HilbertMidpointExists` as a stable wrapper for the neutral theorem `hilbert_midpoint_exists` from `HilbertPlaneTheory`.

Thus the public path is

$$\text{hilbert_midpoint_exists} \longrightarrow \text{HilbertMidpointExists} \longrightarrow \text{euclid_proposition_10}.$$

The middle arrow is packaging: the primitive conclusion

$$\exists E, \quad A - E - B \quad \text{and} \quad AE \cong EB$$

is repackaged as the project predicate `HilbertIsMidpoint`.

10.6 Hilbert Reconstruction

The hidden mathematical engine is Hilbert's Theorem 26 as reconstructed in `hilbert_midpoint_exists`. It does not replay Euclid's equilateral triangle and angle-bisector proof. Instead it constructs two congruent, parallel side segments on opposite sides of the given base and forces their joining segment to cross the base at its midpoint.

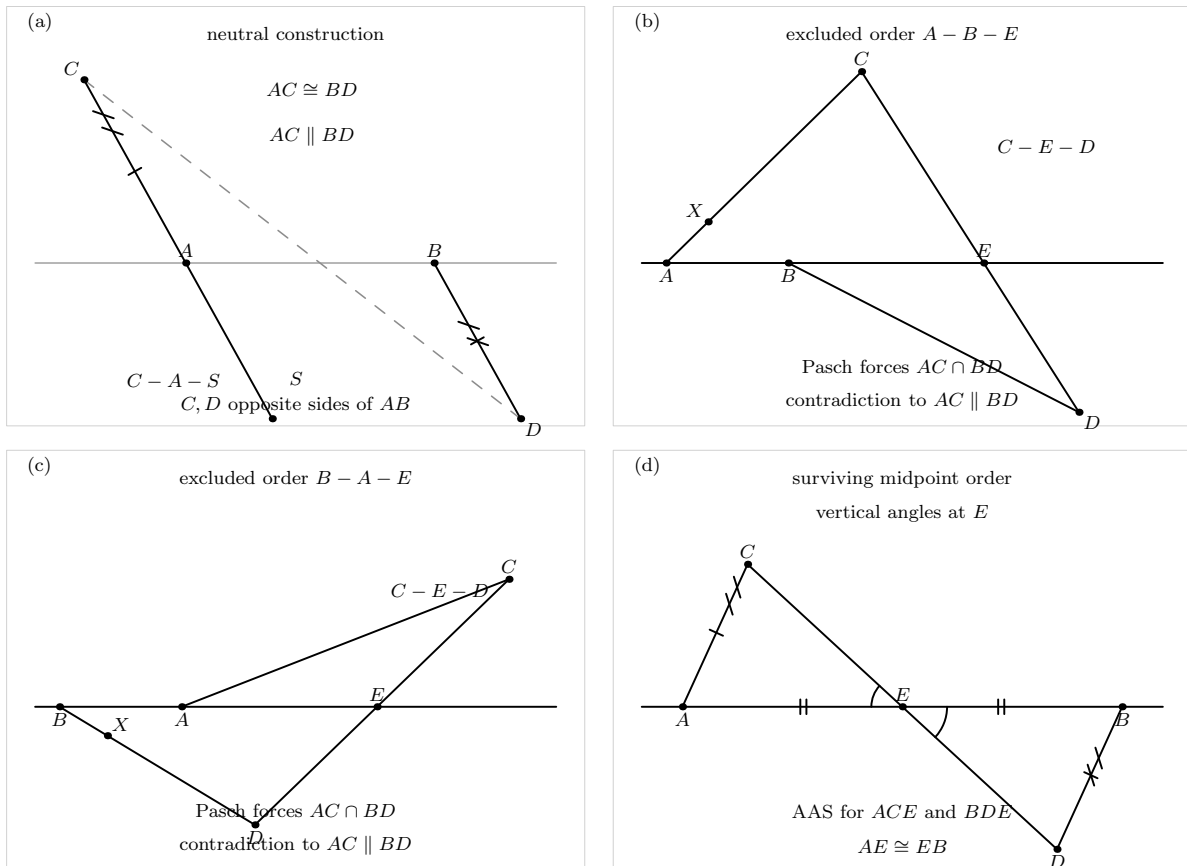


Figure 10.3: Formal Hilbert geometry underlying Proposition I.10. Panel (a) records the neutral construction of congruent parallel sides on opposite sides of AB . Panels (b) and (c) show the two exterior orders for the crossing point E that are excluded by Pasch together with disjointness of the parallel carriers. Panel (d) is the surviving order $A - E - B$; vertical angles at E and AAS for $\triangle ACE$ and $\triangle BDE$ yield $AE \cong EB$.

10.6.1 Choose an exterior point and the opposite side

Let `base` be the line through A and B . The proof chooses a point C outside this line. It then extends the segment CA beyond A to a point S :

$$C - A - S.$$

Consequently C and S lie on opposite sides of `base`. The role of S is purely orientational: it selects the side on which the angle at B is to be constructed.

10.6.2 Copy the angle and lay off a congruent side

Hilbert III.4 constructs a ray at B on the side of AB containing S so that its angle with BA is congruent to $\angle BAC$. The source code first names a representative point D_0 on this ray. Hilbert III.1 then lays off on the same ray a point D satisfying

$$BD \cong AC.$$

Because D remains on the selected side, C and D are on opposite sides of the base.

The representative D_0 is not given a separate diagram panel. Its sole role is to create the ray; after segment construction, all stable geometric data are carried by D and the same-ray relation between D_0 and D .

10.6.3 Obtain the neutral parallel pair

A point T between A and B is chosen to normalize the two directions of the transversal. Same-ray transport converts the copied angle into a pair of equal alternate angles. The neutral converse theorem for alternate angles then yields

$$AC \parallel BD.$$

This implication uses no Euclidean parallel axiom. The proof of `hilbert_midpoint_exists` is explicitly placed before the declaration of the Hilbert Group IV class and states that only Groups I–III are used.

The point T is another normalization witness rather than a persistent new state of the construction, so it is also omitted from the final diagram. Panel (a) of Figure 10.3 records the stable result: $AC \cong BD$, $AC \parallel BD$, and C, D on opposite sides of AB .

10.6.4 Construct the crossing point

Since C and D are on opposite sides of `base`, the definition of `HilbertOppositeSide` supplies a point E with

$$C - E - D$$

and E on `base`. At this stage the proof knows that A, B, E are collinear, but it has not yet established the desired order $A - E - B$.

The parallel lines through AC and BD are represented by their incidence carriers. Their disjointness is used repeatedly below to forbid a second intersection of the transversal CD with either parallel side.

10.6.5 Exclude the exterior order $A - B - E$

Assume temporarily that

$$A - B - E.$$

The open segment AE then meets the carrier BD at B . Pasch is applied to the triangle AEC . The other candidate side EC cannot meet BD : such an intersection would force a second common point of the crossing line and BD , and hence eventually contradict the disjointness of AC and BD . Pasch therefore forces the remaining side AC to meet BD , directly contradicting the parallelism.

This is the geometric content of panel (b). It is a genuine branch of the proof, not merely a tactic artifact, because it assumes a different strict betweenness order of the actual crossing point E .

10.6.6 Exclude the exterior order $B - A - E$

The second possible exterior order

$$B - A - E$$

is eliminated symmetrically. Now the segment BE meets the carrier AC at A , and Pasch is applied to the triangle BED . The side ED cannot meet AC a second time, so Pasch would force BD to meet AC , again contradicting parallelism.

Panel (c) records this second genuine order branch.

10.6.7 The surviving order

The three points A, B, E are distinct and collinear. Hilbert's betweenness trichotomy therefore leaves only three possible strict orders. The first two have just been excluded, so

$$A - E - B.$$

This is exactly the positional half of the midpoint conclusion.

10.6.8 Vertical angles and AAS

It remains to prove equality of the two halves. Since

$$A - E - B \quad \text{and} \quad C - E - D,$$

the angles at the crossing point are vertical:

$$\angle AEC \cong \angle BED.$$

The copied-angle construction, transported along the rays AE and BE , gives the second angle congruence needed for the triangles ACE and BDE . Together with

$$AC \cong BD,$$

the reusable AAS theorem `hilbert_aas_sides` yields the remaining side congruence. After reversing the second segment representation where needed, the conclusion is

$$AE \cong EB.$$

Combining this with $A - E - B$ gives the required strict midpoint.

10.7 Lean Formalization

The public theorem itself remains intentionally small. It should be read as an API endpoint, not as a claim that midpoint existence is primitive:

```
exact
  HilbertMidpointExists
    Geo A B hAB
```

The interface theorem in turn packages the lower result. The important architectural distinction is therefore:

- `euclid_proposition_10`: Book I-facing wrapper;
- `HilbertMidpointExists`: stable interface wrapper;
- `hilbert_midpoint_exists`: mathematical proof.

No proposition-local construction occurs in `Proposition10.lean`, but a substantial construction does occur in the imported Hilbert layer. The documentation now records both facts instead of collapsing them into the statement that “no auxiliary construction is required.”

10.8 Relationship with Earlier Propositions

10.8.1 Proposition I.1

In Euclid’s proof, I.1 supplies the equilateral triangle on AB . This is a classical dependency only. The current `euclid_proposition_10` does not call `euclid_proposition_1`.

10.8.2 Proposition I.9

I.9 supplies the classical angle bisector at the apex of the equilateral triangle. Again, it belongs to the Euclidean historical DAG, not the direct Lean call graph of Proposition I.10.

10.8.3 Proposition I.4

SAS is the decisive final comparison in Euclid’s proof. The lower Hilbert midpoint theorem follows a different route and finishes with the already derived AAS theorem `hilbert_aas_sides` rather than by calling the Book I proposition.

10.8.4 Transition to Proposition I.11

Midpoint infrastructure is used immediately in the formal development of right-angle existence underlying Proposition I.11. This makes I.10 and I.11 a useful adjacent pair for understanding the project architecture: both have very short public endpoints while their reusable Hilbert-layer existence theorems contain the substantial geometry.

10.9 Hilbert Components Used

10.9.1 Strict midpoint representation

The target predicate is

$$\text{HilbertIsMidpoint } Geo\ M\ A\ B,$$

which packages

$$A - M - B \quad \text{and} \quad AM \cong MB.$$

The order component is therefore intrinsic to the result.

10.9.2 Incidence and order

The construction uses a line through A, B , a point outside that line, segment extension, plane separation, betweenness trichotomy, and uniqueness properties of incidence lines. These operations are not coordinates; they are consequences of Hilbert's incidence and order groups.

10.9.3 Congruence constructions

Hilbert III.4 copies the angle at A onto the selected side at B , while Hilbert III.1 lays off $BD \cong AC$. Same-ray transport ensures that replacing the initial angle-ray representative by the laid-off point does not change the constructed angle.

10.9.4 Neutral parallel criterion

The equality of alternate angles is used in the converse direction to prove $AC \parallel BD$. This direction is neutral and does not require Hilbert's Euclidean uniqueness-of-parallels axiom.

10.9.5 Pasch and plane separation

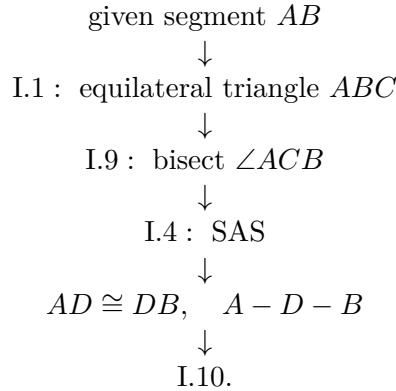
Pasch is the key tool for excluding the two exterior orders of E . In each branch, one side of a triangle is known to meet one of the parallel carriers; another side is proved not to meet it, so Pasch forces the third side to meet the supposedly disjoint parallel carrier. The contradiction is entirely synthetic.

10.9.6 Vertical angles and AAS

The final side congruence uses the vertical-angle theorem at the intersection E and the reusable AAS theorem for the two triangles. Thus the formal Hilbert proof and Euclid's classical proof both finish by triangle congruence, but they reach the comparison through different constructions and different congruence criteria.

10.10 Dependency Summary

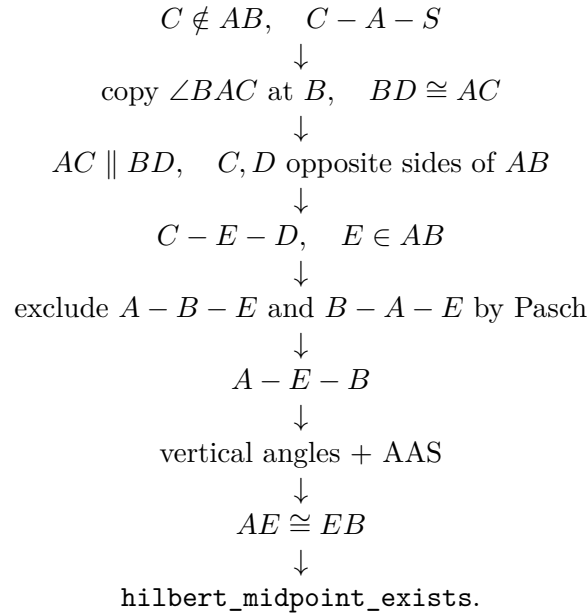
The classical architecture is



The public formal architecture is

$$\text{HilbertMidpointExists} \longrightarrow \text{euclid_proposition_10.}$$

The hidden mathematical architecture is



The final API layering is therefore

$$\text{hilbert_midpoint_exists} \longrightarrow \text{HilbertMidpointExists} \longrightarrow \text{euclid_proposition_10.}$$

No numerical length, coordinate system, continuity assumption, or Hilbert Group IV parallel axiom is introduced in this chain.

10.11 Diagrammatic Audit

The preserved Book1R chapter already had two useful classical pictures: the input segment and the completed Euclidean construction. Both are retained, now as Asymptote figures so that the chapter follows the same graphics policy as the later audited propositions.

The old documentation stopped the formal diagram analysis at `HilbertMidpointExists`. That was too shallow. The current source states that this interface theorem merely packages `hilbert_midpoint_exists`; the latter contains real geometric states that are not visible in the classical figure.

Figure 10.3 therefore adds a four-panel formal map. The first panel records the stable neutral construction after the angle-ray and segment-layoff witnesses have been normalized. The next two panels correspond to the two strict exterior betweenness branches actually eliminated in the proof. The final panel records the surviving order and triangle comparison.

No separate panel is added for D_0 , for the internal point T used in angle normalization, or for the final congruence-reversal operation. These steps do not create new persistent geometric configurations. The audited figure set is therefore

3 figures, 6 panels in total.

10.12 Conclusion

Proposition I.10 is simultaneously one of the shortest public Book I theorems and the endpoint of a substantial earlier construction. Euclid reaches the midpoint through an equilateral triangle, an angle bisector, and SAS. The current Hilbert layer reaches it through a different neutral configuration: copy an angle on the opposite side, lay off a congruent side, obtain parallel carriers from alternate angles, force their transversal to meet the base, exclude the two exterior orders by Pasch, and finish with vertical angles and AAS.

The public Lean theorem hides this work deliberately behind a reusable interface. The correct documentation must therefore distinguish theorem size from mathematical dependency depth. Once that distinction is made, I.10 becomes a particularly clear illustration of the library's layered architecture and prepares the reader for the same phenomenon in Proposition I.11.

Chapter 11

Proposition I.11

11.1 Introduction

Proposition I.11 states:

To draw a straight line at right angles to a given straight line from a prescribed point on it.

Let AB be the given straight line and let C be the prescribed point on AB .

Euclid's construction is concrete. Choose a point D on one side of C , lay off on the opposite ray a point E with

$$CD \cong CE,$$

construct the equilateral triangle DFE , and join C to F . SSS for $\triangle DCF$ and $\triangle ECF$ gives equality of the adjacent angles at C . Since $D - C - E$, Euclid's Definition 10 identifies those equal adjacent angles as right angles.

The current Hilbert reconstruction proves the same existence statement by a different synthetic mechanism. The public theorem in `Proposition11.lean` is only an interface theorem; the substantial geometry lives in the Book Zero theorem `hilbert_right_angle_exists_nondegenerate`.

11.2 The Classical Configuration

The classical configuration is

$$D - C - E, \quad CD \cong CE, \quad DF \cong FE,$$

with DFE equilateral and CF joined. The proof compares

$$\triangle DCF \quad \text{and} \quad \triangle ECF.$$

11.3 Statement

The synthetic content of Proposition I.11 is the existence of a right angle through a prescribed point of a straight line.

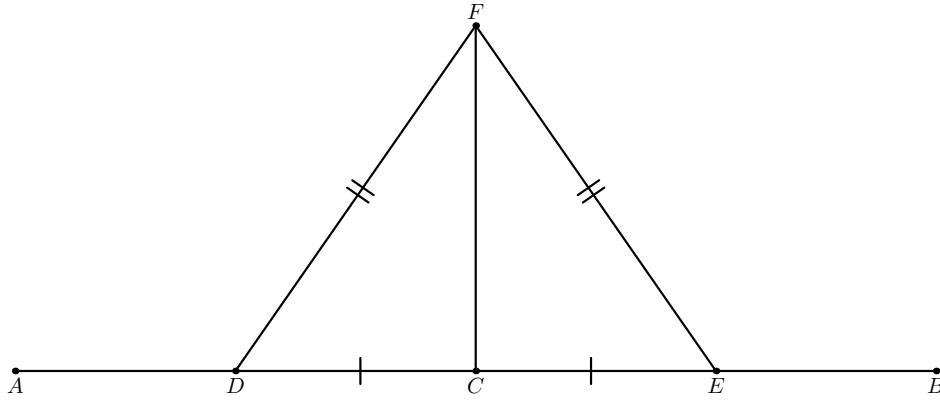


Figure 11.1: Classical Euclid–Wilkins construction for Proposition I.11. The points D, C, E are collinear with C between D and E and $CD \cong CE$; DFE is equilateral. The segment CF is the candidate perpendicular.

Theorem 11 (Euclid I.11). *Let A, C, B satisfy*

$$A * C * B.$$

Then there exists a point X such that

$$\text{HilbertRightAngle}(A, C, X).$$

The right-angle predicate is the synthetic adjacent-angle formulation:

$$\text{HilbertRightAngle}(A, O, B)$$

means that there exists a point C' such that

$$A * O * C'$$

and

$$\angle AOB \cong \angle BOC'.$$

No numerical value such as 90° is used.

The current proposition file has the form

```
theorem euclid_proposition_11
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A C B : Geo.Point)
  (hACB : Geo.Between A C B) :
  Exists (fun X : Geo.Point =>
    HilbertRightAngle Geo A C X) := by
exact
  hilbert_right_angle_exists
    Geo A C B hACB
```

Thus the theorem carrying Euclid's name is a thin wrapper.

11.4 Classical Proof

Choose D on one side of C on the given line. By Proposition I.3, lay off on the opposite ray a point E such that

$$CD \cong CE.$$

Hence

$$D - C - E.$$

By Proposition I.1 construct the equilateral triangle DFE and join C to F ; see Figure 11.1. Then

$$DF \cong EF, \quad CF \cong CF.$$

Together with $CD \cong CE$, Proposition I.8 gives

$$\angle DCF \cong \angle ECF.$$

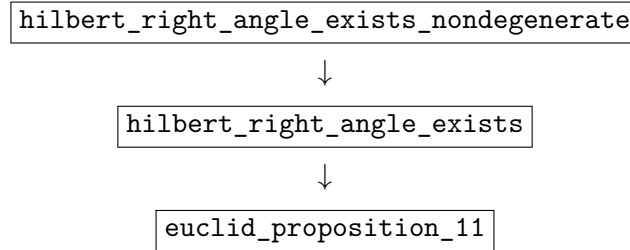
Since CD and CE are opposite rays, the two congruent angles are adjacent angles on a straight line. By Euclid's Definition 10 they are right angles. Therefore CF is perpendicular to the given line at C .

The classical dependency pattern is therefore

$$\text{I.3} \longrightarrow \text{segment layoff} \longrightarrow \text{I.1} \longrightarrow \text{I.8} \longrightarrow \text{equal adjacent angles} \longrightarrow \text{I.11}.$$

11.5 Current Formal Architecture

The current production architecture has three theorem levels:



The first theorem proves the stronger conclusion

$$\exists X, \quad \neg \text{Collinear}(A, C, X) \quad \wedge \quad \text{HilbertRightAngle}(A, C, X).$$

The second theorem discards the explicit noncollinearity component and returns only the right-angle witness. The proposition theorem is then a direct wrapper over that result.

This corrects an older description in which `hilbert_right_angle_exists` was presented as the theorem that carried the entire construction itself.

11.6 Hilbert Reconstruction

Assume

$$A * C * B.$$

Let b be the line through A and B , so $C \in b$.

11.6.1 Off-line point and angle construction

Choose an arbitrary point $D \notin b$. Hilbert III.4 is then applied at C to copy the angle ACD to the ray based on CB , on the side of b selected by D . It produces a point E with

$$\text{SameSide}(E, D; b)$$

and

$$\angle ACD \cong \angle BCE.$$

The current source also returns the uniqueness field of the angle construction theorem, but that field is not consumed by the active proof of I.11. The proof uses the side condition and the angle congruence.

At this point the proof performs its first genuine geometric case split:

$$\text{Collinear}(C, E, D) \quad \text{or} \quad \neg \text{Collinear}(C, E, D).$$

The four formal states needed for the audit are collected in Figure 11.2.

11.6.2 Collinear branch

Suppose C, E, D are collinear. Because E and D lie on the same side of the original base line, they cannot lie on opposite rays from C . The proof obtains

$$\text{SameRay}(C; E, D).$$

Hence the copied angle congruence transports to

$$\angle ACD \cong \angle BCD$$

and then, by reversal of the second angle,

$$\angle ACD \cong \angle DCB.$$

Since $A - C - B$, the point B supplies the extension witness in the definition of a right angle. Thus this branch ends immediately with

$$\text{HilbertRightAngle}(A, C, D).$$

In particular, the output point in this branch is simply $X = D$.

11.6.3 Noncollinear branch: layoff and midpoint

Now suppose

$$\neg \text{Collinear}(C, E, D).$$

On ray CD , segment construction produces F with

$$CF \cong CE.$$

Then midpoint existence produces X satisfying

$$F - X - E, \quad FX \cong XE.$$

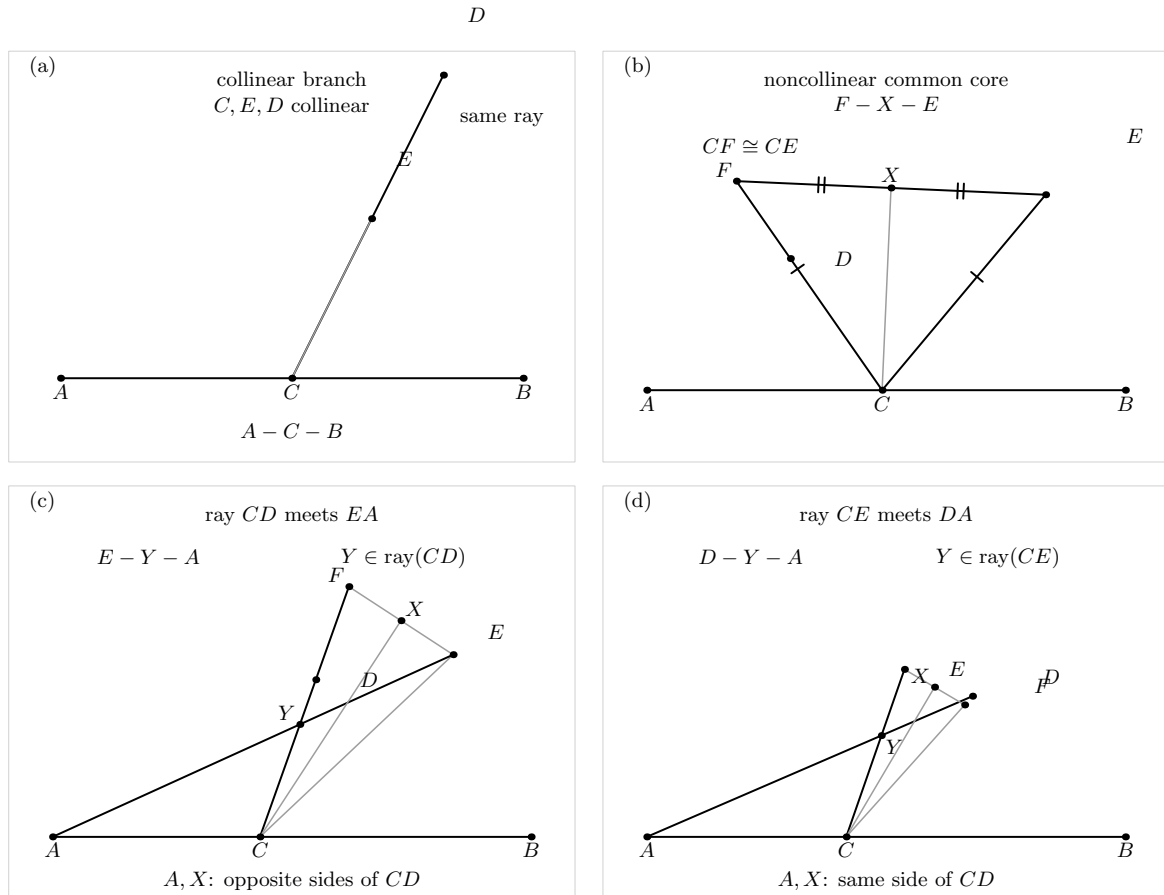


Figure 11.2: Formal Hilbert geometry used by the current I.11 proof. (a) If C, E, D are collinear, the same-side information forces CE and CD to be the same ray, and D itself is returned as the right-angle point. (b) In the noncollinear branch, F is laid off on ray CD with $CF \cong CE$, and X is the midpoint of FE . Panels (c) and (d) are the two subsequent ray-order cases: ray CD meets the open segment EA , or ray CE meets the open segment DA . The relative locations are qualitative representatives of the proved incidence and side data; no additional point order is asserted beyond the labelled betweenness relations.

Hilbert SSS applied to

$$\triangle CFX \quad \text{and} \quad \triangle CEX$$

gives

$$\angle FCX \cong \angle ECX.$$

Since F and D are on the same ray from C , this becomes

$$\angle DCX \cong \angle ECX.$$

Together with the angle-copy relation

$$\angle ACD \cong \angle BCE,$$

these are the two component congruences to be assembled by angle addition.

11.6.4 Ray-order case split

The noncollinear configuration still does not determine automatically how the component angles are arranged. The proof therefore invokes the ray-order theorem and obtains exactly two possibilities:

$$\text{RayMeetsSegment}(CD, EA)$$

or

$$\text{RayMeetsSegment}(CE, DA).$$

In the first case there is a point Y with

$$E - Y - A$$

and Y on ray CD . The SameSide/OppositeSide calculus proves that A and X are on opposite sides of line CD , while B and X are on opposite sides of line CE .

In the second case there is a point Y with

$$D - Y - A$$

and Y on ray CE . Here the corresponding pairs are on the same sides of their reference lines.

Both branches therefore establish the invariant

$$\text{SameSide}(A, X; CD) \iff \text{SameSide}(B, X; CE).$$

This is precisely the side configuration required by the synthetic angle-addition theorem.

11.6.5 Angle addition and right angle

Angle addition combines

$$\angle ACD \cong \angle BCE$$

and

$$\angle DCX \cong \angle ECX$$

to obtain

$$\angle ACX \cong \angle BCX.$$

After reversing the second angle,

$$\angle ACX \cong \angle XCB.$$

The original hypothesis $A - C - B$ means that the existing point B is already the extension witness required by `HilbertRightAngle`. No further geometric construction is needed at the endpoint.

11.7 Lean Formalization

The current formal route can be summarized as follows:

```

hilbert_right_angle_exists_nondegenerate
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A C B : Geo.Point)
  (hACB : Geo.Between A C B) :
  Exists (fun X : Geo.Point =>
    And (Not (PrimCollinear Geo A C X))
        (HilbertRightAngle Geo A C X))

```

The theorem constructs the line through A, B , an off-line point D , and an angle-copy point E , and then splits on

```

by_cases hCED : Collinear Geo C E D

```

The collinear branch ends with D as the right-angle point. Only the noncollinear branch constructs F and midpoint X and later executes

```

rcases hRayOrder with hCaseD | hCaseE

```

The two ray-order branches establish the same side-configuration biconditional and then share the final angle-addition step.

The next theorem is only a projection. In ASCII-only Lean notation, the logical content can be written equivalently as

```

theorem hilbert_right_angle_exists ... := by
  exact Exists.elim
    (hilbert_right_angle_exists_nondegenerate
      Geo A C B hACB)
    (fun X h => Exists.intro X h.2)

```

Finally, `euclid_proposition_11` calls `hilbert_right_angle_exists` directly and introduces no new geometry.

11.8 Relationship with Earlier Propositions

11.8.1 Proposition I.1

Euclid uses I.1 to construct the equilateral triangle DFE . The current Hilbert proof does not call `euclid_proposition_1`; its auxiliary geometry is obtained from Hilbert angle and segment construction and midpoint existence.

11.8.2 Proposition I.3

Euclid uses I.3 for the layoff $CE \cong CD$ on the opposite ray. The formal reconstruction uses the generic Hilbert segment-construction machinery instead of calling the proposition theorem.

11.8.3 Proposition I.8

Euclid uses I.8 for SSS on DCF and ECF . The current formal proof also contains an SSS step, but in the different pair of triangles CFX and CEX , and it invokes the generic `HilbertSSS` interface rather than `euclid_proposition_8`.

11.8.4 Transition to Proposition I.12

I.11 constructs a perpendicular at a point already on the given line. I.12 solves the complementary problem from a point outside the line. Together they form the basic perpendicular-construction block used by the subsequent straight-angle propositions.

11.9 Hilbert and Book Zero Components Used

The active proof of `hilbert_right_angle_exists_nondegenerate` uses, among other things:

- line construction and an off-line point;
- Hilbert III.4 angle construction;
- SameSide/OppositeSide and same-ray transport;
- segment construction on a ray;
- midpoint existence;
- HilbertSSS;
- the same-side ray-order theorem;
- line uniqueness and plane-separation lemmas;
- synthetic angle addition.

The angle-construction uniqueness field is returned by the API but is not used in the current active I.11 proof. It should therefore not be shown as a dependency edge of this proposition.

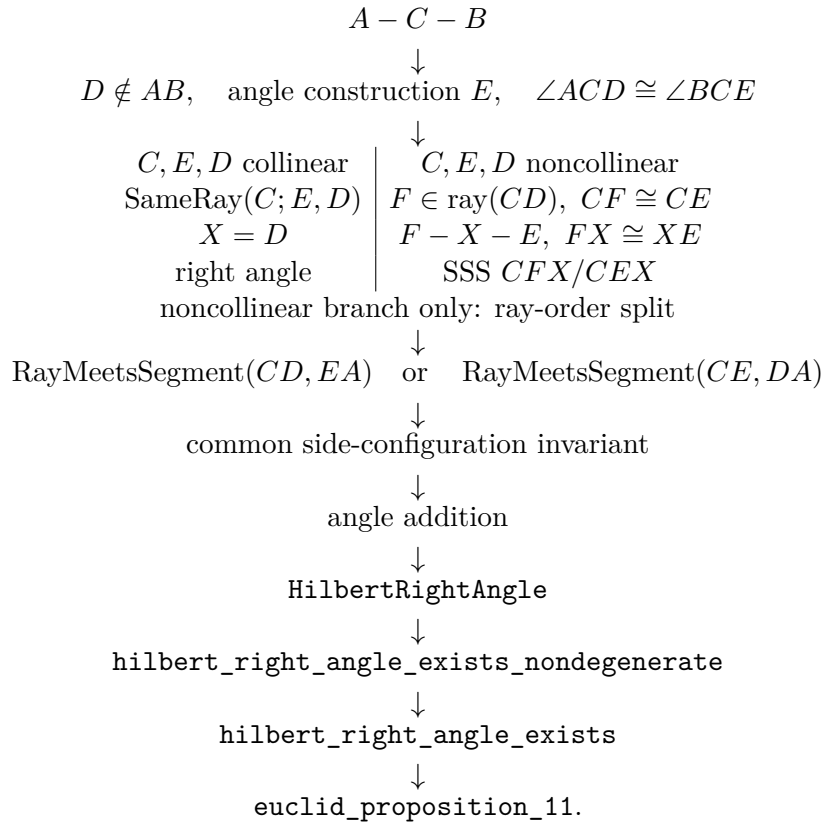
11.10 Dependency Summary

The classical and formal architectures should be kept distinct.

For Euclid:

$$\text{I.3} \longrightarrow \text{I.1} \longrightarrow \text{I.8} \longrightarrow \text{equal adjacent angles} \longrightarrow \text{I.11}.$$

For the current Hilbert reconstruction:



11.11 Diagrammatic Audit

The preserved Book1R chapter contained one numbered geometric figure: the classical Euclid–Wilkins construction. That figure is retained as Figure 11.1.

The current production Lean, however, contains two genuine geometric case splits that the preserved diagram set did not record. The first split distinguishes whether the angle-copy point E is collinear with C, D . The second, inside the noncollinear branch, distinguishes which of the two rays CD and CE meets the opposite open segment. These are not tactic-only alternatives: they are different incidence and side-of-line configurations. Figure 11.2 therefore records them in one four-panel formal map.

No separate figure is added for the final angle-addition step because no new point, line, betweenness, intersection, or side relation appears there. Likewise the uniqueness field returned by angle construction is not diagrammed because it is not consumed by the current proof.

Thus the audited set is

$$1 \text{ classical panel} \quad + \quad 4 \text{ formal panels} \quad = \quad 5 \text{ panels in 2 figures.}$$

11.12 Conclusion

Proposition I.11 is a clear example of reconstruction rather than literal transcription. Euclid obtains the perpendicular from a symmetric equilateral-triangle construction and I.8. The Hilbert library proves a stronger reusable nondegenerate right-angle existence theorem through angle

construction, segment layoff, midpoint geometry, SSS, plane separation, ray order, and angle addition; two thin wrappers then expose the result as Euclid I.11.

The audited diagrams now reflect both layers without mixing them: one classical construction and one formal four-panel map of the actual current case structure.

Chapter 12

Proposition I.12

12.1 Introduction

Euclid’s Proposition I.12 asks for the construction of a perpendicular to a given straight line from a prescribed point outside that line.

The classical proof uses a circle. A point is chosen on the side of the given line opposite the prescribed point; the circle centered at the prescribed point through that auxiliary point meets the given line in two points. Proposition I.10 bisects the segment joining those intersections, and Proposition I.8 proves that the line from the external point to the midpoint is perpendicular to the given line.

The current Hilbert reconstruction follows a different synthetic route. It constructs a second point on the opposite side of the base so that two distinct points of the base are equidistant from the original point and the constructed point. The base is then treated as a perpendicular-bisector axis. No circle-line intersection theorem is used.

The current production file contains the general construction theorem

```
hilbert_perpendicular_from_point_exists
```

and the public Euclidean theorem

```
euclid_proposition_12
```

which is a thin wrapper around it.

12.2 The Classical Configuration

Let AB be the given straight line and let C be the prescribed point outside it. Euclid’s construction produces a point H on AB such that

$$CH \perp AB.$$

The classical construction is summarized in Figure [12.1](#). The initial-data-only state from the preserved Book1R chapter is not shown as a separate panel: no new geometry has yet been created at that stage.

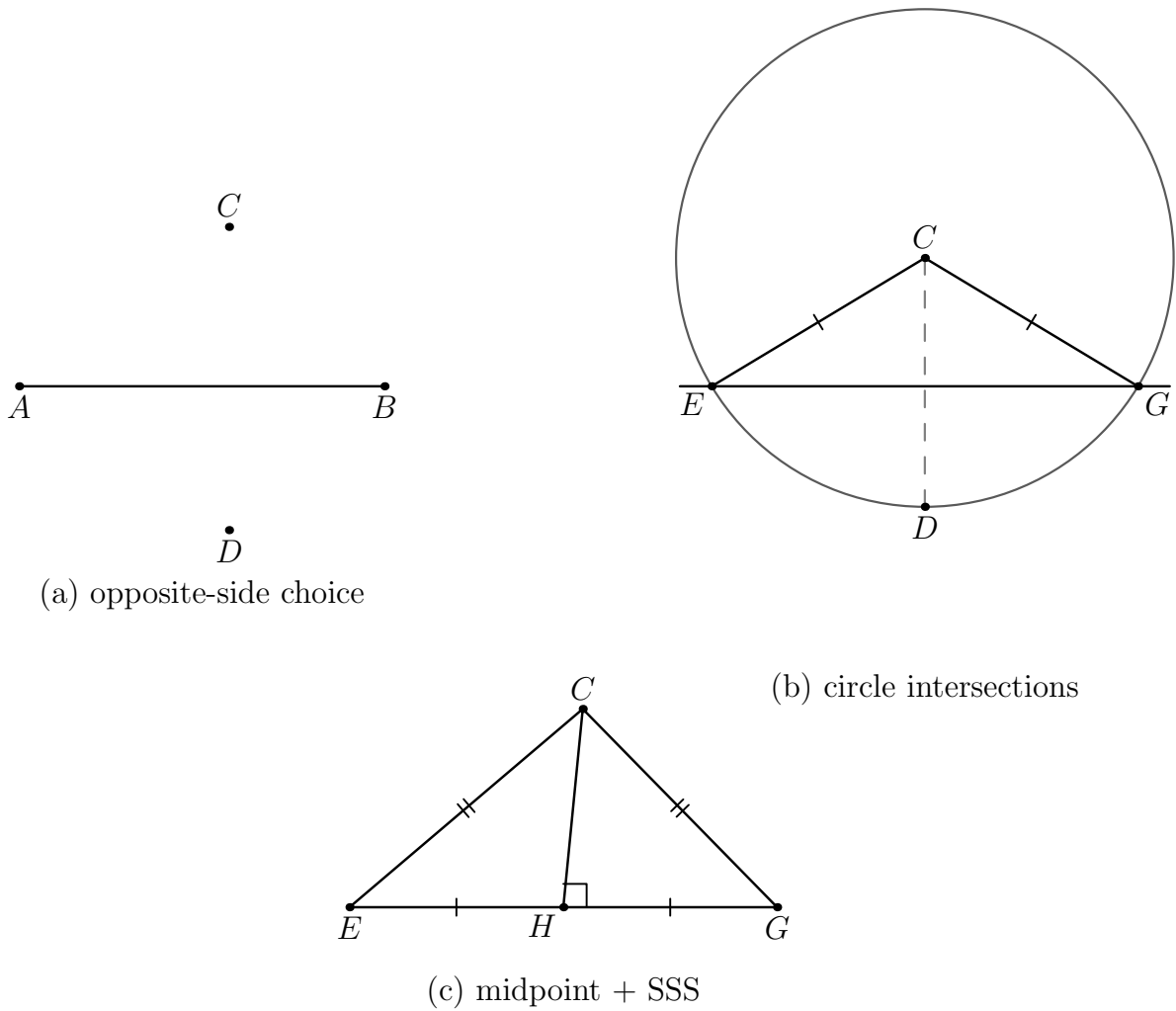


Figure 12.1: Classical Euclid–Wilkins construction, retaining only genuinely new geometric states. (a) A point D is chosen on the side of the line AB opposite the external point C . (b) The circle centered at C through D meets the given line at E and G , so $CE \cong CG$. (c) Proposition I.10 gives the midpoint H of EG ; after joining CH , $EH \cong HG$, $CE \cong CG$, and the common side CH form the SSS configuration used by Proposition I.8.

12.2.1 Classical Construction Algorithm

Step 1: given line and external point. The data are the straight line AB and a point C not on it. This is only the initial state and is therefore not assigned a separate figure under the present audit standard.

Step 2: choose an opposite-side point. Choose a point D on the side of AB opposite C ; this is Figure 12.1(a).

Step 3: draw the circle centered at C . Draw the circle centered at C through D , and let it meet the given line at two distinct points E and G . Then

$$CE \cong CG.$$

This is Figure 12.1(b).

The existence of the two intersections is a genuine geometric existence issue. The diagram records the configuration after those intersections have been obtained; it does not prove that they exist. This is precisely the kind of hidden classical assumption that should not be replaced by visual intuition.

Step 4: bisect EG . Apply Proposition I.10 to obtain a midpoint H :

$$E - H - G, \quad EH \cong HG.$$

The former midpoint-only snapshot is not retained as a separate panel, because the stable next state immediately joins CH and uses the same points.

Step 5: join CH . The completed proof state is Figure 12.1(c). The triangles EHC and GHC satisfy

$$EH \cong HG, \quad HC \cong HC, \quad CE \cong CG.$$

By Proposition I.8,

$$\angle EHC \cong \angle GHC.$$

Since E, H, G are collinear, these are congruent adjacent angles and are therefore right angles. Hence

$$CH \perp AB.$$

12.3 Classical Proof Architecture

The direct classical dependency chain is

$$\begin{array}{c}
 \text{choose opposite-side } D \\
 \downarrow \\
 \text{circle centered at } C \text{ through } D \\
 \downarrow \\
 E, G \in AB, \quad CE \cong CG \\
 \downarrow \\
 \text{I.10} \\
 \downarrow \\
 E - H - G, \quad EH \cong HG \\
 \downarrow \\
 \text{I.8 / SSS} \\
 \downarrow \\
 \angle EHC \cong \angle GHC \\
 \downarrow \\
 CH \perp AB.
 \end{array}$$

Thus the proposition-level dependencies of the classical proof include I.10 and I.8. The circle-line intersection itself also carries a classical existence burden not represented by either proposition.

12.4 Formal Architecture

The current production file imports only

CGJteamLab.HilbertBookZero

and defines two proposition-specific theorems:

```
hilbert_perpendicular_from_point_exists
euclid_proposition_12
```

The second theorem is a thin wrapper. All geometric work is performed by the first theorem together with reusable Hilbert and Book Zero infrastructure.

The main theorem has the form

```
theorem hilbert_perpendicular_from_point_exists
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B P : Geo.Point)
  (base : Geo.Line)
  (hAB : A != B)
  (hAbase : HilbertIncidence.OnLine A base)
  (hBbase : HilbertIncidence.OnLine B base)
  (hPbase : not HilbertIncidence.OnLine P base) :
  exists H R : Geo.Point,
    HilbertIncidence.OnLine H base /\
    HilbertIncidence.OnLine R base /\
    HilbertRightAngle Geo R H P
```

Here H is the foot of the required perpendicular. The point R is a second point of the base used to specify the base direction in the predicate `HilbertRightAngle`.

12.5 Hilbert Reconstruction

The formal construction is summarized in Figure [12.2](#).

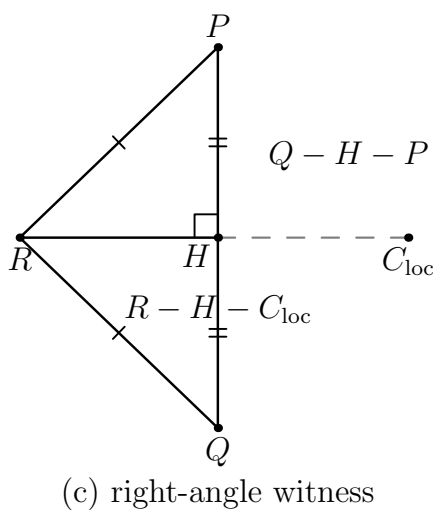
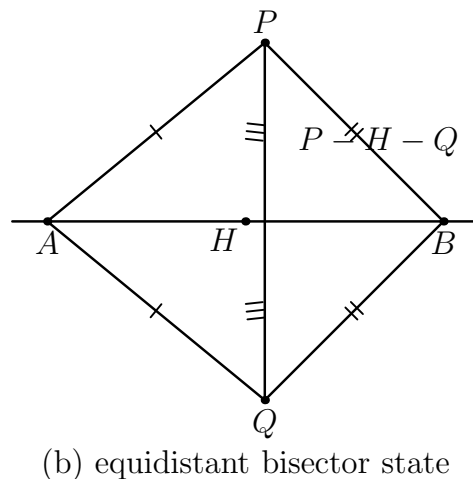
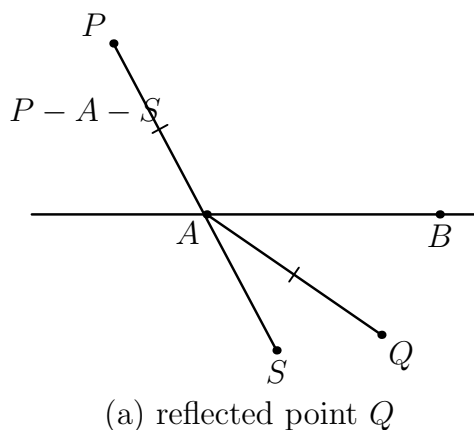


Figure 12.2: Formal Hilbert construction. (a) The line PA is extended through A to S ; Hilbert angle construction supplies a direction and segment layoff produces Q with $AQ \cong AP$. The intermediate point Q_0 used only to determine that direction is intentionally omitted. (b) SAS gives $BP \cong BQ$; because P and Q are on opposite sides of the base, $P-H-Q$ for an intersection point H , and the reusable perpendicular-bisector theorem gives $PH \cong HQ$. The displayed order of A, H, B is schematic only: the Lean hypotheses assert only that all three lie on the base. (c) After choosing $R \in \text{base}$ with $R \neq H$ and $RP \cong RQ$, the local continuation point C_{loc} with $R-H-C_{\text{loc}}$ packages the final right-angle witness.

12.5.1 Constructing the opposite-side point Q

First extend PA through A :

$$P - A - S.$$

Since P is off the base and A lies on it, P and S are on opposite sides of the base.

Hilbert angle construction at A produces an auxiliary direction point Q_0 on the side containing S . Segment construction then lays off a point Q on the same ray so that

$$AQ \cong AP.$$

The current proof subsequently transports the side information from Q_0 to Q , obtaining

$$\text{OppositeSide}(P, Q; \text{base}).$$

The point Q_0 is a genuine construction witness, but it is only a direction scaffold. The stable proposition-specific geometry uses Q , and Figure 12.2(a) therefore does not draw Q_0 as a separate state. The uniqueness field returned by this particular angle-construction call is also not consumed later in the current proof.

12.5.2 The two equidistance relations

After replacing the auxiliary ray AQ_0 by the ray AQ in the angle congruence, SAS is applied to the triangles APB and AQB . Together with

$$AP \cong AQ, \quad AB \cong AB,$$

it gives

$$BP \cong BQ.$$

The opposite-side relation provides an intersection point H with

$$P - H - Q, \quad H \in \text{base}.$$

The resulting stable state is represented by Figure 12.2(b).

12.6 The Perpendicular-Bisector Helper

The reusable theorem

`hilbert_equidistant_line_bisects_segment`

is more structured than the preserved Book1R dependency sketch suggested.

Its inputs include

$$AP \cong AQ, \quad BP \cong BQ, \quad P - H - Q,$$

with A, B, H on the base and P, Q on opposite sides.

First, the current implementation invokes

`hilbert_theorem_17`

and obtains the angle symmetry

$$\angle ABP \cong \angle ABQ.$$

It then performs a genuine positional split on the already existing point H :

The three branches are:

- if $H = A$, the midpoint congruence follows directly from $AP \cong AQ$;
- if $H = B$, it follows directly from $BP \cong BQ$;
- if $H \neq A$ and $H \neq B$, the proof uses `hilbert_collinear_angle_transport`, followed by SAS on the triangles BHP and BHQ .

Thus `hilbert_collinear_angle_transport` is a genuine transitive dependency of I.12, but it is not called directly from `Proposition12.lean`; it occurs only in the generic branch of this reusable Book Zero theorem.

The three cases do not require three separate geometric panels. They do not introduce new points, lines, intersections, or side-of-line configurations; they only determine whether the already constructed intersection H coincides with one of the two base points.

The conclusion is

$$PH \cong HQ.$$

12.7 The Final Right-Angle Construction

The current main theorem does not stop at the midpoint relation. It defines a local final step for any point R satisfying

$$R \in \text{base}, \quad R \neq H, \quad RP \cong RQ.$$

To package a right angle in the exact formal orientation, Lean extends RH through H to a fresh local point, named $\mathbb{C}$ in the source. To avoid collision with the classical external point C in this chapter, Figure 12.2(c) labels it C_{loc} :

$$R - H - C_{\text{loc}}.$$

SSS is applied to the triangles

$$\triangle RHP \quad \text{and} \quad \triangle RHQ$$

using

$$RH \cong RH, \quad HP \cong HQ, \quad RP \cong RQ.$$

This gives congruent angles at H . Since

$$Q - H - P,$$

vertical-angle congruence converts the result to the adjacent-angle orientation required by

`HilbertRightAngle Geo R H P.`

This local continuation point is a genuine hidden construction and is therefore included in the audited formal diagram.

12.8 Why the theorem returns a second base point

The main proof finally decides which original base point may be used as R .

It performs

```
by_cases hHA : H = A
```

If $H = A$, then $B \neq H$ and the proof takes $R = B$, using $BP \cong BQ$. Otherwise $A \neq H$ and it takes $R = A$, using $AP \cong AQ$.

This is why the public conclusion existentially returns a second base point R instead of fixing A or B in advance. The two branches do not create distinct final geometric states and therefore do not receive separate panels.

12.9 Lean Formalization

The public Euclidean theorem is

```
theorem euclid_proposition_12
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B P : Geo.Point)
  (base : Geo.Line)
  (hAB : A != B)
  (hAbase : HilbertIncidence.OnLine A base)
  (hBbase : HilbertIncidence.OnLine B base)
  (hPbase : not HilbertIncidence.OnLine P base) :
  exists H R : Geo.Point,
    HilbertIncidence.OnLine H base /\
    HilbertIncidence.OnLine R base /\
    HilbertRightAngle Geo R H P := by
  exact
    hilbert_perpendicular_from_point_exists
      Geo A B P base hAB hAbase hBbase hPbase
```

Thus `euclid_proposition_12` is a thin interface wrapper. It does not duplicate the construction.

12.10 Relationship with Earlier Propositions

12.10.1 Proposition I.8

Euclid uses Proposition I.8 at the final stage of the classical proof. The Hilbert reconstruction also uses an SSS argument, but it is not a call to `euclid_proposition_8`. The SSS configuration is the formal pair of triangles RHP and RHQ .

12.10.2 Proposition I.10

In Euclid's construction, I.10 bisects EG . The current formal proof does not call I.10. Instead,

$$PH \cong HQ$$

is obtained from `hilbert_equidistant_line_bisects_segment`.

12.10.3 Proposition I.11

I.11 constructs a perpendicular at a point already on a line. I.12 solves the complementary construction problem, with the prescribed point outside the line. I.11 is not a direct theorem-level dependency of the current I.12 proof.

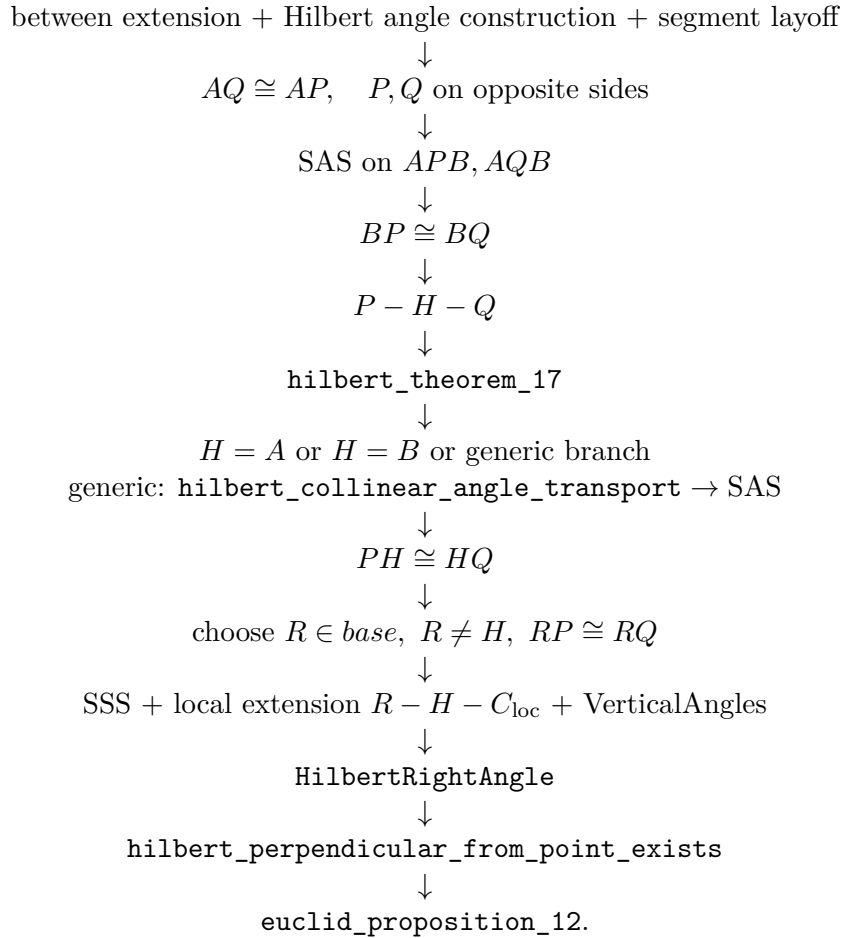
12.11 Dependency Summary

The two architectures should be kept separate.

For Euclid,

$$\text{circle construction} \longrightarrow \text{I.10} \longrightarrow \text{I.8} \longrightarrow \text{I.12}.$$

For the current reconstruction, a more faithful dependency summary is



This is the current logical DAG at the level relevant to the proposition and the reusable perpendicular-bisector helper. Internal incidence, ray, side-of-line, and congruence transport lemmas are not expanded into separate diagrammatic states unless they create new geometry.

12.12 Diagrammatic Audit

The preserved Book1R chapter contained six numbered geometric figures: input data, the opposite-side point, the circle intersections, the completed classical configuration, the formal reflected-point construction, and the formal perpendicular-bisector configuration.

The audit retains all six meaningful geometric stages but reorganizes them into two multi-panel figures.

- The initial-data-only classical figure is removed because it records no new construction.
- The classical opposite-side choice, circle-intersection state, and final midpoint/SSS state become the three panels of Figure 12.1.
- The two preserved formal states become panels (a) and (b) of Figure 12.2.
- A new panel (c) records the hidden local continuation $R - H - C_{\text{loc}}$ used by the current source to package the final `HilbertRightAngle` witness.

The formal point Q_0 is intentionally not assigned its own panel: it serves only to choose the construction ray on which Q is subsequently laid off. Likewise, the endpoint-coincidence cases $H = A$ and $H = B$ do not create new geometric objects and are described in the proof DAG rather than duplicated as figures.

A further audit correction concerns Figure 12.2(b). The current theorem states only that A , B , and H lie on the same base line. It does not establish a betweenness relation among those three points. Their displayed left-to-right order is therefore explicitly schematic and must not be read as a Lean hypothesis.

The final audited set is consequently

2 figures /6 panels .

The organization changes, but the document does not collapse genuinely different geometric states into tactic bookkeeping.

12.13 Conclusion

Proposition I.12 is an especially clear example of reconstruction rather than literal transcription. Euclid uses a circle, I.10, and I.8. The current Hilbert proof instead constructs an opposite-side point, recognizes an equidistant-line configuration, obtains the midpoint of PQ , and finishes with SSS and vertical angles.

The public Euclidean theorem remains deliberately thin. Its geometric content is carried by `hilbert_perpendicular_from_point_exists` and by reusable Hilbert/Book Zero infrastructure. The audited diagrams now reflect that architecture: the classical circle mechanism and the formal perpendicular-bisector mechanism are visually separated, while every panel records a genuine change of geometry rather than a change of proof state.

Chapter 13

Proposition I.13

13.1 Introduction

Euclid's Proposition I.13 concerns the two adjacent angles formed when one straight line stands on another straight line.

Let C, B, D lie on one straight line, with B between C and D , and let A be a point outside that line. The two adjacent angles are

$$\angle CBA \quad \text{and} \quad \angle ABD.$$

Euclid states that these angles are either two right angles or are together equal to two right angles.

The classical proof distinguishes two cases. If the adjacent angles are equal, then by Definition 10 they are both right angles. If they are not equal, Euclid erects a perpendicular at B using Proposition I.11 and compares the resulting angle decompositions.

The reconstruction over the Hilbert/Book Zero layer is much shorter. Book Zero already contains an explicit relation expressing that one angle is supplementary to another:

`BookZeroSupplement`

For the configuration of Proposition I.13, this relation contains exactly the required geometric data. Consequently, the formal theorem does not reconstruct Euclid's case distinction or introduce an arithmetic operation on angles. It identifies the two adjacent angles directly as a supplementary pair.

Thus Proposition I.13 provides a particularly clear example of the compression obtained from the Book Zero language: a classical geometric argument becomes essentially an instance of an already available synthetic relation.

13.2 The Classical Configuration

Proposition I.13 is not a construction theorem. The geometric configuration is already given.

Let C, B, D lie on one straight line with

$$C - B - D,$$

and let A lie outside the straight line CD .

The straight line BA stands on the straight line CD as shown in Figure 13.1(a) and forms the adjacent angles

$$\angle CBA \quad \text{and} \quad \angle ABD.$$

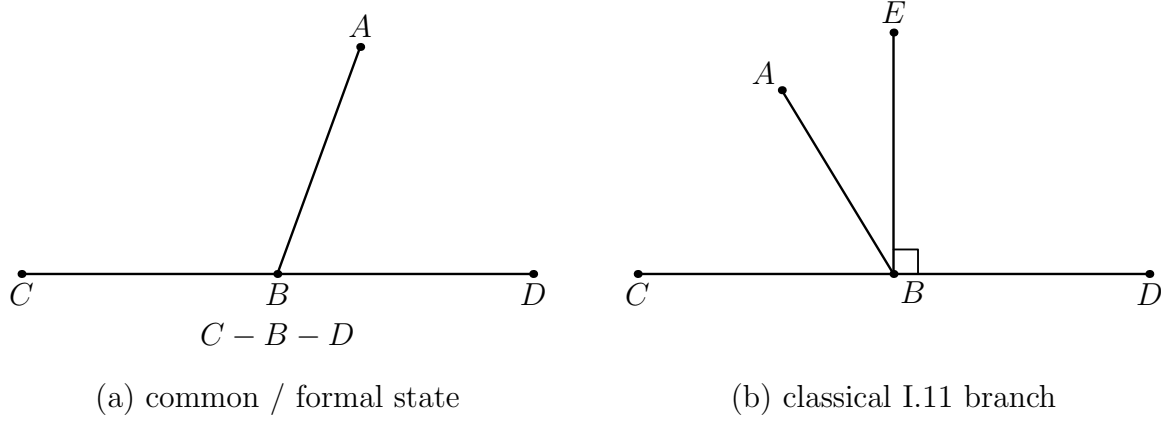


Figure 13.1: Two genuinely different geometric states associated with Proposition I.13. Panel (a) is the common straight-line configuration $C - B - D$ with the standing ray BA ; it is also the complete proposition-specific geometry used by the current Lean theorem. Panel (b) is the representative classical non-equal-angle branch, where Proposition I.11 erects $BE \perp CD$. The placement of A in panel (b) records one representative orientation only and does not encode numerical angle data.

13.3 Statement

Assume

$$C - B - D$$

and let A be a point distinct from B .

Then the adjacent angles

$$\angle CBA \quad \text{and} \quad \angle ABD$$

form a supplementary pair.

In Euclid's formulation, they are therefore either two right angles or together equal to two right angles.

In the Book Zero language this conclusion is represented by

$$\text{BookZeroSupplement}(C, B, A, A, D).$$

By definition,

$$\text{BookZeroSupplement}(C, B, A, A, D)$$

means

$$\text{SameRay}(B, A, A)$$

together with

$$C - B - D.$$

The first condition expresses the trivial fact that the ray BA coincides with itself, while the second is precisely the given straight-line configuration.

There is an important difference between this formal statement and the classical configuration drawn in Figure 13.1(a). Euclid understands the standing line BA as genuinely distinct from CD . The current production theorem assumes only $B \neq A$; it does not assume that A lies off the carrier CD . This is not an additional geometric branch. It reflects the fact that the formal target is the structural relation **BookZeroSupplement**, whose two fields are already meaningful under these weaker hypotheses.

13.4 Classical Proof

Euclid proves Proposition I.13 by distinguishing two cases.

13.4.1 Case 1. The Adjacent Angles Are Equal

Suppose first that

$$\angle CBA \cong \angle ABD.$$

Since C, B, D lie on one straight line, the two angles are adjacent angles formed by the straight line BA standing on CD .

By Euclid's Definition 10, when a straight line standing on another straight line makes the adjacent angles equal, each of those angles is a right angle.

Therefore

$$\angle CBA \quad \text{and} \quad \angle ABD$$

are two right angles.

This is the first alternative in Proposition I.13.

13.4.2 Case 2. The Adjacent Angles Are Not Equal

Suppose now that

$$\angle CBA \not\cong \angle ABD.$$

By Proposition I.11, construct through B a perpendicular BE to the straight line CD ; see Figure 13.1(b).

Thus

$$\angle CBE \quad \text{and} \quad \angle EBD$$

are right angles.

Since BE is perpendicular to CD ,

$$\angle CBE + \angle EBD$$

is equal to two right angles.

The original ray BA divides one of these right-angle configurations. In the arrangement shown in Figure 13.1(b),

$$\angle CBE = \angle CBA + \angle ABE.$$

Adding $\angle EBD$ gives

$$\angle CBE + \angle EBD = \angle CBA + \angle ABE + \angle EBD.$$

But

$$\angle ABE + \angle EBD = \angle ABD.$$

Hence

$$\angle CBE + \angle EBD = \angle CBA + \angle ABD.$$

The left-hand side consists of two right angles. Therefore

$$\angle CBA + \angle ABD$$

is equal to two right angles.

This is the second alternative in Proposition I.13.

The two panels of Figure 13.1 exhaust the genuinely distinct geometric states relevant to this chapter. The equal-angle branch introduces no point or auxiliary line, while the unequal-angle branch adds precisely the perpendicular BE . The two states are therefore kept in one compact float.

Thus, whether the two original adjacent angles are equal or unequal, they are either themselves two right angles or are together equal to two right angles.

13.5 Proof Architecture of the Reconstruction

Euclid's proof proceeds through the geometry of right angles and angle addition:

Euclid

C-B-D and standing ray BA

are the adjacent angles equal?

+-----+-----+

yes no

Definition 10 erect perpendicular BE

right angles angle decomposition

+-----+-----+

two right angles

Hilbert / Book Zero

```

C-B-D
+
SameRay(B,A,A)
|
v
BookZeroSupplement(C,B,A,A,D)
|
v
euclid_proposition_13

```

The two architectures should not be represented by additional geometric figures. The difference is logical rather than spatial: Euclid proves a statement about angle sums, whereas the Book Zero theorem recognizes the same straight-angle configuration directly.

The contrast is substantial. Euclid proves the proposition by introducing a perpendicular and reasoning about decompositions and combinations of angles. At the Book Zero level, the same geometric configuration has already been abstracted as the supplementary-angle relation.

No new geometric construction is therefore required, and Proposition I.11 disappears from the formal dependency path.

13.6 Hilbert Reconstruction

The Hilbert reconstruction of Proposition I.13 does not reproduce Euclid's proof.

The essential observation is that the conclusion of the proposition has already been isolated in Book Zero as the relation

BookZeroSupplement

For five points A, B, C, D, F , this relation is defined by

$$\text{BookZeroSupplement}(A, B, C, D, F)$$

if and only if

$$\text{SameRay}(B, C, D)$$

and

$$A - B - F.$$

Thus the relation records synthetically the configuration in which one side of an angle is continued through its vertex while the other side is preserved along the same ray.

For Proposition I.13 we instantiate this relation as

$$\text{BookZeroSupplement}(C, B, A, A, D).$$

Expanding the definition gives exactly two conditions:

$$\text{SameRay}(B, A, A)$$

and

$$C - B - D.$$

The second condition is one of the hypotheses of Proposition I.13.

The first condition says only that the ray BA is the same ray as itself. Since $A \neq B$, it follows immediately from reflexivity of the same-ray relation.

Consequently,

$$\text{BookZeroSupplement}(C, B, A, A, D)$$

follows directly.

No perpendicular is constructed. No case distinction between equal and unequal adjacent angles is required. No decomposition or addition of angles occurs in the proof.

This does not mean that Euclid's geometric argument has been discarded. Rather, the geometric notion expressed by that argument has already been incorporated into the language of Book Zero. Proposition I.13 therefore becomes a recognition theorem: the given configuration is an instance of an existing synthetic relation.

13.7 Lean Formalization

The Lean theorem is correspondingly short. Its explicit typeclass requirements are only `HilbertIncidence` and `HilbertOrder`; no `HilbertCongruence` instance is required:

```
theorem euclid_proposition_13
  [HilbertIncidence Geo]
  [HilbertOrder Geo]
  (C B D A : Geo.Point)
  (hCBD : Geo.Between C B D)
  (hBA : B != A) :
  BookZeroSupplement Geo C B A A D := by

  constructor

  · exact
    hilbert_sameRay_refl
    Geo B A hBA.symm

  · exact hCBD
```

The proof follows the definition of

`BookZeroSupplement`

literally.

The first component of the conclusion is

$$\text{SameRay}(B, A, A).$$

It is obtained from

`hilbert_sameRay_refl`

using $A \neq B$, which is supplied by the symmetric form of the hypothesis $B \neq A$.

The second component is

$$C - B - D,$$

which is exactly the hypothesis

`hCBD`

Thus the formal proof contains no intermediate geometric construction and requires no theorem corresponding to Euclid's use of Proposition I.11.

13.8 Discussion

Proposition I.13 is a particularly clear test of the role assigned to Book Zero in the reconstruction.

In Euclid's original development, the statement requires an actual proof. The argument distinguishes whether the two adjacent angles are equal. In the nontrivial case, Proposition I.11 is invoked to erect a perpendicular, and the conclusion is obtained by decomposing and recombining angles.

At the Hilbert level, one could reconstruct this argument explicitly. The existing theory already contains the necessary machinery for right angles, supplementary angles, and angle addition.

That reconstruction is unnecessary, however, once the Book Zero language is used.

The relation

`BookZeroSupplement`

was introduced independently as a reusable description of a standard synthetic configuration. Proposition I.13 is precisely an instance of that configuration.

The resulting compression is therefore structural rather than tactical. Lean is not discovering a shorter proof of Euclid's argument. The formal language has been organized at a level at which the conclusion of I.13 can already be stated directly.

This produces an unusually short dependency path:

betweenness + same-ray reflexivity $\longrightarrow$ BookZeroSupplement $\longrightarrow$ Euclid I.13.

In particular, Proposition I.11, although essential to Euclid’s classical proof, is not a dependency of the reconstructed theorem.

This distinction is important. The objective of the reconstruction is not to force every Euclidean proposition to retain the dependency graph of the original text. The objective is to express Euclidean geometry over a stable synthetic interface derived from the Hilbert foundation.

Earlier propositions exhibited several possible relationships with this interface. Some consumed already available Book Zero results, while others exposed missing reusable geometry that had to be added to the interface.

Proposition I.13 exhibits a third and especially simple situation: the proposition is already expressed by the vocabulary of Book Zero.

Schematically,

Euclid	:	case distinction + I.11 + angle composition
Hilbert/Book Zero	:	recognition of a supplementary configuration
Lean	:	same-ray reflexivity + given betweenness

The four-line proof is therefore not merely an accidental shortening of Proposition I.13. It is evidence that the Book Zero layer is functioning as intended: elementary synthetic arguments can be encapsulated once and subsequently used as part of the language in which later geometry is formulated.

13.9 Relationship with Earlier Propositions

13.9.1 Proposition I.11

Proposition I.11 constructs a perpendicular to a given straight line at a given point on it. In the classical development this supplies the canonical right-angle configuration against which the adjacent angles in I.13 can be compared.

The formal reconstruction can package the same geometry through the Hilbert and Book Zero angle machinery, but the conceptual role remains the same: a perpendicular provides the reference configuration for two right angles.

13.9.2 Proposition I.12

Proposition I.12 extends the perpendicular construction to a point outside a line. It is not the principal dependency of I.13, but together with I.11 it completes the elementary perpendicular infrastructure that precedes the straight-angle block.

13.9.3 Transition to Proposition I.14

Proposition I.13 is the direct theorem: when a straight line stands on another straight line, the adjacent angles are two right angles or are together equal to two right angles.

Proposition I.14 reverses this logic. From the corresponding supplementary-angle condition it recovers the straight-line configuration.

Thus I.13 and I.14 form a natural theorem/converse pair and should be read as one local unit in the development of straight-angle geometry.

13.10 Hilbert and Book Zero Components Used

The reconstruction of Proposition I.13 relies on the Book Zero relation

$\text{BookZeroSupplement}$

defined by

$$\text{BookZeroSupplement}(A, B, C, D, F)$$

if and only if

$$\text{SameRay}(B, C, D)$$

and

$$A - B - F.$$

Geometrically, this represents a pair of supplementary angles. The angle determined by the rays BA and BC has as its supplement the angle determined by the rays BD and BF , where BD lies on the same ray as BC and BF is the opposite extension of BA .

For Proposition I.13 the relation is instantiated as

$$\text{BookZeroSupplement}(C, B, A, A, D).$$

Thus the required conditions become

$$\text{SameRay}(B, A, A)$$

and

$$C - B - D.$$

The first is immediate from same-ray reflexivity, while the second is the given betweenness hypothesis.

The proof also uses

`hilbert_sameRay_refl`

which expresses reflexivity of a nondegenerate ray:

$$A \neq B \implies \text{SameRay}(A, B, B).$$

13.11 Formalization Notes

The formal reconstruction is considerably shorter than Euclid's classical proof because the supplementary-angle configuration has already been packaged in Book Zero.

The theorem does not reproduce Euclid's case distinction and does not construct the perpendicular from Proposition I.11. Instead, it proves

$$\text{BookZeroSupplement}(C, B, A, A, D)$$

directly from the two components required by the definition:

$$\text{SameRay}(B, A, A) \quad \text{and} \quad C - B - D.$$

The first component follows from

`hilbert_sameRay_refl`

using the nondegeneracy hypothesis $A \neq B$. The second component is exactly the given betweenness hypothesis.

Thus no numerical angle measure, angle addition, right-angle case analysis, or auxiliary perpendicular is needed in the Lean proof. The formalization records the same synthetic content at a more abstract level: Proposition I.13 is recognized as an instance of the already established Book Zero notion of supplementary angles.

This is also the form needed by Proposition I.14, which uses the same supplementary-angle structure in the converse direction.

13.12 Dependency Summary

The current production theorem has an unusually short direct dependency path:

$$\begin{array}{c}
B \neq A \\
\downarrow \\
\text{hilbert_sameRay_refl} \\
\downarrow \\
\text{SameRay}(B, A, A) \\
C - B - D \\
\downarrow \\
\text{BookZeroSupplement}(C, B, A, A, D) \\
\downarrow \\
\text{euclid_proposition_13.}
\end{array}$$

The constructor proof uses only same-ray reflexivity and the given betweenness hypothesis. Proposition I.11, Definition 10, the classical right-angle comparison, angle decomposition, and the equal/unequal case split belong to Euclid's proof DAG, not to the direct formal DAG of the current `Proposition13.lean`.

13.13 Diagrammatic Audit

The preserved Book1R chapter contained two separate geometric floats: the initial configuration and the perpendicular introduced in the non-equal-angle branch. Both are genuine geometric states, but the first is literally contained in the second. The audited chapter therefore combines them into one two-panel figure rather than retaining two separate floats.

Panel (a) records the common input and simultaneously the complete proposition-specific geometric state used by Lean. Panel (b) records the one new classical construction, the perpendicular BE supplied by I.11. No figure is added for the equal-angle branch, for `SameRay`(B, A, A), or for expansion of `BookZeroSupplement`: none of these creates new geometry.

The displayed off-line position of A is classical. The current Lean theorem requires only $B \neq A$, so the figure must not be read as an additional formal noncollinearity hypothesis.

13.14 Conclusion

Proposition I.13 begins the straight-angle block of Book I.

Its familiar classical statement looks almost arithmetical: the two adjacent angles made by a line standing on another line amount to two right angles. The formal reconstruction shows that no numerical angle measure is required to express or prove this fact.

Instead, the theorem is recovered directly from incidence/order data already packaged by the Book Zero vocabulary. The classical equal/unequal-angle case distinction is not replayed in the current Lean proof, and no local right-angle or angle-congruence argument is required.

The proposition is also structurally important. I.14 supplies its converse, and I.15 completes the local theory by proving the congruence of vertical angles. I.13 therefore marks the beginning of a coherent three-proposition development of the geometry of intersecting straight lines.

Chapter 14

Proposition I.14

14.1 Introduction

Euclid's Proposition I.14 is the converse in spirit of the preceding Proposition I.13.

Proposition I.13 starts from a straight-line configuration and proves that the adjacent angles are together equal to two right angles. Proposition I.14 reverses this direction: if two adjacent angles are together equal to two right angles, and their outer rays lie on opposite sides of the common straight line, then those outer rays form one straight line.

In Euclid's original proof the argument is indirect. The ray opposite to one of the given rays is constructed, Proposition I.13 is applied, and equality of the resulting angles forces the constructed ray to coincide with the remaining given ray.

The Hilbert reconstruction follows the same geometric idea at a more structural level. The condition "equal to two right angles" is encoded without numerical angle measure, using an opposite ray and angle congruence. The decisive step is then supplied by the uniqueness clause of Hilbert's angle-construction axiom.

Proposition I.14 is therefore an instructive example of the interaction between Euclid's angle language, Hilbert plane separation, angle construction, and the ray calculus developed in Book Zero.

14.2 The Classical Configuration

Let AB be a straight line, and let two further straight lines BC and BD meet it at the common point B , with C and D lying on opposite sides of AB .

The two adjacent angles are therefore

$$\angle ABC \quad \text{and} \quad \angle ABD.$$

The hypothesis of Proposition I.14 is that these two angles are together equal to two right angles. The conclusion is that the two outer rays BC and BD are opposite rays and hence form one straight line.

14.3 Statement

Theorem 12 (Euclid I.14). *Let AB be a straight line, and let the straight lines BC and BD meet it at the point B on opposite sides of AB .*

If the adjacent angles

$$\angle ABC \quad \text{and} \quad \angle ABD$$

are together equal to two right angles, then BC and BD form one straight line.

Equivalently,

$$C - B - D.$$

The proposition may therefore be viewed as a converse to the straight-line angle relation established in Proposition I.13.

The geometric content of the conclusion is stronger than mere collinearity. The point B lies between C and D , so the rays BC and BD are opposite rays with common endpoint B .

14.4 Classical Proof

Euclid argues by extending one of the two outer rays through the common vertex; the auxiliary configuration is shown in Figure 14.1.

Extend CB through B to a point E . Then

$$C - B - E,$$

so BC and BE form one straight line.

By Proposition I.13, the adjacent angles

$$\angle ABC \quad \text{and} \quad \angle ABE$$

are together equal to two right angles.

But by hypothesis,

$$\angle ABC \quad \text{and} \quad \angle ABD$$

are also together equal to two right angles.

Therefore the remaining angles must be equal:

$$\angle ABE = \angle ABD.$$

But if BE and BD were distinct, one of these two angles would be a proper part of the other. A proper part cannot be equal to the whole. Hence the rays BE and BD cannot be distinct.

Therefore BE coincides with BD . Since

$$C - B - E,$$

it follows that

$$C - B - D.$$

Thus BC and BD form one straight line.

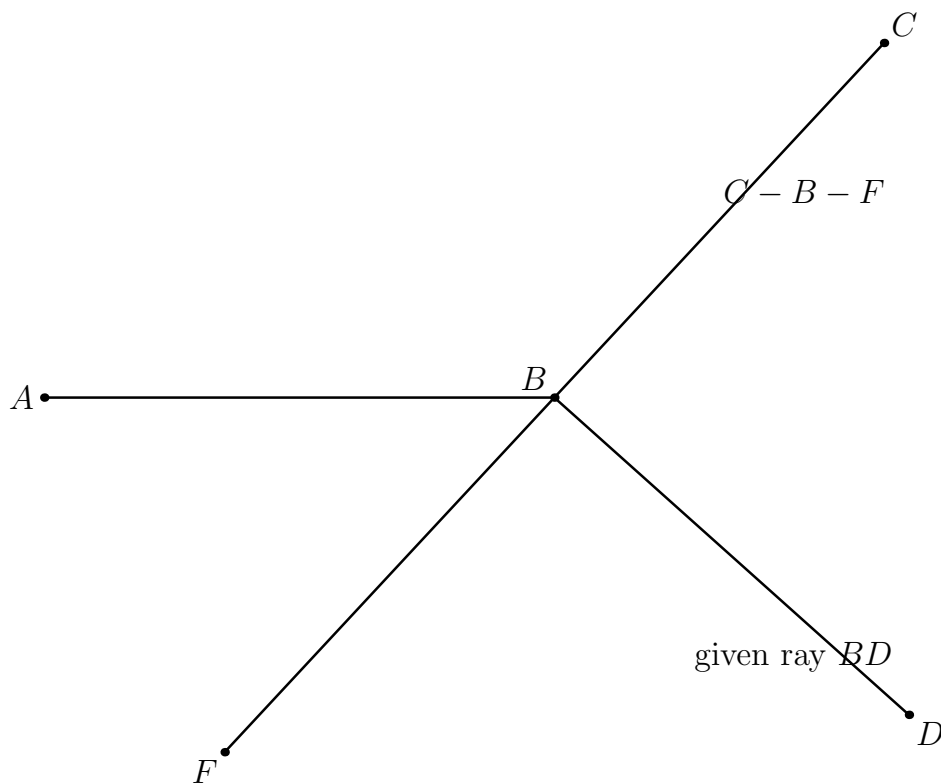


Figure 14.1: Classical Wilkins pre-coincidence configuration. The straight line CBF is the constructed extension through B , while BD is the given ray. The point F is Wilkins's classical auxiliary point; the formal Lean witness for the same supplementary pattern is called E . The drawing does not assume that BF and BD already coincide.

The essential structure of Euclid's argument is therefore

extension $\longrightarrow$ I.13 $\longrightarrow$ equality of the remaining angles $\longrightarrow$ coincidence of the rays.

The classical auxiliary diagram is genuinely useful, but it must be read as a *pre-coincidence* configuration. It would be misleading to draw BD as a visibly different ray and simultaneously treat $\angle ABF = \angle ABD$ as already geometrically realized. The equality is precisely what forces the two rays to coincide. Wilkins makes the missing uniqueness step explicit by using the fact that D and F lie on the same side of AB .

14.5 Proof Architecture of the Reconstruction

The Hilbert reconstruction preserves the central geometric idea of Euclid's proof but reorganizes its logical dependencies.

Euclid extends CB through B to a point E and uses Proposition I.13 to obtain the supplementary angle relation. He then compares this relation with the hypothesis of Proposition I.14 and concludes that the rays BD and BE must coincide.

In the formal reconstruction, the point E is incorporated directly into the synthetic representation of the hypothesis. We require

$$C - B - E$$

together with

$$\angle ABD \cong \angle ABE.$$

Thus the intermediate appeal to Proposition I.13 is no longer needed: the supplementary configuration required by Euclid's argument is already present in the formal hypothesis.

The remaining problem is to prove that BD and BE determine the same ray from B .

The current source also derives

$$\text{OppositeSide}(C, E; \text{base})$$

from $C - B - E$ and the fact that B lies on the base line. This is a valid geometric consequence, but in the present production proof the local fact `hOppCE` is not used after it is established. It is therefore not a dependency edge of the subsequent argument and does not receive a separate figure.

At this point the reconstruction divides into two cases according to whether C, D, E are collinear.

Hilbert hypothesis

C-B-E

angle ABD ~= angle ABE

C and D opposite sides of base AB

|

v

derive: C and E opposite sides of base AB

|

v

case split: are C,D,E collinear?

|

+-----+-----+

|

yes

|

line

uniqueness

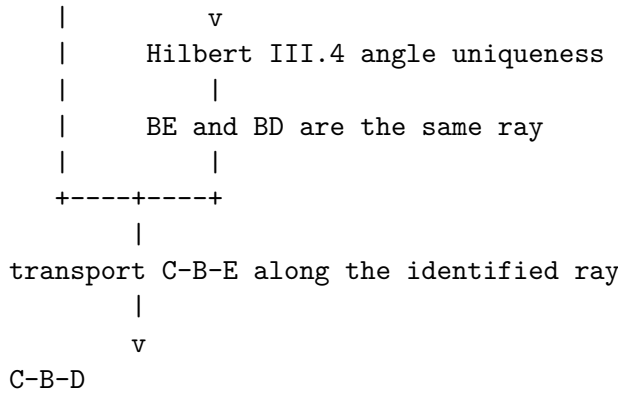
|

no

|

D and E are on the same side of AB

|



The audited presentation keeps only the auxiliary classical configuration. The earlier stand-alone statement picture added no new geometric state: the same rays and base line are already present in the extension diagram. The formal proof, however, contains a genuine geometric split according to whether C, D, E are collinear. Those two states are recorded together in Figure 14.2.

In the collinear case, incidence and line uniqueness are sufficient.

In the noncollinear case, plane separation first places D and E on the same side of the base line. The uniqueness clause of Hilbert’s angle-construction axiom then identifies the rays BD and BE . Ray calculus and transport of betweenness finally yield

$$C - B - D.$$

Schematically, the two proof architectures are

$$\begin{array}{ll} \text{Euclid} & : \text{extension} \rightarrow \text{I.13} \rightarrow \text{angle comparison} \\ & \rightarrow \text{ray coincidence,} \\ \text{Hilbert reconstruction} & : \text{encoded supplement} \rightarrow \text{case split} \\ & \rightarrow \text{line/angle uniqueness} \rightarrow \text{betweenness.} \end{array}$$

14.6 Hilbert Reconstruction

The Hilbert reconstruction begins by replacing Euclid’s phrase “together equal to two right angles” by a purely synthetic configuration.

No numerical measure of angles and no operation of angle addition are introduced. Instead, the supplementary relation is represented by an opposite ray together with angle congruence.

14.6.1 Encoding the Angle Hypothesis

Choose a point E on the continuation of CB beyond B , so that

$$C - B - E.$$

The rays BC and BE are therefore opposite. Consequently, $\angle ABE$ is the angle supplementary to $\angle ABC$.

The hypothesis that

$$\angle ABC \quad \text{and} \quad \angle ABD$$

are together equal to two right angles is represented by requiring

$$\angle ABD \cong \angle ABE.$$

The same supplementary pattern is visible in the classical Figure 14.1, where Wilkins calls the extension point F . In the Lean statement the corresponding witness is E . The formal case split generated from this witness is shown separately in Figure 14.2.

In Lean this condition is packaged as

```
def HilbertAnglesEqualTwoRightAngles
  [HilbertIncidence Geo]
  (A B C D : Geo.Point) : Prop :=
  exists E : Geo.Point,
  Geo.Between C B E /\
  Geo.AngleCongruent
  A B D
  A B E
```

Thus the witness E contains exactly the two pieces of geometric information needed by the reconstruction:

$$C - B - E$$

and

$$\angle ABD \cong \angle ABE.$$

This definition should not be interpreted as a numerical definition of the sum of two angles. It is a synthetic encoding of the particular supplementary configuration occurring in Proposition I.14.

14.6.2 The Collinear Case

Assume first that C , D , and E are collinear, as in Figure 14.2(a).

From the definition of `HilbertAnglesEqualTwoRightAngles` we already have

$$C - B - E.$$

Hence B lies on the line determined by C and E .

Since C and D lie on opposite sides of the base line AB , the segment CD meets that line. Let X be a point of intersection, so that

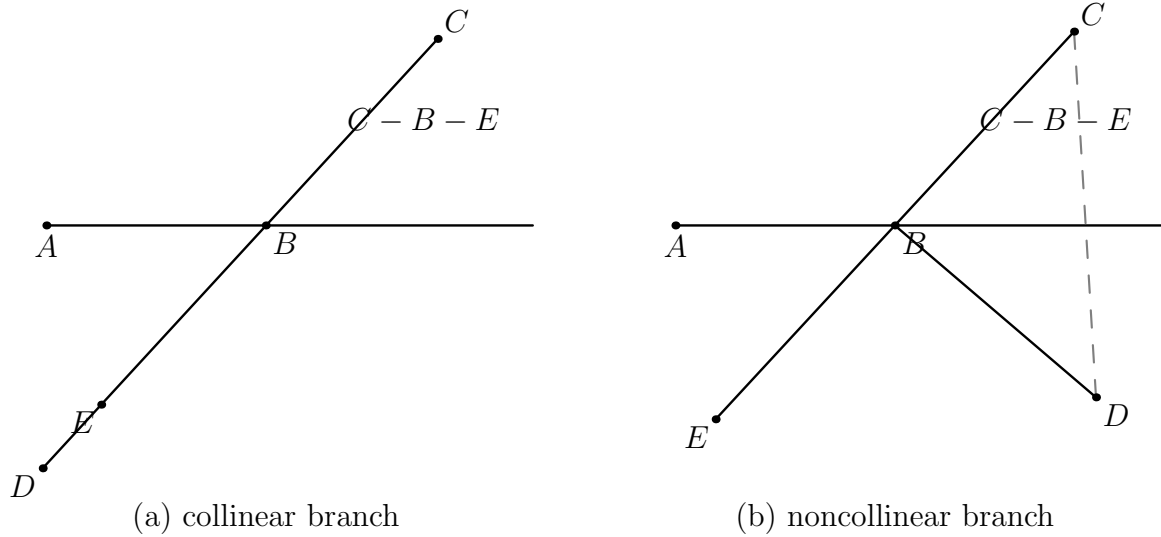


Figure 14.2: The genuine geometric case split in the current Lean proof. (a) In the collinear branch, C, D, E lie on one carrier and the line-uniqueness argument identifies the crossing with B , yielding $C - B - D$. The displayed placement of D relative to E is only representative; no order between D and E , and not even their distinctness, is assumed. (b) In the noncollinear branch, $C - B - E$ and plane separation give $\text{SameSide}(D, E; \text{base})$; the dashed segment records the opposite-side crossing of CD with the base line.

$$C - X - D$$

and X lies on the base line.

No separate figure is needed for this branch. The point X is only an incidence witness: if the line through C, D, E meets the base line at X , then line uniqueness forces $X = B$. Drawing X as a second visible point near B is more confusing than informative, because the proof immediately shows that the two points coincide.

Both B and X now lie on two lines: the base line AB and the line containing C, D, E .

Suppose that

$$X \neq B.$$

By uniqueness of the line through two distinct points, the two lines would then have to coincide. In particular, C would lie on the base line.

This contradicts the hypothesis that C and D lie on opposite sides of the base line, since a point belonging to the line itself cannot be one of the two opposite-side points.

Therefore

$$X = B.$$

Substituting this equality into

$$C - X - D$$

gives

$$C - B - D.$$

Thus the collinear branch requires no angle argument at all. Once C, D, E are known to be collinear, the conclusion follows entirely from betweenness, plane separation, incidence, and uniqueness of the line through two distinct points.

14.6.3 The Noncollinear Case

Assume now that C, D , and E are not collinear.

The segment CD meets the base line because C and D lie on opposite sides of it. The segment CE also meets the base line, namely at B , since

$$C - B - E.$$

Because C, D, E form a genuine triangle, the Hilbert plane-separation theory implies that the remaining endpoints D and E lie on the same side of the base line; this is the configuration shown in Figure 14.2(b). Thus

$$\text{SameSide}(D, E; AB).$$

We now use Hilbert's angle-construction axiom.

The angle $\angle ABE$ is nondegenerate. Construct, on the side of the base line containing D , a ray BZ satisfying

$$\angle ABZ \cong \angle ABE.$$

Hilbert III.4 includes a uniqueness clause: on the prescribed side of the base ray there is only one ray realizing the prescribed angle. In the current proof this uniqueness closure is the operative output of the construction. The returned facts named `hZDSame` and `hAngleZ` are not used later; the proof retains `hUnique` and applies it twice.

Since E lies on the same side of the base line as D , and

$$\angle ABE \cong \angle ABE,$$

the uniqueness clause gives

$$\text{SameRay}(B, Z, E).$$

On the other hand, the encoded hypothesis of Proposition I.14 gives

$$\angle ABD \cong \angle ABE,$$

and hence, by symmetry,

$$\angle ABE \cong \angle ABD.$$

The point D lies on the side on which BZ was constructed. Applying the same uniqueness clause again gives

$$\text{SameRay}(B, Z, D).$$

We therefore have two same-ray relations with the common reference ray BZ :

$$\text{SameRay}(B, Z, E), \quad \text{SameRay}(B, Z, D).$$

The Book Zero ray calculus now yields

$$\text{SameRay}(B, E, D).$$

Finally, from the encoded supplementary configuration we already know

$$C - B - E.$$

Replacing E by another point D on the same ray from B preserves the betweenness relation. Transport of betweenness along equal rays therefore gives

$$C - B - D.$$

This completes the noncollinear branch and hence the Hilbert reconstruction of Proposition I.14.

14.7 Lean Formalization

The reconstructed proposition is formalized by the theorem

```
theorem euclid_proposition_14
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D : Geo.Point)
  (base : Geo.Line)
  (hAB : A != B)
  (hAbase :
    HilbertIncidence.OnLine A base)
  (hBbase :
    HilbertIncidence.OnLine B base)
  (hOppCD :
    HilbertOppositeSide Geo C D base)
  (hTwoRight :
    HilbertAnglesEqualTwoRightAngles
    Geo A B C D) :
  Geo.Between C B D := by
```

The formal conclusion

$\text{Between}(C, B, D)$

expresses directly that B lies between C and D , and hence that the rays BC and BD form one straight line.

The principal dependency structure of the proof is

```

HilbertAnglesEqualTwoRightAngles
  |
  +-- C-B-E
  +-- angle ABD ~= angle ABE
  |
HilbertOppositeSide(C,D; base AB)
  |
case split on Collinear(C,D,E)
  |
  +-----+-----+
  |               |
collinear         noncollinear
  |               |
line uniqueness   SameSide(D,E; base AB)
  |               |
X = B             Hilbert III.4 uniqueness
  |               |
  |               same ray BE, BD
  |               |
  +-----+-----+
  |
hilbert_between_transport_sameRays
  |
  v
C-B-D

```

The proof introduces no new geometric axiom. The new predicate

`HilbertAnglesEqualTwoRightAngles`

serves only as an interface representation of the angle hypothesis. The actual derivation uses the existing incidence, order, plane separation, angle-construction, and ray machinery.

A notable difference from Euclid's dependency graph is that the Lean theorem does not invoke Proposition I.13. The supplementary configuration needed in the classical proof has already been encoded in the hypothesis, and the identification of the relevant rays is obtained directly from Hilbert III.4. Likewise Proposition I.4 is historical/foundational background rather than a direct theorem-level dependency of `euclid_proposition_14`.

The current source computes the additional local fact `hOppCE`, asserting that C and E are on opposite sides of the base, but no subsequent expression depends on it. The audited DAG therefore records it as a source-level by-product rather than as a causal step.

14.8 Relationship with Earlier Propositions

14.8.1 Proposition I.13

Proposition I.13 gives the direct statement: when one straight line stands on another, the two adjacent angles are either two right angles or together equal to two right angles.

Proposition I.14 is the converse. Starting from the supplementary-angle condition, it proves that the two outer rays form one straight line.

Thus I.13 and I.14 form a natural direct/converse pair.

14.8.2 Proposition I.4 as Foundational Background

The wider congruence infrastructure has classical antecedents including SAS, but the current `Proposition14.lean` does not call `euclid_proposition_4`. The direct mechanism relevant here is Hilbert III.4, which supplies uniqueness of angle construction on a prescribed side of a ray.

This uniqueness principle plays the role of a rigidity theorem: once two rays on the same side make congruent angles with the same initial ray, they must coincide.

14.8.3 Transition to Proposition I.15

Propositions I.13 and I.14 establish the supplementary and straight-line geometry surrounding an intersection. Proposition I.15 then turns to the opposite angles produced by two intersecting lines and proves their congruence.

The three propositions therefore form a coherent local block: straight-angle structure, its converse, and vertical angles.

14.9 Hilbert and Book Zero Components Used

The proof of Proposition I.14 uses several previously established components of the Hilbert and Book Zero layers. Their roles in the reconstruction are recorded here explicitly.

14.9.1 Collinear branch: incidence and line uniqueness

When C, D, E are collinear, the opposite-side hypothesis supplies a point X with $C - X - D$ on the base line. Since $C - B - E$, the point B also lies on the carrier of C, D, E . If $X \neq B$, the two distinct common points X, B would force that carrier to be the base line by `HilbertPlaneIncidence.line_unique`, contradicting that C is off the base. Hence $X = B$, and the branch closes as $C - B - D$.

The witness X is not drawn in Figure 14.2(a): its only purpose is to be identified with the already visible point B .

14.9.2 Third-side endpoints on the same side

In the noncollinear branch, the points C, D, E form a genuine triangle. The base line meets the interior of CD , because C and D lie on opposite sides of it, and it meets the interior of CE at B , since

$$C - B - E.$$

The theorem

`hilbert_third_side_endpoints_sameSide`

then shows that the remaining endpoints D and E lie on the same side of the base line.

Schematically,

$$\begin{array}{l} C - X - D, \\ C - B - E, \\ X, B \in AB \end{array} \quad \implies \quad \text{SameSide}(D, E; AB).$$

This is the plane-separation step that makes the uniqueness clause of Hilbert's angle-construction axiom applicable to both D and E .

14.9.3 Hilbert III.4: Uniqueness of Angle Construction

Hilbert's angle-construction axiom provides not only the existence of a copy of a given angle but also uniqueness on a prescribed side of the base ray.

In Proposition I.14 a ray BZ is constructed so that

$$\angle ABZ \cong \angle ABE.$$

Both E and D lie on the prescribed side of the base line. The uniqueness clause can therefore be applied twice, yielding

$$\text{SameRay}(B, Z, E)$$

and

$$\text{SameRay}(B, Z, D).$$

This is the formal replacement for Euclid's final inference that equal angles on the same side determine the same straight line.

14.9.4 Book Zero ray composition

The Book Zero theorem

`bookZero_36_ray3`

is a ray-composition principle.

In the present proof,

$$\text{SameRay}(B, Z, E)$$

and

$$\text{SameRay}(B, Z, D)$$

imply

$$\text{SameRay}(B, E, D).$$

Thus the auxiliary ray BZ introduced by Hilbert III.4 can be eliminated, leaving the geometrically relevant statement that BE and BD are the same ray.

14.9.5 Betweenness transport along same rays

The final step uses

`hilbert_between_transport_sameRays`

to transport strict betweenness along corresponding rays.

We know

$$C - B - E$$

and have established

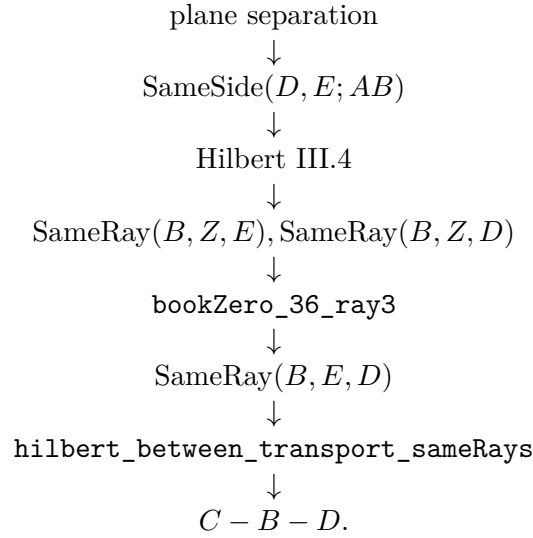
$$\text{SameRay}(B, E, D).$$

The ray from B toward C is unchanged, while E is replaced by the point D on the same ray from B . Therefore the theorem gives

$$C - B - D.$$

This is precisely the formal conclusion of Proposition I.14.

The dependency path of the noncollinear branch can therefore be summarized as



14.10 Formalization Notes

Proposition I.14 is a good example of a classical converse whose formal proof is governed by uniqueness rather than by numerical angle arithmetic.

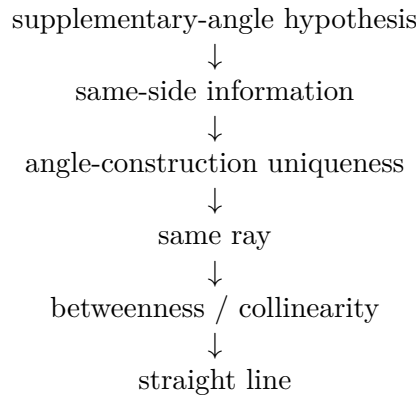
The hypothesis says that two adjacent angles have the same total effect as a straight angle. In a traditional proof this is often read directly from the diagram. In the Hilbert reconstruction, the decisive step is to compare rays using the uniqueness clause of angle construction.

The proof naturally separates into two cases.

If C, D, E are collinear, the desired straight-line conclusion follows from order, incidence, and line uniqueness.

If they are not collinear, plane separation first puts D and E on the same side of the base. Hilbert III.4 then forces the relevant rays to coincide. The resulting same-ray information is transported back to betweenness, yielding the same conclusion $C - B - D$.

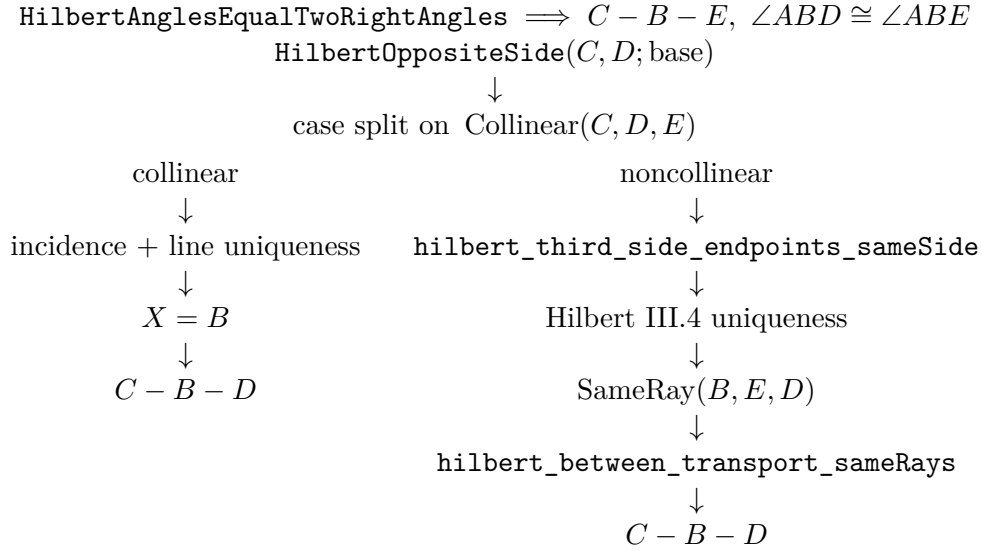
Schematically,



This is structurally different from simply writing an equation such as $\alpha + \beta = 180^\circ$. No numerical angle measure is introduced. The straight-line conclusion is recovered entirely from incidence, order, angle congruence, side-of-line information, and uniqueness.

14.11 Dependency Summary

The current formal dependency structure is branch-sensitive:



The source-level fact `hOppCE` is derived before the split but is not consumed by either branch. It is therefore intentionally omitted as a dependency arrow.

14.12 Diagrammatic Audit

The preserved Book1R chapter contained three geometric floats:

1. the initial configuration with BC and BD on opposite sides of the base;
2. Wilkins's auxiliary extension $C - B - F$;
3. the noncollinear formal branch.

The first float is redundant under the current audit standard. The Wilkins configuration already contains the same initial rays and adds the genuine extension used by the classical proof. It is therefore retained as the sole classical figure.

The current production Lean proof, however, performs a genuine geometric case split on whether C, D, E are collinear. The preserved chapter drew only the noncollinear branch. Figure 14.2 now records both branches in one two-panel formal figure. The intersection witness X in the collinear branch and the angle-copy witness Z in the noncollinear branch are not drawn separately: X is immediately proved equal to B , while Z exists only to activate Hilbert III.4 uniqueness and is immediately identified at the ray level with both E and D .

Thus the audited diagram set records stable geometric states rather than theorem-call or tactic state.

14.13 Conclusion

Proposition I.14 completes the converse half of the supplementary-angle theory begun in I.13.

Its formal reconstruction is especially instructive because it avoids introducing angle addition or numerical angle measures. The proof instead uses Hilbert's side-of-line and angle-construction uniqueness machinery to show that the geometry of the rays is rigid enough to force a straight line.

The proposition therefore illustrates a recurring feature of the project: statements that look algebraic in modern notation can often be recovered synthetically from incidence, order, congruence, and uniqueness.

Together with I.13, it prepares the local theory of intersecting lines used immediately in Proposition I.15.

Chapter 15

Proposition I.15

15.1 Introduction

Euclid's Proposition I.15 establishes the equality of vertical angles formed by two intersecting straight lines.

If two straight lines intersect at a point, each pair of opposite angles is equal. In modern terminology, these are the vertical-angle relations.

Euclid proves the proposition by applying Proposition I.13 to two adjacent linear pairs and subtracting a common angle from two sums that are each equal to two right angles.

The Hilbert reconstruction is considerably shorter. The corresponding vertical-angle theorem has already been derived in the Hilbert layer as

`hilbert_vertical_angles`

from Hilbert's theory of adjacent angles. Proposition I.15 therefore becomes an application of an existing synthetic theorem rather than a new geometric derivation.

This proposition provides another example in which a classical Euclidean proof is compressed by the intermediate Hilbert interface: the mathematical content required by Euclid has already been isolated as a reusable theorem.

15.2 The Classical Configuration

Let the straight lines AB and CD intersect at the point E , with

$$A - E - B$$

and

$$C - E - D.$$

The intersection determines four angles around E .

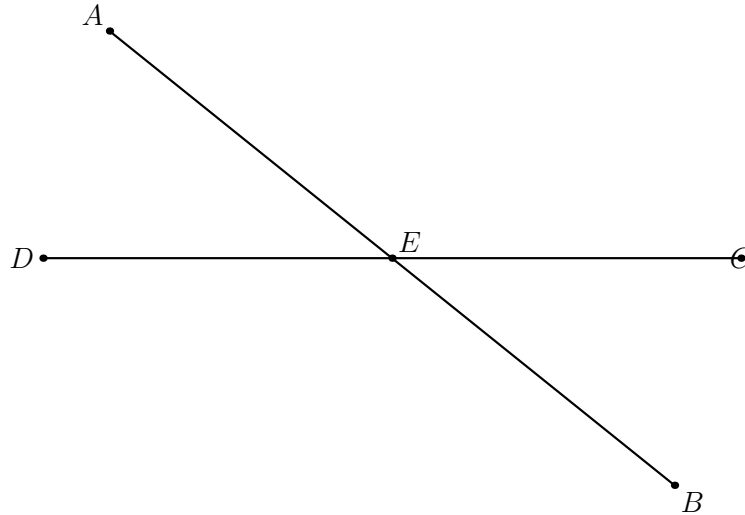


Figure 15.1: Classical and formal crossing configuration for Proposition I.15. The strict betweenness relations $A - E - B$ and $C - E - D$ encode the two pairs of opposite rays at E .

The opposite pairs are

$$\angle AEC \quad \text{and} \quad \angle BED,$$

and

$$\angle CEB \quad \text{and} \quad \angle DEA.$$

Proposition I.15 states that each pair is equal.

The betweenness relations

$$A - E - B, \quad C - E - D$$

express synthetically that EA and EB are opposite rays and that EC and ED are opposite rays. Thus the classical configuration is already naturally adapted to Hilbert's order geometry.

15.3 Statement

Euclid's Proposition I.15 states:

If two straight lines cut one another, they make the vertical angles equal to one another.

For the configuration

$$A - E - B, \quad C - E - D,$$

the proposition gives the two vertical-angle congruences

$$\angle AEC \cong \angle BED$$

and

$$\angle CEB \cong \angle DEA.$$

In the Hilbert reconstruction we assume, in addition, that the two lines are genuinely distinct. This is expressed by

$$\neg \text{Collinear}(A, E, C).$$

The formal statement is therefore

$$\begin{array}{l} A - E - B, \\ C - E - D, \\ \neg \text{Collinear}(A, E, C) \end{array} \implies \begin{cases} \angle AEC \cong \angle BED, \\ \angle CEB \cong \angle DEA. \end{cases}$$

The noncollinearity hypothesis is the synthetic expression of a proper intersection of two distinct straight lines. It also guarantees that the angles occurring in the proposition are nondegenerate.

Thus the Lean formulation makes explicit a condition that is implicit in Euclid's phrase "two straight lines cut one another." Strict betweenness specifies the two pairs of opposite rays, while noncollinearity excludes the case in which all five points lie on a single straight line.

15.4 Classical Proof

Euclid first proves the equality of one pair of vertical angles.

Since the straight line AE stands upon the straight line CD , Proposition I.13 gives

$$\angle CEA + \angle AED$$

equal to two right angles.

Again, since the straight line DE stands upon the straight line AB , Proposition I.13 gives

$$\angle AED + \angle DEB$$

equal to two right angles.

Therefore

$$\angle CEA + \angle AED = \angle AED + \angle DEB.$$

Subtracting the common angle $\angle AED$ from both sides, by Common Notion 3, gives

$$\angle CEA = \angle DEB.$$

Thus one pair of vertical angles is equal.

In the same way,

$$\angle CEB = \angle AED.$$

Therefore, when two straight lines cut one another, they make the vertical angles equal to one another.

15.5 Proof Architecture of the Reconstruction

The formal reconstruction and Euclid’s original proof establish the same geometric fact at different levels of the theory.

Classically, Proposition I.15 is derived from Proposition I.13 twice. Two adjacent linear pairs are each equal to two right angles, and Common Notion 3 removes the common angle. Thus

$$\text{I.13} \longrightarrow \text{two linear-pair equalities} \longrightarrow \text{Common Notion 3} \longrightarrow \text{I.15}.$$

The current Hilbert reconstruction does not reproduce angle addition or subtraction locally. The corresponding result has already been packaged as

`hilbert_vertical_angles.`

The first vertical pair is obtained by a direct call to that theorem. The second pair is obtained from the *same crossing*: Lean reverses the first betweenness relation, transports the required noncollinearity to the reordered triple, and calls `hilbert_vertical_angles` again.

Schematically,

$$\begin{array}{c}
 A - E - B, \quad C - E - D, \quad \neg \text{Collinear}(A, E, C) \\
 \downarrow \\
 \text{hilbert_vertical_angles} \\
 \downarrow \\
 \angle AEC \cong \angle BED \\
 \text{reorient the same crossing and transport noncollinearity} \\
 \downarrow \\
 \text{hilbert_vertical_angles} \\
 \downarrow \\
 \angle CEB \cong \angle DEA.
 \end{array}$$

The formal proof therefore introduces no new geometric point and no new diagrammatic case. Its extra work is structural: the ordered predicates must be normalized so that the reusable theorem applies in the second orientation.

15.6 Hilbert Reconstruction

The Hilbert reconstruction starts directly from the synthetic data of the intersection:

$$A - E - B, \quad C - E - D,$$

together with

$$\neg \text{Collinear}(A, E, C).$$

Unlike Euclid's proof, no explicit notion of a sum of angles equal to two right angles is required. The relevant geometric content has already been established in the Hilbert layer as the vertical-angle theorem.

15.6.1 The First Pair of Vertical Angles

The theorem

$$\text{hilbert_vertical_angles}$$

has precisely the form needed for the first pair.

Applied to

$$A - E - B, \quad C - E - D, \quad \neg \text{Collinear}(A, E, C),$$

it gives directly

$$\angle AEC \cong \angle BED.$$

Thus the first conclusion of Proposition I.15 requires no auxiliary construction.

15.6.2 The Second Pair of Vertical Angles

For the second pair we view the same intersection with the two straight lines exchanged.

From

$$A - E - B$$

the symmetry of strict betweenness gives

$$B - E - A.$$

We also require the noncollinearity condition

$$\neg \text{Collinear}(C, E, B).$$

Suppose, for contradiction, that

$$\text{Collinear}(C, E, B).$$

Since $A - E - B$, the points A, E, B are collinear. Permuting the first collinearity relation and using transitivity of collinearity through the distinct points E and B would then imply

$$\text{Collinear}(A, E, C),$$

contrary to the original hypothesis.

Hence

$$\neg \text{Collinear}(C, E, B).$$

We may therefore apply `hilbert_vertical_angles` a second time, now to

$$C - E - D, \quad B - E - A.$$

This yields

$$\angle CEB \cong \angle DEA.$$

Combining the two applications gives the complete conclusion

$$\angle AEC \cong \angle BED, \quad \angle CEB \cong \angle DEA.$$

The Hilbert reconstruction is therefore substantially shorter than Euclid's original argument. The classical use of Proposition I.13 and Common Notion 3 has already been absorbed into the previously proved vertical-angle theorem; the only additional formal work is the permutation of the intersection data required for its second application.

15.7 Lean Formalization

The reconstructed proposition is formalized by the theorem

```
theorem euclid_proposition_15
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D E : Geo.Point)
  (hAEB : Geo.Between A E B)
  (hCED : Geo.Between C E D)
  (hAEC : not PrimCollinear Geo A E C) :
  Geo.AngleCongruent A E C B E D /\
  Geo.AngleCongruent C E B D E A := by
```

The conclusion records both pairs of vertical angles:

$$\angle AEC \cong \angle BED$$

and

$$\angle CEB \cong \angle DEA.$$

The first pair is obtained directly:

```
have hFirst :
Geo.AngleCongruent A E C B E D :=
hilbert_vertical_angles
Geo A E C B D
hAEB
hCED
hAEC
```

For the second pair, the first line is read in the opposite direction:

```
have hBEC : Geo.Between B E A :=
(HilbertOrder.between_incidence
A E B hAEB).2.2.2.2
```

The required noncollinearity condition is then derived by contradiction. If C, E, B were collinear, cyclic permutation gives E, B, C collinear. Since A, E, B are already collinear and $E \neq B$, transitivity would imply that A, E, C are collinear, contradicting **hAEC**.

In Lean this is expressed by

```
have hCEB :
not PrimCollinear Geo C E B := by
intro h

have hEBC :
PrimCollinear Geo E B C :=
PrimCollinearCycle Geo C E B h

have hAEBcol :
PrimCollinear Geo A E B :=
(HilbertOrder.between_incidence
A E B hAEB).2.2.2.1

have hEB : E != B :=
(HilbertOrder.between_incidence
A E B hAEB).2.1

have hAECcol :
PrimCollinear Geo A E C :=
hilbert_primCollinear_trans
Geo A E B C
```

```

hEB
hAEBcol
hEBC

exact hAEC hAECcol

```

The second pair of vertical angles is now another direct application:

```

have hSecond :
  Geo.AngleCongruent C E B D E A :=
  hilbert_vertical_angles
  Geo C E B D A
  hCED
  hBEC
  hCEB

exact <hFirst, hSecond>

```

Thus the formal dependency structure is

$$\begin{array}{c}
 \text{hAEB, hCED, hAEC} \\
 \downarrow \\
 \text{hilbert_vertical_angles (first orientation)} \\
 \downarrow \\
 \text{betweenness symmetry and collinearity transport} \\
 \downarrow \\
 \text{hilbert_vertical_angles (second orientation)} \\
 \downarrow \\
 \angle AEC \cong \angle BED, \quad \angle CEB \cong \angle DEA.
 \end{array}$$

No new geometric axiom or auxiliary geometric construction is introduced in the formalization. Proposition I.15 is obtained entirely from the existing Hilbert incidence, order, collinearity, and vertical-angle theory.

15.8 Relationship with Earlier Propositions

15.8.1 Proposition I.13

Proposition I.13 establishes the supplementary-angle configuration created when a straight line stands on another straight line. This is the classical dependency used twice in Euclid’s proof of I.15. The current `Proposition15.lean`, however, does not call `euclid_proposition_13`; that earlier content has already been absorbed into the reusable Hilbert vertical-angle theorem.

15.8.2 Proposition I.14

Proposition I.14 gives the converse straight-line criterion for a supplementary configuration. Together I.13 and I.14 complete the local straight-angle infrastructure around an intersection. I.15 extracts a different invariant of the same local geometry: opposite, or vertical, angles are congruent. Proposition I.14 is therefore contextual rather than a direct theorem-level dependency of the current I.15 proof.

15.8.3 Transition to Proposition I.16

Vertical-angle congruence becomes an explicit transport tool in Proposition I.16. The current I.16 production file calls the generic interface theorem **VerticalAngles** directly when it must identify two differently oriented exterior angles. Thus I.15 closes the elementary intersecting-lines block while the same geometric content immediately becomes infrastructure for strict angle comparison.

15.9 Hilbert and Book Zero Components Used

Proposition I.15 is formally short because its main geometric content has already been isolated in the Hilbert layer.

The proof uses one principal theorem together with elementary order and collinearity machinery required to apply it in the two orientations of the intersection.

15.9.1 hilbert_vertical_angles

The central result is

$$\text{hilbert_vertical_angles}$$

which states that if

$$C - E - B, \quad D - E - F,$$

and

$$\neg \text{Collinear}(C, E, D),$$

then

$$\angle CED \cong \angle BEF.$$

This theorem is the immediate vertical-angle corollary developed in the Hilbert layer after the theorem on adjacent congruent angles.

In Proposition I.15 it is used twice.

The first application gives

$$\angle AEC \cong \angle BED.$$

The second application, after reversing the first straight line and transporting noncollinearity, gives

$$\angle CEB \cong \angle DEA.$$

Thus essentially all of the geometric content of Euclid's Proposition I.15 is already contained in `hilbert_vertical_angles`.

15.9.2 `HilbertOrder.between_incidence`

The theorem

`HilbertOrder.between_incidence`

extracts the elementary order and incidence consequences of strict betweenness.

From

$$A - E - B$$

the proof obtains both

$$B - E - A$$

and the collinearity relation

$$\text{Collinear}(A, E, B).$$

It also supplies the inequality

$$E \neq B,$$

which is needed for the transitivity of collinearity.

Thus a single betweenness hypothesis provides all of the auxiliary order data required for the second application of the vertical-angle theorem.

15.9.3 `PrimCollinearCycle`

The permutation theorem

`PrimCollinearCycle`

cyclically reorders a collinearity relation.

In the contradiction argument,

$$\text{Collinear}(C, E, B)$$

is transformed into

$$\text{Collinear}(E, B, C).$$

This places the points in exactly the orientation required by the transitivity theorem used in the next step.

15.9.4 hilbert_primCollinear_trans

The theorem

`hilbert_primCollinear_trans`

expresses the transitivity of collinearity when two distinct common points determine the same line.

In Proposition I.15 we combine

$$\text{Collinear}(A, E, B)$$

with

$$\text{Collinear}(E, B, C)$$

and the inequality

$$E \neq B.$$

The result is

$$\text{Collinear}(A, E, C),$$

contradicting the original noncollinearity hypothesis.

Hence

$$\neg \text{Collinear}(C, E, B),$$

which is precisely the hypothesis needed for the second invocation of `hilbert_vertical_angles`.

The complete formal dependency may therefore be summarized as

$$\begin{array}{c}
A - E - B, \quad C - E - D, \quad \neg \text{Collinear}(A, E, C) \\
\downarrow \\
\text{hilbert_vertical_angles} \\
\downarrow \\
\angle AEC \cong \angle BED \\
\text{betweenness symmetry + collinearity transport} \\
\downarrow \\
\neg \text{Collinear}(C, E, B) \\
\downarrow \\
\text{hilbert_vertical_angles} \\
\downarrow \\
\angle CEB \cong \angle DEA.
\end{array}$$

No genuinely new Book Zero theorem is required for Proposition I.15. The proposition is therefore primarily an interface theorem over the already established Hilbert vertical-angle theory.

15.10 Formalization Notes

Proposition I.15 is classically almost immediate from the supplementary-angle theory, but its formal reconstruction highlights a different issue: orientation. At an intersection the geometry is symmetric, whereas the formal predicates are written with ordered point triples. Consequently the proof must recover the appropriate primitive collinearities from betweenness, cycle them into the required orientation, and then invoke the reusable vertical-angle theorem.

The essential pattern is

$$\begin{array}{c}
\text{betweenness} \\
\downarrow \\
\text{primitive collinearity} \\
\downarrow \\
\text{orientation normalization} \\
\downarrow \\
\text{hilbert_vertical_angles} \\
\downarrow \\
\text{angle congruence.}
\end{array}$$

No new geometric point, extension, intersection, or case split is introduced in `Proposition15.lean`. The current source contains one theorem, `euclid_proposition_15`, and its geometric work is delegated to the existing Hilbert interface.

15.11 Diagrammatic Audit

The preserved Book1R chapter already made the correct diagrammatic choice: a single crossing is sufficient. Figure 15.1 records the only proposition-specific geometric state,

$$A - E - B, \quad C - E - D,$$

together with the proper-intersection hypothesis expressed textually by

$$\neg \text{Collinear}(A, E, C).$$

The two conclusions

$$\angle AEC \cong \angle BED, \quad \angle CEB \cong \angle DEA$$

are two readings of the same crossing, not two geometric configurations. Reversing $A - E - B$ to $B - E - A$, cycling collinearity, and deriving $\neg \text{Collinear}(C, E, B)$ likewise introduce no new geometry. They are orientation and incidence bookkeeping required by the formal API.

The recovered older TEX contained additional TikZ flowcharts for proof architecture and dependency structure. Those diagrams record logical flow rather than geometry and are not retained as proposition figures. The audited chapter therefore has exactly one Asymptote figure and one panel. A figure records geometry, not tactic state.

15.12 Dependency Summary

The direct formal dependency structure can be summarized as

$$\begin{array}{c}
 A - E - B, \quad C - E - D, \quad \neg \text{Collinear}(A, E, C) \\
 \downarrow \\
 \text{hilbert_vertical_angles} \\
 \downarrow \\
 \angle AEC \cong \angle BED \\
 \text{HilbertOrder.between_incidence} \\
 + \text{PrimCollinearCycle} \\
 + \text{hilbert_primCollinear_trans} \\
 \downarrow \\
 \neg \text{Collinear}(C, E, B) \\
 \downarrow \\
 \text{hilbert_vertical_angles} \\
 \downarrow \\
 \angle CEB \cong \angle DEA \\
 \downarrow \\
 \text{euclid_proposition_15.}
 \end{array}$$

The module imports `HilbertBookZero`, but the theorem body does not call Proposition I.13, Proposition I.14, or another proposition-specific Euclidean theorem. The classical dependency on I.13 and Common Notion 3 is represented formally by the already proved generic vertical-angle theorem.

15.13 Conclusion

Proposition I.15 completes the elementary theory of angles formed by intersecting straight lines. Its classical content is the familiar theorem that vertical angles are congruent. In the current Hilbert reconstruction, essentially all of this geometric content is already contained in `hilbert_vertical_angles`; the local proof mainly manages orientation and collinearity so that this reusable result can be applied twice.

The current `Proposition15.lean` contains no local `sorry` and no local axiom declaration. This documentation audit did not run `lake build` or `#print axioms`, so no fresh compilation or axiomatic-status claim is made here.

Chapter 16

Proposition I.16

16.1 Introduction

Euclid’s Proposition I.16 is the exterior-angle inequality for a triangle. If a side of a triangle is extended beyond one of its vertices, the exterior angle formed at that vertex is greater than either of the two remote interior angles.

For a nondegenerate triangle ABC , extend BC through C to a point D , so that

$$B - C - D.$$

The proposition asserts

$$\angle BAC < \angle ACD \quad \text{and} \quad \angle ABC < \angle ACD.$$

Unlike Proposition I.15, this proposition is not merely a thin application of one previously packaged theorem. Its reconstruction requires a substantial piece of Hilbert order-and-congruence geometry: an interior ray must be transported between congruent angles, and the resulting angle comparison must be expressed without numerical angle measure.

The formal development therefore isolates the geometric machinery in the Hilbert interface and leaves the Euclidean proposition itself comparatively short. In particular, the final Lean file contains three theorems:

- `euclid_proposition_16_first`,
- `euclid_proposition_16_second`,
- `euclid_proposition_16`.

16.2 The Classical Configuration

Let ABC be a nondegenerate triangle and let the side BC be produced through C to D :

$$B - C - D.$$

The angle $\angle ACD$ is exterior to the triangle. The two interior opposite angles are $\angle BAC$ and $\angle ABC$.

The formal hypotheses are

$$\neg \text{Collinear}(A, B, C) \quad \text{and} \quad B - C - D.$$

The noncollinearity hypothesis makes the triangle nondegenerate, while strict betweenness records that D lies on the continuation of BC beyond C .

16.3 Statement

Euclid states Proposition I.16 as follows:

If one side of a triangle is produced, the exterior angle is greater than either of the two interior opposite angles.

In the Hilbert reconstruction, strict comparison of angles is represented by `HilbertAngleLess`. Thus the formal conclusion is

$$\text{HilbertAngleLess}(BAC, ACD) \quad \wedge \quad \text{HilbertAngleLess}(ABC, ACD).$$

More explicitly,

$$\neg \text{Collinear}(A, B, C), \quad B - C - D \implies \begin{cases} \angle BAC < \angle ACD, \\ \angle ABC < \angle ACD. \end{cases}$$

The relation `HilbertAngleLess` is synthetic. It does not compare numerical measures. An angle is smaller than another when a congruent copy of the first can be realized as a proper interior subangle of the second.

16.4 Euclid's Classical Proof

Euclid proves the first inequality by a midpoint construction.

Let E be the midpoint of AC . Join BE and extend it beyond E to a point F such that

$$BE = EF.$$

Join CF .

Since $AE = EC$, $BE = EF$, and the vertical angles $\angle AEB$ and $\angle CEF$ are equal by Proposition I.15, the triangles AEB and CEF are congruent by Proposition I.4. Hence

$$\angle BAE = \angle ECF.$$

Since E lies on AC , this gives

$$\angle BAC = \angle ACF.$$

But the ray CF lies inside the exterior angle ACD . Therefore

$$\angle BAC < \angle ACD.$$

For the second opposite angle Euclid repeats the same argument after extending another side of the triangle. The equality of the corresponding vertical exterior angles then identifies the resulting exterior angle with $\angle ACD$.

Thus the classical architecture is

$$\text{midpoint construction} \rightarrow \text{I.15} \rightarrow \text{SAS (I.4)} \rightarrow \text{interior subangle} \rightarrow \text{I.16}.$$

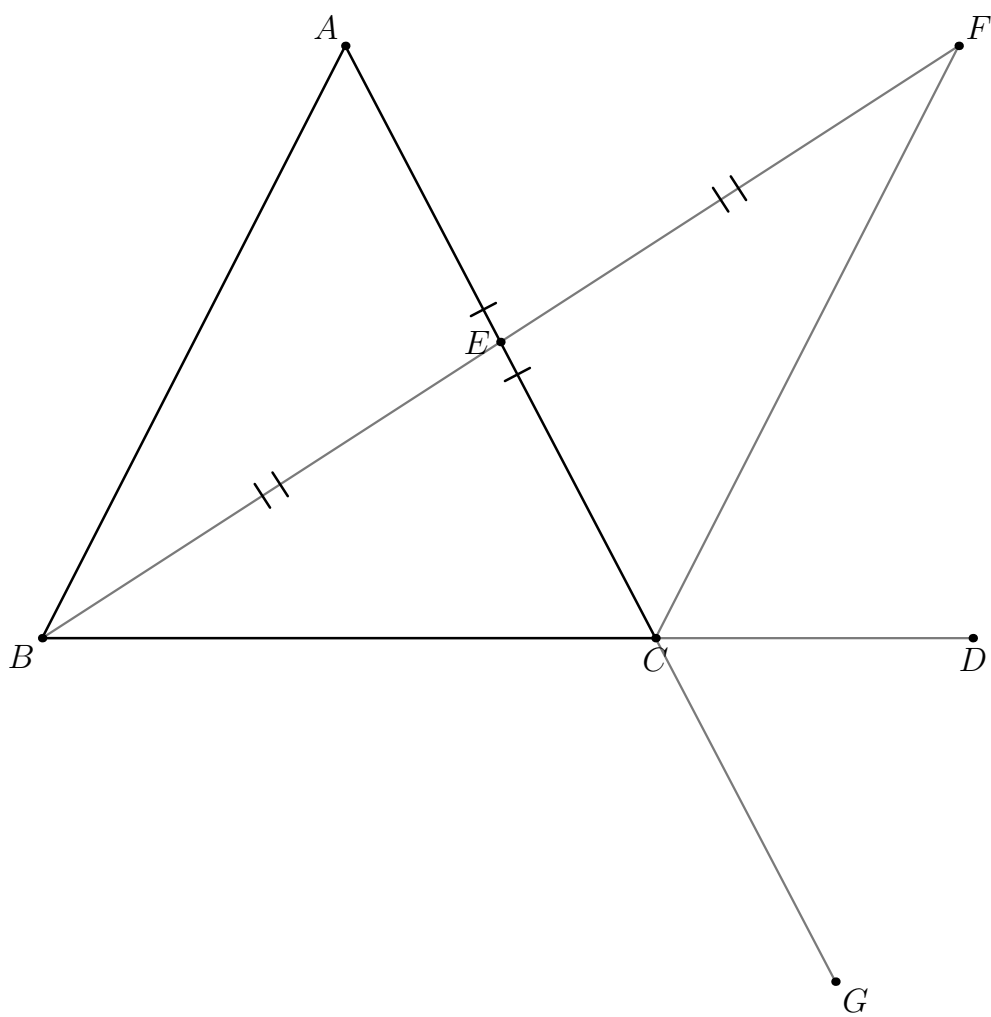


Figure 16.1: Classical Wilkins configuration for the first inequality. The midpoint construction uses $A - E - C$, $AE \cong EC$; BE is produced through E to F with $BE \cong EF$; BC is produced through C to D , and AC through C to G . The ray CF lies in the exterior angle only after the required side/order argument.

16.5 Proof Architecture of the Reconstruction

The Hilbert reconstruction retains the synthetic structure of Euclid’s argument but packages the difficult order-theoretic steps into reusable interface theorems.

For the first opposite angle the auxiliary construction produces points M and E with

$$\begin{aligned} A - M - C, \quad & AM \cong MC, \\ B - M - E, \quad & BM \cong ME, \end{aligned}$$

and

$$\angle BAC \cong \angle MCE.$$

These are the exact geometric outputs unpacked from `hilbert_exterior_angle_aux`. From these data the interface proves that AB is parallel to CE . The theorem `hilbert_exterior_angle_less` then concludes directly that

$$\angle BAC < \angle ACD.$$

Schematically:

$$\begin{array}{c} \neg \text{Collinear}(A, B, C) \\ \downarrow \\ \text{hilbert_exterior_angle_aux} \\ \downarrow \\ A - M - C, B - M - E, AM \cong MC, BM \cong ME, \angle BAC \cong \angle MCE \\ \downarrow \\ \text{hilbert_exterior_angle_aux_parallel} \\ \downarrow \\ AB \parallel CE \\ \downarrow \\ \text{hilbert_exterior_angle_less} \\ \downarrow \\ \angle BAC < \angle ACD. \end{array}$$

The second inequality is deliberately reduced to the first. Extend AC through C to F . Apply the first part to triangle BAC to obtain

$$\angle ABC < \angle BCF.$$

The lines ACF and BCD intersect at C . Classically this is the vertical-angle content of Proposition I.15; the current Lean proof calls the generic theorem `VerticalAngles` directly and obtains, after symmetry and ray reversal,

$$\angle BCF \cong \angle ACD.$$

Transporting the right-hand side of the strict angle comparison across this congruence yields

$$\angle ABC < \angle ACD.$$

Thus the second half has the compact architecture

$$\text{first half of I.16} \rightarrow \text{VerticalAngles} \rightarrow \text{transport of } < \rightarrow \text{second half of I.16}.$$

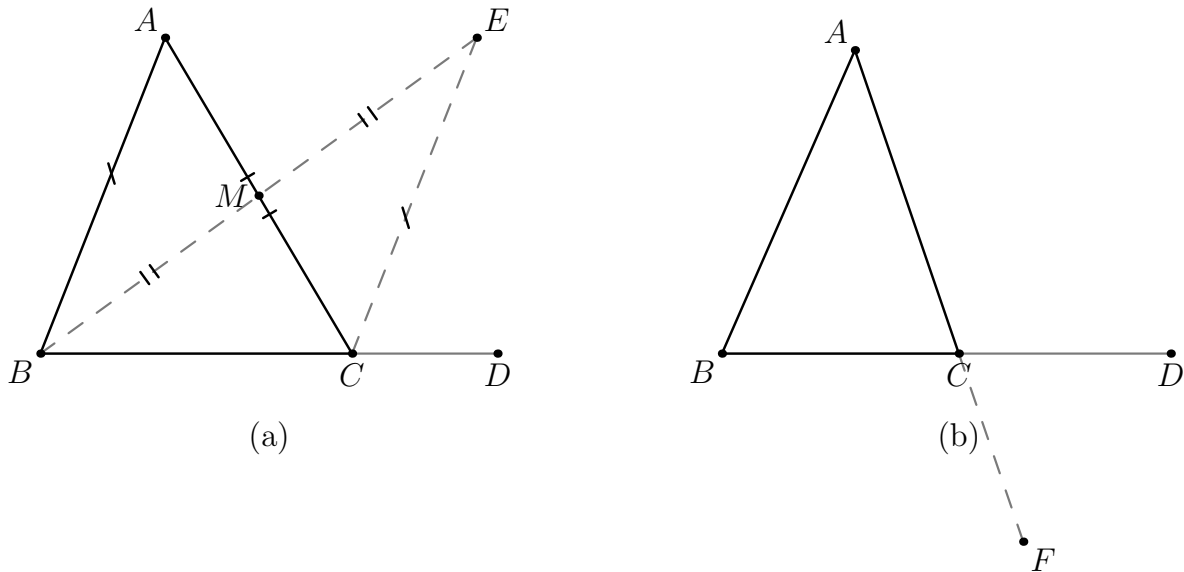


Figure 16.2: Formal Lean geometry. (a) The first branch unpacks M, E with $A - M - C$, $AM \cong MC$, $B - M - E$, $BM \cong ME$; the same helper data yield $\angle BAC \cong \angle MCE$ and then $AB \parallel CE$. (b) For the second branch Lean constructs $A - C - F$; together with $D - C - B$ this gives the vertical-angle configuration $\angle BCF \cong \angle ACD$.

16.6 Hilbert Reconstruction

16.6.1 The First Opposite Interior Angle

The theorem `euclid_proposition_16_first` begins from

$$\neg \text{Collinear}(A, B, C), \quad B - C - D.$$

It invokes

`hilbert_exterior_angle_aux`

to obtain points M, E and the auxiliary configuration required for the exterior-angle argument.

The same configuration is passed to

`hilbert_exterior_angle_aux_parallel,`

which proves

$$AB \parallel CE.$$

Finally,

`hilbert_exterior_angle_less`

combines the auxiliary construction, the extension $B - C - D$, and the parallelism to establish

$$\angle BAC < \angle ACD.$$

The Euclidean theorem itself therefore contains no unresolved construction. The substantial work has been moved into the Hilbert interface, where it can be reused independently of Proposition I.16.

16.6.2 The Second Opposite Interior Angle

For the second inequality, the proof first derives $A \neq C$ from the noncollinearity of A, B, C . Hilbert's order axiom then extends AC through C to a point F :

$$A - C - F.$$

The triangle is viewed in the order BAC . Applying `euclid_proposition_16_first` to this reordered triangle gives

$$\angle ABC < \angle BCF.$$

It remains to identify $\angle BCF$ with the desired exterior angle $\angle ACD$. From $B - C - D$ we obtain the reversed betweenness relation

$$D - C - B.$$

The two straight lines ACF and BCD therefore form a pair of vertical angles at C . The current Lean proof calls `VerticalAngles` directly (the generic theorem corresponding to the classical content of I.15) and, after the required symmetry and reversal operations, obtains

$$\angle BCF \cong \angle ACD.$$

The theorem

`hilbert_angleLess_transport_right`

then transports the comparison across the congruent right-hand angle:

$$\angle ABC < \angle BCF, \quad \angle BCF \cong \angle ACD \implies \angle ABC < \angle ACD.$$

16.7 Lean Formalization

The first half is formalized as follows:

```

theorem euclid_proposition_16_first
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D : Geo.Point)
  (hABC : not PrimCollinear Geo A B C)
  (hBCD : Geo.Between B C D) :
  HilbertAngleLess Geo B A C A C D := by

rcases
  hilbert_exterior_angle_aux
    Geo A B C hABC with
  <M, E,
  hAMC,
  hAMMC,
  hBME,
  hBMME,
  hAngle>

have hParallel :
  Geo.Parallel A B C E :=

```

```

hilbert_exterior_angle_aux_parallel
  Geo A B C M E
  hABC
  hAMC
  hBME
  hAngle

exact
hilbert_exterior_angle_less
  Geo A B C D M E
  hABC
  hAMC
  hBME
  hBCD
  hAngle
  hParallel

```

The second half deliberately reuses the first:

```

have hLessBCF :
  HilbertAngleLess Geo A B C B C F :=
  euclid_proposition_16_first
  Geo
  B A C F
  hBACnc
  hACF

```

The vertical-angle relation is then constructed at C and normalized to the required orientation:

```

have hVerticalRaw :
  Geo.AngleCongruent A C D F C B :=
  VerticalAngles
  Geo
  A C D F B
  hACF
  hDCA
  hACDnc

have hVerticalSymm :
  Geo.AngleCongruent F C B A C D :=
  Geo.angle_congruent_symmetry
  A C D
  F C B
  hVerticalRaw

have hVertical :
  Geo.AngleCongruent B C F A C D :=
  (Geo.angle_congruent_reverse_first
   F C B A C D).mp
  hVerticalSymm

```

Finally the strict inequality is transported:

```

exact
hilbert_angleLess_transport_right
  Geo

```



```

A B C
B C F
A C D
hLessBCF
hACDnc
hVertical

```

The complete proposition simply combines the two parts:

```

theorem euclid_proposition_16
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D : Geo.Point)
  (hABC : not PrimCollinear Geo A B C)
  (hBCD : Geo.Between B C D) :
  HilbertAngleLess Geo B A C A C D /\
  HilbertAngleLess Geo A B C A C D := by

  have hFirst :=
    euclid_proposition_16_first
      Geo A B C D hABC hBCD

  have hSecond :=
    euclid_proposition_16_second
      Geo A B C D hABC hBCD

  exact <hFirst, hSecond>

```

16.8 Relationship with Earlier Propositions

16.8.1 Proposition I.10

Euclid’s classical proof uses a midpoint construction as part of the auxiliary configuration. Proposition I.10 is therefore one of the constructional ingredients behind the traditional argument. In the present Lean file, however, no call to `euclid_proposition_10` occurs: the midpoint-style geometry has already been packaged inside the Hilbert interface helper `hilbert_exterior_angle_aux`.

16.8.2 Proposition I.15

Classically, vertical-angle equality is Proposition I.15. The second formal branch uses exactly that geometry after constructing $A - C - F$ and reversing $B - C - D$ to $D - C - B$. The current production file does not import `Proposition15` or call `euclid_proposition_15`; it calls the generic interface theorem `VerticalAngles` directly. Thus I.15 is the classical correspondence, while `VerticalAngles` is the direct formal dependency.

16.8.3 Transition to Proposition I.17

Proposition I.16 supplies the strict exterior-angle comparisons used immediately in I.17. Once a side is extended, I.17 can represent “two angles together less than two right angles” by comparing

an interior angle with the supplementary exterior angle. I.16 therefore begins the strict angle-order block that continues through I.17–I.19.

16.9 Hilbert and Book Zero Components Used

16.9.1 HilbertAngleLess

The proposition requires a genuinely synthetic notion of strict angle comparison. The relation `HilbertAngleLess` expresses that a congruent copy of one angle occurs as a proper interior subangle of another. This avoids introducing numerical angle measure and keeps Proposition I.16 inside pure incidence, order, and congruence geometry.

16.9.2 Auxiliary Exterior-Angle Construction

This theorem supplies the auxiliary points used in the first exterior-angle construction. It packages the midpoint-style configuration that underlies Euclid’s original proof and returns the betweenness and congruence data required by the subsequent argument.

16.9.3 Derived Parallelism

From the auxiliary configuration, this theorem proves

$$AB \parallel CE.$$

The parallel relation is not an additional Euclidean postulate here; it is a derived relation inside the neutral Hilbert development used to organize the exterior-angle argument.

16.9.4 Exterior-Angle Strict Comparison

This is the principal interface theorem used by the first half of I.16. It converts the auxiliary construction and its parallelism into the strict angle comparison

$$\angle BAC < \angle ACD.$$

Its proof depends on the interior-subangle machinery developed in the Hilbert layer.

16.9.5 Interior-subangle transport

A substantial part of the work required for Proposition I.16 lies below the final Euclidean file. The Hilbert interface establishes that an interior ray can be transported between congruent angles while preserving the corresponding subangle.

The completed infrastructure includes the segment and crossbar mechanisms needed for that transport, in particular the ability to transport strict segment comparison through congruence, transport an interior point between congruent segments, and preserve ray–segment intersection when the endpoints are moved along the same two rays.

This is the part of the reconstruction where the formal proof exposes geometric structure that is largely implicit in the paper proof.

16.9.6 VerticalAngles

The second half uses the generic theorem **VerticalAngles**, which is the formal interface counterpart of the vertical-angle content of Proposition I.15. After AC is extended through C to F , the angles

$$\angle BCF \quad \text{and} \quad \angle ACD$$

are vertical. Their congruence allows the second inequality to be reduced to the first.

16.9.7 Right-Hand Angle Transport

This theorem expresses invariance of strict angle comparison under congruence of the right-hand angle. In the final step,

$$\angle ABC < \angle BCF$$

and

$$\angle BCF \cong \angle ACD$$

imply

$$\angle ABC < \angle ACD.$$

16.10 Formalization Notes

Proposition I.16 is the transition from angle congruence to strict angle comparison. The production theorem uses **HilbertAngleLess**; no numerical angle measure is introduced. The first inequality is delegated to reusable exterior-angle infrastructure, while the second reuses the first after a genuinely new extension $A - C - F$ and a vertical-angle transport.

The two branches are not geometric cases: both conclusions hold simultaneously. Their asymmetry is representational and proof-theoretic. The first branch exposes the hidden midpoint-style configuration M, E ; the second creates a new extension and normalizes the resulting vertical angle to the statement orientation.

16.11 Diagrammatic Audit

The preserved Book1R chapter contained two figures: a statement-only extension $B - C - D$ and the classical Wilkins construction. The current production Lean reveals two additional proposition-level geometric states. The audited figure set therefore records the geometry as follows.

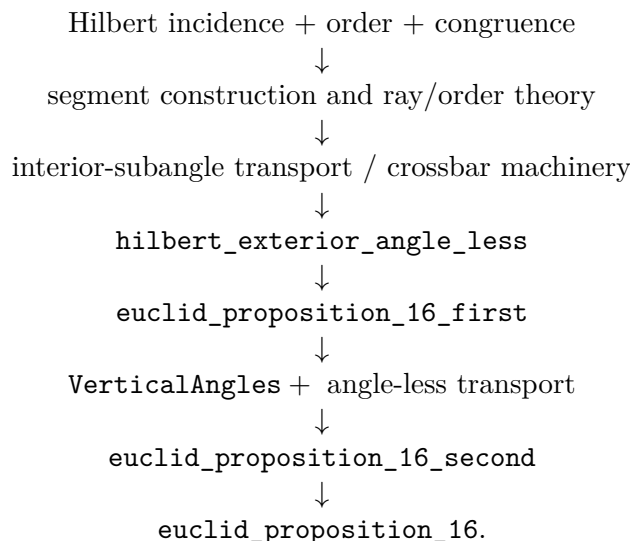
- Figure 16.1 preserves the classical Wilkins construction. It is kept separate from the formal helper because the point names and proof architecture are not the same: classical E, F correspond qualitatively to formal M, E , but the two layers must not be conflated.
- Figure 16.2(a) records the exact output of **hilbert_exterior_angle_aux**: $A - M - C$, $AM \cong MC$, $B - M - E$, $BM \cong ME$, together with the auxiliary ray CE . The derived relation $AB \parallel CE$ is marked on the same panel because it adds no new points or incidence state.
- Figure 16.2(b) records the genuinely new extension $A - C - F$ used in **euclid_proposition_16_second**. Together with $D - C - B$ it forms the vertical-angle configuration at C .

- The former statement-only figure $B - C - D$ is not retained as a separate float: that geometry is present in both substantive figures. Likewise, congruence symmetry, angle reversal, and `hilbert_angleLess_transport_right` change the logical reading of an existing configuration rather than creating new geometry.
- Internal witnesses used below `hilbert_exterior_angle_less` or inside the representation of `HilbertAngleLess` are reusable interface witnesses. They are described in the dependency discussion but are not proposition-specific diagram states.

Thus the audited chapter uses two figures with three panels: one classical panel and a two-panel formal figure. A figure records geometry, not tactic state.

16.12 Dependency Summary

The formal dependency structure can be summarized as



The final Euclidean theorem is therefore compact, but its compactness is meaningful: the difficult geometry has been factored into reusable Hilbert-interface results rather than hidden in an axiom or a local `sorry`.

16.13 Conclusion

Proposition I.16 marks a significant step beyond the immediately preceding propositions. It introduces a strict comparison of angles and requires a robust synthetic notion of interiority. In Euclid's presentation this is handled by a midpoint construction, SAS, and the observation that one angle is a proper part of another. In the formal Hilbert reconstruction the same content is decomposed into explicit reusable mechanisms for rays, betweenness, congruence, crossbar behavior, and transport of strict angle comparison.

The final Lean theorem is short precisely because those mechanisms have been established independently. The current `Proposition16.lean` contains no local `sorry` or local axiom declaration. This documentation audit did not run `lake build` or `#print axioms`, so no fresh compilation or axiomatic-status claim is made here.

Chapter 17

Proposition I.17

17.1 Introduction

Euclid’s Proposition I.17 states that in every triangle any two interior angles are together less than two right angles.

For a nondegenerate triangle ABC , the three conclusions are

$$\angle BAC + \angle ABC < 2R,$$

$$\angle BAC + \angle ACB < 2R,$$

and

$$\angle ABC + \angle ACB < 2R.$$

In Euclid’s classical proof, Proposition I.16 supplies an exterior-angle inequality and Proposition I.13 identifies an adjacent pair of angles with two right angles. The informal argument then adds the same angle to both sides of an inequality.

The formal reconstruction takes a different route. No numerical angle measure and no binary operation of angle addition are introduced. Instead, the phrase “two angles are together less than two right angles” is represented synthetically: the first angle is required to be strictly smaller than the supplementary angle of the second. This follows the same design principle already used in the reconstruction of Proposition I.14, where “equal to two right angles” was encoded by comparison with a supplementary angle.

The final Lean development consists of a new interface predicate, three pairwise forms of Proposition I.17, and one wrapper theorem combining them.

17.2 The Classical Configuration

Let ABC be a nondegenerate triangle. Euclid proves a representative case by extending one side of the triangle. In Wilkins’s presentation, BC is produced beyond C to a point D :

$$B - C - D.$$

Proposition I.16 gives

$$\angle BAC < \angle ACD \quad \text{and} \quad \angle ABC < \angle ACD.$$

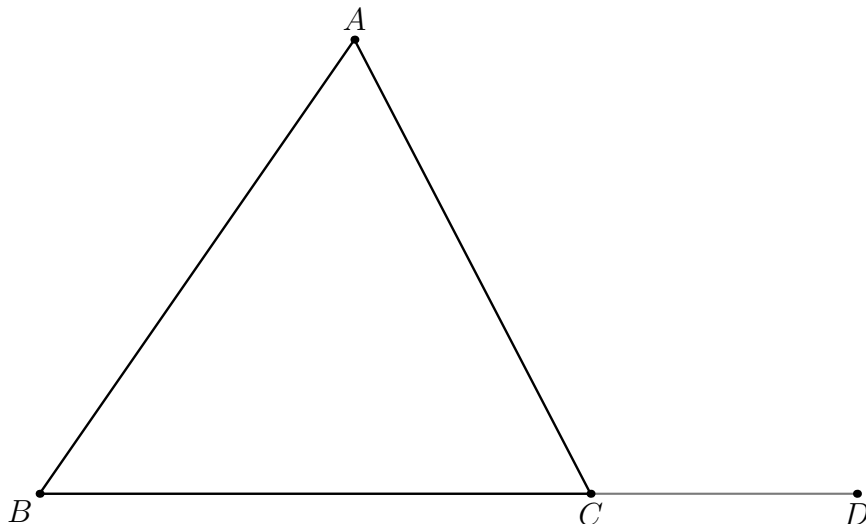


Figure 17.1: The classical extension used for the pair $\angle ABC, \angle ACB$: BC is produced beyond C .

Since CB and CD are opposite rays, $\angle ACB$ and $\angle ACD$ form a supplementary pair. Proposition I.13 therefore gives the classical conclusion

$$\angle ABC + \angle ACB < 2R.$$

The formal reconstruction uses the same extension pattern for two of the three pairwise conclusions. I.16 supplies both

$$\angle BAC < \angle ACD \quad \text{and} \quad \angle ABC < \angle ACD,$$

so the geometry $B - C - D$ proves both

$$\angle BAC + \angle ACB < 2R$$

and

$$\angle ABC + \angle ACB < 2R.$$

In the current Lean implementation these two pairwise theorems each call `HilbertOrder.between_extension` independently; the diagram records the common geometric pattern, not identity of their existential witnesses.

For the remaining pair, the formal proof uses the symmetric extension at the other endpoint of the base. Produce CB beyond B to a point E :

$$C - B - E.$$

Applying I.16 to the reordered triangle ACB yields

$$\angle CAB < \angle ABE.$$

After reversing the first angle,

$$\angle BAC < \angle ABE,$$

and $\angle ABE$ is supplementary to $\angle ABC$. Hence

$$\angle BAC + \angle ABC < 2R.$$

The two figures therefore have genuinely different roles. They are not duplicate views of the same proof: the first extension pattern supplies the two pairs containing $\angle ACB$, while the second supplies the remaining pair.

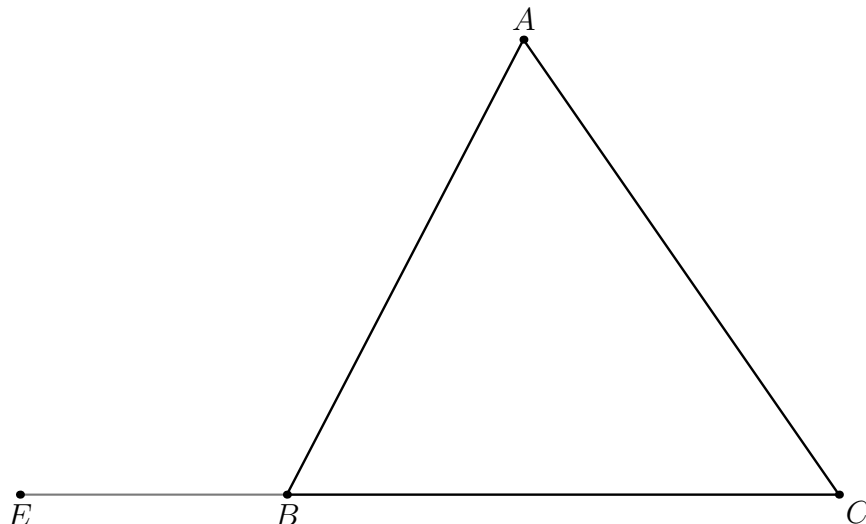


Figure 17.2: The second extension used by the Lean wrapper: CB is produced beyond B .

17.3 Classical Proof

Euclid proves one representative case and then observes that the same argument applies to the other pairs.

Extend BC beyond C to D . By Proposition I.16,

$$\angle ABC < \angle ACD.$$

Add $\angle ACB$ to both sides:

$$\angle ABC + \angle ACB < \angle ACD + \angle ACB.$$

By Proposition I.13, the adjacent angles $\angle ACD$ and $\angle ACB$ are together equal to two right angles. Hence

$$\angle ABC + \angle ACB < 2R.$$

The other two pairs follow in the same way after choosing the appropriate side extension.

The classical dependency pattern is therefore

$$\begin{array}{c}
 \text{I.16 exterior-angle inequality} \\
 \downarrow \\
 \text{add the remaining interior angle} \\
 \downarrow \\
 \text{I.13 supplementary pair} \\
 \downarrow \\
 < 2R.
 \end{array}$$

The formal reconstruction deliberately removes the middle arithmetic step.

17.4 Synthetic Representation of “Less Than Two Right Angles”

The central design choice is the predicate

HilbertAnglesLessThanTwoRightAngles.

For two angles α and β , instead of introducing a sum $\alpha + \beta$, choose a supplementary angle β' to β . Then

$$\alpha + \beta < 2R$$

is represented by

$$\alpha < \beta'.$$

In the formal language, if the second angle is $\angle CPD$, a witness E is chosen with

$$D - P - E.$$

Then PE is the opposite ray to PD , so $\angle CPE$ is supplementary to $\angle CPD$. The definition requires

$$\angle AOB < \angle CPE.$$

The Lean definition is:

```
def HilbertAnglesLessThanTwoRightAngles
  [HilbertIncidence Geo]
  (A O B C P D : Geo.Point) : Prop :=
  exists E : Geo.Point,
    Geo.Between D P E /\
    HilbertAngleLess Geo A O B C P E
```

This representation is entirely synthetic. It records only strict betweenness and strict comparison of angles. No real-valued measure, no degree notation, and no algebraic operation on angles is required.

17.5 Proof Architecture of the Reconstruction

The reconstruction is especially compact because Proposition I.16 already gives exactly the inequalities needed against exterior angles.

For the two pairs containing $\angle ACB$, extend BC beyond C :

$$B - C - D.$$

Then Proposition I.16 gives

$$\angle BAC < \angle ACD$$

and

$$\angle ABC < \angle ACD.$$

Since $\angle ACD$ is the supplement of $\angle ACB$, these are precisely the synthetic forms of

$$\angle BAC + \angle ACB < 2R$$

and

$$\angle ABC + \angle ACB < 2R.$$

For the remaining pair, extend CB beyond B :

$$C - B - E.$$

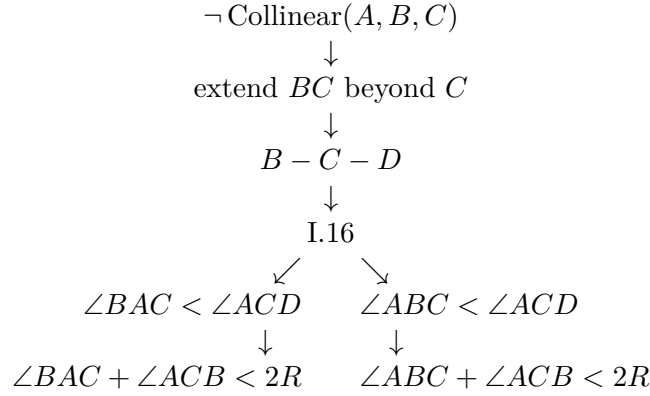
Apply Proposition I.16 to the reordered triangle ACB . This gives

$$\angle CAB < \angle ABE.$$

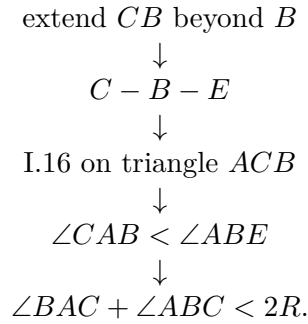
Reversal of the first angle identifies $\angle CAB$ with $\angle BAC$, while $\angle ABE$ is supplementary to $\angle ABC$. Hence

$$\angle BAC + \angle ABC < 2R.$$

Schematically:



and separately



17.6 Diagrammatic Audit

The current production proof contains two genuinely different incidence/order states and no geometric case split.

- Figure 17.1 records the extension pattern $B - C - D$. It supports the two pairwise results whose second angle is $\angle ACB$. The two Lean theorems construct their extension witnesses independently, but no additional diagram is justified because their geometry is identical.
- Figure 17.2 records the distinct extension $C - B - E$ required for the pair $\angle BAC, \angle ABC$ after reordering the triangle.
- The witness X unpacked from `HilbertAngleLess` in the third pair is a representation witness for strict angle comparison. It introduces no new proposition-specific incidence/order state and is therefore not drawn.
- Collinearity symmetry, angle reversal, and the choice between `euclid_proposition_16_first` and `euclid_proposition_16_second` change the logical reading of these configurations, not the configurations themselves.

Thus the final audit retains exactly two figures. A figure records geometry, not theorem-selection or normalization state.

17.7 Lean Formalization

17.7.1 The Pair $\angle BAC$ and $\angle ACB$

The first theorem extends BC through C and uses the first half of Proposition I.16.

```

theorem euclid_proposition_17_BAC_ACB
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C : Geo.Point)
  (hABC : not PrimCollinear Geo A B C) :
  HilbertAnglesLessThanTwoRightAngles
    Geo B A C A C B := by

  have hBCA :
    not PrimCollinear Geo B C A := by
  intro h

  have hACB :
    PrimCollinear Geo A C B :=
    PrimCollinearSymm Geo B C A h

  have hABC' :
    PrimCollinear Geo A B C :=
    PrimCollinearRotate Geo A C B hACB

  exact hABC hABC'

  have hBC : B != C :=
    hilbert_noncollinear_ne_first
      Geo B C A hBCA

  rcases
    HilbertOrder.between_extension
      B C hBC with
    <D, hBCD>

  refine <D, hBCD, ?_>

  exact
    euclid_proposition_16_first
      Geo A B C D hABC hBCD

```

The witness D simultaneously records the supplementary configuration and serves as the exterior-angle target in I.16.

17.7.2 The Pair $\angle ABC$ and $\angle ACB$

The second theorem uses the same extension but invokes the second half of I.16.

```

theorem euclid_proposition_17_ABC_ACB
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C : Geo.Point)
  (hABC : not PrimCollinear Geo A B C) :

```

```

HilbertAnglesLessThanTwoRightAngles
  Geo A B C A C B := by

have hBCA :
  not PrimCollinear Geo B C A := by
intro h

have hACB :
  PrimCollinear Geo A C B :=
  PrimCollinearSymm Geo B C A h

have hABC' :
  PrimCollinear Geo A B C :=
  PrimCollinearRotate Geo A C B hACB

exact hABC hABC'

have hBC : B != C :=
  hilbert_noncollinear_ne_first
  Geo B C A hBCA

rcases
  HilbertOrder.between_extension
    B C hBC with
  <D, hBCD>

refine <D, hBCD, ?_>

exact
  euclid_proposition_16_second
  Geo A B C D hABC hBCD

```

Thus the two inequalities associated with the vertex C differ only in which conclusion of I.16 is selected. Each theorem performs its own call to `extttbetween_extension`; the shared symbol D denotes the same geometric role, not a proof-term-level shared witness.

17.7.3 The Pair $\angle BAC$ and $\angle ABC$

For the remaining pair, extend CB through B to E and apply I.16 to the triangle in the order ACB .

```

theorem euclid_proposition_17_BAC_ABC
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C : Geo.Point)
  (hABC : not PrimCollinear Geo A B C) :
  HilbertAnglesLessThanTwoRightAngles
    Geo B A C A B C := by

have hCBA :
  not PrimCollinear Geo C B A := by
intro h

have hABC' :
  PrimCollinear Geo A B C :=
  PrimCollinearSymm Geo C B A h

```

```

exact hABC hABC'

have hCB : C != B :=
  hilbert_noncollinear_ne_first
  Geo C B A hCBA

rcases
  HilbertOrder.between_extension
    C B hCB with
  <E, hCBE>

refine <E, hCBE, ?_>

have hACB :
  not PrimCollinear Geo A C B := by
intro h

have hABC' :
  PrimCollinear Geo A B C :=
  PrimCollinearRotate Geo A C B h

exact hABC hABC'

have hLessCAB :
  HilbertAngleLess Geo C A B A B E :=
  euclid_proposition_16_first
  Geo A C B E hACB hCBE

rcases hLessCAB with
  <hCABnc, hABEnc, X, hMeet, hAngle>

have hBACnc :
  not PrimCollinear Geo B A C := by
intro h
exact hABC
  (PrimCollinearSwap Geo B A C h)

have hAngle' :
  Geo.AngleCongruent B A C A B X :=
  (Geometry.Geo.angle_congruent_reverse_first
    Geo C A B A B X).mp hAngle

exact
  <hBACnc,
    hABEnc,
    X,
    hMeet,
    hAngle'>

```

The only additional work here is normalization of the orientation of the first angle: I.16 returns $\angle CAB < \angle ABE$, whereas the final interface uses $\angle BAC$. The unordered-angle representation makes this a reversal operation rather than a new geometric argument.

17.7.4 The Final Wrapper

The final proposition simply combines the three pairwise results.

```

theorem euclid_proposition_17
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C : Geo.Point)
  (hABC : not PrimCollinear Geo A B C) :
  HilbertAnglesLessThanTwoRightAngles Geo B A C A B C /\
  HilbertAnglesLessThanTwoRightAngles Geo B A C A C B /\
  HilbertAnglesLessThanTwoRightAngles Geo A B C A C B := by

  have h1 :=
    euclid_proposition_17_BAC_ABC
    Geo A B C hABC

  have h2 :=
    euclid_proposition_17_BAC_ACB
    Geo A B C hABC

  have h3 :=
    euclid_proposition_17_ABC_ACB
    Geo A B C hABC

  exact <h1, h2, h3>

```

17.8 Relationship with Propositions I.13, I.14, and I.16

Proposition I.17 sits at an interesting point in the reconstruction because its classical statement appears to require angle arithmetic, while the existing synthetic interface allows that arithmetic to be avoided.

17.8.1 Proposition I.13

I.13 identifies two adjacent angles on a straight line as a supplementary pair. In the Book Zero reconstruction this is represented structurally by `BookZeroSupplement`, rather than by an equation involving angle sums.

For I.17 we use the same geometric idea but do not need to invoke the theorem I.13 explicitly. The witness produced by `between_extension` already records the opposite ray required to construct the supplementary angle.

17.8.2 Proposition I.14

I.14 introduced the useful design pattern of representing “equal to two right angles” by congruence with a supplementary angle. The I.17 predicate is the strict-inequality analogue of that pattern:

$$\alpha + \beta = 2R \quad \rightsquigarrow \quad \alpha \cong \text{supp}(\beta),$$

while

$$\alpha + \beta < 2R \quad \rightsquigarrow \quad \alpha < \text{supp}(\beta).$$

Thus I.17 extends an already established interface convention rather than introducing a new angle algebra.

17.8.3 Proposition I.16

I.16 is the decisive geometric input. Once a side is extended, it provides strict comparison of each remote interior angle with the exterior angle. Since that exterior angle is automatically supplementary to the adjacent interior angle, the conclusions of I.16 are already in the exact form required by `HilbertAnglesLessThanTwoRightAngles`.

This gives the particularly short dependency path

$$\text{betweenness extension} \longrightarrow \text{I.16} \longrightarrow \text{synthetic} < 2R.$$

17.9 Book Zero and Hilbert Components Used

17.9.1 HilbertAngleLess

The strict comparison relation introduced for I.16 remains the fundamental notion of angle inequality. It is synthetic: one angle is less than another when a congruent copy of the first occurs as a proper interior subangle of the second.

17.9.2 HilbertOrder.between_extension

This construction supplies the points D and E on the required opposite rays:

$$B - C - D \quad \text{or} \quad C - B - E.$$

The witness is simultaneously the side extension used by I.16 and the supplementary-ray witness used by the definition of “less than two right angles.”

17.9.3 Collinearity Symmetry and Rotation

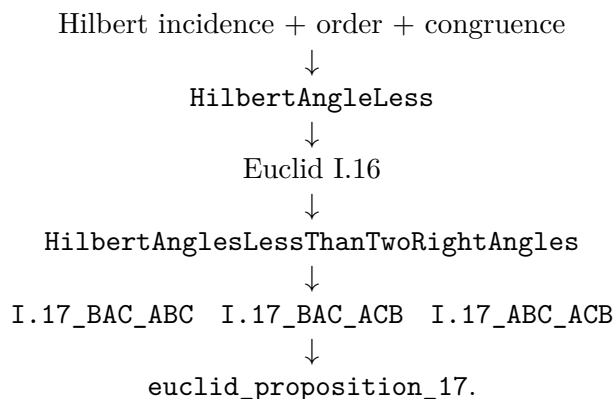
The three applications of I.16 require the triangle vertices in different orders. The small uses of `PrimCollinearSymm`, `PrimCollinearRotate`, and `PrimCollinearSwap` normalize noncollinearity hypotheses without adding any geometry.

17.9.4 Angle Reversal

In the third pair, Proposition I.16 naturally produces the angle $\angle CAB$, while the public form of I.17 uses $\angle BAC$. The theorem `angle_congruent_reverse_first` transfers the congruence across reversal of the first angle.

17.10 Dependency Summary

The formal dependency structure is



No proposition-local axiom declaration appears in the current source. More importantly, no general arithmetic of angles is introduced merely to reproduce Euclid’s phrase “together less than two right angles.” The proposition is expressed entirely in the existing synthetic language of rays, betweenness, supplementary configurations, and strict angle comparison.

17.11 Conclusion

Proposition I.17 is a useful example of the difference between reproducing a classical proof literally and reconstructing its mathematical content over a synthetic interface.

Euclid’s paper proof uses an implicit monotonicity principle for angle addition: from $\alpha < \beta$ one passes to $\alpha + \gamma < \beta + \gamma$. A literal formalization would therefore require an angle-sum operation and an order theory for such sums.

The Hilbert/Book Zero reconstruction avoids that additional layer. The statement “two angles are together less than two right angles” is encoded directly by comparison of one angle with the supplement of the other. Once this representation is chosen, Proposition I.16 supplies exactly the needed inequalities, and Proposition I.17 becomes a short structural consequence of side extension and exterior-angle comparison.

The current `Proposition17.lean` source contains no local `sorry` and no local `axiom` declaration. This documentation audit did not rerun `lake build` or `#print axioms`, so no new compilation or axiomatic-status claim is made here.

Chapter 18

Proposition I.18

18.1 Introduction

Euclid's Proposition I.18 states that in a triangle the greater side is opposite the greater angle.

For a nondegenerate triangle ABC , if

$$AB < AC,$$

then

$$\angle ACB < \angle ABC.$$

The classical proof is short but structurally rich. A copy of the shorter side AB is laid off on the longer side AC . This produces a point D with

$$A - D - C \quad \text{and} \quad AD \cong AB.$$

The auxiliary triangle ABD is therefore isosceles. Proposition I.5 identifies its base angles, while Proposition I.16 compares one of these angles with an angle of the original triangle. Finally, the fact that an interior subangle is smaller than the whole angle completes the comparison.

The formal reconstruction follows this synthetic argument closely. It does not introduce numerical angle measures. Instead, it uses the strict comparison relation `HilbertAngleLess` already developed for Proposition I.16.

The reconstruction also isolates two general facts about strict angle comparison:

$$\text{hilbert_interior_angle_less}$$

and

$$\text{hilbert_angleLess_trans.}$$

These belong naturally to the Hilbert interface rather than to Proposition I.18 itself.

18.2 The Classical Configuration

Let ABC be a nondegenerate triangle and suppose

$$AB < AC.$$

Choose D on AC so that

$$A - D - C \quad \text{and} \quad AD \cong AB.$$

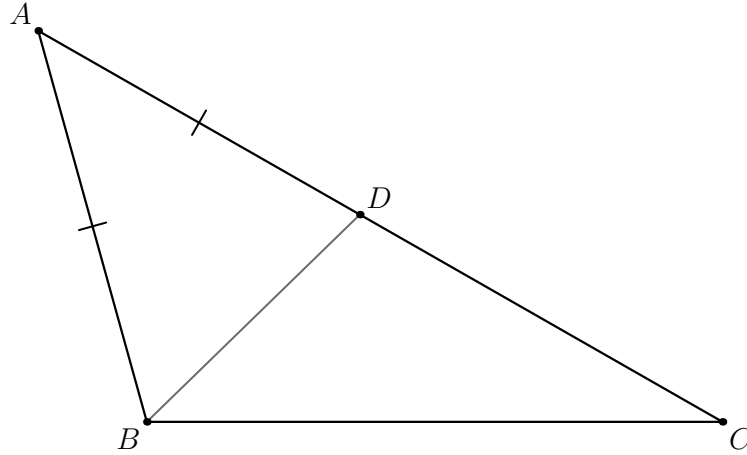


Figure 18.1: Classical and formal configuration for Proposition I.18. The point D lies strictly between A and C , with $AD \cong AB$, and the auxiliary segment BD is joined. In the current Lean theorem, D is not constructed by the proof body: it is the witness already contained in the hypothesis $AB < AC$.

The construction creates two auxiliary triangles.

First, triangle ABD is isosceles because

$$AB \cong AD.$$

Hence Proposition I.5 gives

$$\angle ABD \cong \angle ADB.$$

Second, consider triangle BCD . Since $A - D - C$, the ray DA is the continuation of DC through D . Thus $\angle BDA$ is an exterior angle of triangle BCD .

Proposition I.16 therefore gives

$$\angle BCD < \angle BDA.$$

Because A, D, C are collinear with D between A and C , the rays CD and CA coincide. Hence

$$\angle BCD = \angle BCA.$$

Combining these observations gives

$$\angle BCA < \angle BDA \cong \angle ABD.$$

Finally, the ray BD lies inside $\angle ABC$, since it meets the segment AC at the interior point D . Therefore

$$\angle ABD < \angle ABC.$$

By transitivity,

$$\angle BCA < \angle ABC,$$

which is the desired conclusion.

18.3 Diagrammatic Audit

Figure 18.1 is intentionally the only geometric figure for Proposition I.18. It records the one genuine proposition-specific state:

$$A - D - C, \quad AD \cong AB, \quad BD \text{ joined.}$$

The classical and formal configurations coincide geometrically, but the logical origin of D differs. Euclid lays off the shorter side on the longer side; the current Lean theorem instead unpacks D from the hypothesis `HilbertSegmentLess Geo A B A C`. Thus D is a hidden witness in the formal input rather than a point newly constructed by the proof body.

No additional panel is used for the exterior-angle comparison, the isosceles-angle congruence, the interior-subangle comparison, transitivity, same-ray normalization, or the final reversal from $\angle BCA$ to $\angle ACB$. These steps all read new relations from the same fixed configuration; they do not create new geometry.

The witness named X that appears when the proof unpacks the `HilbertAngleLess` result returned by Proposition I.16 is likewise not drawn. It belongs to the internal representation of strict angle comparison and is not a new proposition-specific construction.

The current production source also contains a duplicated consecutive block defining `hInsideBD` and `hABD_ABC`. The second copy shadows the first. This source-level duplication does not create another geometric state and is therefore not represented diagrammatically.

18.4 Classical Proof

Assume that in triangle ABC ,

$$AC > AB.$$

Choose a point D on AC such that

$$AD = AB.$$

Since $AC > AB$, the point D lies strictly between A and C .

The triangle ABD is isosceles, so by Proposition I.5,

$$\angle ABD = \angle ADB.$$

Now $\angle ADB$ is an exterior angle of triangle BCD . Proposition I.16 gives

$$\angle ADB > \angle BCD.$$

Since D lies on AC ,

$$\angle BCD = \angle BCA.$$

Consequently,

$$\angle ABD > \angle BCA.$$

But BD lies inside $\angle ABC$, so

$$\angle ABC > \angle ABD.$$

Therefore

$$\angle ABC > \angle BCA.$$

The classical dependency pattern is thus

$$\begin{array}{c}
 AB < AC \\
 \downarrow \\
 A - D - C, \quad AB \cong AD \\
 \downarrow \\
 \text{I.5} \\
 \downarrow \\
 \angle ABD \cong \angle ADB \\
 \downarrow \\
 \text{I.16 on } BCD \\
 \downarrow \\
 \angle BCA < \angle ADB \\
 \downarrow \\
 \angle BCA < \angle ABD \\
 \downarrow \\
 \angle ABD < \angle ABC \\
 \downarrow \\
 \angle BCA < \angle ABC.
 \end{array}$$

18.5 Strict Angle Comparison

The formal statement uses the synthetic relation

HilbertAngleLess.

No numerical measure of angles is introduced. Informally,

$$\alpha < \beta$$

means that a congruent copy of α occurs as a proper interior subangle of β .

This relation was already required for Proposition I.16. Proposition I.18 reveals that two further structural properties are needed if strict angle comparison is to behave as a genuine order relation.

The first is that an interior subangle is smaller than the whole angle.

`hilbert_interior_angle_less`

Geometrically, if a ray from the vertex of $\angle AOB$ meets the opposite side AB internally, then the angle cut off by that ray is strictly smaller than $\angle AOB$.

The second property is transitivity.

`hilbert_angleLess_trans`

It expresses the expected implication

$$\alpha < \beta \quad \text{and} \quad \beta < \gamma \quad \implies \quad \alpha < \gamma.$$

Both results are independent of Proposition I.18 and were therefore added to the general Hilbert interface.

18.6 Proof Architecture of the Reconstruction

The formal proof separates naturally into five geometric stages.

18.6.1 Recovering the Interior Point

The hypothesis

$$AB < AC$$

is represented by `HilbertSegmentLess`.

Unpacking it produces a witness D satisfying

$$A - D - C$$

and

$$AB \cong AD.$$

Thus the first step of Euclid's construction is already contained in the definition of strict segment comparison.

18.6.2 The Isosceles Triangle

From

$$AB \cong AD$$

the existing theorem

$$\text{hilbert_isosceles_base_angles}$$

gives

$$\angle ABD \cong \angle ADB.$$

This plays the classical role of Proposition I.5. The current Lean proof does not call `euclid_proposition_5` directly; it calls the reusable theorem `hilbert_isosceles_base_angles`.

18.6.3 The Exterior-Angle Comparison

Since

$$A - D - C,$$

we also have

$$C - D - A.$$

Apply Proposition I.16 to triangle BCD with DA as the extension of DC . This yields

$$\angle BCD < \angle BDA.$$

The ray from C through D is the same ray as the ray from C through A . Hence the first angle can be normalized:

$$\angle BCD = \angle BCA.$$

Thus

$$\angle BCA < \angle BDA.$$

Using the isosceles-angle congruence and the symmetry of the unordered angle representation gives

$$\angle BCA < \angle ABD.$$

18.6.4 The Interior Subangle

The point D lies on the segment AC . Consequently the ray BD meets the interior of that segment.

Therefore BD lies inside $\angle ABC$, and `hilbert_interior_angle_less` gives

$$\angle ABD < \angle ABC.$$

18.6.5 Transitivity

We now have

$$\angle BCA < \angle ABD$$

and

$$\angle ABD < \angle ABC.$$

Applying `hilbert_angleLess_trans` gives

$$\angle BCA < \angle ABC.$$

The final public statement writes the first angle as $\angle ACB$. Since the formal angle representation is insensitive to reversal of its two rays, this is a final orientation normalization rather than an additional geometric argument.

18.7 Lean Formalization

The public theorem has the form

```
theorem euclid_proposition_18
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C : Geo.Point)
  (hABC : not PrimCollinear Geo A B C)
  (hABltAC : HilbertSegmentLess Geo A B A C) :
  HilbertAngleLess Geo A C B A B C := by
```

The hypothesis `hABltAC` is immediately unpacked:

```
rcases hABltAC with
  <D, hADC, hAB_AD>
```

Thus `hADC` records

$$A - D - C,$$

while `hAB_AD` records

$$AB \cong AD.$$

The isosceles part of the argument is then:

```
have hIso :
  Geo.AngleCongruent A B D A D B :=
  hilbert_isosceles_base_angles
  Geo
  A B D
  hABD
  hAB_AD
```

The exterior-angle step uses Proposition I.16:

```
have hBCD_BDA :
  HilbertAngleLess Geo B C D B D A :=
  euclid_proposition_16_second
  Geo B C D A hBCD hCDA
```

Here

$$C - D - A$$

is obtained from the original betweenness relation

$$A - D - C.$$

The ray identity $CD = CA$ is expressed by `HilbertSameRay`:

```
have hRayCDA :
  HilbertSameRay Geo C D A :=
  hilbert_sameRay_of_between
  Geo C D A hCDA
```

This permits normalization of

$$\angle BCD$$

to

$$\angle BCA.$$

After transporting the right-hand angle across the isosceles congruence, the proof obtains

```
have hBCA_ABD :
  HilbertAngleLess Geo B C A A B D := ...
```

which represents

$$\angle BCA < \angle ABD.$$

For the second comparison, the ray BD meets the segment AC at D :

```
have hInsideBD :
  HilbertRayMeetsSegment Geo B D A C :=
  <D, hADC, hRayBDD>
```

Hence

```
have hABD_ABC :
  HilbertAngleLess Geo A B D A B C :=
  hilbert_interior_angle_less
  Geo
  B D A C
  hABC
  hInsideBD
```

which is exactly

$$\angle ABD < \angle ABC.$$

Transitivity now gives

```
have hFinal :
  HilbertAngleLess Geo B C A A B C :=
  hilbert_angleLess_trans
  Geo
  B C A
  A B D
  A B C
  hBCA_ABD
  hABD_ABC
```

The final lines merely reverse the first angle from

$$\angle BCA$$

to

$$\angle ACB,$$

matching the public orientation of the proposition.

18.8 Relationship with Earlier Propositions

18.8.1 Proposition I.5 and the formal base-angle theorem

Classically, Proposition I.5 supplies the equality of the base angles in the auxiliary isosceles triangle ABD :

$$AB \cong AD \implies \angle ABD \cong \angle ADB.$$

In the current Lean source this classical step is discharged directly by `hilbert_isosceles_base_angles`; there is no theorem-level call to `euclid_proposition_5`. This distinction is part of the formal dependency audit.

18.8.2 Proposition I.16

Proposition I.16 supplies the strict inequality. Applied to triangle BCD , it compares the remote interior angle at C with the exterior angle at D :

$$\angle BCD < \angle BDA.$$

Thus I.18 has a genuine proposition-level dependency on I.16: the current source calls `euclid_proposition_16_second` directly.

18.8.3 Proposition I.17

Proposition I.17 is not required by the proof of I.18. The formal dependency follows Euclid's direct argument through I.5 and I.16.

This is useful structurally: I.18 does not rely on an angle-sum theory developed for I.17. Both propositions instead reuse the synthetic strict-angle relation introduced for I.16.

18.9 Hilbert and Book Zero Components Used

18.9.1 HilbertSegmentLess

The hypothesis that one side is shorter than another already contains the point required by Euclid's construction. Unpacking

$$AB < AC$$

produces

$$A - D - C \quad \text{and} \quad AB \cong AD.$$

Thus no separate segment-layoff construction is required inside Proposition I.18.

18.9.2 hilbert_isosceles_base_angles

This theorem supplies the I.5 step for the auxiliary triangle ABD .

18.9.3 HilbertSameRay

Same-ray machinery is used to recognize that, because $A - D - C$, the rays CD and CA determine the same side of the angle at C .

This turns the I.16 result about $\angle BCD$ into the required comparison involving the original angle $\angle BCA$.

18.9.4 hilbert_interior_angle_less

This new interface theorem expresses the elementary principle that a proper interior part of an angle is smaller than the whole angle.

In I.18 it gives

$$\angle ABD < \angle ABC$$

from the fact that BD meets AC internally.

18.9.5 hilbert_angleLess_trans

This new interface theorem provides transitivity of strict angle comparison.

It combines

$$\angle BCA < \angle ABD$$

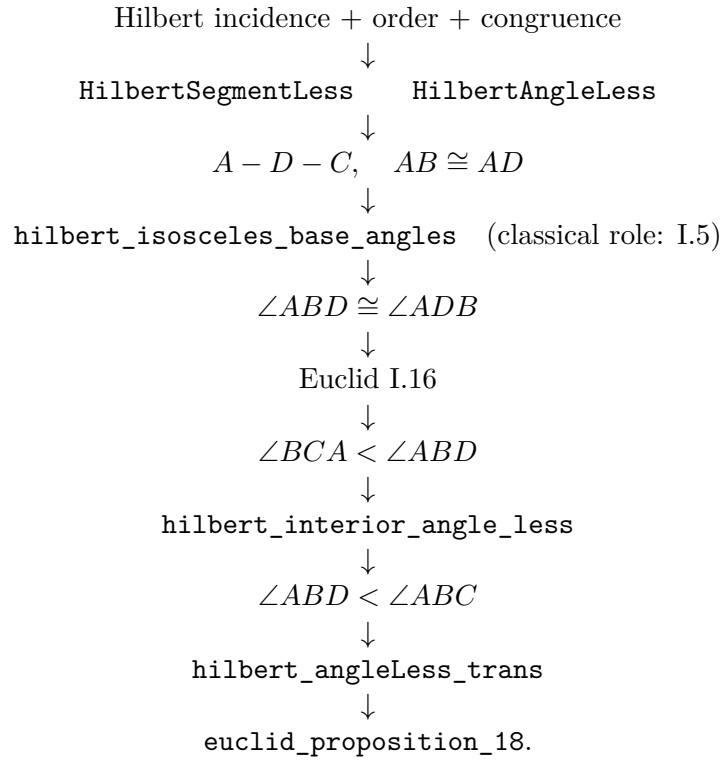
and

$$\angle ABD < \angle ABC$$

into the final inequality.

18.10 Dependency Summary

The formal dependency structure is



No proposition-specific axiom is declared in the current `Proposition18.lean` source. This documentation audit did not run `#print axioms`. The strict-angle results used here are reusable Hilbert-interface theorems rather than assumptions introduced specifically for Proposition I.18.

18.11 Conclusion

Proposition I.18 is a particularly clear example of how Euclid's construction survives formal reconstruction.

The classical proof introduces a point D on the longer side, creates an isosceles triangle, applies the exterior-angle theorem, and finally observes that a part of an angle is smaller than the whole. The Lean proof follows exactly this geometry, with the important formal difference that the point D is recovered by unpacking the strict-segment hypothesis rather than constructed anew.

What changes is not the mathematical argument but the amount of structure that must be made explicit. The informal steps

$$\angle BCA < \angle ABD < \angle ABC$$

require a formal notion of strict angle comparison, a theorem asserting that an interior subangle is smaller than its containing angle, and a proof of transitivity.

Consequently, Proposition I.18 does more than add another theorem of Book I. It strengthens the reusable language of synthetic angle order. The new results `hilbert_interior_angle_less` and `hilbert_angleLess_trans` are available for subsequent propositions independently of I.18.

The preserved Book1R documentation recorded the development as compiling cleanly and containing no `sorry`. In the present diagrammatic audit, `lake build` was not rerun; inspection of the current `Proposition18.lean` source found no `sorry`.

Chapter 19

Proposition I.19

19.1 Introduction

Euclid's Proposition I.19 is the converse of Proposition I.18. In a nondegenerate triangle, the greater angle is opposite the greater side.

For a triangle ABC , if

$$\angle ACB < \angle ABC,$$

then

$$AB < AC.$$

The classical proof is indirect. One compares the two opposite sides AB and AC . Exactly one of the following three possibilities must occur:

$$AB \cong AC, \quad AB < AC, \quad AC < AB.$$

The first case contradicts Proposition I.5, because equal sides would imply congruent base angles. The third case contradicts Proposition I.18, because the greater side AB would force the opposite angle at C to be greater than the angle at B . Hence only

$$AB < AC$$

remains.

The current formal reconstruction makes this trichotomy explicit through the reusable general theorem

`hilbert_segment_trichotomy.`

It also uses the general irreflexivity theorem for strict angle comparison:

`hilbert_angleLess_irrefl.`

These two results are reusable structural components of the Hilbert layer rather than local devices specific to Proposition I.19.

19.2 The Classical Configuration

Let ABC be a nondegenerate triangle and suppose

$$\angle ACB < \angle ABC.$$

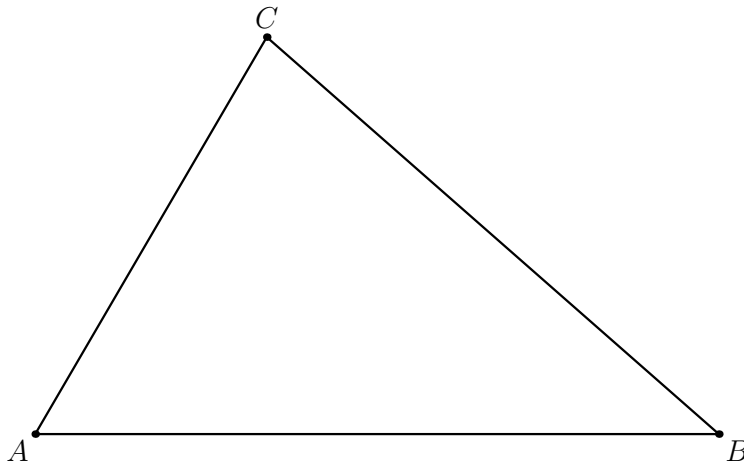


Figure 19.1: Classical and formal geometric configuration for Proposition I.19. The strict angle hypothesis $\angle ACB < \angle ABC$ and the target $AB < AC$ are stated in the text; they are not encoded by the apparent sizes of the drawn angles or sides.

The theorem asserts that the side opposite the larger angle must be larger:

$$AB < AC.$$

No new auxiliary point is required. The proof is entirely comparative: it uses the trichotomy of the two sides and eliminates the two alternatives incompatible with the assumed angle inequality.

19.3 Diagrammatic Audit

The diagrammatic audit retains exactly one geometric figure. Proposition I.19 itself constructs no new point, line, extension, intersection, betweenness relation, or auxiliary triangle. The three alternatives produced by segment trichotomy,

$$AB \cong AC, \quad AB < AC, \quad AC < AB,$$

are logical alternatives for the same fixed triangle ABC ; they are not three geometric states and therefore do not receive three panels.

The reusable theorem `hilbert_segment_trichotomy` has an internal construction involving a copied endpoint on a ray. That geometry belongs to the generic Hilbert comparison theorem, not to the proposition-specific proof of I.19, so it is documented textually rather than duplicated here. Likewise, `hilbert_angleLess_irrefl` is used only as a contradiction endpoint and introduces no new proposition-specific geometry.

Figure 19.1 therefore records only the common triangle. Neither the strict angle inequality nor the desired strict side inequality is inferred from the visual proportions of the drawing.

19.4 Classical Proof

Assume

$$\angle ACB < \angle ABC.$$

Compare the two sides AB and AC .

If

$$AB = AC,$$

then triangle ABC is isosceles. Proposition I.5 gives

$$\angle ABC = \angle ACB,$$

contradicting the hypothesis that

$$\angle ACB < \angle ABC.$$

If instead

$$AC < AB,$$

then Proposition I.18, applied to triangle ACB , yields

$$\angle ABC < \angle ACB,$$

which again contradicts the original hypothesis.

The only remaining possibility is

$$AB < AC.$$

Thus the greater angle is subtended by the greater side.

The classical dependency pattern is therefore

$$\begin{array}{c}
 \angle ACB < \angle ABC \\
 \downarrow \\
 AB \cong AC \vee AB < AC \vee AC < AB \\
 \downarrow \\
 \begin{array}{ccc}
 AB \cong AC & AB < AC & AC < AB \\
 \downarrow & & \downarrow \\
 \text{I.5 contradiction} & \text{desired case} & \text{I.18 contradiction}
 \end{array}
 \end{array}$$

19.5 Segment Trichotomy

The key reusable structural theorem called by the formal proof is

`hilbert_segment_trichotomy.`

Its conclusion has the form

$$AB \cong CD \vee AB < CD \vee CD < AB.$$

The proof does not introduce a new axiom. It is derived from segment construction and the order theory of collinear points.

A copy of AB is laid off on the ray CD , producing a point X such that

$$CX \cong AB.$$

If $X = D$, then

$$AB \cong CD.$$

Otherwise, C, D, X are distinct collinear points. The Hilbert betweenness trichotomy gives three possibilities. The case

$$D - C - X$$

is incompatible with the fact that X lies on the ray CD . The remaining two orders give respectively

$$CD < CX$$

and

$$CX < CD.$$

Transporting these inequalities through

$$CX \cong AB$$

yields the required segment trichotomy.

Thus the theorem packages a general completeness property of strict segment comparison.

19.6 Irreflexivity of Strict Angle Comparison

The second structural theorem used in I.19 is

$$\text{hilbert_angleLess_irrefl.}$$

It states

$$\neg(\alpha < \alpha).$$

In the synthetic definition of `HilbertAngleLess`, the relation

$$\alpha < \beta$$

means that a congruent copy of α occurs as a proper interior subangle of β .

Therefore

$$\alpha < \alpha$$

would imply that an angle is congruent to one of its own proper interior subangles. This is excluded by the previously proved theorem

$$\text{hilbert_interior_subangle_not_congruent_whole.}$$

The irreflexivity result provides exactly the contradiction mechanism needed in both non-target branches of Proposition I.19.

19.7 Proof Architecture of the Reconstruction

The Lean proof divides naturally into three stages.

19.7.1 Nondegeneracy of the Compared Side

From the noncollinearity hypothesis

$$\neg \text{Collinear}(A, B, C)$$

the proof first derives

$$A \neq C.$$

This is the nondegeneracy condition required by the comparison theorem `hilbert_segment_trichotomy`.

19.7.2 Apply Segment Trichotomy

The theorem

$$\text{hilbert_segment_trichotomy}$$

is applied to the segments AB and AC .

This produces three branches:

$$AB \cong AC, \quad AB < AC, \quad AC < AB.$$

The middle branch is exactly the desired conclusion.

19.7.3 Eliminate the Two Impossible Branches

In the congruent-side branch, Proposition I.5 gives

$$\angle ABC \cong \angle ACB.$$

The assumed strict inequality

$$\angle ACB < \angle ABC$$

is transported across that congruence, producing

$$\angle ACB < \angle ACB.$$

This contradicts `hilbert_angleLess_irrefl`.

In the branch

$$AC < AB,$$

Proposition I.18 applied to triangle ACB gives

$$\angle ABC < \angle ACB.$$

Combining this with the original inequality by transitivity yields

$$\angle ACB < \angle ACB,$$

again contradicting irreflexivity.

Only

$$AB < AC$$

remains.

19.8 Lean Formalization

The current production module imports `CGJteamLab.Proposition18`. The public theorem is:

```
theorem euclid_proposition_19
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C : Geo.Point)
  (hABC : not PrimCollinear Geo A B C)
  (hAngle :
    HilbertAngleLess Geo A C B A B C) :
  HilbertSegmentLess Geo A B A C := by
```

The proof first derives the nondegeneracy of AC :

```
have hAC : A != C :=
  hilbert_noncollinear_ne_first
  Geo A C B
  (by
    intro h
    exact hABC
    (PrimCollinearRotate Geo A C B h))
```

Then segment trichotomy is applied:

```
rcases
  hilbert_segment_trichotomy
  Geo
  A B
  A C
  hAC with
  hEq | hABltAC | hACltAB
```

The middle branch closes immediately:

```
exact hABltAC
```

For the congruent-side branch, the generic isosceles-base-angle theorem (the formal counterpart of the classical I.5 step) gives:

```
have hIso :
  Geo.AngleCongruent A B C A C B :=
  hilbert_isosceles_base_angles
```



```

Geo
A B C
hABC
hEq

```

The strict inequality is transported to the congruent target angle:

```

have hSame :
  HilbertAngleLess Geo A C B A C B :=
  hilbert_angleLess_transport_right
  Geo
  A C B
  A B C
  A C B
  hAngle
  hACB
  hIso

```

This contradicts irreflexivity:

```

exact
  False.elim
    ((hilbert_angleLess_irrefl
      Geo A C B)
     hSame)

```

In the final branch, Proposition I.18 gives the opposite strict inequality:

```

have hOpp :
  HilbertAngleLess Geo A B C A C B :=
  euclid_proposition_18
  Geo
  A C B
  hACB
  hACltAB

```

Transitivity then creates an impossible strict self-comparison:

```

have hCycle :
  HilbertAngleLess Geo A C B A C B :=
  hilbert_angleLess_trans
  Geo
  A C B
  A B C
  A C B
  hAngle
  hOpp

```

and finally:

```

exact
  False.elim
    ((hilbert_angleLess_irrefl
      Geo A C B)
     hCycle)

```

19.9 Relationship with Earlier Propositions

19.9.1 Proposition I.5 and the formal isosceles theorem

Classically, I.5 eliminates the equality case: if

$$AB \cong AC,$$

then the base angles are congruent. In the current Lean source, however, `euclid_proposition_5` is not called directly. The proof uses the reusable theorem

`hilbert_isosceles_base_angles`

to obtain

$$\angle ABC \cong \angle ACB.$$

Thus I.5 belongs to the classical dependency DAG, while `hilbert_isosceles_base_angles` is the direct formal dependency for this branch.

19.9.2 Proposition I.18

I.18 is used to eliminate the reversed side inequality. If

$$AC < AB,$$

then applying I.18 to triangle ACB gives

$$\angle ABC < \angle ACB.$$

Together with the hypothesis

$$\angle ACB < \angle ABC,$$

this produces a strict cycle and therefore a contradiction.

19.9.3 Logical Relation between I.18 and I.19

The two propositions form a converse pair:

$$AB < AC \implies \angle ACB < \angle ABC$$

for I.18, and

$$\angle ACB < \angle ABC \implies AB < AC$$

for I.19.

Together they establish the equivalence between ordering of opposite sides and ordering of opposite angles in a nondegenerate triangle.

19.10 Hilbert Components Used

19.10.1 `hilbert_segment_trichotomy`

This theorem provides the complete three-way comparison of two segments and is the main reusable structural component called by I.19.

19.10.2 hilbert_angleLess_irrefl

This theorem excludes strict self-comparison of an angle. It is used in both contradictory branches of the proof.

19.10.3 hilbert_isosceles_base_angles

This is the formal version of Proposition I.5 needed to eliminate the equality case.

19.10.4 hilbert_angleLess_transport_right

This theorem transports a strict angle comparison across congruence of the containing angle. It converts the equality branch into an impossible self-comparison.

19.10.5 hilbert_angleLess_trans

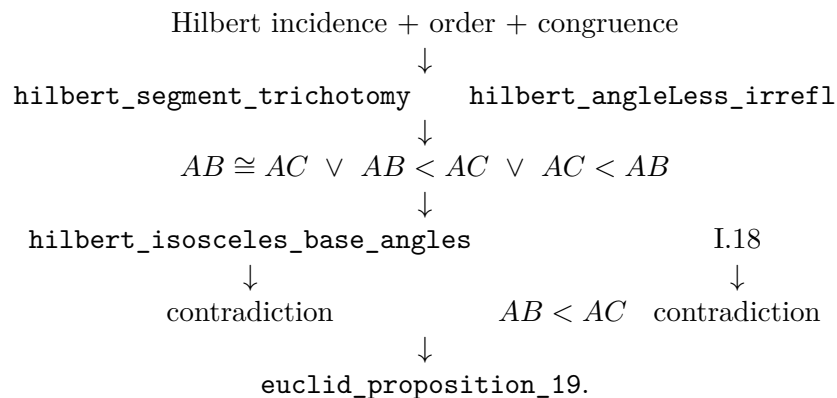
Transitivity combines the original strict inequality with the opposite inequality supplied by I.18.

19.10.6 Direct proposition-level dependency

The only imported Book-I proposition used directly by `euclid_proposition_19` is `euclid_proposition_18`, in the branch $AC < AB$. The equality branch is closed by generic Hilbert infrastructure rather than a direct call to `euclid_proposition_5`.

19.11 Dependency Summary

The formal dependency structure is



No local axiom declaration occurs in the current `Proposition19.lean`; this diagrammatic audit did not run a fresh axiomatic dependency check.

19.12 Conclusion

Proposition I.19 is formally shorter than I.18, but it exposes an important structural issue: the proof requires a complete comparison principle for segments.

Once the theorem `hilbert_segment_trichotomy` is available, the Euclidean argument becomes exactly the expected three-case proof. Equal sides are ruled out by I.5, the reversed inequality is ruled out by I.18, and the remaining case is the desired conclusion.

The proof also clarifies the role of strict angle order. Instead of introducing an explicit asymmetry theorem, the development uses the more fundamental statement `hilbert_angleLess_irrefl` together with transitivity. This keeps the order theory modular and avoids unnecessary duplication.

Thus the current I.19 proof relies on two reusable Hilbert-layer principles: completeness of segment comparison and irreflexivity of strict angle comparison. The Euclidean theorem itself becomes a concise application of those general principles.

The preserved Book1R document recorded a clean Lean development with no `sorry`. In the present diagrammatic audit the production Lean file was read but not rebuilt, and no fresh `#print axioms` check was run.

Chapter 20

Proposition I.20

20.1 Introduction

Euclid's Proposition I.20 is the triangle inequality in purely synthetic form. For a nondegenerate triangle ABC , Euclid proves that any two sides taken together are greater than the third. For one of the three cyclic cases the statement is

$$BA + AC > BC.$$

The Hilbert reconstruction does not introduce a numerical length function and does not define an arithmetic sum of real numbers. Instead, the sum $BA + AC$ is represented geometrically. Extend BA beyond A to a point D such that

$$B - A - D \quad \text{and} \quad AD \cong AC.$$

Then the whole segment BD represents the concatenation of BA with a copy of AC . The required inequality becomes

$$BC < BD.$$

Thus the formal theorem asserts the existence of such a point D together with the strict segment comparison $BC < BD$.

The completed proof remains entirely inside neutral Hilbert geometry. The classical proof uses Proposition I.5, while the current Lean proof calls the generic theorem `hilbert_isosceles_base_angles` directly. Its decisive proposition-level dependency is Proposition I.19. No parallel postulate, numerical angle measure, or numerical segment length is used.

20.2 The Classical Configuration

Let ABC be a nondegenerate triangle. Extend BA beyond A to a point D and choose D so that

$$AD \cong AC.$$

The construction records

$$B - A - D, \quad AD \cong AC.$$

The target is

$$BC < BD.$$

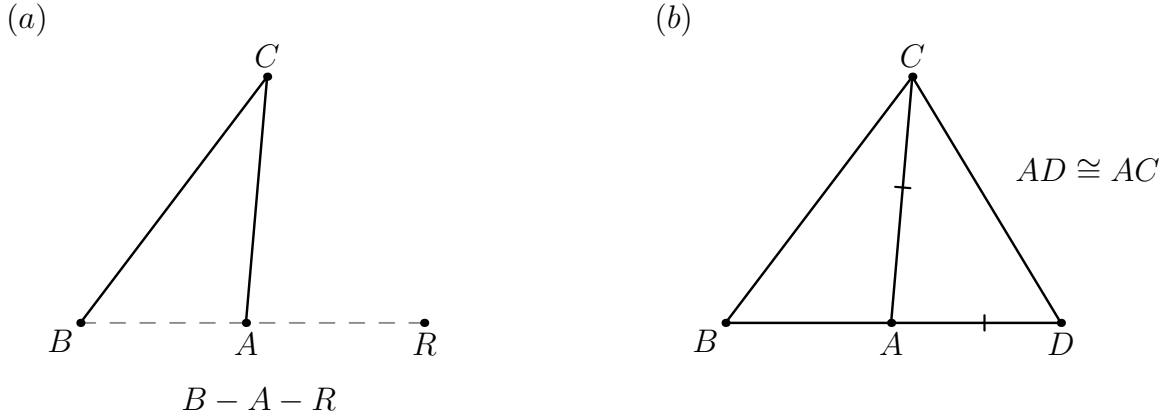


Figure 20.1: Construction and final configuration for Proposition I.20. (a) The current Lean proof first obtains a hidden extension witness R with $B - A - R$. (b) A copy of AC is then laid off on the ray AR , giving the final point D with $B - A - D$ and $AD \cong AC$. The two panels are deliberately separate because the proof does not require an explicit order relation between R and D .

Geometrically, this is exactly the inequality

$$BC < BA + AC.$$

20.3 Euclid's Classical Proof

After extending BA to D so that $AD = AC$, join CD .

The auxiliary triangle ADC is isosceles. Proposition I.5 therefore gives

$$\angle ADC = \angle ACD.$$

Since A lies between B and D , the ray CA lies inside the angle BCD . Hence

$$\angle ACD < \angle BCD.$$

Combining this with the isosceles-triangle equality gives

$$\angle BDC < \angle BCD.$$

Now apply Proposition I.19 to triangle BCD . The greater angle is opposite the greater side, so

$$BC < BD.$$

Finally, because $B - A - D$ and $AD = AC$, the segment BD is the geometric concatenation of BA and AC . Therefore

$$BA + AC > BC.$$

The classical dependency pattern is

$$\begin{array}{c}
 B - A - D, AD \cong AC \\
 \downarrow \\
 \angle ADC \cong \angle ACD \quad (\text{I.5}) \\
 \downarrow \\
 \angle BDC < \angle BCD \\
 \downarrow \quad (\text{I.19}) \\
 BC < BD.
 \end{array}$$

20.4 Synthetic Form of the Triangle Inequality

The formal reconstruction deliberately avoids introducing an arithmetic addition operation on segments. The public result is expressed by an extension point D :

$$\exists D (B - A - D \wedge AD \cong AC \wedge BC < BD).$$

This is the natural Hilbert form of Euclid’s phrase “two sides taken together are greater than the remaining one.”

The representation has two advantages. First, it remains purely geometric. Second, it makes explicit which part of the statement belongs to order and congruence geometry: the “sum” is a concatenated segment, not an externally imposed numerical quantity.

20.5 Proof Architecture of the Reconstruction

The current production module imports only

```
import CGJteamLab.Proposition19
```

and contains one proposition-specific theorem,

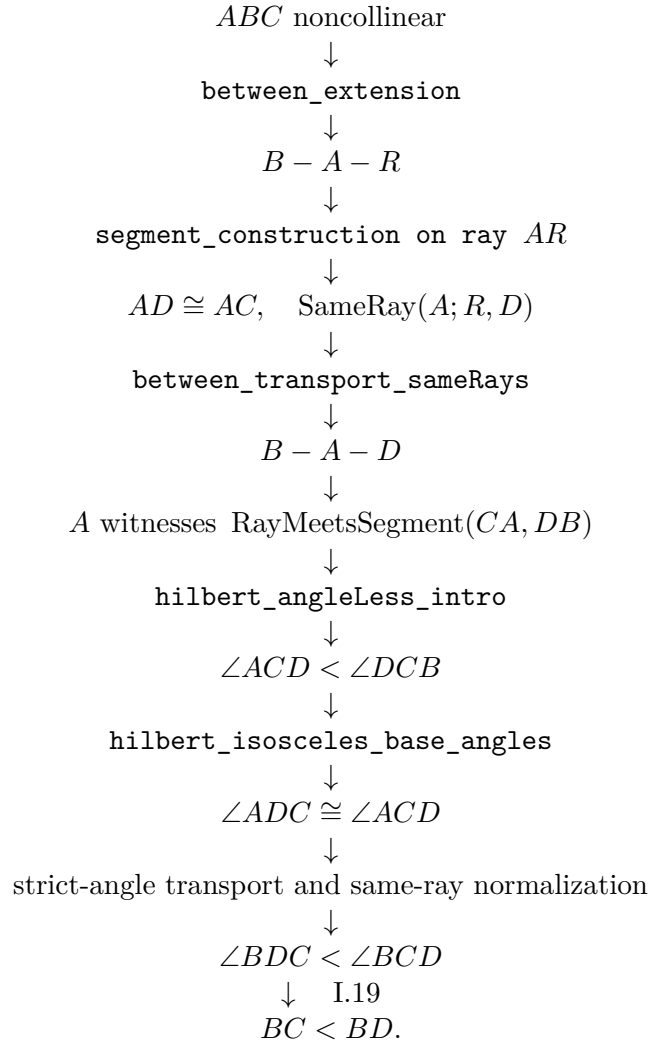
```
euclid_proposition_20
```

```
under
```

```
[HilbertIncidence Geo]
[HilbertCongruence Geo].
```

It is a direct proof rather than a wrapper around a proposition-local helper.

The current formal chain is



A crucial audit correction is that the current proof does *not* invoke Proposition I.16. The strict subangle relation is introduced directly from the interior-ray witness by `hilbert_angleLess_intro`. Likewise, the formal proof does not call a proposition-level I.5 theorem; it uses the reusable isosceles-base-angle theorem directly.

20.6 Diagrammatic Audit

The preserved Book1R chapter had one classical diagram. That final configuration is correct, but it suppresses the genuinely constructed intermediate point R used by the current Lean proof. The audited Figure 20.1 therefore keeps a single float but uses two panels.

Panel (a) records the hidden extension state $B - A - R$. Panel (b) records the final Euclid/Lean state $B - A - D$, $AD \cong AC$, with CD joined. The two points R and D are not placed on one common panel: from the actual theorem state we need only that D lies on the same ray from A as R , followed by transport of betweenness. Drawing a specific order such as $A - R - D$ or $A - D - R$ would add geometric information not required by the proof.

No separate figure is introduced for the witness A in `HilbertRayMeetsSegment`, for angle congruence transport, for same-ray normalization, or for the final I.19 application. These steps

change the logical interpretation of the same final configuration but do not create a new geometric state.

20.7 Construction of the Extension Point

The point D is constructed in two stages.

First, Hilbert’s order axiom extends the line beyond A :

$$B - A - R.$$

In Lean this is obtained from

`HilbertOrder.between_extension.`

Second, a copy of AC is laid off from A on the ray AR using

`HilbertCongruence.segment_construction.`

This produces a point D satisfying

$$AD \cong AC$$

and placing D on the same ray from A as R .

The theorem

`hilbert_between_transport_sameRays`

then transports the original betweenness relation $B - A - R$ to

$$B - A - D.$$

Thus the geometric “sum” segment BD is obtained entirely from existing order and congruence infrastructure.

20.8 The Interior-Angle Step

The key order-theoretic observation is that the ray CA meets the open segment DB at the point A itself. Since

$$D - A - B,$$

the witness for

`HilbertRayMeetsSegment Geo C A D B`

is simply A .

The strict angle comparison is then introduced directly with

`hilbert_angleLess_intro.`

The first angle is $\angle ACD$. The corresponding proper subangle of $\angle DCB$ is represented by $\angle DCA$, which is the same unoriented angle as $\angle ACD$. The representation-level symmetry of angles is supplied by

`angle_congruent_reverse_second`

together with reflexivity of angle congruence.

This yields

$$\angle ACD < \angle DCB.$$

No auxiliary half-plane or perpendicular construction is required. The entire strict comparison follows from the ray meeting the opposite segment.

20.9 Transport Through the Isosceles Triangle

From

$$AD \cong AC$$

and the noncollinearity of A, D, C , Proposition I.5 is available in the form

`hilbert_isosceles_base_angles.`

It gives

$$\angle ADC \cong \angle ACD.$$

For Proposition I.20 we also introduced the reusable theorem

`hilbert_angleLess_transport_left.`

It expresses invariance of strict angle comparison under replacing the smaller angle by a congruent angle.

Applied here, it transports

$$\angle ACD < \angle DCB$$

across

$$\angle ADC \cong \angle ACD,$$

producing

$$\angle ADC < \angle DCB.$$

Since $B - A - D$, the rays DA and DB coincide. The theorem

`hilbert_angle_eq_of_sameRay_first`

therefore identifies the angle $\angle ADC$ with $\angle BDC$. Hence

$$\angle BDC < \angle DCB.$$

Finally, reversal of the two boundary rays at C identifies

$$\angle DCB \cong \angle BCD.$$

The existing theorem

hilbert_angleLess_transport_right

then gives

$$\angle BDC < \angle BCD.$$

20.10 Application of Proposition I.19

At this point the proof has produced exactly the hypothesis needed for Proposition I.19 in triangle BCD :

$$\angle BDC < \angle BCD.$$

Applying

euclid_proposition_19

yields

$$BC < BD.$$

This is the synthetic triangle inequality for the chosen pair of sides. The other two cyclic cases follow by renaming the vertices.

20.11 Lean Formalization

The public theorem is:

```
theorem euclid_proposition_20
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C : Geo.Point)
  (hABC : not PrimCollinear Geo A B C) :
  exists D : Geo.Point,
    Geo.Between B A D /\
    Geo.Congruent A D A C /\
    HilbertSegmentLess Geo B C B D := by
```

The first stage derives $A \neq B$ and extends BA beyond A :

```
have hAB : A != B :=
  hilbert_noncollinear_ne_first
  Geo A B C hABC

have hBA : B != A :=
  hAB.symm

rcases
  HilbertOrder.between_extension
  B A hBA with
  <R, hBAR>
```

A copy of AC is then laid off on the ray AR :

```

have hAR : A != R :=
  (HilbertOrder.between_incidence
    B A R hBAR).2.1

rcases
  HilbertCongruence.segment_construction
    (Geo := Geo)
    A C
    A R
    hAR with
  <D, hRayARD, hAD_AC>

```

Betweenness is transported to the actual constructed point D :

```

have hRayABA :
  HilbertSameRay Geo A B B :=
  hilbert_sameRay_refl
  Geo A B hBA

have hBAD :
  Geo.Between B A D :=
  hilbert_between_transport_sameRays
  Geo
  B A R
  B D
  hBAR
  hRayABA
  hRayARD

```

The theorem then packages the construction and continues with the strict segment comparison:

```

refine <D, hBAD, hAD_AC, ?_>

```

The interior ray is represented directly by the point A :

```

have hDAB :
  Geo.Between D A B :=
  (HilbertOrder.between_incidence
    B A D hBAD).2.2.2.2

have hRayCAA :
  HilbertSameRay Geo C A A :=
  hilbert_sameRay_refl
  Geo C A hAC

have hInside :
  HilbertRayMeetsSegment Geo C A D B :=
  <A, hDAB, hRayCAA>

```

Angle reversal is obtained through the representation-level congruence laws:

```

have hAngleRefl :
  Geo.AngleCongruent A C D A C D :=
  Geometry.Geo.angle_congruent_reflexive
  Geo A C D

```

```

have hAngleSwap :
  Geo.AngleCongruent A C D D C A :=
  (Geo.angle_congruent_reverse_second
   A C D
   A C D).mp
  hAngleRefl

```

This gives the strict subangle relation:

```

have hLessACD_DCB :
  HilbertAngleLess Geo A C D D C B :=
  hilbert_angleLess_intro
  Geo
  A C D
  D C B
  A
  hACD
  hDCB
  hInside
  hAngleSwap

```

Proposition I.5 and left transport give the corresponding comparison with the base angle at D :

```

have hIso :
  Geo.AngleCongruent A D C A C D :=
  hilbert_isosceles_base_angles
  Geo
  A D C
  hADC
  hAD_AC

have hLessADC_DCB :
  HilbertAngleLess Geo A D C D C B :=
  hilbert_angleLess_transport_left
  Geo
  A C D
  A D C
  D C B
  hLessACD_DCB
  hADC
  hIso

```

The ray DA is then replaced by the same ray DB :

```

have hRayDAB :
  HilbertSameRay Geo D A B :=
  hilbert_sameRay_of_between
  Geo D A B hDAB

have hRayDBA :
  HilbertSameRay Geo D B A :=
  hilbert_sameRay_symm
  Geo D A B hRayDAB

have hBDC_ACD :
  Geo.AngleCongruent B D C A C D := by

```

```

unfold Geometry.Geo.AngleCongruent at hIso |-
rw [hilbert_angle_eq_of_sameRay_first
    Geo D B A C hRayDBA]
exact hIso

```

Using left transport again gives

```

have hLessBDC_DCB :
  HilbertAngleLess Geo B D C D C B :=
  hilbert_angleLess_transport_left
  Geo
  A C D
  B D C
  D C B
  hLessACD_DCB
  hBDC
  hBDC_ACD

```

Finally, the right-hand angle is reversed and Proposition I.19 closes the proof:

```

have hAngleRef1DCB :
  Geo.AngleCongruent D C B D C B :=
  Geometry.Geo.angle_congruent_reflexive
  Geo D C B

have hDCB_BCD :
  Geo.AngleCongruent D C B B C D :=
  (Geo.angle_congruent_reverse_second
   D C B
   D C B).mp
  hAngleRef1DCB

have hLessBDC_BCD :
  HilbertAngleLess Geo B D C B C D :=
  hilbert_angleLess_transport_right
  Geo
  B D C
  D C B
  B C D
  hLessBDC_DCB
  hBCD
  hDCB_BCD

exact
  euclid_proposition_19
  Geo
  B C D
  hBCD
  hLessBDC_BCD

```

20.12 Relationship with Earlier Propositions

20.12.1 Classical Proposition I.5 versus the formal isosceles theorem

Euclid’s classical proof uses I.5 after constructing the isosceles triangle ADC . The current Lean theorem does not call `euclid_proposition_5`; it calls `hilbert_isosceles_base_angles` directly. Thus I.5 belongs to the classical proof DAG, while the reusable Hilbert theorem is the direct formal dependency at this step.

20.12.2 Proposition I.16

The preserved Book1R chapter contained a subsection assigning I.16 the strict-angle step. That description does not match the current production proof. No call to `euclid_proposition_16` occurs in `Proposition20.lean`. Instead, the point A itself witnesses that the ray CA meets the open segment DB , and `hilbert_angleLess_intro` yields

$$\angle ACD < \angle DCB.$$

I.16 remains relevant to an ancient alternative proof mentioned in the historical literature, but not to the current formal DAG of I.20.

20.12.3 Proposition I.19

Proposition I.19 is the direct proposition-level dependency that closes the proof. Once strict angle transport has produced

$$\angle BDC < \angle BCD,$$

`euclid_proposition_19` gives

$$BC < BD.$$

20.13 New Reusable Infrastructure

The current I.20 proof uses the reusable Hilbert-interface theorem

$$\text{hilbert_angleLess_transport_left}.$$

Its role is symmetric to

$$\text{hilbert_angleLess_transport_right}.$$

Together these two transport theorems express that strict synthetic angle comparison is invariant under congruent replacement of either compared angle. The preserved Book1R chapter records the left-transport theorem as infrastructure developed for this reconstruction; the present audit does not re-run the historical introduction point.

The proof also reuses the existing construction pattern

$$\text{between_extension} \longrightarrow \text{segment_construction} \longrightarrow \text{between_transport_sameRays},$$

which already occurs elsewhere in the Hilbert development.

20.14 Dependency Summary

The formal dependency structure of Proposition I.20 can be summarized as

$$\begin{array}{c}
 \text{Hilbert incidence} + \text{order} + \text{congruence} \\
 \downarrow \\
 \text{between extension} + \text{segment construction} \\
 \downarrow \\
 B - A - D, \quad AD \cong AC \\
 \downarrow \\
 \text{hilbert_isosceles_base_angles} + \text{HilbertAngleLess} \\
 \downarrow \\
 \angle BDC < \angle BCD \\
 \downarrow \quad \text{I.19} \\
 BC < BD \\
 \downarrow \\
 BA + AC > BC \quad (\text{synthetic interpretation}).
 \end{array}$$

No Euclidean parallel axiom appears in this chain. Proposition I.20 is therefore a theorem of neutral geometry in the present Hilbert reconstruction.

20.15 Conclusion

Proposition I.20 is the first place in this sequence where Euclid’s language of adding lengths is naturally reconstructed without introducing numerical length. The formal statement replaces the expression $BA + AC$ by an explicit extension segment BD satisfying

$$B - A - D, \quad AD \cong AC.$$

The rest of the argument is purely synthetic. Classically I.5 supplies the isosceles base-angle equality; in the current Lean proof this is supplied directly by `hilbert_isosceles_base_angles`. The interior-ray witness produces the strict angle inequality through `hilbert_angleLess_intro`, congruence transport normalizes the two angles, and Proposition I.19 converts the angle inequality back into the desired side inequality.

The resulting Lean proof therefore reproduces the classical geometry while exposing its exact logical structure. The preserved Book1R chapter records no new geometric axiom for this proposition. In this audit the production Lean file was read but no new `lake build` or `#print axioms` check was run.

Chapter 21

Proposition I.21

21.1 Introduction

Euclid's Proposition I.21 concerns two straight lines drawn from the ends of one side of a triangle to a point inside the triangle. Let ABC be nondegenerate. Let D be an interior point and let the line BD meet AC at E , so that

$$A - E - C, \quad B - D - E.$$

Euclid asserts simultaneously

$$BD + DC < BA + AC$$

and

$$\angle BAC < \angle BDC.$$

The present reconstruction deliberately separates these conclusions. The segment comparison is a substantial piece of synthetic segment arithmetic; the angle comparison is an independent exterior-angle argument. They are assembled only in the public theorem.

No numerical length function or numerical angle measure is introduced. Segment sums are represented by ordinary constructed segments, and strict comparisons remain the synthetic relations `HilbertSegmentLess` and `HilbertAngleLess`.

21.2 The Classical Configuration

The classical configuration uses only the original triangle and the intersection point E of the line BD with the side AC .

The single classical diagram supports two logically independent chains. For segments, Proposition I.20 is applied first to triangle BAE and then to triangle CDE . For angles, Proposition I.16 is applied to the same two auxiliary triangles.

21.3 Classical Proof

For the segment part, Proposition I.20 in triangle BAE gives

$$BE < BA + AE.$$

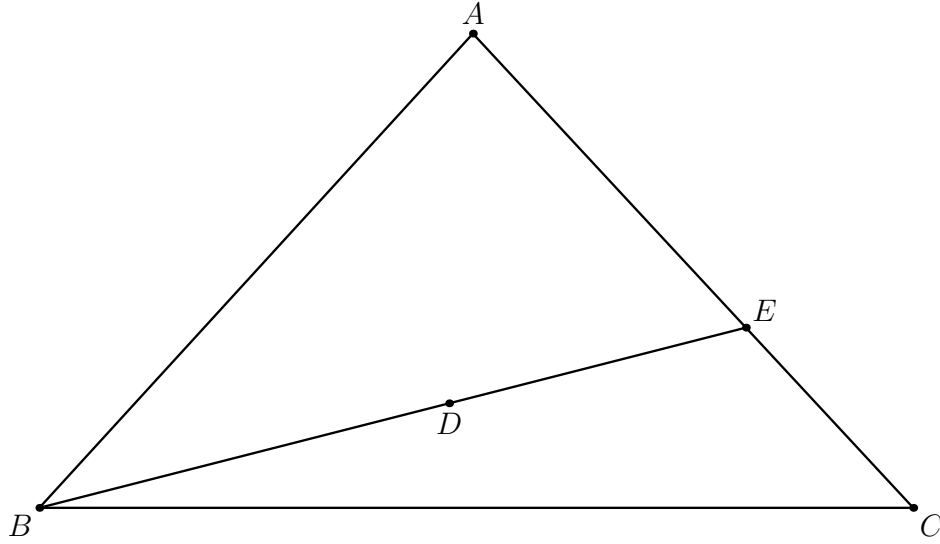


Figure 21.1: Classical configuration for Proposition I.21. The incidences are $A - E - C$ and $B - D - E$. The same geometric state supports both the segment-sum argument and the angle-order argument.

Adding EC geometrically gives

$$BE + EC < BA + AE + EC = BA + AC.$$

Now apply Proposition I.20 to triangle CDE . The required triangle inequality is

$$DC < DE + EC.$$

Since $B - D - E$,

$$BD + DE = BE,$$

and therefore

$$BD + DC < BD + DE + EC = BE + EC < BA + AC.$$

Thus

$$BD + DC < BA + AC.$$

For the angle part, Proposition I.16 in triangle BAE , with AE extended through E to C , gives

$$\angle BAC = \angle BAE < \angle BEC.$$

Proposition I.16 in triangle CED , with ED extended through D to B , gives

$$\angle BEC = \angle CED < \angle CDB = \angle BDC.$$

Hence

$$\angle BAC < \angle BDC.$$

This audit corrects a sentence in the preserved Book1R text which stated the second triangle inequality in the form $CE < CD + DE$. That inequality is true, but the comparison used by the I.21 chain is the different I.20 conclusion

$$CD < CE + ED.$$

The current Lean proof uses exactly this latter form.

21.4 From Euclid's Diagram to the Formal Argument

The classical proof silently performs several kinds of operations. The formal reconstruction separates them:

- a sum such as $BA + AC$ is represented by a concrete collinear segment rather than by a number;
- replacement of a segment by a congruent representative requires explicit transport lemmas;
- addition of unequal segments is represented by betweenness, congruence, and a theorem such as `bookZero_53_lessThanAdditive`;
- the angle proof reconstructs the exact noncollinearity and same-ray data needed by Proposition I.16.

Thus the formal proof remains synthetic, but it exposes the incidence, order, and congruence data suppressed by the classical diagram.

21.5 Synthetic Segment Arithmetic

The public theorem represents the two sums by points X and Y :

$$B - A - X, \quad AX \cong AC,$$

and

$$B - D - Y, \quad DY \cong DC.$$

Hence BX represents $BA + AC$, while BY represents $BD + DC$. The target segment assertion is the ordinary strict comparison

$$BY < BX.$$

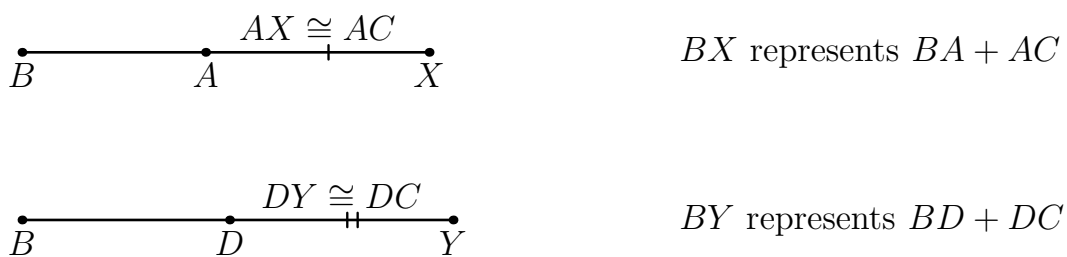


Figure 21.2: The two public synthetic representatives in the Lean statement. The drawing records only betweenness and the copied terminal segments; the strict relation $BY < BX$ is a theorem, not a numerical comparison read from the scale of the picture.

This is an important structural feature of the reconstruction. There is no separate algebraic object for a segment sum. Ordinary geometric points carry the required arithmetic.

21.6 Proof Architecture of the Reconstruction

The current production file imports

```
import CGJteamLab.HilbertBookZero
import CGJteamLab.Proposition20
import CGJteamLab.Proposition16
```

and contains three proposition-specific theorems:

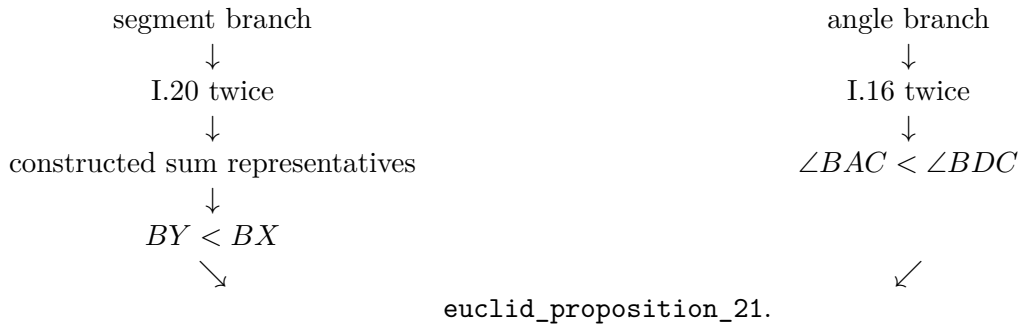
```
euclid_proposition_21_segment_helper
euclid_proposition_21_angle
euclid_proposition_21
```

All three work under

```
[HilbertIncidence Geo]
[HilbertCongruence Geo]
```

with no Euclidean-parallel typeclass.

The logical architecture is



The current segment branch is longer than the abbreviated description in the preserved Book1R chapter. Its actual construction chain is recorded in Figure 21.3.

21.6.1 The first segment-helper application

The main theorem first applies Proposition I.20 to triangle BAE . It obtains H with

$$B - A - H, \quad AH \cong AE, \quad BE < BH.$$

The theorem

```
euclid_proposition_21_segment_helper
```

then constructs, in the actual main theorem, points X and G such that

$$B - A - X, \quad AX \cong AC,$$

$$B - E - G, \quad EG \cong EC,$$

and

$$BG < BX.$$

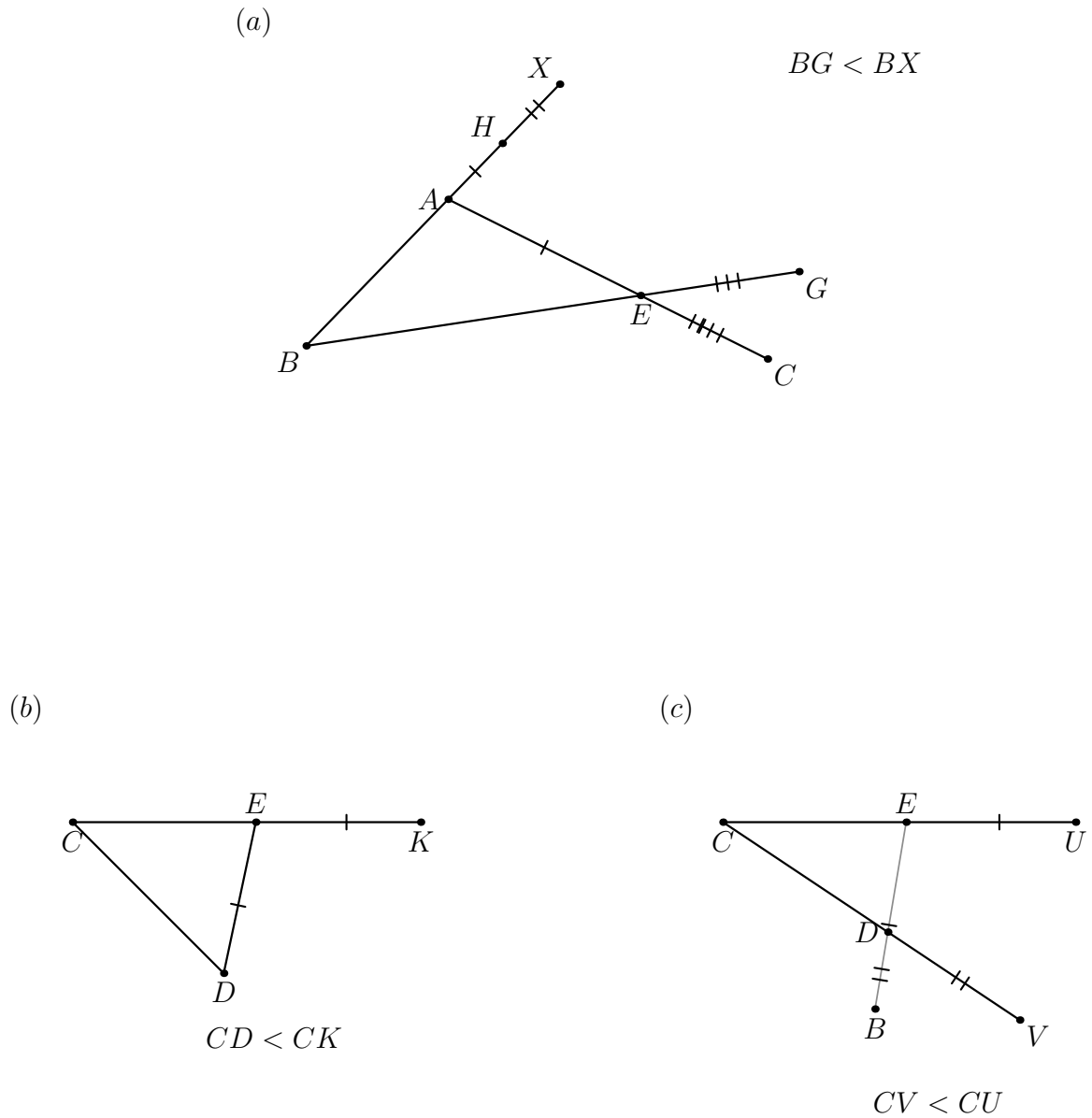


Figure 21.3: The proposition-specific geometric states in the current segment branch. (a) The first I.20 witness H is extended by the first segment-helper application to X and G , yielding $BG < BX$. (b) A second application of I.20 to triangle ECD constructs K with $C - E - K$, $EK \cong ED$, and $CD < CK$. (c) The second segment-helper application constructs U, V with $C - E - U$, $C - D - V$, $EU \cong EB$, $DV \cong DB$, and $CV < CU$. Panels (b) and (c) are deliberately separate: the proof does not need an explicit order relation between K and U .

Inside the generic helper the point later bound as X is named F . From $A - E - C$ one has $AE < AC$. Congruence transport changes this to $AH < AF$, and the same-ray order theorem gives

$$A - H - F.$$

Together with $B - A - H$ this yields the required extended collinear configuration. Difference of parts gives

$$EC \cong HF,$$

and `bookZero_53_lessThanAdditive` extends $BE < BH$ by congruent terminal parts to the final helper conclusion.

A temporary point R is used internally only to create a suitable ray for the construction of G . It is a representation witness and does not define a new proposition-level diagram state.

21.6.2 The second I.20 state and the second helper

The public proof then applies Proposition I.20 a second time, now to triangle ECD . It obtains K with

$$C - E - K, \quad EK \cong ED, \quad CD < CK.$$

The same segment helper is then used with the roles of the points permuted. Its relevant output is a pair U, V satisfying

$$C - E - U, \quad EU \cong EB,$$

$$C - D - V, \quad DV \cong DB,$$

and

$$CV < CU.$$

The two states are geometrically distinct. Both K and U lie on the continuation determined by C, E , but the proof does not require a specific mutual order between them. They are therefore not placed on one artificially ordered diagram.

The current source contains a duplicated source-level invocation of this second helper. The first resulting pair U, V is not used; the same helper is immediately invoked again and the second pair is used in the remainder of the proof. This appears to be an incremental-proof residue rather than a distinct mathematical branch. The documentary DAG records the single logical use and no extra figure is introduced.

21.6.3 Final segment comparison

The proof recognizes the two intermediate sum representatives by congruence. In particular,

$$CU \cong BG.$$

Transport gives

$$CV < BG,$$

and transitivity with $BG < BX$ yields

$$CV < BX.$$

Only at this later stage is the public point Y constructed with

$$B - D - Y, \quad DY \cong DC.$$

Additivity shows that CV and BY represent the same sum, hence

$$CV \cong BY.$$

A final transport through congruence gives

$$BY < BX.$$

This is exactly

$$BD + DC < BA + AC.$$

21.6.4 The angle branch

The theorem

`euclid_proposition_21_angle`

constructs no new geometric point. Its main formal work is to recover the noncollinearity data for triangles BAE and CED and then apply Proposition I.16 twice.

The exact order is

$$\angle BAC < \angle BEC = \angle CED < \angle CDB = \angle BDC.$$

The equalities in this display are implemented through same-ray and unoriented-angle transport rather than by rewriting numerical angle measures.

21.7 Lean Formalization

The public theorem is

```
theorem euclid_proposition_21
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D E : Geo.Point)
  (hABC : Not (PrimCollinear Geo A B C))
  (hAEC : Geo.Between A E C)
  (hBDE : Geo.Between B D E) :
  (exists X Y : Geo.Point,
    Geo.Between B A X /\
    Geo.Congruent A X A C /\
    Geo.Between B D Y /\
    Geo.Congruent D Y D C /\
    HilbertSegmentLess Geo B Y B X) /\
  HilbertAngleLess Geo B A C B D C
```

The existential part is not auxiliary clutter. It is the exact synthetic encoding of the broken-line inequality. The second conjunct is the angle-order assertion in the established strict-angle language.

21.8 Diagrammatic Audit

The preserved Book1R chapter contains two figures, both retained here. Figure 21.1 is the unique classical configuration; it supports both classical proof branches and should not be duplicated merely to highlight different inequalities. Figure 21.2 records the public witnesses X, Y which reify the two sums.

The audit adds exactly one new figure, Figure 21.3, because the current Lean contains three further proposition-specific construction states which are not visible in the preserved chapter: the first I.20/helper state, the second I.20 witness K , and the second helper state U, V .

No separate diagrams are introduced for congruence symmetry, transitivity, difference of parts, inequality transport, or the angle transport steps. Those operations change proof information but do not create new incidence/order geometry. Likewise, the temporary ray witness R and the duplicated unused helper call are not diagram states.

Thus the final diagram set is

3 figures, 5 panels.

21.9 Relationship with Earlier Propositions

21.9.1 Proposition I.16

Proposition I.16 supplies the two strict exterior-angle comparisons in the angle branch. It is the principal earlier proposition behind

$$\angle BAC < \angle BDC.$$

21.9.2 Proposition I.20

Proposition I.20 is used twice in the actual segment branch. The first application to triangle BAE produces H and the inequality $BE < BH$. The second application to triangle ECD produces K and the inequality $CD < CK$. The latter application is essential to the current production DAG and should not be collapsed into a generic phrase such as “construct the remaining representative.”

21.9.3 Cumulative character

I.21 coordinates order, congruence, strict segment comparison, construction on rays, subtraction of parts, addition of inequalities, segment-comparison transitivity, I.16, and I.20. It is therefore a substantial test of Book Zero as a working synthetic proof language.

21.10 Hilbert and Book Zero Components Used

The segment branch makes direct use of, among others,

```
bookZero_30_lessThanCongruence
bookZero_32_lessThanCongruence2
```


bookZero_51_lessThanBetween
 bookZero_3_7a
 bookZero_3_7b
 bookZero_differenceOfParts
 bookZero_53_lessThanAdditive
 bookZero_sumOfParts
 bookZero_52_lessThanTransitive

The angle branch uses the Book Zero collinearity/noncollinearity transport layer together with the Hilbert same-ray and strict-angle transport theorems.

These results play the role that ordinary substitution and elementary segment arithmetic would play in a numerical proof, but they remain entirely synthetic.

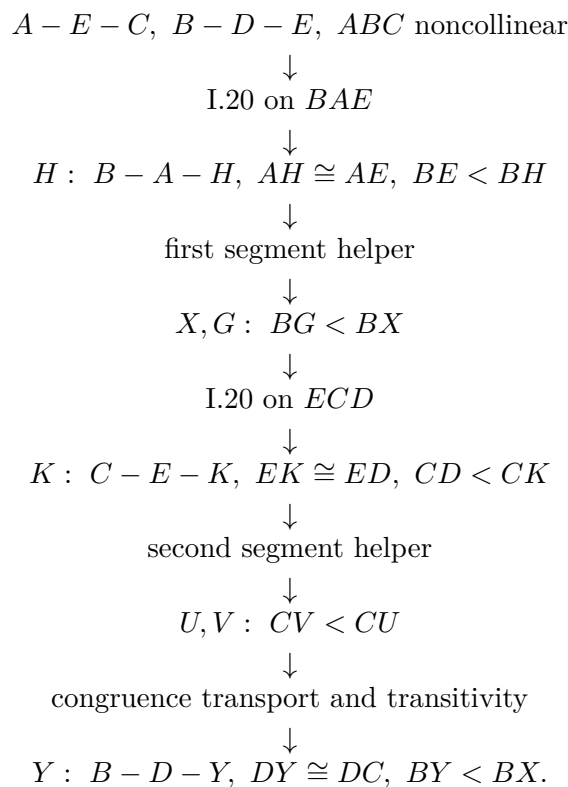
21.11 What I.21 Reveals About Book Zero

The proof shows that an explicit algebra of lengths is not yet needed. Betweenness, congruence, segment construction, and strict comparison already permit finite sums to be represented, compared, extended, and transported.

In particular, the geometric witnesses themselves carry the arithmetic. This is exactly the intended role of Book Zero: not merely to shorten proofs, but to provide the intermediate operations in which ordinary synthetic reasoning can be expressed over the Hilbert primitives.

21.12 Dependency Summary

The current formal segment DAG is more accurately summarized as



The angle DAG is

$$\begin{array}{c}
 A - E - C, B - D - E, ABC \text{ noncollinear} \\
 \downarrow \\
 \text{noncollinearity reconstruction} \\
 \downarrow \\
 \text{I.16 twice} \\
 \downarrow \\
 \text{same-ray and angle transport} \\
 \downarrow \\
 \angle BAC < \angle BDC.
 \end{array}$$

The two outputs are combined only at the public endpoint `euclid_proposition_21`.

21.13 Formalization Notes

The current production source contains no local declaration named `axiom` and no `sorry`. This statement is only a source inspection result: the present documentation audit did not run `#print axioms` and therefore does not claim a fresh axiomatic audit of the endpoint.

The preserved Book1R chapter recorded that the development through I.21 compiled successfully at the time of that document. The present diagrammatic audit did not rerun `lake build`; consequently no new build claim is made here.

The main documentary correction beyond the figures is the complete second half of the segment DAG. The current Lean explicitly performs a second I.20 construction and a second segment-helper construction before building the final public witness Y .

21.14 Conclusion

Euclid I.21 is an important transition point in Book I. One classical diagram supports two different arguments, while the formal reconstruction makes their dependencies explicit.

The angle branch remains close to Euclid’s exterior-angle reasoning. The segment branch exposes a richer synthetic arithmetic of constructed segments: two uses of I.20, two logical uses of the segment helper, several distinct sum representatives, and final transport to the public witnesses X, Y .

The audited diagrams record exactly those changes of geometric state. They do not replace the synthetic proof by coordinates, numerical lengths, or numerical angle measures.

Chapter 22

Proposition I.22

22.1 Introduction

Euclid's Proposition I.22 is the classical existence theorem for a triangle with three prescribed side lengths.

Three nondegenerate segments are given. Euclid assumes that any two of them, taken together, are greater than the remaining one. Under these hypotheses he constructs a triangle whose three sides are congruent to the three given segments.

In the present reconstruction the given segments are denoted

$$Aa, \quad Bb, \quad Cc.$$

The three hypotheses are the strict triangle inequalities

$$Aa < Bb + Cc, \quad Bb < Aa + Cc, \quad Cc < Aa + Bb.$$

The formal conclusion is the existence of a point K such that

$$BK \cong Aa \quad \text{and} \quad bK \cong Cc.$$

Since Bb is already the third prescribed side, the triangle BbK has precisely the required side lengths.

Proposition I.22 is especially important for the formal development because its proof exposes two pieces of infrastructure which classical notation tends to hide:

1. the phrase “two segments taken together are greater than a third” must receive a synthetic, witness-based meaning;
2. the existence of the vertex K is an application of circle-circle continuity.

No numerical length function is introduced. Segment sums, strict inequalities, and the circle intersection are all expressed in the synthetic language of betweenness and congruence.

22.2 The Classical Configuration

Euclid's construction is best understood in the classical ray configuration used by Wilkins. Let the three prescribed lengths be represented by A , B , and C . On one ray choose consecutive

segments

$$DF = A, \quad FG = B, \quad GH = C.$$

Draw the circle centered at F with radius A and the circle centered at G with radius C . Under the three strict triangle inequalities the two circles intersect properly in two points K and L . Either triangle FGK or triangle FGL then has side lengths A, B, C .

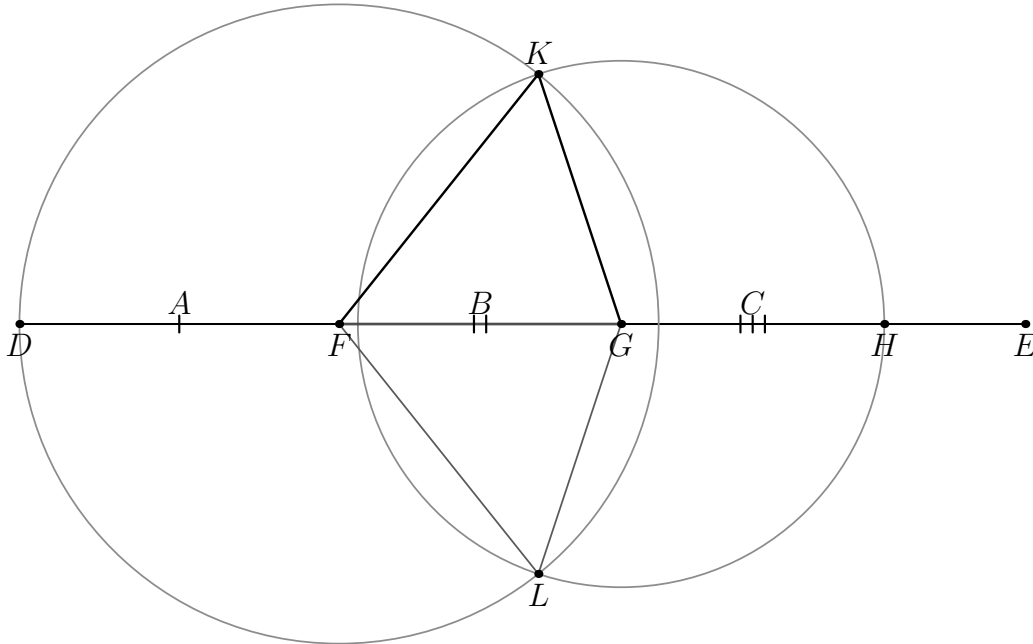


Figure 22.1: Wilkins-style classical configuration for Euclid I.22. The circles centered at F and G have radii represented by DF and GH and intersect in K and L . Either intersection produces a triangle with the three prescribed side lengths.

The strict triangle inequalities are not decorative side conditions. They are exactly what prevents tangency, separation, or containment. The six equality and reversed-inequality boundary states are shown in Figure 22.2.

22.3 Classical Proof

Euclid's proof is a direct ruler-and-compass construction. Let the three prescribed segments be represented by

$$Aa, \quad Bb, \quad Cc,$$

and assume

$$Aa < Bb + Cc, \quad Bb < Aa + Cc, \quad Cc < Aa + Bb.$$

Choose Bb as the base. Draw the circle centered at B with radius congruent to Aa and the circle centered at b with radius congruent to Cc . Let K be an intersection point. By construction,

$$BK \cong Aa, \quad bK \cong Cc,$$

so triangle BbK has the three prescribed side lengths.

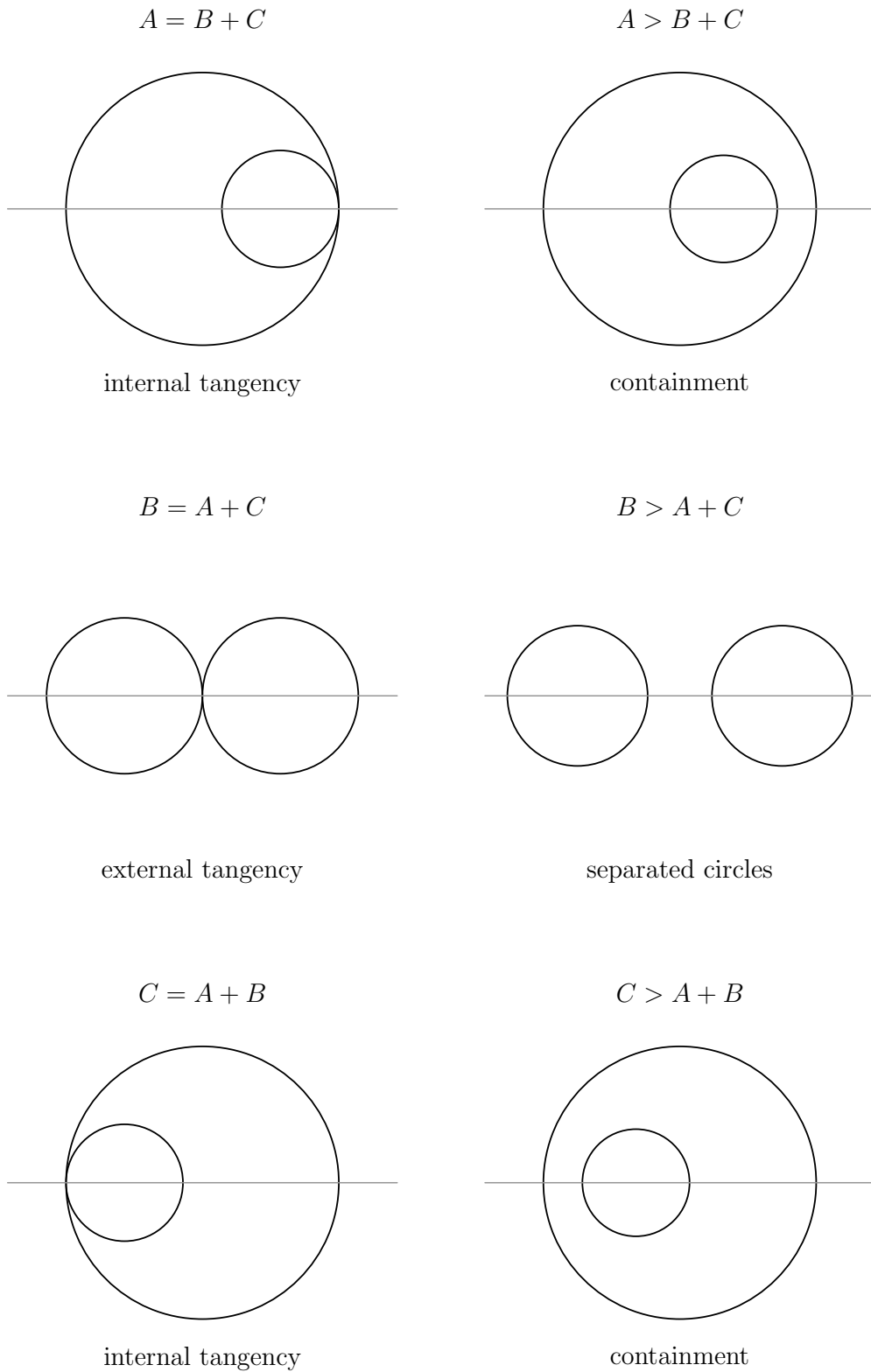


Figure 22.2: Classical boundary and failure configurations for the three triangle inequalities. Equality produces tangency; reversing an inequality destroys proper intersection. These six panels are classical admissibility states, not branches of the Lean proof.

The classical proof may be summarized as

$$\begin{array}{c}
 Aa < Bb + Cc, \quad Bb < Aa + Cc, \quad Cc < Aa + Bb \\
 \downarrow \\
 \text{draw the two circles} \\
 \downarrow \\
 K \text{ is an intersection point} \\
 \downarrow \\
 BK \cong Aa, \quad bK \cong Cc \\
 \downarrow \\
 \triangle BbK.
 \end{array}$$

The point at which the axiomatic reconstruction must say more is the existence of K . In the traditional diagram the circles are simply drawn as intersecting. The formal proof must derive the hypotheses of an intersection theorem from explicit segment-order data.

22.4 Statement

The synthetic statement used in the final Lean theorem is:

```

theorem euclid_proposition_22
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A a B b C c : Geo.Point)
  (hAa : A != a)
  (hBb : B != b)
  (hCc : C != c)
  (hAa_lt_BbCc :
    HilbertSegmentSumGreater Geo B b C c A a)
  (hBb_lt_AaCc :
    HilbertSegmentSumGreater Geo A a C c B b)
  (hCc_lt_AaBb :
    HilbertSegmentSumGreater Geo A a B b C c) :
  exists K : Geo.Point,
    Geo.Congruent B K A a /\
    Geo.Congruent b K C c

```

The three assumptions involving `HilbertSegmentSumGreater` express

$$Bb + Cc > Aa, \quad Aa + Cc > Bb, \quad Aa + Bb > Cc.$$

The conclusion gives the third vertex K of the required triangle:

$$BK \cong Aa, \quad bK \cong Cc,$$

while the base itself is the given segment Bb .

22.5 A Synthetic Predicate for “Two Together Greater”

Proposition I.21 showed that segment sums need not in general be reified as numerical objects. A concrete collinear segment can serve as a representative of a sum.

Proposition I.22 introduces a slightly different requirement. The three triangle inequalities are not merely intermediate steps in the proof: they are part of the public hypotheses of the theorem. Consequently it is useful to package the geometric witness explicitly.

The Book Zero predicate is

```
def HilbertSegmentSumGreater
  [HilbertCongruence Geo]
  (A B C D E F : Geo.Point) : Prop :=
  exists P : Geo.Point,
    Geo.Between A B P /\
    Geo.Congruent B P C D /\
    HilbertSegmentLess Geo E F A P
```

Geometrically, a witness P satisfies

$$A - B - P, \quad BP \cong CD.$$

Therefore AP represents the sum

$$AB + CD.$$

The final condition

$$EF < AP$$

says precisely

$$AB + CD > EF.$$

For example,

`HilbertSegmentSumGreater Geo B b C c A a`

means that there exists P with

$$B - b - P, \quad bP \cong Cc, \quad Aa < BP.$$

Thus BP is a synthetic representative of $Bb + Cc$.

The public theorem contains three such hidden witnesses. Their incidence/order content is recorded in Figure 22.3; the points are ordinary geometric points, not numerical sum objects.

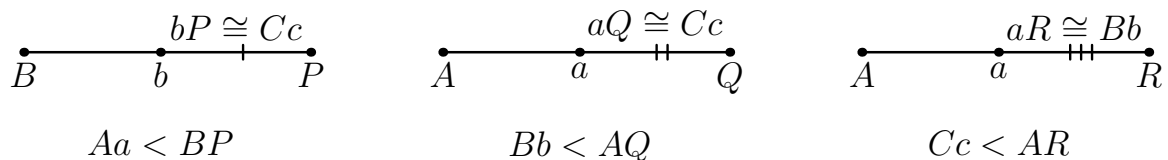


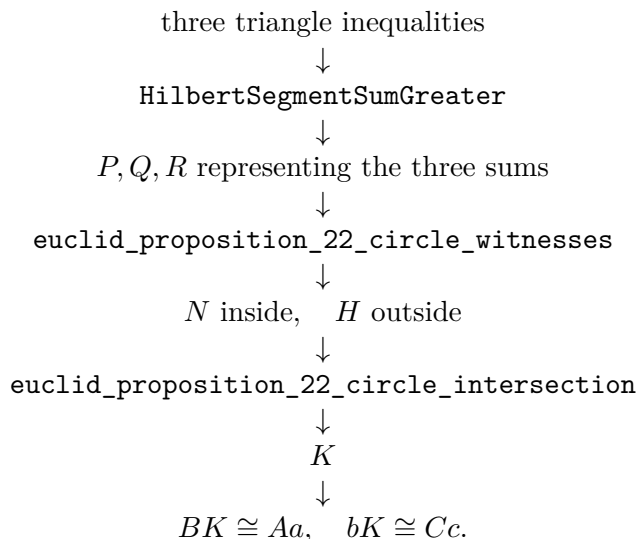
Figure 22.3: The three witnesses unpacked from the public hypotheses `HilbertSegmentSumGreater`: $B - b - P$ with $bP \cong Cc$ and $Aa < BP$; $A - a - Q$ with $aQ \cong Cc$ and $Bb < AQ$; and $A - a - R$ with $aR \cong Bb$ and $Cc < AR$. They are three independent realizations of the three synthetic sums.

This definition does not introduce an algebra of lengths. It merely packages an ordinary geometric construction together with the strict comparison already provided by `HilbertSegmentLess`.

22.6 Proof Architecture of the Reconstruction

The reconstruction is intentionally layered. The final theorem is short because the difficult circle geometry is isolated in reusable helpers.

The overall dependency is



The principal intermediate results are discussed below.

22.6.1 Transporting the outside inequality

The helper

`euclid_proposition_22_outside_helper`

handles the synthetic arithmetic behind the outside point.

Suppose BP represents

$$Bb + Cc,$$

and another collinear segment FH is assembled from pieces

$$FG \cong Bb, \quad GH \cong Cc.$$

Using `bookZero_sumOfParts`, the proof obtains

$$BP \cong FH.$$

Hence the triangle inequality

$$Aa < BP$$

can be transported to

$$Aa < FH$$

by `bookZero_30_lessThanCongruence`.

This is the formal replacement for the classical step “equal sums may be substituted in an inequality.”

22.6.2 Same-ray trichotomy

A point constructed on a ray can lie before, at, or beyond the reference point. The helper

`euclid_proposition_22_sameRay_trichotomy`

makes this order distinction explicit.

For points G, F, N on the same ray from G , it proves

$$G - N - F \quad \vee \quad N = F \quad \vee \quad G - F - N.$$

In the main I.22 construction one substitutes $G := b$ and $F := B$. The exact three geometric branches are therefore shown in Figure 22.4.

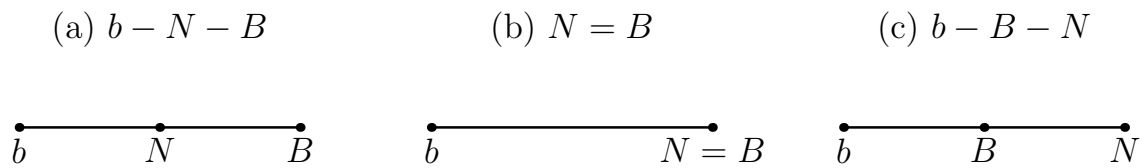


Figure 22.4: The same-ray trichotomy used for the inside witness N : $b - N - B$, $N = B$, or $b - B - N$. The first and third branches are handled by `euclid_proposition_22_inside_left` and `euclid_proposition_22_inside_right`; the middle branch is the center case.

The proof uses strict segment trichotomy together with `bookZero_51_lessThanBetween` and layoff uniqueness.

This case distinction is essential when the point N on the second circle is compared with the first circle.

22.6.3 Cancellation for strict segment inequalities

The helper

`euclid_proposition_22_lessThan_cancel_right`

is a synthetic cancellation principle.

If

$$A - B - E, \quad C - D - F,$$

the terminal parts satisfy

$$BE \cong DF,$$

and

$$AE < CF,$$

then

$$AB < CD.$$

Classically this is an immediate cancellation of equal addends. Formally it requires a proof from the developed strict-order and congruence theory.

This helper is one of the clearest examples in I.22 of Book Zero providing arithmetic-like behavior without numerical lengths.

22.6.4 The inside point

The proof constructs a point N on the ray from b through B such that

$$bN \cong Cc.$$

The two possible order configurations are handled by

```
euclid_proposition_22_inside_left
euclid_proposition_22_inside_right
euclid_proposition_22_inside_helper
```

and the same-ray trichotomy theorem.

The goal is not merely to locate N on the second circle. One must also prove that it is inside, or on the center of, the first circle:

$$N = B \quad \vee \quad BN < Aa.$$

The relevant triangle inequalities are used to derive this comparison in each possible collinear configuration.

22.6.5 Constructing explicit sum representatives

The helper

```
euclid_proposition_22_extend_with_segment
```

constructs the ordinary geometric representative of a sum.

Starting from a nondegenerate segment AB and another segment CD , it extends the ray through B to a point P satisfying

$$A - B - P, \quad BP \cong CD.$$

Hence

$$AP$$

represents

$$AB + CD.$$

This is the generic geometric operation of adjoining a congruent terminal segment. In the current public theorem, however, the witnesses P, Q, R are *not* constructed by this helper: they are unpacked directly from the three hypotheses of type `extttHilbertSegmentSumGreater`. On the actual main DAG, `exttteuclid_proposition_22_extend_with_segment` is called inside `exttteuclid_proposition_22_circle_witnesses` to construct the outside witness H with

$$B - b - H, \quad bH \cong Cc.$$

22.6.6 Circle witnesses

The theorem

`euclid_proposition_22_circle_witnesses`

is the main preparatory result.

From the three represented triangle inequalities it constructs points H and N on the circle centered at b with radius congruent to Cc , with the following properties:

$$bH \cong Cc, \quad Aa < BH,$$

and

$$bN \cong Cc, \quad N = B \vee BN < Aa.$$

Thus H is outside the circle centered at B with radius Aa , while N is inside it or coincides with its center.

This is precisely the configuration required by circle-circle continuity.

22.7 Circle-Circle Continuity

The central existence step is isolated in

`euclid_proposition_22_circle_intersection`

The geometric input and output of the continuity step are shown in Figure 22.5. Panel (a) depicts the strict-inside branch $BN < Aa$; the alternative center case $N = B$ has already been recorded in Figure 22.4. Panel (b) records the genuinely new intersection witness K .

The Hilbert circle-continuity theorem does not use the radius Aa directly as a numerical quantity. The proof first lays off a congruent segment BT :

$$BT \cong Aa.$$

The first circle is therefore represented by center B and radius segment BT .

The second circle has center b and radius bN , with

$$bN \cong Cc.$$

The already constructed points satisfy

$$bH \cong bN,$$

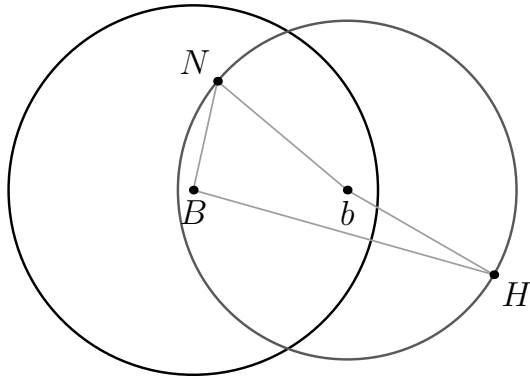
so H lies on the second circle, and

$$Aa < BH.$$

After transport through $BT \cong Aa$, this becomes

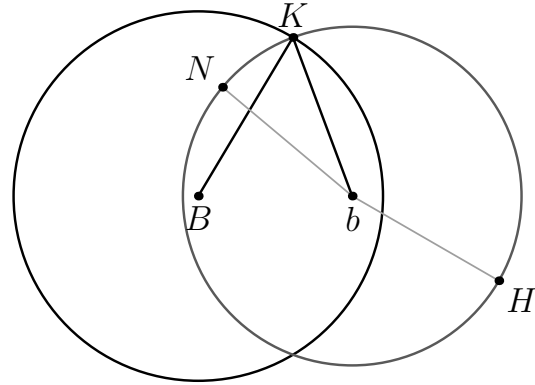
$$BT < BH,$$

(a) inside/outside witnesses



$$BN < Aa \quad Aa < BH$$

(b) circle-circle continuity



$$BK \cong Aa \quad bK \cong Cc$$

Figure 22.5: Formal circle-continuity chain. Both N and H lie on the circle centered at b with radius congruent to Cc . In the strict-inside branch, $BN < Aa$ while $Aa < BH$; circle-circle continuity then supplies a common point K , after which congruence transport yields $BK \cong Aa$ and $bK \cong Cc$.

showing that H is outside the first circle.

Similarly,

$$N = B \quad \vee \quad BN < Aa$$

is transported to

$$N = B \quad \vee \quad BN < BT,$$

showing that N is inside the first circle.

The theorem

`hilbert_circle_circle_intersection`

then yields a point K satisfying

$$BK \cong BT \quad \text{and} \quad bK \cong bN.$$

Finally congruence transitivity returns to the original given segments:

$$BK \cong Aa, \quad bK \cong Cc.$$

This is the required vertex.

22.8 The Core and the Public Theorem

Once the circle witnesses and the circle-intersection theorem are available, the remaining layers become very short.

The theorem

`euclid_proposition_22_core`

accepts explicit representatives P, Q, R of the three sums:

$$BP = Bb + Cc, \quad AQ = Aa + Cc, \quad AR = Aa + Bb,$$

together with

$$Aa < BP, \quad Bb < AQ, \quad Cc < AR.$$

It invokes the two successive helpers

`euclid_proposition_22_circle_witnesses`
`euclid_proposition_22_circle_intersection`

The intermediate wrapper

`euclid_proposition_22_from_sum_witnesses`

packages the same construction when all three sum witnesses are supplied explicitly.

The public theorem finally removes P, Q, R from its interface. It opens the three witnesses stored in `HilbertSegmentSumGreater` and passes them to the internal construction.

Thus the public API states exactly the geometric theorem rather than the implementation details of its proof.

22.9 Lean Formalization

The current production theorem exposes only the three prescribed segments, their nondegeneracy, the three witness-bearing triangle inequalities, and the required vertex:

```

theorem euclid_proposition_22
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A a B b C c : Geo.Point)
  (hAa : A != a)
  (hBb : B != b)
  (hCc : C != c)
  (hAa_lt_BbCc :
    HilbertSegmentSumGreater Geo B b C c A a)
  (hBb_lt_AaCc :
    HilbertSegmentSumGreater Geo A a C c B b)
  (hCc_lt_AaBb :
    HilbertSegmentSumGreater Geo A a B b C c) :
  Exists fun K : Geo.Point =>
    Geo.Congruent B K A a /\
    Geo.Congruent b K C c := by
  rcases hAa_lt_BbCc with
    <P, hBbP, hbP_Cc, hAa_BP>
  rcases hBb_lt_AaCc with
    <Q, hAaQ, haQ_Cc, hBb_AQ>
  rcases hCc_lt_AaBb with

```

```

    <R, hAaR, haR_Bb, hCc_AR>
  exact
  euclid_proposition_22_from_sum_witnesses
    Geo A a B b C c P Q R
    hAa hBb hCc
    hAa_BP hBb_AQ hCc_AR
    hBbP hbP_Cc
    hAaQ haQ_Cc
    hAaR haR_Bb

```

Thus the public endpoint is a wrapper over a substantial proposition-specific construction rather than over a single pre-existing interface theorem. The main internal chain is

```

    P, Q, R unpacked from the three sum hypotheses
      ↓
    euclid_proposition_22_circle_witnesses
      ↓
    H, N with outside/inside data
      ↓
    euclid_proposition_22_circle_intersection
      ↓
    K
      ↓
    euclid_proposition_22.

```

The module also still contains two public declarations with historical test names,

```

euclid_proposition_22_outside_transport_test
euclid_proposition_22_inside_point_test

```

which isolate earlier infrastructure checks. They are not on the final dependency path from the public theorem to K .

No local `axiom` or `sorry` declaration occurs in the current `Proposition22.lean`. This audit did not run `lake build` or `#print axioms`; therefore it makes no new global axiom-audit claim beyond the source-level observation and the already documented use of circle-circle continuity.

22.10 Relationship with Earlier Propositions

22.10.1 Proposition I.20

I.20 proves the triangle inequality from an already existing triangle. I.22 uses the converse pattern as an existence hypothesis: three nondegenerate segments satisfying all three strict triangle inequalities are required to determine a constructible triangle. The logical directions are therefore different.

22.10.2 Proposition I.21

I.21 showed that segment sums can be handled synthetically by ordinary constructed collinear segments. I.22 promotes that technique to the public interface through the witness-bearing predi-

cate `HilbertSegmentSumGreater`. The resulting witnesses P, Q, R are geometric representatives of the three sums used as hypotheses.

22.10.3 Proposition I.23

In Euclid's historical dependency graph, I.23 uses I.22 essentially to build an auxiliary triangle before applying SSS. In the Hilbert reconstruction, I.23 does not theorem-depend on I.22 because Hilbert III.4 already supplies angle construction. I.22 therefore marks an early and useful separation between the historical Book I DAG and the DAG induced by the Hilbert foundation.

22.11 Hilbert and Book Zero Components Used

22.11.1 HilbertSegmentLess

The relation

`HilbertSegmentLess`

is the synthetic strict order on segments.

It supplies the comparisons needed to distinguish points lying inside and outside a circle and supports the transport and transitivity operations used throughout the proof.

22.11.2 bookZero_49_layoff

The theorem

`bookZero_49_layoff`

copies a given nondegenerate segment onto a chosen ray.

In I.22 it is used to represent the radius Aa by a segment based at the center B :

$$BT \cong Aa.$$

22.11.3 bookZero_sumOfParts

The theorem

`bookZero_sumOfParts`

recognizes congruent wholes assembled from congruent adjacent parts.

It is used when a segment representing $Bb + Cc$ must be identified with another segment built from a copy of Bb followed by a copy of Cc .

22.11.4 Congruence transport for inequalities

The theorems

`bookZero_30_lessThanCongruence`
`bookZero_32_lessThanCongruence2`

transport strict segment comparisons through congruent representatives.

They replace the informal algebraic operation of substituting equal lengths into inequalities.

22.11.5 `bookZero_31_trichotomy1`

This theorem supplies the equality case when neither of two nondegenerate segments is strictly shorter than the other.

Together with layoff uniqueness it provides the middle case

$$N = F$$

in the same-ray trichotomy used by I.22.

22.11.6 `bookZero_51_lessThanBetween`

The theorem

`bookZero_51_lessThanBetween`

converts strict segment comparison on a common ray into strict betweenness.

It is the bridge between the order relation on segment lengths and the incidence-order geometry needed to locate the constructed circle points.

22.11.7 Circle-circle intersection

The theorem

`hilbert_circle_circle_intersection`

is the decisive existence principle.

Given a point of one circle lying inside a second circle and another point of the same first circle lying outside the second, it produces a common point of the two circles.

Proposition I.22 is therefore not merely an application of elementary segment arithmetic. It is the point where that arithmetic is used to establish the hypotheses of a genuine continuity theorem.

22.12 Comparison with Euclid's Original Organization

Euclid presents I.22 as an explicit ruler-and-compass construction. He places copies of the three given segments on a straight line and draws two circles whose intersection gives the required triangle.

The Hilbert reconstruction preserves the same geometric idea but changes its logical organization.

First, the base Bb is already available as one of the given segments, so there is no need to reproduce Euclid's preliminary line solely to manufacture a base of the correct length.

Second, the three assumptions "two together greater than the remaining one" are exposed through `HilbertSegmentSumGreater`. Their witnesses give concrete segments representing the required sums.

Third, the existence of the intersection is not treated as visually obvious. Explicit points N and H are constructed on one circle, and they are proved to lie respectively inside and outside the other. Only then is circle-circle continuity invoked.

Thus the formal proof separates three issues which Euclid's diagram compresses:

$$\text{segment arithmetic} \quad + \quad \text{order on rays} \quad + \quad \text{continuity}.$$

The resulting proof is longer, but the additional length identifies the exact logical content required for the construction.

22.13 What Proposition I.22 Adds to Book Zero

I.21 suggested that explicit segment-sum objects were unnecessary: inside a proof, sums could be represented directly by constructed segments.

I.22 refines that conclusion rather than contradicting it.

The new predicate

`HilbertSegmentSumGreater`

does not turn segment sums into algebraic objects. It is a witness-bearing proposition:

$$AB + CD > EF$$

means that a point exists which geometrically realizes $AB + CD$, and that the resulting whole segment is strictly greater than EF .

This distinction is important.

The project still has no numerical length map and no binary addition operation on lengths. What has been added is a convenient *interface proposition* for a classical phrase that occurs as a hypothesis of Euclid's theorem.

Proposition I.22 also shows that the Book Zero layer is now strong enough to prepare the exact inside/outside data required by a continuity theorem. The layer is therefore doing more than abbreviating elementary congruence arguments: it is connecting synthetic segment arithmetic with topological existence principles.

22.14 Dependency Summary

The public assumptions are

$$\begin{aligned} Aa \neq 0, \quad Bb \neq 0, \quad Cc \neq 0, \\ Aa < Bb + Cc, \\ Bb < Aa + Cc, \\ Cc < Aa + Bb. \end{aligned}$$

In Lean the three inequalities are encoded by

```
HilbertSegmentSumGreater Geo B b C c A a,
HilbertSegmentSumGreater Geo A a C c B b,
HilbertSegmentSumGreater Geo A a B b C c.
```

The complete proof architecture is

```
triangle inequalities
↓
P, Q, R sum witnesses
↓
segment transport + same-ray order
↓
N = B ∨ BN < Aa    Aa < BH
↓
bN ≅ Cc,    bH ≅ Cc
↓
hilbert_circle_circle_intersection
↓
BK ≅ Aa,    bK ≅ Cc
↓
euclid_proposition_22.
```

No new geometric axiom is introduced in Proposition I.22. The proof uses the already available circle-circle continuity theorem together with the developed Hilbert and Book Zero segment theory.

22.15 Formalization Notes

Several features of I.22 are easy to miss in an informal proof.

First, a circle radius is represented by an ordinary segment, not by a real number. Even when the intended radius is Aa , the circle-continuity theorem is applied using a constructed representative BT with

$$BT \cong Aa.$$

Second, “inside a circle” is expressed synthetically by a strict segment comparison such as

$$BN < BT,$$

with the center case $N = B$ handled separately.

Third, “outside a circle” is likewise a strict segment comparison:

$$BT < BH.$$

Fourth, the fact that N lies on the ray from b through B is not enough to determine its order relative to B . The proof requires an explicit three-way analysis of the possible positions.

Fifth, cancellation of equal terminal pieces from strict inequalities is not automatic. It is isolated as a theorem and proved from the existing segment-order machinery.

Finally, the public theorem is deliberately cleaner than the internal construction. The implementation works with many auxiliary points, but the theorem exposes only the three original segments, their three triangle inequalities, and the required vertex.

22.16 Diagrammatic Audit

The preserved `Book1R.pdf` contained two figures for I.22, both classical. Figure 22.1 recorded the Wilkins-style proper-intersection construction, while Figure 22.2 recorded the six tangency/separation/containment boundary states. Both are retained because they represent genuinely different geometric configurations.

The audit adds three formal figures required by the current Lean DAG:

- Figure 22.3 records the hidden witnesses P, Q, R carried by the three public sum-greater hypotheses;
- Figure 22.4 records the exact three-way order split for N on the ray from b through B ;
- Figure 22.5 records the inside/outside circle-continuity state and the subsequent appearance of the intersection witness K .

No separate figure is added for the radius representative T , for congruence/inequality transport, for `bookZero_sumOfParts`, for the cancellation helper, or for the wrapper layers. Those steps change the formal representation or logical information without creating a new proposition-specific incidence/order configuration. In particular, T is an important formal witness for the radius segment based at B , but drawing it as an additional geometric state would duplicate the same first circle rather than record new geometry.

The final diagram set therefore consists of five figures: two classical and three formal. The six panels of the classical boundary map are not Lean branches, while the three panels of the same-ray figure are an actual exhaustive formal case split.

22.17 Conclusion

Proposition I.22 is a major structural milestone in the reconstruction of Book I.

Its classical content is elementary: three admissible lengths determine a constructible triangle. Formalizing that statement reveals, however, a precise interaction between three layers of synthetic geometry.

The first layer is segment arithmetic. Betweenness, congruence, strict segment comparison, addition of adjacent parts, cancellation, and transport are used to express the three triangle inequalities and to manipulate their witnesses.

The second layer is ray and order geometry. Constructed points must be located explicitly on rays, and their possible orders must be distinguished.

The third layer is continuity. Once one point of the second circle is proved inside the first and another outside it, `hilbert_circle_circle_intersection` supplies the required common point.

The final theorem therefore has a particularly clear conceptual form: the triangle inequalities are exactly the hypotheses needed to build the inside/outside witnesses for two circles.

No coordinates, numerical lengths, or angle measures enter the proof. The reconstruction remains entirely synthetic, and the complete development through Euclid I.22 compiles successfully.

Chapter 23

Proposition I.23

23.1 Introduction

Euclid's Proposition I.23 asks for the construction, at a prescribed point on a prescribed straight line, of an angle congruent to a given rectilinear angle.

In Euclid's development this is a genuine construction theorem. The angle is not transported directly. Euclid chooses a triangle containing the given angle, reconstructs a second triangle from its three side lengths using Proposition I.22, and then invokes Proposition I.8 (SSS) to identify the required angle.

In the Hilbert reconstruction used by CGJteamLab the logical status is different. Hilbert's congruence axiom III.4 already provides angle construction on a prescribed side of a prescribed ray, together with uniqueness of the resulting ray. Consequently the formal Proposition I.23 is a thin wrapper over a stronger foundational principle.

This contrast is mathematically important. At I.23 the textual order of Euclid's propositions and the dependency order induced by the Hilbert foundation visibly diverge.

23.2 The Classical Configuration

Euclid's construction is best displayed after the auxiliary triangles have actually been constructed. Let the given angle be represented by triangle CDE . On the prescribed straight line through A , choose G , and construct a second triangle AFG so that

$$AG \cong CE, \quad AF \cong CD, \quad FG \cong DE.$$

The point G lies on the prescribed line through A .

The figure records the actual constructional state, not merely two already-congruent angles.

23.3 Classical Proof

Euclid derives angle transport from earlier propositions rather than taking it as primitive. Choose points on the two arms of the given angle, thereby obtaining a triangle and three side lengths. The triangle inequalities needed for the construction are supplied in the classical dependency chain by Proposition I.20. Proposition I.22 then constructs a triangle with the same three side

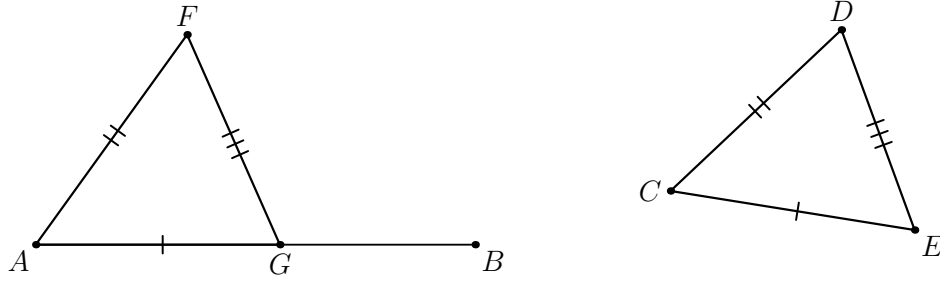


Figure 23.1: Classical Euclidean auxiliary construction for I.23. Proposition I.22 supplies triangle AFG with the three side congruences corresponding to triangle CDE . Proposition I.8 then identifies the transported angle.

lengths on the prescribed line. Proposition I.8 gives SSS congruence, hence equality of the corresponding included angles.

In the notation of Figure 23.1,

$$AG \cong CE, \quad AF \cong CD, \quad FG \cong DE$$

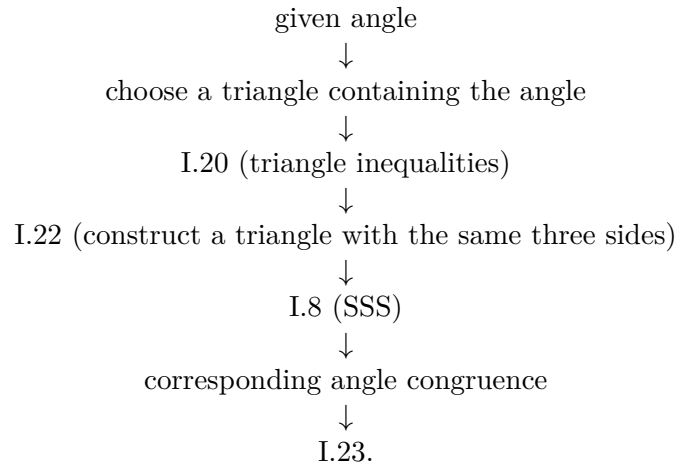
implies

$$\triangle AFG \cong \triangle CDE,$$

and therefore

$$\angle FAG \cong \angle DCE.$$

The classical dependency pattern is therefore



Thus Euclid reduces angle transport to segment construction, circle intersection, and triangle congruence.

23.4 Hilbert III.4

Hilbert reverses the foundational emphasis. Instead of deriving angle transport from triangle construction, the congruence axioms contain a primitive angle-construction principle. In the present formal layer the relevant operation is

`HilbertCongruence.angle_construction`

Conceptually, its input consists of

- a nondegenerate source angle;
- a prescribed initial ray;
- the supporting line of that ray;
- a choice of one side of the supporting line.

It returns a ray on the selected side forming an angle congruent to the source angle. The Hilbert operation is stronger than Euclid I.23 because it also contains a uniqueness assertion for that ray.

23.5 Proof Architecture of the Reconstruction

23.5.1 The Source Angle

The hypothesis

$$\neg \text{PrimCollinear}(A, B, C)$$

ensures that $\angle ABC$ is nondegenerate.

23.5.2 The Prescribed Ray

The points D and E represent the initial ray of the new angle, with vertex E and initial ray ED . The formal hypotheses say that

$$D \neq E$$

and that both points lie on a common supporting line **base**. Thus the formal input specifies an oriented ray rather than only an unoriented line.

23.5.3 Selecting the Side

A point S not lying on **base** selects one of the two sides of that line. The constructed point F must satisfy

$$\text{HilbertSameSide}(F, S; \mathbf{base}).$$

This datum has no explicit counterpart in Euclid's informal formulation: the intended side is normally supplied by the diagram. In an axiomatic reconstruction it must be stated.

23.5.4 Applying Angle Construction

Hilbert III.4 supplies the new point F directly. The formal geometric state is shown in Figure 23.2.

The uniqueness component supplied by III.4 is not needed for the Euclidean proposition and is discarded by the public wrapper.

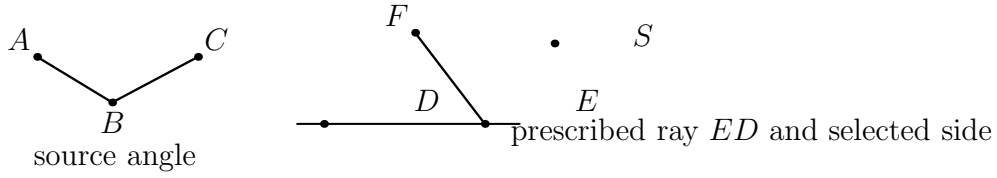


Figure 23.2: Hilbert III.4 configuration. The source angle is $\angle ABC$; ED is the prescribed initial ray on the supporting line; S selects a side of that line; and the constructed point F lies on the selected side with $\angle ABC \cong \angle DEF$.

23.6 Lean Formalization

The current production theorem is

```

theorem euclid_proposition_23
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D E S : Geo.Point)
  (hABC : Not (PrimCollinear Geo A B C))
  (hDE : D != E)
  (base : Geo.Line)
  (hDbase : HilbertIncidence.OnLine D base)
  (hEbase : HilbertIncidence.OnLine E base)
  (hSbase : Not (HilbertIncidence.OnLine S base)) :
  Exists fun F : Geo.Point =>
    HilbertSameSide Geo F S base /\
    Geo.AngleCongruent A B C D E F := by
rcases
  HilbertCongruence.angle_construction
    (Geo := Geo)
    A B C D E S
    hABC hDE base hDbase hEbase hSbase with
  <F, hFSame, hAngle, _hUnique>
exact <F, hFSame, hAngle>

```

The proof is therefore a direct interface extraction:

$$\begin{array}{c}
 \text{source angle} + \text{prescribed ray} + \text{side selector} \\
 \downarrow \\
 \text{HilbertCongruence.angle_construction} \\
 \downarrow \\
 F, \text{ SameSide}(F, S; \text{base}), \quad \angle ABC \cong \angle DEF \\
 \downarrow \\
 \text{euclid_proposition_23.}
 \end{array}$$

There are no proposition-local helper theorems, no geometric case split, and no auxiliary construction beyond the returned point F .

23.7 The Euclidean Route and the Hilbert Route

23.7.1 The Euclidean Route

Euclid reduces angle transport to triangle construction:

$$\text{I.22} + \text{I.8} \longrightarrow \text{I.23},$$

with Proposition I.20 entering the expanded classical route to establish the triangle inequalities required by I.22.

23.7.2 The Hilbert Route

In the Hilbert foundation the dependency is simply

$$\text{III.4} \longrightarrow \text{I.23}.$$

Angle transport has already been placed in the congruence layer.

23.7.3 Foundational Interpretation

This is not merely a shorter proof of the same dependency graph. Euclid treats the ability to copy an angle as something constructed from more elementary segment operations. Hilbert treats it as part of the primitive congruence structure. The same visible Book I proposition therefore occupies two different logical levels in the two foundations.

23.8 Relationship with Earlier Propositions

23.8.1 Proposition I.8

I.8 is the final congruence theorem in Euclid's proof: once the auxiliary triangle has the required three side lengths, SSS identifies the transported angle. The current Hilbert proof does not call I.8.

23.8.2 Proposition I.22

Euclid uses I.22 to construct the auxiliary triangle with prescribed sides. The current Hilbert proof does not call the theorem `euclid_proposition_22`. The source module does import `Proposition22`, but that module import must not be confused with a direct theorem-level dependency of the proof body. No import-removal test was performed in this audit.

23.8.3 Book I Order versus Formal Dependency

The file for I.23 naturally follows I.22 in the Book I sequence, but the formal proof graph contains no theorem-level edge

$$\text{I.22} \longrightarrow \text{I.23}.$$

This is one of the clearest early examples where textual order and formal dependency order diverge.

23.9 Hilbert and Book Zero Components Used

23.9.1 HilbertCongruence.angle_construction

This is the direct engine of the proof. It supplies existence, the same-side condition, angle congruence, and a uniqueness property stronger than the public conclusion.

23.9.2 HilbertSameSide

The same-side relation makes explicit which of the two mirror-image constructions is intended. The point S is therefore genuine geometric input, not bookkeeping.

23.9.3 HilbertSameRay

The uniqueness clause returned by III.4 is naturally expressed through ray equality / same-ray information. The public theorem does not expose this stronger result.

23.9.4 No New Book Zero Lemma

Proposition I.23 exposes no missing low-level operation. Unlike several preceding propositions, the exact construction already exists in the Hilbert congruence layer. No new proposition-specific Book Zero lemma is introduced by the current proof.

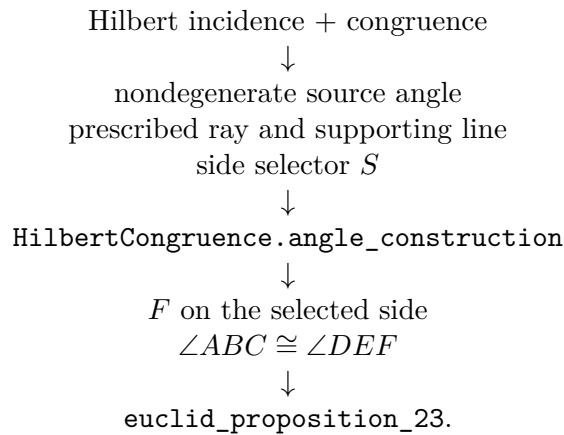
23.10 Uniqueness and the Strength of the Hilbert Statement

Euclid asks only for a construction. Hilbert III.4 gives existence together with uniqueness on the selected side of the supporting line. If another point F' on that side produces an angle congruent to the same source angle, the corresponding ray must coincide with the ray through F .

The public theorem intentionally forgets this information. This is an interface-design choice: the Book I theorem states the Euclidean conclusion, while the stronger foundational theorem remains available separately.

23.11 Dependency Summary

The current formal dependency structure is



No direct theorem-level use is made of Proposition I.22 or Proposition I.8, and no new continuity argument is introduced by I.23.

23.12 Formalization Notes

The point S is additional explicit data relative to Euclid’s informal statement. Without it there are two mirror-image rays satisfying the angle-congruence requirement.

The conditions that D and E lie on **base**, together with $D \neq E$, specify the nondegenerate initial ray ED with vertex E . The noncollinearity of A, B, C guarantees a nondegenerate source angle. The theorem returns a point F , not an abstract ray, matching the point-based ray representation of the Hilbert layer.

The short Lean proof is not evidence that the proposition has become geometrically empty. Its mathematical content has been moved into the axiomatic congruence interface.

23.13 Diagrammatic Audit

The preserved Book1R chapter contained three figures. The audit retains only geometric states that add new construction data.

- The old Figure 23.1 showed only the initial segment and the given angle. It is removed as a separate float because the completed classical construction already contains these data together with the genuinely new auxiliary triangle.
- The old Figure 23.2 is retained, renumbered as Figure 23.1. It records the actual Euclidean construction: a new triangle and three side congruences.
- The old Figure 23.3 is retained in role but corrected in orientation, and renumbered as Figure 23.2. The current Lean target is $\angle DEF$, whose vertex is E ; therefore the prescribed initial ray is ED , not a ray based at D . The corrected figure records the supporting line, side selector S , and the constructed point F .

- No separate uniqueness diagram is added. The uniqueness result changes logical information about the already constructed ray but introduces no new geometric state.

Thus the final audited chapter has two figures: one for Euclid's derived construction and one for Hilbert's primitive angle transport.

23.14 Conclusion

Proposition I.23 provides a particularly clean example of the difference between historical construction order and formal axiomatic dependency.

Euclid constructs the transported angle indirectly through triangle construction and SSS. The Hilbert reconstruction invokes III.4 directly and then forgets the extra uniqueness information. The audited diagrams now mirror exactly those two geometric routes, without a redundant float for the initial data alone.

Chapter 24

Proposition I.24

24.1 Introduction

Euclid's Proposition I.24 is the classical hinge theorem. Two triangles have two corresponding pairs of congruent sides. If the included angle of the first triangle is greater than the included angle of the second, then the side opposite the greater angle is greater than the corresponding side of the second triangle.

In the notation of the current formal development, the hypotheses are

$$AB \cong DE, \quad AC \cong DF,$$

and

$$\angle EDF < \angle BAC.$$

The conclusion is

$$EF < BC.$$

The reconstruction is entirely synthetic. No numerical length function, coordinate system, or numerical angle measure is introduced. Proposition I.24 is nevertheless substantially more delicate than its statement suggests. The traditional diagram suppresses a genuine positional issue, and the Lean reconstruction makes the required order information explicit.

There are two distinct three-way analyses in the documentation, and they must not be identified:

- the *historical* three configurations concern the position of the classical auxiliary points relative to the line GF ;
- the *formal* three cases concern only the order of the points H and Y on one ray from A .

24.2 The Classical Configuration

Let $\triangle ABC$ and $\triangle DEF$ be nondegenerate triangles with

$$AB \cong DE, \quad AC \cong DF, \quad \angle EDF < \angle BAC.$$

Euclid constructs at D an angle congruent to the larger angle at A and lays off a segment DG congruent to AC and hence to DF . Thus

$$\angle EDG \cong \angle BAC, \quad DG \cong AC \cong DF.$$

By SAS,

$$\triangle ABC \cong \triangle DEG,$$

so that

$$BC \cong EG.$$

It remains to prove $EF < EG$.

The historical difficulty is best seen by displaying all three configurations distinguished in the classical commentary tradition. Following the preserved Book1R discussion, the distinction is made by the positions of D and E relative to the line through G and F .

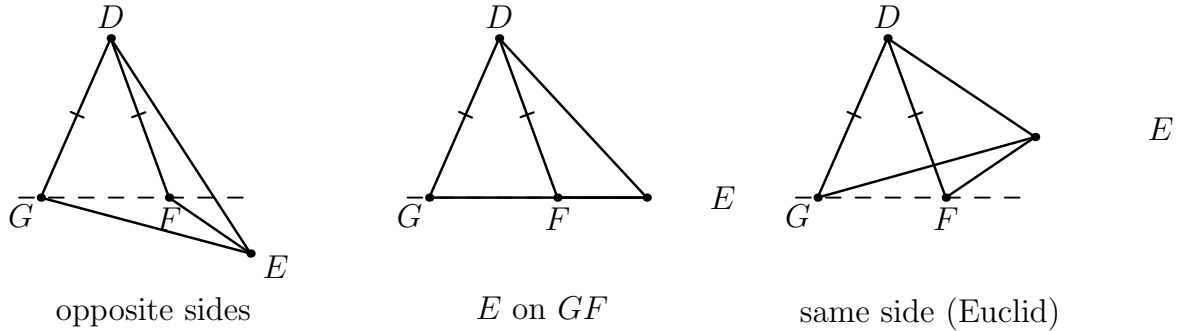


Figure 24.1: The three configurations distinguished in the classical discussion of Proposition I.24: D and E on opposite sides of the line GF , E on GF , and D, E on the same side. Euclid's printed proof treats the third. The equal marks on DG and DF record the isosceles triangle used in the classical argument. These are historical configurations, not the three Lean hinge branches.

24.3 Classical Proof and the Configuration Gap

Euclid's argument proceeds as follows. Since

$$AB \cong DE, \quad AC \cong DG, \quad \angle BAC \cong \angle EDG,$$

Proposition I.4 gives

$$BC \cong EG.$$

Also $DF \cong DG$, so triangle DFG is isosceles. Proposition I.5 therefore supplies the corresponding equality of its base angles. In the configuration displayed by Euclid, this angle equality together with the position of E yields a strict comparison between the two angles of triangle EFG . Proposition I.19 then gives

$$EF < EG.$$

Since $EG \cong BC$, one obtains $EF < BC$.

The classical dependency pattern is therefore

```

angle EDF < angle BAC
|
construct G: DG ~= AC, angle EDG ~= angle BAC      I.23
|
I.4
|
EG ~= BC

```

```

      |
    DG ~= DF
      |
    I.5
      |
angle comparison in triangle EFG
      |
    I.19
      |
    EF < EG
      |
    EF < BC

```

The problem is not the congruence part of this argument. The problem is the unproved assumption about the relative position of the auxiliary points. Ancient and later commentary already recognized the omitted configurations; the Hilbert reconstruction supplies an axiomatic language in which such position information cannot be inferred silently from one preferred picture.

In Euclid's third configuration the useful angle-comparison chain may be written schematically as

$$\angle EFG > \angle DFG = \angle DGF > \angle EGF.$$

The preserved Book1R PDF displayed this same configuration a second time with explanatory coloured angle regions. No new point, line, betweenness relation, or construction appears there, so the audited version keeps the comparison in the text rather than retaining a duplicate figure.

24.4 Strict Angle and Segment Comparison

Two synthetic order relations are central to the reconstruction. `HilbertAngleLess` expresses strict comparison of angles without a numerical angle measure. Informally, $\alpha < \beta$ means that a congruent copy of α occurs as a proper interior subangle of β .

Similarly, `HilbertSegmentLess` expresses strict comparison of segments through betweenness and congruence, without introducing a real-valued length function. Proposition I.24 bridges these two order theories: its hypothesis is a strict angle inequality and its conclusion is a strict segment inequality.

24.5 Formal Architecture of the Current Lean Proof

The current production file imports

```

import CGJteamLab.HilbertBookZero
import CGJteamLab.Proposition19
import CGJteamLab.Proposition23

```

and contains the proposition-specific theorem chain

```

euclid_proposition_24_construct_aux
euclid_proposition_24_sas_aux
euclid_proposition_24_hinge_case1
euclid_proposition_24_hinge_case3

```

```
euclid_proposition_24_hinge_aux
euclid_proposition_24
```

All these declarations use the Hilbert incidence and congruence typeclasses. No Euclidean-plane typeclass is required by I.24.

24.5.1 The Interior-Ray Construction

The input

$$\angle EDF < \angle BAC$$

is already a witness-bearing synthetic statement. The theorem `euclid_proposition_24_construct_aux` extracts a point X whose ray from A meets the open segment BC . Unpacking that meeting relation provides a point Y with

$$B - Y - C, \quad \text{SameRay}(A; X, Y).$$

The proof then lays off on the ray AX a point H satisfying

$$\text{SameRay}(A; X, H), \quad AH \cong AC, \quad \angle BAH \cong \angle EDF.$$

The point X is an important representation witness, but it is not given a separate diagram. The formal statements do not determine a special order of X relative to H and Y , and placing it arbitrarily on a figure would add positional information not supplied by the theorem.

24.5.2 The SAS Construction

The theorem `euclid_proposition_24_sas_aux` packages the congruence part of the argument. From

$$AH \cong AC \cong DF, \quad AB \cong DE, \quad \angle BAH \cong \angle EDF,$$

SAS applied to $\triangle ABH$ and $\triangle DEF$ gives

$$BH \cong EF.$$

The original target $EF < BC$ is therefore reduced to the hinge comparison

$$BH < BC.$$

24.5.3 The Hinge Configuration

The interior ray through H meets the open segment BC at Y . Same-ray information does not determine the order of H and Y on the ray from A . The current proof invokes `euclid_proposition_22_sameRay_trichotomy` and obtains exactly

$$A - H - Y, \quad H = Y, \quad A - Y - H.$$

This is the central geometric case split of the current Lean proof. It is not the historical three-way split of Figure 24.1.

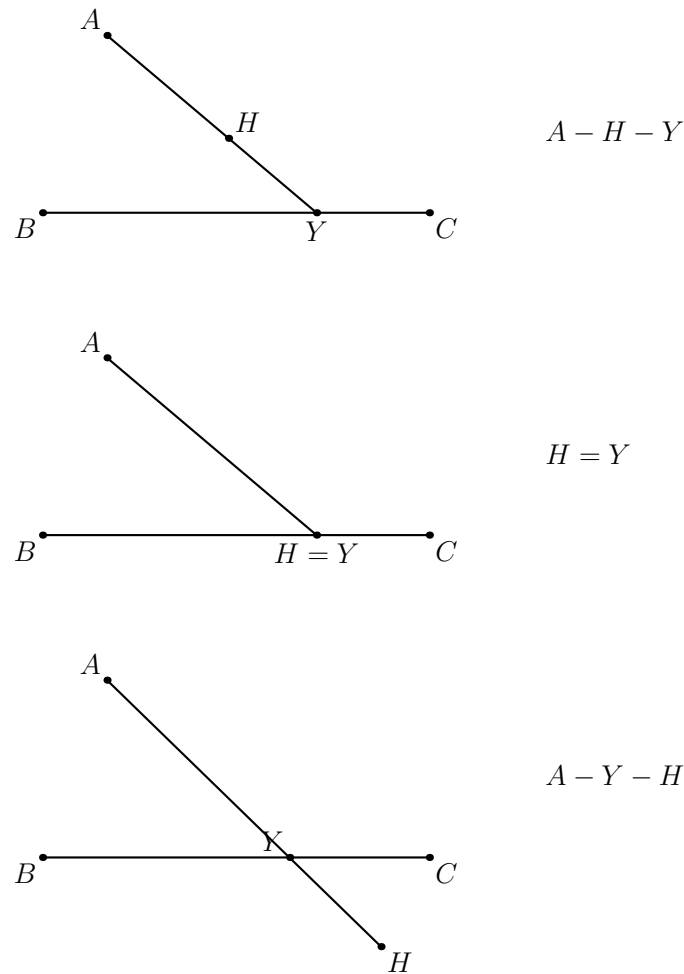


Figure 24.2: The three order configurations used by `euclid_proposition_24_hinge_aux`. In all three panels Y lies in the open segment BC ; the split concerns only the order of H and Y on their common ray from A . The earlier SAS helper also supplies $AH \cong AC$ and $BH \cong EF$, but those congruence facts do not create additional order configurations.

24.5.4 Case 1: $A - H - Y$

This is the longer nontrivial branch. After reconstructing the required noncollinearity facts, the proof uses Proposition I.16 to obtain strict exterior-angle comparisons. Since

$$AH \cong AC,$$

triangle AHC is isosceles, allowing the relevant angles at H and C to be transported across congruence. The incidence of the ray through Y with the opposite side is converted into strict angle order by `hilbert_interior_angle_less`. After the required same-ray normalizations and transitivity steps, the branch obtains

$$\angle BCH < \angle BHC.$$

The current code then calls `euclid_proposition_19` directly and concludes

$$BH < BC.$$

24.5.5 Case 2: $H = Y$

The middle branch is genuinely different and much simpler. After substituting $H = Y$, the already available betweenness

$$B - Y - C$$

becomes $B - H - C$. The witness-based definition of strict segment order immediately supplies

$$BH < BC.$$

No angle comparison is required.

24.5.6 Case 3: $A - Y - H$

This branch has different incidence geometry. At C , the ray through Y meets the open segment HA , producing a proper interior-subangle comparison. The betweenness $B - Y - C$ identifies the relevant ray at C ; $AH \cong AC$ transfers the comparison across the isosceles triangle AHC . At H , the ray through Y meets the open segment CB , giving a second interior-angle comparison. Since $H - Y - A$, the ray HY agrees with the ray HA . The transported inequalities again yield

$$\angle BCH < \angle BHC,$$

and a direct application of Proposition I.19 gives $BH < BC$.

24.5.7 The Hinge Helper and Completion

The theorem `euclid_proposition_24_hinge_aux` packages the complete order analysis:

```

ray through H meets BC at Y
|
same-ray trichotomy for A,Y,H
|
+-----+-----+
|       |       |
A-H-Y  H=Y    A-Y-H
|       |       |
case1  direct case3
+-----+-----+
|
BH < BC

```

The public theorem then has the short endpoint architecture

```

euclid_proposition_24_sas_aux
|  BH ~= EF
v
euclid_proposition_24_hinge_aux
|  BH < BC
v
bookZero_32_lessThanCongruence2
|
v
EF < BC

```

Thus the final strict segment comparison is transported explicitly by the Book Zero theorem

```
bookZero_32_lessThanCongruence2
```

24.6 Diagrammatic Audit

The preserved Book1R PDF contained three figures for I.24:

1. a three-panel map of the historical classical configurations;
2. the Euclid same-side configuration repeated with explanatory angle regions;
3. a three-panel map of the formal hinge trichotomy.

The audited version retains all six genuine geometric states but reduces the chapter to two figures. Figure 24.1 records the three historically distinct positions. Figure 24.2 records the three formally distinct orders of H and Y .

The former Figure 24.2 is removed because it repeated the third panel of the historical map and changed only the logical interpretation of angles. No new construction or order relation appeared there. The comparison is therefore kept in prose rather than as a separate float.

No additional figure is introduced for the witness X , for SAS, for angle-congruence transport, or for the final call to `bookZero_32_lessThanCongruence2`. Those steps add no new proposition-specific geometric state. This implements the audit rule: *a figure should record geometry, not tactic state*.

24.7 Relationship with Earlier Propositions

24.7.1 Proposition I.5

The isosceles base-angle theorem is used after $AH \cong AC$. It is the main congruence bridge between the angles at H and C in the formal hinge triangle. In the classical proof I.5 is likewise applied to the isosceles triangle DGF .

24.7.2 Proposition I.16

Case 1 uses the exterior-angle theorem to obtain strict angle comparisons which are not available directly from the interior-ray configuration.

24.7.3 Proposition I.19

Both nontrivial formal branches end with

$$\angle BCH < \angle BHC,$$

and call `euclid_proposition_19` directly to conclude $BH < BC$. In the classical proof I.19 converts the angle comparison in $\triangle EFG$ into $EF < EG$.

24.7.4 Proposition I.23

The production module imports `Proposition23`, and `I.23` belongs to the classical dependency picture as the construction of an angle equal to the given larger angle. The current `I.24` proof body, however, does not call `euclid_proposition_23` directly; its auxiliary angle is obtained through the more general witness machinery already packaged in `HilbertAngleLess`.

24.8 Hilbert and Book Zero Components Used

The main reusable infrastructure is:

- `HilbertAngleLess`, carrying the interior-ray witness of the strict input inequality;
- `HilbertSegmentLess`, the witness-based target relation;
- `HilbertSameRay` and the `I.22` same-ray trichotomy;
- `hilbert_interior_angle_less`;
- angle-congruence and angle-order transport;
- Book Zero collinearity, order, and strict-segment transport lemmas.

The current module itself contains no declaration of `axiom` or `sorry`. This documentation audit did not rerun `lake build` or `#print axioms`; therefore no new build or transitive axiom-audit claim is made here.

24.9 Historical Note

Proposition `I.24` is a useful example of the distinction between a classical geometric argument and a complete axiomatic treatment of its diagram. The traditional text presents one configuration; ancient commentary already supplied arguments for the omitted configurations, and later editors reorganized or restricted the construction in different ways. Hilbert’s role is different: incidence, betweenness, ray identity, and separation must be stated and justified instead of being silently read from a preferred picture.

Accordingly, the three-way split in the present Lean proof is not a proof-assistant artifact. It is a precise order-theoretic analysis of the auxiliary hinge configuration. At the same time, it must not be confused with the three historical configurations in the classical discussion.

24.10 Dependency Summary

The formal dependency structure can be summarized as

```
Hilbert incidence + order + congruence
|
HilbertAngleLess / HilbertSegmentLess / HilbertSameRay
|
angle construction + interior-ray theory + I.22 trichotomy
```

```

      |
I.5 + I.16 + I.19 + Book Zero transport
      |
construct_aux -> sas_aux
      |
hinge_case1 / direct middle case / hinge_case3
      |
hinge_aux
      |
bookZero_32_lessThanCongruence2
      |
euclid_proposition_24

```

No numerical metric, coordinate system, numerical angle measure, or new hinge axiom is introduced by the proposition.

24.11 Conclusion

Proposition I.24 is one of the clearest examples of what formal reconstruction can reveal about a classical synthetic proof. The metric core is short: the SAS construction gives $BH \cong EF$, and the hinge comparison gives $BH < BC$. The real difficulty is positional: the hypotheses do not force a single order of the auxiliary point H and the intersection point Y .

The audited documentation therefore keeps exactly the geometry that matters. It retains a three-panel historical map and a separate three-panel formal hinge map, while removing the duplicate coloured angle figure. The resulting two-figure presentation preserves all genuine geometric states and separates the classical and formal DAGs explicitly.

Chapter 25

Proposition I.25

25.1 Introduction

Euclid's Proposition I.25 is the converse comparison theorem to Proposition I.24.

Suppose two triangles have two corresponding sides congruent:

$$AB \cong DE, \quad AC \cong DF.$$

If the third side of the first triangle is greater than the third side of the second,

$$EF < BC,$$

then the included angle of the first triangle is greater:

$$\angle EDF < \angle BAC.$$

The formal reconstruction is strikingly short. Unlike Proposition I.24, no auxiliary geometric construction is needed inside the proposition itself. The proof is an elimination argument based on the previously established trichotomy for strict angle comparison.

The difficult geometric work needed for I.25 has already been absorbed into the Hilbert interface, in particular into `angle_trichotomy`. Once that theorem is available, Proposition I.25 becomes almost purely logical.

25.2 The Classical Configuration

Let $\triangle ABC$ and $\triangle DEF$ satisfy

$$AB \cong DE, \quad AC \cong DF,$$

and suppose

$$EF < BC.$$

The conclusion is

$$\angle EDF < \angle BAC.$$

No new point is constructed by the proof body of Proposition I.25. The same two triangles are compared throughout the trichotomy argument. The strict segment-order hypothesis, however, contains its own hidden geometric witness; this is separated below from proposition-specific construction.

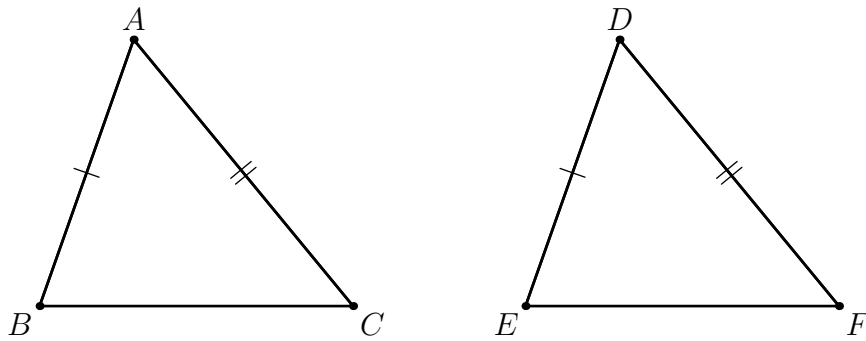


Figure 25.1: The classical configuration for Proposition I.25. The two pairs of corresponding sides $AB \cong DE$ and $AC \cong DF$ are marked. The hypothesis $EF < BC$ is not encoded by the apparent drawn lengths; strict segment order is a synthetic relation and is expanded explicitly in Figure 25.2.

25.3 Diagrammatic Audit

The preserved Book1R PDF contains one figure for I.25. It correctly records the two triangles and the two side-congruence classes, but the strict hypothesis $EF < BC$ is visible only through the accidental scale of the drawing. Such scale is not part of the synthetic proof language.

In the current formalization, `HilbertSegmentLess Geo E F B C` means that there exists a point H with

$$B - H - C, \quad EF \cong BH.$$

Thus the formal input contains a genuine hidden witness even though the proof of `euclid_proposition_25` does not construct a new point. Figure 25.2 records this witness explicitly.

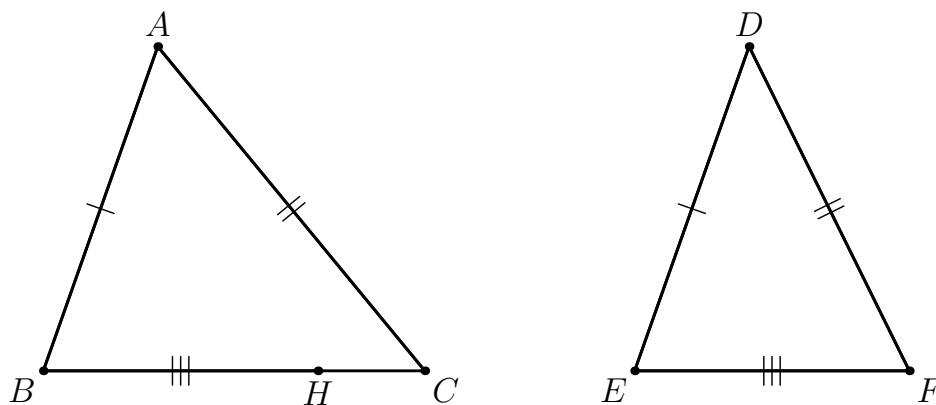


Figure 25.2: The exact geometric witness hidden in the strict segment-order hypothesis. The point H satisfies $B - H - C$ and $BH \cong EF$. Together with the original side congruences this is a synthetic realization of $EF < BC$; no numerical length comparison is used.

No additional figure is introduced for the three branches of `angle_trichotomy`. Those branches change only the logical relation between the same two included angles; they introduce no new point, line, betweenness relation, side relation, or auxiliary construction. Likewise, the internal geometry of Proposition I.24 is not repeated here: it belongs to the diagrammatic audit of I.24.

25.4 Classical Proof

Euclid excludes the two alternatives incompatible with the required conclusion.

First suppose

$$\angle BAC \cong \angle EDF.$$

Then Proposition I.4 (SAS), applied to the two pairs of congruent sides, gives

$$BC \cong EF,$$

contradicting $EF < BC$.

Next suppose

$$\angle BAC < \angle EDF.$$

Apply Proposition I.24 with the two triangles interchanged. Since the included angle of $\triangle DEF$ is greater, I.24 gives

$$BC < EF,$$

again contradicting $EF < BC$.

Therefore the only remaining possibility is

$$\angle EDF < \angle BAC.$$

25.5 Angle Trichotomy

The decisive interface result is `angle_trichotomy`. For two nondegenerate angles it provides exactly the three alternatives

$$\angle BAC \cong \angle EDF, \quad \angle BAC < \angle EDF, \quad \angle EDF < \angle BAC.$$

This theorem was established in the Hilbert layer immediately before the formalization of I.25. Its proof uses Hilbert angle construction, same-side information, the order of rays in a half-plane, and the principle that two collinear points lying on the same side of a line through the vertex determine the same ray.

Thus angle trichotomy is not an additional axiom introduced for I.25. It is a derived structural theorem about the already defined relation `HilbertAngleLess`.

25.6 Proof Architecture of the Reconstruction

The formal proof is a three-branch elimination argument. Its complete architecture is short enough to display directly in ASCII:

```
angle_trichotomy
|
+-- angles congruent
|   -> SAS
|   -> BC ~= EF
|   -> contradiction with EF < BC
```



```

|
+--- BAC < EDF
|   -> Proposition I.24
|   -> BC < EF
|   -> contradiction with EF < BC
|
+--- EDF < BAC
      -> goal

```

This is not merely a summary of the Lean tactics. It is the actual mathematical dependency structure of the reconstructed proof.

The first branch is eliminated by triangle congruence. The second is eliminated by the direct hinge theorem I.24. The third branch is already the desired statement.

Consequently, after angle trichotomy has been established, the proof body of I.25 contains no proposition-specific construction and no additional geometric case analysis. The only hidden point relevant to the theorem statement is the witness already packaged inside the input relation $EF < BC$.

25.7 Hilbert Reconstruction

25.7.1 The Strict-Segment Input Witness

The hypothesis

HilbertSegmentLess Geo E F B C

is an existential synthetic statement. Unfolded, it supplies a point H such that

$$B - H - C, \quad EF \cong BH.$$

This witness is part of the input data rather than a construction made by `euclid_proposition_25`. The public proof never needs to choose it explicitly because the generic segment-order lemmas `hilbert_segmentLess_not_congruent` and `hilbert_segmentLess_asymm` consume the strict-order relation directly.

25.7.2 The Congruent-Angle Branch

Assume

$$\angle BAC \cong \angle EDF.$$

Together with

$$AB \cong DE, \quad AC \cong DF,$$

Hilbert SAS yields

$$BC \cong EF.$$

The interface theorem `hilbert_segmentLess_not_congruent` states that a strictly shorter segment cannot be congruent to the longer one. Hence this branch is impossible.

25.7.3 The Wrong Strict-Inequality Branch

Assume

$$\angle BAC < \angle EDF.$$

Interchange the roles of the triangles and reverse the side congruences:

$$DE \cong AB, \quad DF \cong AC.$$

Then Proposition I.24 gives

$$BC < EF.$$

But the original hypothesis is

$$EF < BC.$$

The asymmetry theorem `hilbert_segmentLess_asymm` excludes these simultaneous strict inequalities.

25.7.4 The Desired Branch

The remaining case is exactly

$$\angle EDF < \angle BAC,$$

so no further argument is needed.

25.8 Lean Formalization

The public theorem has the form

```
theorem euclid_proposition_25
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D E F : Geo.Point)
  (hABC : not (PrimCollinear Geo A B C))
  (hDEF : not (PrimCollinear Geo D E F))
  (hAB_DE : Geo.Congruent A B D E)
  (hAC_DF : Geo.Congruent A C D F)
  (hEF_BC : HilbertSegmentLess Geo E F B C) :
  HilbertAngleLess Geo E D F B A C
```

The proof begins by invoking `angle_trichotomy` on the two included angles.

In the equality branch, `SAS` produces a `TriangleCongruenceResult`; its field `sideBC` gives $BC \cong EF$. After reversing the congruence, the contradiction is discharged by `hilbert_segmentLess_not_congruent`.

In the second branch, Proposition I.24 is called with the two triangles interchanged. This gives $BC < EF$, contradicting the hypothesis through `hilbert_segmentLess_asymm`.

The third branch is returned directly.

25.9 Relationship with Earlier Propositions

25.9.1 Proposition I.4

Proposition I.4 supplies the SAS criterion used to eliminate the equality branch:

$$\angle BAC \cong \angle EDF, \quad AB \cong DE, \quad AC \cong DF \quad \implies \quad BC \cong EF.$$

25.9.2 Proposition I.24

Proposition I.24 is the central direct dependency of I.25. I.24 states that, under two pairs of congruent sides, a greater included angle implies a greater third side. I.25 reverses the comparison: a greater third side implies a greater included angle.

The formal proof of I.25 uses I.24 exactly once, to eliminate the wrong direction of angle inequality.

Together they give, under

$$AB \cong DE, \quad AC \cong DF,$$

the order correspondence

$$\angle EDF < \angle BAC \quad \iff \quad EF < BC.$$

25.10 Hilbert and Book Zero Components Used

25.10.1 `angle_trichotomy`

This is the principal structural result exposed by I.25. It converts the comparison of two nondegenerate angles into an exhaustive three-way split: congruent, smaller in one direction, or smaller in the other direction.

25.10.2 SAS

The existing SAS theorem converts equality of the included angles and the two side congruences into congruence of the third sides.

25.10.3 Strict order versus congruence

This theorem expresses incompatibility between strict segment order and segment congruence. It eliminates the first trichotomy branch.

25.10.4 Asymmetry of strict segment order

This theorem expresses asymmetry of strict segment comparison:

$$AB < CD \quad \implies \quad \neg(CD < AB).$$

It eliminates the second branch after I.24 produces the opposite inequality.

25.11 Formalization Notes

Proposition I.25 provides a useful contrast with I.24.

I.24 required substantial geometric work. Its formal reconstruction had to expose point-order cases hidden by Euclid's diagram and develop the corresponding hinge argument in each configuration.

I.25 does not repeat any of that proposition-specific geometry. Its strict segment-order hypothesis already packages a betweenness witness, but the proof itself does not construct or unpack that witness. Instead, the reconstruction first strengthens the reusable Hilbert interface by proving angle trichotomy. Once this has been done, I.25 is a direct logical consequence of

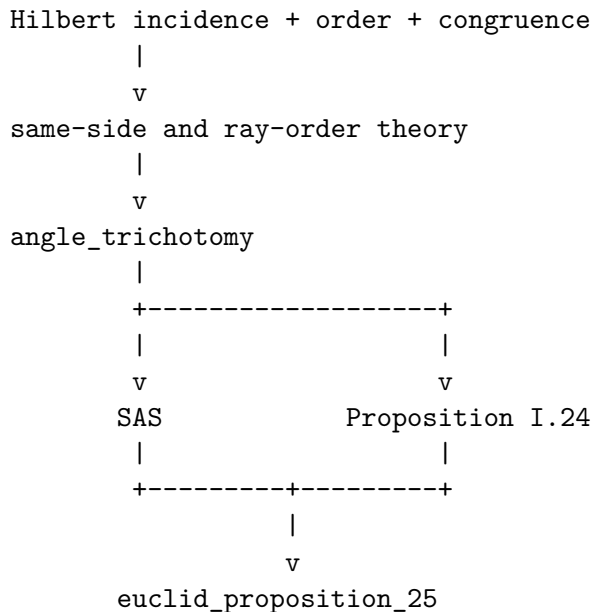
$$\text{angle trichotomy}, \quad \text{SAS}, \quad \text{I.24}.$$

This separation is important for the architecture of the project. A long foundational or order-theoretic argument should not be buried inside every later Euclidean proposition that needs it. Once identified, it becomes part of the intermediate synthetic language and later proofs become correspondingly transparent.

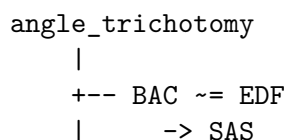
I.25 is therefore a particularly clean example of the role of the Hilbert interface: difficult geometric infrastructure is proved once, after which a classical theorem can recover its natural short proof.

25.12 Dependency Summary

The underlying dependency graph is



At the level of the final theorem:



```

|      -> BC ~= EF
|      -> contradiction with EF < BC
|
+--- BAC < EDF
|      -> I.24
|      -> BC < EF
|      -> contradiction with EF < BC
|
+--- EDF < BAC
      -> exact

```

No coordinate system, numerical segment length, or numerical angle measure is introduced. The current production file contains no local `axiom` declaration and no `sorry`; this documentation audit does not claim a fresh transitive axiom audit because neither `lake build` nor `#print axioms` was run here.

25.13 Conclusion

Proposition I.25 is the converse comparison theorem to Proposition I.24.

Its formal reconstruction is exceptionally concise, but the concision is not accidental. The theorem becomes short only after the underlying angle-order theory has been completed.

The proof begins with angle trichotomy. Congruence of the included angles is excluded by SAS, because it would force congruence of the third sides. The wrong strict inequality is excluded by Proposition I.24, because it would force the opposite strict comparison of the third sides. The remaining branch is precisely the desired conclusion.

Thus I.25 gives a particularly clear instance of the architecture

foundational Hilbert theory $\longrightarrow$ reusable synthetic interface $\longrightarrow$ short Euclidean theorem.

Chapter 26

Proposition I.26

26.1 Introduction

Euclid's Proposition I.26 contains two triangle-congruence criteria.

For two nondegenerate triangles ABC and DEF , assume the two corresponding angles

$$\angle ABC \cong \angle DEF, \quad \angle BCA \cong \angle EFD.$$

Euclid then considers two possibilities for the given corresponding side.

1. In the ASA case,

$$BC \cong EF.$$

2. In the SAA case,

$$AB \cong DE.$$

In either case the conclusion is full triangle congruence:

$$AB \cong DE, \quad BC \cong EF, \quad AC \cong DF,$$

together with

$$\angle BAC \cong \angle EDF.$$

The two parts have very different formal behavior.

The ASA part is almost entirely absorbed by the existing Hilbert congruence theory. The SAA part is not. Its proof still requires the essential contradiction argument of Euclid: a hypothetical strict inequality of the corresponding sides is converted into an auxiliary SAS configuration and then contradicted by Proposition I.16.

Thus Proposition I.26 provides a useful boundary case. One half is already available as interface-level geometry; the other still exposes genuine Book I proof structure.

The current production file imports `HilbertBookZero` and `Proposition16` and exposes three proposition-specific theorems:

```
euclid_proposition_26_asa
euclid_proposition_26_saa_not_less
euclid_proposition_26_saa
```

It requires `HilbertIncidence` and `HilbertCongruence`, but no `HilbertEuclideanPlane` instance.

26.2 Classical Configurations

Wilkins emphasizes that Proposition I.26 establishes both the ASA and the SAA congruence rules. The same basic pair of triangles is used for both arguments, but the auxiliary point lies on a different side in the two cases.

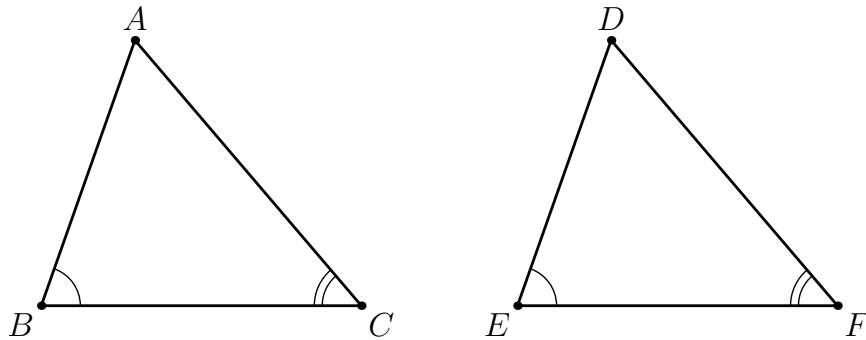


Figure 26.1: The common angle data of Proposition I.26: $\angle ABC \cong \angle DEF$ and $\angle BCA \cong \angle EFD$. No side congruence is marked: the ASA clause assumes $BC \cong EF$, whereas the SAA clause assumes $AB \cong DE$.

26.2.1 ASA auxiliary configuration

Assume first

$$BC \cong EF.$$

Euclid supposes, for contradiction, that the corresponding sides AB and DE are unequal. If AB is the greater side, choose a point G between A and B such that

$$GB \cong DE,$$

and join G to C .

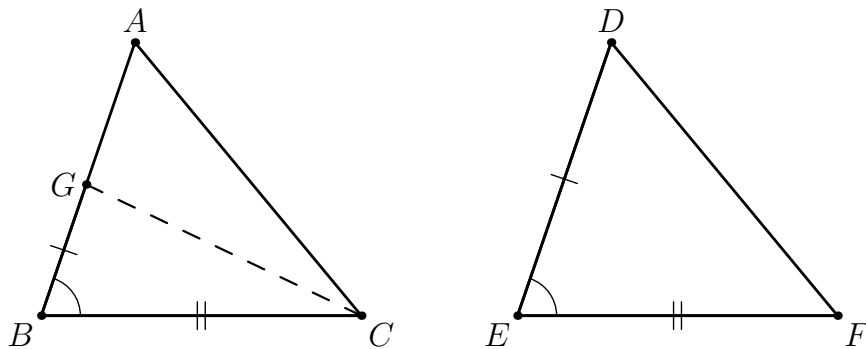


Figure 26.2: Euclid's classical ASA reductio configuration. The point G lies between B and A . The diagram records exactly the local SAS data $GB \cong DE$, $BC \cong EF$, and $\angle GBC \cong \angle DEF$. The dashed segment GC is auxiliary. This point G is not constructed by the current Lean ASA theorem.

SAS gives

$$\angle BCG \cong \angle EFD.$$

Together with the original hypothesis

$$\angle EFD \cong \angle BCA,$$

this would imply

$$\angle BCG \cong \angle BCA.$$

But G lies inside the side AB , so $\angle BCG$ is a proper part of $\angle BCA$. This contradiction forces

$$AB \cong DE.$$

A final SAS application to the original triangles gives the remaining side and angle congruences.

26.2.2 SAA auxiliary configuration

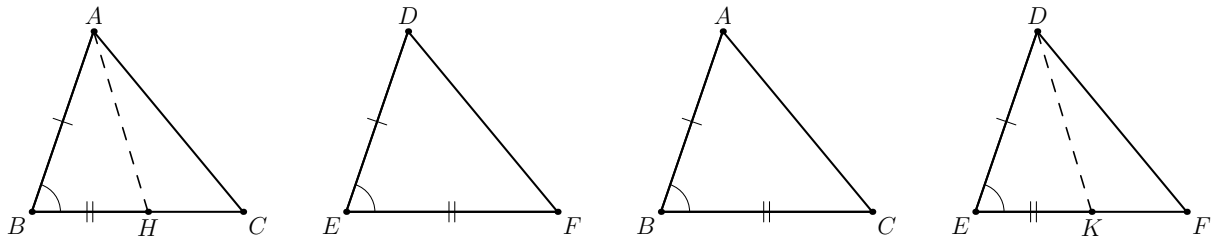
Now assume instead

$$AB \cong DE.$$

Suppose, for contradiction, that $EF < BC$. Then there is a point H between B and C such that

$$BH \cong EF.$$

Join A to H .



(a) Assume $EF < BC$: choose $B - H - C$ with $BH \cong EF$.

(b) After swapping the triangles, assume $BC < EF$: choose $E - K - F$ with $EK \cong BC$.

Figure 26.3: The two strict-inequality exclusions used by the current SAA proof. Panel (a) is the configuration explicitly constructed by `euclid_proposition_26_saa_not_less`. Panel (b) is the same helper applied with the two triangles interchanged. The witnesses H and K belong to alternative reductio arguments and are not simultaneous points of one construction. Dashed segments are auxiliary.

Since B, H, C are collinear on the same ray from B ,

$$\angle ABH \cong \angle ABC.$$

Hence SAS on $\triangle BAH$ and $\triangle EDF$ gives

$$\angle BHA \cong \angle EFD.$$

Using the original angle hypothesis,

$$\angle EFD \cong \angle BCA,$$

we obtain

$$\angle BHA \cong \angle BCA.$$

On the other hand, $C - H - B$, so $\angle BHA$ is an exterior angle of triangle AHC . Proposition I.16 gives the strict inequality

$$\angle BCA < \angle BHA,$$

contradicting their congruence.

Thus

$$\neg(EF < BC).$$

Interchanging the two triangles and repeating the same argument gives

$$\neg(BC < EF).$$

Segment trichotomy therefore yields

$$BC \cong EF.$$

The SAA case has now been reduced to the ASA case.

26.3 Diagrammatic Audit

The audited chapter uses three figure floats containing four geometric panels.

Figure 26.1 records only the geometry common to the two clauses: the same pair of triangles and the two corresponding angle congruences. The preserved `Book1R.pdf` additionally marked $BC \cong EF$ in this figure, thereby privileging the ASA clause inside a figure described as “common input.” The audited figure removes that side mark. The ASA and SAA side hypotheses are stated in the text instead.

Figure 26.2 retains Euclid’s genuine ASA reductio state. The auxiliary point G and segment GC are historically part of the classical proof. They are not constructed by the current Lean ASA theorem, `euclid_proposition_26_asa`; the reusable theorem `hilbert_asa_sides` has absorbed that argument.

The SAA half behaves differently. The current production proof really does construct an interior witness when excluding a strict side inequality. Panel (a) of Figure 26.3 records the first call to `euclid_proposition_26_saa_not_less`: from $EF < BC$ it obtains $B - H - C$ and $BH \cong EF$. The public SAA theorem then calls the same helper with the two triangles interchanged. Panel (b) records that second geometric state, written with a fresh documentary name K : from $BC < EF$ one obtains $E - K - F$ and $EK \cong BC$.

The two SAA panels are alternative reductio states, not successive stages of one construction. No separate figures are introduced for same-ray transport, congruence reversal, angle-less transport, segment trichotomy, SSS, or the final reduction of SAA to ASA. Those steps change the formal information attached to an existing configuration but do not create new geometry.

Thus the audit keeps the classical ASA witness, preserves the first SAA witness from the historical chapter, and adds the second witness that is present in the current Lean DAG but was only described textually in the preserved PDF.

26.4 Hilbert Reconstruction

The two halves of Proposition I.26 behave differently over the present Hilbert interface.

26.4.1 ASA

The theorem `hilbert_asa_sides` already contains the central mathematical content of the ASA case.

The given side in Euclid's formulation is

$$BC \cong EF.$$

The theorem is applied after cyclically viewing the triangles as

$$\triangle BCA \quad \text{and} \quad \triangle EFD.$$

After the required angle and endpoint reorientations it gives

$$BA \cong ED, \quad CA \cong FD.$$

These are transported to

$$AB \cong DE, \quad AC \cong DF.$$

At that point all three pairs of sides are congruent. The existing `HilbertSSS` theorem supplies the remaining corresponding angle:

$$\angle BAC \cong \angle EDF.$$

Thus the formal ASA proof does not reconstruct Euclid's reductio.

26.4.2 SAA

No permutation of `hilbert_asa_sides` directly solves the SAA problem. With the side $AB \cong DE$ fixed, ASA would require the angles at A and B , while the hypotheses supply the angles at B and C .

The formal reconstruction therefore isolates Euclid's essential contradiction argument in the helper

```
euclid_proposition_26_saa_not_less
```

The helper proves

$$\neg(EF < BC).$$

Its architecture is

```
EF < BC
|
witness H with B-H-C and BH ~= EF
|
```

```

same-ray transport:
angle ABH ~= angle ABC
  |
SAS on BAH and EDF
  |
angle BHA ~= angle EFD ~= angle BCA
  |
I.16 on triangle ACH
  |
angle BCA < angle BHA
  |
contradiction

```

The same helper is then called again with the two triangles interchanged, producing

$$\neg(BC < EF).$$

Book Zero segment trichotomy yields

$$BC \cong EF,$$

and the public SAA theorem finishes by calling the already proved ASA case.

26.5 Proof Architecture of the Reconstruction

The two formal paths can be summarized separately.

I.26 ASA

```

two angle congruences
+ BC ~= EF
  |
  v
hilbert_asa_sides
on BCA and EFD
  |
  v
AB ~= DE
AC ~= DF
  |
  v
HilbertSSS
  |
  v
angle BAC ~= angle EDF

```

I.26 SAA

```

two angle congruences

```

```

+ AB ~= DE
  |
  +-----+
  |               |
  v               v
not (EF < BC)    not (BC < EF)
SAS + I.16       same helper,
                  triangles swapped
  |               |
  +-----+-----+
                |
                v
        bookZero_31_trichotomy1
                |
                v
                BC ~= EF
                |
                v
                I.26 ASA case
                |
                v
AC ~= DF
angle BAC ~= angle EDF

```

The contrast with Euclid is precise.

For ASA, the Hilbert interface has already absorbed the reductio into a general theorem.

For SAA, Proposition I.16 remains a genuine formal dependency.

26.6 Lean Formalization

The complete implementation is available in the repository. Only the decisive interface steps are reproduced here.

For ASA, the first substantial call is

```

have hASA :=
  hilbert_asa_sides
    Geo
    B C A
    E F D
    hBCA
    hEFD
    hBC
    hAngleB'
    hAngleC

```

After endpoint reversal this gives the two missing sides. The final angle is then obtained from

```

have hSSS :=
  HilbertSSS
    Geo

```

```

A B C
D E F
hABC
hAB
hBC
hAC

```

For SAA, the public proof first excludes both strict segment inequalities:

```

have hNotEF_BC :=
  euclid_proposition_26_saa_not_less
  Geo A B C D E F
  hABC hDEF hAngleB hAngleC hAB

```

The same helper is applied with the two triangles interchanged. Segment trichotomy then produces the missing side congruence, and the theorem reduces to the already established ASA case.

This organization keeps the long exterior-angle argument inside one named helper rather than duplicating it in the public theorem.

26.7 Relationship with Earlier Propositions

26.7.1 Proposition I.4

Euclid uses SAS in both halves of I.26.

In the ASA proof, SAS is used after cutting off a segment congruent to the corresponding side of the second triangle.

In the SAA proof, SAS is applied to the auxiliary triangle BAH and $\triangle EDF$.

The formal SAA helper preserves this use of SAS directly.

26.7.2 Proposition I.16

Proposition I.16 is the decisive difference between the ASA and SAA arguments.

The SAA helper requires the exterior-angle inequality

$$\angle BCA < \angle BHA.$$

This strict inequality contradicts the angle congruence obtained from SAS. Without I.16 the helper would not close.

26.7.3 Segment trichotomy

The formal SAA proof separates the two possible strict segment inequalities:

$$EF < BC \quad \text{and} \quad BC < EF.$$

After both are excluded, `bookZero_31_trichotomy1` returns the remaining possibility

$$BC \cong EF.$$

Thus Book Zero supplies the final order-theoretic bridge from the two contradiction arguments to the ASA case.

26.8 Hilbert and Book Zero Components Used

26.8.1 `hilbert_asa_sides`

This is the main interface theorem for the ASA half. It converts a common corresponding side together with the two adjacent angle congruences into the remaining two side congruences.

Its availability explains why the formal ASA proof is much shorter than Euclid's proof.

26.8.2 `HilbertSSS`

Once all three corresponding side congruences have been recovered, SSS provides the remaining angle congruence required by the public statement.

26.8.3 `euclid_proposition_26_saa_not_less`

This proposition-specific helper is the formal core of the SAA proof. It packages the classical argument

$$\text{strict side inequality} \longrightarrow \text{SAS} \longrightarrow \text{angle congruence} \longrightarrow \text{I.16 contradiction}.$$

The helper is reusable under interchange of the two triangles, so the symmetric strict-inequality branch does not duplicate the proof.

26.8.4 `bookZero_31_trichotomy1`

This theorem converts the exclusion of both strict comparisons into segment congruence.

It is the precise formal replacement for the classical conclusion that, if neither segment is greater than the other, the two segments are equal.

26.9 Formalization Notes

Proposition I.26 is a particularly useful diagnostic theorem.

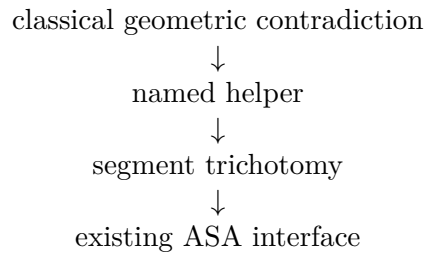
The ASA part shows what happens when a classical Euclidean proof has already been absorbed into the lower Hilbert interface. The Book I wrapper is then short and need not reproduce the historical reductio.

The SAA part shows the opposite situation. The available ASA theorem cannot be reached merely by permuting the vertices, because the given side is not the side between the two supplied angles. A genuine argument is still required.

The formal proof therefore mirrors an important part of Euclid's historical structure: it constructs an interior point, applies SAS, and invokes the exterior-angle theorem I.16.

At the same time, the proof is reorganized to expose the reusable logical unit. Instead of proving two symmetric strict-inequality branches independently, one helper is applied twice.

This gives the architecture



No new geometric axiom, numerical length, coordinate system, or numerical angle measure is introduced.

26.10 Dependency Summary

At the classical level:

```

ASA:
I.3-style segment cut
  |
I.4 (SAS)
  |
proper-angle contradiction
  |
equal corresponding side
  |
I.4
  |
triangle congruence

```

```

SAA:
segment cut
  |
I.4 (SAS)
  |
I.16
  |
contradiction
  |
equal corresponding side

```

```

      |
I.4
      |
triangle congruence

```

At the formal level:

ASA:

```

hilbert_asa_sides
      |
      v
two missing sides
      |
      v
HilbertSSS
      |
      v
euclid_proposition_26_asa

```

SAA:

```

SAS + Proposition I.16
      |
      v
euclid_proposition_26_saa_not_less
      |
      +--- exclude EF < BC
      |
      +--- exclude BC < EF
      |
      v
bookZero_31_trichotomy1
      |
      v
BC ~= EF
      |
      v
euclid_proposition_26_asa
      |
      v
euclid_proposition_26_saa

```

Thus Proposition I.26 exhibits both modes of the current reconstruction in a single theorem: one half collapses to established interface geometry, while the other still requires a substantial Euclidean argument.

26.11 Conclusion

Proposition I.26 completes the elementary triangle-congruence theory of the first part of Book I by establishing the ASA and SAA criteria.

The formal reconstruction makes the distinction between the two cases sharper than the classical presentation.

ASA is already essentially available in the Hilbert interface. The Book I proof only has to orient the triangles correctly, recover the missing sides, and use SSS for the remaining angle.

SAA cannot be reduced by permutation alone. Its proof requires the exterior-angle theorem I.16 to eliminate a hypothetical strict inequality of the corresponding sides. Once both strict inequalities are excluded, Book Zero segment trichotomy reduces SAA to ASA.

The proposition therefore gives a compact example of the layered architecture of the project:

Hilbert interface $\longrightarrow$ Book Zero order theory $\longrightarrow$ Euclidean congruence theorem.

Chapter 27

Proposition I.27

27.1 Introduction

Proposition I.27 is the first proposition in Book I whose conclusion is parallelism. Its mathematical content is the converse direction of the familiar alternate-angle criterion:

$$\text{equal alternate interior angles} \implies \text{parallel straight lines.}$$

No form of Euclid's Fifth Postulate is required. The proposition belongs to neutral geometry: Euclid proves it by showing that any hypothetical intersection of the two lines would contradict Proposition I.16.

The present formalization is especially interesting because the Hilbert development already contains the alternate-angle implication as a general theorem. At the same time, translating the classical transversal diagram into the hypotheses of that theorem exposes additional incidence, order, ray-orientation, and side-of-line information that is suppressed in the traditional picture.

For this reason the Lean file deliberately retains two formulations of Proposition I.27. Their precise proof-theoretic relation is left visible rather than being eliminated by premature refactoring.

27.2 Classical Statement

Let a straight line meet the straight lines AB and CD at E and F , respectively. Suppose that the alternate interior angles

$$\angle AEF \quad \text{and} \quad \angle EFD$$

are equal. Then

$$AB \parallel CD.$$

The basic configuration is shown in Figure [27.1](#).

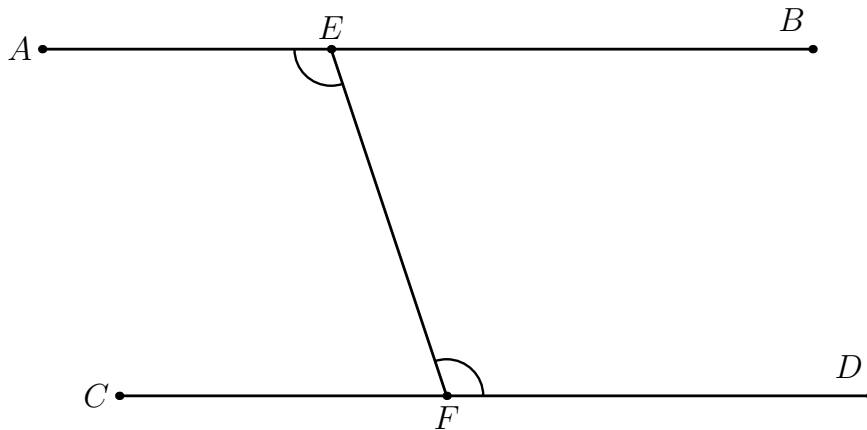


Figure 27.1: The classical input configuration of Proposition I.27. The transversal meets the two candidate lines at E and F , and the alternate interior angles $\angle AEF$ and $\angle EFD$ are congruent. The conclusion $AB \parallel CD$ is deliberately not marked.

27.3 Euclid's Proof

The proof is by contradiction.

Suppose first that the straight lines AB and CD , when produced beyond B and D , meet at a point G . Then $\angle AEF$ is an exterior angle of the triangle EFG . By Proposition I.16,

$$\angle EFD < \angle AEF.$$

This contradicts the hypothesis

$$\angle AEF \cong \angle EFD.$$

The first reductio configuration is shown in the left panel of Figure 27.2.

There is a second possible direction of intersection. Suppose that the lines, when produced beyond A and C , meet at a point H . Then $\angle EFD$ is an exterior angle of the triangle EFH . Again Proposition I.16 gives

$$\angle AEF < \angle EFD,$$

which contradicts the same angle-congruence hypothesis.

Both possible directions of intersection therefore lead to contradictions. Hence the two straight lines do not meet:

$$AB \parallel CD.$$

This is exactly the classical content of Proposition I.27.

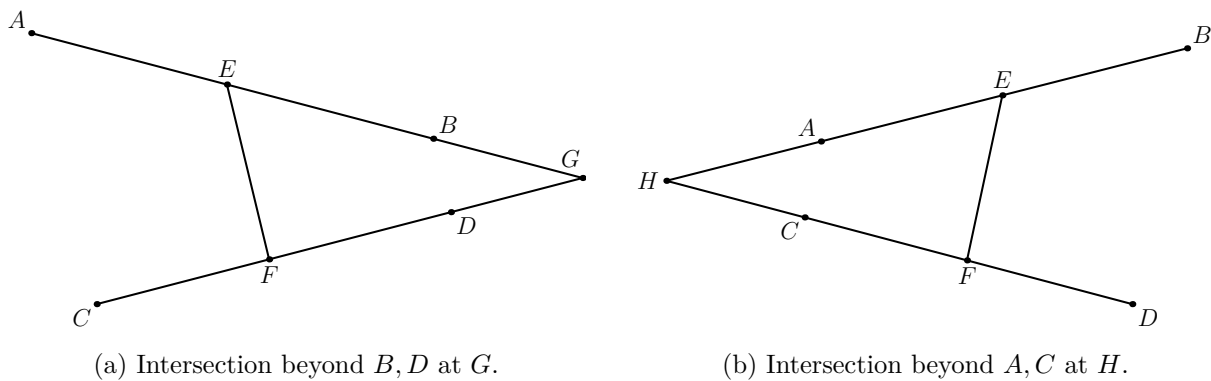


Figure 27.2: Euclid’s two reductio configurations. In panel (a), Proposition I.16 makes $\angle AEF$ strictly greater than $\angle EFD$. In panel (b), it makes $\angle EFD$ strictly greater than $\angle AEF$. The two panels are alternative hypothetical intersection states, not simultaneous construction stages.

27.4 Diagrammatic Audit

The audited chapter uses three figure floats containing four geometric panels.

Figure 27.1 records the classical hypothesis: a transversal and a pair of congruent alternate interior angles. The two possible Euclidean intersections are genuinely different geometric states, so both are retained. They are placed together in Figure 27.2: panel (a) records the hypothetical intersection beyond B, D , while panel (b) records the hypothetical intersection beyond A, C . These panels are reductio states and are not simultaneously realizable with the original angle-congruence hypothesis.

The current Lean proof contains one further proposition-specific geometric state that was absent from the preserved figures. In the explicit transversal theorem an auxiliary point M is chosen with $E - M - F$. This state is recorded separately in Figure 27.3. It is the point at which the classical transversal arms are replaced by the explicit same-ray representatives required by the Hilbert interface.

No separate figure is introduced for the subsequent reversal of angle congruence, collinear carrier transport, symmetry of parallelism, or the final conclusion $AB \parallel CD$. Those operations change the formal representation or the information known about the same geometry; they do not introduce a new geometric state.

Thus the audit preserves every genuine Euclidean reductio configuration and adds the hidden Lean witness M , while avoiding diagrams that merely record tactic state.

27.5 Hilbert Reconstruction

The Hilbert development reaches the mathematical core of I.27 before the Euclidean sequence reaches Proposition I.27.

At the synthetic-interface level the implication

$$\text{equal alternate angles} \implies \text{parallel lines}$$

is already available through

```
parallel_from_equal_angles
```

The first Lean formulation therefore exposes I.27 directly at this level. Its proof is only an application of the existing theorem.

This is not an appeal to the Euclidean parallel postulate. The alternate-angle implication is a theorem of the neutral Hilbert development. The logical direction is important: I.27 proves a sufficient condition for parallelism; the converse direction used in Euclidean geometry appears later, in Proposition I.29, where the Fifth Postulate enters.

27.6 Two Formal Formulations

The current Lean file intentionally retains two formulations.

27.6.1 The interface-level formulation

The theorem

```
euclid_proposition_27
```

uses the ordered configuration already expected by `parallel_from_equal_angles`. Its proof is:

```
exact
  parallel_from_equal_angles
    Geo
    A C D
    B E F
    hADC
    hCEB
    hDEF
    hCED
    hAngle
```

At this level the central mathematical content of Proposition I.27 has already been packaged by the Hilbert theory. The Book I theorem merely exposes that result in the Euclidean sequence.

27.6.2 The explicit transversal formulation

The second theorem is

```
euclid_proposition_27_transversal
```

and begins with a configuration closer to the classical diagram:

$$A - E - B, \quad C - F - D,$$

with E and F on an explicitly supplied transversal *trans*. It also records that the selected interior arms EA and FD lie on opposite sides of the transversal and assumes

$$\angle AEF \cong \angle EFD.$$

This version reveals information that the traditional phrase “alternate interior angles” leaves implicit.

27.7 Normalization of the Transversal Configuration

The explicit proof first chooses a point M between E and F :

$$E - M - F.$$

This provides explicit representatives for the two directions of the transversal. Betweenness gives the corresponding same-ray relations. After reversing the first angle and transporting its transversal arm along these rays, the original congruence

$$\angle AEF \cong \angle EFD$$

is normalized to

$$\angle MEA \cong \angle MFD.$$

The theorem

`hilbert_parallel_of_alternate_angles_oppositeSide_lines`

then yields the local parallelism

$$EA \parallel FD.$$

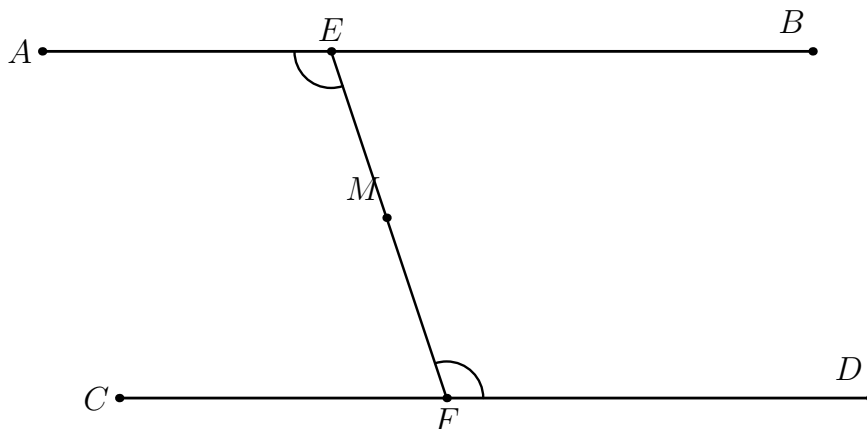


Figure 27.3: Formal transversal normalization in the current Lean proof. The new auxiliary point M satisfies $E - M - F$. It supplies the two `SameRay` representatives used to normalize the classical angle hypothesis to $\angle MEA \cong \angle MFD$. The selected arms EA and FD are on opposite sides of the supplied transversal.

The remainder of the proof is transport along the two original straight lines. Since

$$A - E - B$$

the first local line is extended from EA to AB . Since

$$C - F - D,$$

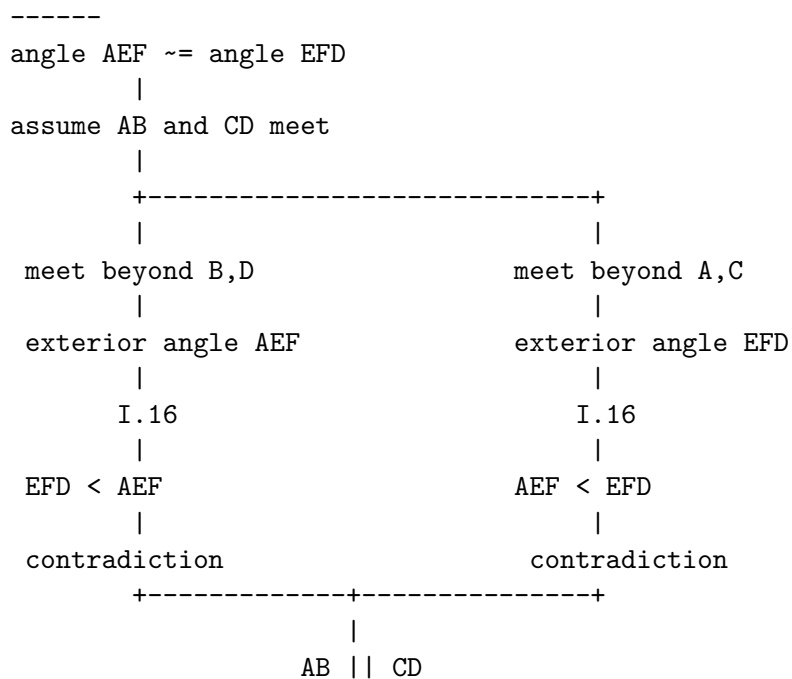
the second is transported from FD to CD . Thus

$$AB \parallel CD.$$

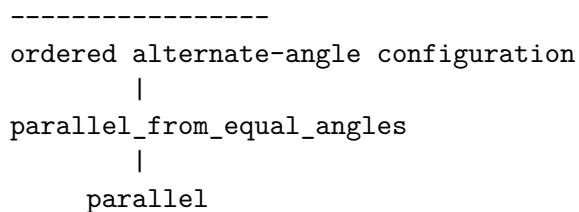
27.8 Proof Architecture

The classical and formal routes may be displayed schematically as follows.

Euclid



Hilbert interface



Explicit transversal formalization

```

-----
A-E-B, C-F-D
+ transversal EF
+ opposite-side information
+ angle AEF ~= angle EFD
  |
choose M with E-M-F
  |
same-ray and angle normalization
  |
angle MEA ~= angle MFD
  |
hilbert_parallel_of_alternate_angles_oppositeSide_lines
  |
EA || FD
  |
transport along A-E-B and C-F-D
  |
AB || CD

```

27.9 Proof-Theoretic Status

The coexistence of the two Lean formulations is deliberate.

The short formulation makes the mathematical core of I.27 unusually visible: the proposition coincides closely with an already established general theorem of the Hilbert layer.

The explicit transversal formulation answers a different question: what formal information is required to pass from the classical diagram to that general theorem? Its additional work concerns incidence, betweenness, same rays, side-of-line data, orientation of angles, and transport of parallelism.

At present we do not force a final interpretation of this distinction. It may eventually be preferable to regard the short theorem as the canonical Book I interface and the transversal theorem as a normalization result. It may instead prove useful to retain the explicit configuration as part of the public Euclidean layer.

Keeping both versions preserves this question for later comparison with the proof corpus and with the dependency graph of the Hilbert reconstruction.

27.10 Relationship with Proposition I.16

Euclid's classical proof depends directly on Proposition I.16. Each possible intersection converts one of the supposedly equal alternate angles into an exterior angle of a triangle, and I.16 makes that angle strictly greater than the remote interior angle.

The present Lean theorem does not invoke `euclid_proposition_16` directly at the Book I level because the corresponding alternate-angle theorem has already been established inside the Hilbert theory.

Thus the difference is one of proof organization:

Euclid : I.16 $\longrightarrow$ I.27,

Hilbert reconstruction : order and angle theory $\longrightarrow$ alternate-angle parallelism $\longrightarrow$ I.27.

The Book I order and the formal dependency order therefore need not coincide.

27.11 Neutral Geometry and the Fifth Postulate

Proposition I.27 is a useful logical boundary marker.

It establishes

equal alternate interior angles $\implies$ parallel lines

without using the Fifth Postulate. This direction is valid in neutral geometry.

The converse statement,

parallel lines $\implies$ equal alternate interior angles,

is part of Proposition I.29 and requires Euclidean parallelism.

Consequently I.27 should not be classified as a theorem depending on Hilbert’s Euclidean parallel axiom merely because its conclusion contains the word “parallel”. Its proof uses the definition of parallelism as nonintersection, together with the neutral theory developed before it.

27.12 Lean Formalization

The complete implementation is contained in `Proposition27.lean`. The public file currently records both the interface-level theorem and the explicit transversal theorem.

The latter proof has the following principal stages:

1. choose M with $E - M - F$;
2. obtain the two same-ray relations along the transversal;
3. normalize the given angle congruence;
4. apply the Hilbert alternate-angle parallelism theorem;
5. transport the resulting local parallelism along the collinear triples $A - E - B$ and $C - F - D$.

No Euclidean parallel axiom is used.

This diagrammatic audit did not rerun `lake build` or a fresh `#print axioms` check. The current production source was treated as the authority for theorem statements, dependencies, and proof structure.

27.13 Conclusion

Proposition I.27 is formally small at the level of the developed Hilbert interface but conceptually revealing.

Euclid proves parallelism by excluding the two possible intersection directions through Proposition I.16. The Hilbert development has already abstracted this reasoning into a reusable alternate-angle theorem, so the core Book I formulation becomes a thin wrapper.

The explicit transversal theorem shows what that compression hides: the classical diagram silently carries order, orientation, side-of-line, and ray information that must be made explicit before the abstract theorem can be applied.

For the moment both formalizations are retained. Their coexistence is not duplication to be removed immediately, but a useful record of the difference between the mathematical content of I.27 and the formal normalization of its classical diagram.

Chapter 28

Proposition I.28

28.1 Introduction

Proposition I.28 gives two sufficient angle conditions for parallelism. A transversal meets two straight lines. Parallelism follows either from an equality of corresponding angles or from the condition that the two interior angles on the same side are together equal to two right angles.

The central structural fact is that neither branch introduces a new parallelism mechanism. Both normalize their angle data to the alternate interior angle hypothesis of Proposition I.27:

$$\text{angle data} \longrightarrow \angle AGH \cong \angle GHD \longrightarrow \text{I.27} \longrightarrow AB \parallel CD.$$

This direction is neutral. The current production Lean file requires only incidence and congruence, not `HilbertEuclideanPlane`. The logical reversal occurs in Proposition I.29.

28.2 Classical Configuration

Let the straight line EF meet AB at G and CD at H . The common geometric configuration is

$$A - G - B, \quad C - H - D, \quad E - G - H - F.$$

The two clauses of the proposition differ only in the angle condition placed on this same configuration.

28.2.1 Corresponding-angle criterion

Assume

$$\angle EGB \cong \angle GHD.$$

Since $A - G - B$ and $E - G - H$, the angles $\angle EGB$ and $\angle AGH$ are vertical. Proposition I.15 gives

$$\angle AGH \cong \angle EGB.$$

By transitivity,

$$\angle AGH \cong \angle GHD,$$

and Proposition I.27 yields

$$AB \parallel CD.$$

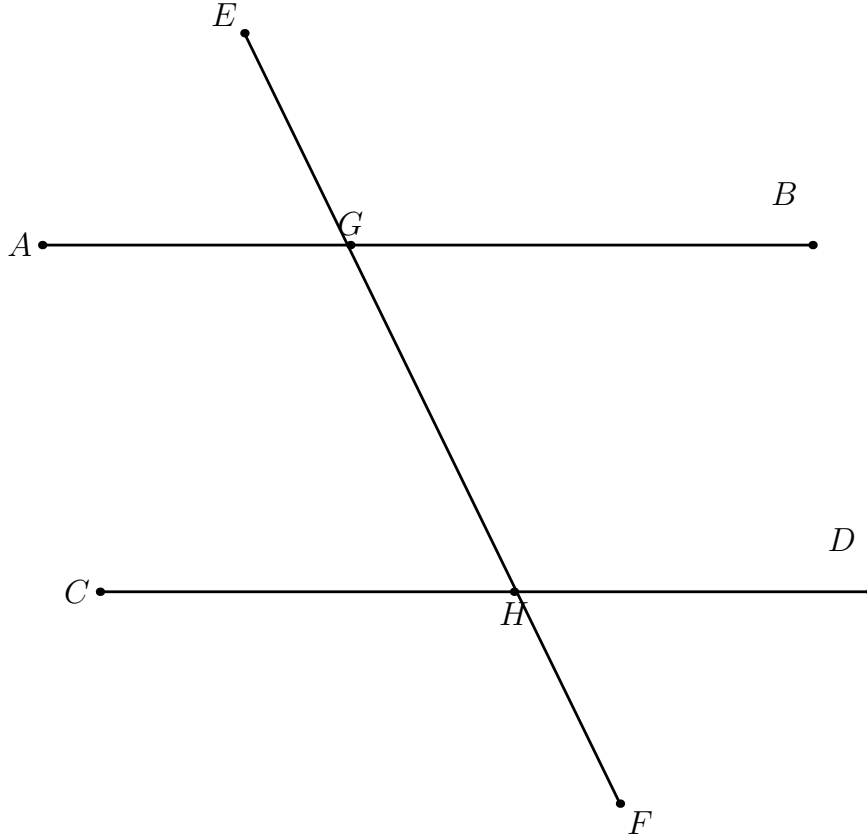


Figure 28.1: The common transversal configuration for Proposition I.28. The selected arms through A and D lie on opposite sides of the transversal in the explicit formal statement. No angle equality and no final parallelism conclusion is encoded in the drawing; the two clauses of I.28 are different hypotheses on this same geometric state.

Thus the classical dependency chain of the first clause is

$$\text{corresponding angles} \longrightarrow \text{I.15} \longrightarrow \text{alternate angles} \longrightarrow \text{I.27}.$$

28.2.2 Same-side interior-angle criterion

Assume instead that

$$\angle BGH \quad \text{and} \quad \angle GHD$$

are together equal to two right angles. Because $A - G - B$, the angles $\angle AGH$ and $\angle BGH$ are supplementary. Euclid uses Proposition I.13 and Common Notion 3 to cancel the common angle and obtain

$$\angle AGH \cong \angle GHD.$$

Again I.27 gives $AB \parallel CD$. The classical dependency chain is therefore

$$\text{I.13} + \text{C.N.3} \longrightarrow \text{alternate angles} \longrightarrow \text{I.27}.$$

28.3 Current Lean Architecture

The production module imports only `Proposition27` and exposes two public theorem entrypoints:

euclid_proposition_28_corresponding
euclid_proposition_28_same_side

There is no proposition-local helper theorem and no geometric case split. The two theorem entrypoints formalize the two alternative clauses of the classical proposition over the same transversal geometry.

Both statements use the explicit line `trans` through G, H and include the orientation condition

`HilbertOppositeSide Geo A D trans`

needed by the explicit transversal form of Proposition I.27.

28.3.1 Corresponding-angle branch

The first theorem receives

$$A - G - B, \quad C - H - D, \quad E - G - H, \quad \angle EGB \cong \angle GHD.$$

The current proof first reverses the transversal betweenness to obtain $H - G - E$. It then invokes

`VerticalAngles`

to prove a congruence equivalent to

$$\angle AGH \cong \angle EGB.$$

After endpoint reversal on the second angle and transitivity with the hypothesis, the exact alternate-angle input is obtained:

$$\angle AGH \cong \angle GHD.$$

The branch ends by calling

`euclid_proposition_27_transversal`

with the same intersection points G, H , the same transversal, and the same opposite-side orientation.

Schematically,

$$\begin{array}{c} \angle EGB \cong \angle GHD \\ \downarrow \text{VerticalAngles} \\ \angle AGH \cong \angle EGB \\ \downarrow \text{transitivity} \\ \angle AGH \cong \angle GHD \\ \downarrow \text{euclid_proposition_27_transversal} \\ AB \parallel CD. \end{array}$$

28.3.2 Same-side branch

The second theorem does not introduce numerical angle measure or an angle addition operation. Its hypothesis is the synthetic predicate

HilbertAnglesEqualTwoRightAnglesWithSupplement
 Geo G H D H G B A

which supplies directly

$$B - G - A \quad \text{and} \quad \angle GHD \cong \angle HGA.$$

Betweenness symmetry recovers $A - G - B$. Symmetry and endpoint reversal of angle congruence then normalize the second component to

$$\angle AGH \cong \angle GHD.$$

The proof again terminates in `euclid_proposition_27_transversal`.

Hence the actual formal branch is

$$\begin{array}{c} \text{HilbertAnglesEqualTwoRightAnglesWithSupplement} \\ \downarrow \\ B - G - A, \quad \angle GHD \cong \angle HGA \\ \downarrow \text{ symmetry/reversal } \\ \angle AGH \cong \angle GHD \\ \downarrow \text{ euclid_proposition_27_transversal } \\ AB \parallel CD. \end{array}$$

The classical use of I.13 and Common Notion 3 is therefore represented in Lean by the shape of the synthetic supplementary-angle hypothesis rather than replayed as angle arithmetic inside this theorem.

28.4 Downstream Geometry in Proposition I.27

Both I.28 branches delegate their final geometric work to the explicit transversal theorem of Proposition I.27. That theorem chooses an interior point between its two transversal intersections, obtains two same-ray relations, normalizes the alternate angles, applies the neutral line-level alternate-angle criterion, and transports the resulting local parallelism back to the original carriers.

After substituting the I.28 intersection points, this hidden point lies between G and H . It is not constructed by either I.28 theorem itself. Consequently it belongs diagrammatically to the audit of I.27 and is not repeated as a new I.28 figure.

This distinction is useful: the direct architecture of I.28 is angle normalization followed by a call to I.27; the deeper geometry of the parallelism endpoint is a dependency, not a new state introduced by I.28.

28.5 Neutral Geometry

Both theorem statements require

[HilbertIncidence Geo]
 [HilbertCongruence Geo]

and neither requires `HilbertEuclideanPlane`. Thus I.28 stays on the neutral side of the boundary:

angle condition $\longrightarrow$ parallel lines.

I.29 reverses this implication and is where the Euclidean parallel axiom enters the Book I sequence.

28.6 Diagrammatic Audit

The preserved Book1R chapter contained three separate floats:

1. the common transversal configuration;
2. the same configuration with arcs marking the corresponding-angle branch;
3. the same configuration with arcs marking the same-side branch.

Only the first records a geometric state. Figures 28.2 and 28.3 changed only which angles of the already drawn configuration were being discussed. They introduced no point, extension, intersection, betweenness relation, side relation, auxiliary line, or geometric case.

The audited chapter therefore keeps one figure only. The two angle criteria remain fully documented in the prose and proof DAGs, but they do not receive separate diagrams. This is an intentional application of the principle that a figure should record geometry rather than tactic state or a change of logical interpretation.

The formal orientation condition A, D on opposite sides of the transversal is stated in the caption and in the Lean architecture section. It does not require a second picture because it describes the same visible configuration.

There is also no geometric case split in the production file. The two clauses are two public theorem entrypoints with different hypotheses, not two positional cases discovered inside one proof.

28.7 Conclusion

Proposition I.28 is a pair of neutral normalization theorems over one common transversal geometry. The corresponding-angle clause uses vertical angles; the same-side clause receives the supplementary relation directly in synthetic form. Both produce the same alternate-angle congruence and then delegate parallelism to Proposition I.27.

Diagrammatically, one common figure is therefore sufficient. The reduction from three historical floats to one audited figure removes duplicate pictures without suppressing any genuine geometric state.

Chapter 29

Proposition I.29

29.1 Introduction

Proposition I.29 is the first proposition in Book I whose proof genuinely uses Euclidean parallelism. Propositions I.27 and I.28 establish neutral implications of the form

angle conditions $\longrightarrow$ parallel lines,

whereas I.29 reverses the direction:

parallel lines $\longrightarrow$ angle relations.

For a transversal meeting two parallel lines, Euclid records three conclusions: alternate interior angles are equal, the exterior angle equals the interior and opposite angle, and the interior angles on the same side are together equal to two right angles.

The current Lean development makes a useful distinction. The genuinely Euclidean content is concentrated in the first, alternate-angle conclusion. The corresponding-angle and same-side conclusions are then neutral consequences of that result.

29.2 Classical Configuration

Let the transversal EF meet the parallel lines AB and CD at G and H , respectively:

$$A - G - B, \quad C - H - D, \quad E - G - H - F,$$

with

$$AB \parallel CD.$$

29.3 Euclid's Proof Architecture

Euclid's proof is not the same as the Hilbert reconstruction used by the project. In the classical argument the Fifth Postulate is used through the same-side interior-angle condition. Together with Proposition I.13 it rules out an inequality between the relevant alternate angles: if one were larger, the resulting pair of same-side interior angles would be less than two right angles, forcing the two supposedly parallel lines to meet on that side. Thus the alternate angles are equal.

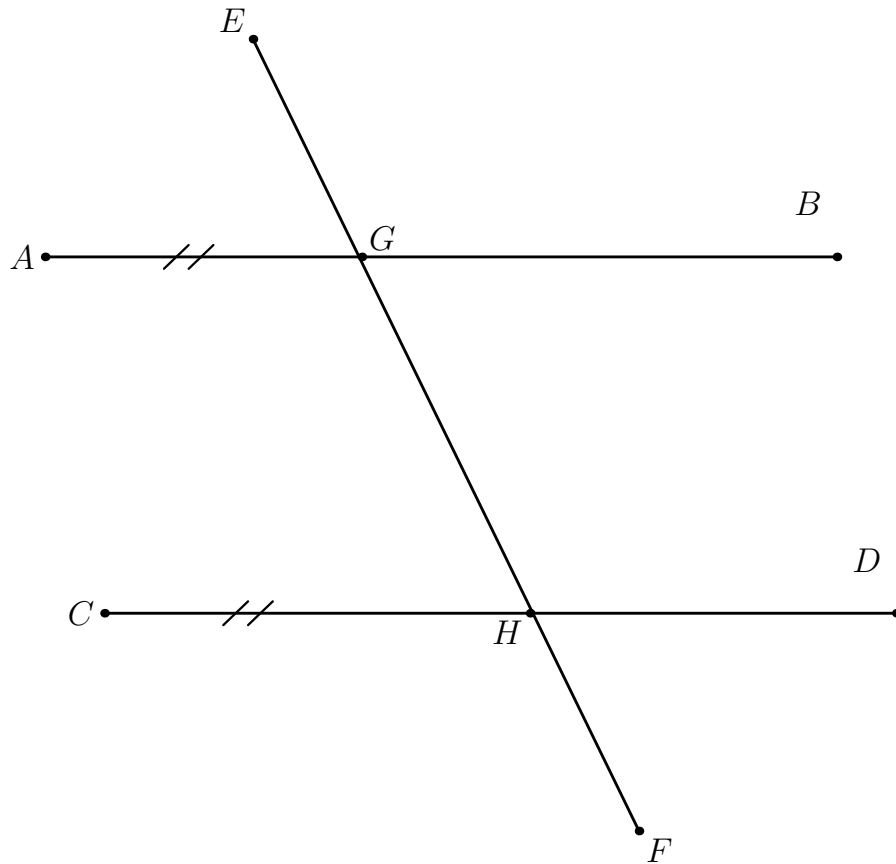


Figure 29.1: The classical transversal configuration for Proposition I.29. The lines AB and CD are given parallel; EF is the transversal. This single configuration supports all three conclusions of the proposition.

Schematically, the classical core is

$$\begin{array}{c}
 AB \parallel CD \\
 \downarrow \\
 \text{assume the alternate angles are unequal} \\
 \downarrow \\
 \text{I.13 converts the inequality into a same-side sum } < 2R \\
 \downarrow \text{ Fifth Postulate} \\
 \text{the two lines would meet} \\
 \downarrow \\
 \perp \\
 \downarrow \\
 \angle AGH \cong \angle GHD.
 \end{array}$$

The remaining classical conclusions use only the same configuration. Vertical angles give the corresponding-angle statement, while the straight angle at G supplies the supplementary relation needed for the same-side interior angles.

29.4 Current Lean Architecture

The production file imports only

`CGJteamLab.HilbertInterface`

and contains five public theorems.

29.4.1 Interface-level wrapper

The theorem

`euclid_proposition_29`

is a thin wrapper over

`equal_angles_from_parallel`

for the interface-level six-point configuration. The reusable theorem in turn delegates to the Euclidean alternate-angle theorem in the Hilbert interface. Thus the construction and the use of Group IV are no longer implemented locally in `Proposition29.lean`.

29.4.2 Explicit transversal theorem

The theorem

`euclid_proposition_29_transversal`

uses the explicit configuration

$$A - G - B, \quad C - H - D,$$

an incidence line `trans` through G and H , the orientation hypothesis

$$\text{OppositeSide}(A, D; \text{trans}),$$

and the given parallelism

$$AB \parallel CD.$$

It proves

$$\angle AGH \cong \angle GHD.$$

The proof first transports the global parallel relation to the local angle arms:

$$AB \parallel CD \quad \longrightarrow \quad GA \parallel HD.$$

This transport changes representation but not geometry.

The first genuinely new formal object is an interior point M chosen with

$$G - M - H.$$

The line-level Hilbert theorem then yields a congruence with M as the transversal-arm endpoint. Same-ray transport at G and H converts that raw result into the requested angles $\angle AGH$ and $\angle GHD$.

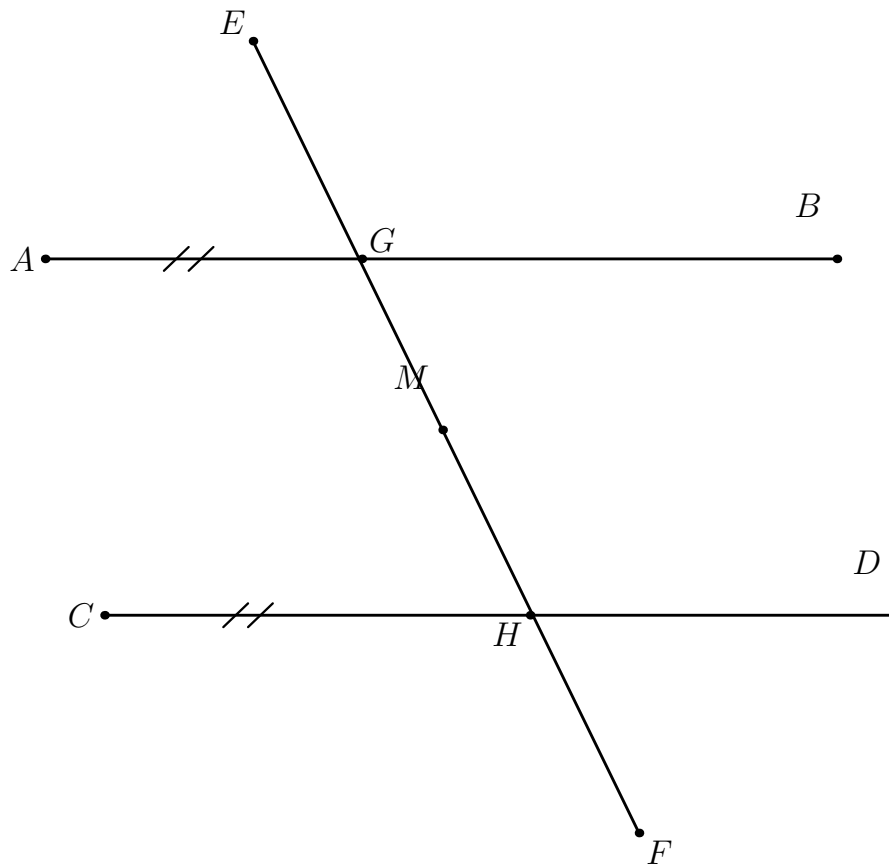


Figure 29.2: The formal transversal state used by `euclid_proposition_29_transversal`. The hidden witness M satisfies $G - M - H$. The selected points A and D are required to lie on opposite sides of the transversal. The point M provides the two same-ray normalizations used to pass from the line-level alternate-angle theorem to the endpoint formulation of I.29.

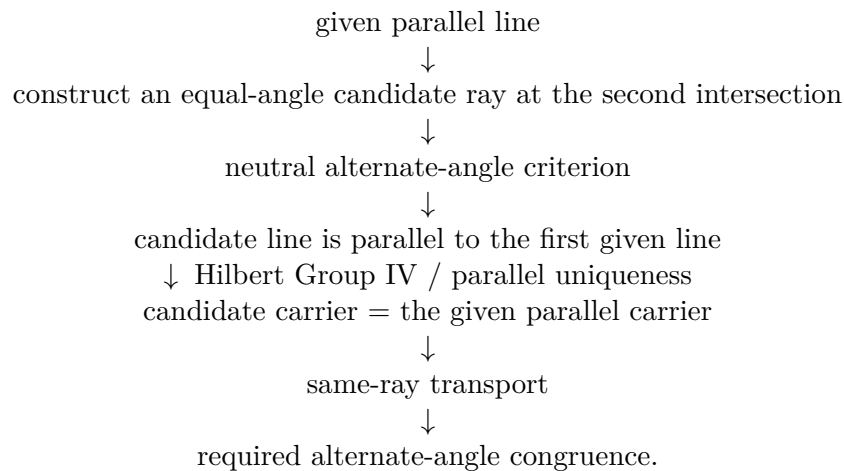
29.5 Where Group IV Enters

The Euclidean step is inside the reusable theorem

```
hilbert_alternate_angles_of_parallel_oppositeSide_lines
```

not after that theorem has returned.

Its internal structure is:



This existence–uniqueness decomposition is important. Neutral geometry can construct a line through the second intersection with the desired angle and can prove it parallel by the I.27 direction. The Euclidean axiom is needed to identify that constructed parallel with the already given parallel line.

The lower-level theorem constructs a candidate endpoint on the required ray. The final proof establishes that this endpoint and the already supplied endpoint determine the same ray. It does not need to prove that the two points are distinct. Therefore no separate diagram is drawn with two visibly distinct candidate endpoints: such a picture would assert more geometry than the theorem provides.

29.6 The Two Neutral Corollary Branches

29.6.1 Corresponding angles

The theorem

`euclid_proposition_29_corresponding`

first calls `euclid_proposition_29_transversal`. It then uses the order data $A - G - B$ and $E - G - H$ to obtain the vertical-angle congruence

$$\angle EGB \cong \angle AGH.$$

Transitivity with the alternate-angle result gives

$$\angle EGB \cong \angle GHD.$$

No new point, line, intersection, or side relation is introduced here.

29.6.2 Same-side interior angles

The theorem

`euclid_proposition_29_same_side`

again reuses the alternate-angle theorem. The order relation $A-G-B$ provides the supplementary ray to GB , and the result is packaged in the synthetic predicate

`HilbertAnglesEqualTwoRightAnglesWithSupplement`

rather than in a numerical angle-sum language.

Finally,

`euclid_proposition_29_all`

packages the alternate, corresponding, and same-side conclusions. It adds no new geometry.

29.7 Formal Dependency Graph

At the level of the current production file the dependency structure is

$$\begin{array}{c} \text{euclid_proposition_29} \\ \downarrow \\ \text{equal_angles_from_parallel} \\ \downarrow \\ \text{Hilbert Euclidean alternate-angle theorem} \end{array}$$

and, for the explicit transversal API,

$$\begin{array}{c} AB \parallel CD \\ \downarrow \\ GA \parallel HD \\ \downarrow \\ G - M - H \\ \downarrow \\ \text{hilbert_alternate_angles_of_parallel_oppositeSide_lines} \\ \downarrow \\ \angle MGA \cong \angle MHD \\ \downarrow \text{ same-ray normalization} \\ \angle AGH \cong \angle GHD \end{array}$$

followed by two neutral branches:

$$\begin{array}{ccc} & \angle AGH \cong \angle GHD & \\ \swarrow & & \searrow \\ \text{vertical angles} & & \text{supplementary configuration} \\ \downarrow & & \downarrow \\ \angle EGB \cong \angle GHD & & \text{same-side interior angles equal } 2R \text{ synthetically.} \end{array}$$

There is no geometric case split in `Proposition29.lean`.

29.8 Diagrammatic Audit

The preserved Book1R chapter used three figures. All three showed the same basic geometry. The first recorded the transversal configuration, while the second and third only changed which angle arcs were highlighted.

Under the present audit rule, those are not three geometric states. The final documentation therefore uses only two figures:

1. the classical transversal configuration, shared by all three of Euclid’s conclusions;
2. the formal transversal-normalization state with the hidden witness M and the explicit opposite-side orientation required by the Lean API.

The old separate alternate-angle and corresponding-angle pictures are not retained as separate floats because no new point, construction, intersection, or order/side configuration appears between them. They record a change of logical interpretation, not a change of geometry.

The deeper candidate-ray witness used inside the reusable Hilbert theorem is also not drawn separately. The proof ultimately establishes only that its ray agrees with the given target ray; drawing two distinct candidate points would add an unproved inequality.

29.9 Logical Position of Proposition I.29

The block I.27–I.29 has a clean logical boundary:

- I.27 : equal alternate angles $\longrightarrow$ parallel lines,
- I.28 : other angle conditions $\longrightarrow$ parallel lines,
- I.29 : parallel lines $\longrightarrow$ angle relations.

The first two directions are neutral. The third requires

[HilbertEuclideanPlane Geo]

and hence the project-level form of Hilbert’s Group IV.

This explains why I.29 becomes a central transport theorem for the next part of Book I: it feeds I.30, supplies the uniqueness side of the discussion around I.31, is used in I.32, and provides the parallel-to-angle interface used in later parallelogram propositions.

29.10 Conclusion

Proposition I.29 is the first genuinely Euclidean parallel-to-angle theorem in the Book I reconstruction. Its formal core is the alternate-angle implication. The current reusable Hilbert proof exposes the exact role of Group IV: neutral geometry constructs a candidate parallel direction, while parallel uniqueness identifies that direction with the line already present in the hypotheses. The explicit Book I theorem then uses the hidden point M only to normalize the line-level result along the transversal.

The corresponding-angle and same-side conclusions introduce no additional geometry. For that reason the audited chapter needs two figures rather than three: one for Euclid's common transversal configuration and one for the formal hidden geometry actually present in the Lean proof.

Chapter 30

Proposition I.30

30.1 Introduction

Proposition I.30 states a structural property of Euclidean parallelism. If AB and CD are both parallel to the same straight line EF , then AB and CD are parallel to one another.

In Euclid's proof the result is obtained indirectly through angle relations: a transversal is introduced, Proposition I.29 is applied twice, angle congruence is transported by transitivity, and Proposition I.27 closes the argument. The current Hilbert reconstruction exposes a different structure. It proves the same statement directly from uniqueness of a parallel through a point, that is, from the Euclidean parallel layer represented by Hilbert's Group IV.

A formal subtlety must be stated explicitly. In the project, `Geo.Parallel` is strict: the two point-lines are nondegenerate, distinct, and disjoint. Consequently the formal theorem contains the additional hypothesis

$$\text{PointLine}(A, B) \neq \text{PointLine}(C, D).$$

In the classical picture this distinctness is implicit in the fact that the three straight lines are drawn as separate lines.

30.2 Formal Statement and Public Architecture

The current production theorem is

```
import CGJteamLab.HilbertInterface

namespace Geometry

universe u

variable {Geo : Geometry.Geo.{u}}

theorem euclid_proposition_30
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D E F : Geo.Point)
  (hAB_EF : Geo.Parallel A B E F)
  (hCD_EF : Geo.Parallel C D E F)
```



```

(hDistinct :
  Geo.PointLine A B != Geo.PointLine C D) :
Geo.Parallel A B C D := by

exact
  hilbert_parallel_transitive_distinct
    Geo
    A B C D E F
    hAB_EF
    hCD_EF
    hDistinct
end Geometry

```

Thus `euclid_proposition_30` is a thin wrapper. It contains no proposition-local helper theorem, no proposition-local geometric construction, and no case split. Its direct dependency is

$$\text{euclid_proposition_30} \longrightarrow \text{hilbert_parallel_transitive_distinct}.$$

The public typeclasses are

```

[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]

```

so, unlike I.31, Proposition I.30 lives explicitly in the Euclidean parallel layer.

30.3 Classical Configuration

Let AB , CD , and EF be three distinct straight lines with

$$AB \parallel EF, \quad CD \parallel EF.$$

Euclid introduces one transversal meeting the three lines at G , H , and K . This is the first genuinely new geometric state in the classical proof.

The two applications of I.29 give the chain

$$\angle AGK \cong \angle GHF, \quad \angle GHF \cong \angle GKD.$$

By transitivity,

$$\angle AGK \cong \angle GKD.$$

These are alternate interior angles for the first and third lines, so I.27 gives

$$AB \parallel CD.$$

The classical dependency graph is therefore

$$\begin{array}{ccc}
 AB \parallel EF & & CD \parallel EF \\
 \downarrow \text{I.29} & & \downarrow \text{I.29} \\
 \angle AGK \cong \angle GHF & & \angle GHF \cong \angle GKD \\
 & \searrow \quad \downarrow \quad \swarrow & \\
 & \angle AGK \cong \angle GKD & \\
 & \downarrow \text{I.27} & \\
 & AB \parallel CD. &
 \end{array}$$

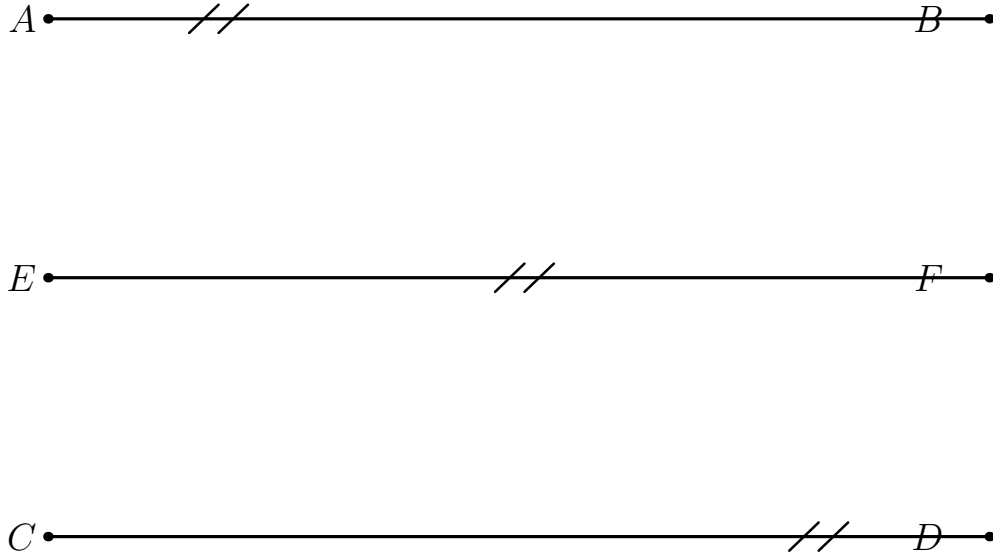


Figure 30.1: The classical configuration for Proposition I.30. The two given parallel relations are encoded separately: $AB \parallel EF$ and $CD \parallel EF$. Euclid introduces the dashed transversal GHK . The diagram deliberately does not mark $AB \parallel CD$, since that is the conclusion of the proposition.

Thus the literal Euclidean route is angle-theoretic even though the theorem's structural content is a transitivity property of parallelism.

30.4 Hilbert Reconstruction

The reusable theorem is

```

theorem hilbert_parallel_transitive_distinct
  [HilbertEuclideanPlane Geo]
  (A B C D E F : Geo.Point)
  (hAB_EF : Geo.Parallel A B E F)
  (hCD_EF : Geo.Parallel C D E F)
  (hDistinct :
    Geo.PointLine A B != Geo.PointLine C D) :
  Geo.Parallel A B C D

```

The proof first chooses incidence carriers

$$\ell_{AB}, \quad \ell_{CD}, \quad \ell_{EF}$$

through the three endpoint pairs. The two parallel hypotheses are converted into disjointness statements

$$\ell_{AB} \cap \ell_{EF} = \emptyset, \quad \ell_{CD} \cap \ell_{EF} = \emptyset.$$

To prove $AB \parallel CD$, the proof unfolds the disjointness of the first and third project point-lines. It assumes a hypothetical common point P :

$$P \in \text{PointLine}(A, B), \quad P \in \text{PointLine}(C, D).$$

Membership is transported to incidence on the selected Hilbert lines, so $P \in \ell_{AB}$ and $P \in \ell_{CD}$.

Since ℓ_{AB} is disjoint from ℓ_{EF} , one has

$$P \notin \ell_{EF}.$$

Now both ℓ_{AB} and ℓ_{CD} pass through the same point P outside ℓ_{EF} , and each is disjoint from ℓ_{EF} . The call

`HilbertEuclideanPlane.parallel_unique`

therefore gives

$$\ell_{AB} = \ell_{CD}.$$

The equality of Hilbert incidence carriers is then converted back to the project-level extensional carriers by

`hilbert_pointLine_eq_of_points_on_line,`

yielding

$$\text{PointLine}(A, B) = \text{PointLine}(C, D),$$

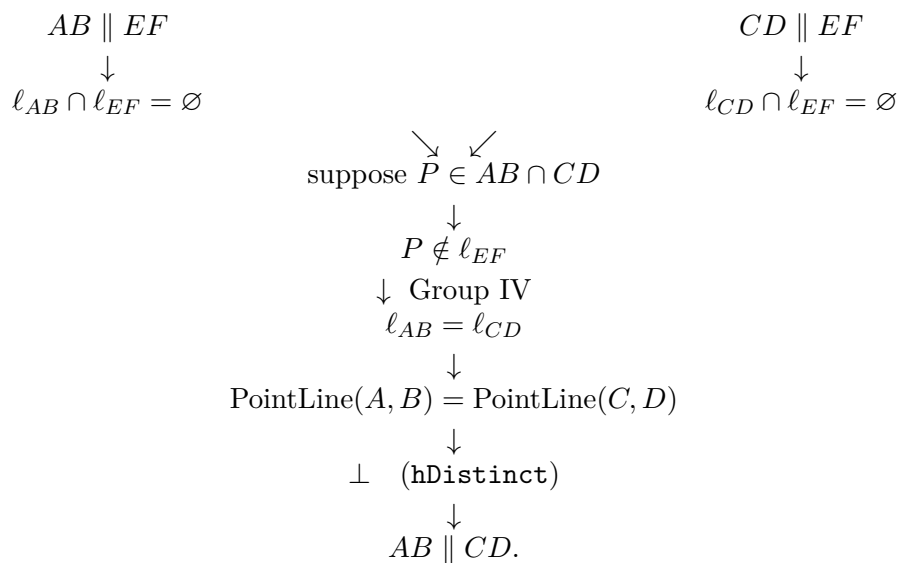
contrary to the explicit hypothesis `hDistinct`.

Hence the assumed common point cannot exist and the two project point-lines are disjoint. This is exactly the desired strict parallelism.

There is no geometric case split. The point P appears only under the contradictory assumption that the first and third point-lines meet.

30.5 Actual Lean Dependency Graph

The current formal architecture can be summarized as



The formal proof therefore does not call I.29 or I.27. Those propositions are part of Euclid's historical route, whereas the reusable Hilbert theorem exposes the uniqueness principle directly.

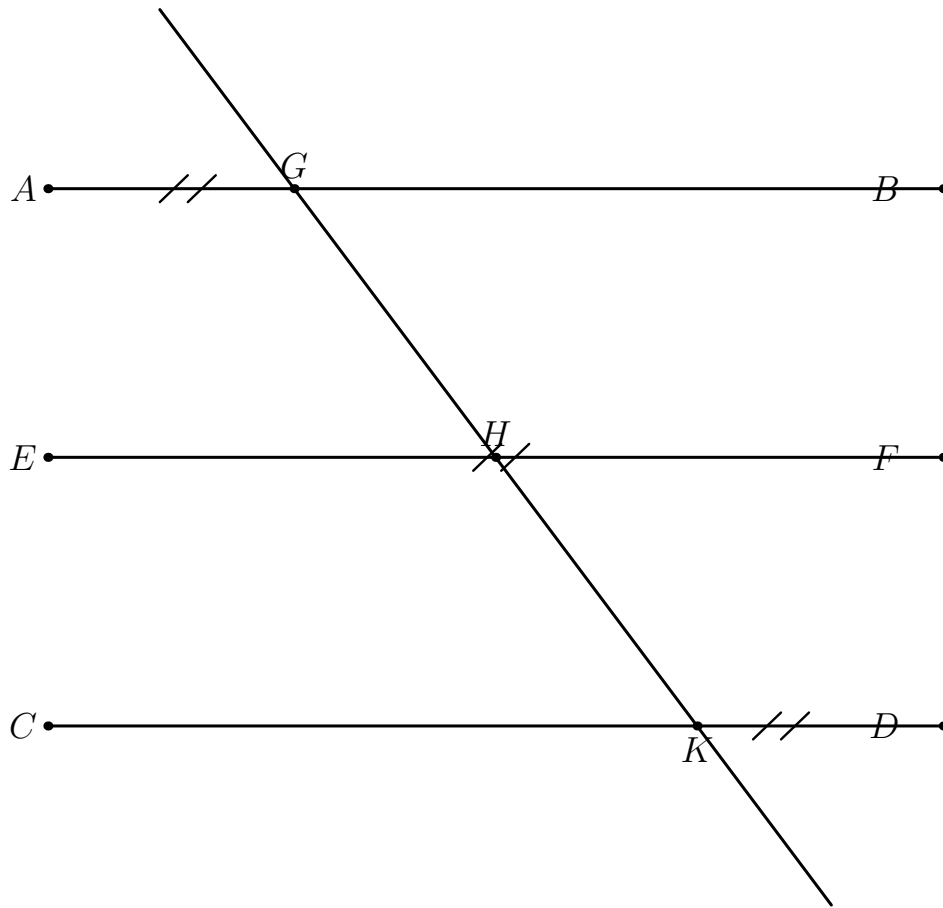


Figure 30.2: The hidden contradiction state in the current Hilbert proof. One supposes that the two distinct outer point-lines have a common point P , while the hypotheses still assert $AB \parallel EF$ and $CD \parallel EF$. This cannot be realized as a genuine Euclidean configuration; the diagram records only the counterfactual intersection witness. Group IV identifies the two lines through P that are disjoint from EF , contradicting **hDistinct**.

30.6 Diagrammatic Audit

The preserved Book1R chapter contained two separate diagrammatic ideas, but they were not correctly matched to their captions.

The old initial-data figure showed only the three horizontal lines. As a standalone state it was redundant: the classical proof immediately introduces a transversal, and the richer transversal picture already contains the initial geometry. Moreover, using one identical parallel mark on all three lines visually suggested the conclusion $AB \parallel CD$ before it had been proved.

The preserved Figure 30.2 had a more serious mismatch. Its caption described the Hilbert contradiction argument with a common point P , while the picture actually showed the classical transversal through G, H, K . The caption and the geometry therefore belonged to different proof architectures.

The audited chapter uses exactly two figures:

1. a classical Euclid figure with the transversal GHK , with the two input parallel relations marked separately and without marking the conclusion;
2. a formal counterfactual figure showing the hidden intersection witness P used in the proof

by contradiction.

No separate final figure is added for $AB \parallel CD$. After the contradiction no new point, line, intersection, side relation, or construction is introduced; only the logical status of the existing two point-lines changes. A figure should record geometry, not tactic state.

30.7 Logical Position of Proposition I.30

The sequence I.27–I.31 makes the logical boundary in the parallel theory particularly clear:

- I.27 : equal alternate angles $\implies$ parallel,
- I.28 : other angle conditions $\implies$ parallel,
- I.29 : parallel $\implies$ angle relations,
- I.30 : parallel to the same line $\implies$ parallel to one another,
- I.31 : constructive existence of a parallel.

The first two implications are neutral. I.29 crosses into the Euclidean layer because the reverse angle transport requires uniqueness of parallels. I.30 is another direct manifestation of that same principle. Its current formal proof uses `parallel_unique` exactly at the point where a hypothetical intersection would create two different parallels through one point to the same given line.

In this sense the current reconstruction makes the structural content of I.30 more explicit than the classical angle proof: transitivity for distinct parallels is a consequence of uniqueness.

30.8 Conclusion

Proposition I.30 has two genuinely different proof pictures. Euclid introduces a transversal and proves the result through I.29, angle transitivity, and I.27. The current Lean development instead assumes a common point of the two outer lines and uses Hilbert’s Euclidean parallel uniqueness to force the carriers to coincide, contradicting their explicit distinctness.

The audited documentation therefore keeps the two architectures separate. It uses one classical transversal figure and one formal contradiction figure, and it removes the former mismatch in which a Hilbert caption was attached to a classical diagram.

Chapter 31

Proposition I.31

31.1 Introduction

Proposition I.31 is a construction theorem: given a straight line and a point outside it, construct a straight line through the point parallel to the given line.

At the level of Euclid's sequence the proposition follows I.29 and I.30, but its logical status is different. The construction itself does not require a Euclidean uniqueness principle for parallels. It transfers a direction by copying an angle and then invokes the neutral alternate-angle criterion of I.27.

In the present Lean development the Book I theorem is correspondingly small. The substantial geometry is already isolated in the reusable interface theorem `hilbert_parallel_through_point_exists`. Proposition I.31 is a thin wrapper around that result.

31.2 Classical Configuration

Euclid's literal configuration uses different labels from the project-level statement. Let A be the given external point and let BC be the given straight line. Choose a point D on BC and join AD . At A construct, by I.23, an angle equal to $\angle ADC$, and extend the constructed ray to a straight line EAF .

Thus the classical dependency is

$$\text{I.23} \longrightarrow \text{equal alternate angles} \xrightarrow{\text{I.27}} EAF \parallel BC.$$

The construction uses no right angle. The copied angle may have any nondegenerate size; its numerical measure is irrelevant.

31.3 Why No Right Angle Is Needed

A familiar modern construction of a parallel can proceed through two perpendiculars, but this is not Euclid's proof of I.31. The relevant mathematical mechanism is simply direction transfer by angle congruence.

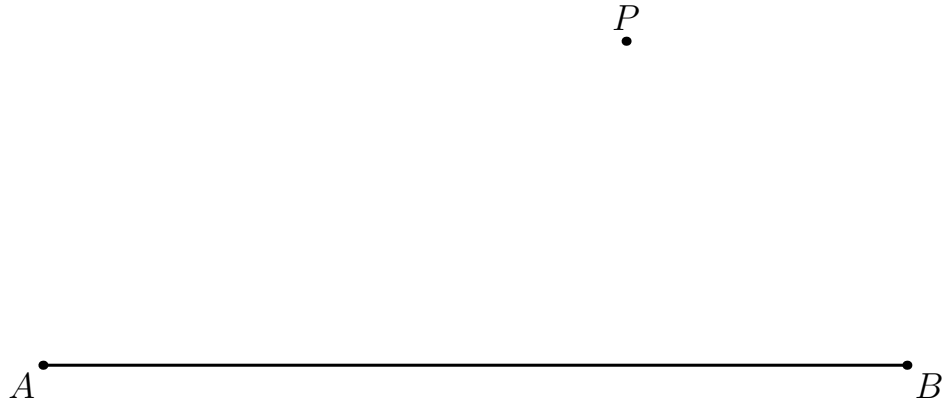


Figure 31.1: The classical configuration of Proposition I.31. The point D is chosen on the given line BC , the transversal AD is joined, and the line EAF through the external point A is constructed so that $\angle EAD \cong \angle ADC$. Proposition I.27 then identifies EAF as parallel to BC .

The point is also important for the logical organization of the Hilbert reconstruction. Existence of a parallel through an external point can be obtained from angle construction and the neutral converse alternate-angle theorem. Uniqueness of that parallel is additional Euclidean content.

Consequently the natural schematic distinction is

$$\boxed{\text{parallel existence in neutral geometry}} \\ \text{versus} \\ \boxed{\text{parallel uniqueness in the Euclidean layer}}.$$

31.4 The Hidden Formal Configuration

The project-level theorem uses the labels A, B, P for the given line and the external point. Its statement asks for a point Q such that

$$P \neq Q, \quad AB \parallel PQ.$$

The reusable proof contains more order and side information than this final statement displays.

First, the line AP is used as the transversal. An interior point E is chosen with

$$A - E - P.$$

The proof then introduces a second auxiliary point S by extending BA beyond A :

$$B - A - S.$$

Because A lies on the transversal AP , while B and S lie on opposite sides of it, the proof obtains

$$\text{OppositeSide}(B, S; AP).$$

This point S is a genuine hidden witness in the current Lean proof. It is not present in the public theorem statement, but it is the geometric device that selects the correct side on which the copied angle is constructed.

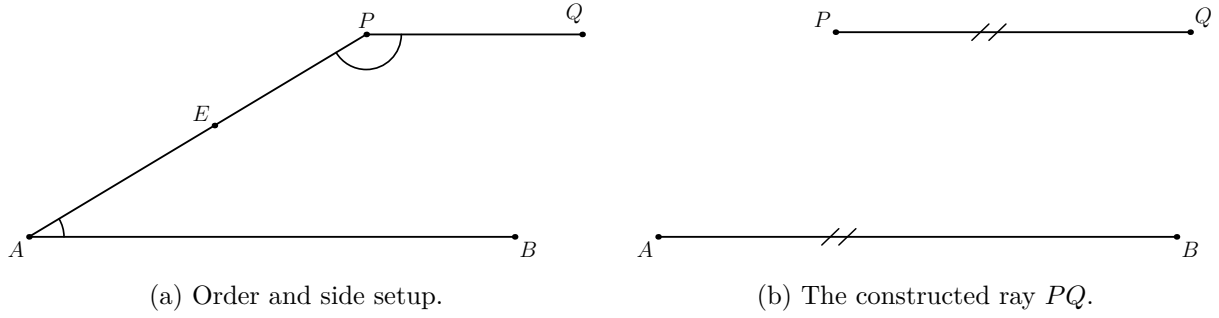


Figure 31.2: The formal geometry hidden behind the thin Book I wrapper. In panel (a), $A - E - P$ fixes the transversal rays and $B - A - S$ places the auxiliary point S on the side of AP opposite to B . In panel (b), angle construction produces Q on the same side of AP as S with $\angle EAB \cong \angle EPQ$. Hence B and Q are on opposite sides of AP , which is exactly the orientation needed by the alternate-angle parallel criterion.

31.5 Hilbert Reconstruction

The substantial theorem is

```

theorem hilbert_parallel_through_point_exists
  [HilbertCongruence Geo]
  (A B P : Geo.Point)
  (hAB : A != B)
  (hABP : not Collinear Geo A B P) :
  exists Q : Geo.Point,
    P != Q /\
    Geo.Parallel A B P Q

```

Its current proof has the following geometric architecture:

$$\begin{array}{c}
 AB, \quad P \notin AB \\
 \downarrow \\
 \text{construct the transversal } AP \\
 \downarrow \\
 A - E - P \\
 \downarrow \\
 B - A - S \\
 \downarrow \\
 \text{OppositeSide}(B, S; AP) \\
 \downarrow \\
 \text{angle construction at } P \\
 \downarrow \\
 \angle EAB \cong \angle EPQ, \quad \text{SameSide}(S, Q; AP) \\
 \downarrow \\
 \text{OppositeSide}(B, Q; AP) \\
 \downarrow \\
 \text{neutral alternate-angle criterion} \\
 \downarrow \\
 AB \parallel PQ.
 \end{array}$$

The side transport is therefore not decorative bookkeeping. It converts the side selected for the

temporary witness S into the orientation data for the actual constructed point Q .

There is no geometric case split in this proof. The construction proceeds through a single sequence of witnesses and side relations.

31.6 Lean Formalization

The current production file is

```
import CGJteamLab.HilbertInterface

namespace Geometry

universe u

variable {Geo : Geometry.Geo.{u}}

theorem euclid_proposition_31
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B P : Geo.Point)
  (hAB : A != B)
  (hABP : not Collinear Geo A B P) :
  exists Q : Geo.Point,
    P != Q /\
    Geo.Parallel A B P Q := by

exact
  hilbert_parallel_through_point_exists
    Geo
    A B P
    hAB
    hABP

end Geometry
```

Thus `euclid_proposition_31` is a genuine thin wrapper. It contains no proposition-local helper theorem and introduces no additional geometric witness beyond those already produced by the interface theorem.

The direct dependency of the Book I theorem is therefore

$$\text{euclid_proposition_31} \longrightarrow \text{hilbert_parallel_through_point_exists}.$$

The deeper construction uses incidence for the transversal, order for E and S , Hilbert angle construction for Q , side-of-line transport, and the neutral alternate-angle criterion.

The public typeclasses are

```
[HilbertIncidence Geo]
[HilbertCongruence Geo]
```

and no `HilbertEuclideanPlane` instance occurs in the proposition statement.

The current source of `Proposition31.lean` contains no `sorry` and no proposition-local `axiom`.

No fresh `lake build` or `#print axioms` audit was run as part of this diagrammatic review, so no stronger compilation or axiomatic-status claim is made here.

31.7 Neutral Existence Versus Euclidean Uniqueness

Proposition I.31 proves existence:

$$\exists PQ \text{ through } P, \quad PQ \parallel AB.$$

It does not prove uniqueness; that belongs to the Euclidean parallel layer. This also clarifies I.29 structurally: an I.31-type construction produces a candidate parallel neutrally, while a uniqueness principle can identify it with an already given parallel. This is not an extra step in Euclid’s proof of I.31.

31.8 Diagrammatic Audit

The preserved Book1R chapter used three separate figures: initial data, the copied-angle construction, and the final parallel pair. In the revised audit the first and third do not count as new geometric states: the initial picture adds no construction, while I.27 changes only the logical status of the already constructed line PQ .

The revised figure set instead records the two genuinely different layers:

1. the literal classical Euclid/Heath construction;
2. the formal Hilbert construction, split into the order/side state and the subsequent angle-construction state.

The hidden witness S is included because it is a real auxiliary point and encodes the side choice used by the current Lean proof. It was absent from the preserved diagrams even though the text discussed the side condition.

Accordingly the final count is two figures with three panels. No extra panel is used for $AB \parallel PQ$, because that conclusion introduces no new geometry.

31.9 Conclusion

Proposition I.31 transfers direction by constructing an equal alternate angle and invoking I.27. The Hilbert reconstruction exposes the order and side data suppressed by the classical picture, most notably the hidden witness S that selects the correct side of the transversal. The Book I theorem `euclid_proposition_31` is a thin wrapper around the reusable neutral existence theorem `hilbert_parallel_through_point_exists`; uniqueness of the parallel belongs to the later Euclidean structure.

Chapter 32

Proposition I.32

32.1 Introduction

Proposition I.32 is the first proposition in Book I that applies the Euclidean parallel theory developed in I.29–I.31 directly to a triangle.

Let ABC be a nondegenerate triangle and extend BC through C to D . Euclid asserts first that the exterior angle at C is equal to the two remote interior angles:

$$\angle ACD = \angle BAC + \angle ABC.$$

He then concludes that the three interior angles of the triangle are together equal to two right angles.

In the present reconstruction no numerical angle measure is introduced. Thus the familiar modern formula

$$\angle BAC + \angle ABC + \angle ACB = 180^\circ$$

is not the literal Lean statement. Instead the proposition is represented synthetically: a ray inside the exterior angle decomposes it into two angles congruent to the two remote interior angles, while $B - C - D$ records the straight-angle configuration at C .

This distinction is mathematically important. The formal proof establishes the synthetic content of the 180° theorem without introducing degrees or an arithmetic operation of angle addition.

32.2 Classical Statement

Let ABC be a triangle and let BC be produced to D . Proposition I.32 has two conclusions.

First,

$$\angle ACD = \angle BAC + \angle ABC.$$

Second,

$$\angle BAC + \angle ABC + \angle ACB = 2R,$$

where $2R$ denotes two right angles.

The classical Euclidean proof constructs through C a line parallel to AB . Proposition I.29 then identifies the two parts of the exterior angle with the two remote interior angles.

32.3 The Formal Configuration

The Lean proof refines the classical diagram because the formal argument must record the orientation of rays and the incidence data needed by the parallel-angle theorem.

The principal points satisfy

$$A - M - C, \quad B - M - E, \quad A - R - D, \quad B - C - D.$$

The auxiliary construction gives

$$AB \parallel CE.$$

Moreover E and R determine the same ray from C . Thus the ray CR lies inside the exterior angle ACD and decomposes it into the two angles used in the formal statement.

For the explicit application of Proposition I.29 the proof also extends the two parallel lines:

$$A - B - X, \quad E - C - F.$$

Hence

$$AX \parallel EF.$$

32.4 Synthetic Representation of the Exterior-Angle Equality

The first conclusion is represented by the predicate

`HilbertExteriorAngleEqualsRemoteAngles`

defined by

```
def HilbertExteriorAngleEqualsRemoteAngles
  [HilbertIncidence Geo]
  (A B C D : Geo.Point) : Prop :=
  exists R : Geo.Point,
    Geo.Between A R D /\
    Geo.AngleCongruent B A C A C R /\
    Geo.AngleCongruent A B C R C D
```

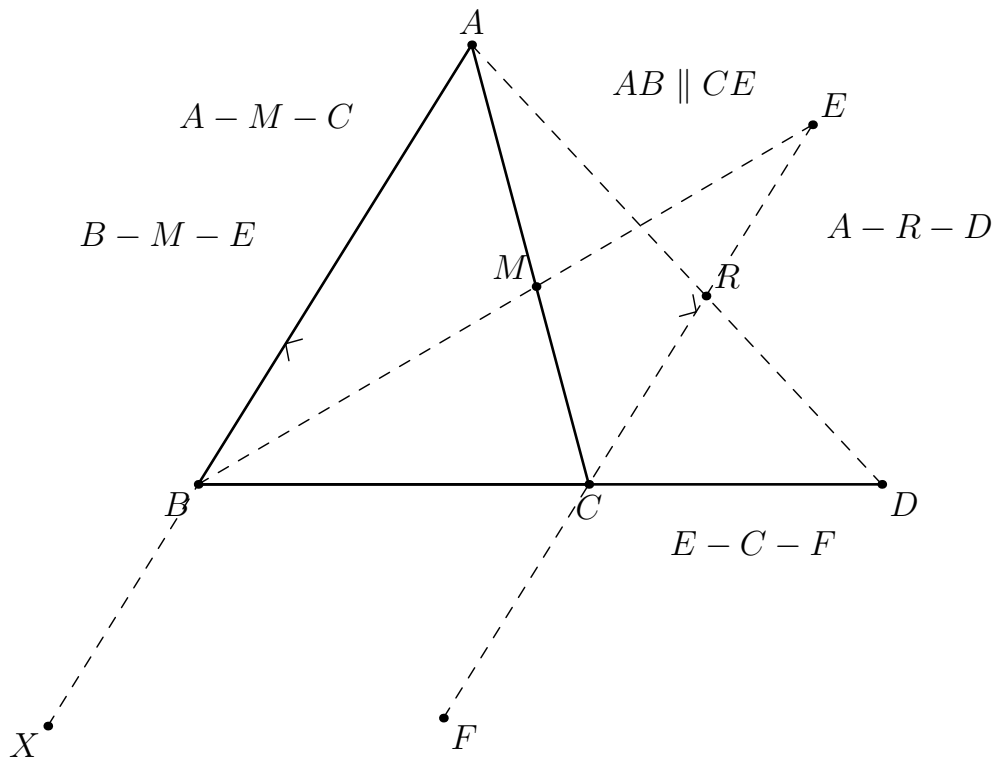


Figure 32.1: The formal configuration of Proposition I.32. The essential parallel relation is $AB \parallel CE$. The points X and F are technical extensions used to put the configuration into the explicit transversal form of Proposition I.29.

Thus the equation

$$\angle ACD = \angle BAC + \angle ABC$$

is encoded without angle addition. Instead there exists a point R with

$$A - R - D$$

such that

$$\angle BAC \cong \angle ACR$$

and

$$\angle ABC \cong \angle RCD.$$

The ray CR is therefore the synthetic “division point” of the exterior angle.

32.5 Construction of the Correctly Oriented Parallel Ray

A naive use of Proposition I.31 would produce a line through C parallel to AB , but a line alone does not determine which of its two rays lies inside the exterior angle ACD .

This orientation issue is genuine in the formal proof.

The reconstruction therefore uses the already established auxiliary configuration

`hilbert_exterior_angle_aux`

which supplies points M, E satisfying

$$A - M - C, \quad B - M - E,$$

together with

$$\angle BAC \cong \angle MCE.$$

The theorem

`hilbert_exterior_angle_aux_parallel`

then yields

$$AB \parallel CE.$$

The important point is that E is not an arbitrary point on the parallel through C . Its orientation is inherited from the auxiliary exterior-angle construction.

The helper

`hilbert_exterior_angle_inside`

uses this oriented configuration, together with $B - C - D$, to show that the ray CE meets the open segment AD . Hence there is a point R such that

$$A - R - D$$

and E, R lie on the same ray from C .

This is the formal crossbar step of the proof.

32.6 First Remote Interior Angle

From

$$A - M - C$$

the points M and A lie on the same ray from C . Therefore

$$\angle MCE = \angle ACE.$$

Since E and R lie on the same ray from C ,

$$\angle ACE = \angle ACR.$$

Combining these ray transports with the congruence supplied by the auxiliary construction gives

$$\boxed{\angle BAC \cong \angle ACR}.$$

In Lean this is the intermediate result

```
hFirst :
  Geo.AngleCongruent B A C A C R
```

and it establishes the first half of the exterior-angle decomposition.

32.7 Preparing Proposition I.29

The second congruence is obtained from the parallel relation, but the explicit theorem

```
euclid_proposition_29_transversal
```

expects the two intersections to lie between endpoints of the parallel lines.

The proof therefore extends AB beyond B :

$$A - B - X,$$

and extends EC beyond C :

$$E - C - F.$$

Parallelism is transported along the corresponding collinear triples to obtain

$$AX \parallel EF.$$

The transversal is the line BD , with intersection points B and C .

There remains an orientation condition. Since $A - R - D$, the points A and R lie on the same side of BD . From the same-ray relation between E and R , together with $E - C - F$, the proof obtains

$$R - C - F.$$

Thus R and F lie on opposite sides of BD . Transporting this relation through the same-side relation between A and R gives

$$A \text{ and } F \text{ on opposite sides of } BD.$$

At this point all hypotheses of the explicit transversal form of I.29 are available.

32.8 Second Remote Interior Angle

Proposition I.29 applied to

$$AX \parallel EF$$

with transversal BD gives

$$\angle ABC \cong \angle BCF.$$

At C the two straight lines are represented by

$$B - C - D \quad \text{and} \quad E - C - F.$$

Proposition I.15, through the theorem **VerticalAngles**, therefore gives

$$\angle BCF \cong \angle ECD.$$

By transitivity,

$$\angle ABC \cong \angle ECD.$$

Finally E and R lie on the same ray from C , so the first arm of the second angle may be transported from E to R :

$$\boxed{\angle ABC \cong \angle RCD}.$$

This is the Lean result

`hSecond :`

`Geo.AngleCongruent A B C R C D`

and completes the exterior-angle theorem.

32.9 Proof Architecture of the Exterior-Angle Part

The actual formal dependency structure is

$$\begin{array}{c}
 ABC \text{ noncollinear}, \quad B - C - D \\
 \downarrow \\
 \text{hilbert_exterior_angle_aux} \\
 \downarrow \\
 A - M - C, \quad B - M - E, \quad \angle BAC \cong \angle MCE \\
 \downarrow \\
 AB \parallel CE \\
 \downarrow \\
 \text{hilbert_exterior_angle_inside} \\
 \downarrow \\
 A - R - D, \quad \text{SameRay}(C, E, R)
 \end{array}$$

and then the proof splits:

$$\begin{array}{ccc}
 & A - R - D, \quad AB \parallel CE & \\
 \swarrow & & \searrow \\
 \text{same-ray transport} & & \text{extend the two parallels} \\
 \downarrow & & \downarrow \\
 \angle BAC \cong \angle ACR & & AX \parallel EF \\
 & & \downarrow \\
 & & \text{I.29 + vertical angles} \\
 & & \downarrow \\
 & & \angle ABC \cong \angle RCD.
 \end{array}$$

The two branches are then packaged as

$$\exists R, \quad A - R - D, \quad \angle BAC \cong \angle ACR, \quad \angle ABC \cong \angle RCD.$$

32.10 Where the Euclidean Parallel Axiom Enters

The construction and order-theoretic parts of the proof use the Hilbert incidence, order, congruence, side-of-line and crossbar machinery.

The specifically Euclidean step is the use of Proposition I.29:

$$AX \parallel EF \longrightarrow \angle ABC \cong \angle BCF.$$

As established in the reconstruction of I.29, this direction depends on Hilbert's Group IV, or equivalently on uniqueness of the parallel through a point.

Thus I.32 inherits its Euclidean character from I.29.

This is structurally different from I.27 and I.28, where angle conditions imply parallelism in neutral geometry. In I.32 the proof needs the converse direction:

parallelism $\longrightarrow$ angle congruence.

32.11 The Triangle-Angle-Sum Statement

The second formal predicate is

`HilbertTriangleAnglesEqualTwoRightAngles`

defined by

```
def HilbertTriangleAnglesEqualTwoRightAngles
  [HilbertIncidence Geo]
  (A B C : Geo.Point) : Prop :=
  exists D : Geo.Point,
    Geo.Between B C D /\
    HilbertExteriorAngleEqualsRemoteAngles
      A B C D
```

The theorem

`euclid_proposition_32`

is consequently short. From noncollinearity one obtains $B \neq C$, extends BC through C to a point D , and applies

`euclid_proposition_32_exterior`

to that extension.

The proof architecture is simply

$$\begin{array}{c}
 ABC \text{ noncollinear} \\
 \downarrow \\
 B \neq C \\
 \downarrow \\
 \text{extend } BC \text{ through } C \\
 \downarrow \\
 B - C - D \\
 \downarrow \\
 \text{euclid_proposition_32_exterior} \\
 \downarrow \\
 \text{HilbertTriangleAnglesEqualTwoRightAngles.}
 \end{array}$$

32.12 What “Equal to Two Right Angles” Means Here

There is an important distinction between the classical mathematical consequence and the literal formal statement.

In ordinary angle-measure notation, I.32 yields

$$\angle BAC + \angle ABC + \angle ACB = 180^\circ.$$

The present Hilbert layer, however, does not assign numerical measures to angles. Nor does it define a general arithmetic operation

$$\alpha + \beta + \gamma.$$

Instead the formalization records the synthetic configuration from which the two-right-angles conclusion follows:

$$B - C - D$$

makes the rays CB and CD opposite, so $\angle ACB$ and $\angle ACD$ form a straight-angle configuration. The exterior angle is then decomposed by CR into angles congruent to $\angle BAC$ and $\angle ABC$.

Consequently the formal theorem proves the geometric content

$$\angle BAC + \angle ABC + \angle ACB = 2R$$

without translating $2R$ into the numerical value 180° .

Strictly speaking, the current predicate `HilbertTriangleAnglesEqualTwoRightAngles` stores the extension $B - C - D$ and the exterior-angle decomposition; the supplementary interpretation of the straight line at C is implicit in this data rather than represented by a separate angle-addition object.

This is not merely a coding detail. It identifies exactly what must be formalized in a synthetic geometry before the familiar numerical 180° formulation can even be stated.

32.13 Lean Formalization

The principal theorem is

`euclid_proposition_32_exterior`

with hypotheses

```
(A B C D : Geo.Point)
(hABC : Not (PrimCollinear Geo A B C))
(hBCD : Geo.Between B C D)
```

and conclusion

HilbertExteriorAngleEqualsRemoteAngles A B C D

The proof uses the following main components:

```

hilbert_exterior_angle_aux
hilbert_exterior_angle_aux_parallel
hilbert_exterior_angle_inside
hilbert_angle_eq_of_sameRay_first
hilbert_angle_eq_of_sameRay_second
hilbert_between_points_sameSide_transversal
euclid_proposition_29_transversal
VerticalAngles

```

The second theorem,

euclid_proposition_32

is a corollary obtained by extending BC and invoking the exterior-angle theorem.

The typeclass

[HilbertEuclideanPlane Geo]

appears because the proof ultimately uses Proposition I.29 and hence the Euclidean parallel axiom.

32.14 Relation to the Classical Proof

At the level of a paper proof, I.32 is short:

construct a parallel through $C \longrightarrow$ use I.29 twice $\longrightarrow$ add the adjacent angles.

The Lean reconstruction exposes several pieces hidden by this description.

First, a parallel *line* through C is insufficient: the correct orientation of the ray must be established.

Second, the ray used to split the exterior angle must be shown to meet the segment AD . This is the crossbar component.

Third, the explicit I.29 theorem carries side-of-line hypotheses that must be proved from the betweenness configuration.

Fourth, the proof does not “add angles” numerically. It records a synthetic decomposition of the exterior angle.

Thus the expanded formal proof is not evidence that Euclid’s argument is mathematically complicated. It is an audit of the incidence, order and orientation information suppressed by the classical diagram.

32.15 Conclusion

Proposition I.32 combines the parallel theory of I.29–I.31 with the synthetic theory of triangles. Its essential Euclidean step is

$$AB \parallel CE \longrightarrow \text{the required angle congruence,}$$

which is supplied by I.29.

The formal proof then shows that the correctly oriented parallel ray through C cuts the exterior angle into two parts congruent to the remote interior angles:

$$\angle BAC \cong \angle ACR, \quad \angle ABC \cong \angle RCD.$$

Together with the straight-line relation $B - C - D$, this is the synthetic content of the classical theorem

$$\boxed{\angle BAC + \angle ABC + \angle ACB = 180^\circ}.$$

The reconstruction therefore reaches the familiar 180° theorem without introducing degrees: the numerical formulation appears only as the modern interpretation of a purely synthetic Hilbert-style configuration.

Chapter 33

Proposition I.33

33.1 Introduction

Euclid's Proposition I.33 states that the straight lines joining the corresponding ends of equal and parallel straight lines in the same directions are themselves equal and parallel.

In the present reconstruction the proposition is a compact meeting point of three pieces of geometry already developed in Book I:

$$\text{parallelism} \longrightarrow \text{alternate-angle congruence} \longrightarrow \text{SAS} \longrightarrow \text{parallelism}.$$

The first implication is Euclidean and uses the parallel axiom through the I.29 infrastructure. The final implication is the neutral converse alternate-angle theorem represented by I.27.

33.2 Classical Proof Architecture

In the classical configuration, let AB and CD be equal and parallel, with the joining segments AC and BD .

A diagonal is used to compare two triangles. Parallelism supplies a pair of congruent alternate angles, the given equality supplies one pair of corresponding sides, and the diagonal is common. SAS therefore gives the remaining side equality and the second required angle equality. The converse alternate-angle criterion then proves the second pair of lines parallel.

Schematically,

$AB \parallel CD \implies \text{alternate angles} \implies \text{SAS} \implies AC \cong BD \text{ and alternate angles} \implies AC \parallel BD.$
--

The Lean proof preserves this architecture. What becomes explicit is the orientation information that a classical diagram carries silently.

33.3 Explicit Crossing Formulation

The lower-level theorem uses a crossing point M to encode the configuration:

$$A - M - D, \quad B - M - C.$$

Its documentation-level statement is

```

theorem euclid_proposition_33_crossing
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D M : Geo.Point)
  (hParallel : Geo.Parallel A B C D)
  (hCongruent : Geo.Congruent A B C D)
  (hAMD : Geo.Between A M D)
  (hBMC : Geo.Between B M C) :
  Geo.Parallel A C B D /\
  Geo.Congruent A C B D := by
  -- proof in Proposition33.lean

```

The two betweenness assumptions determine which sides of the transversal contain the four endpoints. From them the proof reconstructs the SameSide/OppositeSide and SameRay information required by the Hilbert angle interface.

33.4 Configuration Normalization

The main formal issue in I.33 is not SAS itself but orientation.

From

$$B - M - C$$

the proof obtains the incidence data for the transversal through B, M, C . Since $AB \parallel CD$, the endpoints A and D cannot lie on that transversal. Together with

$$A - M - D$$

this yields the required opposite-side relation.

An interior point of BC is then used to normalize the rays entering the angle statements. This makes it possible to apply the Euclidean alternate-angle theorem in exactly the orientation needed for SAS.

Thus the formal preparation is

$$\text{betweenness} \longrightarrow \text{incidence} \longrightarrow \text{side-of-line data} \longrightarrow \text{same-ray normalization}.$$

No additional Euclidean content is introduced in this stage.

33.5 The I.29 Step

The Euclidean direction of the parallel-angle theory is invoked through the established interface theorem for alternate angles of parallel lines. Conceptually it supplies

$$AB \parallel CD \implies \angle BAC \cong \angle DCB$$

after the required ray and side normalizations.

This is the only essentially Euclidean step in the proof. It is precisely where Hilbert Group IV enters.

33.6 The SAS Step

The two triangles determined by the diagonal are compared by SAS.

The proof uses:

- the given segment congruence $AB \cong CD$;
- the common diagonal;
- the angle congruence obtained from parallelism.

The resulting triangle-congruence object provides two pieces of information needed for the conclusion.

First,

$$AC \cong BD.$$

Second, it supplies a further angle congruence. After same-ray normalization, this is the alternate-angle statement required to prove

$$AC \parallel BD.$$

33.7 The I.27 Step

The second parallelism is obtained from the converse alternate-angle theorem.

This part is neutral: once the appropriate alternate angles are known congruent and the side configuration is normalized, no use of the Euclidean parallel axiom is required.

The architecture can therefore be written more sharply as

configuration normalization $\longrightarrow$ I.29 $\longrightarrow$ SAS $\longrightarrow$ I.27.
--

33.8 Relationship with Earlier Propositions

Proposition I.33 combines the two directions of the parallel-angle theory developed immediately before it.

The forward direction

$$\text{parallel lines} \implies \text{congruent alternate angles}$$

belongs to the Euclidean layer and is represented by I.29.

The converse direction

$$\text{congruent alternate angles} \implies \text{parallel lines}$$

is represented by I.27 and belongs to the neutral layer.

I.33 is therefore a useful architectural checkpoint: a Euclidean theorem is used to enter a triangle congruence argument, and the conclusion is completed by a neutral converse theorem.

33.9 Hilbert and Book Zero Components Used

No new axiom is introduced for Proposition I.33.

The proof uses established infrastructure for:

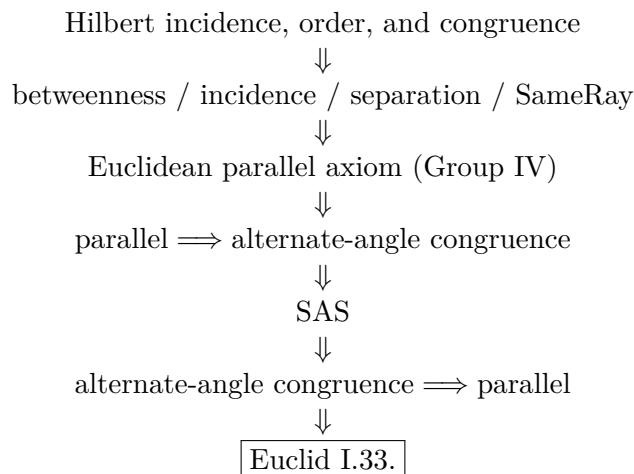
- incidence and line carriers;
- betweenness and its incidence consequences;
- separation by a line;
- SameSide and OppositeSide relations;
- SameRay transport;
- segment-congruence symmetries;
- non-collinearity;
- SAS triangle congruence;
- the Euclidean alternate-angle theorem;
- the converse alternate-angle theorem.

The API-level formulation also uses the existing parallelogram infrastructure, including the one-pair-parallel-and-congruent criterion and congruence of opposite sides of a parallelogram.

No I.33-specific helper is added to `HilbertInterface` or to Book Zero.

33.10 Dependency Summary

The formal dependency structure is



New axiom:	no
New proposition-specific helper:	no
Continuity:	no
Euclidean parallel axiom:	yes
Build status:	clean
sorry:	no

33.11 Conclusion

Euclid I.33 is mathematically short but formally revealing. Once the orientation hidden in the classical picture is exposed, the original proof mechanism survives intact:

$$\boxed{\text{I.29} \longrightarrow \text{SAS} \longrightarrow \text{I.27.}}$$

The proposition therefore confirms that the developed Hilbert interface can express not only individual classical constructions, but also the interaction between the Euclidean and neutral layers of Book I.

Chapter 34

Proposition I.34

34.1 Introduction

Euclid's Proposition I.34 collects the fundamental metric and angular properties of a parallelogram. For a parallelogram $ABCD$, it asserts that opposite sides are equal, opposite angles are equal, and a diagonal bisects the parallelogram.

The current CGJteamLab reconstruction separates these assertions according to the language actually present in the Hilbert development. Opposite sides and opposite angles are packaged directly by segment and angle congruence. The classical statement about the diagonal is represented separately by congruence of the two triangles cut off by that diagonal. No numerical area function is introduced in Proposition I.34, and no `HilbertScissors` term occurs in the production file.

This distinction is important for the later area propositions. At I.34 the project uses triangle congruence as a sufficient synthetic representative of the diagonal bisection statement; beginning with I.35 the documentation must also distinguish geometric congruence from the more general scissors/content calculus.

34.2 The Classical Configuration

Let $ABCD$ be a parallelogram, with

$$AB \parallel CD, \quad BC \parallel DA.$$

Draw the diagonal AC . It separates the parallelogram into the triangles ABC and CDA .

The required metric and angular conclusions are

$$AB \cong CD, \quad BC \cong DA,$$

$$\angle DAB \cong \angle BCD, \quad \angle ABC \cong \angle CDA.$$

In the present formalization the diagonal assertion is represented by

$$\triangle ABC \cong \triangle CDA.$$

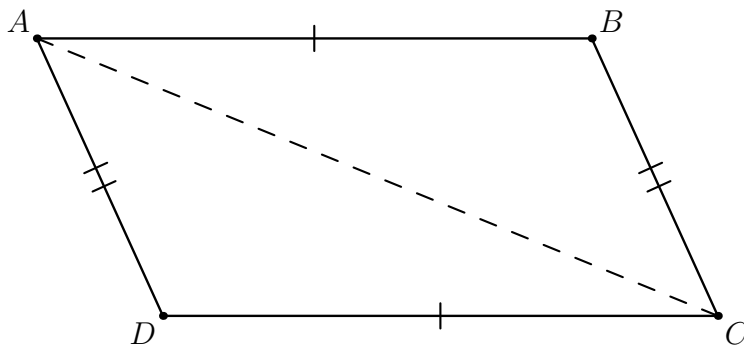


Figure 34.1: The configuration for Proposition I.34. The single and double chevrons mark the two *parallel* pairs $AB \parallel CD$ and $BC \parallel DA$; they do not assert the opposite-side congruences that I.34 is meant to prove. The dashed segment AC is the diagonal used in both the classical and formal arguments.

34.3 Classical Proof

Euclid's original lettering is different: he writes the parallelogrammic area as $ACDB$ and uses BC as its diameter. After relabelling to the present $ABCD$ picture with diagonal AC , the proof has four logically distinct steps.

First, Proposition I.29 is applied to the two pairs of parallels. With AC as transversal it gives

$$\angle BAC \cong \angle ACD, \quad \angle BCA \cong \angle CAD.$$

Second, Proposition I.26 is applied to the two diagonal triangles. The two angle equalities and the common side AC give

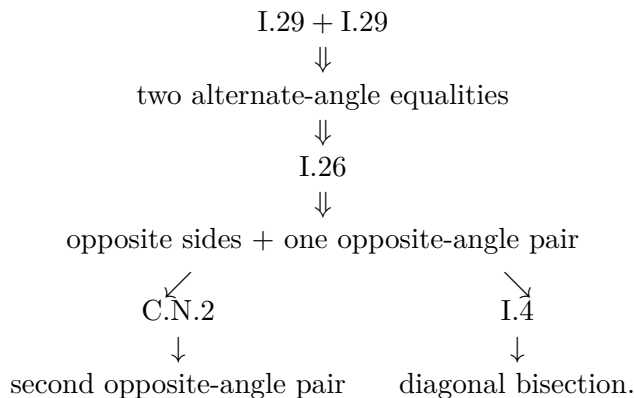
$$AB \cong CD, \quad BC \cong DA, \quad \angle ABC \cong \angle CDA.$$

Third, Euclid does not obtain the other pair of whole opposite angles by a single application of I.26. He combines the two alternate-angle equalities by Common Notion 2: the two parts of $\angle DAB$ are respectively equal to the two parts of $\angle BCD$, hence

$$\angle DAB \cong \angle BCD.$$

Finally, the statement that the diameter bisects the parallelogram is proved as an equality of rectilinear figures by Proposition I.4. In the present lettering one may use $BC \cong DA$, the common side AC , and the included angle equality $\angle BCA \cong \angle CAD$. Euclid therefore obtains the equality of the two triangles on the two sides of the diagonal.

A compact classical DAG is



This corrects the earlier compressed description. I.26 supplies corresponding sides and the remaining angle; Euclid’s final “bisects the areas” clause is an additional magnitude statement and is closed explicitly by I.4.

34.4 The Main Formalization Issue: Reusing Parallelogram Theory

The production file `Proposition34.lean` does not replay the classical I.29–I.26 route. By the time Proposition I.34 is reached, the developed `HilbertInterface` already contains a Euclidean theory of parallelograms. In particular, the public theorem reuses

```
ParallelogramOppositeSidesCongruent
ParallelogramOppositeAnglesCongruent
```

and is therefore a thin wrapper.

34.4.1 Where the Euclidean work actually lives

The important hidden fact in the formal DAG is that `ParallelogramOppositeSidesCongruent` is not a primitive congruence rule. Its core is the reusable theorem

```
parallelogram_second_pair_congruent
```

which proves $BC \cong DA$ from `IsParallelogram Geo A B C D`. The proof constructs a genuine auxiliary point X by segment construction: X lies on the ray from D through A and

$$DX \cong BC.$$

Collinearity transports $AD \parallel BC$ to $XD \parallel BC$. The proof then builds the side-orientation data required by `OnePairParallelCongruentCriterion`; this makes $XBCD$ a parallelogram and hence gives

$$XB \parallel CD.$$

Now both XB and AB are parallels to CD through B . Hilbert Group IV, through

```
HilbertEuclideanPlane.parallel_unique
```

identifies their carriers. Since X also lies on the carrier DA , incidence uniqueness at the intersection of the carriers AB and DA forces

$$X = A.$$

Substitution gives $BC \cong DA$.

The other side pair is obtained by applying the same theorem to the cyclically relabelled parallelogram $BCDA$, followed by symmetry of segment congruence. This is the precise point at which the Euclidean parallel axiom is consumed in the reusable side theorem.

34.5 Proof Architecture of the Reconstruction

The actual reusable side-congruence DAG is

$$\begin{array}{c}
 AB \parallel CD, \quad BC \parallel DA \\
 \Downarrow \\
 \text{IsParallelogram } ABCD \\
 \Downarrow \\
 \text{parallelogram_second_pair_congruent} \\
 \text{segment construction of } X + \text{orientation} + \text{Hilbert IV} \\
 \Downarrow \\
 \text{ParallelogramOppositeSidesCongruent} \\
 \Downarrow \\
 AB \cong CD, \quad BC \cong DA.
 \end{array}$$

For one pair of opposite angles the interface then uses the diagonal:

$$\begin{array}{c}
 AB \cong CD, \quad BC \cong DA, \quad AC \cong CA \\
 \Downarrow \text{HilbertSSS} \\
 \triangle ABC \cong \triangle CDA \\
 \Downarrow \\
 \angle ABC \cong \angle CDA.
 \end{array}$$

The theorem `ParallelogramOppositeAnglesCongruent` applies the one-pair result to $ABCD$ and to the cyclic relabelling $BCDA$, then uses symmetry of angle congruence to package both opposite-angle conclusions.

No geometric case split occurs in `Proposition34.lean`. The only contradiction branch in the diagonal theorem proves non-collinearity, while the hidden $X = A$ argument belongs to the reusable interface theorem above.

34.6 Lean Formalization

The production module imports only

```
import CGJteamLab.HilbertInterface
```

and exposes two public theorems. There are no proposition-local helper theorems in the file.

The main theorem is deliberately thin:

```
theorem euclid_proposition_34
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D : Geo.Point)
  (hParallelogram : IsParallelogram Geo A B C D) :
  And
    (OppositeSidesCongruent Geo A B C D)
    (OppositeAnglesCongruent Geo A B C D)
```

Its two branches are discharged by

```
ParallelogramOppositeSidesCongruent
ParallelogramOppositeAnglesCongruent
```

The third classical assertion is represented separately:

```
theorem euclid_proposition_34_diagonal
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D : Geo.Point)
  (hParallelogram : IsParallelogram Geo A B C D) :
  TriangleCongruenceResult Geo A B C C D A := by
```

The diagonal proof first derives $\neg \text{Collinear}(A, B, C)$ from the disjointness of the parallel point-lines AB and CD . It then takes the two opposite-side congruences, uses reflexivity plus endpoint reversal for the common diagonal, and applies SSS:

```
have hSides :
  OppositeSidesCongruent Geo A B C D :=
  ParallelogramOppositeSidesCongruent
  Geo A B C D hParallelogram

have hAC_CA :
  Geo.Congruent A C C A :=
  CongruentSwapSecond
  Geo A C A C
  (hilbert_congruent_reflexive Geo A C)

have hSSS :=
  HilbertSSS
  Geo
  A B C
  C D A
  hABC
  hSides.1
  hSides.2
  hAC_CA

exact hSSS.2
```

No area object occurs in either theorem. The diagonal theorem proves a triangle-congruence result; this is sufficient for this particular bisection statement but is not a general formal theory of equal content.

34.7 Relationship with Earlier Propositions

Classically, I.34 directly uses I.29 twice and I.26, then Common Notion 2 and I.4. The production I.34 file calls none of those proposition names directly. The Euclidean work has instead been moved into reusable Hilbert-level parallelogram theorems.

The relation with I.33 remains close. I.33 is a recognition theorem: from one pair of opposite sides that are parallel and congruent, together with the required orientation data, it obtains a parallelogram. The lower-level side proof used by I.34 invokes the reusable analogue `OnePairParallelCongruentCriterion` for the auxiliary quadrilateral $XBCD$.

34.8 Hilbert and Book Zero Components Used

At proposition level the visible dependencies are compact:

- `IsParallelogram`;
- `ParallelogramOppositeSidesCongruent`;
- `ParallelogramOppositeAnglesCongruent`;
- `HilbertSSS`;
- incidence uniqueness and extensional point-lines, used in the diagonal theorem to prove A, B, C non-collinear;
- reflexivity and endpoint reversal of segment congruence.

The side-congruence theorem in `HilbertInterface` additionally uses:

- Hilbert segment construction, producing the auxiliary point X ;
- `SameRay` and collinearity transport along DA ;
- `SameSide` data for the one-pair parallelogram criterion;
- `OnePairParallelCongruentCriterion`;
- uniqueness of a Euclidean parallel through a point;
- uniqueness of an incidence line through two distinct points.

No new Book Zero theorem is declared in `Proposition34.lean`.

34.9 Formalization Notes

34.9.1 Axiomatic status

Both public theorems assume

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

The Euclidean strength is used lower in the actual DAG when the two parallels through B to the carrier CD are identified. This is the Hilbert IV step inside `parallelogram_second_pair_congruent`.

The current source contains no proposition-local `axiom` declaration and no `sorry`. This diagrammatic audit did *not* rerun `lake build` or `#print axioms`; historical build statements in the preserved Book1R PDF are therefore not presented here as fresh audit results.

34.9.2 Hidden configuration data

There are two different levels of hidden data.

At proposition level, `euclid_proposition_34_diagonal` must prove explicitly that A, B, C are not collinear before SSS can be applied. If C lay on AB , it would lie on both disjoint parallel carriers AB and CD .

At reusable-interface level, the stronger hidden witness is the point X constructed on the ray DA . The proof uses genuine `SameRay` and `SameSide` geometry to turn $XBCD$ into an auxiliary parallelogram and then proves $X = A$ by parallel and incidence uniqueness. These side/order facts do not appear in the public I.34 theorem because the interface theorem has already discharged them.

34.9.3 Geometry, congruence, content, and representation

The layers should not be conflated:

- **Incidence/order/parallel geometry:** the parallelogram carriers, the diagonal AC , and lower in the reusable DAG the point X with its ray and side data.
- **Congruence:** the opposite-side congruences, opposite-angle congruences, and the SSS comparison of the diagonal triangles.
- **Content:** no general content object or scissors-equivalence relation occurs in Proposition I.34.
- **Representation:** the classical bisection clause is represented here by congruence of the two diagonal triangles. This is stronger than merely saying that they have equal content, but it is not the later `HilbertScissorsEquicomplementable` language.

34.9.4 Reusable interface helpers

The current I.34 DAG uses

```
parallelogram_second_pair_congruent
ParallelogramOppositeSidesCongruent
ParallelogramOppositeAngleCongruent
ParallelogramOppositeAnglesCongruent
```

The first contains the auxiliary construction and Hilbert-IV uniqueness step. The second packages both side pairs by cyclic relabelling. The third uses SSS across the diagonal for one opposite-angle pair. The fourth packages both angle pairs by cyclic relabelling and symmetry.

34.9.5 Diagrammatic audit

The preserved I.34 figure had the correct quadrilateral and diagonal but used one and two transverse tick marks on opposite sides. Those marks visually asserted

$$AB \cong CD, \quad BC \cong DA,$$

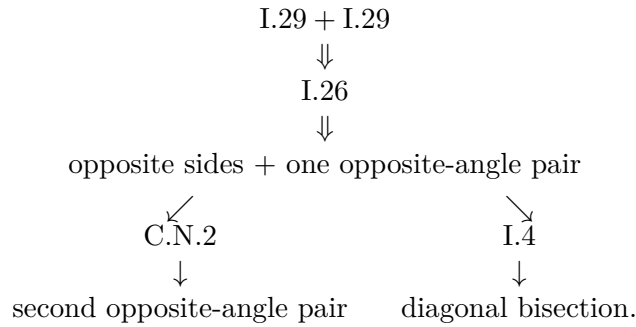
which are conclusions of Proposition I.34 rather than hypotheses. The audited figure therefore replaces them by conventional single and double *parallel* chevrons. The input geometry now records exactly $AB \parallel CD$ and $BC \parallel DA$.

No second figure is introduced for the hidden point X . The reusable proof constructs X and subsequently proves $X = A$. A panel that drew X as a separate final vertex while simultaneously displaying all the parallel data would depict a configuration excluded by the uniqueness theorem being used; drawing X coincident with A would add no new geometric state. This is therefore proof-state bookkeeping, not an additional proposition-level figure.

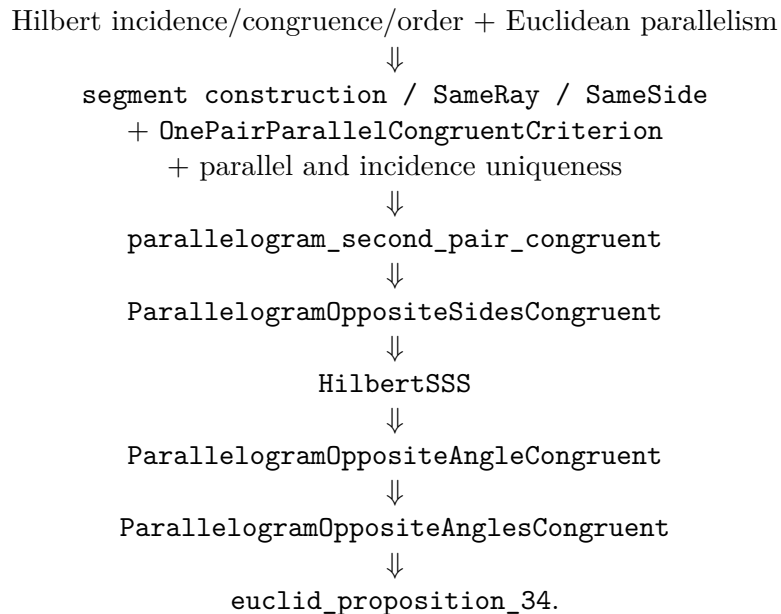
Likewise no figure is added for diagonal reflexivity, SSS, cyclic relabelling, segment reversal, or angle-congruence symmetry. These steps change the logical interpretation of the same configuration but do not create a new geometric state. The final audited figure count is one.

34.10 Dependency Summary

The classical dependency is



The actual production DAG is



The separate diagonal endpoint is

$$\begin{array}{c} \text{ParallelogramOppositeSidesCongruent} + AC \cong CA + \neg \text{Collinear}(A, B, C) \\ \Downarrow \\ \text{HilbertSSS} \\ \Downarrow \\ \text{euclid_proposition_34_diagonal.} \end{array}$$

Production import:	CGJteamLab.HilbertInterface
Public theorems in <code>Proposition34.lean</code> :	two
Proposition-local helper theorem:	no
Proposition-local <code>axiom</code> declaration:	no
<code>sorry</code> in <code>Proposition34.lean</code> :	no
New Book Zero declaration in this file:	no
Geometric case split in this file:	no
Hidden constructed point in reusable DAG:	yes, X
Hilbert Group IV used in reusable DAG:	yes
Numerical area semantics:	no
Scissors/content term in I.34:	no
<code>lake build</code> rerun for this audit:	no
<code>#print axioms</code> rerun for this audit:	no

34.11 Conclusion

Proposition I.34 is short at the proposition-file level because the developed Euclidean theory of parallelograms already contains its main metric work. The public theorem is a thin wrapper, but the actual dependency graph is not trivial: opposite-side congruence ultimately passes through an auxiliary segment construction, orientation data, a one-pair parallelogram criterion, Hilbert-IV parallel uniqueness, and incidence uniqueness.

Euclid's historical DAG is different. He obtains alternate angles by I.29, uses I.26 for corresponding sides and one opposite-angle pair, Common Notion 2 for the other whole opposite angle, and I.4 for equality of the two diagonal triangles as rectilinear figures. The current Lean development instead reuses the established side theorem and applies SSS across the same diagonal.

The diagrammatic audit leaves exactly one figure. Its role is now precise: it records only the parallelogram hypotheses and the auxiliary diagonal. It no longer displays the theorem's opposite-side conclusions as if they were input data.

Chapter 35

Proposition I.35

35.1 Introduction

Euclid’s Proposition I.35 is the first proposition in Book I in which the word “equal” no longer refers merely to congruent segments or congruent angles. The proposition states that two parallelograms on the same base and in the same parallels are equal to one another. The two parallelograms need not be congruent. What Euclid uses is a cut-and-paste comparison of rectilinear figures.

This makes I.35 a genuine transition point in the reconstruction. Up to I.34, the main formal objects are points, lines, betweenness, segment congruence, angle congruence, and parallelism. At I.35 one must decide what “equal figures” means before the classical argument can even be stated faithfully.

The present development does not introduce numerical area. Instead it introduces a small formal scissors calculus. Triangles are the basic pieces, finite formal sums of triangles represent decompositions, and a generated relation records the operations of splitting a triangle, replacing it by a congruent triangle, adding decompositions, and composing such steps.

The final Lean theorem therefore concludes not a numerical area equality but formal equicomplementability:

```
theorem euclid_proposition_35
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D E F : Geo.Point)
  (hABCD : IsParallelogram Geo A B C D)
  (hEBCF : IsParallelogram Geo E B C F)
  (hADF : Geo.Between A D F)
  (hAEF : Collinear Geo A E F) :
  HilbertScissorsEquicomplementable Geo
    (hilbertParallelogramTerm Geo A B C D)
    (hilbertParallelogramTerm Geo E B C F)
```

The extra order and collinearity hypotheses are not cosmetic. They make explicit part of the configuration which is left implicit by the classical diagram and are essential to a complete formal case analysis.

35.2 The Classical Configuration

Let $ABCD$ and $EBCF$ be parallelograms on the common base BC . Their opposite sides AD and EF lie on one straight line which is parallel to BC . Euclid's statement is therefore

$$ABCD \text{ and } EBCF \text{ on the same base } BC \implies ABCD = EBCF$$

in Euclid's language of equality of rectilinear figures.

A complete synthetic proof has to distinguish three possible positions of the two upper sides. Wilkins isolates exactly these configurations:

1. the upper sides overlap, with order A, E, D, F ;
2. the upper sides meet at a common endpoint, so $E = D$;
3. the upper sides are disjoint, with order A, D, E, F .

Only the third configuration is treated explicitly in Euclid's proof. The first two are not exceptional from the formal point of view: they are genuine cases which a theorem prover cannot recover from the picture.

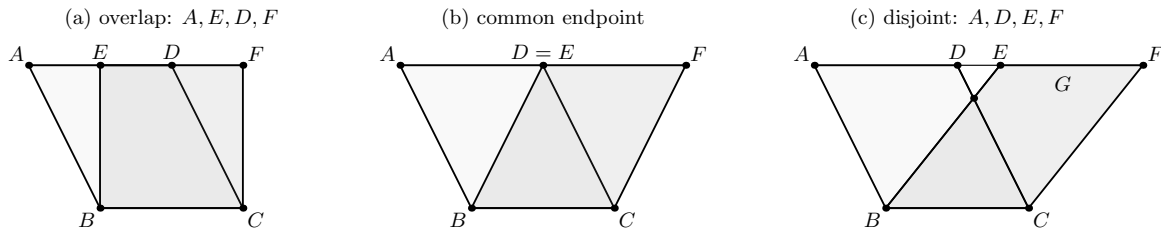


Figure 35.1: The three upper-side configurations relevant to Proposition I.35. The point G is required only in the disjoint case.

The current Lean theorem uses $A - D - F$ as an outer container:

$$A - D - F, \quad A, E, F \text{ collinear}.$$

From the parallelogram hypotheses and this data, the reconstruction first proves $A - E - F$. The relative order of the two interior points D and E then yields precisely the three cases shown in Figure 35.1.

35.3 Classical Proof

Euclid's explicit proof is the disjoint case. Since $ABCD$ and $EBCF$ are parallelograms, Proposition I.34 gives

$$AD = BC = EF, \quad AB = DC.$$

Because DE is common to the two longer upper segments, Common Notions 1 and 2 yield

$$AE = DF.$$

The lines AB and DC are parallel, and the upper line AF is a transversal. Proposition I.29 therefore gives

$$\angle EAB = \angle FDC.$$

Euclid then uses Proposition I.4 to conclude that the outer triangles are equal:

$$\triangle EAB = \triangle FDC.$$

In the disjoint configuration the lines EB and DC meet at G . Euclid subtracts the common triangle DGE from the two equal outer triangles, obtaining equal trapezia, and then adds the common triangle GBC . This produces the two parallelograms.

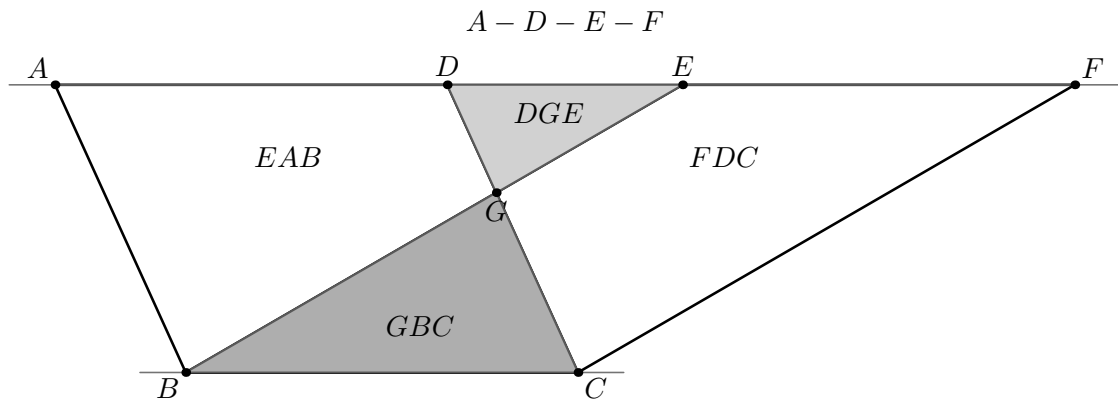


Figure 35.2: The classical disjoint configuration of Proposition I.35. The outer triangles EAB and FDC are compared. Their crossing sides determine G ; Euclid subtracts the common triangle DGE and then adds the common triangle GBC .

The classical dependency graph is therefore

$$\begin{array}{c}
 \text{I.34} + \text{I.29} \\
 \Downarrow \\
 AE = DF, \quad AB = DC, \quad \angle EAB = \angle FDC \\
 \Downarrow \quad \text{I.4} \\
 \triangle EAB = \triangle FDC \\
 \Downarrow \quad \text{C.N.3} \\
 \text{equal remainders} \\
 \Downarrow \quad \text{C.N.2} \\
 ABCD = EBCF.
 \end{array}$$

The overlap and common-endpoint cases use the same outer-triangle comparison but a different addition pattern. Wilkins' three-case analysis is therefore a useful completion of Euclid's single drawn configuration.

35.4 Equidecomposition and Equicomplementability

The decisive modern control source for I.35 is Hilbert’s theory of plane area. In Chapter IV, Section 18 of the *Foundations of Geometry*, Hilbert defines two polygons to be equidecomposable when they can be triangulated into finitely many pairwise congruent triangles. He then defines equicomplementability by allowing pairs of equidecomposable polygons to be adjoined before comparing the resulting figures by equidecomposition.

In Section 19 Hilbert states Theorem 44: two parallelograms with the same bases and the same altitudes are equicomplementable. This is the Hilbertian counterpart of the Euclidean area argument which begins at I.35.

The distinction between the two notions is essential. Equidecomposability is a direct finite scissors congruence. Equicomplementability is weaker: the figures may become equidecomposable only after suitable complements have been adjoined. Hilbert explicitly notes that in non-Archimedean geometry figures with equal bases and altitudes may be equicomplementable without being equidecomposable.

Hartshorne makes the same distinction under the name *equal content*. His definition requires figures P, Q and P', Q' to be nonoverlapping, requires Q and Q' to be equidecomposable, and requires the two enlarged figures to be equidecomposable. His Example 22.1.2 identifies Euclid I.35 as an example of equal content.

This matters for the present Lean development because the formal calculus is not yet a complete model of polygonal regions. It records formal sums of triangles, but it does not yet carry an interior or a nonoverlap predicate. Consequently

`HilbertScissorsEquicomplementable`

should be read exactly as its source comment says: a *term-level analogue* of Hilbert’s equicomplementability relation. The theorem proved in I.35 is therefore a faithful algebraic scissors certificate, but a later semantic layer would still be needed to identify every formal sum with an actual nonoverlapping polygonal union.

35.4.1 Why numerical area is deliberately avoided

A modern proof could assign an area to each parallelogram and write

$$\text{area} = \text{base} \cdot \text{height}.$$

That would be the wrong logical direction here. Numerical area is developed later than the elementary scissors theory in both Hilbert and Hartshorne. Using it at I.35 would replace Euclid’s synthetic argument by a stronger and logically downstream theorem.

The present reconstruction therefore stays at the level of finite cutting, congruence, and recombination.

35.5 The Hilbert Scissors Calculus

The file `HilbertScissors.lean` introduces two formal types:

```

abbrev HilbertTriangleTerm :=
  Multiset Geo.Point

abbrev HilbertScissorsTerm :=
  Multiset (HilbertTriangleTerm Geo)

```

A triangle is represented by an unordered multiset of its three vertices. A scissors term is an unordered multiset of triangle terms. This choice has two useful formal consequences:

- the order of the three names used to write a triangle is not geometric data;
- the order in which the triangular pieces are listed is not geometric data either.

Thus much of the bookkeeping which would otherwise be expressed by explicit list permutations disappears at the representation level.

The central generated relation is

```

inductive HilbertScissorsEq
  (Geo : Geometry.Geo) :
  HilbertScissorsTerm Geo ->
  HilbertScissorsTerm Geo ->
  Prop

| refl
  (P : HilbertScissorsTerm Geo) :
  HilbertScissorsEq Geo P P

| symm
  {P Q : HilbertScissorsTerm Geo}
  (h : HilbertScissorsEq Geo P Q) :
  HilbertScissorsEq Geo Q P

| trans
  {P Q R : HilbertScissorsTerm Geo}
  (hPQ : HilbertScissorsEq Geo P Q)
  (hQR : HilbertScissorsEq Geo Q R) :
  HilbertScissorsEq Geo P R

| add
  {P Q R S : HilbertScissorsTerm Geo}
  (hPQ : HilbertScissorsEq Geo P Q)
  (hRS : HilbertScissorsEq Geo R S) :
  HilbertScissorsEq Geo
    (P + R)
    (Q + S)

| split
  (A B C X : Geo.Point)
  (hBXC : Geo.Between B X C) :
  HilbertScissorsEq Geo
    (hilbertScissorsTriangle Geo A B C)
    (hilbertScissorsTriangle Geo A B X +
     hilbertScissorsTriangle Geo A X C)

| congruent
  (A B C D E F : Geo.Point)

```



```

(hCong :
  TriangleCongruenceResult
    Geo A B C D E F) :
HilbertScissorsEq Geo
  (hilbertScissorsTriangle Geo A B C)
  (hilbertScissorsTriangle Geo D E F)

```

The two geometric generators are therefore exactly the expected ones:

- **split**: cut a triangle at a point lying between two vertices;
- **congruent**: replace one triangle by a congruent triangle.

The remaining constructors make this relation an equivalence relation compatible with formal addition.

For formal equicomplementability, the implementation asks for formal complements R and S such that

$$R \sim S \quad \text{and} \quad P + R \sim Q + S,$$

where $\sim$ denotes `HilbertScissorsEq`. This is the algebraic pattern of Hilbert's definition, while the geometric nonoverlap conditions remain outside the present term language.

35.5.1 Triangulation lemmas

Two reusable lemmas are particularly important:

```

parallelogram_two_triangulations
crossing_quadrilateral_two_triangulations

```

The first proves that the two diagonal triangulations of a parallelogram are scissors-equivalent. It uses the existing theorem that the diagonals of a Euclidean parallelogram meet internally.

The second is more general. If the diagonals AC and BD of a quadrilateral meet internally at M , then

$$ABC + ACD \sim ABD + BCD.$$

The scissors move is easier to read as a three-stage refinement. The first diagonal triangulation is split at the internal crossing point M ; the same four elementary triangles are then regrouped along the other diagonal.

Its proof is pure scissors algebra after the two betweenness witnesses are provided. This lemma is what makes the overlap case short: the common trapezoid $EBCD$ can be flipped from one diagonal triangulation to the other without introducing a special area theorem.

35.6 Proof Architecture of the Reconstruction

The actual Lean proof is not a mechanical transcription of Euclid's proof. It has two logically distinct stages.

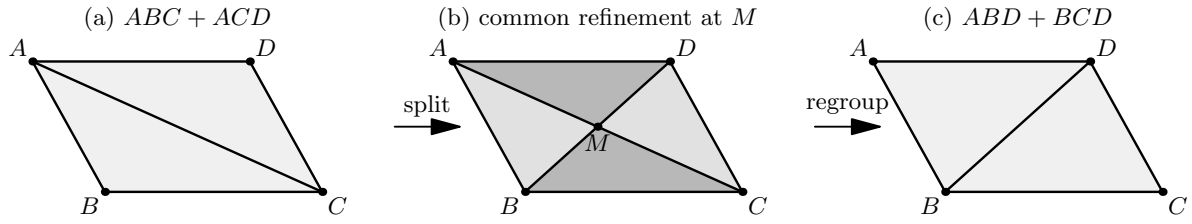


Figure 35.3: The generic diagonal-flip lemma used by the I.35 scissors layer. The two diagonal triangulations pass through one common four-triangle refinement at the internal intersection point M .

35.6.1 Stage 1: recover the upper-line order

The final theorem is given

$$A - D - F \quad \text{and} \quad A, E, F \text{ collinear.}$$

The theorem `i35_upper_order` proves

$$A - E - F.$$

The proof uses opposite-side congruence for the two parallelograms to obtain $AD \cong EF$. Hilbert's betweenness trichotomy on A, E, F then gives three possible orders. The two bad orders are excluded separately.

The first exclusion,

$$\text{i35_not_EAF,}$$

is a segment-order argument. From $A - D - F$ one gets $AD < AF$; congruence transports this to $EF < AF$. The bad order $E - A - F$ would imply the reverse strict inequality, contradicting asymmetry.

The second exclusion,

$$\text{i35_not_AFE,}$$

is genuinely planar. The helper `i35_cross_intersection` first constructs a point where BF crosses AC . The diagonal theorem for the second parallelogram gives another point where BF crosses EC . Under the bad order $A - F - E$, the same line would also meet the third side AE of the triangle AEC . The line-separation theorem `hilbert_line_avoids_third_triangle_side` forbids one line from meeting all three sides in the required interior configuration.

Once $A - E - F$ is known, the pure order theorem `i35_right_cases_of_common_container` compares the two interior points D and E and produces exactly

$$A - E - D - F, \quad E = D, \quad \text{or} \quad A - D - E - F.$$

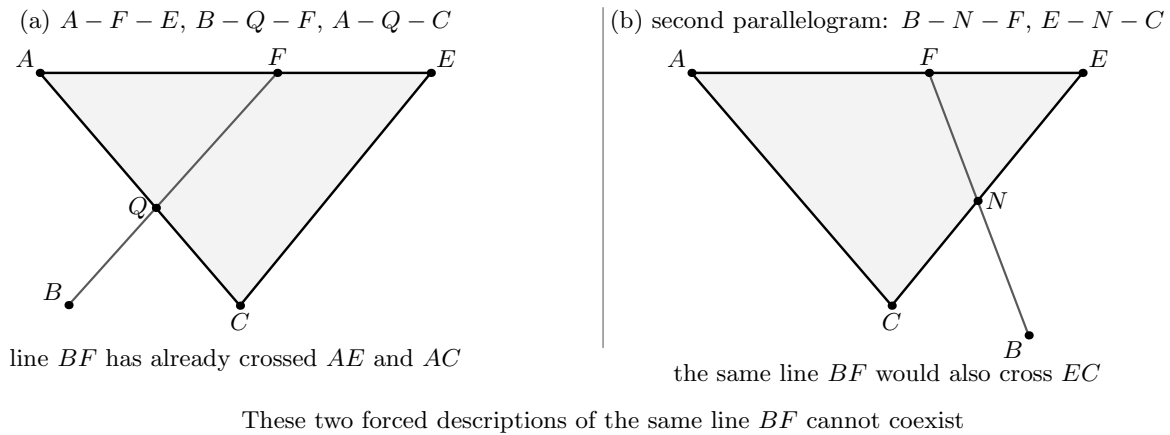


Figure 35.4: The planar contradiction used to exclude the bad order $A - F - E$. The first parallelogram forces $B - Q - F$ with $A - Q - C$; line separation in triangle AEC then forbids the same line BF from meeting EC . The second parallelogram, however, forces an internal point N with $B - N - F$ and $E - N - C$. The two panels display the incompatible geometric requirements, not two simultaneous realizations of one configuration.

35.6.2 Stage 2: solve the three scissors cases

The three cases are dispatched by `i35_from_three_cases`.

Overlap case

The outer triangles EAB and FDC are shown congruent. The common trapezoid $EBCD$ is then re-triangulated by `crossing_quadrilateral_two_triangulations`. No nonzero complement is needed. The two parallelogram terms are directly scissors-equivalent, so they are equicomplementable with zero complements.

Common-endpoint case

The two parallelograms share the upper endpoint. Again no auxiliary point G is required. The proof changes the triangulations and replaces the two outer triangles by congruent ones. The resulting parallelogram terms are directly scissors-equivalent, hence equicomplementable with zero complements.

Disjoint case

This is Euclid's explicit configuration. Here the point G is essential. The helper

`i35_disjoint_intersection`

constructs G with

$$E - G - B, \quad D - G - C.$$

The existence proof is not read off from the diagram. It uses Hilbert order and Pasch-type intersection machinery, followed by parallel/order transport to place the intersection on both required segments.

Instead of literally subtracting DGE as Euclid does, the formal proof uses the equivalent completion pattern. The four genuine geometric states are shown together in Figure 35.5: adjoin DGE , refine the left augmented figure, replace the congruent outer triangle, and reverse the refinement on the right.

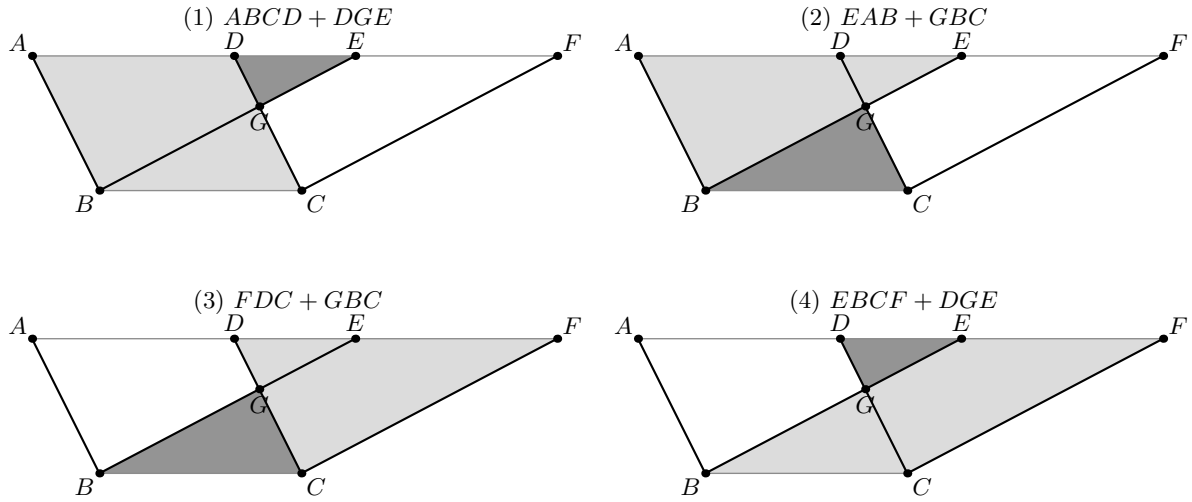


Figure 35.5: The exact four-stage scissors witness in the disjoint case. The formal proof passes from $ABCD + DGE$ to $EAB + GBC$, replaces the congruent outer triangle, and regroups to $EBCF + DGE$. The nonzero complement is the same geometric triangle DGE on both sides.

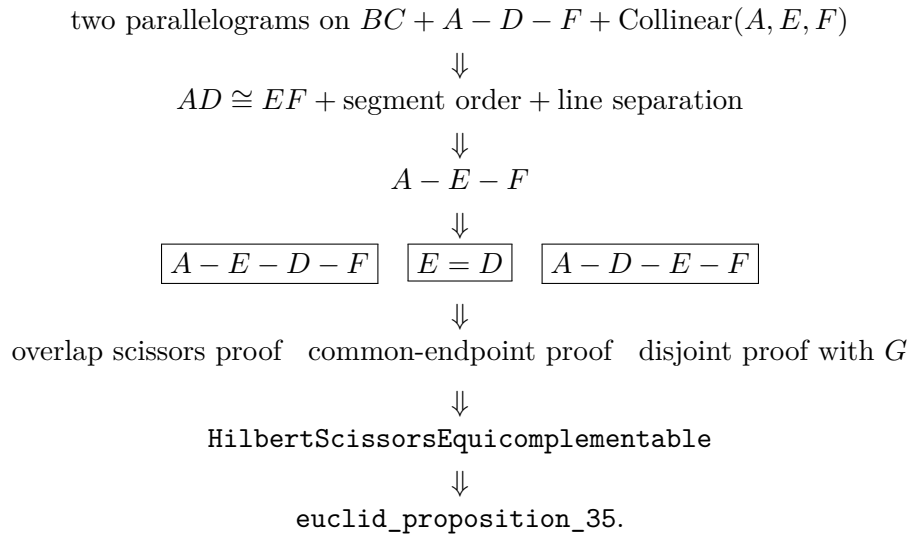
Thus the complete formal chain is

$$\begin{aligned}
 ABCD + DGE &\sim EAB + GBC \\
 &\sim FDC + GBC \\
 &\sim EBCF + DGE.
 \end{aligned}$$

This is exactly the formal equicomplementability witness needed for I.35.

35.6.3 The complete Lean DAG

The reconstruction can be summarized as



35.7 Lean Formalization

The current module boundary is explicit. The generic scissors calculus lives in

```
import CGJteamLab.HilbertScissors
```

while the proposition-specific Hilbert geometry and the complete three-case analysis are exposed directly in `Proposition35.lean`.

The key proposition-local declarations are

```
i35_left_augmented_decomposition
i35_right_augmented_decomposition
i35_outer_triangles_congruent
i35_equicomplementable_core
i35_disjoint_intersection
i35_common_endpoint
i35_overlap_trapezoid_intersection
i35_overlap
I35RightCases
i35_from_three_cases
i35_right_cases_of_common_container
i35_not_EAF
i35_cross_intersection
i35_not_AFE
i35_upper_order
i35_ordered
```

Thus the scissors module is no longer a hiding place for the proof of I.35. It supplies the reusable term language, splitting and congruence generators, triangulation lemmas, and equicomplementability operations. The geometry specific to the proposition—upper-line order, the three positional cases, the construction of G , and the final recombination—is visible in the `Proposition` file itself.

The public theorem is currently

```
theorem euclid_proposition_35
```

```

[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
(A B C D E F : Geo.Point)
(hABCD : IsParallelogram Geo A B C D)
(hEBCF : IsParallelogram Geo E B C F)
(hADF : Geo.Between A D F)
(hAEF : Collinear Geo A E F) :
HilbertScissorsEquicomplementable Geo
  (hilbertParallelogramTerm Geo A B C D)
  (hilbertParallelogramTerm Geo E B C F) := by
exact
  i35_ordered
    Geo
    A B C D E F
    hABCD
    hEBCF
    hADF
    hAEF

```

At the top level the formal route is therefore

```

i35_upper_order
  ↓↓
i35_right_cases_of_common_container
  ↓↓
i35_from_three_cases
  ↓↓
i35_ordered
  ↓↓
euclid_proposition_35.

```

35.8 Relationship with Earlier Propositions

35.8.1 Proposition I.29

Euclid uses I.29 explicitly to identify the corresponding angles $\angle EAB$ and $\angle FDC$. The current Lean proof does not replay this exact SAS route at the top level. Parallelism enters through the developed Hilbert theory and through the parallelogram infrastructure.

35.8.2 Proposition I.34

I.34 is the immediate metric input. Opposite sides of the two parallelograms are congruent, so

$$AD \cong BC \cong EF$$

and the second pair of opposite sides provides the remaining outer-triangle side congruence. The Lean helper `ParallelogramOppositeSidesCongruent` packages this information.

The current reconstruction then proves the outer triangles by the already available Hilbert SSS theorem rather than by Euclid's SAS argument. This is another example of the project deliberately reusing a stronger developed Hilbert API rather than mechanically replaying the historical DAG.

35.8.3 Transition to Proposition I.36

I.36 is a direct generalization from the same base to equal bases. Euclid’s classical proof uses I.33 to construct an intermediate parallelogram and then applies I.35 twice. For the formal development this means that I.35 is not an isolated experiment: the scissors/equicomplementability language is now a real dependency of the next area propositions.

35.9 Hilbert and Book Zero Components Used

The I.35 reconstruction uses a broad part of the developed Hilbert interface. After the module refactor this dependence is visible directly in `Proposition35.lean`, rather than being hidden behind a thin wrapper.

- `IsParallelogram` and the Euclidean parallelogram theory;
- `ParallelogramOppositeSidesCongruent`;
- `ParallelogramDiagonalIntersectionExists`;
- the derived `HilbertSSS` triangle-congruence theorem;
- Hilbert betweenness trichotomy and inner/outer transitivity;
- synthetic segment comparison and transport under congruence;
- Pasch/intersection machinery for the disjoint case;
- line separation through `hilbert_line_avoids_third_triangle_side`;
- transport of betweenness from collinearity and parallelism;
- the newly introduced formal scissors layer in `HilbertScissors.lean`.

No new Book Zero theorem is introduced by Proposition I.35. The new reusable material is concentrated in the generic scissors module, while the declarations whose names begin with `i35_` remain proposition-specific in `Proposition35.lean`.

The source hierarchy also changes emphasis at this point. Borsuk remains a useful control source for order, congruence, and parallelogram geometry, but the new notion of equality of figures is controlled more directly by Hilbert’s Chapter IV and Hartshorne’s theory of content. The Beeson proof corpus remains useful primarily as an audit of hidden incidence and order conditions rather than as the architecture of the Lean proof.

35.10 Formalization Notes

35.10.1 Axiomatic status

The final theorem assumes

```
[HilbertEuclideanPlane Geo]
```

and is therefore Euclidean in the current project API. This is consistent with Hilbert's Chapter IV treatment, which assumes the parallel axiom in addition to incidence, order, and congruence. The present file does not claim a sharper minimal axiom boundary for every internal helper.

The Euclidean strength is already visible in the reusable parallelogram results. Opposite-side congruence and the developed intersection/parallel machinery are called through the Euclidean-plane instance used by the project. Once the relevant congruences and betweenness data are available, individual scissors rewrites themselves are purely formal.

35.10.2 Hidden configuration data

I.35 exposes more hidden diagrammatic information than the immediately preceding propositions. The formal proof must explicitly manage:

- the outer order $A - D - F$;
- collinearity of A, E, F ;
- the derived order $A - E - F$;
- the trichotomy between overlap, common endpoint, and disjoint upper sides;
- existence and exact betweenness position of the point G in the disjoint case;
- internal diagonal intersections used for re-triangulation;
- non-collinearity needed for Pasch and line-separation arguments.

The point G is especially instructive. It is not a universal witness for all forms of I.35. It appears only in the disjoint configuration. Trying to force one G -based proof onto every configuration would obscure the geometry and create unnecessary order obligations.

35.10.3 The scissors language is formal, not yet semantic

The nested `Multiset` representation is a formal algebra of triangle terms. It intentionally forgets vertex order and piece order, but it also forgets geometric interiors. Therefore formal addition is total even when two actual geometric pieces would overlap.

This is the main semantic limitation of the present module. A future polygonal semantics could add a nonoverlap predicate, partial composition, or a realization theorem connecting valid formal scissors derivations to nonoverlapping rectilinear figures. None of that is required to complete the current I.35 certificate, but the distinction is documented here so that the formal relation is not silently identified with a stronger geometric notion.

35.10.4 Reusable helpers introduced

The most obviously reusable scissors helpers are

```
triangle_swap12
triangle_swap23
triangle_cycle
scissors_triangle_swap12
```



```

scissors_triangle_swap23
scissors_triangle_cycle
parallelogram_two_triangulations
crossing_quadrilateral_two_triangulations
equicomplementable_refl
equicomplementable_symm

```

The remaining helpers whose names begin with `i35_` are deliberately left proposition-specific for now. Some may later reveal a more general Hilbert statement, but no architectural promotion is required for I.35.

Diagrammatic audit. The audited chapter uses five figures, each with one geometric role. Figure 35.1 is retained as the global map of the overlap, common-endpoint, and disjoint upper-side configurations. Figure 35.2 is new and records Euclid’s actual disjoint construction, including the crossing point G and the two common triangles used in subtraction and addition. Figure 35.3 consolidates the former three separate diagonal-flip pictures into one three-panel geometric refinement; no state is lost, but the reusable move is now read as one operation. Figure 35.4 exposes the deepest hidden positional argument in `i35_not_AFE` without pretending that the contradictory requirements form a realizable single diagram. Finally Figure 35.5 consolidates the former four disjoint scissors pictures into one four-panel completion sequence.

No separate figure is added for `i35_not_EAF`, which is a purely one-dimensional strict-segment comparison. The common-endpoint branch is already visible in the case overview and introduces no new point or carrier. Likewise SSS congruence, triangle-name permutations, multiset reassociation, conversion to equicomplementability, and final transitivity are deliberately left without diagrams: they change the proof representation, not the geometry.

35.11 Dependency Summary

The classical dependency is

$$\text{I.34} + \text{I.29} + \text{I.4} + \text{Common Notions 1–3} \longrightarrow \text{I.35.}$$

The actual Lean dependency is better represented as

Hilbert incidence + order + congruence + Euclidean parallel theory
 $\Downarrow$
 HilbertInterface
 $\Downarrow$
 parallelogram congruence and diagonal theory
 + betweenness / Pasch / line separation
 $\Downarrow$
 HilbertScissors.lean : generic formal scissors calculus
 $\Downarrow$
 Proposition35.lean :
 i35_upper_order
 i35_right_cases_of_common_container
 i35_overlap i35_common_endpoint i35_equicomplementable_core
 $\Downarrow$
 i35_ordered
 $\Downarrow$
 euclid_proposition_35.

New axiom:	no
New formal framework:	yes — HilbertScissors.lean
Proposition-local geometry:	yes — Proposition35.lean
Full lake build re-run in this documentation pass:	no
sorry in Proposition35.lean:	no

35.12 Conclusion

Proposition I.35 is a structural turning point in the formal reconstruction of Book I. The geometry is still elementary, but the proposition introduces an equality of figures which cannot be represented faithfully by segment or angle congruence alone.

Hilbert’s distinction between equidecomposability and equicomplementability provides the correct conceptual guide. The present Lean development follows that guide by introducing a formal scissors calculus rather than a numerical area function. The calculus is intentionally modest: it is a term-level algebra of triangular decompositions, not yet a full semantic theory of polygonal regions.

The proof itself also exposes a classical omission which is invisible on the usual diagram. There are three upper-side configurations, and the auxiliary intersection point G belongs only to the disjoint one. Once this case split is made at the correct level, the proof decomposes naturally into order geometry, triangle congruence, and formal scissors recombination.

The resulting module boundary is now explicit. The generic cut-and-paste calculus lives in the scissors module, while the substantial Hilbertian geometry of I.35 is exposed in `Proposition35.lean`. This is the form needed for the next area propositions, beginning with I.36: they can reuse the scissors language without obscuring their own proposition-specific geometry.

Chapter 36

Proposition I.36

36.1 Introduction

Proposition I.36 is the first immediate reuse of the content machinery introduced at I.35. Euclid passes from parallelograms on the same base to parallelograms on equal bases. The classical proof constructs an intermediate parallelogram and then applies I.35 twice.

The present reconstruction keeps the same content architecture but changes the geometric route to the intermediate parallelogram. The final Lean theorem does not import Proposition I.33 and does not require an explicit crossing point. Instead it reconstructs the missing pair of parallel sides from the developed Hilbert theory.

The conclusion remains synthetic. No numerical area is introduced. The two parallelogram terms are proved formally equicomplementable in the scissors calculus already used for I.35.

36.2 The Classical Configuration

Let $ABCD$ and $EFGH$ be parallelograms on equal bases BC and FG , lying between the same two parallels. In the standard ordered configuration the points satisfy

$$A - D - E - H, \quad B - C - F - G.$$

The bases satisfy

$$BC \cong FG.$$

The current Lean theorem makes this order explicit through four strict betweenness hypotheses:

$$A - D - E, \quad D - E - H, \quad B - C - F, \quad C - F - G.$$

Thus the formal statement is configuration-faithful to the standard ordered diagram, but it has stronger positional hypotheses than Euclid's order-free wording.

36.3 Classical Proof

Euclid first uses Proposition I.34 on the parallelogram $EFGH$:

$$EH \cong FG.$$

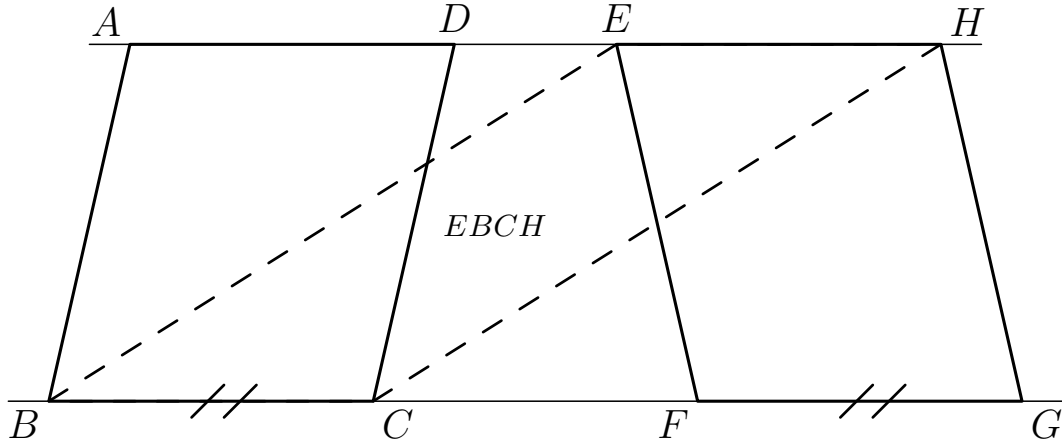


Figure 36.1: The ordered configuration used by the Lean theorem. The dashed quadrilateral $EBCH$ is the intermediate parallelogram constructed internally by the reconstruction.

Since the bases are equal,

$$BC \cong FG,$$

Common Notion 1 gives

$$BC \cong EH.$$

The segments BC and EH are also parallel. Proposition I.33 is then used to conclude that the joining segments EB and HC are equal and parallel, hence $EBCH$ is a parallelogram.

Now $ABCD$ and $EBCH$ are parallelograms on the same base BC and in the same parallels, so Proposition I.35 gives

$$ABCD = EBCH.$$

Similarly, $EFGH$ and $EBCH$ are equal by I.35. Common Notion 1 therefore yields

$$ABCD = EFGH.$$

The classical dependency graph is

$$\begin{array}{c}
 \text{I.34} + BC \cong FG \\
 \Downarrow \\
 BC \cong EH \\
 \Downarrow \quad \text{I.33} \\
 EBCH \text{ is a parallelogram} \\
 \Downarrow \quad \text{I.35 twice} \\
 ABCD = EBCH = EFGH \\
 \Downarrow \quad \text{C.N.1} \\
 ABCD = EFGH.
 \end{array}$$

This is exactly the role of I.36 in the classical sequence: it is a generalization of I.35 obtained by inserting one intermediate parallelogram.

36.4 The Main Issue: Constructing the Intermediate Parallelogram

The difficult part of the formal reconstruction is not the content transitivity. Once $EBCH$ has been obtained, I.35 already supplies the required equality of figures. The real issue is geometric:

construct $EBCH$ from the ordered I.36 data.

The final proof does this without Proposition I.33 and without introducing a crossing witness.

36.4.1 The upper and lower parallel sides

The helper

```
i36_equal_parallel_EH_BC
```

extracts two facts:

$$EH \parallel BC, \quad EH \cong BC.$$

Parallelism is transported from the side FG of $EFGH$ to the collinear carrier BC . Congruence comes from opposite-side congruence in the parallelogram together with the hypothesis $BC \cong FG$.

36.4.2 The shifted bottom segments

From

$$B - C - F - G, \quad BC \cong FG,$$

the helper

```
i36_bottom_shift
```

derives

$$BF \cong CG$$

by segment additivity. This is the metric relation needed for the later SAS comparison.

36.4.3 The corresponding angle

The helper

```
i36_corresponding_angle
```

proves

$$\angle EFB \cong \angle HGC.$$

This is where the formal proof exposes the side-of-line data hidden by the ordinary diagram. A point X is constructed with

$$E - F - X.$$

The line through F and G is used as a carrier. The proof establishes the required SameSide and OppositeSide relations, transports $EF \parallel GH$ to $FX \parallel GH$, and then applies the developed Hilbert theorem

```
hilbert_alternate_angles_of_parallel_
oppositeSide_lines
```

to obtain an alternate-angle congruence. Same-ray normalization and vertical angles convert that statement to the required corresponding-angle form.

36.4.4 SAS across the two bases

The helper

```
i36_cross_triangles_congruent
```

compares the triangles

$$\triangle FEB \quad \text{and} \quad \triangle GHC.$$

The three inputs are

$$FE \cong GH, \quad FB \cong GC, \quad \angle EFB \cong \angle HGC.$$

SAS therefore produces a full `TriangleCongruenceResult`. In particular,

$$\angle FBE \cong \angle GCH.$$

36.4.5 Recovering the second pair of parallel sides

The helper

```
i36_cross_parallel
```

uses the angle obtained from SAS to prove

$$EB \parallel HC.$$

Choose T with

$$B - T - C$$

and extend HC beyond C to a point Y :

$$H - C - Y.$$

The bottom carrier gives the required line-incidence data. Because E and H lie on the same side of that carrier, while H and Y lie on opposite sides, one obtains that E and Y are on opposite sides.

The angle from SAS is normalized to the rays through T and Y . Vertical angles at C give

$$\angle TBE \cong \angle TCY.$$

The neutral converse alternate-angle criterion

```
hilbert_parallel_of_alternate_angles_
oppositeSide_lines
```

then yields

$$BE \parallel CY.$$

Finally parallelism is transported along the collinear triple H, C, Y :

$$EB \parallel HC.$$

The helper

```
i36_intermediate_EBCH_no_crossing
```

combines

$$EH \parallel BC \quad \text{and} \quad EB \parallel HC$$

and returns

$$\text{IsParallelogram}(E, B, C, H).$$

36.5 Proof Architecture of the Reconstruction

After separating geometry from content, the final proof has a short top-level shape.

The geometric branch is

$$\begin{array}{c} BC \cong FG, \quad EFGH, \quad B - C - F - G \\ \Downarrow \\ EH \parallel BC, \quad EH \cong BC \\ \Downarrow \\ BF \cong CG \\ \Downarrow \\ \angle EFB \cong \angle HGC \\ \Downarrow \text{SAS} \\ \angle FBE \cong \angle GCH \\ \Downarrow \\ EB \parallel HC \\ \Downarrow \\ EBCH \text{ is a parallelogram.} \end{array}$$

The content branch is

$$\begin{array}{c} ABCD, EBCH, EFGH \\ \Downarrow \text{I.35} \\ ABCD \sim EBCH \\ \Downarrow \text{I.35} \\ EBCH \sim EFGH \\ \Downarrow \text{transitivity} \\ ABCD \sim EFGH. \end{array}$$

Here $\sim$ denotes the formal `HilbertScissorsEquicomplementable` relation.

This separation is significant. The geometry of I.36 is new local work, but the equality-of-figures layer is inherited from I.35.

36.6 Lean Formalization

The current file imports only Proposition I.35:

```
import CGJteamLab.Proposition35
```

The public theorem is:

```
theorem euclid_proposition_36
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D E F G H : Geo.Point)
  (hABCD : IsParallelogram Geo A B C D)
  (hEFGH : IsParallelogram Geo E F G H)
  (hBCFG : Geo.Congruent B C F G)
  (hADE : Geo.Between A D E)
  (hDEH : Geo.Between D E H)
  (hBCF : Geo.Between B C F)
  (hCFG : Geo.Between C F G) :
  HilbertScissorsEquicomplementable Geo
    (hilbertParallelogramTerm Geo A B C D)
    (hilbertParallelogramTerm Geo E F G H)
```

The local helper sequence is:

```
i36_equal_parallel_EH_BC
i36_bottom_shift
i36_corresponding_angle
i36_cross_triangles_congruent
i36_cross_parallel
i36_intermediate_EBCH_no_crossing
i36_from_intermediate
euclid_proposition_36
```

The final theorem first derives the outer order information

$$A - E - H, \quad A - D - H,$$

and

$$B - F - G, \quad B - C - G$$

from the four explicit betweenness hypotheses. It then constructs $EBCH$ through the no-crossing helper and sends the result to the content helper.

The helper

```
i36_from_intermediate
```

is the exact content bridge. It applies I.35 once to $ABCD$ and $EBCH$. For the second application it reverses the parallelogram representations so that $EBCH$ and $EFGH$ are presented on the common base EH . The formal reversal is converted into scissors equivalence and then into equicomplementability. Two transitivity steps finish the proof.

36.7 Relationship with Earlier Propositions

36.7.1 Proposition I.33

I.33 is part of Euclid’s classical DAG. After obtaining $BC \cong EH$ and $BC \parallel EH$, Euclid uses I.33 to conclude that the joining segments form the intermediate parallelogram $EBCH$.

The final Lean proof does not import Proposition I.33. Instead it proves $EB \parallel HC$ directly by the SAS and alternate-angle route described above. Thus I.33 remains historically relevant but is not a formal dependency of the final file.

36.7.2 Proposition I.34

Classically, I.34 supplies

$$EH \cong FG.$$

The current Lean proof accesses this content through the developed parallelogram API, in particular:

```
ParallelogramOppositeSidesCongruent
```

Thus the mathematical input of I.34 is present even though the Book I theorem is not called by name.

36.7.3 Proposition I.35

I.35 is the decisive content dependency. Once $EBCH$ has been constructed, the final result is obtained by two applications of I.35 plus transitivity of equicomplementability.

This agrees with the classical organization and with modern content-based treatments of I.36.

36.8 Hilbert and Book Zero Components Used

The proof uses three layers of developed infrastructure.

First, the Hilbert geometry layer supplies:

- parallelogram opposite-side congruence;
- transport of parallelism along collinear carriers;
- betweenness extension and inner/outer transitivity;
- SameRay normalization of angles;
- SameSide and OppositeSide transport;
- vertical angles;
- SAS triangle congruence;
- the Euclidean theorem giving equal alternate angles from parallel lines;

- the neutral converse criterion giving parallel lines from equal alternate angles.

Second, the Book Zero layer supplies elementary order and congruence infrastructure used by the developed Hilbert interface. Proposition I.36 does not introduce a new Book Zero theorem.

Third, the scissors layer inherited from I.35 supplies:

- parallelogram terms;
- representation reversal;
- conversion from scissors equivalence to equicomplementability;
- transitivity of equicomplementability.

No polygonal realization layer is introduced. The existing `HilbertScissorsTerm` representation is sufficient.

36.9 Formalization Notes

36.9.1 Geometry, content, and representation

I.36 is a useful example of the three-way separation introduced after I.35.

Geometry. The new work is the intrinsic construction of $EBCH$ from equal bases, parallel carriers, and the ordered configuration.

Content. No new equality-of-figures principle is required. I.35 and transitivity are sufficient.

Representation. The only extra bookkeeping is reversal of parallelogram terms before the second application of I.35. This is handled by existing scissors-equivalence helpers.

36.9.2 Hidden configuration data

The formal theorem exposes several facts left implicit by the classical diagram:

- the upper order $A - D - E - H$;
- the lower order $B - C - F - G$;
- the carrier line through F and G ;
- SameSide information for E and H relative to that carrier;
- the extension point X with $E - F - X$;
- the interior point T with $B - T - C$;
- the extension point Y with $H - C - Y$;
- the required non-collinearity conditions for genuine angles;

- the OppositeSide condition used by the alternate-angle criterion.

These data are derived or constructed internally except for the four betweenness hypotheses in the theorem statement.

36.9.3 Axiomatic status

The public theorem assumes

```
[HilbertEuclideanPlane Geo]
```

and is therefore Euclidean in the current project API.

The local geometric route is informative about where the parallel axiom is used. The helper proving the corresponding angle invokes

```
hilbert_alternate_angles_of_parallel_
oppositeSide_lines
```

which is the Euclidean direction

parallel lines $\implies$ equal alternate angles.

By contrast, the final step that recovers $EB \parallel HC$ invokes

```
hilbert_parallel_of_alternate_angles_
oppositeSide_lines
```

which is the neutral converse direction.

Hilbert's Chapter IV theory of plane area works over Groups I–IV and does not require the continuity group. The formal reconstruction likewise introduces no continuity axiom at I.36.

36.9.4 Axiom audit

The final audit is:

```
'Geometry.euclid_proposition_36' depends on axioms:
[propext, Classical.choice, bookZero_nullSegment1, Quot.sound]
```

The geometric helpers have the smaller audit:

```
i36_cross_parallel:
[propext, Classical.choice, Quot.sound]

i36_intermediate_EBCH_no_crossing:
[propext, Classical.choice, Quot.sound]
```

The local project axiom

```
bookZero_nullSegment1
```

enters only through Proposition I.35 and hence through the content helper. The new I.36 geometry introduces no additional local axiom.

36.9.5 Source audit

The source audit gives a consistent picture.

Euclid/Joyce and Wilkins give the classical DAG

$$\text{I.34} \rightarrow \text{I.33} \rightarrow \text{I.35 twice} \rightarrow \text{C.N.1.}$$

Hilbert's Chapter IV, Section 19, Theorem 44 identifies equicomplementability of parallelograms with equal bases and equal altitudes as the correct synthetic area-theoretic target. It also warns against silently strengthening the result to unconditional equidecomposability in non-Archimedean settings.

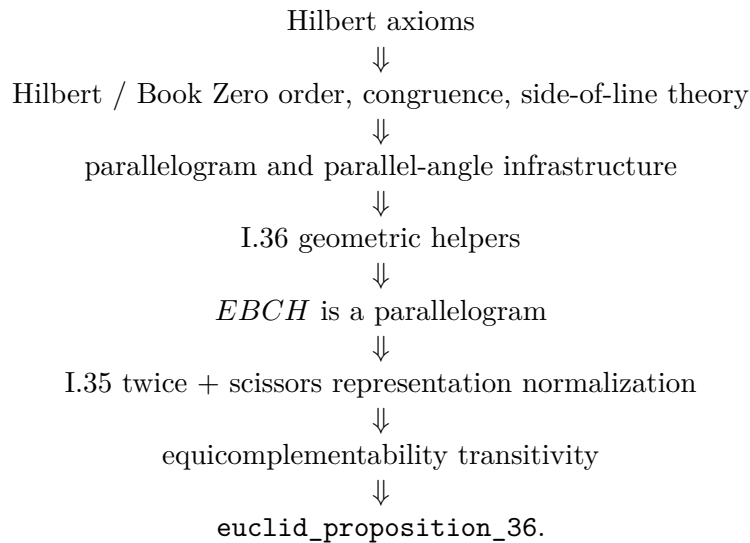
Hartshorne's Section 22 states that I.36 follows from transitivity of equal content. The additional property commonly denoted (Z) is not required for I.36; it first becomes relevant in the discussion of I.37.

The Beeson proof corpus was used as an audit of hidden orientation and crossing data. The final Lean proof deliberately does not reproduce its case split: the intermediate parallelogram is derived intrinsically instead.

Borsuk-Szmielew and Greenberg remain control sources for order, parallelograms, neutral geometry, and the neutral/Euclidean boundary, but they do not determine the content architecture of the final proof.

36.10 Dependency Summary

The complete formal dependency chain is



The short comparison is

Euclid : $\text{I.34} \rightarrow \text{I.33} \rightarrow \text{I.35 twice}$,

Lean : $\text{equal-base geometry} \rightarrow \text{SAS} \rightarrow \text{alternate-angle parallelism} \rightarrow \text{\textit{EBCH}} \rightarrow \text{I.35 twice}$.

New axiom:	no
New proposition-local helpers:	yes
New Book Zero theorem:	no
New polygon/content representation:	no
Build clean:	yes
sorry:	no

36.11 Conclusion

Proposition I.36 confirms that the scissors layer introduced at I.35 is already reusable. No new notion of area, content, or polygon is needed. The content argument is exactly the expected transitivity pattern through an intermediate parallelogram.

The principal formal contribution lies instead in the geometry of that intermediate figure. The final reconstruction removes both an explicit crossing witness and the Book I dependency on I.33. Equal-base information, parallel-line angle theory, SAS, side-of-line control, and a neutral alternate-angle criterion are sufficient to recover $EBC H$ directly.

Thus I.36 separates cleanly into a new geometric normalization stage followed by an already established I.35 content stage.

Chapter 37

Proposition I.37

37.1 Introduction

Proposition I.37 is the first triangle theorem in Euclid’s area block. Two triangles have the same base BC , while their third vertices A and D lie on one line parallel to BC . Euclid concludes that the two triangles are “equal.” At this point of Book I, as already in I.35 and I.36, this is an equality of content rather than congruence of triangles.

The formal reconstruction keeps that conclusion synthetic. It introduces no numerical area and does not strengthen the result to unconditional finite equidecomposability. The public Lean theorem concludes `HilbertScissorsEquicomplementable`.

The principal change from Euclid’s proof is the treatment of the final “halving” step. Euclid surrounds each triangle by a parallelogram, proves the two parallelograms equal by I.35, and then appeals to the fact that halves of equals are equal. The present reconstruction does not add a general cancellation or division-by-two principle. Instead it follows Hilbert’s Chapter IV organization: each triangle is first converted, by the midpoint construction of Hilbert’s Theorem 45, into an equidecomposable parallelogram; those two parallelograms are then compared by the already formalized I.35 scissors machinery.

This makes I.37 a particularly clear example of the three layers that now have to be kept distinct:

$$\text{geometry} \longrightarrow \text{representation} \longrightarrow \text{content}.$$

37.2 The Classical Configuration

Let ABC and DBC be triangles on the same base BC . Their top vertices A and D lie on a line parallel to BC :

$$AD \parallel BC.$$

Euclid says that the two triangles are in the same parallels AD and BC .

The current Lean theorem encodes the configuration with the single strict parallelism hypothesis

$$\text{Parallel}(A, D; B, C).$$

Because the project’s parallel relation is nondegenerate, this already gives $A \neq D$ and $B \neq C$, and the disjointness of the two carrier lines forces both A and D off the base line. Thus the two triangles are internally proved noncollinear; nondegeneracy is not an extra theorem hypothesis.

The formal proof then constructs four midpoint witnesses:

$$\begin{array}{ll} M \text{ midpoint of } BA, & N \text{ midpoint of } CA, \\ P \text{ midpoint of } BD, & Q \text{ midpoint of } CD. \end{array}$$

Hilbert's Theorem 45 is applied to the two triangles and produces points F and G such that

$$MBCF \quad \text{and} \quad PBCG$$

are parallelograms, while the original triangles are scissors-equivalent to these parallelogram terms.

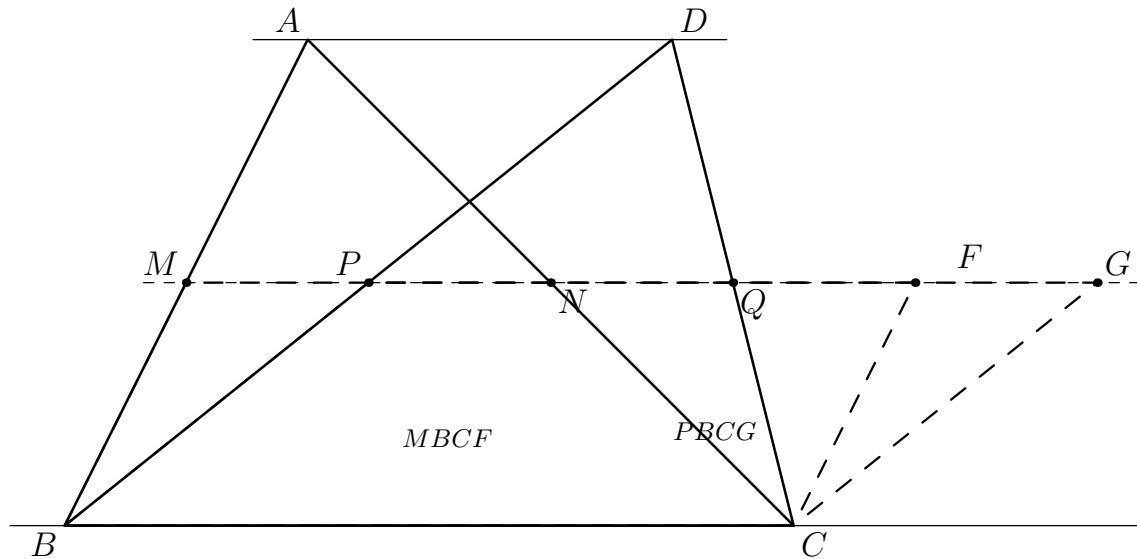


Figure 37.1: The geometric reduction used by the Lean proof. The solid triangles ABC and DBC are the figures in the public theorem. The points M, N, P, Q are the four midpoint witnesses used by two applications of Hilbert's Theorem 45; the dashed figures $MBCF$ and $PBCG$ are the resulting parallelograms. The displayed order $M - P - F - G$ is one representative strict case only. The proof derives the common upper carrier and handles all possible orders rather than assuming the order shown in the picture.

37.3 Classical Proof

Euclid extends the top line AD in both directions to points E and F . Through B he draws a line parallel to CA , and through C a line parallel to BD , using I.31. This produces two parallelograms

$$EBCA \quad \text{and} \quad DBCF.$$

They have the same base BC and lie in the same parallels, so I.35 gives

$$EBCA = DBCF.$$

By I.34 the diagonal AB bisects the first parallelogram into two congruent triangles, and the diagonal DC bisects the second. Hence

$$\triangle ABC \text{ is half of } EBCA, \quad \triangle DCB \text{ is half of } DBCF.$$

Euclid then concludes that the two halves are equal.

The classical dependency graph is therefore

$$\begin{array}{c}
 AD \parallel BC \\
 \Downarrow \text{ I.31 twice} \\
 EBCA, DBCF \text{ parallelograms} \\
 \Downarrow \text{ I.35} \\
 EBCA = DBCF \\
 \Downarrow \text{ I.34 twice} \\
 \triangle ABC, \triangle DBC \text{ are corresponding halves} \\
 \Downarrow \text{ "halves of equals are equal"} \\
 \triangle ABC = \triangle DBC.
 \end{array}$$

The final step is the important foundational point. In Heath’s presentation the principle “halves of equal things are equal” is bracketed: it is not one of Euclid’s explicitly stated Common Notions. A formal development must therefore either justify a suitable cancellation principle or avoid the step. The present reconstruction takes the second route.

37.4 Hilbert’s Theorems 44–46 and the Halving Issue

Hilbert’s Chapter IV gives a direct synthetic organization of exactly the material needed here.

Theorem 44. Parallelograms with equal bases and equal altitudes are *equicomplementable*.

Theorem 45. Every triangle is *equidecomposable* with a parallelogram having an equal base and half the altitude. Hilbert’s construction bisects two sides of the triangle and extends the segment joining the midpoints by an equal amount. The two small triangles created by the construction are congruent, so the triangle can be rearranged into the parallelogram.

Theorem 46. From Theorems 44 and 45, together with transitivity of the content relation, Hilbert obtains that triangles with equal bases and equal altitudes are equicomplementable.

This is the correct control theorem for I.37. Hilbert also explicitly warns that the stronger assertion of finite equidecomposability is not available in full generality without an Archimedean hypothesis: in non-Archimedean geometries triangles of equal base and altitude can be equicomplementable without being equidecomposable.

Hartshorne makes the same logical issue visible in the language of equal content. In Section 22 he observes that Euclid’s original proof of I.37 uses “halves of equals are equal,” a property depending on his additional property (Z). He therefore gives a midpoint/parallelogram proof of I.37 which avoids (Z).

The Lean reconstruction follows the same conceptual escape from cancellation, but its exact proof graph is Hilbertian: Theorem 45 is formalized directly in the scissors calculus, and I.35 supplies the parallelogram content comparison. No general theorem of the form

$$2P \sim 2Q \implies P \sim Q$$

is introduced.

37.5 Proof Architecture of the Reconstruction

37.5.1 Step 1: nondegeneracy from the parallel hypothesis

The helper

```
i37_nondegenerate_triangles
```

derives

$$A, B, C \text{ noncollinear}, \quad D, B, C \text{ noncollinear}, \quad B, A, D \text{ noncollinear}.$$

The proof uses the disjoint carriers contained in the strict parallelism hypothesis. Thus the ordinary diagrammatic statement that the top vertices are off the base is converted into explicit Hilbert incidence information.

37.5.2 Step 2: four midpoint witnesses and Hilbert 45 twice

The helper

```
i37_half_parallelgrams
```

constructs M, N, P, Q as the four midpoints shown in Figure 37.1. It then applies

```
hilbert_triangle_to_parallelogram
```

twice.

For triangle ABC the internal call is made in the cyclic presentation BCA and returns a point F with

$$BCFM \text{ a parallelogram}$$

and a formal scissors equality between the triangle and that parallelogram. For triangle DBC the analogous construction returns G and the parallelogram $BCGP$.

This theorem is the formal version of Hilbert's Theorem 45. No area value is computed. The result is a finite triangle-rearrangement certificate in `HilbertScissorsEq`.

37.5.3 Step 3: the common upper carrier

The midpoint theorem is applied in triangle BAD . Since M is the midpoint of BA and P is the midpoint of BD ,

$$MP \parallel AD.$$

The public hypothesis gives $AD \parallel BC$, and the developed Euclidean parallel theory yields

$$MP \parallel BC.$$

The two Hilbert-45 parallelograms also give

$$FM \parallel BC, \quad GP \parallel BC.$$

Uniqueness/transitivity of parallels then identifies their carriers with the carrier MP . Hence

$$M, F, P \text{ are collinear,} \quad M, P, G \text{ are collinear.}$$

This is packaged by

```
i37_upper_line
```

and is one of the most important hidden configuration facts of the proof.

37.5.4 Step 4: equal upper sides

Opposite-side congruence in the two parallelograms gives

$$MF \cong BC, \quad PG \cong BC.$$

Therefore

$$MF \cong PG.$$

The helper

```
i37_upper_sides_congruent
```

packages this short metric step.

37.5.5 Step 5: representation normalization

Hilbert 45 naturally returns parallelograms in the forms $BCFM$ and $BCGP$. Proposition I.35 is most conveniently applied to the cyclic presentations $MBCF$ and $PBCG$, in which BC is the common base and the upper sides are MF and PG .

The helper

```
i37_parallelogram_reduction
```

performs precisely this normalization. The geometric parallelograms are rotated by

```
ParallelogramRotateOne
```

and the corresponding formal scissors terms are transported by

```
parallelogram_term_rotateOne
```

This step changes no geometry; it only changes the representation expected by the downstream API.

At the end of the reduction we have points M, P, F, G satisfying

$$\begin{aligned} &MBCF, PBCG \text{ parallelograms,} \\ &MP \parallel BC, \\ &M, F, P \text{ collinear,} \quad M, P, G \text{ collinear,} \\ &MF \cong PG, \end{aligned}$$

together with scissors equivalences

$$\triangle ABC \sim MBCF, \quad \triangle DBC \sim PBCG.$$

Here $\sim$ in this display denotes `HilbertScissorsEq`. The stronger content relation used later is

`HilbertScissorsEquicomplementable`

37.5.6 Step 6: order normalization on the common upper line

Proposition I.35 is sensitive to the actual order of the four upper points. The formal proof therefore does not read that order from Figure 37.1. It first handles two boundary configurations:

$$M = P, \quad F = P.$$

If $M = P$, uniqueness of the fourth vertex of a parallelogram gives $F = G$, so the two reduced parallelograms coincide. This is handled by

`i37_finish_same_left_vertex`

using the reusable theorem

`ParallelogramFourthVertexUnique`

If $F = P$, the two parallelograms have a common upper endpoint, and the existing I.35 common-endpoint helper closes the content comparison. This is

`i37_finish_common_upper_endpoint`

In the remaining case, Hilbert's betweenness trichotomy gives exactly three strict orders of M, F, P :

$$M - F - P, \quad F - M - P, \quad M - P - F.$$

The helper

`i37_upper_trichotomy`

packages this split.

37.5.7 Step 7: the orientation lemma for the strict cases

The case $M - F - P$ contains the deepest proposition-local order argument. The helper

`i37_order_MFP_implies_FPG`

proves

$$M - F - P \implies F - P - G.$$

The proof is genuinely synthetic. In triangle PMB , the line CF enters through the side PM at F . Since $CF \parallel MB$, it cannot leave through MB , so Pasch forces an intersection with PB . The

existing helper `hilbert_collinear_between_of_parallel` locates this crossing on FC . The second parallelogram supplies SameSide information for C and G relative to the line PB . Hence F and G are on opposite sides of PB . Their segment must cross PB , and incidence uniqueness identifies that crossing point with P . Thus $F - P - G$.

This is exactly the sort of positional statement that is obvious in a well-drawn affine picture but cannot be taken from the picture in a Hilbert formalization.

The case $F - M - P$ is closed by reversing the two parallelogram presentations and applying I.35 in the reversed configuration. The helper is

```
i37_finish_order_FMP
```

For the case $M - P - F$, the helper

```
i37_order_MPF_implies_PFG
```

uses the congruence $MF \cong PG$, strict segment comparison, and a reuse of the previous orientation lemma to eliminate the two impossible orders of P, F, G . The surviving order is

$$P - F - G.$$

The final content step for this branch is packaged in

```
i37_finish_order_MPF
```

37.5.8 Step 8: Proposition I.35 and transport back to the triangles

Once the correct upper order is known, Proposition I.35 compares the two reduced parallelograms:

$$MBCF \quad \text{and} \quad PBCG.$$

The resulting relation is `HilbertScissorsEquicomplementable`.

Finally

```
equicomplementable_transport
```

uses the two Hilbert-45 scissors equivalences to move the conclusion from the parallelogram representatives back to the original triangles. This is the content stage of the proof; the preceding midpoint, parallel, Pasch, and order arguments belong to the geometry stage.

37.6 Lean Formalization

The current file imports

```
import CGJteamLab.Proposition35
import CGJteamLab.MidsegmentParallel
```

Thus I.35 is the direct Book I content dependency, while the midpoint theorem is imported as developed Hilbert infrastructure. Proposition I.36 is not a formal dependency of I.37.

The public theorem is

```

theorem euclid_proposition_37
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D : Geo.Point)
  (hAD_BC : Geo.Parallel A D B C) :
  HilbertScissorsEquicomplementable Geo
    (hilbertScissorsTriangle Geo A B C)
    (hilbertScissorsTriangle Geo D B C)

```

The proposition-local proof path is represented by the following declarations:

```

i37_nondegenerate_triangles
i37_half_parallelograms
i37_upper_line
i37_upper_sides_congruent
i37_parallelogram_reduction
i37_finish_from_right_cases
i37_finish_same_left_vertex
i37_finish_common_upper_endpoint
i37_upper_trichotomy
i37_finish_order_FMP
i37_finish_order_MFP
i37_order_MFP_implies_FPG
i37_order_MPF_implies_PFG
i37_finish_order_MPF
euclid_proposition_37

```

At the top level the final theorem is intentionally short. It first calls

```
i37_parallelogram_reduction
```

then separates the two boundary cases, uses

```
i37_upper_trichotomy
```

for the three strict cases, and dispatches each case to its corresponding finishing helper.

The crucial point is that the top-level theorem contains no hidden arithmetic area operation and no cancellation axiom. The only equality-of-figures relations in the statement and final transport are the formal scissors relations already introduced at I.35.

37.7 Relationship with Earlier Propositions

37.7.1 Proposition I.31

I.31 belongs to Euclid's literal construction. Euclid draws two parallels in order to surround the two triangles by parallelograms. The final Lean proof does not call Proposition I.31 by name. Its construction is replaced by Hilbert's midpoint theorem and Hilbert 45.

37.7.2 Proposition I.34

Classically I.34 supplies the fact that a diagonal bisects a parallelogram and therefore that each original triangle is one half of its surrounding parallelogram. This is precisely the source of the final halving issue.

The Lean reconstruction does not use I.34 as a Book I theorem to divide equal parallelograms by two. Its relevant parallelogram congruence content is already available in the developed interface, while Hilbert 45 provides a direct scissors equivalence between each triangle and a smaller parallelogram.

37.7.3 Proposition I.35

I.35 is the decisive earlier proposition in the actual Lean dependency graph. After both triangles have been converted to parallelogram terms and their upper-order configuration has been normalized, I.35 proves those parallelograms equicomplementable. The result is then transported back to the triangles.

Thus the common core of Euclid and Lean is still I.35, but the bridge from triangles to parallelograms is different.

37.7.4 Proposition I.36

I.36 is historically adjacent but is not used by the final formal proof. It compares parallelograms on equal bases, whereas I.37 has a common base from the start. The two propositions nevertheless confirm the same architectural lesson: the content machinery belongs in a reusable scissors layer, while the new work in each proposition is geometric normalization.

37.8 Hilbert and Book Zero Components Used

The proof uses a broad but well-separated collection of reusable results.

37.8.1 Midpoint and midsegment infrastructure

The midpoint layer supplies

- midpoint existence;
- midpoint symmetry;
- conversion of midpoint data into strict betweenness and congruence;
- `hilbert_midpoints_noncollinear`;
- `MidsegmentTheorem`.

These results produce the four midpoint witnesses and the parallel line MP .

37.8.2 Hilbert Theorem 45 in the scissors layer

The reusable theorem

```
hilbert_triangle_to_parallelogram
```

is the central new content bridge. Its geometric construction is provided by

```
hilbert_triangle_parallelogram_data
```

and its scissors conclusion states that the triangle term is `HilbertScissorsEq` to the corresponding parallelogram term.

This theorem was introduced while reconstructing I.37 but belongs in the reusable scissors/interface layer rather than in the proposition-local file.

37.8.3 Parallel and parallelogram infrastructure

The proof uses

- transport of parallelism along collinear carriers;
- Euclidean transitivity/uniqueness of parallels;
- `ParallelogramOppositeSidesCongruent`;
- `ParallelogramFourthVertexUnique`;
- `ParallelogramRotateOne`;
- representation-level parallelogram rotations in the scissors layer.

37.8.4 Order, Pasch, and segment comparison

The strict-case normalization uses

- Hilbert betweenness trichotomy;
- inner and outer betweenness transitivity;
- Pasch's axiom through the derived intersection machinery;
- `SameSide` and `OppositeSide` transport;
- incidence uniqueness of the intersection of two distinct lines;
- strict segment comparison and its transport under congruence.

No new Book Zero theorem is introduced by the final Proposition file. The proof instead exercises the existing Book Zero/order language and adds several reusable helpers to the Interface and Scissors layers.

37.9 Formalization Notes

37.9.1 Geometry, representation, and content

I.37 makes the three-layer architecture especially explicit.

Geometry. The geometric work constructs midpoint data, obtains two parallelograms, proves that their upper sides lie on one common carrier, and determines the possible point orders on that carrier. Pasch and side-of-line arguments occur only here.

Representation. The same geometric parallelogram can appear as *BCFM*, *MBCF*, *FCBM*, and other cyclic or reversed presentations. Lean cannot silently identify these argument lists. Helpers such as `ParallelogramRotateOne`, `parallelogram_term_rotateOne`, and the reversal lemmas normalize the presentation without adding geometric content.

Content. Once the representatives are aligned, I.35 and `equicomplementable_transport` are sufficient. No new area function, polygon semantics, or cancellation rule is introduced.

37.9.2 Hidden configuration data

The proof explicitly derives or manages facts that the classical diagram suppresses:

- noncollinearity of both original triangles;
- four strict midpoint betweenness relations;
- the midpoint parallel $MP \parallel AD$;
- the derived common upper carrier $MP \parallel BC$;
- collinearity of M, F, P, G on that carrier;
- congruence $MF \cong PG$;
- the boundary cases $M = P$ and $F = P$;
- the three strict orders $M - F - P$, $F - M - P$, $M - P - F$;
- Pasch data in triangle PMB ;
- SameSide/OppositeSide information relative to PB ;
- the exact crossing point used to prove $F - P - G$;
- strict segment comparisons used to eliminate impossible orders in the $M - P - F$ branch.

None of these order relations is inferred merely from the chosen drawing.

37.9.3 Axiomatic status

The public theorem assumes

```
[HilbertEuclideanPlane Geo]
```

and is therefore Euclidean in the current project API.

The Euclidean parallel axiom enters through already developed infrastructure, not through a new I.37-specific axiom. In particular, the proof uses the midsegment module and Euclidean transitivity/uniqueness of parallels, and it calls Proposition I.35, whose parallelogram/content infrastructure is also exposed under the Euclidean-plane instance.

This documentation does not claim that every internal sublemma has the minimal possible axiom signature. The relevant project-level statement is that the completed I.37 theorem is derived from the existing `HilbertEuclideanPlane` hierarchy and introduces no new geometric axiom. No continuity axiom is added at I.37.

37.9.4 Why the conclusion is equicomplementability

The conclusion is deliberately

```
HilbertScissorsEquicomplementable
```

rather than `HilbertScissorsEq`. This matches Hilbert's Theorem 46 and avoids an invalid strengthening in non-Archimedean settings. The individual Hilbert-45 reductions are scissors equivalences, but the comparison of the two resulting parallelograms comes from I.35 and therefore has the weaker, correct content relation.

37.9.5 Reusable helpers introduced around I.37

The formalization produced reusable material beyond the proposition-local case analysis.

In `HilbertInterface`:

```
hilbert_triangle_parallelogram_data
hilbert_midpoints_noncollinear
ParallelogramFourthVertexUnique
ParallelogramRotateOne
```

In `HilbertScissors`:

```
hilbert_triangle_to_parallelogram
equicomplementable_transport
parallelogram_term_rotateOne
```

The declarations beginning with `i37_` remain proposition-local because they package the particular reduction and order analysis required by I.37.

37.9.6 Source audit

The source audit separates four functions of the literature.

Euclid, Heath, and Wilkins control the historical statement and classical DAG: I.31 constructs the two parallelograms, I.35 compares them, I.34 identifies the triangles as halves, and the proof ends with the unstated halving principle.

Hilbert, Chapter IV, Sections 18–19, controls the semantics and the preferred formal architecture. Theorems 44–46 distinguish equidecomposability from equicomplementability and supply the midpoint route from triangles to parallelograms.

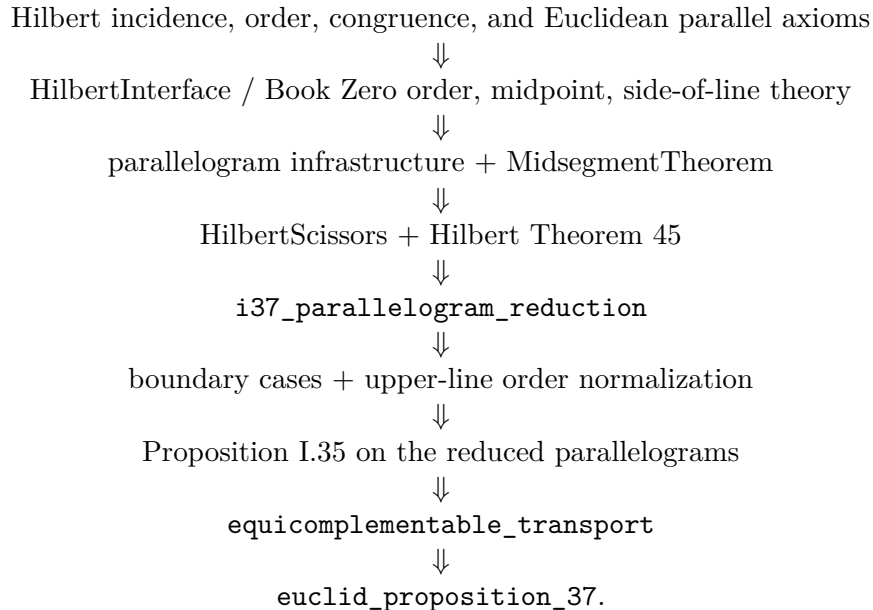
Hartshorne, Section 22, independently identifies the halving step as depending on property (Z) and gives a proof of I.37 that avoids (Z) by using midpoints and parallelograms. His treatment also confirms that equal content must not be silently replaced by numerical area or by unconditional finite equidecomposability.

Borsuk–Szmielew and Greenberg remain control sources for the underlying order, midpoint, parallelogram, and Euclidean parallel infrastructure. Neither is the primary source for the equal-content semantics at this stage: that role is filled more directly by Hilbert and Hartshorne.

The Beeson proof corpus remains useful as an audit of hidden configuration conditions and Euclidean gaps, but the final I.37 architecture is not organized around Tarski-style tactics or a literal reproduction of Beeson’s proof.

37.10 Dependency Summary

The actual formal dependency chain is



The historical and formal DAGs can be compared succinctly as

Euclid : I.31 twice $\rightarrow$ I.35 $\rightarrow$ I.34 twice $\rightarrow$ halving,
 Lean : midpoints $\rightarrow$ Hilbert 45 twice $\rightarrow$ order geometry $\rightarrow$ I.35 $\rightarrow$ transport.

New geometric axiom:	no
New proposition-local helpers:	yes
New reusable Interface/Scissors helpers:	yes
New Book Zero theorem:	no
New numerical area or polygon semantics:	no
Public theorem compiles:	yes
Full <code>lake build</code> re-run in this documentation pass:	no
sorry in <code>Proposition37.lean</code> :	no

37.11 Conclusion

Proposition I.37 is not merely I.35 with triangles substituted for parallelograms. It is the point where the classical phrase “half of an equal figure” becomes a genuine foundational obligation.

The reconstruction resolves that obligation without adding a division or cancellation axiom. Hilbert’s Theorem 45 converts each triangle into an appropriate parallelogram by a finite congruent-piece rearrangement. The midsegment theorem and Euclidean parallel theory place the two resulting parallelograms on one common upper carrier. A substantial but localized order analysis then supplies the precise configuration needed by I.35. Finally the existing scissors transport returns the equicomplementability result to the original triangles.

The proof therefore makes the architecture of the area block more precise: geometry determines where the pieces and carriers lie, representation helpers put the same figures into the argument order required by the API, and the scissors layer alone carries equality of content. That separation is likely to be the reusable pattern for the propositions immediately following I.37.

Chapter 38

Proposition I.38

38.1 Introduction

Proposition I.38 continues the triangle part of Euclid’s area block. The two triangles no longer share a base: ABC is on BC and DEF is on EF . The bases are congruent, the four base points are collinear, and the two top vertices lie on a line parallel to the common base carrier. Euclid concludes that the triangles are “equal.” As in I.35–I.37, the relevant equality is synthetic equality of content, not congruence and not a numerical area value.

The formal statement therefore concludes `HilbertScissorsEquicomplementable`. It does not introduce an area function and it does not strengthen the conclusion to unconditional finite scissors equivalence. Hilbert’s Chapter IV is the controlling source for this choice: his Theorem 46 states that triangles with equal bases and equal altitudes are equicomplementable, while he explicitly warns that the stronger equidecomposability assertion cannot be obtained in full generality without an Archimedean hypothesis.

The proof architecture, however, is not a literal formalization of Euclid’s proof. The classical route constructs two surrounding parallelograms, invokes I.36, uses I.34 to identify the triangles as halves, and then cancels the common factor two. The Lean proof deliberately avoids this final cancellation. Instead it constructs a triangle GEF congruent to ABC , proves that G lies on the same upper parallel as A and D , applies I.37 to GEF and DEF , and finishes by transitivity of equicomplementability.

Thus I.38 is a particularly clean realization of Hartshorne’s observation that, once the equal-content theory and I.37 are available, I.38 follows by transitivity. The work hidden by that sentence is geometric: one must build the correct congruent copy and prove that its third vertex has the required position.

38.2 The Classical Configuration

Let ABC and DEF be triangles on equal bases BC and EF . In the configuration used by the formal theorem the base points occur in the strict order

$$B - C - E - F,$$

encoded by

$$\text{Between}(B, C, E), \quad \text{Between}(C, E, F).$$

The equality of bases is represented by segment congruence

$$BC \cong EF.$$

The top vertices A and D determine a line parallel to the base carrier:

$$AD \parallel BC.$$

These hypotheses are stronger than the informal phrase “on equal bases in the same parallels” because they expose the point order needed by the later synthetic argument. In particular, the Lean proof does not read the order of B, C, E, F from a drawing.

Figure 38.1 displays four stages of the proof. Panel (a) is the public configuration. Panel (b) shows the congruent-copy construction. Panel (c) displays the extension points and parallelogram that are hidden in the formal proof of the copy’s position. Panel (d) is the nondegenerate branch in which I.37 is applied. The separate boundary case $G = D$ is handled directly in Lean and is not represented as a second picture.

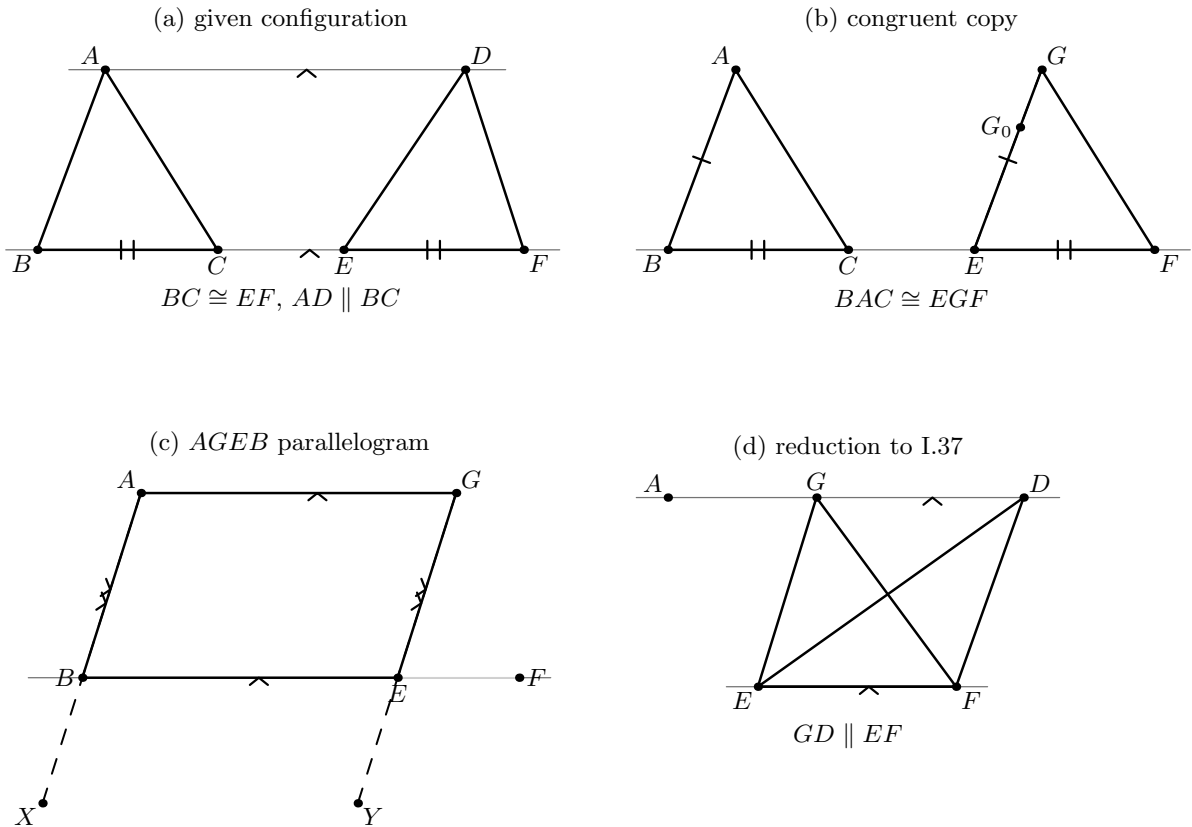


Figure 38.1: The principal geometric stages of the Lean reconstruction of I.38. (a) The given triangles have ordered collinear bases BC and EF , with $BC \cong EF$ and the top carrier parallel to the base carrier. (b) The angle at B is copied at E on the side containing A , and $EG \cong AB$ is laid off on the constructed ray; SAS gives $BAC \cong EGF$. The temporary point G_0 names the ray used for the layoff. (c) Extensions $A - B - X$ and $G - E - Y$ provide the oriented transversal configuration used with I.28; the resulting parallel and congruence data give the parallelogram $AGEB$. (d) The top carrier is identified with AD ; in the branch $G \neq D$ one obtains $GD \parallel EF$, so I.37 compares GEF and DEF .

38.3 Classical Proof

The classical proof follows the same pattern as I.37 but first uses I.36 to compare parallelograms on equal bases. In the standard configuration one constructs parallelograms around the two triangles so that a diagonal of each parallelogram cuts it into two congruent triangles. Wilkins describes the parallelograms as $GBCA$ and $DEFH$.

Because their bases BC and EF are equal and they lie in the same parallels, I.36 gives equality of the two parallelograms. Proposition I.34 then says that each diagonal divides its parallelogram into two congruent triangles. Thus each of the given triangles is one half of one of the equal parallelograms, and the classical conclusion is that the two halves are equal.

The dependency graph is therefore

$$\begin{array}{c}
 BC \cong EF, \quad \text{same parallels} \\
 \Downarrow \quad \text{construct two parallelograms} \\
 GBCA, DEFH \\
 \Downarrow \quad \text{I.36} \\
 GBCA = DEFH \\
 \Downarrow \quad \text{I.34 twice} \\
 \triangle ABC, \triangle DEF \text{ are corresponding halves} \\
 \Downarrow \quad \text{“halves of equals are equal”} \\
 \triangle ABC = \triangle DEF.
 \end{array}$$

This is exactly the route visible in the Beeson proof corpus: the trace for `Prop38.prf` calls Proposition 36, Proposition 34 twice, converts triangle congruence to equality of figures, and ends with an explicit `halvesofequals` axiom. That trace is useful as a dependency audit, but it is not adopted as the architecture of the present proof.

38.4 Equal Bases by Congruent Copy and Transitivity

Hartshorne’s Section 22 gives the key conceptual simplification. After proving a version of I.37 that avoids the problematic halving principle, he states that I.38 follows by transitivity. This is possible because equal content is an equivalence relation and because equidecomposable figures have equal content.

In an informal affine picture, one can see the transitivity route immediately: translate a congruent copy of ABC onto the equal base EF . Its third vertex must lie on the same upper parallel, so I.37 compares that copied triangle with DEF . A Hilbert formalization cannot simply say “translate the triangle,” because translation is not a primitive operation in the project. The copy has to be reconstructed from angle layoff, segment layoff, side-of-line data, SAS, and parallel uniqueness.

The Lean DAG is therefore

$$\begin{array}{c}
BC \cong EF, B - C - E - F, AD \parallel BC \\
\Downarrow \\
\text{construct } G \text{ with } BAC \cong EGF \\
\Downarrow \\
AGEB \text{ is a parallelogram} \\
\Downarrow \\
AG \parallel BC \text{ and } AG \text{ has the same carrier as } AD \\
\Downarrow \\
G = D \quad \text{or} \quad GD \parallel EF \\
\Downarrow \\
\triangle ABC \sim \triangle GEF \\
\triangle GEF \approx \triangle DEF \quad \text{by I.37} \\
\Downarrow \quad \text{transitivity} \\
\triangle ABC \approx \triangle DEF.
\end{array}$$

Here $\sim$ denotes the stronger relation `HilbertScissorsEq` produced by triangle congruence, while $\approx$ denotes `HilbertScissorsEquicomplementable`. The distinction is important: the first step is a genuine congruent replacement, while I.37 supplies only the weaker content relation that is correct in the non-Archimedean setting.

Hilbert’s Theorem 46 supplies a broader control theorem: equal-base, equal-altitude triangles are equicomplementable. I.38 is exactly a Euclidean same-parallel instance of that geometry. The current library does not invoke Theorem 46 as a monolithic theorem; it realizes the equal-base case through the already completed I.37 and a congruent-copy reduction.

38.5 Proof Architecture of the Reconstruction

38.5.1 Step 1: recovering one common base carrier

The helper

`i38_base_carrier`

starts from $AD \parallel BC$ and uses `parallel_endpoints_sameSide` to obtain an actual Hilbert line `base` carrying B and C , together with `SameSide` information for A and D . The two strict betweenness hypotheses then place E and F on that same carrier by incidence transport.

Mathematically, the result packages the statement that the apparently separate segments BC and EF belong to one lower line, while both top vertices are off that line and lie on the same side of it. This is the first place where the formal proof exposes data that the classical phrase “the same parallels” suppresses.

The helper remains proposition-local. It is a useful normalization for this ordered six-point configuration, but it is not promoted to the general interface because its statement is tailored to $B - C - E - F$.

38.5.2 Step 2: nondegeneracy is derived, not assumed

The helper

`i38_nondegenerate`

proves

$$A, B, C \text{ noncollinear}, \quad D, E, F \text{ noncollinear}.$$

Since B, C, E, F lie on the base carrier and SameSide data place A, D off that carrier, both conclusions are incidence consequences of the configuration.

This helper is logically small but important for the next stage. Hilbert's angle construction and SAS require explicit noncollinearity witnesses. The classical diagram treats these as evident; Lean requires them as proof terms. No new geometric hypothesis is added to the public theorem.

38.5.3 Step 3: constructing the congruent copy on the equal base

The central construction is

`i38_equal_base_copy`.

It first uses Hilbert's angle-construction axiom to copy the angle ABC at E , with the fixed ray directed through F and with the new ray chosen on the side of the base containing A . This creates a temporary point G_0 and an angle congruence of the form

$$\angle ABC \cong \angle FEG_0.$$

Next, Hilbert's segment construction lays off on the ray EG_0 a point G satisfying

$$EG \cong AB.$$

The relation `HilbertSameRay` between G_0 and G is then used twice: first to preserve the side-of-line choice, and second to replace G_0 by G in the copied angle. Thus the construction retains the orientation selected by the angle axiom instead of merely obtaining an unoriented congruent angle.

With the public hypothesis $BC \cong EF$, SAS is applied to the triangles in the formal orientation

$$BAC \quad \text{and} \quad EGF.$$

The result is

$$\text{TriangleCongruenceResult}(BAC, EGF).$$

Panel (b) of Figure 38.1 records this stage. The point G_0 is not part of the final geometry; it exists solely to name the constructed ray on which the final segment is laid off.

This helper is proposition-local, but it reveals a reusable proof pattern: angle layoff + segment layoff + SameRay transport + SAS gives a controlled oriented copy of a triangle on a prescribed congruent base.

38.5.4 Step 4: proving that the copy is a parallelogram

The helper

`i38_copy_parallelogram`

contains the most substantial hidden positional argument of I.38. The desired conclusion is

$AGEB$ is a parallelogram.

At an affine level this is the statement that the congruent copy on EF is a translation of the original triangle. The project does not use translation as a primitive notion, so the claim is rebuilt synthetically.

First GE is extended through E to a point Y , giving $G - E - Y$, and AB is extended through B to a point X , giving $A - B - X$. From the common base carrier and the SameSide choice of G , the proof derives the required SameSide/OppositeSide relations for the transversal configuration. The lower order $B - C - E - F$ also yields the derived order $F - E - B$.

The copied angle is normalized from the ray BC to the ray BE using `HilbertSameRay`; then the unordered angle representation is reoriented to exactly the corresponding-angle shape expected by I.28. Proposition I.28 is applied in its neutral corresponding-angle form to obtain parallelism of the extended side lines. Collinearity transport then gives

$$AB \parallel GE.$$

SAS had already supplied the corresponding side congruence

$$AB \cong GE.$$

Together with the fact that A and G lie on the same side of the lower carrier, these data satisfy `OnePairParallelCongruent`. The reusable criterion `OnePairParallelCongruentCriterion` therefore yields

$AGEB$ parallelogram.

Panel (c) of Figure 38.1 shows the extension witnesses X, Y that make this argument formally oriented.

This helper is local to I.38. The underlying one-pair parallelogram criterion is reusable Interface infrastructure and corresponds closely to the stronger parallelogram theorem isolated in the Borsuk–Szmielw development.

38.5.5 Step 5: identifying the upper carrier

The helper

`i38_copy_upper_position`

turns the parallelogram conclusion into the exact hypothesis needed by I.37. From $AGEB$ it obtains

$$AG \parallel EB.$$

Because E, B, C are collinear, parallel transport gives

$$AG \parallel BC.$$

But the public hypothesis already says

$$AD \parallel BC.$$

The crucial point is now uniqueness of the Euclidean parallel. If the extensional point-lines AG and AD were distinct, the developed transitivity result `hilbert_parallel_transitive_distinct` would make them parallel to one another. This is impossible because they both contain A . Hence the two point-line carriers are equal, and G lies on the upper carrier through A, D .

The proof then separates a boundary case that a drawing normally ignores:

$$G = D$$

or

$$G \neq D.$$

If $G = D$, the copied triangle already has the desired top vertex. Otherwise, parallelism is transported along the common upper and lower carriers to obtain

$$GD \parallel EF.$$

Panel (d) of Figure 38.1 depicts this nondegenerate branch.

This step is the clearest point at which the Euclidean uniqueness theory of parallels becomes indispensable. The earlier angle construction, segment construction, side-of-line transport, SAS, and I.28 criterion are neutral.

38.5.6 Step 6: the content proof is now short

The final theorem converts the triangle congruence certificate into `HilbertScissorsEq` using `scissors_congruent`. Because triangle terms are represented as unordered multisets of vertices, two applications of `scissors_triangle_swap12` normalize

$$BAC \sim EGF$$

to

$$ABC \sim GEF.$$

This is a representation step only; it changes no geometry.

The stronger scissors equality is embedded into equicomplementability by the existing helper

$$\text{equicomplementable_of_scissorsEq}.$$

The two position cases then close as follows.

If $G = D$, the copied triangle is already DEF , so the congruent replacement finishes the proof. If $G \neq D$, the parallelism $GD \parallel EF$ is exactly the hypothesis of Proposition I.37 for the two triangles GEF and DEF . Thus

$$GEF \approx DEF.$$

Finally `equicomplementable_trans` composes

$$ABC \approx GEF, \quad GEF \approx DEF$$

to obtain the public conclusion.

This is the content stage of the proof. All of the preceding line, ray, SameSide, extension, and parallel arguments exist solely to put the copied triangle into the configuration in which I.37 can be called.

38.6 Lean Formalization

The current Proposition file imports

```
import CGJteamLab.Proposition28
import CGJteamLab.Proposition37
```

Thus I.28 is the direct neutral parallel criterion used inside the geometric copy argument, while I.37 is the direct content theorem used at the end.

The public theorem is

```
theorem euclid_proposition_38
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D E F : Geo.Point)
  (hAD_BC : Geo.Parallel A D B C)
  (hBCE : Geo.Between B C E)
  (hCEF : Geo.Between C E F)
  (hBC_EF : Geo.Congruent B C E F) :
  HilbertScissorsEquicomplementable Geo
    (hilbertScissorsTriangle Geo A B C)
    (hilbertScissorsTriangle Geo D E F) := by
  ...
```

The ellipsis here is documentation notation only; the repository theorem is fully proved and contains no `sorry`.

The proposition-local proof path is represented by

```
i38_base_carrier
i38_nondegenerate
i38_equal_base_copy
i38_copy_parallelogram
i38_copy_upper_position
euclid_proposition_38
```

The relatively short list should not be mistaken for a trivial proof. The middle three helpers package the substantial geometry: orientation of the copied triangle, the I.28 extension configuration, construction of the parallelogram, and identification of the upper carrier.

At the top level the final theorem is intentionally small. It consumes the geometric package returned by `i38_copy_upper_position`, converts the congruent copy to a scissors equivalence, and dispatches the two position cases to either reflexive transport or I.37 plus transitivity.

38.7 Relationship with Earlier Propositions

38.7.1 Proposition I.28

I.28 is a direct formal dependency. The helper `i38_copy_parallelogram` uses the theorem

`euclid_proposition_28_corresponding`

after explicitly constructing the extension points and the opposite-side configuration required by the transversal theorem. This is a neutral use: equal corresponding angles imply parallelism without invoking the Euclidean uniqueness axiom.

38.7.2 Proposition I.33

The final I.38 file does not call `euclid_proposition_33` by name, but its mathematical role is present in reusable form. The Interface theorem `OnePairParallelCongruentCriterion` says that one oriented pair of parallel and congruent opposite sides determines a parallelogram. This is the structural content needed to turn $AB \parallel GE$ and $AB \cong GE$ into the parallelogram $AGEB$.

38.7.3 Proposition I.34

I.34 belongs to Euclid’s classical proof: it supplies the diagonal congruence showing that each surrounding parallelogram is double its triangle. The Lean proof does not call I.34. Avoiding that step is precisely what removes the need for a general “halves of equals” principle.

38.7.4 Proposition I.36

I.36 is the decisive earlier proposition in the classical DAG. It compares the two constructed parallelograms on the equal bases BC and EF . Nevertheless, it is not an actual dependency of the final Lean proof. The formal route replaces the I.36-plus-halving argument by a congruent-copy reduction to I.37.

This difference is structurally important. It confirms that the formal Book I dependency graph is not forced to reproduce Euclid’s proposition order even when the public statement remains faithful to Euclid.

38.7.5 Proposition I.37

I.37 is the decisive content dependency in the actual Lean proof. Once the copy point G has been placed on the upper carrier and the nondegenerate branch gives $GD \parallel EF$, I.37 applies immediately to the triangles GEF and DEF on the common base EF .

Thus the formal relationship is

$$\text{congruent replacement} \quad + \quad \text{I.37} \quad + \quad \text{transitivity}.$$

This is the exact formal meaning of Hartshorne’s compressed statement that I.38 follows by transitivity.

38.8 Hilbert and Book Zero Components Used

38.8.1 Incidence, order, and side-of-line infrastructure

The proof uses the developed Hilbert language for

- strict betweenness and its incidence consequences;

- propagation of collinearity to a fixed carrier line;
- noncollinearity from a point lying off a line;
- line existence and uniqueness through two distinct points;
- SameRay transport;
- SameSide and OppositeSide symmetry and transitivity;
- extension of a segment beyond an endpoint.

These mechanisms are responsible for most of the positional detail absent from the classical diagram.

38.8.2 Congruence construction and SAS

Hilbert’s congruence layer supplies two genuine constructions: `angle_construction` creates the oriented ray at E , and `segment_construction` lays off $EG \cong AB$ on that ray. The theorem `hilbert_sameRay_points_sameSide` ensures that replacing G_0 by the final layoff point G preserves the chosen half-plane.

SAS then gives the full triangle-congruence package

$$BAC \cong EGF.$$

No circle construction or continuity axiom is used in this stage.

38.8.3 Parallel and parallelogram infrastructure

The geometric reduction consumes

- I.28 corresponding-angle parallelism;
- parallel transport along collinear carriers;
- `OnePairParallelCongruentCriterion`;
- Euclidean transitivity and uniqueness of parallels;
- the extensional `PointLine` representation used to identify the upper carrier.

The distinction between neutral existence/angle criteria and Euclidean uniqueness is visible here rather than hidden in a generic “parallel” tactic.

38.8.4 Scissors and content infrastructure

No new scissors constructor is needed for I.38. The proof uses only established operations:

- `scissors_congruent`;
- triangle-term permutation lemmas;

- `equicomplementable_of_scissorsEq`;
- `equicomplementable_trans`;
- Proposition I.37.

This is a strong confirmation that the term-level content interface introduced at I.35 is adequate for I.38.

38.9 Formalization Notes

38.9.1 Geometry, representation, and content

I.38 again separates cleanly into three layers.

Geometry. The proof constructs the copied ray and segment, proves the copied triangle congruent, derives the parallelogram $AGEB$, identifies the common upper carrier, and transports the lower carrier from BC to EF .

Representation. Angles have to be reversed or transported along `SameRay` relations to match the argument order required by I.28 and SAS. At the end, the congruent triangles appear as BAC and EGF , while the public scissors terms are ABC and GEF . The triangle swap lemmas normalize this representation without adding geometric content.

Content. Only after the geometry is complete does the proof enter the scissors layer. Congruence gives a strong `HilbertScissorsEq`; I.37 gives `HilbertScissorsEq``equicomplementable`; transitivity composes them. No subtraction, cancellation, numerical area, or new polygon semantics is needed.

38.9.2 Hidden configuration data

The formal proof explicitly establishes the following facts instead of reading them from Figure 38.1:

- B, C, E, F lie on one Hilbert line;
- A, D lie off that line and on the same side of it;
- both original triangles are noncollinear;
- the copied point G_0 is on the chosen side of the base;
- G is on the same ray from E as G_0 ;
- moving from G_0 to G preserves `SameSide` data and the copied angle;
- $G - E - Y$ and $A - B - X$ are strict extensions;
- the extension point Y is on the opposite side of the base from G and therefore from A ;
- the order $B - C - E - F$ implies the transversal order $F - E - B$;

- the exact angle orientation required by I.28;
- equality of the extensional carriers AG and AD ;
- the boundary case $G = D$ separately from the strict case $G \neq D$.

The last two items are particularly significant: a standard affine drawing makes G look automatically located on the upper line, but in Hilbert geometry that position is a theorem obtained from Euclidean parallel uniqueness.

38.9.3 Axiomatic status

The public theorem assumes

[HilbertEuclideanPlane Geo]

and is therefore Euclidean in the current project API.

The proof nevertheless contains a substantial neutral prefix. Incidence, strict order, Same-Side/OppositeSide reasoning, angle construction, segment construction, SameRay transport, SAS, and the corresponding-angle direction of I.28 do not require the Euclidean uniqueness of parallels.

Group IV enters when the copied side data are converted into the full parallelogram structure and, most visibly, when the proof identifies the two upper carriers through Euclidean transitivity/uniqueness of parallels. The final call to I.37 also inherits the Euclidean-plane instance.

No new parallel axiom is introduced in I.38. No continuity or Archimedean axiom is added. The construction of G is an existence argument inside the neutral congruence/order layer; the specifically Euclidean force lies in the uniqueness and structural consequences of parallelism.

38.9.4 Why the conclusion remains equicomplementability

Hilbert's Theorem 46 is the decisive semantic control. Equal-base, equal-altitude triangles are equicomplementable. Hilbert explicitly notes that without the Archimedean axiom one cannot in general strengthen this to finite equidecomposability.

The Lean proof reflects that distinction exactly. The replacement $ABC \rightsquigarrow GEF$ is a direct scissors equality because the two triangles are congruent. The subsequent comparison $GEF \rightsquigarrow DEF$ comes from I.37 and is only equicomplementability. Therefore the final theorem correctly returns the weaker relation.

38.9.5 Reusable helpers and proposition-local helpers

I.38 introduces no new reusable theorem in `HilbertInterface`, `Book Zero`, or `HilbertScissors`. Its new declarations are all prefixed `i38_` and remain proposition-local.

Their mathematical roles are nevertheless worth recording:

- `i38_base_carrier`: ordered lower-line normalization;

- `i38_nondegenerate`: noncollinearity consequences;
- `i38_equal_base_copy`: oriented congruent-copy construction;
- `i38_copy_parallelogram`: corresponding-angle and one-pair parallelogram argument;
- `i38_copy_upper_position`: Euclidean carrier identification and reduction to the I.37 configuration.

If the congruent-copy pattern recurs in later propositions, the third and fourth helpers are candidates for refactoring. At I.38 itself there is not yet enough evidence to enlarge the general Interface merely to shorten one proposition.

38.9.6 Source audit

The source audit assigns distinct roles to the standard references.

Euclid’s classical proof, as recorded by Wilkins, uses two constructed parallelograms, I.36, I.34 twice, and the concluding halving principle. The Beeson proof corpus confirms the same dependency at a formal trace level and makes the final `halvesofequals` assumption explicit.

Hilbert, Chapter IV, Section 19, supplies the semantic control. Theorems 44–46 distinguish equidecomposability from equicomplementability and culminate in the theorem that equal-base, equal-altitude triangles are equicomplementable.

Hartshorne, Section 22, is the most direct architectural source for the chosen proof. After replacing Euclid’s I.37 halving argument by a proof valid for equal content without property (*Z*), he states that I.38 follows by transitivity. The present Lean proof makes the geometric content of that transitivity step explicit by constructing *GEF*.

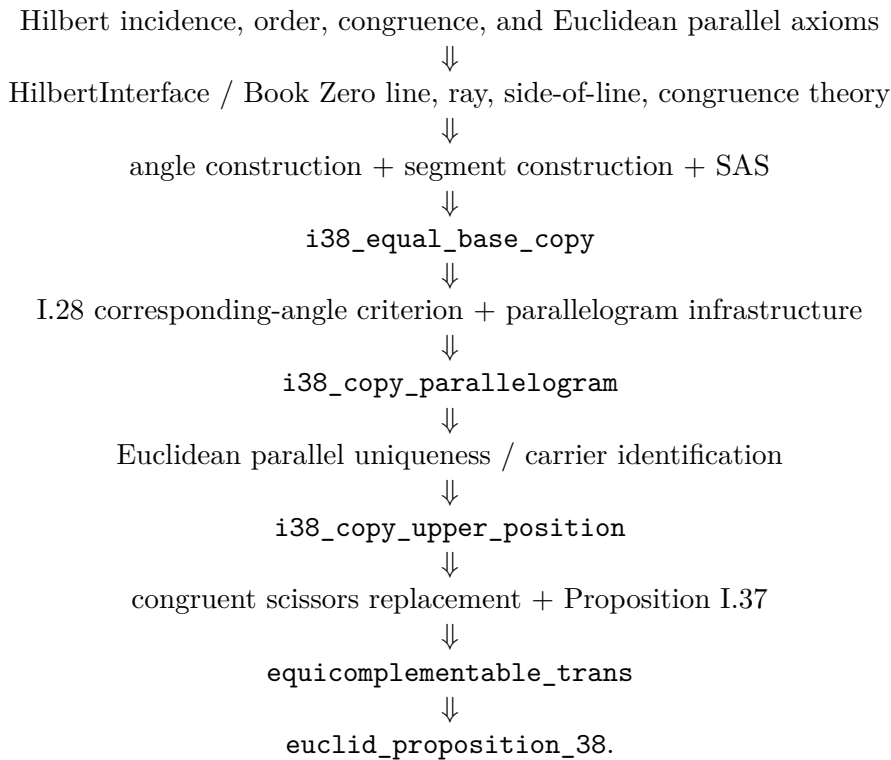
Borsuk–Szmielew are not the source for the equal-content semantics at this stage, but their Euclidean parallelogram theory is directly relevant to the one-pair parallel-plus-congruence criterion consumed by the proof. Their organization supports keeping this criterion in reusable interface infrastructure.

Greenberg remains a control source for the semantic boundary. He distinguishes numerical area from equal content and points to Hartshorne for the full synthetic theory rather than developing it himself. This supports the decision not to replace the scissors relation by a numerical area function.

The Euclid/Joyce edition and the project Book I source compilation remain statement and historical cross-checks; neither is used to force the Lean proof into the classical proposition DAG.

38.10 Dependency Summary

The actual formal dependency chain is



The historical and formal DAGs are therefore sharply different:

Euclid: construct parallelograms $\rightarrow$ I.36 $\rightarrow$ I.34 twice $\rightarrow$ halving,
Lean: congruent copy $\rightarrow$ upper-carrier proof $\rightarrow$ I.37 $\rightarrow$ transitivity.

New geometric axiom:	no
New proposition-local helpers:	yes
New reusable Interface/Scissors helper:	no
New Book Zero theorem:	no
New numerical area or polygon semantics:	no
Public theorem compiles:	yes
Full lake build:	clean
sorry in Proposition38.lean:	no

38.11 Conclusion

Proposition I.38 is classically presented as the equal-base analogue of I.37, but its formal reconstruction exposes a more useful structure. The content part of the theorem is almost entirely inherited from I.37 and transitivity. The new work is geometric: construct an oriented congruent copy on the second base, prove that the copy is related to the original triangle by a parallelogram, and use Euclidean parallel uniqueness to place its top vertex on the required upper carrier.

This explains why a short classical proposition can generate a substantial Lean development. The extra code is not a hidden area calculation. It is the formal cost of orientation, side-of-line data,

ray replacement, extension points, carrier transport, and uniqueness of parallels – all information that a conventional diagram suppresses.

At the same time, I.38 is evidence that the content architecture introduced at I.35 has stabilized. No new scissors operation, polygon semantics, numerical area, or cancellation principle is required. Congruence supplies the strong local replacement $ABC \sim GEF$; I.37 supplies the equal-content comparison $GEF \approx DEF$; and transitivity closes the theorem. In that precise sense, the final proof realizes Hartshorne's concise diagnosis: I.38 follows by transitivity, once the geometry needed to make the intermediate triangle explicit has been proved.

Chapter 39

Proposition I.39

39.1 Introduction

Proposition I.39 is the first genuine converse theorem in Euclid’s area block. Propositions I.35–I.38 proceed in the forward direction: geometric information about bases and parallels is converted into equality of content. Proposition I.39 reverses that direction. Two triangles on the same base are assumed to be equal, and from this equality one must recover a geometric conclusion about the position of their third vertices.

In the present formalization, Euclid’s equality of the two triangles is represented by the relation

```
HilbertScissorsEquicomplementable
```

Thus the theorem starts from a synthetic content relation rather than from numerical areas. The conclusion is also geometric rather than numerical: the two top vertices lie on a common Hilbert incidence line disjoint from the base line.

This converse direction exposes a new foundational issue. The formal scissors calculus developed for I.35–I.38 supports congruent replacement, splitting, addition, and equicomplementability, but it does not by itself express the strict statement that a whole figure cannot have the same content as one of its proper parts. Hartshorne isolates exactly this issue as property (Z), and he notes that Euclid’s proof of I.39 uses the principle that the whole is greater than the part.

The reconstruction therefore does not import a numerical area function and does not assume Hilbert’s Theorem 48 wholesale. Instead it isolates a configuration-free positivity principle for the scissors calculus: adjoining a nondegenerate triangle to an arbitrary scissors term cannot leave the content unchanged. A triangle split along an interior point of one side then derives, as a theorem, the specialized De Zolt statement actually used by I.39.

The result is a useful architectural checkpoint. The geometric part of the proof is derived from the existing Hilbert theory and Proposition I.37. The specialized proper-part statement is no longer an axiom. The remaining new foundational debt is the single positivity assumption `hilbert_scissors_triangle_positive`.

39.2 The Classical Configuration

Let ABC and DBC be triangles on the common base BC . The top vertices A and D lie on the same side of the line carrying BC , and the two triangles are assumed equal in content. The conclusion is that A and D lie on a line parallel to the base.

The classical *reductio* draws through A a line parallel to BC . If AD is not already this parallel, the new line meets the carrier of BD in a point E . There are two strict configurations compatible with A and D lying on the same side of the base:

$$B - E - D, \quad \text{or} \quad B - D - E.$$

In the first case EBC is a proper subtriangle of DBC . In the second case DBC is a proper subtriangle of EBC . The third collinear order

$$E - B - D$$

would place E and D on opposite sides of the base line, because the open segment ED would meet the base at B . The formal proof therefore excludes this third order from SameSide/OppositeSide theory rather than from the drawing.

Figure 39.1 records the two strict configurations that actually drive the classical *reductio*. The public input configuration is already contained in each panel: the two triangles share BC , while the auxiliary parallel through A meets the carrier of BD at E .

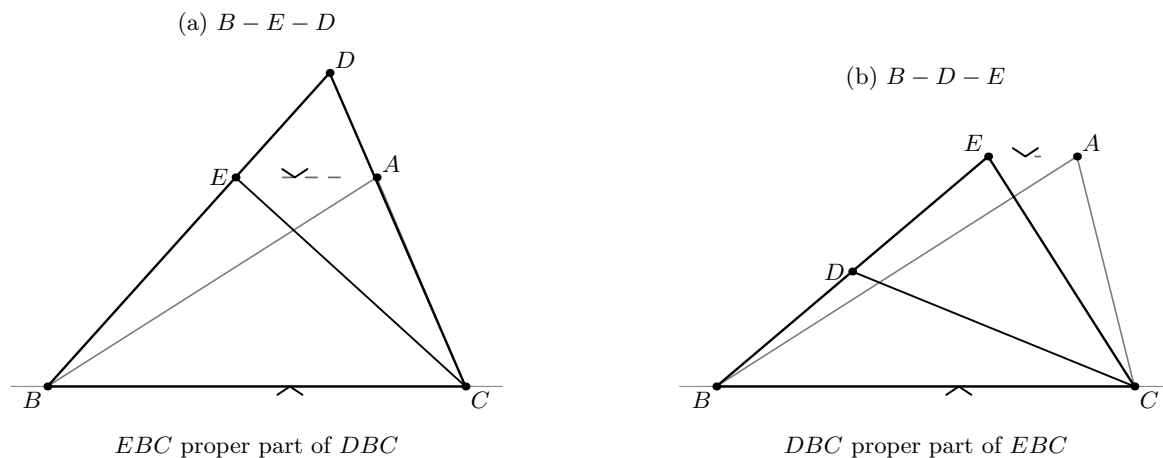


Figure 39.1: The two genuine proper-part configurations. In panel (a), $B - E - D$, so EBC is a proper subtriangle of DBC . In panel (b), $B - D - E$, so DBC is a proper subtriangle of EBC . The dashed carrier through A and E is parallel to the base BC . The original triangle ABC is drawn in restrained gray so that the black proper-part relation is visually primary. These are the two cases retained after the formal order normalization; the excluded order $E - B - D$ is documented separately because it is incompatible with the SameSide data rather than being another realizable branch of the *reductio*.

39.3 Classical Proof

Wilkins emphasizes that a complete proof must treat two positional configurations, although Euclid explicitly discusses only one of them. In each configuration the same auxiliary idea is used.

Draw through A a straight line parallel to the common base BC . If this line already contains D , the conclusion is immediate. Otherwise let E be its intersection with the straight line BD , or with the continuation of BD beyond D .

The auxiliary triangle EBC has the same base BC as ABC , and its third vertex E lies on the parallel through A . Hence an earlier equal-content result shows that

$$ABC = EBC$$

in Euclid's area language. Wilkins' Study Note labels this step as Proposition I.36. The standard Book I dependency, and the Euclid/Joyce dependency table for I.39, identify the relevant triangle theorem as I.37: triangles on the same base and in the same parallels are equal. The present Lean development uses Proposition I.37 directly.

Now compare EBC with DBC . If

$$B - E - D,$$

then EBC is properly contained in DBC . If

$$B - D - E,$$

then DBC is properly contained in EBC . In either case the hypothesis that the original triangles have equal content, together with the equality $ABC = EBC$, would force a whole triangle to be equal to one of its proper parts.

This contradicts Euclid's Common Notion 5, that the whole is greater than the part. Therefore the assumption that D does not lie on the parallel through A is impossible. Hence the top vertices A and D lie on one straight line parallel to the common base.

The classical dependency is therefore schematically

$$\begin{array}{c}
 \text{construct the parallel through } A \\
 \Downarrow \\
 E \in BD \text{ or its extension} \\
 \Downarrow \\
 ABC = EBC \text{ by an earlier equal-content theorem} \\
 \Downarrow \\
 EBC \text{ is a proper part of } DBC \text{ or conversely} \\
 \Downarrow \text{ C.N.5} \\
 \text{contradiction} \\
 \Downarrow \\
 AD \parallel BC.
 \end{array}$$

The point of the reconstruction is not to suppress this argument but to make its hidden logical requirement explicit.

39.4 Property (Z), De Zolt, and the Converse Content Problem

39.4.1 Why I.39 is different from I.38

Hartshorne's analysis of Euclid's area theory makes the boundary especially clear. Once an adequate equal-content relation is available, I.38 follows by transitivity. I.39 is its converse, and the converse uses the claim that a proper part cannot have the same content as the whole. Hartshorne calls the relevant strictness condition property (Z).

Thus the implication

same parallels $\longrightarrow$ equal content

belongs to the forward scissors theory, while

equal content $\longrightarrow$ same parallels

requires an additional strictness principle.

This is exactly what the current formalization detects. The operations already present in `HilbertScissors` do not yield a contradiction merely from the fact that one triangle term arises from a geometric proper subtriangle of another. The term calculus intentionally has no primitive notion of geometric interior or nonoverlap.

39.4.2 Hilbert’s Theorem 48

Hilbert reaches the converse by a different route. In Chapter IV he first develops segment arithmetic and an area function. In Section 21, equicomplementable polygons are shown to have equal areas. For two equicomplementable triangles with equal base g and altitudes h, h' , Hilbert writes the equality

$$\frac{1}{2}gh = \frac{1}{2}gh'$$

and cancels the common nonzero factor to obtain

$$h = h'.$$

This is Theorem 48.

That theorem would certainly be strong enough for I.39. It is not, however, the theorem formalized here. Importing it directly would hide the exact foundational obstacle encountered by the present scissors language and would force the development prematurely into multiplication of segment classes and the later area theory.

39.4.3 Positivity and the derived De Zolt principle

Hilbert later states a De Zolt theorem asserting, in effect, that after a rectilinear figure has been decomposed into triangular pieces, omitting one piece cannot leave enough material to reconstruct the whole. The current formalization isolates the strictness behind that phenomenon at a more general algebraic boundary.

The one new temporary content assumption is:

```
axiom hilbert_scissors_triangle_positive
  [HilbertIncidence Geo]
  (P : HilbertScissorsTerm Geo)
  (A B C : Geo.Point)
  (hNoncol : Not (Collinear Geo A B C)) :
  Not (HilbertScissorsEquicomplementable Geo
    (P + hilbertScissorsTriangle Geo A B C)
    P)
```

This statement contains no betweenness, parallelism, or proposition-specific configuration. It says only that adjoining a nondegenerate triangular piece strictly changes content.

The geometric proper-part statement needed in I.39 is then derived. First,

```
scissors_split_side
```

uses the primitive scissors split to prove that, when $B - X - A$,

$$ABC \sim XBC + AXC,$$

where $\sim$ denotes `HilbertScissorsEq`. The proof also establishes that the cut-off triangle AXC is nondegenerate. If one assumed

$$ABC \approx XBC,$$

then transport across the split would give

$$XBC + AXC \approx XBC,$$

contradicting positivity.

Consequently the former I.39 axiom is now a theorem:

```
theorem hilbert_scissors_triangle_proper_part
  [HilbertIncidence Geo]
  [HilbertOrder Geo]
  (A X B C : Geo.Point)
  (hBXA : Geo.Between B X A)
  (hBC : Ne B C)
  (hNoncol : Not (Collinear Geo B C A)) :
  Not (HilbertScissorsEquicomplementable Geo
    (hilbertScissorsTriangle Geo A B C)
    (hilbertScissorsTriangle Geo X B C))
```

The argument should not be read as a proof that positivity is equivalent, in the present Lean API, to Hilbert's Theorem 48. What has been established formally is narrower and cleaner: positivity implies the specialized proper-part theorem, and that theorem is sufficient for I.39. A future concrete additive valuation, nonzero on nondegenerate triangles, will discharge the positivity axiom directly.

39.5 Proof Architecture of the Reconstruction

The Lean proof separates the geometry from the one missing content principle.

39.5.1 Step 1: the coincident-vertex boundary case

If

$$A = D,$$

there is no need for a reductio. The point A lies off the base because the `SameSide` hypothesis contains this information. The existing parallel-through- point theorem constructs a line through A disjoint from the base, and the same incidence fact is used for both top vertices.

This boundary case is invisible in the usual diagram but must be handled explicitly because the strict point-line parallel relation requires two distinct points.

39.5.2 Step 2: the already-parallel branch

Assume now that $A \neq D$. The proof tests whether

$$AD \parallel BC.$$

If so, the helper

```
proposition39_parallel_points_to_top_line
```

converts point-line parallelism into the incidence-line form used by the public conclusion. It constructs the line through A, D and proves that it is disjoint from the given incidence line **base** through B, C .

39.5.3 Step 3: construction of the auxiliary point

The only nontrivial branch is

$$A \neq D, \quad AD \nparallel BC.$$

The helper

```
proposition39_auxiliary_point
```

draws the parallel through A to BC and proves that the carrier of BD must meet it. If the two incidence lines were disjoint, Euclidean uniqueness of the parallel through B would force the carrier BD to coincide with the base, contradicting the fact that D lies off the base.

The resulting point E satisfies

$$\begin{aligned} &B, E, D \text{ collinear,} \\ &E \neq B, \quad E \neq D, \\ &E \notin \text{base,} \\ &\text{SameSide}(A, E; \text{base}), \end{aligned}$$

and either

$$E = A$$

or

$$AE \parallel BC.$$

The last disjunction is exactly what is needed for Proposition I.37. If $E = A$, the content comparison is reflexive; otherwise the nondegenerate parallel $AE \parallel BC$ gives it directly.

39.5.4 Step 4: normalization of the order on BD

The helper

```
proposition39_auxiliary_order
```

takes the collinearity of B, E, D and proves

$$B - E - D \quad \text{or} \quad B - D - E.$$

The raw betweenness trichotomy has a third branch, $E - B - D$. The proof excludes it by side-of-line theory. Because B lies on the base, $E - B - D$ would make E and D opposite sides of the base. But transitivity and symmetry of *SameSide*, using the common point A , show that E and D are on the same side. The two statements are incompatible.

This is one of the main pieces of hidden configuration data revealed by the formalization.

Figure 39.2 collects the positional cases that the ordinary classical picture suppresses. They are not three further branches of Euclid’s reduction. The first two are genuine boundary cases of the current API; the third is the raw betweenness branch that the side-of-line argument proves impossible.

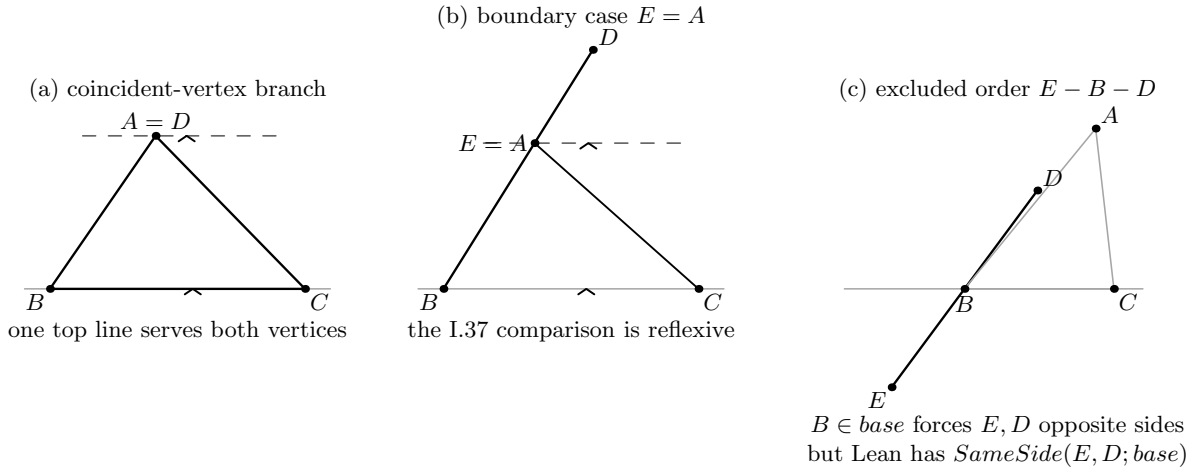


Figure 39.2: Formal positional cases exposed by the Lean proof. (a) If $A = D$, one incidence line through the single top point serves both occurrences in the public conclusion. (b) In the auxiliary-point construction the intersection may satisfy $E = A$; then the comparison $ABC \approx EBC$ is reflexive, so no nondegenerate point-line parallel $AE \parallel BC$ is required. The representative drawing uses the valid order $B - A - D$ only to display the coincidence $E = A$; no such order is part of the helper’s output. (c) The raw order $E - B - D$ would put E and D on opposite sides of the base because $B \in \mathbf{base}$ lies between them. This contradicts the independently proved $\text{SameSide}(E, D; \mathbf{base})$ and is therefore an excluded configuration, not a realizable proof branch.

39.5.5 Step 5: the I.37 content comparison

The auxiliary parallel gives

$$ABC \approx EBC,$$

where $\approx$ denotes *HilbertScissorsEquicomplementable*. The original hypothesis gives

$$ABC \approx DBC.$$

Symmetry and transitivity therefore yield

$$DBC \approx EBC.$$

No subtraction, cancellation, numerical area, or new scissors constructor is used here. This part is entirely inside the content interface already established before I.39.

39.5.6 Step 6: the proper-part contradiction

There are now exactly two order branches.

If

$$B - E - D,$$

then EBC is a proper subtriangle of the nondegenerate triangle DBC . The derived proper-part theorem says

$$DBC \not\approx EBC,$$

contradicting the comparison obtained above.

If

$$B - D - E,$$

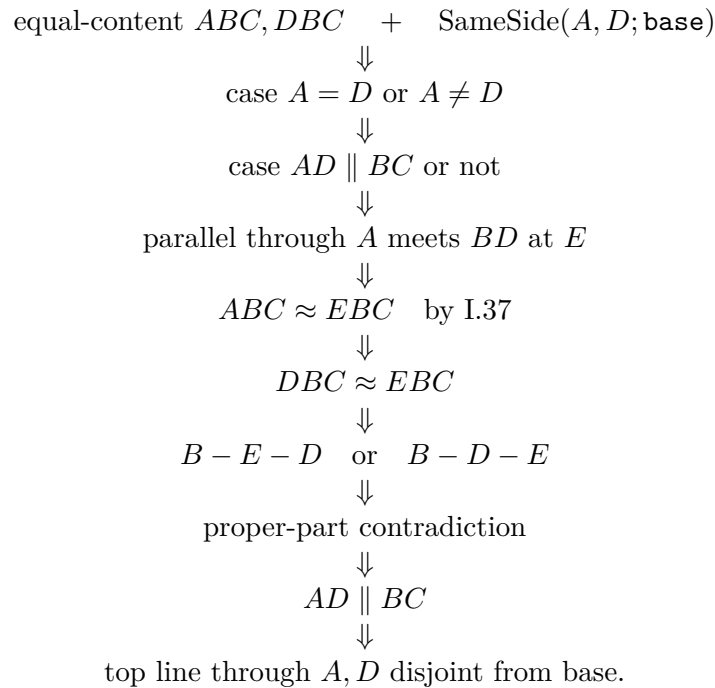
then DBC is a proper subtriangle of the nondegenerate triangle EBC . After symmetry of equicomplementability, the same derived theorem gives the opposite contradiction.

Thus the assumption

$$AD \parallel BC$$

has no possible realization. The remaining branch is $AD \not\parallel BC$, which is converted into the required incidence line through A and D disjoint from the base.

The formal DAG can be summarized as



39.6 Lean Formalization

The current file imports

```
import CGJteamLab.Proposition37
import CGJteamLab.HilbertScissorsPositivity
```

The positivity module in turn imports `HilbertScissorsInvariant`, which contains the generic valuation interface intended for the eventual discharge of the positivity assumption.

The public theorem is:

```
theorem euclid_proposition_39
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D : Geo.Point)
  (base : Geo.Line)
  (hBC : Ne B C)
  (hBbase : HilbertIncidence.OnLine B base)
  (hCbase : HilbertIncidence.OnLine C base)
  (hSame : HilbertSameSide Geo A D base)
  (hEqual :
    HilbertScissorsEquicomplementtable
      Geo
      (hilbertScissorsTriangle Geo A B C)
      (hilbertScissorsTriangle Geo D B C)) :
  Exists fun top : Geo.Line =>
    HilbertIncidence.OnLine A top /\
    HilbertIncidence.OnLine D top /\
    HilbertLinesDisjoint Geo top base := by
  ...
```

The ellipsis is documentation notation only. The repository theorem is fully proved and contains no `sorry`. Unicode logical symbols from the source signature have been normalized above to ASCII forms for the documentation listing; declaration names and argument structure are unchanged.

The mathematically significant proposition-local declarations are

```
proposition39_auxiliary_point
proposition39_auxiliary_order
proposition39_parallel_points_to_top_line
```

The content module contributes the chain

```
scissors_split_side
hilbert_scissors_triangle_positive
hilbert_scissors_triangle_proper_part
```

where the middle declaration is the axiom and the last declaration is a theorem derived from the first two.

The final axiom audit reported by Lean is:

```
'Geometry.euclid_proposition_39' depends on axioms:
[propept,
 Classical.choice,
 bookZero_nullSegment1,
 hilbert_scissors_triangle_positive,
 Quot.sound]
```

Here `propept`, `Classical.choice`, and `Quot.sound` are logical foundations inherited by the Lean development. The pre-existing Book Zero debt shown in the audit is inherited transitively.

The only new I.39-specific content assumption is the positivity declaration shown there.

The full project command

```
lake build
```

is clean with the current `Proposition39.lean`.

39.7 Relationship with Earlier Propositions

39.7.1 Proposition I.31

Classically, the proof requires drawing through A a line parallel to the base. Euclid would cite the parallel construction of I.31. The Lean proof does not call I.31 by name. The construction is already available as

```
hilbert_parallel_through_point_exists
```

in the developed Hilbert interface.

This is another example of the distinction between the historical proposition DAG and the actual dependency graph of the reconstruction.

39.7.2 Proposition I.37

I.37 is the decisive earlier proposition in the actual Lean proof. Once E has been constructed on the parallel through A , the two triangles ABC and EBC have the same base and lie between the same parallels. Therefore

$$ABC \approx EBC.$$

This use of I.37 is particularly important because the project’s proof of I.37 was deliberately constructed without a general “halves of equals” or property-(Z) principle. Consequently there is no circularity: I.37 provides the forward equal-content theorem, while I.39 is exactly the later converse that first introduces a strict proper-part assumption.

39.7.3 Proposition I.38

I.38 is the immediate conceptual contrast. Its content conclusion is still forward and can be obtained from a congruent-copy construction, I.37, and transitivity. No strict comparison of contents is needed.

I.39 starts with equal content and concludes a geometric relation. This is why the existing I.35–I.38 scissors interface is almost sufficient but not quite: the missing ingredient is not another equality operation but a strictness principle.

39.7.4 Proposition I.40

The clean project build confirms that the public theorem name `euclid_proposition_39` is now available to Proposition I.40. Thus I.39 is not an isolated experimental theorem; it is part of the

continuing Book I dependency chain.

39.8 Hilbert and Book Zero Components Used

39.8.1 Incidence and order

The auxiliary construction uses the Hilbert incidence API for:

- the line through two distinct points;
- uniqueness of a line through two distinct points;
- conversion between point-line carriers and incidence lines;
- collinearity induced by a common incidence line;
- betweenness trichotomy for three distinct collinear points;
- transport of betweenness to incidence information.

These tools are responsible for proving that the constructed point E is actually on the carrier BD and for reducing its position to the two strict orders needed by the reductio.

39.8.2 SameSide and OppositeSide

The public hypothesis

$$\text{SameSide}(A, D; \text{base})$$

contains more information than a conventional drawing usually records. It implies that both top vertices lie off the base. After the auxiliary point is constructed, the proof establishes

$$\text{SameSide}(A, E; \text{base})$$

and hence, by symmetry and transitivity,

$$\text{SameSide}(E, D; \text{base}).$$

This is used against

```
hilbert_oppositeSide_not_sameSide
```

to eliminate the order $E - B - D$.

39.8.3 Euclidean parallelism

The reconstruction is Euclidean in the current project API. Parallelism is used in two logically distinct ways.

First, the proof constructs a parallel through A to the base. Second, Euclidean uniqueness is used in the contradiction showing that the carrier BD cannot be disjoint from this new parallel unless it coincides with the base, which would place D on the base.

The helper

```
collinear_parallel_trans
```

then transports the auxiliary parallel along a collinear carrier so that, when $E \neq A$, the exact hypothesis

$$AE \parallel BC$$

required by I.37 is available.

39.8.4 Scissors content

The content stage uses only the already established equivalence operations:

- `equicomplementable_refl`;
- `equicomplementable_symm`;
- `equicomplementable_trans`;
- Proposition I.37.

No new scissors constructor is introduced. The existing `split` constructor is packaged by `scissors_split_side`. The specialized proper-part principle is derived from that `split` and the single new positivity axiom.

39.8.5 Book Zero

No new Book Zero theorem is introduced for I.39. The axiom audit nevertheless shows the inherited dependency

```
bookZero_nullSegment1
```

through earlier project infrastructure. This dependency predates I.39 and is not part of the new converse-content argument.

39.9 Formalization Notes

39.9.1 Geometry, representation, and content

From I.35 onward it is important to keep three layers separate.

Geometry. The proof constructs the parallel through A , proves that it meets the carrier BD , obtains the point E , establishes its side-of-line position, classifies the order of B, E, D , and finally proves the required parallelism.

Representation. The public theorem speaks about an explicit incidence line `base` and returns an explicit incidence line `top`. Proposition I.37, in contrast, uses point-line parallelism. The dedicated top-line packaging helper bridges these two representations. Its exact declaration name is

```
proposition39_parallel_points_to_top_line
```

This is a representation issue, not an additional geometric axiom.

At the triangle-term level no special permutation machinery is central to I.39: all three triangles are written on the common ordered base BC .

Content. The equality information is entirely synthetic. The formal theorem never assigns a number to the area of a triangle. It derives

$$DBC \approx EBC$$

by symmetry and transitivity of equicomplementability and then invokes one strict-content principle to contradict the proper inclusion.

Thus the proof does not identify `HilbertScissorsEquicomplementable` with numerical equality of area. It uses only the operations explicitly available in the formal content layer.

39.9.2 Hidden configuration data

The Lean proof establishes explicitly:

- A and D are off the base line;
- $B \neq D$ because B is on the base and D is off it;
- the parallel through A is disjoint from the base;
- the carrier BD must meet that parallel in the nonparallel branch;
- the intersection point E is off the base;
- $E \neq B$ and $E \neq D$;
- B, E, D are collinear;
- A and E are on the same side of the base;
- therefore E and D are on the same side of the base;
- the order $E - B - D$ is impossible;
- in the remaining two orders the larger triangle is nondegenerate;
- the proper-subtriangle relation is therefore strict rather than degenerate.

None of these facts is read from the figures as an assumption. Figures 39.1 and 39.2 are summaries of conclusions proved inside the geometry layer; in particular, the excluded $E - B - D$ panel is explicitly a contradiction schematic rather than a model satisfying all hypotheses.

39.9.3 Axiomatic status

The public theorem assumes

[HilbertEuclideanPlane Geo].

The proof uses incidence, order, congruence-derived side-of-line theory, and Euclidean parallelism. It introduces no continuity axiom and no Archimedean axiom.

There is, however, one new non-geometric assumption:

```
hilbert_scissors_triangle_positive
```

This should be read as an explicit positivity axiom for content, not as part of Hilbert Group IV. Its role is analogous to the strictness isolated by Hartshorne’s property (Z) and to the De Zolt phenomenon in the classical theory of area. The proposition-specific proper-part declaration is now a theorem:

```
hilbert_scissors_triangle_proper_part
```

The current theorem is therefore conditional in a very precise sense: all geometry of I.39 is proved from the established Hilbert/Euclidean infrastructure, while the missing semantic strength of the term-level scissors calculus is exposed as one configuration-free positivity assumption.

39.9.4 Why Hilbert Theorem 48 is not an imported axiom

An earlier experimental route packaged the converse as an equal-altitude statement close to Hilbert Theorem 48. That route was abandoned for the final I.39 module because it concealed too much of the real dependency.

Hilbert’s own proof of Theorem 48 is downstream from segment arithmetic and the area function. Reproducing it faithfully would require substantially more infrastructure than I.39 itself needs. The present route instead follows the classical *reductio* and isolates the exact strict-content principle at the point where the current formal scissors calculus stops.

This makes the remaining debt cleaner and easier to study independently. The module `HilbertScissorsInvariant` already provides the generic valuation interface and proves that equicomplementable terms have equal valuation. What is still missing is a concrete additive valuation whose value on every nondegenerate triangle is nonzero. Such a valuation immediately discharges `hilbert_scissors_triangle_positive`, after which the derived proper-part theorem and I.39 require no change.

39.9.5 Source audit

The sources play different roles in the reconstruction.

Wilkins supplies the clearest complete audit of the classical configuration. His Study Note explicitly distinguishes the two positions of the intersection point E and notes that Euclid discusses only one of them. This two-case analysis matches the two strict betweenness branches retained by the Lean proof.

Hartshorne supplies the semantic diagnosis. Section 22 distinguishes forward equal-content arguments from converse arguments and explicitly states that I.39 uses “the whole is greater than the part,” hence property (Z). This is the principal source for interpreting the new formal positivity requirement.

Hilbert supplies the broader theory of plane area. Theorems 44–46 justify equicomplementability as the correct synthetic target for the forward equal-base/equal-altitude theorems. Section 21 proves Theorem 48 only after the development of area, while Theorem 52 records a De Zolt-type impossibility of reconstructing a whole after omitting a triangular part. These results explain both why the converse requires strictness and why it is reasonable to isolate a positivity principle from which the proper-part statement can be derived.

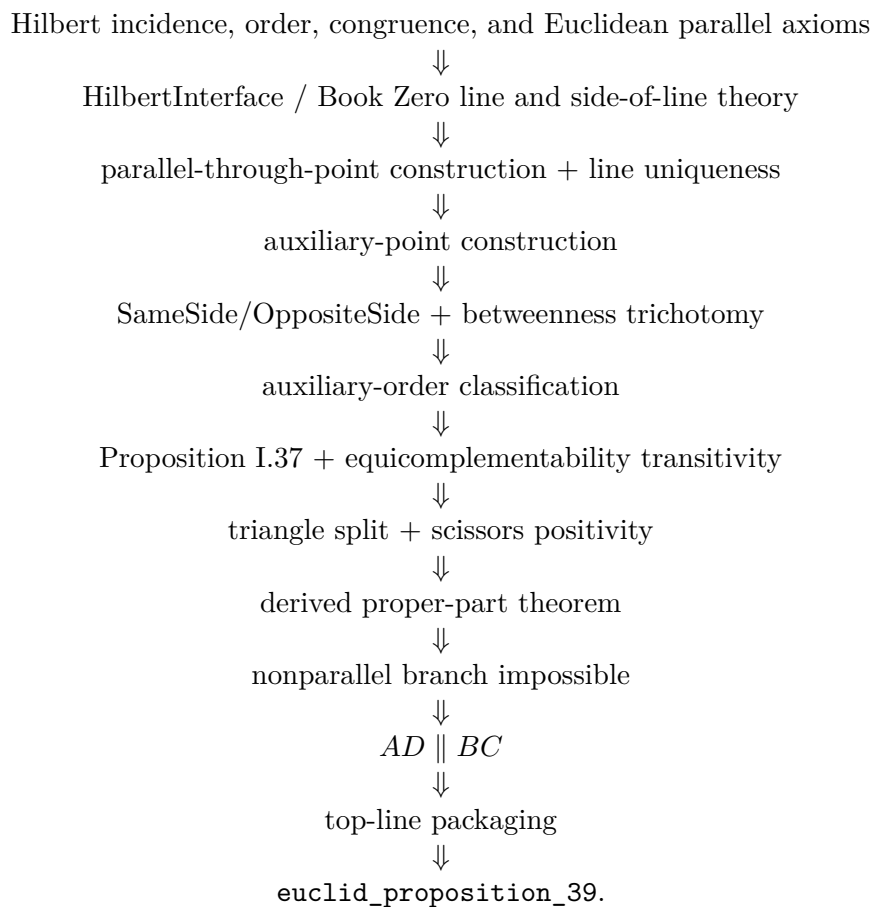
The Beeson proof corpus remains a control source for hidden positional and order obligations, but it is not used as the architecture of the final Lean proof. In particular, the project does not import a Tarski-style proof trace or reorganize the Hilbert interface around Beeson tactics.

Borsuk–Szmielew and Greenberg remain control sources for incidence, order, parallelism, and the neutral/Euclidean boundary. They are not used here to replace the content relation by a numerical area theory.

The Euclid/Joyce edition controls the classical statement and Common Notion 5. The project Book I compilation is used as a consistency check against the documentation of I.35–I.38. Neither source overrides the current Lean module, which is authoritative for the formal dependency graph.

39.10 Dependency Summary

The actual formal dependency chain is



The historical and formal DAGs are therefore related but not identical:

Euclid/Wilkins:	parallel through $A \rightarrow E \rightarrow$ earlier equal-area theorem $\rightarrow$ C.N.5,
Lean:	parallel through $A \rightarrow E \rightarrow$ I.37 $\rightarrow$ derived proper-part theorem.

New geometric axiom:	no
New content axiom:	yes — scissors positivity
New proposition-local helpers:	yes
New reusable Scissors theorems:	yes — split-side and proper-part
New Book Zero theorem:	no
New numerical area or polygon semantics:	no
Public theorem compiles:	yes
Full lake build:	clean
sorry in Proposition39.lean:	no

Diagrammatic audit. The former single three-panel figure mixed the public input, the two classical proper-part alternatives, and the formal order analysis. It has been replaced by two figures with distinct roles. Figure 39.1 contains only the two realizable strict orders $B - E - D$ and $B - D - E$ that support the classical proper-part contradiction. Figure 39.2 records the three positional facts that the standard Euclidean picture suppresses: the coincident-vertex case $A = D$, the auxiliary-intersection boundary case $E = A$, and the excluded raw order $E - B - D$.

No separate figure is added for the already-parallel branch, because it is the geometric conclusion rather than a new construction. Likewise, symmetry and transitivity of equicomplementability, the incidence-line packaging helper, and the weakening/transport operations in the scissors layer do not change the configuration. The split underlying `hilbert_scissors_triangle_proper_part` is already visible in the two proper-part panels, so duplicating it as a third scissors diagram would record the same geometry twice.

39.11 Conclusion

Proposition I.39 marks a real change in the logical direction of Euclid’s area theory. The preceding propositions show that appropriate geometric configurations imply equality of content. I.39 asks equal content to force a geometric conclusion. That reversal is exactly where a strict content principle becomes unavoidable.

The formal reconstruction makes the boundary unusually transparent. All of the geometry needed for the classical reductio – the parallel through A , its intersection with BD , the SameSide analysis, the collinear order trichotomy, and the reduction to Proposition I.37 – is derived inside the existing Hilbert/Euclidean infrastructure. No new geometric axiom is needed.

What remains is one explicit content assumption: adjoining a nondegenerate triangle to an arbitrary scissors term cannot leave the content unchanged. Rather than encoding the I.39 configuration itself as an axiom, the development records this configuration-free positivity statement in the declaration

```
hilbert_scissors_triangle_positive
```

The triangle split then proves the proposition-specific proper-part statement:

```
hilbert_scissors_triangle_proper_part
```

The axiom audit confirms that positivity, not the proper-part statement, is the only new I.39-specific foundational debt.

This is the main structural result of the proposition. I.39 itself is now stable. A future positive additive valuation can discharge the remaining positivity assumption without reopening either the proper-part derivation or the geometric proof of I.39.

Chapter 40

Proposition I.40

40.1 Introduction

Proposition I.40 is the equal-base companion to Proposition I.39. The four triangle propositions at the end of the first area block therefore form a natural pattern:

- I.37 : same base + same parallels $\longrightarrow$ equal content,
- I.38 : equal bases + same parallels $\longrightarrow$ equal content,
- I.39 : same base + equal content $\longrightarrow$ same parallels,
- I.40 : equal bases + equal content $\longrightarrow$ same parallels.

The present Lean theorem keeps the same synthetic content language used since I.35. Equality of triangles is represented by

```
HilbertScissorsEquicomplementable
```

rather than by a numerical area function. Its conclusion is geometric: the two top vertices lie on a Hilbert incidence line disjoint from the common base carrier.

The main architectural fact is that I.40 does not need a second converse-area argument. The difficult same-base converse has already been isolated in I.39. The only additional task is to reduce the equal-base configuration to a same-base one. For that reduction the formal proof reuses the congruent-copy construction developed for I.38.

This reuse was not merely cosmetic. The first draft of I.40 repeated the angle construction, segment layoff, SAS argument, oriented parallel argument, and parallelogram criterion already present in I.38. Inspection of I.40 showed that two I.38 helpers had been stated with unnecessary proposition-specific data. Their signatures were generalized, the full project was rebuilt successfully, and I.40 now calls that common construction directly. Thus the proposition records a genuine API improvement discovered by a later proof.

No new content principle is introduced in I.40. The theorem depends on I.39, and therefore inherits the positivity assumption already isolated there, but I.40 itself adds no new axiom, no numerical area, and no new polygon semantics.

40.2 Statement and Formal Ordered Configuration

Euclid’s statement concerns two equal triangles on equal bases lying in one straight line and on the same side of that line. In the current formal statement the triangles are

$$ABC \quad \text{and} \quad DEF,$$

with equal bases

$$BC \cong EF.$$

The four base points are placed in the strict order

$$B - C - E - F,$$

encoded by

$$\text{Between}(B, C, E), \quad \text{Between}(C, E, F).$$

The points B and C are given on an incidence line **base**. The betweenness hypotheses then force E and F onto that same carrier. The top vertices A and D satisfy

$$\text{SameSide}(A, D; \text{base}).$$

This is a deliberate specialization of the fully general Euclidean statement. Wilkins’ Study Note points out that a complete treatment of arbitrary equal bases on one line requires six positional configurations. These arise from the possible relative positions of the two base segments together with the position of the second top vertex relative to the auxiliary parallel through the first. Euclid explicitly discusses only one of those cases.

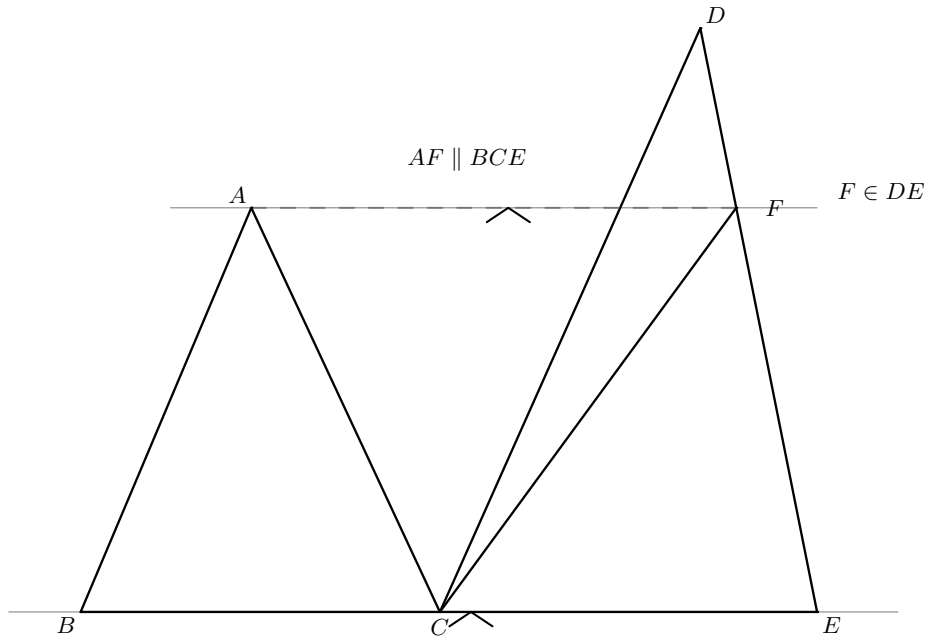
The present Lean theorem does *not* claim to formalize all six configurations. It works with the ordered separated-base arrangement $B - C - E - F$, the same carrier normalization already used by Proposition I.38. The advantage is that the equal-base geometry can be reused without reopening a large order case split. A future fully order-free wrapper could normalize other base arrangements before calling this theorem; that normalization is not part of the current proposition.

The formal theorem begins with this ordered separated-base configuration. It is displayed at the start of the formal reduction below. Euclid’s adjacent- base reductio is geometrically different and is shown separately in the next section.

40.3 Classical Proof

In Euclid’s displayed configuration the two equal bases share an endpoint: the triangles are ABC and CDE , with bases BC and CE . Join the two top vertices A and D . The aim is to prove that AD is parallel to the base line.

Assume that AD is not the required parallel. Through A draw another line parallel to the base. Let it meet the side of the second triangle, or its extension, at an auxiliary point F , and join F to the appropriate base endpoint. Proposition I.38 applies to the original triangle and the auxiliary triangle because their bases are equal and they lie in the same parallels. Hence the auxiliary triangle is equal to the first triangle. By hypothesis the first triangle is equal to the second, so transitivity of equality forces the auxiliary triangle to be equal to the second triangle. But one of these is geometrically a proper part of the other. This is the “greater equals the less” contradiction used by Euclid.



FCE is a proper part of DCE

Figure 40.1: Euclid's displayed adjacent-base case. The equal bases are BC and CE . Under the *reductio* assumption, a line through A parallel to the base meets DE at the auxiliary point F ; joining FC produces the triangle FCE . Proposition I.38 compares ABC with FCE , while the hypothesis compares ABC with DCE , forcing the impossible equality of FCE with the larger triangle DCE . The classical point F in this figure is unrelated to the endpoint named F in the formal ordered theorem.

The classical dependency in the displayed case is therefore

equal triangles on equal bases
 $\Downarrow$
 draw a parallel through the first top vertex
 $\Downarrow$
 auxiliary triangle in the same parallels
 $\Downarrow$ I.38
 auxiliary triangle = first triangle
 $\Downarrow$ C.N.1
 auxiliary triangle = second triangle
 $\Downarrow$
 proper-part contradiction
 $\Downarrow$
 the top vertices lie on the required parallel.

The historical proof and the current Lean proof therefore share the idea that I.40 should be reduced to an earlier equal-base theorem, but they reduce in opposite directions. Euclid uses the forward theorem I.38 inside a *reductio*. The Lean development first uses the I.38 *construction* to move to a common base and then invokes the already completed converse I.39.

Wilkins' expanded analysis is important here. Once the two bases are allowed to be adjacent, separated, or interlocking, and once both possible positions of the second top vertex relative to

the auxiliary parallel are considered, six cases must be ruled out. The present formal theorem avoids claiming this full classical case analysis: its explicit $B - C - E - F$ hypotheses select one ordered base arrangement from the outset.

40.4 Equal Bases by Reduction to the Same-Base Converse

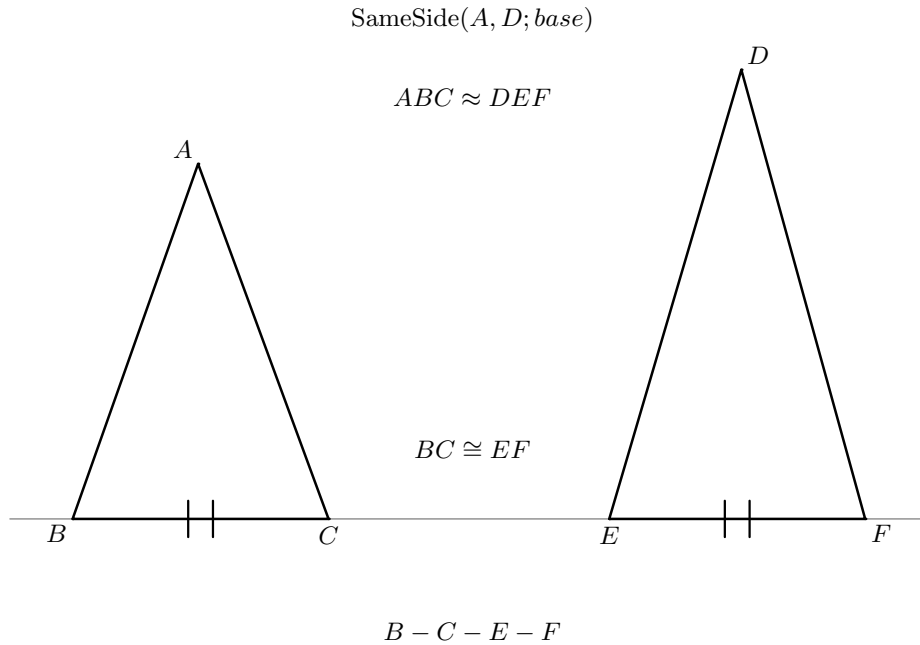


Figure 40.2: The actual input configuration of the production theorem. The base points satisfy $B - C - E - F$, the segments $BC \cong EF$, and the triangles ABC and DEF are equicomplementable. The top vertices A and D are only assumed to lie on the same side of the base; no collinearity of A and D is available at this stage.

The central structural observation is that the passage from I.39 to I.40 is the converse analogue of the passage from I.37 to I.38.

For the forward direction, I.38 constructs a congruent copy of one triangle on the other equal base and then compares the copied triangle with the target by I.37. For the converse direction, I.40 uses the same copy construction, but now the copied triangle and the target triangle are equal in content and share *the same base*. Proposition I.39 can therefore be applied directly.

Schematically,

$$\begin{array}{ccc}
 \text{I.37} & \xrightarrow{\text{equal-base copy}} & \text{I.38} \\
 \Updownarrow & & \Updownarrow \\
 \text{I.39} & \xrightarrow{\text{same equal-base copy}} & \text{I.40}
 \end{array}$$

where the vertical symbols indicate only the forward/converse relationship, not a formal biconditional theorem in Lean.

This observation led to a small but significant refactor of `Proposition38.lean`. The helpers

```
i38_equal_base_copy
i38_copy_parallelogram
```

had originally accepted the second top vertex D and a hypothesis of the form $AD \parallel BC$. Inspection showed that these data were not intrinsic to the copy itself. What the construction actually needs is:

- one explicit base carrier;
- B, C, E, F on that carrier;
- the ordered-base data $B - C - E - F$;
- $BC \cong EF$;
- the point A off the base.

After the signatures were generalized, I.38 continued to build cleanly and I.40 could call the construction without repeating its proof.

The resulting Lean DAG is

$$\begin{array}{c}
 ABC \approx DEF, \quad BC \cong EF, \quad B - C - E - F, \quad \text{SameSide}(A, D; \mathbf{base}) \\
 \Downarrow \\
 E, F \in \mathbf{base} \\
 \Downarrow \quad \text{i38_copy_parallelogram} \\
 \exists G : \quad BAC \cong EGF, \quad \text{SameSide}(A, G; \mathbf{base}), \quad AG \parallel BC \\
 \Downarrow \\
 ABC \sim GEF \\
 \Downarrow \quad \text{content symmetry/transitivity} \\
 GEF \approx DEF \\
 \Downarrow \quad \text{I.39 on base } EF \\
 G, D \in \mathbf{top}, \quad \mathbf{top} \cap \mathbf{base} = \emptyset \\
 \Downarrow \quad AG \parallel BC \\
 \text{carrier}(AG) \cap \mathbf{base} = \emptyset \\
 \Downarrow \quad \text{Euclidean parallel uniqueness through } G \\
 A, G, D \in \mathbf{top} \\
 \Downarrow \\
 \text{euclid_proposition_40.}
 \end{array}$$

Here $\sim$ denotes the stronger relation

```
HilbertScissorsEq
```

while $\approx$ denotes

```
HilbertScissorsEquicomplementtable
```

40.5 Proof Architecture of the Reconstruction

40.5.1 Step 1: putting the second base on the common carrier

The first proposition-local helper is

```
i40_base_endpoints
```


It starts from $B, C \in \mathbf{base}$ together with

$$B - C - E, \quad C - E - F,$$

and proves

$$E, F \in \mathbf{base}.$$

The proof is pure incidence/order transport. Betweenness supplies both collinearity and endpoint distinctness, and line uniqueness transports the new points to the already chosen carrier.

This helper is intentionally local. It packages the particular ordered four-point configuration used by I.40; it does not introduce a new general notion of collinear base segments.

40.5.2 Step 2: reusing the I.38 congruent-copy construction

The second helper is

```
i40_copy_parallel
```

Its output is exactly the geometric bridge needed by the converse proof:

$$\exists G : \text{SameSide}(A, G; \mathbf{base}), \quad BAC \cong EGF, \quad AG \parallel BC.$$

The helper does not rebuild this geometry. It calls

```
i38_copy_parallelogram
```

which returns the stronger package

$$\text{SameSide}(A, G; \mathbf{base}), \quad AGE B \text{ is a parallelogram}, \quad BAC \cong EGF.$$

From the parallelogram one gets $AG \parallel EB$. Since B, C, E are collinear, parallelism is transported from the carrier EB to BC . The result is $AG \parallel BC$.

This is the point at which the I.38 refactor pays off. The substantial angle, ray, SAS, extension, and I.28 machinery remains proved once, in I.38. I.40 uses only its mathematical output.

40.5.3 Step 3: transporting equal content to the copied triangle

Triangle congruence gives the strong scissors relation

$$BAC \sim EGF.$$

The scissors representation of a triangle is insensitive to exchanging its first two vertices, so the proof rewrites this as

$$ABC \sim GEF.$$

The relation `HilbertScissorsEq` implies `HilbertScissorsEquicomplementable`, hence

$$ABC \approx GEF.$$

By symmetry and transitivity with the hypothesis

$$ABC \approx DEF,$$

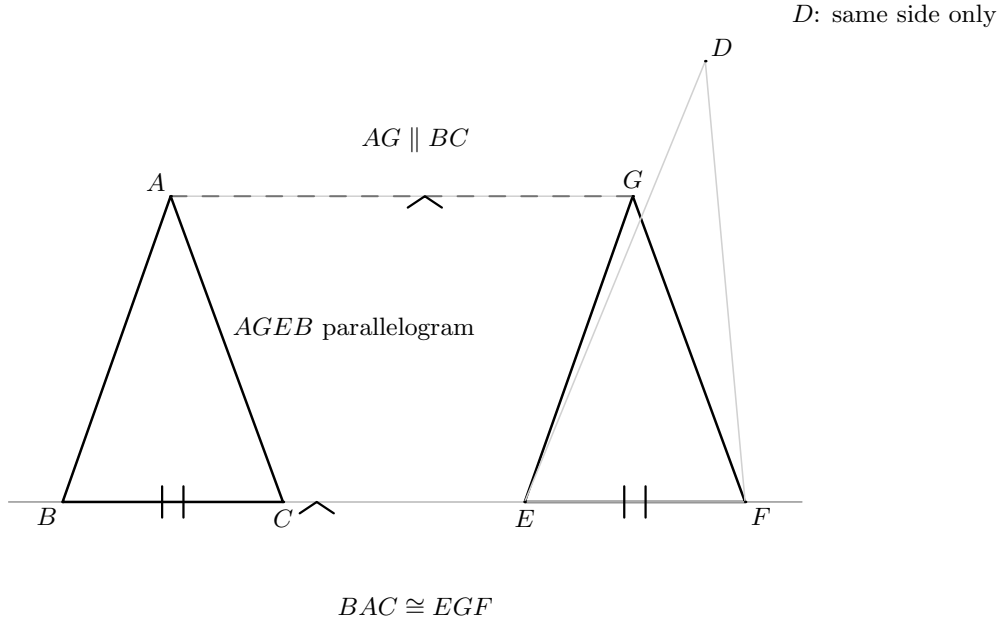


Figure 40.3: The geometric output consumed from the generalized I.38 construction. The new point G gives the parallelogram $AGEB$, the congruence $BAC \cong EGF$, and $AG \parallel BC$. At this point D is known only to be on the same side of the base as A ; it is deliberately drawn off the carrier AG . Its later collinearity with A, G is a conclusion, not an input to the copy step.

we obtain

$$GEF \approx DEF.$$

No subtraction, cancellation, strict content comparison, or numerical area is used in this stage. It is ordinary equivalence transport in the existing scissors layer.

40.5.4 Step 4: reducing to I.39 on the common base EF

The copy construction puts G on the same side of **base** as A , while the public hypothesis puts D on that side as well. Symmetry and transitivity of **HilbertSameSide** therefore yield

$$\text{SameSide}(G, D; \text{base}).$$

The strict betweenness $C - E - F$ gives $E \neq F$. All hypotheses of I.39 are now available for the triangles GEF and DEF on the same base EF .

Proposition I.39 returns a line **top** such that

$$G, D \in \text{top}, \quad \text{HilbertLinesDisjoint}(\text{top}, \text{base}).$$

This is the only converse-content step in I.40. Consequently all strictness needed to turn equal content back into geometry is inherited from I.39 rather than reintroduced locally.

40.5.5 Step 5: identifying the final **top** line

It remains to place A on the line returned by I.39. The helper `i40_copy_parallel` already supplied $AG \parallel BC$. Let **topAG** be the incidence line through A and G . Since BC lies on **base**, point-line parallelism implies that **topAG** and **base** are disjoint.

Now both **topAG** and **top**

- pass through G , and
- are disjoint from **base**.

The Euclidean uniqueness of a parallel through an external point therefore forces

$$\text{top}AG = \text{top}.$$

Since $A \in \text{top}AG$, it follows that $A \in \text{top}$. Together with $D \in \text{top}$ and disjointness from the base, this is exactly the public conclusion.

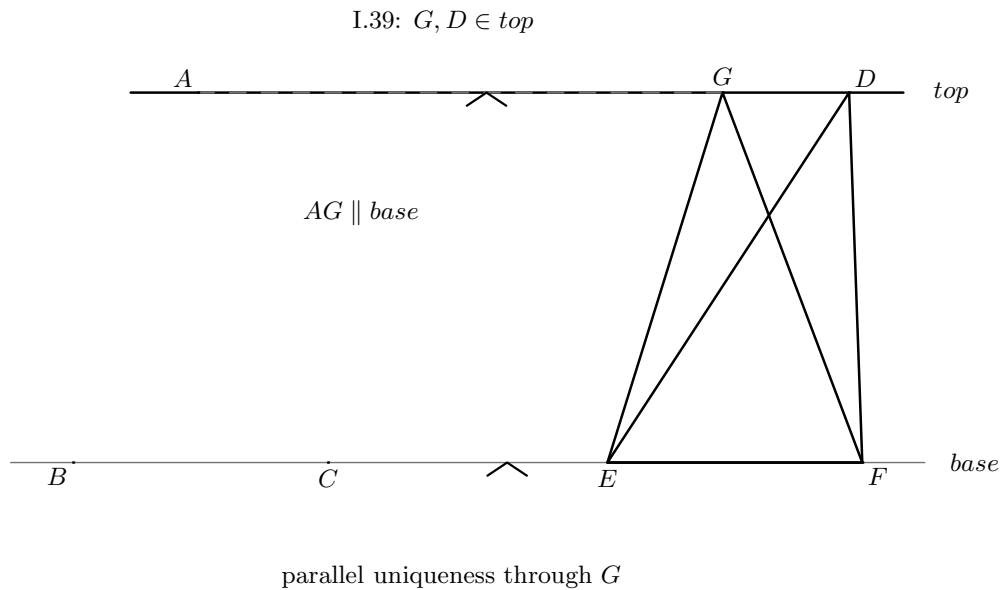


Figure 40.4: After content transport, I.39 places G and D on a line **top** disjoint from the base. Independently, $AG \parallel BC$ makes the carrier through A, G disjoint from the same base. Both candidate top carriers pass through G , so Euclidean parallel uniqueness identifies them. Only here is the final collinearity of A, G, D justified.

40.6 Lean Formalization

The current file imports the two propositions that carry its geometry and its converse-content step:

```
import CGJteamLab.Proposition38
import CGJteamLab.Proposition39
```

The public theorem is:

```
theorem euclid_proposition_40
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D E F : Geo.Point)
  (base : Geo.Line)
  (hBCE : Geo.Between B C E)
  (hCEF : Geo.Between C E F)
  (hBbase : HilbertIncidence.OnLine B base)
  (hCbase : HilbertIncidence.OnLine C base)
```

```

(hBC_EF : Geo.Congruent B C E F)
(hSame : HilbertSameSide Geo A D base)
(hEqual :
  HilbertScissorsEquicomplementable
    Geo
    (hilbertScissorsTriangle Geo A B C)
    (hilbertScissorsTriangle Geo D E F)) :
Exists fun top : Geo.Line =>
  HilbertIncidence.OnLine A top /\
  HilbertIncidence.OnLine D top /\
  HilbertLinesDisjoint Geo top base := by
...

```

The ellipsis is documentation notation only. The repository theorem is fully proved. Unicode logical symbols from the Lean source have been normalized to ASCII forms in the listing above.

The two proposition-local helpers are:

```

i40_base_endpoints
i40_copy_parallel

```

The second of these calls the generalized I.38 helper

```

i38_copy_parallelogram

```

rather than repeating its proof.

The scissors part of the final theorem uses the following existing operations:

```

scissors_congruent
scissors_triangle_swap12
equicomplementable_of_scissorsEq
equicomplementable_symm
equicomplementable_trans

```

The line-placement part then uses:

```

euclid_proposition_39
HilbertPlaneIncidence.line_through
hilbert_mem_pointLine_iff_onLine
HilbertEuclideanPlane.parallel_unique

```

40.7 Relationship with Earlier Propositions

40.7.1 Proposition I.38

I.38 supplies the reusable equal-base geometry. Its original formal proof had to construct G by angle layoff and segment layoff, establish triangle congruence by SAS, recover the appropriate orientation through extensions and I.28, and package the result as the parallelogram $AGEB$.

I.40 revealed that this construction was more general than its original signature. The helpers `i38_equal_base_copy` and `i38_copy_parallelogram` were therefore generalized to accept the base carrier and off-line data directly. The public I.38 theorem continues to use them, and I.40

now reuses the same package. This is a concrete example of a later proposition improving the reusable boundary of an earlier proof.

40.7.2 Proposition I.39

I.39 is the decisive converse theorem. It says that equal-content triangles on one common base and on the same side lie between the same parallels. Its proof is where the strict content issue appears: the proper-part contradiction ultimately depends on the positivity principle isolated in the scissors layer.

I.40 does not reproduce that argument. Once the copied triangle GEF has been made equal in content to DEF , I.39 is applied directly on the base EF . Thus the foundational debt of I.40 is inherited from I.39 rather than multiplied.

40.7.3 Proposition I.37

I.37 is not called directly by `Proposition40.lean`, but it remains in the transitive dependency graph through I.39. It is the forward same-base theorem used inside the I.39 reductio. The four propositions I.37–I.40 therefore form a coherent block rather than four independent area arguments.

40.8 Hilbert and Book Zero Components Used

The proposition sits on several already established layers.

40.8.1 Incidence and order

Strict betweenness provides endpoint distinctness and collinearity. The helper

```
hilbert_collinear_on_line
```

transports E and F onto the chosen base carrier. The theorem also uses incidence lines through distinct points and the equivalence between membership in a point-line carrier and incidence on a chosen Hilbert line.

40.8.2 Side-of-line geometry

The public statement exposes

```
HilbertSameSide Geo A D base
```

explicitly. The copied point G comes with a second `SameSide` relation from I.38. Symmetry and transitivity then produce the exact `SameSide` hypothesis required by I.39. This information is invisible in the short classical phrase “on the same side” but is a first-class proof object in Lean.

40.8.3 Parallelism and parallelograms

The I.38 construction returns the parallelogram $AGEB$. The existing parallel transport lemmas convert $AG \parallel EB$ into $AG \parallel BC$ using the collinearity of B, C, E . At the final stage the Euclidean uniqueness of the parallel through G identifies the carrier through A, G with the line through G, D returned by I.39.

Thus Group IV enters through the Euclidean parallel layer. No continuity axiom is used by the proposition.

40.8.4 Scissors content

The content calculation is intentionally small. Triangle congruence gives a strong scissors equality; representation transport reorders the triangle vertices; and the existing equivalence operations convert and compose the relations. No new scissors constructor or theorem is introduced in I.40.

The strict content principle needed by the converse direction is already encapsulated upstream in I.39 through

```
hilbert_scissors_triangle_positive
```

I.40 introduces no second positivity or cancellation assumption.

40.9 Source Audit

40.9.1 Euclid and Joyce

The classical Proposition I.40 is explicitly the equal-base converse of I.38. In Euclid’s displayed proof, a parallel is drawn through one top vertex; I.38 makes the corresponding auxiliary triangle equal to the first triangle, and transitivity of equality then produces the impossible equality of a greater triangle with a lesser one. Joyce also notes that the transmitted statement is less precise than it should be: the bases must lie on one straight line, and the proof should not depend on their sharing an endpoint.

40.9.2 Wilkins

Wilkins makes the hidden configuration problem explicit. A complete proof for two arbitrary equal bases on one line must exclude six configurations; only one appears explicitly in Euclid’s proof. This is an important control on the formal statement. The current Lean theorem does not silently claim those six cases: it assumes the concrete order $B - C - E - F$.

40.9.3 Hilbert and Hartshorne

Hilbert’s area theory and Hartshorne’s Section 22 remain the controlling sources for the meaning of equal content. The present reconstruction keeps `HilbertScissorsEquicomplementable` distinct from numerical area and from unconditional finite equidecomposability.

For the converse direction, I.39 has already isolated the proper-part strictness issue. I.40 therefore needs no new area semantics. Its genuinely new work is only the geometric reduction from equal

bases to one common base.

40.9.4 Borsuk–Szmielw and Greenberg

Borsuk–Szmielw remain useful for the incidence, order, side-of-line, congruence, and parallelogram infrastructure, but they are not used as the primary source for equal-content semantics. Greenberg similarly provides a control on the Euclidean/neutral boundary and distinguishes equal content from numerical area, while referring the detailed content theory elsewhere.

40.9.5 Beeson

The Beeson proof corpus is retained as a configuration audit rather than as the architectural source of the present proof. No Tarski-style proof trace is copied into I.40. In particular, the final Lean architecture is determined by the already developed Hilbert/I.38/I.39 interfaces.

40.10 Formalization Notes

40.10.1 Geometry

The geometric data actually used by the theorem are:

- the strict order $B - C - E - F$;
- one common base carrier;
- equal base segments $BC \cong EF$;
- SameSide data for A and D ;
- the reusable congruent-copy/parallelogram package from I.38;
- Euclidean uniqueness of the parallel through a point.

The theorem does not infer all possible relative positions of equal base segments. Those alternatives are outside its current statement.

40.10.2 Content

The public content hypothesis is exactly `HilbertScissorsEquicomplementable`. The local copy step uses the stronger `HilbertScissorsEq` because congruent triangles can be replaced directly. The proof then weakens this relation and uses only symmetry and transitivity.

There is no local subtraction, cancellation, comparison relation, numerical area, or whole-greater-than-part argument in I.40. The only strict content principle in the dependency graph is the positivity axiom already used to prove I.39.

40.10.3 Representation

The congruence theorem returned by I.38 is oriented as

$$BAC \cong EGF,$$

while the content theorem is stated for ABC and GEF . The formal proof therefore uses

```
scissors_triangle_swap12
```

twice. This is representational transport, not additional geometry.

Likewise, the conclusion of I.39 is expressed by an actual incidence line `top`, whereas $AG \parallel BC$ is initially expressed by point-line carriers. The final proof explicitly bridges these two representations before applying parallel uniqueness.

40.10.4 Axiomatic status

The public theorem assumes `HilbertEuclideanPlane`; it is therefore an Euclidean theorem in the current hierarchy. The final carrier identification uses the uniqueness of the parallel through G , and the reused I.38 construction also lives in the Euclidean layer. No continuity assumption is introduced.

On the content side, I.40 adds no axiom. It imports I.39 and therefore inherits the previously isolated scissors positivity assumption. This inheritance should not be confused with a new I.40-specific foundational principle.

40.11 Dependency Summary

The actual formal dependency chain is

```

Hilbert incidence, order, congruence, and Euclidean parallel axioms
      ↓
HilbertInterface / Book Zero line, side-of-line, and parallel transport
      ↓
generalized I.38 equal-base copy + parallelogram construction
      ↓
i40_base_endpoints + i40_copy_parallel
      ↓
ABC ~ GEF
      ↓
GEF ≈ DEF
      ↓
Proposition I.39 on the common base EF
      ↓
G, D ∈ top, top ∩ base = ∅
      ↓
parallel uniqueness through G
      ↓
A, D ∈ top, top ∩ base = ∅
      ↓
euclid_proposition_40.
```


The historical and formal DAGs are therefore related but distinctly organized:

Euclid : I.31 $\rightarrow$ I.38 $\rightarrow$ proper-part contradiction $\rightarrow$ same parallels,
 Lean : I.38 copy geometry $\rightarrow$ content transport $\rightarrow$ I.39 $\rightarrow$ parallel uniqueness.

New geometric axiom in I.40:	no
New content axiom in I.40:	no
Inherited content axiom:	yes — scissors positivity through I.39
New proposition-local helpers:	yes — two
New reusable declaration in I.40:	no
Reusable refactor required:	yes — generalized I.38 copy helpers
New Interface/Scissors theorem:	no
New Book Zero theorem:	no
New numerical area or polygon semantics:	no
Public theorem compiles:	yes
Full <code>lake build</code> :	clean
sorry in <code>Proposition40.lean</code> :	no

Diagrammatic audit. The former single three-panel figure has been replaced by four figures with distinct roles. Figure 40.1 records Euclid’s actual adjacent-base reductio, while Figures 40.2–40.4 record the three genuinely new geometric states in the current Lean proof: the ordered input $B - C - E - F$, the reused I.38 copy with its new point G , and the final carrier obtained from I.39 and parallel uniqueness. The copy figure intentionally keeps D off the carrier AG , because the Lean proof has not yet established that incidence relation at that stage.

No separate figure is added for `i40_base_endpoints`: it only transports E and F onto the already drawn base carrier. Likewise, `scissors_triangle_swap12`, conversion from exact scissors equality to equicomplementability, symmetry, and transitivity are representation or content bookkeeping on an unchanged geometry. Wilkins’ six positional cases are retained in the prose as a statement-scope warning, but they are not branches of the present production theorem and therefore are not promoted to formal proof-state figures.

40.12 Conclusion

Proposition I.40 is a useful test of whether the preceding area block has produced reusable mathematics rather than proposition-specific code. Once I.39 supplies the same-base converse and I.38 supplies an equal-base congruent copy, the new theorem does not need another proper-part proof. It transports the original equal-content hypothesis to a copied triangle on the base EF , applies I.39, and finishes by uniqueness of the parallel through the copied top vertex.

The most important development result occurred just before the final proof was written. I.40 exposed that the equal-base copy construction in I.38 had been stated too narrowly. Generalizing the existing I.38 helpers eliminated a large duplicate proof while leaving I.38 itself unchanged. The clean rebuild of the entire project confirmed that the refactor preserved the earlier proposition and produced a genuinely reusable boundary for I.40.

The content architecture also remains stable. No numerical area, new scissors operation, or new strictness axiom is introduced. I.40 inherits the single positivity debt already made explicit by I.39 and otherwise consists of geometry, representation transport, equivalence transport, and Euclidean parallel uniqueness.

Finally, the formal statement remains intentionally narrower than the most general classical proposition: it treats the explicit ordered carrier $B - C - E - F$. Wilkins' six-configuration analysis therefore remains a useful record of what a future order-normalizing wrapper would have to cover. The current theorem is complete for its stated configuration, builds cleanly, and fits the architecture established by I.35–I.39.

Chapter 41

Proposition I.41

41.1 Introduction

Proposition I.41 begins a new use of the area block rather than a new area foundation. The preceding propositions I.37–I.40 compare triangles. I.41 combines the same-base theorem I.37 with the diagonal theorem I.34 and concludes that a parallelogram is “double” a triangle on the same base and between the same parallels.

The present formalization keeps this statement entirely synthetic. It does not introduce a numerical area function and it does not interpret “double” as multiplication by the real number 2. Instead the conclusion is expressed in the existing scissors language: the formal parallelogram term is equicomplementable with the formal sum of two copies of the triangle term. Thus the relevant target has the form

$$P \approx (T + T),$$

where $\approx$ abbreviates the project’s term-level

```
HilbertScissorsEquicomplementable
```

relation.

I.41 is therefore a forward content theorem. No content relation is inverted, no proper-part comparison is required, and no positivity, cancellation, or whole-greater-than-part principle is introduced. This sharply separates I.41 from the converse propositions I.39 and I.40.

The reconstruction also exposed one missing reusable law of the scissors calculus. Equicomplementability must be compatible with formal addition. The initial proposition-local “doubling” helper was consequently replaced by the general theorem

```
equicomplementable_add
```

in `HilbertScissors.lean`. The full project rebuild remained clean, and the same reusable theorem now also replaces the former I.41-specific helper in Proposition I.47.

41.2 The Classical Configuration

Let $ABCD$ be a parallelogram and let EBC be a triangle with the same base BC . Both figures lie between the same two parallel carriers. In Euclid’s notation the upper carrier passes through

A and E , so

$$AE \parallel BC.$$

The parallelogram has the same lower base BC , and its upper side AD lies on the same upper carrier in the classical picture.

The public Lean theorem encodes this configuration economically by the two hypotheses

```
(hParallelogram : IsParallelogram Geo A B C D)
(hAE_BC : Geo.Parallel A E B C)
```

There is no additional hypothesis asserting explicitly that A, D, E are collinear. The proof does not need such a witness. The parallelogram structure is used by I.34, while the parallel pair AE, BC is used independently by I.37. This is an example where the formal statement contains exactly the data needed by the proof rather than reproducing every incidence relation visible in the classical diagram.

Figure 41.1 records the single classical geometric state needed for the proposition. The point E is drawn at one representative position on the common upper carrier. Its displayed order relative to A and D is not a hypothesis of the Lean theorem.

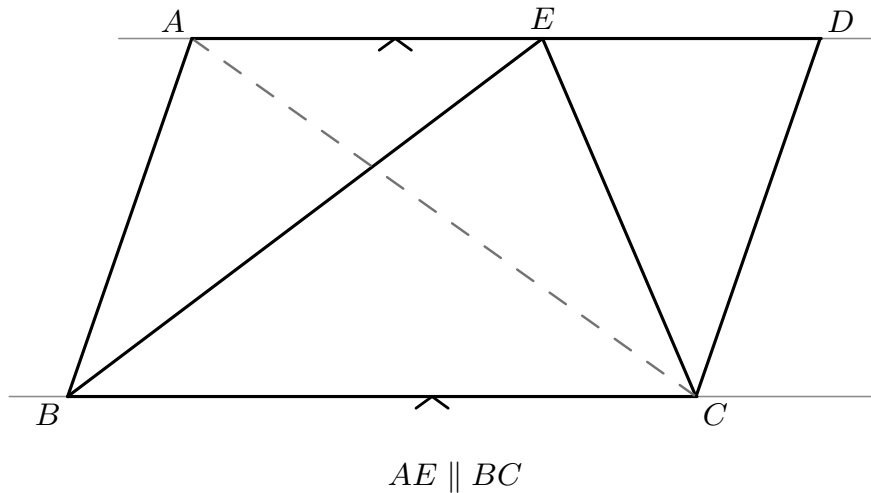


Figure 41.1: The classical I.41 configuration. The parallelogram $ABCD$ and triangle EBC have the common base BC and lie between the same parallel carriers. Euclid joins the auxiliary diagonal AC , which is the diagonal consumed by I.34. The displayed placement of E on the upper carrier is representative only: the formal theorem assumes $AE \parallel BC$ and does not assume a betweenness order among A, D, E .

41.3 Classical Proof

Euclid's proof is one of the shortest in the area block.

Join A to C . Proposition I.37 applies to the triangles ABC and EBC : they stand on the same base BC and lie in the same parallels AE and BC . Hence the two triangles are equal in Euclid's content sense.

Next apply Proposition I.34 to the parallelogram. Its diagonal AC bisects the parallelogram, so the parallelogram is double the triangle ABC . Since ABC is equal to EBC , the parallelogram is also double the triangle EBC .

The classical dependency graph is therefore

$$\begin{array}{c}
 ABCD \text{ parallelogram, } AE \parallel BC \\
 \Downarrow \\
 \triangle ABC = \triangle EBC \quad \text{by I.37} \\
 \Downarrow \\
 ABCD = 2\triangle ABC \quad \text{by I.34} \\
 \Downarrow \\
 ABCD = 2\triangle EBC.
 \end{array}$$

Joyce’s text makes exactly this sequence explicit: join AC , use I.37 for the same-base triangles, and use I.34 for the diagonal bisection of the parallelogram. Unlike I.39 and I.40, there is no *reductio* and no proper-part argument.

41.4 Doubling in the Scissors Calculus

The word “double” needs a precise synthetic interpretation in the current library. The type

```
HilbertScissorsTerm
```

is a formal multiset sum of triangle terms. Thus the term-level double of a triangle T is simply

$$T + T.$$

No scalar multiplication is introduced.

The crucial reusable statement is

```

theorem equicomplementable_add
  {P1 Q1 P2 Q2 : HilbertScissorsTerm Geo}
  (h1 : HilbertScissorsEquicomplementable Geo P1 Q1)
  (h2 : HilbertScissorsEquicomplementable Geo P2 Q2) :
  HilbertScissorsEquicomplementable Geo
    (P1 + P2) (Q1 + Q2)

```

Conceptually, if $P_1 \approx Q_1$ and $P_2 \approx Q_2$, then

$$P_1 + P_2 \approx Q_1 + Q_2.$$

I.41 uses the same relation twice, obtaining

$$ABC + ABC \approx EBC + EBC.$$

This theorem belongs to `HilbertScissors.lean`, not to `Proposition41.lean`. Its proof is pure algebra of the existing scissors relation: the complements for the two hypotheses are added, the two whole-term relations are combined with `HilbertScissorsEq.add`, and associativity and commutativity of multiset addition rearrange the resulting terms.

This is stronger architecturally than an I.41-specific “double” lemma. The law is closed under arbitrary pairs of equicomplementable terms and can be reused whenever a later proposition adds equal-content components.

41.5 Proof Architecture of the Reconstruction

The Lean proof is short enough to be displayed as four conceptual stages.

41.5.1 Step 1: replace the second diagonal triangle by the first

Proposition I.34 supplies

```
euclid_proposition_34_diagonal
```

with result

$$ABC \cong CDA.$$

The theorem

```
scissors_congruent
```

turns this triangle congruence into a scissors equality. A cyclic permutation of the triangle representation identifies CDA with ACD , giving

$$ABC \sim ACD,$$

where $\sim$ denotes `HilbertScissorsEq`.

The formal parallelogram term is defined by the diagonal AC :

$$\text{Par}(ABCD) = ABC + ACD.$$

Adding reflexivity on the first triangle to the reverse of the relation above yields

$$\text{Par}(ABCD) \sim ABC + ABC.$$

This is the formal counterpart of the classical statement that the diagonal bisects the parallelogram.

41.5.2 Step 2: apply Proposition I.37 on the common base

The hypothesis

$$AE \parallel BC$$

is already in exactly the form expected by Proposition I.37. The call

```
euclid_proposition_37 Geo A B C E hAE_BC
```

returns

$$ABC \approx EBC.$$

No new carrier construction, SameSide witness, midpoint construction, or order case split appears in I.41 itself. All geometric normalization required by the same-base theorem is encapsulated in the already proved I.37.

41.5.3 Step 3: add the equal-content relation to itself

The new reusable theorem is applied twice to the same hypothesis:

```
equicomplementable_add
  Geo
  hABC_EBC
  hABC_EBC
```

This gives

$$ABC + ABC \approx EBC + EBC.$$

This is the precise term-level operation corresponding to Euclid’s use of “double.”

41.5.4 Step 4: transport from the doubled triangle to the parallelogram

The final call

```
equicomplementable_transport
```

uses the scissors equality

$$\text{Par}(ABCD) \sim ABC + ABC$$

on the left and reflexivity on $EBC + EBC$ on the right. It therefore transports the doubled I.37 conclusion to the public target

$$\text{Par}(ABCD) \approx EBC + EBC.$$

The complete Lean DAG is

$$\begin{array}{c}
 \text{IsParallelogram}(A, B, C, D) \quad + \quad AE \parallel BC \\
 \Downarrow \text{I.34} \\
 ABC \cong CDA \\
 \Downarrow \text{scissors_congruent} \\
 \text{Par}(ABCD) \sim ABC + ABC \\
 AE \parallel BC \\
 \Downarrow \text{I.37} \\
 ABC \approx EBC \\
 \Downarrow \text{equicomplementable_add} \\
 ABC + ABC \approx EBC + EBC \\
 \Downarrow \text{equicomplementable_transport} \\
 \text{Par}(ABCD) \approx EBC + EBC \\
 \Downarrow \\
 \text{euclid_proposition_41.}
 \end{array}$$

41.6 Lean Formalization

The current proposition file imports exactly the two earlier propositions used by the mathematical proof:

```
import CGJteamLab.Proposition34
import CGJteamLab.Proposition37
```

The public theorem is:

```

theorem euclid_proposition_41
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D E : Geo.Point)
  (hParallelogram : IsParallelogram Geo A B C D)
  (hAE_BC : Geo.Parallel A E B C) :
  HilbertScissorsEquicomplementable Geo
    (hilbertParallelogramTerm Geo A B C D)
    (hilbertScissorsTriangle Geo E B C +
      hilbertScissorsTriangle Geo E B C) := by
  ...

```

The ellipsis is documentation notation only. The repository theorem is fully proved and the full project build is clean.

The mathematically significant declarations called directly by the proof are:

```

euclid_proposition_34_diagonal
scissors_congruent
scissors_triangle_cycle
euclid_proposition_37
equicomplementable_add
equicomplementable_transport

```

The remaining operations are reflexivity, symmetry, and formal addition in `HilbertScissorsEq`. There are no proposition-local helper theorems in the final I.41 file.

41.7 Relationship with Earlier Propositions

41.7.1 Proposition I.34

I.34 supplies the diagonal congruence of the parallelogram. In the classical proof this is stated as bisection of the parallelogram by its diagonal. In the formal proof the stronger triangle-congruence result is converted directly into scissors equality and then used to rewrite

$$ABC + ACD$$

as

$$ABC + ABC.$$

Thus I.34 is the representation bridge from the geometric parallelogram to the formal doubled triangle.

41.7.2 Proposition I.37

I.37 is the direct content theorem. It states that triangles on the same base and between the same parallels are equicomplementable. With the substitution $D := E$, its public hypothesis is exactly the I.41 hypothesis $AE \parallel BC$. The result is therefore available without any new geometric construction.

I.37 is also the reason I.41 needs no halving or cancellation theorem. The formal proof proceeds in the forward direction

$$ABC \approx EBC \implies ABC + ABC \approx EBC + EBC,$$

not in the converse direction from doubled figures back to their halves.

41.7.3 Propositions I.35–I.40

I.35–I.38 established the forward equal-content machinery; I.39 and I.40 then showed where converse reasoning requires strict content information. I.41 returns to the forward direction. Consequently it uses only equivalence transport and addition in the scissors calculus. The positivity issue isolated for I.39 is not needed here.

This position in the Book I sequence is important: I.41 is not another test of proper-part semantics. It is a test of whether the already established content relation behaves compositionally under addition.

41.8 Hilbert and Book Zero Components Used

41.8.1 Triangle congruence and parallelogram infrastructure

The direct geometric input from I.34 ultimately rests on the established Hilbert congruence and parallelogram theory. I.41 itself does not reopen SSS, angle transport, or diagonal incidence. It consumes the packaged theorem

```
euclid_proposition_34_diagonal
```

as a black box.

41.8.2 Same-base triangle content

All of the substantial geometry behind equal-content triangles on a common base is encapsulated by I.37. Its Hilbertian reconstruction uses the midpoint and parallelogram machinery developed earlier in the project, but none of those witnesses is reconstructed in I.41.

41.8.3 Scissors representation

Three levels must remain distinct:

- `hilbertParallelogramTerm` is the chosen formal triangulation of a parallelogram along AC ;
- `HilbertScissorsEq` is the stronger scissors-equivalence relation;
- `HilbertScissorsEquicomplementable` is the equal-content relation used in the public theorem.

I.34 first produces the stronger relation; I.37 produces the weaker content relation; I.41 combines them by addition and transport.

No new Book Zero theorem is introduced. The only new reusable declaration in the scissors layer is

```
equicomplementable_add
```

41.9 Source Audit

41.9.1 Euclid and Joyce

The classical source gives exactly the short proof reflected in the present architecture: join AC , apply I.37 to ABC and EBC , then use I.34 to say that the parallelogram is double the diagonal triangle. Joyce also notes that I.41 generalizes the diagonal statement I.34 and that an equal-base version would be slightly more general.

41.9.2 Hartshorne

Hartshorne’s Section 22 is the main semantic control for the area block. His classification explicitly places I.41–I.45 among the remaining Book I results that are valid for equal content. This is consistent with the present proof: I.41 never uses equal content as a hypothesis to recover metric or incidence information and therefore needs no new property (Z)-style strictness principle.

41.9.3 Hilbert, Borsuk–Szmielew, Greenberg, and Beeson

For I.41 these sources play their established control roles rather than supplying a new proposition-specific theorem. Hilbert remains the architectural source for the separation between congruence, equidecomposition, and equicomplementability. Borsuk–Szmielew and Greenberg control the developed incidence/congruence/parallel background. The Beeson corpus remains a check on formal side conditions, but no Tarski-style local construction is imported into the final Lean proof.

The current Lean source is authoritative for the exact formal claim. In particular, the documentation does not strengthen its conclusion to numerical area equality or to an unconditional geometric dissection statement.

41.10 Formalization Notes

41.10.1 Geometry

The public theorem exposes only two geometric inputs:

- the parallelogram structure of $ABCD$;
- the strict parallel relation $AE \parallel BC$.

There are no explicit Between, SameRay, SameSide, OppositeSide, Pasch, or extension hypotheses in I.41. Any lower-level nondegeneracy needed by I.34 or I.37 is handled inside those earlier theorems.

The absence of an explicit A, D, E collinearity hypothesis is intentional. It is not needed by the actual proof DAG: I.34 consumes the parallelogram, and I.37 consumes $AE \parallel BC$.

41.10.2 Content

The content direction is purely forward:

$$ABC \approx EBC \implies ABC + ABC \approx EBC + EBC.$$

Accordingly there is no subtraction, cancellation, positivity, strict content order, or whole-greater-than-part step. The only new content fact is the closure of equicomplementability under formal addition.

41.10.3 Representation

The diagonal theorem returns the triangle orientation $ABC \cong CDA$, while the parallelogram term is written as $ABC + ACD$. Two applications of

```
scissors_triangle_cycle
```

identify the cyclic presentations. This is representation transport only; it contains no new geometric argument.

The final statement should therefore be read literally at the term level:

$$\text{parallelogram term} \approx \text{triangle term} + \text{triangle term}.$$

It is the current scissors representation of Euclid’s “double,” not a claim about a real-valued area function.

41.10.4 Axiomatic status

The public theorem assumes

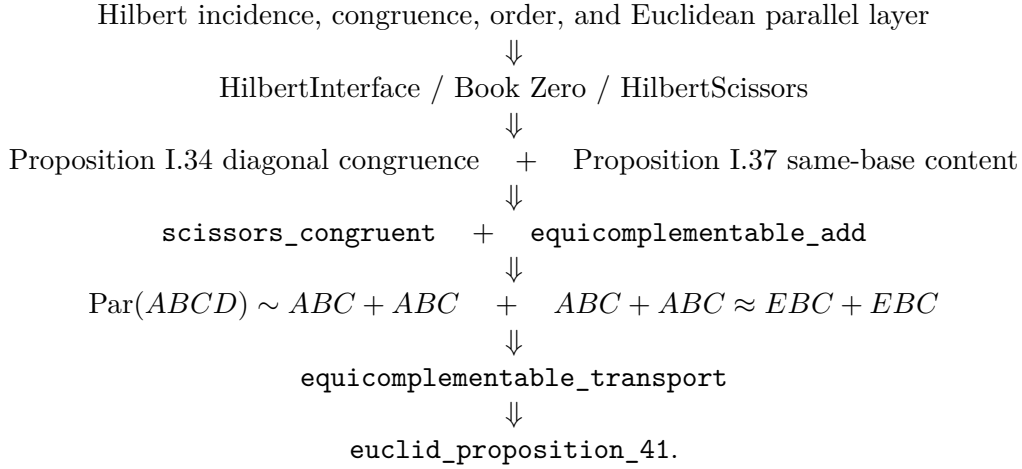
```
HilbertEuclideanPlane Geo
```

and is therefore Euclidean in the current hierarchy. I.41 introduces no new parallel construction and does not call parallel uniqueness directly; its Euclidean dependence is inherited through the already proved I.34/I.37 layer, most importantly the same-parallel content theorem I.37.

No continuity axiom is introduced. No new geometric or content axiom is added by Proposition I.41.

41.11 Dependency Summary

The actual formal dependency chain is



The historical and formal DAGs are unusually close:

$$\begin{array}{ll}
 \text{Euclid} & : \text{I.37} + \text{I.34} \longrightarrow \text{I.41}, \\
 \text{Lean} & : \text{I.37} + \text{I.34} + \text{scissors addition/transport} \longrightarrow \text{I.41}.
 \end{array}$$

The extra Lean layer is representational rather than geometric.

New geometric axiom in I.41:	no
New content axiom in I.41:	no
Inherited strict-content/positivity axiom:	no
New proposition-local helper:	no
New reusable declaration:	yes — <code>equicomplementtable_add</code>
Reusable refactor required:	yes — helper moved to <code>HilbertScissors</code>
New Interface/Scissors theorem:	yes
New Book Zero theorem:	no
New numerical area or polygon semantics:	no
Public theorem compiles:	yes
Full lake build:	clean
sorry in <code>Proposition41.lean</code> :	no

Diagrammatic audit. The revised documentation retains one figure only: the classical configuration of the parallelogram $ABCD$ and the triangle EBC on the common base, together with the auxiliary diagonal AC actually joined in Euclid’s proof. The former second panel has been removed because it repeated exactly the same geometry and only overlaid the content relations supplied by I.34, I.37, and the scissors calculus. No separate figures are added for `scissors_congruent`, `equicomplementtable_add`, cyclic triangle normalization, or `equicomplementtable_transport`: these steps change the formal representation or content relation but introduce no new points, lines, order, side, or intersection state. The figure caption also makes explicit that the drawn order of A, D, E is representative rather than a formal hypothesis.

41.12 Conclusion

Proposition I.41 is a compact confirmation that the area architecture developed since I.35 is compositional. Its geometry is already contained in two earlier results: I.34 identifies the two diagonal halves of a parallelogram, and I.37 compares triangles on the same base and between the same parallels. The new formal ingredient is not another geometric construction but the algebraic fact that equicomplementability is preserved by formal addition.

That observation led directly to the reusable theorem

```
equicomplementable_add
```

in `HilbertScissors.lean`. Once this theorem is available, the I.41 proof becomes almost the same DAG as Euclid's: replace the second diagonal triangle by the first, apply I.37, double the content relation by addition, and transport back to the parallelogram term.

The proposition therefore introduces no new axiom, no numerical area, no strict content order, and no new polygon semantics. The full project rebuild is clean, and the general addition theorem is already reusable beyond I.41.

Chapter 42

Proposition I.42

42.1 Introduction

Proposition I.42 is the first construction in Book I whose target is not merely a line, angle, triangle, or parallel relation, but a new figure with prescribed *content*. Given a triangle and a rectilineal angle, Euclid constructs a parallelogram equal to the triangle and having an angle equal to the prescribed one. In the traditional sequence this is the opening move of the “application of areas” block that continues through I.44 and I.45.

The present reconstruction preserves that synthetic character. There is no real-valued area function. Equality of content is represented by

```
HilbertScissorsEquicomplementable
```

and the constructed figure is represented by

```
hilbertParallelogramTerm
```

The final theorem therefore asserts the existence of points E, F, G such that $FECG$ is a parallelogram, its angle FEC is congruent to the prescribed angle, and the triangle term ABC is equicomplementable with the parallelogram term $FECG$.

I.42 is also a useful test of the separation between three layers that has become essential since I.35:

- **geometry**: midpoint construction, angle layoff, side-of-line information, intersection of a ray carrier with a parallel, and completion of a parallelogram;
- **content**: equal-content comparison by I.38 and the “double” relation supplied by I.41;
- **representation**: carrier identification, ray orientation, and the use of a collinearity statement rather than an invalid strict parallel statement in the boundary case $F = A$.

The classical proof is short. The formal proof is longer because one configuration fact suppressed by the classical diagram becomes decisive: the copied angle ray must actually meet the parallel through A , and the meeting point is allowed to be A itself. The final Lean file proves this rather than assuming it.

42.2 The Classical Configuration

Let ABC be the given triangle and let the prescribed angle be represented formally by the noncollinear triple XYZ . Euclid bisects BC at E , so $BE = EC$. He then constructs at E an angle CEF equal to the prescribed angle, choosing the new ray on the side of the line BC that contains A . Through A he draws a line parallel to EC , meeting the constructed ray at F . Through C a line parallel to EF completes the parallelogram $FECG$.

The intended final configuration is therefore

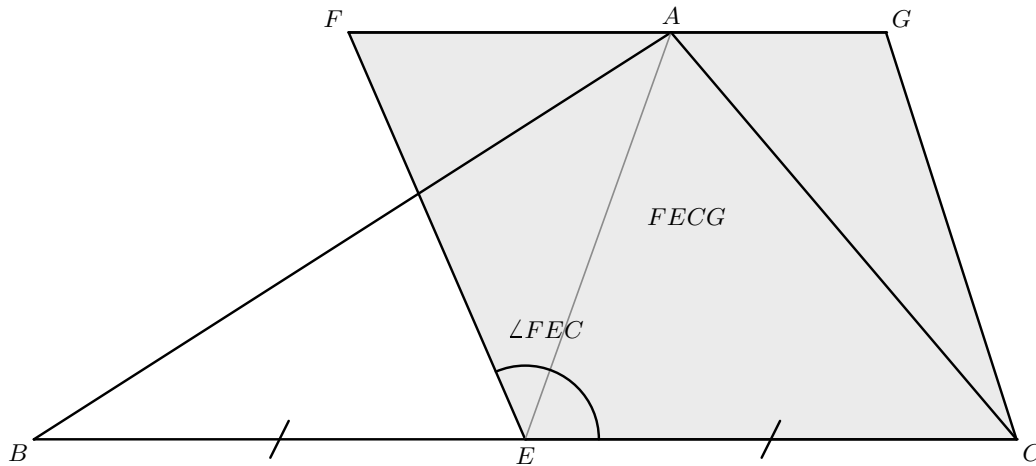
$$B - E - C, \quad F, A, G \text{ on one upper carrier}, \quad FG \parallel EC,$$

with $FECG$ a parallelogram and

$$\angle FEC \cong \angle XYZ.$$

The point A is not required to be strictly between F and G , and the formal construction must not exclude the legitimate case $F = A$.

Figure 42.1 records only the classical configuration. It uses the customary generic placement of A on the upper side of the constructed parallelogram. This placement is illustrative: the formal theorem asserts only that F, A, G are collinear and does not assert any strict order among these three points.



representative classical position

Figure 42.1: The classical configuration of Proposition I.42. The point E bisects BC , the prescribed angle is copied as $\angle FEC$, and the upper side of the parallelogram $FECG$ passes through A . The displayed position of A between F and G is a representative classical drawing only; no such betweenness statement is part of the formal theorem.

42.3 Classical Proof

Euclid's dependency graph is compact.

First bisect BC at E by I.10 and join AE . Since $BE = EC$, Proposition I.38 gives

$$\triangle ABE = \triangle AEC$$

in Euclid’s equality-of-figures sense: the triangles stand on the equal bases BE and EC and between the same parallels. Hence

$$\triangle ABC = 2\triangle AEC.$$

Next construct at E the angle CEF equal to the given angle by I.23. Draw through A the line parallel to EC by I.31 and complete the parallelogram $FECG$. Proposition I.41 then gives

$$FECG = 2\triangle AEC.$$

Thus both the original triangle and the constructed parallelogram are double the same triangle, so Common Notion 1 yields

$$\triangle ABC = FECG.$$

The angle FEC has the prescribed value by construction.

The classical DAG can be displayed as

$$\begin{array}{c}
 E \text{ midpoint of } BC \quad (\text{I.10}) \\
 \Downarrow \\
 ABE = AEC \quad (\text{I.38}) \\
 \Downarrow \\
 ABC = 2AEC \\
 \angle FEC \cong \angle XYZ \quad (\text{I.23}), \quad AG \parallel EC \quad (\text{I.31}) \\
 \Downarrow \\
 FECG \text{ parallelogram} \\
 \Downarrow \\
 FECG = 2AEC \quad (\text{I.41}) \\
 \Downarrow \\
 ABC = FECG.
 \end{array}$$

The historical proof is therefore a forward construction. Equal content is never assumed in order to recover a metric or incidence conclusion. This will matter for the axiomatic status below.

42.4 Application of Areas and the Hidden Intersection

I.42 marks a change in the use of the area block. Propositions I.35–I.41 mainly compare already given figures. I.42 uses those comparison theorems as an engine for *constructing* a new figure with prescribed content and a prescribed angle. This is the first of the Book I “application of areas” constructions that will feed I.44 and I.45.

There is, however, a geometric step hidden by the ordinary diagram. After constructing the ray at E and the line through A parallel to EC , Euclid names their intersection F without separately proving that the two carriers meet. Wilkins makes the intended orientation explicit by directing the ray into the side of BC containing A . The Beeson proof trace goes further: it explicitly stops to prove that the angle ray meets the upper parallel. Its case analysis also contains the equality branch in which the intersection is A itself.

That observation changes the correct formal API. An earlier draft attempted to encode the upper-carrier condition as

`Geo.Parallel F A E C`

This is too strong because the project's parallel relation is strict. If $F = A$, the expression would ask the degenerate carrier AA to be parallel to EC . The final theorem instead returns

```
Collinear Geo F A G
```

for the completed parallelogram $FECG$. This says exactly what the geometry needs: A lies on the upper carrier FG , including the boundary case $F = A$.

This is not a cosmetic change of representation. It is an example of a hidden configuration condition changing the mathematically correct formal statement of an intermediate construction.

42.5 Proof Architecture of the Reconstruction

The final Lean proof is organized into seven conceptual stages. The local helpers are proposition-specific because they package the exact geometry of I.42 rather than a new general synthetic theory.

42.5.1 Step 1: midpoint and equal-content halves

The public theorem first obtains a strict midpoint E of BC . The helper

```
i42_median_bisects_area
```

proves

$$ABE \approx AEC,$$

where $\approx$ denotes `HilbertScissorsEquicomplementable`.

The proof is deliberately not a numerical same-altitude argument. It creates one remote comparison triangle DUV . A line through A and D is parallel to the base carrier. The base is extended so that the ordered points contain $B - E - C - U - V$, and a segment UV is laid off with

$$UV \cong BE \cong EC.$$

Proposition I.38 is then applied twice:

$$ABE \approx DUV, \quad AEC \approx DUV.$$

Transitivity gives the desired midpoint-halves relation.

This reduction is a consequence of the current I.38 API: I.38 is expressed using two separated equal bases, so a common remote triangle supplies a single comparison target for both halves.

42.5.2 Step 2: copy the angle on the correct side and draw the parallel

The helper

```
i42_angle_parallel_setup
```

constructs the carrier EC , uses Hilbert's angle-construction axiom/theorem to copy XYZ at E , and selects the side containing A . It returns a point R satisfying

$$R \text{ and } A \text{ are on the same side of } EC, \quad \angle CER \cong \angle XYZ.$$

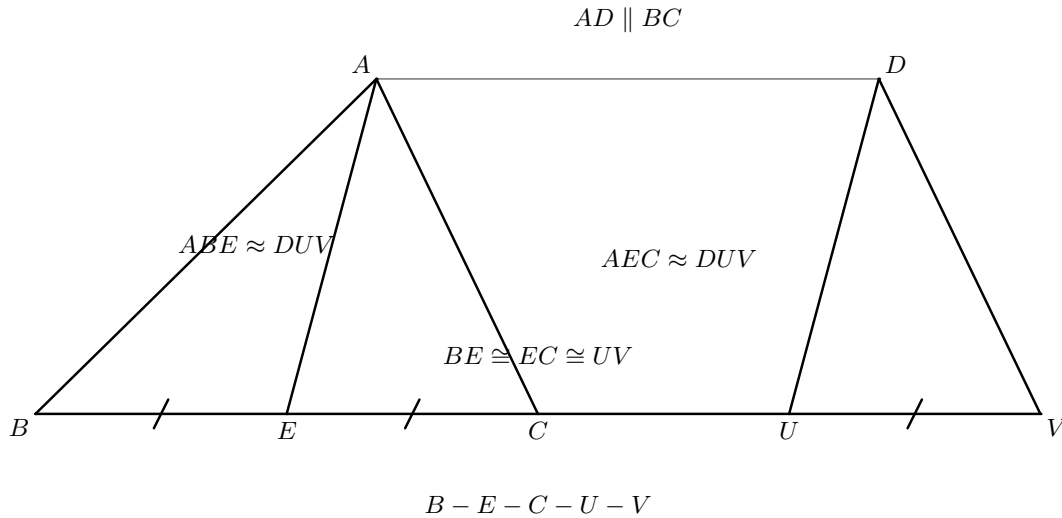


Figure 42.2: The auxiliary configuration used by `i42_median_bisects_area`. The base is extended in the strict order $B - E - C - U - V$; the remote triangle DUV lies on a carrier through D parallel to the original base; and $BE \cong EC \cong UV$. Proposition I.38 compares both ABE and AEC with the same triangle DUV .

It also constructs a point Q such that

$$EC \parallel AQ.$$

Thus the two objects that must meet – the ray carrier ER and the upper parallel carrier AQ – are both explicit Lean data.

42.5.3 Step 3: prove that the two carriers meet

The helper

```
i42_angle_parallel_intersection
```

is the central Euclidean existence argument of the proposition. Suppose the lines ER and AQ were disjoint. The bridge

```
intersection_test_parallel_of_lines_disjoint
```

would convert this incidence-level disjointness into strict parallelism

$$ER \parallel AQ.$$

But $EC \parallel AQ$ as well. If the carriers ER and EC were distinct, Euclidean transitivity of parallels would imply

$$ER \parallel EC,$$

which is impossible because both contain E . If the two carriers were the same, then R would lie on the base EC , contradicting the `SameSide` output of angle construction. Therefore ER and AQ meet, and their intersection is named F .

This is exactly the step that the classical text suppresses.

42.5.4 Step 4: orient the intersection onto the correct ray

An intersection of the two *lines* is not yet enough. I.42 needs F on the chosen ray from E , not on the opposite ray. The helper

```
i42_angle_parallel_oriented_intersection
```

proves this using side-of-line data.

First, A and F lie on the same side of the base. If $F = A$, this is simply SameSide reflexivity. Otherwise the carrier AF is the carrier AQ , hence is parallel to EC , and its endpoints lie on the same side of EC . Since R and A were already constructed on the same side, transitivity gives that R and F are on the same side of EC .

Now E, R, F are collinear. They cannot satisfy $R - E - F$, because then the base line through E would separate R and F . Hence

```
HilbertSameRay Geo E R F
```

and the angle can be transported from the auxiliary ray ER to the actual intersection ray EF . The result is

$$\angle FEC \cong \angle XYZ.$$

Figure 42.3 separates the three distinct geometric states hidden by the usual single Euclidean sketch: construction of the two carriers, existence and orientation of their intersection, and the boundary branch in which the intersection is A itself.

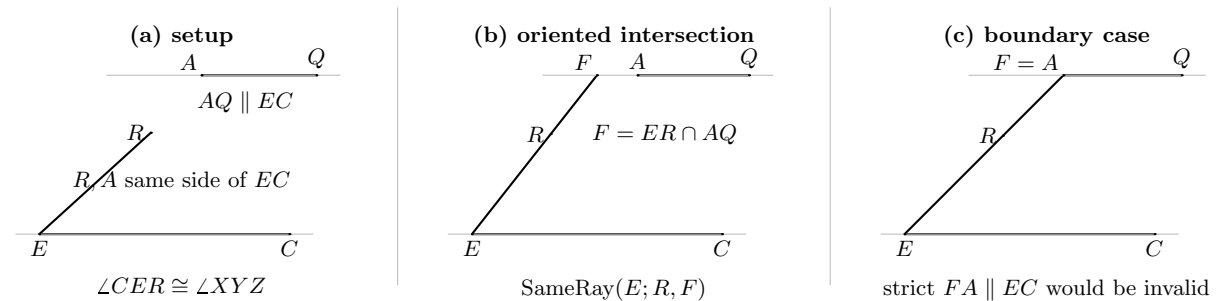


Figure 42.3: The formal angle/parallel intersection sequence. (a) Angle construction gives a ray carrier ER on the same side of EC as A , while $AQ \parallel EC$; no intersection witness has yet been selected. (b) The two carriers are proved to meet at F , and side-of-line information yields $\text{SameRay}(E; R, F)$; the drawing does not assert a strict order between R and F . (c) The legitimate boundary branch $F = A$ is shown literally. This branch is why the final API must not require the degenerate pair FA to be strictly parallel to EC .

42.5.5 Step 5: complete the parallelogram and identify its upper carrier

The helper

```
i42_construct_parallelogram
```

first proves that F is off the base EC : it lies on AQ , and AQ is parallel to EC . Thus E, F, C are noncollinear. The reusable theorem

`hilbert_parallelgram_fourth_vertex_exists`

then completes EFC to a parallelogram $EFGC$, which is reversed to the orientation $FECG$ used by I.42.

The upper side GF and the constructed carrier AQ are both parallel to EC and share F . Euclidean parallel uniqueness/transitivity identifies their carriers. Therefore G also lies on AQ , and the helper returns

`Collinear Geo F A G`

rather than the invalid strict relation $FA \parallel EC$.

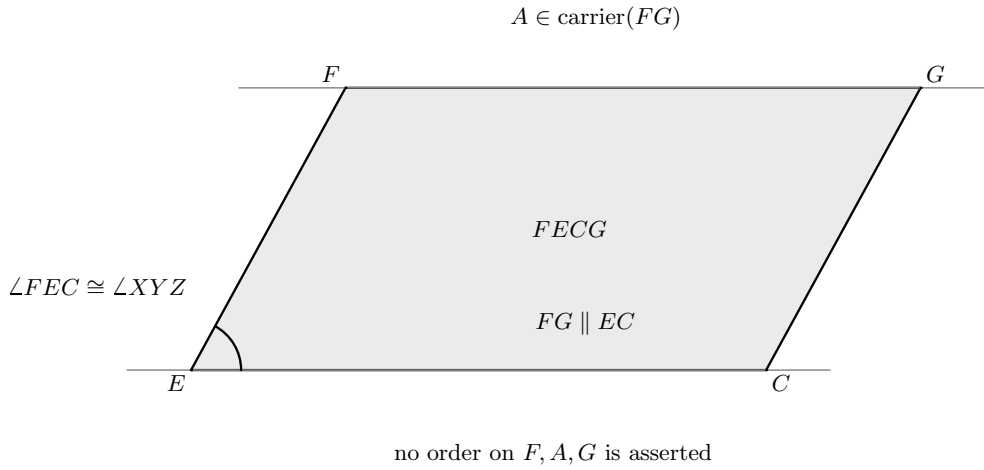


Figure 42.4: The geometric output of `i42_construct_parallelgram`: $FECG$ is a genuine parallelogram, $FG \parallel EC$, and the given point A lies on the carrier of FG . The figure deliberately records $A \in \text{carrier}(FG)$ without assigning any betweenness relation among F, A, G .

42.5.6 Step 6: use I.41 on a genuine side and transport along the carrier

The carrier formulation requires a small wrapper around I.41:

`i42_parallelgram_double_triangle_on_carrier`

The current public I.41 expects a strict parallel pair. Instead of using the possibly degenerate pair FA , the wrapper uses the genuine parallelogram side FG and applies I.41 with apex G :

$$\text{Par}(FECG) \approx GEC + GEC.$$

It then compares GEC with AEC . If $G = A$, this is reflexivity. Otherwise G, A, F are collinear, so the carrier GA is the same upper carrier as GF ; therefore $GA \parallel EC$, and I.37 gives

$$GEC \approx AEC.$$

Using `equicomplementtable_add` twice yields

$$GEC + GEC \approx AEC + AEC,$$

and transitivity gives

$$\text{Par}(FECG) \approx AEC + AEC.$$

42.5.7 Step 7: assemble the content equality

The top-level theorem uses the scissors splitting identity

$$ABC \sim ABE + AEC,$$

where $\sim$ denotes `HilbertScissorsEq`. From Step 1 and reflexivity, `equicomplementable_add` gives

$$ABE + AEC \approx AEC + AEC.$$

Transport across the split yields

$$ABC \approx AEC + AEC.$$

Step 6 gives the same doubled triangle as the content of the constructed parallelogram. A final symmetry and transitivity step closes

$$ABC \approx \text{Par}(FECG).$$

The full formal DAG is therefore

$$\begin{array}{c}
 E \text{ midpoint of } BC \\
 \Downarrow \\
 \text{I.38 twice via } DUV \\
 \Downarrow \\
 ABE \approx AEC \\
 \Downarrow \\
 ABC \approx AEC + AEC \\
 \text{angle construction} + EC \parallel AQ \\
 \Downarrow \\
 ER \cap AQ = \{F\} \quad \text{and} \quad E, R, F \text{ same ray} \\
 \Downarrow \\
 FECG \text{ parallelogram, } F, A, G \text{ collinear} \\
 \Downarrow \\
 \text{I.41 at } G \quad + \quad \text{I.37 along the upper carrier} \\
 \Downarrow \\
 \text{Par}(FECG) \approx AEC + AEC \\
 \Downarrow \\
 ABC \approx \text{Par}(FECG).
 \end{array}$$

42.6 Lean Formalization

The current file imports the two earlier content propositions used directly by the reconstruction:

```
import CGJteamLab.Proposition38
import CGJteamLab.Proposition41
```

The public theorem is the following. The snippet uses the project's preferred ASCII notation for documentation while preserving the exact theorem data.

```
theorem euclid_proposition_42
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
```

```

(A B C X Y Z : Geo.Point)
(hABC : Not (Collinear Geo A B C))
(hXYZ : Not (Collinear Geo X Y Z)) :
exists E F G : Geo.Point,
  IsParallelogram Geo F E C G /\
  Geo.AngleCongruent F E C X Y Z /\
  HilbertScissorsEquicomplementable Geo
    (hilbertScissorsTriangle Geo A B C)
    (hilbertParallelogramTerm Geo F E C G) := by
-- proof in Proposition42.lean

```

The six mathematically significant proposition-local helpers are:

```

i42_median_bisects_area
i42_angle_parallel_setup
i42_angle_parallel_intersection
i42_angle_parallel_oriented_intersection
i42_construct_parallelogram
i42_parallelogram_double_triangle_on_carrier

```

They are not six unrelated technical conveniences. They compress the proof into the six geometric/content transitions described above.

42.6.1 The midpoint-content helper

Its public shape is

```

theorem i42_median_bisects_area
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C E : Geo.Point)
  (hABC : Not (Collinear Geo A B C))
  (hE : HilbertIsMidpoint Geo E B C) :
  HilbertScissorsEquicomplementable Geo
    (hilbertScissorsTriangle Geo A B E)
    (hilbertScissorsTriangle Geo A E C) := by
-- I.38 twice through one remote triangle DUV

```

The helper is proposition-local because its remote-triangle construction is a way of adapting the current I.38 API to the midpoint geometry of I.42. A future general same-altitude theorem might subsume it, but the present file does not introduce such a new interface theorem.

42.6.2 The construction helper

The decisive output is

```

theorem i42_construct_parallelogram
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A E C X Y Z : Geo.Point)
  (hAEC : Not (Collinear Geo A E C))
  (hXYZ : Not (Collinear Geo X Y Z)) :
  exists F G : Geo.Point,

```

```

IsParallelogram Geo F E C G /\
Geo.AngleCongruent F E C X Y Z /\
Collinear Geo F A G := by
-- angle ray, intersection, orientation, and fourth vertex

```

The third conjunct is the important final API decision. It records the common upper carrier without imposing strict distinctness on F and A .

42.6.3 The carrier form of I.41

The last local helper has the shape

```

theorem i42_parallelogram_double_triangle_on_carrier
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (F E C G A : Geo.Point)
  (hParallelogram : IsParallelogram Geo F E C G)
  (hFAG : Collinear Geo F A G) :
  HilbertScissorsEquicomplementable Geo
    (hilbertParallelogramTerm Geo F E C G)
    (hilbertScissorsTriangle Geo A E C +
     hilbertScissorsTriangle Geo A E C) := by
-- I.41 at G, then I.37 from GEC to AEC

```

This wrapper is not a new area principle. It is a geometry/API adapter that keeps the strictness of I.41 on the genuine side FG and uses I.37 to move the triangle apex along the same upper carrier.

42.7 Relationship with Earlier Propositions

42.7.1 Proposition I.10

Classically, I.10 supplies the midpoint E of BC . The Lean theorem does not call I.10 by proposition name. It uses the developed Hilbert midpoint existence theorem directly. This is a typical Hilbert reorganization: the construction is reusable foundational infrastructure rather than a historical Book I dependency.

42.7.2 Propositions I.23 and I.31

Euclid uses I.23 to reproduce the given angle and I.31 to draw the parallel through A . The reconstruction again consumes the corresponding Hilbert operations directly:

```

HilbertCongruence.angle_construction
hilbert_parallel_through_point_exists

```

The formal proof then adds information absent from the classical invocation: the chosen side of the base, the line carriers, and the proof that the two constructed carriers meet.

42.7.3 Proposition I.37

I.37 enters only after the parallelogram has already been constructed. The carrier wrapper for I.41 first obtains the double of GEC , then uses the fact that G and A lie on the same upper carrier to obtain

$$GEC \approx AEC.$$

This is a forward use of same-base equal content.

42.7.4 Proposition I.38

I.38 supplies the equal-content halves cut by the median. The current I.42 proof applies I.38 twice through the auxiliary triangle DUV rather than restating a new theorem about equal altitudes. This is the main content input before the final application of I.41.

42.7.5 Proposition I.41

I.41 states that a parallelogram on the same base and in the same parallels is double the corresponding triangle. In I.42 the strict public API is applied with G as the apex because FG is a genuine side of the parallelogram. I.37 then transports from GEC to AEC . This avoids any need to assume $F \neq A$.

42.7.6 Propositions I.44 and I.45

Historically, I.42 is immediately reusable: I.44 starts by constructing a parallelogram equal to a given triangle in a prescribed angle, and I.45 builds a parallelogram equal to a general rectilinear figure by repeated application of the triangle construction. The current project follows the same broad dependency direction: Proposition I.45 imports Proposition I.42.

Cleaning the final I.42 file also exposed one dependency-hygiene issue in I.45. I.45 had been using an old proposition-local helper for adding an unchanged scissors term to an equicomplementability relation. That call was replaced by the already reusable

```
equicomplementable_add
```

from the scissors layer. No helper had to be restored to I.42, and the full project build remained clean.

42.8 Hilbert and Book Zero Components Used

42.8.1 Midpoint, betweenness, and segment construction

The midpoint data are strict Hilbert data: $B - E - C$ together with $BE \cong EC$. The proof uses betweenness incidence facts, base extensions, segment construction on a ray, and transitivity/symmetry of segment congruence to build the auxiliary equal base UV .

These operations belong to the existing Hilbert/Book Zero language. I.42 introduces no new midpoint or segment-arithmetic theorem.

42.8.2 Angle construction, SameSide, and SameRay

Hilbert angle construction is stronger as an API than the classical phrase “construct an equal angle”: it also records the chosen side of a carrier. That SameSide witness becomes essential later. It is combined with the parallel carrier AQ to prove that the line intersection F is on the correct ray, expressed by

```
HilbertSameRay Geo E R F
```

The angle is then transported along that ray.

This is a good example of hidden order data becoming a first-class part of the formal proof rather than a diagrammatic convention.

42.8.3 Euclidean parallel and intersection infrastructure

The central intersection proof uses the established bridge from disjoint incidence lines to strict point-line parallelism, together with Euclidean transitivity of distinct parallels and the fact that parallel carriers cannot share a point. Carrier identification later uses the same Euclidean uniqueness principle.

Thus the parallel axiom enters I.42 not merely because a parallel line is drawn, but because the proof must control uniqueness and intersection of the constructed carriers.

42.8.4 Parallelogram infrastructure

Once E, F, C are shown noncollinear, the theorem

```
hilbert_parallelogram_fourth_vertex_exists
```

constructs the fourth vertex. The representation is normalized by

```
ParallelogramReverse
```

and the upper side is identified with the already constructed carrier through A . No new parallelogram axiom is introduced.

42.8.5 Scissors content infrastructure

The content layer uses only previously established operations:

```
equicomplementable_add
equicomplementable_refl
equicomplementable_symm
equicomplementable_trans
equicomplementable_transport
```

The triangle split itself is a `HilbertScissorsEq` identity. At no point is a real-valued area, scalar multiplication, subtraction, or content order introduced.

42.9 Source Audit

42.9.1 Euclid, Heath, and Joyce

The classical text gives the exact short DAG used as the historical reference: bisect BC , construct the prescribed angle at E , draw the parallel through A , complete the parallelogram, use I.38 for the two half-triangles, and use I.41 for the parallelogram. Joyce also identifies I.42 as the beginning of the application-of-areas phase of Book I.

This source controls the statement and the classical dependency graph. It does not by itself resolve the hidden existence proof for the intersection point F .

42.9.2 Wilkins

Wilkins makes one orientation choice explicit that is important for the formal reconstruction: the newly constructed ray from E is directed into the side of the line BC containing A . His account then describes the parallel through A as meeting that ray at F and completes the parallelogram. This is the correct classical configuration control for the SameSide/SameRay argument in Lean.

42.9.3 Beeson

The Beeson formal proof corpus is especially useful as a gap audit here. Its I.42 trace explicitly interrupts the Euclidean construction to prove that the angle ray meets the upper parallel, noting that this meeting is omitted in Euclid’s prose. More importantly, the trace includes a branch in which the meeting point is A itself.

The final Lean architecture is not copied from the Tarski-style Beeson proof, but these observations directly validate two design decisions: intersection must be proved, and the intermediate API must allow $F = A$.

42.9.4 Hilbert

Hilbert’s Chapter IV remains the semantic control for the content block. The distinction between congruent-piece equivalence and equicomplementability is maintained throughout the project. In I.42 the final proof does not call Hilbert’s Theorems 45–46 directly; instead their content architecture is already embodied in the previously proved I.37–I.41 layer.

For the geometric construction, the Hilbert organization is even more direct: angle layoff, side-of-line data, line incidence, and the Euclidean uniqueness of parallels are reusable operations rather than historical Book I propositions.

42.9.5 Hartshorne

Hartshorne’s Section 22 is the main semantic check on the forward content argument. He distinguishes the results that require the extra strictness property (Z) from those that remain valid for equal content. The remaining Book I results I.41–I.45 are in the latter class.

That classification matches the actual Lean DAG. I.42 never starts from an equal-content hypothesis and concludes a metric or incidence fact. It only constructs geometry and then

combines forward equal-content statements. Accordingly no cancellation, positivity, whole-greater-than-part principle, or property (Z) is needed.

Hartshorne also places I.42–I.45 in the “algebra of areas” or application of areas: synthetic content is manipulated constructively without assigning a numerical area value to each figure.

42.9.6 Borsuk–Szmielew and Greenberg

Borsuk–Szmielew are strongest here as a control source for the underlying synthetic API. Their organization emphasizes angle layoff on a chosen side, midpoint data with positional information, and – after the Euclidean axiom – parallel uniqueness, parallel transitivity, and parallelogram theory. These are precisely the kinds of components used by the geometric half of I.42.

They do not supply the equal-content semantics of the I.35–I.45 block. As in the preceding documentation, that role remains with Hilbert and Hartshorne. Greenberg likewise serves as a control on the Euclidean parallel boundary and on the distinction between synthetic content and numerical area rather than as the source of an I.42-specific proof.

42.10 Formalization Notes

42.10.1 Geometry

The formal geometry contains substantially more data than the classical picture displays:

- strict noncollinearity for the input triangle and prescribed angle;
- strict midpoint betweenness $B - E - C$;
- a concrete carrier line for EC ;
- the fact that the copied angle point R and A lie on the same side of that carrier;
- a concrete carrier AQ parallel to EC ;
- a proof that the ray carrier ER meets AQ ;
- a proof that the resulting intersection F lies on the selected ray;
- noncollinearity of E, F, C before the fourth vertex can be constructed;
- identification of the completed upper side FG with the carrier through A .

No Pasch step is introduced locally in the top-level theorem, but the proof uses the developed side-of-line and intersection theory whose lower layers are built from Hilbert order/separation infrastructure.

42.10.2 Content

The content argument is entirely forward:

$$ABE \approx AEC \implies ABC \approx AEC + AEC,$$

and

$$\text{Par}(FECG) \approx GEC + GEC \implies \text{Par}(FECG) \approx AEC + AEC.$$

Transitivity then compares the original triangle with the constructed parallelogram.

There is no inverse “halving” theorem. There is no cancellation of a common summand and no deduction from equal content to segment or angle congruence. Thus I.42 stays on the safe forward side of the content semantics established since I.35.

42.10.3 Representation

Three representational choices deserve explicit mention.

First, the angle-construction helper initially returns the angle in the order CER and the final theorem wants FEC . `SameRay` transport and the symmetry of angle representation normalize this without changing the geometry.

Second, the parallelogram fourth-vertex theorem naturally returns $EFGC$, while the Euclid statement is expressed as $FECG$. The theorem `ParallelogramReverse` performs this normalization.

Third, and most importantly, the upper-carrier datum is

```
Collinear Geo F A G
```

rather than a strict parallel using the pair FA . This is the correct representation because the pair may degenerate when $F = A$ even though the upper carrier itself is perfectly nondegenerate through F and G .

42.10.4 Axiomatic status

The public theorem assumes

```
HilbertEuclideanPlane Geo
```

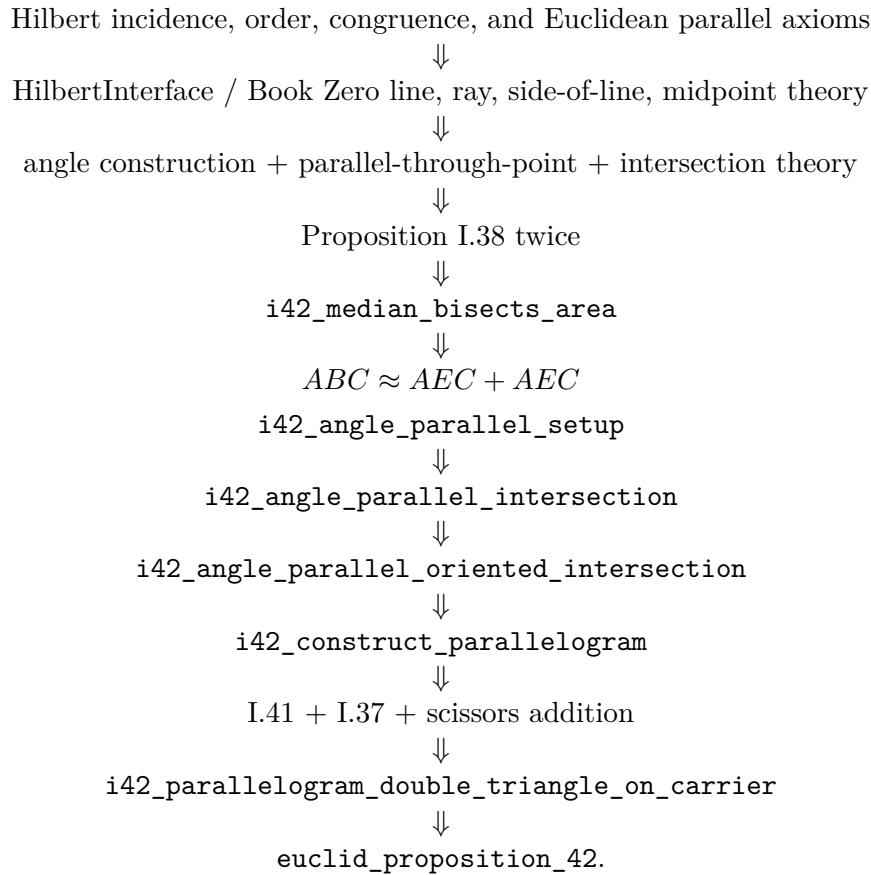
and is therefore Euclidean in the current hierarchy.

The Euclidean parallel axiom is used essentially in the developed theory of parallel uniqueness/transitivity. The most visible direct use in I.42 is the proof that the angle-ray carrier and the parallel through A cannot remain disjoint: two distinct carriers parallel to the same carrier are forced into the Euclidean parallel relation, which contradicts their shared point when appropriate. The same uniqueness theory identifies the final upper carrier.

Angle construction and segment construction are congruence/order operations; they do not require a continuity axiom in this proof. No circle intersection or completeness principle is introduced. Proposition I.42 adds no new geometric axiom and no new content axiom.

42.11 Dependency Summary

The actual formal dependency chain is



The historical and formal DAGs have the same content skeleton but different geometric packaging:

Euclid : I.10 → I.38 → I.23/I.31 → I.41 → I.42,
 Lean : Hilbert midpoint → I.38 twice → angle/parallel intersection → I.41+I.37 → I.42.

New geometric axiom in I.42:	no
New content axiom in I.42:	no
New proposition-local helpers:	yes — six
New reusable Interface/Scissors theorem:	no
New Book Zero theorem:	no
Property (Z) / content cancellation:	no
New numerical area semantics:	no
Continuity axiom introduced:	no
Boundary case $F = A$ supported:	yes
Public theorem compiles:	yes
Full lake build:	clean
sorry in Proposition42.lean:	no

Diagrammatic audit. The revised diagram layer deliberately separates the classical and formal configurations. Figure 42.1 is the historical construction and therefore permits a generic visual placement of A on the upper side. Figure 42.2 records the genuinely new auxiliary geometry

of the midpoint-content helper. Figure 42.3 records the three successive angle/parallel states, including the essential boundary case $F = A$. Figure 42.4 records the actual output of the formal parallelogram construction without inventing an order on F, A, G .

No additional figure is introduced for the exact split $ABC \sim ABE + AEC$, for the applications of I.41 and I.37, or for the final symmetry, addition, transport, and transitivity steps. Those operations change the scissors or logical interpretation of geometry already displayed; they do not introduce a new geometric configuration.

42.12 Conclusion

Proposition I.42 is a construction theorem built on the content theory of I.35–I.41, but the main formal difficulty is geometric rather than algebraic. The equal-content calculation itself is simple once the right configuration has been produced: the midpoint divides the original triangle into two equal content pieces, I.41 makes the new parallelogram double a triangle on its base, and I.37 moves that triangle along the common upper carrier.

What the formal reconstruction adds is the configuration data that the classical diagram suppresses. The copied angle ray is placed on the correct side, its carrier is proved to meet the upper parallel, the intersection is proved to lie on the correct ray, and the completed upper side is identified with the carrier through A . The boundary case $F = A$ is not discarded; it is the reason the final construction returns collinearity on the upper carrier instead of a strict parallel relation involving FA .

The proposition therefore illustrates the mature architecture of the Book I reconstruction. Geometry constructs and orients the carriers; the scissors layer carries equal content; representation lemmas normalize the same figure without changing its meaning. No new axiom, numerical area, cancellation principle, or continuity assumption is needed. The final full project build is clean, and I.42 is available as the construction input required by the later application-of-areas propositions.

Chapter 43

Proposition I.43

43.1 Introduction

Proposition I.43 returns from the construction problem of I.42 to a purely relational statement about an already given parallelogram. A diagonal of the large parallelogram is divided at a point K . Two smaller parallelograms are placed “about the diameter”, one on each part of the diagonal. The two remaining quadrilateral regions are the classical *complements*, and Euclid proves that they are equal.

The formal reconstruction keeps the argument entirely synthetic. It does not introduce numerical area, scalar multiplication, a content order, or a cancellation axiom. Instead, it turns Euclid’s Common Notion 3 step into a direct witness for the already defined relation `HilbertScissorsEquicomplementable`.

This makes I.43 a particularly clean test of the scissors language introduced at I.35. The proof is short enough that the historical and formal dependency graphs remain visibly related, but the formal representation makes explicit which triangular pieces are removed and which remain.

43.2 The Classical Configuration

Let $ABCD$ be a parallelogram with diagonal AC . Let K lie strictly on the diagonal, and choose points

$$A - G - B, \quad A - E - D, \quad B - F - C, \quad D - H - C,$$

so that $AGKE$ and $KFCH$ are parallelograms. The two quadrilateral regions left over are

$$GBFK \quad \text{and} \quad EKHD.$$

These are the two complements in the statement of I.43.

The Lean theorem records the strict order data explicitly:

$$A - K - C, \quad A - G - B, \quad A - E - D, \quad B - F - C, \quad D - H - C.$$

The two small parallelograms are also hypotheses rather than constructions. Thus I.43 is not a construction theorem in the current development; it is a content theorem for a given configuration.

Figure 43.1 records the classical configuration only. The diameter AC is part of the proposition, while the additional diagonals BK and KD used later by the Lean scissors representation are deliberately absent from this figure.

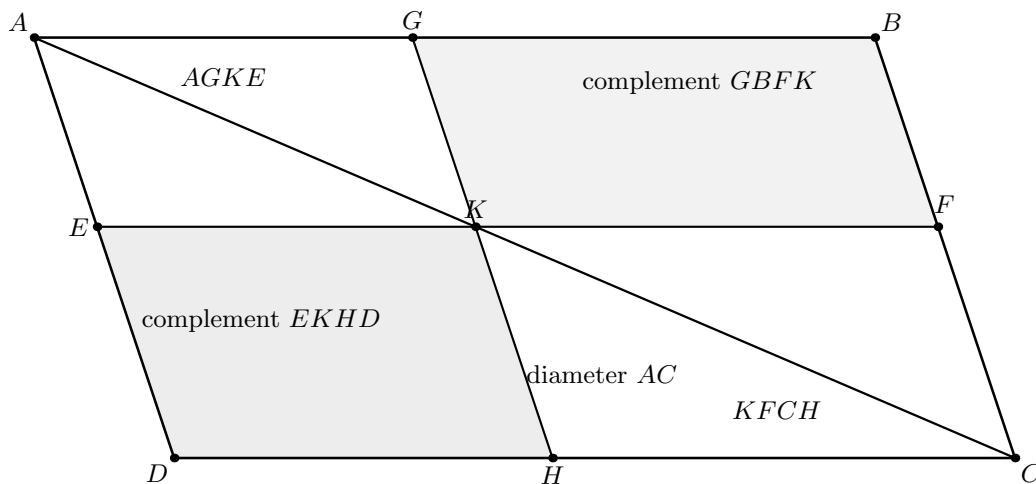


Figure 43.1: The classical configuration of Proposition I.43. The large parallelogram is $ABCD$ with diameter AC and $A - K - C$. The parallelograms $AGKE$ and $KFCH$ lie about the two parts of the diameter, leaving the complements $GBFK$ and $EKHD$. No formal triangulation diagonal is shown here.

43.3 Classical Proof

Euclid's proof uses Proposition I.34 three times. Since the diagonal of a parallelogram bisects it into equal triangles,

$$\triangle ABC = \triangle ACD.$$

The same theorem applied to the two smaller parallelograms gives

$$\triangle AGK = \triangle AEK, \quad \triangle KFC = \triangle KHC.$$

Equivalently, in Euclid's usual lettering, these are the two equal pairs of triangles belonging to the parallelograms about the diameter.

Common Notion 2 combines the two small equalities. The corresponding pairs of corner triangles therefore have equal total content. Since the two large diagonal triangles are equal, Common Notion 3 removes equal content from equal content and leaves the two complements equal.

The classical dependency graph is therefore

$$\begin{aligned}
 &ABCD, AGKE, KFCH \text{ parallelograms} \\
 &\quad \Downarrow \text{ I.34 three times} \\
 &ABC = ACD, \quad AGK = AEK, \quad KFC = KHC \\
 &\quad \Downarrow \text{ C.N.2} \\
 &AGK + KFC = AEK + KHC \\
 &\quad \Downarrow \text{ C.N.3} \\
 &GBFK = EKHD.
 \end{aligned}$$

The historical proof is therefore a forward content argument. No equal-content hypothesis is used to recover a metric, incidence, or parallelism conclusion. This places I.43 on the opposite side of the logical boundary exposed by I.39–I.40.

43.4 Complements and Equicomplementability

The central formal point is that Euclid’s phrase “subtract equals from equals” need not be encoded as a general cancellation rule

$$P + R \sim Q + R \implies P \sim Q.$$

No such theorem is required here.

The existing scissors relation defines formal equicomplementability by the existence of two equidecomposable complements R, S such that

$$R \sim S, \quad P + R \sim Q + S.$$

Thus I.43 can be proved by exhibiting the pieces removed by Euclid as the witnesses R and S .

For the two complement terms put

$$\begin{aligned} P &= GBK + KBF, \\ Q &= EDK + KDH, \end{aligned}$$

and choose

$$\begin{aligned} R &= AGK + KFC, \\ S &= AEK + KHC. \end{aligned}$$

Proposition I.34 gives $R \sim S$. The two full-triangle decompositions and the large diagonal equality give

$$P + R \sim Q + S.$$

These two facts are already exactly the data required by `HilbertScissorsEquicomplementable`.

This is the key difference between an abstract cancellation theorem and the present proof. The formalization does not infer equality after deleting an unrepresented common summand. It records the deleted pieces explicitly as part of the certificate.

43.5 Proof Architecture of the Reconstruction

The working theorem `i43_raw_complements` has five mathematical stages.

43.5.1 Step 1: decompose the first large diagonal triangle

Using $A - K - C$, the triangle ABC is split at K . The resulting AK -piece is split at G using $A - G - B$, and the KC -piece is split at F using $B - F - C$. The result is

$$ABC \sim (AGK + GBK) + (KBF + KFC).$$

In the Lean file this is the local identity `hT1`.

43.5.2 Step 2: decompose the second large diagonal triangle

The mirror construction uses $A - E - D$ and $D - H - C$. After orienting the first betweenness relation as $D - E - A$, the triangle ACD is decomposed as

$$ACD \sim (EDK + AEK) + (KDH + KHC).$$

This is **hT2**.

Figure 43.2 displays these two exact scissors states separately. The dashed segments BK and KD are introduced only at this formal decomposition stage; they are not additional hypotheses of the classical I.43 configuration.

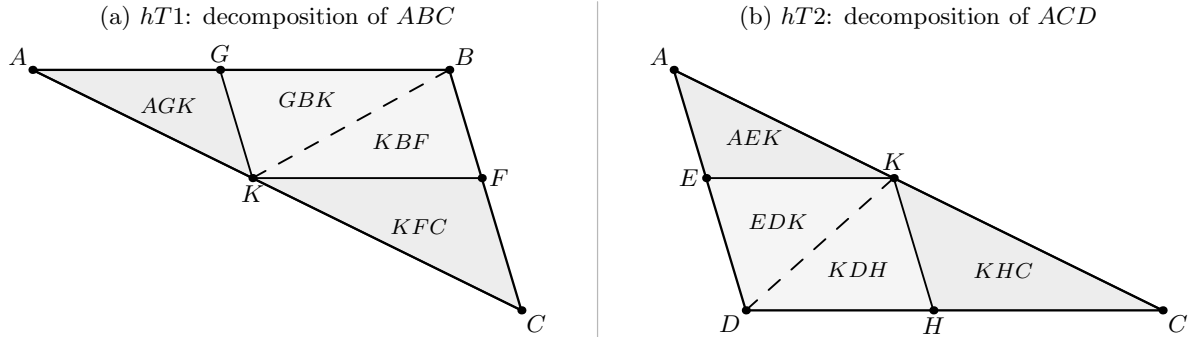


Figure 43.2: The two four-piece decompositions used by **hT1** and **hT2**. Panel (a) decomposes ABC into $AGK + GBK + KBF + KFC$, with the formal diagonal BK . Panel (b) decomposes ACD into $EDK + AEK + KDH + KHC$, with the formal diagonal KD . The segments already belonging to the parallelograms about the diameter remain solid; only the two extra triangulation diagonals are dashed.

43.5.3 Step 3: obtain the three I.34 equalities

The big parallelogram gives

$$ABC \sim ACD.$$

The two parallelograms about the diagonal give

$$AGK \sim AEK, \quad KFC \sim KHC.$$

These are the local facts **hT3**, **hT4**, and **hT5**. The implementation obtains all three through the already proved theorem `euclid_proposition_34_diagonal` and the congruent-triangle generator of the scissors calculus.

43.5.4 Step 4: assemble and regroup the full decompositions

Transitivity gives one four-piece relation

$$(AGK + GBK) + (KBF + KFC) \sim (EDK + AEK) + (KDH + KHC).$$

The code then uses only associativity and commutativity of multiset addition to regroup this as

$$(GBK + KBF) + (AGK + KFC) \sim (EDK + KDH) + (AEK + KHC).$$

No geometric theorem is hidden in this regrouping.

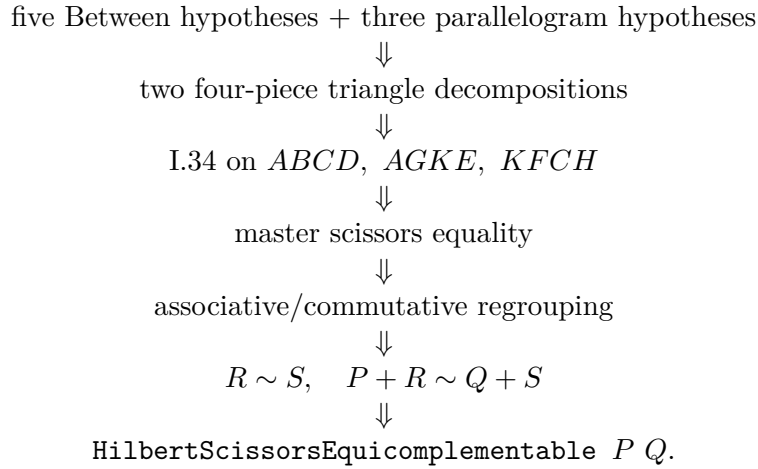
43.5.5 Step 5: package the equicomplementability witness

Finally **hT4** and **hT5** are combined with `HilbertScissorsEq.add`, giving

$$AGK + KFC \sim AEK + KHC.$$

Together with the regrouped master relation, these are exactly the two fields required by the definition of equicomplementability.

The complete Lean DAG is therefore



43.6 Lean Formalization

The file imports only the already established diagonal theorem for parallelograms and the scissors calculus:

```
import CGJteamLab.Proposition34
import CGJteamLab.HilbertScissors
```

The substantial proof is isolated in the proposition-local helper `i43_raw_complements`. Its public shape is the following.

```
theorem i43_raw_complements
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D K G E F H : Geo.Point)
  (hParallelogram : IsParallelogram Geo A B C D)
  (hAKC : Geo.Between A K C)
  (hAGB : Geo.Between A G B)
  (hAED : Geo.Between A E D)
  (hBFC : Geo.Between B F C)
  (hDHC : Geo.Between D H C)
  (hPar1 : IsParallelogram Geo A G K E)
  (hPar2 : IsParallelogram Geo K F C H) :
  HilbertScissorsEquicomplementable Geo
    (hilbertScissorsTriangle Geo G B K +
     hilbertScissorsTriangle Geo K B F)
    (hilbertScissorsTriangle Geo E D K +
     hilbertScissorsTriangle Geo K D H) := by
  -- proof in Proposition43.lean
```

The final Book I endpoint is deliberately thin:

```
theorem euclid_proposition_43
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
```

```

(A B C D K G E F H : Geo.Point)
(hParallelogram : IsParallelogram Geo A B C D)
(hAKC : Geo.Between A K C)
(hAGB : Geo.Between A G B)
(hAED : Geo.Between A E D)
(hBFC : Geo.Between B F C)
(hDHC : Geo.Between D H C)
(hPar1 : IsParallelogram Geo A G K E)
(hPar2 : IsParallelogram Geo K F C H) :
HilbertScissorsEquicomplementable Geo
  (hilbertScissorsTriangle Geo G B K +
   hilbertScissorsTriangle Geo K B F)
  (hilbertScissorsTriangle Geo E D K +
   hilbertScissorsTriangle Geo K D H) := by
exact
i43_raw_complements
  Geo
  A B C D K G E F H
  hParallelogram
  hAKC
  hAGB
  hAED
  hBFC
  hDHC
  hPar1
  hPar2

```

The final certificate returned by the raw helper uses

```

hilbertScissorsTriangle Geo A G K +
hilbertScissorsTriangle Geo K F C

```

and

```

hilbertScissorsTriangle Geo A E K +
hilbertScissorsTriangle Geo K H C

```

as the two equidecomposable complements.

43.7 Relationship with Earlier Propositions

43.7.1 Proposition I.34

I.34 is the only earlier Book I proposition used directly by I.43. It enters three times through `euclid_proposition_34_diagonal`: once for the large parallelogram and once for each parallelogram about the diagonal. The current proof does not replay the parallel-angle or congruence argument behind I.34; it consumes the already established diagonal triangle congruence.

43.7.2 Propositions I.35–I.41

These propositions established and exercised the synthetic content language, but I.43 does not call them by proposition name. Their architectural legacy is `HilbertScissors`: finite triangle

sums, splitting, congruent replacement, addition, transitivity, and equicomplementability.

In particular, I.43 is not a new converse theorem of the I.39 type. It needs no proper-part contradiction and no positivity principle.

43.7.3 Proposition I.42

I.42 is a construction theorem: it creates a parallelogram with prescribed content and angle. I.43 is complementary in character. The entire geometric configuration is supplied as input, and the theorem proves a relation between the two remaining regions.

43.7.4 Proposition I.44

The next proposition uses I.43 immediately. Historically, I.44 first creates a parallelogram of the required content and angle and then uses the complement theorem to transfer that content to a parallelogram applied to the prescribed segment. Thus I.43 is not an isolated scissors identity; it is the bridge from I.42 to the application-of-areas construction in I.44.

43.8 Hilbert and Book Zero Components Used

43.8.1 Betweenness and incidence

The five strict betweenness hypotheses are the hidden configuration data needed for the formal triangle splits. They guarantee that K, G, E, F, H lie on the intended segments, not merely on the corresponding carrier lines. Apart from reversing $A - E - D$ to $D - E - A$ for one split, no new order theorem is introduced.

43.8.2 Parallelogram diagonal congruence

The metric input is entirely packaged by I.34. The current proof therefore uses the developed Euclidean parallelogram layer rather than reconstructing the three congruence proofs locally.

43.8.3 Scissors content infrastructure

The proof uses only existing operations:

```
HilbertScissorsEq.split
HilbertScissorsEq.congruent
HilbertScissorsEq.add
HilbertScissorsEq.symm
HilbertScissorsEq.trans
scissors_congruent
scissors_triangle_swap12
scissors_triangle_swap23
scissors_triangle_cycle
```

Together with associativity and commutativity of multiset addition, these are sufficient to reconstruct Common Notions 2 and 3 in the present configuration.

No new theorem is added to `HilbertInterface`, `HilbertBookZero`, or `HilbertScissors`.

43.9 Source Audit

43.9.1 Euclid, Heath, and Joyce

The classical sources agree on the short proof: I.34 gives equality of the two large diagonal triangles and of the two pairs of small diagonal triangles; C.N.2 combines the small equalities; C.N.3 leaves the two complements equal. Joyce also emphasizes the operational meaning of “parallelograms about the diameter” and notes that the next proposition uses I.43 to change the shape of a parallelogram without changing its content.

43.9.2 Wilkins

Wilkins is used here as a control on the classical dependency graph and the configuration around the diagonal. No additional hidden case split is needed in the current Lean theorem because all five segment-order relations are already explicit hypotheses.

43.9.3 Hilbert

The closest Hilbertian source is Chapter IV, Section 18. Immediately before Hilbert’s Theorem 43, the general theory states that if equidecomposable polygons are removed from equidecomposable polygons, the remaining polygons are equicomplementable. Hilbert’s Theorem 43 then supplies the corresponding transitivity-style properties for equidecomposability and equicomplementability.

The numerical coincidence is worth avoiding: Hilbert’s Theorem 43 is not Euclid’s Proposition I.43. The connection is structural, not numerical. The present Lean proof specializes the general removal pattern directly at the term level by exhibiting the removed pieces as the two complements in the definition of `HilbertScissorsEquicomplementable`.

43.9.4 Hartshorne

Hartshorne’s area theory is a semantic control on the direction of the argument. I.41–I.45 belong to the forward equal-content part of Book I; the extra strictness property used by the converse propositions I.39–I.40 is not needed. This supports the present choice to remain entirely inside synthetic cut-and-paste content rather than numerical area.

43.9.5 Beeson, Borsuk–Szmielw, and Greenberg

These sources add no separate I.43-specific theorem to the current architecture. Beeson remains a control on omitted formal conditions, while Borsuk–Szmielw and Greenberg control the underlying Euclidean parallel and parallelogram infrastructure. The actual proof is already expressed more directly by I.34 plus the existing scissors calculus.

43.10 Formalization Notes

43.10.1 Geometry

The formal theorem assumes more explicit positional data than Euclid writes in the proposition statement:

- K is strictly between A and C ;
- G is strictly between A and B ;
- E is strictly between A and D ;
- F is strictly between B and C ;
- H is strictly between D and C ;
- the large and two small quadrilaterals are explicitly `IsParallelogram` hypotheses.

These hypotheses are not auxiliary constructions introduced by the proof. They are the formal unpacking of the classical diagram required by the splitting constructor.

43.10.2 Content

The content argument is entirely forward. There is no appeal to `HilbertScissorsPositivity`, no whole-greater-than-part theorem, and no cancellation principle. The two deleted collections of corner triangles are retained as explicit witnesses of equicomplementability.

This is exactly why I.43 remains inexpensive after I.39: Common Notion 3 is not being used to infer geometry from equality of content. It is being realized as a constructive scissors certificate.

43.10.3 Representation

The public theorem does not return the two complements as `hilbertParallelogramTerm`. Instead it returns the exact triangulated terms used by the proof:

$$GBK + KBF \quad \text{and} \quad EDK + KDH.$$

This is faithful to the current term-level semantics of `HilbertScissors`, which represents finite sums of triangles rather than polygonal interiors. The theorem therefore establishes the equal-content relation needed for I.43 without claiming a stronger polygon-semantics result that the library does not yet encode.

No nonoverlap predicate, rectilinear-region type, or numerical area function is introduced.

43.10.4 Axiomatic status

The public theorem is stated under

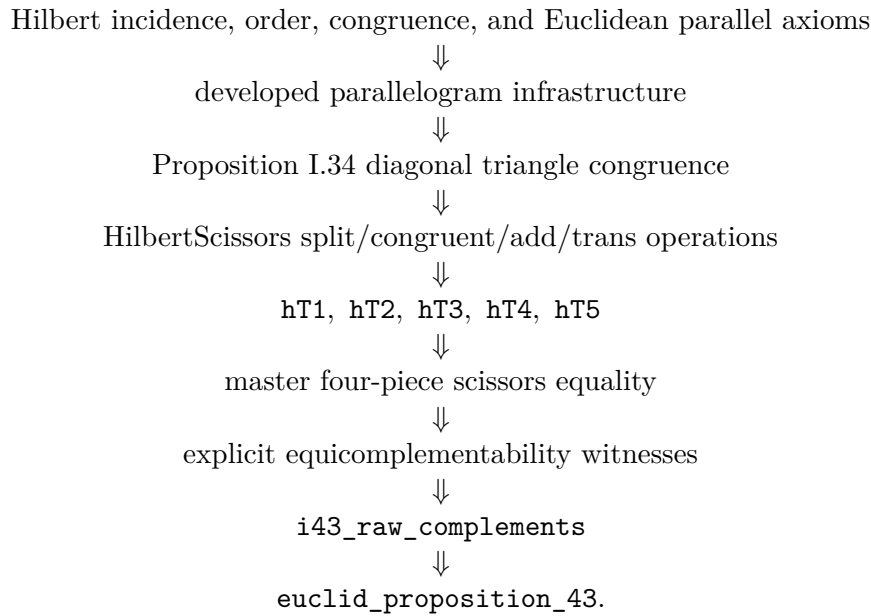
```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

because its I.34 dependency belongs to the developed Euclidean parallelogram layer. No continuity axiom is used locally, no Archimedean principle is used, and I.43 adds no new geometric or content axiom.

Diagrammatic audit. The revised documentation uses exactly two figures. Figure 43.1 records the classical complement configuration and therefore shows the genuine diameter AC but omits the representation-only cuts BK and KD . Figure 43.2 is reserved for the formal scissors layer and displays the two four-piece decompositions $\mathbf{hT1}$ and $\mathbf{hT2}$ with those extra diagonals made explicit. No additional figure is introduced for the three I.34 congruences, the associative/commutative regrouping, or the final equicomplementability packaging: each of those steps reuses the same geometry and changes only the proof or term representation.

43.11 Dependency Summary

The actual formal dependency chain is



New geometric axiom:	no
New content axiom:	no
New proposition-local helper:	yes — <code>i43_raw_complements</code>
New reusable helper:	no
New Book Zero theorem:	no
New Interface/Scissors theorem:	no
Numerical area introduced:	no
Polygon semantics introduced:	no
Full lake build:	clean
sorry in <code>Proposition43.lean</code> :	no

43.12 Conclusion

Proposition I.43 is a compact confirmation that the scissors architecture built from I.35 onward has reached the right level of abstraction. Euclid's proof uses I.34 together with addition and

subtraction of equal figures. The Lean proof preserves the same mathematical structure without numerical area and without introducing a general cancellation theorem.

The decisive formal move is to keep the pieces that Euclid “subtracts” as explicit data. The two collections of corner triangles are proved scissors-equivalent, and the two complements become equidecomposable after those collections are adjoined. That is precisely the existing definition of equicomplementability.

Thus I.43 introduces no new foundational debt. Its one local helper packages a configuration-specific decomposition, while the reusable mathematics remains where it already belongs: I.34 for the diagonal congruences and **HilbertScissors** for finite cutting, congruent replacement, addition, and transitivity. This also leaves the project in the natural position for I.44, where the complement theorem becomes an active construction tool.

Chapter 44

Proposition I.44

44.1 Introduction

Proposition I.44 is the first Book I theorem that takes the synthetic content machinery developed in I.35–I.43 and applies it to a prescribed segment. The input consists of a nondegenerate segment AB , a nondegenerate triangle PQR , and a nondegenerate angle XYZ . The output is a parallelogram on the given side AB , with the prescribed angle at B , and with the same content as the given triangle.

The current Lean statement is deliberately synthetic. It does not introduce a numerical area function. The target parallelogram is represented by `hilbertParallelogramTerm`, and equality of content is represented by `HilbertScissorsEquicomplementable`. Thus the theorem constructs points M, L such that

$$ABML \text{ is a parallelogram,} \quad \angle ABM \cong \angle XYZ,$$

and

$$\triangle PQR \approx ABML,$$

where $\approx$ denotes the project's formal equicomplementability relation.

I.44 is also the first proposition in the application-of-areas block where a small phrase in the classical proof hides a substantial formal obligation. Euclid first invokes I.42 to obtain a parallelogram of the correct content and angle, and then says that one side of this parallelogram is to be placed in a straight line with the given segment. The final Lean proof does not treat this rigid placement as a primitive operation. It is reconstructed from segment construction, angle construction, SAS, SSS, and completion of a parallelogram.

A second hidden obligation occurs later. In the classical diagram two produced lines meet at a point K . Euclid justifies this by an angle argument using I.29 and Postulate 5. The Lean proof reaches the same intersection by a different Euclidean route: if the two carriers were disjoint, they would be parallel; uniqueness and transitivity of parallels would then force two lines through a common point to be parallel, which is impossible. The conclusion is the same, but the formal DAG is not a mechanical transcription of the classical angle-sum proof.

The proof therefore has three sharply separated layers.

- **Geometry:** rigid placement, extensions, intersections, parallel carriers, SameSide/OppositeSide arguments, strict order, and completion of several parallelograms.

- **Content:** a congruent copy preserves scissors content; I.43 identifies the two complements; transitivity then connects the I.42 parallelogram to the final one.
- **Representation:** the I.43 complement terms are sums of triangles, while I.44 exposes whole parallelogram terms. The proof therefore uses alternate triangulations and triangle permutations to bridge the two representations.

The main architectural result of the reconstruction is that I.44 closes with no new geometric axiom and no new content axiom. In particular, the former provisional transport axiom has been eliminated. The public theorem now rests on the previously developed Hilbert and scissors infrastructure together with proposition-local lemmas that expose the geometry suppressed by the classical diagram.

44.2 The Classical Configuration

Let AB be the given straight segment, let PQR be the given triangle, and let XYZ represent the prescribed rectilinear angle. Euclid first uses I.42 to construct a parallelogram of the required content and angle. In the lettering used by the reconstruction, after rigid placement this auxiliary parallelogram is $BEFG$, with

$$A - B - E \quad \text{and} \quad \angle EBG \cong \angle XYZ.$$

The side FG is produced beyond G to H . Through A a line is drawn parallel to GB . The formal order relation is

$$H - G - F, \quad AH \parallel GB.$$

The carrier HB then meets the carrier FE at K , and the proof recovers the strict relations

$$H - B - K, \quad F - E - K.$$

Through K a line is drawn parallel to BE . Its intersections with the produced carriers HA and GB are denoted by L and M . The final order data are

$$H - A - L, \quad G - B - M, \quad L - M - K.$$

These relations are not decorative labels. They are the exact strict-order facts used by the formal proof.

The resulting large figure is the parallelogram $HLKF$, with diameter HK . The two parallelograms about the diameter are

$$HABG \quad \text{and} \quad BMKE.$$

The two complements are precisely the old placed parallelogram $BEFG$ and the target parallelogram $ABML$. This is the configuration to which Proposition I.43 applies.

Figure 44.1 records the classical Euclidean configuration only. It shows the placed I.42 parallelogram, the produced carriers, the intersection K , the parallel through K , and the final I.43 parallelogram. The stricter order information required by Lean is deliberately reserved for the later formal figures.

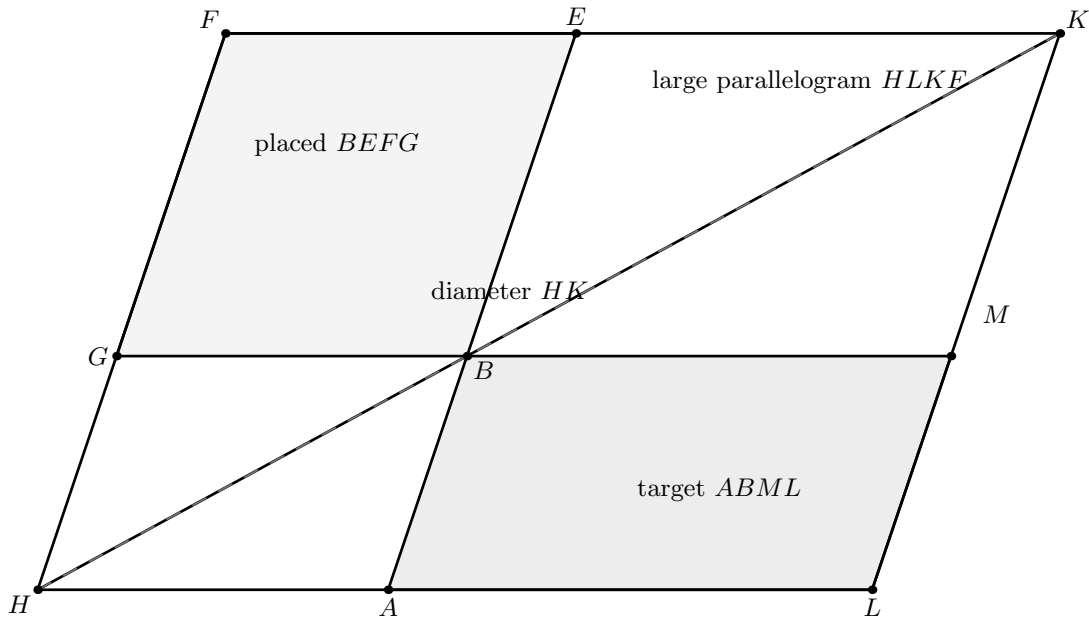


Figure 44.1: The classical Euclidean configuration. The auxiliary parallelogram $BEFG$ is placed on the produced carrier of AB ; the relevant produced lines meet at K ; and the parallel through K completes the large parallelogram $HLKF$. The target parallelogram is the complement $ABML$. No additional strict-order claim is encoded by this historical figure.

44.3 Classical Proof

The classical proof has a compact dependency graph but several diagrammatic steps.

First apply Proposition I.42 to construct a parallelogram $BEFG$ equal to the given triangle and having

$$\angle EBG = \angle XYZ.$$

Place it so that BE is in the same straight line as the prescribed segment AB , with B between A and E . Euclid then produces FG to H and draws AH parallel to BG .

Because $AH \parallel EF$, the transversal HF creates the supplementary angle configuration used in Euclid's text. With I.29 and the angle facts already available from the parallelogram, the interior angles made by the produced lines HB and FE are less than two right angles. Postulate 5 therefore forces the two lines to meet; call their intersection K .

Through K draw a line parallel to EA and produce HA and GB until they meet this new line at L and M . The quadrilateral $HLKF$ is a parallelogram, HK is its diameter, and $HABG$ and $BMKE$ are the two parallelograms about the diameter. Proposition I.43 now gives equality of the complements:

$$ABML = BEFG$$

in Euclid's equality-of-figures sense.

By I.42, $BEFG$ is equal to the given triangle. Common Notion 1 therefore gives the required equality between $ABML$ and the triangle. Finally, $A - B - E$ and $G - B - M$ make $\angle ABM$ and $\angle EBG$ vertical angles, so I.15 gives

$$\angle ABM = \angle EBG = \angle XYZ.$$

Thus the required parallelogram has been applied to AB in the prescribed angle.

The historical dependency graph is therefore

$$\begin{array}{c}
 \triangle PQR, AB, \angle XYZ \\
 \Downarrow \text{I.42} \\
 BEFG \text{ with the required content and angle} \\
 \Downarrow \text{place } BE \text{ on the produced carrier of } AB \\
 A - B - E \\
 \Downarrow \text{I.31, I.29, Postulate 5} \\
 H - G - F, H - B - K, F - E - K \\
 \Downarrow \text{parallel through } K \\
 HLKF, HABG, BMKE \\
 \Downarrow \text{I.43} \\
 ABML = BEFG \\
 \Downarrow \text{C.N.1 and I.15} \\
 ABML = \triangle PQR, \quad \angle ABM = \angle XYZ.
 \end{array}$$

Two points in this graph are especially important for formalization. The phrase “place BE in a straight line with AB ” is not a theorem reference in the classical text, and the meeting of HB and FE is justified through a specifically Euclidean angle argument. Both steps are reconstructed explicitly in Lean.

44.4 Application of Areas and Rigid Placement

The proposition-specific novelty of I.44 is not the existence of a parallelogram with prescribed content and angle; I.42 already supplies that. The new problem is to put such a parallelogram on a *specified segment* without changing either its shape or its content.

The helper

```
i44_place_parallelogram_shape
```

solves the geometric part of this problem. Suppose $STUV$ is a given parallelogram and $A \neq B$. The proof first extends AB beyond B . On the resulting ray it lays off BE congruent to TS . It then copies the angle STU at B with first arm BE , lays off BG congruent to TU on the constructed ray, and completes BEG to the parallelogram $BEFG$. The output contains

$$A - B - E, \quad BE \cong TS, \quad BG \cong TU, \quad \angle EBG \cong \angle STU.$$

Figure 44.2 shows only this rigid-placement stage. It is separate from Figure 44.6 because the two diagrams solve mathematically different problems: the first transports the shape to the prescribed base, while the second performs Euclid’s I.43 complement construction.

The companion helper

```
i44_placed_parallelogram_content
```

proves that this rigid copy preserves the scissors content of the original parallelogram. The argument is stronger than a bare appeal to an undefined notion of “same area”. It triangulates the two parallelograms and proves congruence of corresponding triangles.

The triangle BEG is congruent to TSU by SAS, since the two sides and their included angle

$$BE \cong TS, \quad BG \cong TU$$

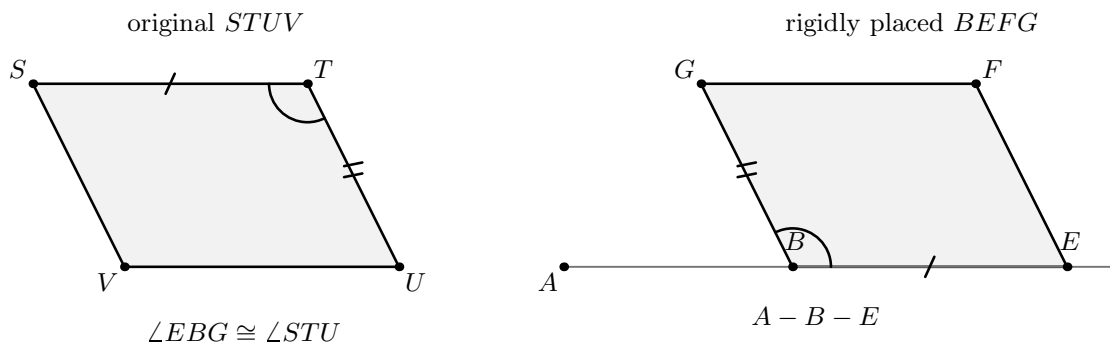


Figure 44.2: The rigid-placement helper. The original parallelogram $STUV$ is copied to $BEFG$ with $A - B - E$, $BE \cong TS$, $BG \cong TU$, and $\angle EBG \cong \angle STU$. The construction uses segment layoff, angle construction on a chosen side, ray orientation, and completion of a parallelogram. No content theorem is used in the geometric placement itself.

were copied explicitly. This also gives the diagonal congruence

$$EG \cong SU.$$

Opposite-side congruence in the two parallelograms supplies the remaining side relations needed for SSS, yielding

$$\triangle EFG \cong \triangle SVU.$$

The two congruences are converted into `HilbertScissorsEq` relations. The existing helper

```
parallelogram_two_triangulations
```

then identifies their sums with the whole parallelogram terms. Hence the original and placed parallelograms are scissors-equivalent, and therefore equicomplementable.

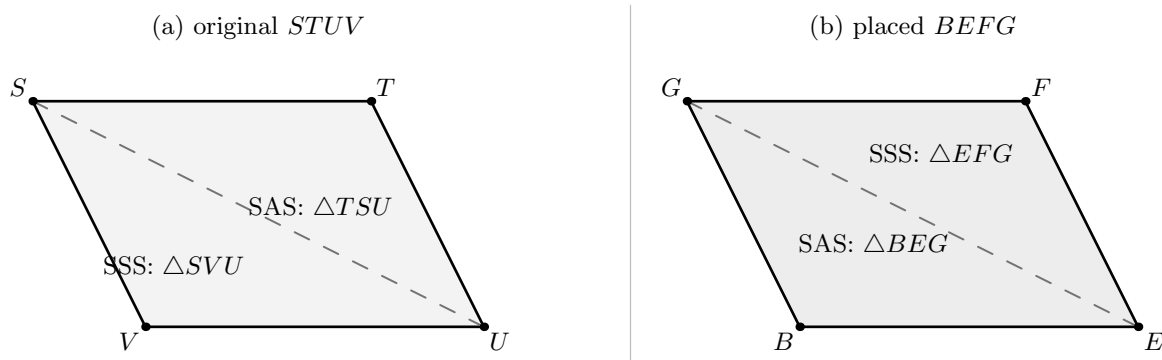


Figure 44.3: The exact two-piece content certificate used by `i44_placed_parallelogram_content`. The original parallelogram $STUV$ is triangulated along SU and the placed copy $BEFG$ along EG . SAS gives $\triangle BEG \cong \triangle TSU$ and the resulting diagonal congruence supports the SSS comparison $\triangle EFG \cong \triangle SVU$. Adding these two congruent triangle pairs produces the required scissors equality of the parallelograms.

This separation between *shape placement* and *content preservation* is important. The first helper belongs to synthetic geometry; the second is a certificate that the rigid copy has not changed the formal content. Neither helper introduces numerical area.

44.5 Proof Architecture of the Reconstruction

The final Lean proof can be compressed into seven mathematical stages. The source file contains more helper declarations, but many of them are local normalization lemmas for order or carrier orientation. The stages below are the mathematically significant transitions.

44.5.1 Step 1: obtain a content-and-angle parallelogram from I.42

The public theorem begins by calling

```
euclid_proposition_42
```

on the triangle PQR and the angle XYZ . It obtains points E, F, G such that $FERG$ is a parallelogram,

$$\angle FER \cong \angle XYZ,$$

and

$$\triangle PQR \approx FERG.$$

At this point the parallelogram has the correct content and angle but is not located on the prescribed segment AB .

44.5.2 Step 2: rigidly place the I.42 parallelogram beyond B

The theorem `i44_transport_onto_line` treats the I.42 output as a general parallelogram $STUV$. The helper `i44_place_parallelogram_shape` constructs the placed copy $BEFG$ with $A-B-E$ and with two adjacent sides and the included angle copied from $STUV$.

This is the formal replacement for the classical phrase that the side of the I.42 parallelogram is to be placed in the same straight line as the given segment. In an earlier provisional architecture this step was packaged as a local transport axiom. The current file derives it completely.

44.5.3 Step 3: certify the content of the rigid copy

The helper `i44_placed_parallelogram_content` proves

$$STUV \approx BEFG.$$

Its proof uses SAS, opposite-side congruence of parallelograms, SSS, and the two triangulations of a parallelogram. Thus the content transport is reduced to finite congruent-piece geometry rather than postulated as an area law.

44.5.4 Step 4: construct the positioned Euclid configuration

Starting with $A-B-E$ and the placed parallelogram $BEFG$, the helper

```
i44_positioned_content
```

constructs the full I.43 configuration. The first key result is

```
i44_construct_H_ordered
```

which returns H with

$$H - G - F, \quad AH \parallel GB.$$

The incidence part first constructs the parallel through A and proves that it meets the carrier FG . The order part then uses line separation to show that the intersection lies beyond G rather than in another collinear position.

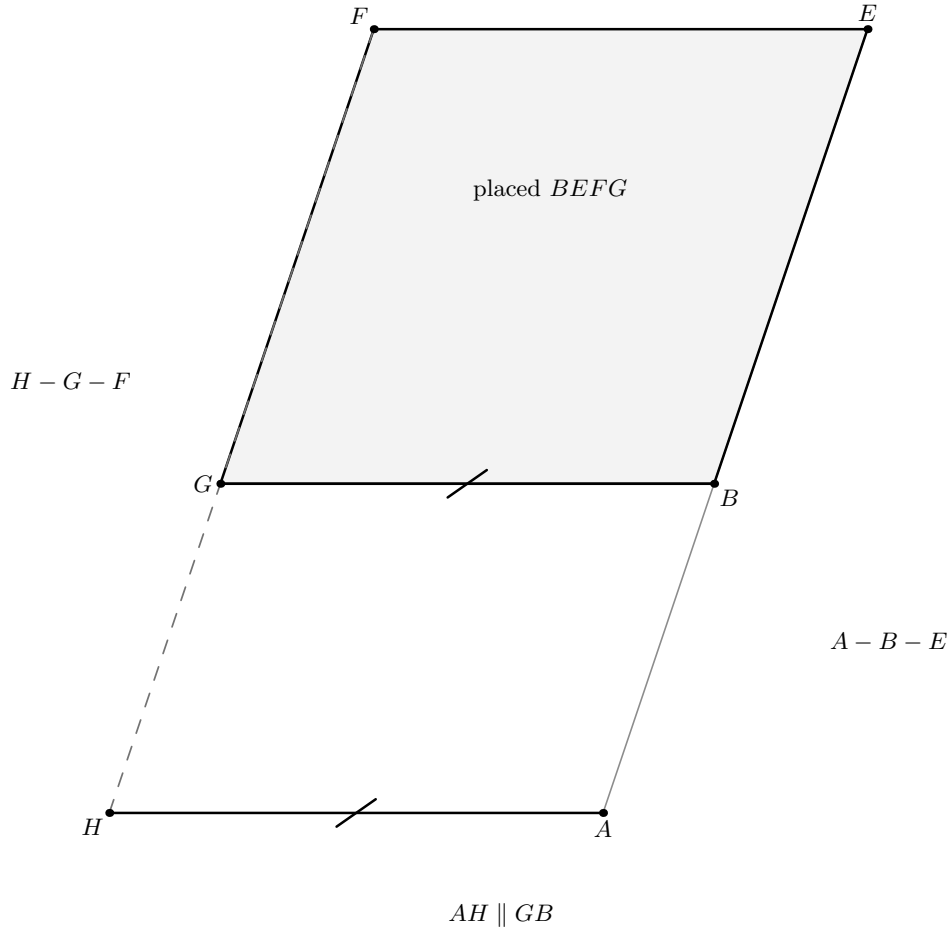


Figure 44.4: The first new formal state after rigid placement. The carrier FG is produced to an intersection H with the parallel through A , and the proof establishes the strict relation $H - G - F$ together with $AH \parallel GB$. The position $H - G - F$ is proved by line separation; it is not read from the drawing.

The next key result is

```
i44_construct_K_ordered
```

which returns K with

$$H - B - K, \quad F - E - K.$$

This is where the formal proof departs most visibly from Euclid's literal argument. Instead of reproducing the sum-of-interior-angles calculation, it supposes that the carriers HB and EF are disjoint. Disjointness gives $HB \parallel EF$. But the old parallelogram already gives $GB \parallel EF$. Euclidean transitivity/uniqueness of parallels would then force HB and GB to be parallel, impossible because they share B . Hence the carriers meet. SameSide/OppositeSide arguments then identify the correct strict order on both lines.

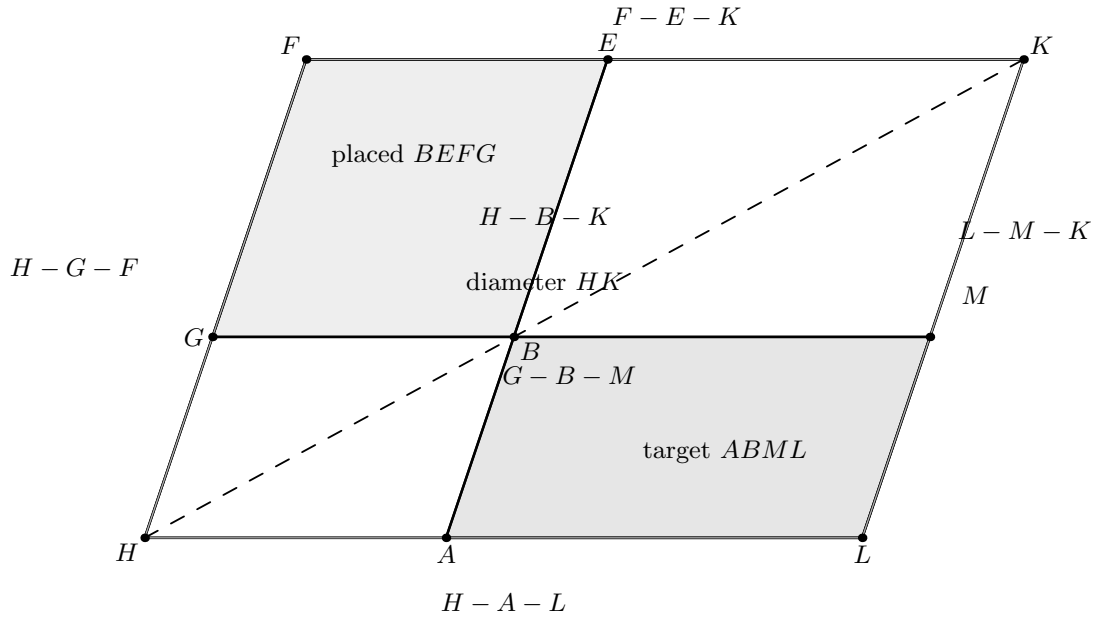


Figure 44.6: The complete formal configuration immediately before Proposition I.43 is invoked. The large parallelogram is $HLKF$ with diameter HK ; the parallelograms about the diameter are $HABG$ and $BMKE$; and the complements are the placed parallelogram $BEFG$ and the target $ABML$. The displayed strict relations $H - A - L$, $H - G - F$, $H - B - K$, $F - E - K$, $G - B - M$, and $L - M - K$ are exactly the order data established by the construction and order helpers.

44.5.6 Step 6: use I.43 and transfer the angle

The helper

```
i44_i43_content_bridge
```

feeds the constructed data to `euclid_proposition_43`. The point mapping is

I.43 name	A	B	C	D	K	G	E	F	H
I.44 point	H	L	K	F	B	A	G	M	E .

Thus the I.43 large parallelogram is $HLKF$, its diameter point is B , and the two parallelograms about the diameter are $HABG$ and $BMKE$.

The public I.43 theorem returns its complements as sums of triangle terms, whereas I.44 wants whole parallelogram terms. The bridge uses

```
parallelogram_two_triangulations
```

together with triangle permutation identities to prove

$$ABML \approx BEFG.$$

This is a representation bridge, not a new content axiom.

For the angle, $A - B - E$ and $G - B - M$ expose vertical angles. The helper

```
i44_angle_transfer
```

first proves that ABM is nondegenerate from the target parallelogram, then invokes the existing vertical-angle theorem and angle transitivity to obtain

$$\angle ABM \cong \angle EBG \cong \angle XYZ.$$

44.5.7 Step 7: chain the two content transports

The derived transport theorem now has

$$STUV \approx BEFG \quad \text{and} \quad BEFG \approx ABML.$$

Transitivity gives

$$STUV \approx ABML.$$

At the top level, I.42 already supplied

$$\triangle PQR \approx STUV,$$

so one last transitivity step proves

$$\triangle PQR \approx ABML.$$

The actual formal DAG is therefore

$$\begin{array}{c}
 PQR, AB, XYZ \\
 \Downarrow \text{I.42} \\
 STUV \text{ with prescribed content and angle} \\
 \Downarrow \text{segment and angle construction} \\
 A - B - E, BEFG \text{ rigid copy of } STUV \\
 \Downarrow \text{SAS + SSS + parallelogram triangulations} \\
 STUV \approx BEFG \\
 \Downarrow \text{parallel/intersection/order construction} \\
 H - G - F, H - B - K, F - E - K, H - A - L, L - M - K, G - B - M \\
 \Downarrow \\
 HLKF, HABG, BMKE, ABML \text{ parallelograms} \\
 \Downarrow \text{I.43 + representation bridge} \\
 ABML \approx BEFG \\
 \Downarrow \text{I.15 / VerticalAngles} \\
 \angle ABM \cong \angle XYZ \\
 \Downarrow \text{content transitivity} \\
 \triangle PQR \approx ABML.
 \end{array}$$

44.6 Lean Formalization

The current file imports exactly the two immediately preceding construction and complement results used by the public theorem:

```
import CGJteamLab.Proposition42
import CGJteamLab.Proposition43
```

The public theorem, written in the project's documentation-friendly ASCII notation, is

```

theorem euclid_proposition_44
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B P Q R X Y Z : Geo.Point)
  (hAB : Not (A = B))
  (hPQR : Not (Collinear Geo P Q R))
  (hXYZ : Not (Collinear Geo X Y Z)) :
  exists M L : Geo.Point,
    IsParallelogram Geo A B M L /\
    Geo.AngleCongruent A B M X Y Z /\
    HilbertScissorsEquicomplementable Geo
    (hilbertScissorsTriangle Geo P Q R)
    (hilbertParallelogramTerm Geo A B M L) := by
  -- proof in Proposition44.lean

```

The decisive derived transport theorem has the following shape:

```

theorem i44_transport_onto_line
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B S T U V X Y Z : Geo.Point)
  (hAB : Not (A = B))
  (hParallelogram : IsParallelogram Geo S T U V)
  (hAngle : Geo.AngleCongruent S T U X Y Z) :
  exists M L : Geo.Point,
    IsParallelogram Geo A B M L /\
    Geo.AngleCongruent A B M X Y Z /\
    HilbertScissorsEquicomplementable Geo
    (hilbertParallelogramTerm Geo S T U V)
    (hilbertParallelogramTerm Geo A B M L) := by
  -- rigid placement + positioned I.43 construction

```

This theorem is the formal content of the phrase “apply the parallelogram to the given line”. It is no longer an axiom.

The two helpers that isolate rigid placement and its content certificate are

```

theorem i44_place_parallelogram_shape
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B S T U V : Geo.Point)
  (hAB : Not (A = B))
  (hParallelogram : IsParallelogram Geo S T U V) :
  exists E F G : Geo.Point,
    Geo.Between A B E /\
    IsParallelogram Geo B E F G /\
    Geo.AngleCongruent E B G S T U /\
    Geo.Congruent B E T S /\
    Geo.Congruent B G T U := by
  -- segment layoff, angle copy, ray control, fourth vertex

```

and

```

theorem i44_placed_parallelogram_content
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (B E F G S T U V : Geo.Point)

```

```

(hOriginal : IsParallelogram Geo S T U V)
(hPlaced   : IsParallelogram Geo B E F G)
(hAngle    : Geo.AngleCongruent E B G S T U)
(hBE_TS    : Geo.Congruent B E T S)
(hBG_TU    : Geo.Congruent B G T U) :
HilbertScissorsEquicomplementable Geo
  (hilbertParallelogramTerm Geo S T U V)
  (hilbertParallelogramTerm Geo B E F G) := by
-- SAS + SSS + the two parallelogram triangulations

```

The positioned core is

```

theorem i44_positioned_content
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B E F G : Geo.Point)
  (hABE : Geo.Between A B E)
  (hPar : IsParallelogram Geo B E F G) :
exists M L : Geo.Point,
  IsParallelogram Geo A B M L /\
  Geo.Between G B M /\
  HilbertScissorsEquicomplementable Geo
    (hilbertParallelogramTerm Geo A B M L)
    (hilbertParallelogramTerm Geo B E F G) := by
-- H, K, L, M construction + I.43

```

The mathematically significant local declarations can be grouped by role.

- `i44_place_parallelogram_shape`: rigid placement on the prescribed base;
- `i44_placed_parallelogram_content`: congruent-piece certificate for the rigid copy;
- `i44_construct_H_ordered`: construct $H - G - F$ with $AH \parallel GB$;
- `i44_construct_K_ordered`: construct $H - B - K$ and $F - E - K$;
- `i44_construct_LM`: construct the parallel through K and its two intersections;
- the three order helpers for $H - A - L$, $L - M - K$, and $G - B - M$: recover strict order from carrier data;
- `i44_i43_content_bridge`: adapt I.43 to whole parallelogram terms;
- `i44_positioned_content`: assemble the full positioned construction;
- `i44_angle_transfer`: transfer the prescribed angle by vertical angles;
- `i44_transport_onto_line`: package the reusable high-level transport theorem.

The remaining proposition-local helpers mostly recognize parallelograms, transport parallelism along collinear carriers, or normalize line orientation. They are essential to compilation but do not represent independent new mathematical principles.

44.7 Relationship with Earlier Propositions

44.7.1 Proposition I.15

The last angle step is exactly the vertical-angle theorem. Once the order lemmas have established

$$A - B - E \quad \text{and} \quad G - B - M,$$

I.15 identifies $\angle ABM$ with $\angle EBG$. In Lean this appears through the already exposed helper `VerticalAngles`; no new angle theorem is proved in I.44.

44.7.2 Proposition I.31 and the Euclidean parallel layer

Classically, I.31 is invoked to draw the parallels through A and K . The Lean file does not call the historical proposition by name. It uses the Hilbert-level construction

```
hilbert_parallel_through_point_exists
```

and the developed parallel-carrier infrastructure.

More importantly, the existence of the required intersections is proved by Euclidean uniqueness/transitivity of parallels. Thus the formal proof has a clear Euclidean dependency even where it does not replay the literal I.29 plus Postulate-5 angle argument.

44.7.3 Proposition I.42

I.42 is the first direct Book I dependency of the final theorem. It converts the given triangle and angle into a parallelogram with the desired content and angle. I.44 adds the missing location constraint: the parallelogram must be applied to a specified segment.

The final architecture therefore treats I.42 as a content-and-shape producer, not as a theorem that already knows where the resulting parallelogram is to be placed.

44.7.4 Proposition I.43

I.43 is the central content theorem inside the positioned construction. Once $HLKF$, $HABG$, and $BMKE$ have been recognized, I.43 states that the two remaining complements have equal synthetic content. The local bridge `i44_i43_content_bridge` does not strengthen I.43. It only converts its triangle-sum representation of the complements into the parallelogram terms $ABML$ and $BEFG$ used by I.44.

44.7.5 Propositions I.34–I.41

These propositions are not replayed directly by the public I.44 theorem, but they remain present through its two major dependencies. I.42 already rests on the forward content theory of I.37–I.41, and I.43 rests on the diagonal parallelogram theorem I.34 together with the scissors calculus. I.44 is therefore the first point where those two branches are combined in an active construction.

44.7.6 Proposition I.45

I.45 generalizes the application-of-areas idea from a triangle to an arbitrary rectilinear figure. The transport theorem isolated in I.44 is therefore a natural reusable unit for the next stage: once a suitable parallelogram of the required total content is available, it can be applied to a prescribed segment with a prescribed angle.

44.8 Hilbert and Book Zero Components Used

44.8.1 Betweenness, extensions, and carrier incidence

The formal proof uses strict betweenness much more extensively than the classical text writes. The core order relations are

$$A - B - E, \quad H - G - F, \quad H - B - K, \quad F - E - K, \quad H - A - L, \quad L - M - K, \quad G - B - M.$$

They are obtained by segment extension, line intersection, and plane-separation arguments. Carrier identification repeatedly uses incidence uniqueness for two distinct points.

The proof also makes heavy use of the bridge between a Hilbert incidence line and the extensional `Geo.PointLine` representation. This is what lets the code move safely between line-uniqueness arguments and the strict `Geo.Parallel` relation.

44.8.2 SameSide and OppositeSide

SameSide/OppositeSide data are the main device for turning a bare line intersection into the correct order statement. For example, `i44_construct_H_ordered` uses the line GB as a separator to prove that the intersection on FG lies beyond G . The K , L , and M order helpers use analogous arguments.

The proof of $G - B - M$ is particularly revealing. It shows that H and G are on the same side of BE , while K and M are on the same side of BE . Since $H - B - K$, the points H and K lie on opposite sides. Transporting through the two SameSide relations gives G and M on opposite sides, and uniqueness of the intersection with BE identifies the crossing point with B .

These side-of-line facts are invisible in the short Euclidean proof but are central to the formal reconstruction.

44.8.3 Segment construction, SameRay, and angle construction

Rigid placement uses Hilbert segment construction twice: once to copy the first side of the original parallelogram onto the extension of AB , and once to copy the second side onto the constructed angle ray. The first copy must remain on the ray beyond B , and the second must remain on the copied angle ray. SameRay machinery records these orientation facts.

Angle construction is used with an explicit carrier and a chosen side. This produces more information than the classical phrase “construct an equal angle”, and that information is exactly what is needed to prove that the new point G is off the line BE and that BEG is a genuine triangle.

44.8.4 Euclidean parallelism

The theorem is genuinely Euclidean. The class assumption is

```
[HilbertEuclideanPlane Geo]
```

and the Euclidean input is used in several places:

- construction and uniqueness of a parallel through a point;
- transitivity of distinct parallel carriers;
- proving that the constructed parallel through A meets FG ;
- proving that HB meets EF ;
- identifying carriers that are both parallel to the same line and share a point;
- completing and recognizing the several parallelograms in the I.43 configuration.

This is the formal counterpart of Euclid's use of I.29 and Postulate 5. In neutral geometry the relevant uniqueness/intersection conclusions are not available in this form.

44.8.5 Congruence and parallelogram infrastructure

The rigid-copy content proof uses SAS and SSS. It also uses

```
ParallelogramOppositeSidesCongruent  
hilbert_parallelogram_fourth_vertex_exists
```

Together they show that a shape copied by two adjacent sides and the included angle is congruent piecewise after triangulation, and that the copied three vertices determine a genuine parallelogram.

No new congruence axiom or parallelogram axiom is introduced in I.44.

44.8.6 Scissors content infrastructure

The content layer uses only the existing scissors operations, principally

```
scissors_congruent  
HilbertScissorsEq.add  
HilbertScissorsEq.trans  
equicomplementable_of_scissorsEq  
equicomplementable_symm  
equicomplementable_trans  
equicomplementable_transport  
parallelogram_two_triangulations
```

I.44 neither defines numerical area nor introduces subtraction or a content order. The only equality-of-figures conclusion is the already established synthetic relation of equicomplementability.

44.9 Source Audit

44.9.1 Euclid, Heath, and Joyce

The classical sources agree on the main construction. I.42 supplies a parallelogram equal to the given triangle in the prescribed angle. One side is placed on the produced carrier of the given segment. The side FG is produced, a parallel through A is drawn, and the two produced lines are forced to meet by the Fifth Postulate. A parallel through the intersection creates the large parallelogram and the two parallelograms about its diameter. I.43 identifies the complements, and I.15 supplies the final vertical-angle identity.

These sources control the historical statement and the classical DAG. They also expose the two formal pressure points: the unreferenced placement step and the specifically Euclidean meeting step.

44.9.2 Wilkins

Wilkins gives the same dependency pattern with explicit attention to the produced lines and the I.43 configuration. His account is useful as a control on point order and on which quadrilaterals are intended to be the two complements. The current Lean diagram follows those order constraints rather than relying on a visually plausible but logically weaker collinearity sketch.

44.9.3 Hilbert and Borsuk–Szmielw

Hilbert’s Chapter IV and the Borsuk–Szmielw treatment control the semantics of content. Their plane-area theory distinguishes equidecomposability from equicomplementability and develops the forward parallelogram/triangle content theorems without requiring the continuity group. In particular, the non-Archimedean distinction warns against replacing the current target by an unjustified global equidecomposability statement.

The same sources reorganize the geometry away from Euclid’s proposition numbers. Segment layoff, angle construction, line order, and parallel uniqueness are reusable parts of the axiomatic geometry. This is why the Lean proof can derive the placement and intersection steps without introducing a new I.44-specific axiom.

44.9.4 Hartshorne

Hartshorne places I.35–I.48 in the area part of Book I and describes I.44 as the construction of a parallelogram with a given side and angle equal to a given triangle. His discussion of “equal content” is also the correct semantic control for the present documentation: Euclid has not introduced a real-valued area function, so equality of figures in this block should not be silently rewritten as a numerical formula.

Hartshorne’s analysis of the parallel postulate also supports the formal classification. The Fifth Postulate and Playfair uniqueness are equivalent over the developed neutral theory. The Lean proof chooses the uniqueness form of the Euclidean input when proving the hidden intersections.

44.9.5 Greenberg

Greenberg supplies an independent modern control on the same Euclidean boundary: Euclid's Fifth Postulate and Hilbert/Playfair parallel uniqueness are equivalent in the neutral setting. This explains why the formal replacement of Euclid's less-than-two-right-angles argument by a uniqueness-of-parallels contradiction does not change the axiomatic strength of the proposition.

44.9.6 Beeson proof corpus

The Beeson/Tarski corpus remains useful in the project as a gap audit for classical constructions, especially where a traditional diagram names an intersection without isolating its existence or orientation. The final I.44 proof, however, is not organized around Tarski tactics. Its actual DAG is the Hilbert-style one documented above: explicit line carriers, SameSide/OppositeSide order recovery, Euclidean parallel uniqueness, and the project's own scissors representation.

44.10 Formalization Notes

44.10.1 Geometry

The geometric cost of I.44 is substantially higher than the length of the classical text suggests. The formal proof must establish all of the following, rather than reading them from a picture:

- the first copied side lies beyond B , giving $A - B - E$;
- the copied angle ray produces a point genuinely off the carrier BE ;
- the fourth vertex completing the placed parallelogram exists;
- the parallel through A actually meets the produced carrier FG ;
- that intersection is ordered as $H - G - F$;
- the carriers HB and EF actually meet;
- their intersection satisfies both $H - B - K$ and $F - E - K$;
- the parallel through K meets the two required carriers;
- those intersections satisfy $H - A - L$, $L - M - K$, and $G - B - M$;
- the four quadrilaterals used by I.43 satisfy the strict `IsParallelogram` relation.

No continuity axiom is used to obtain these intersections. Their existence comes from the developed Euclidean parallel theory plus incidence and plane-separation arguments.

44.10.2 Content

There are two content transports and they have different proofs.

First, the arbitrary I.42 parallelogram is rigidly copied to $BEFG$. Here the proof actually produces a direct scissors equality by congruent triangulation: SAS for one triangle, SSS for the other, and addition of the two congruent pieces.

Second, *BEFG* is transformed into the target *ABML* by the I.43 complement construction. Here the conclusion is equicomplementability, exactly matching the public content strength of I.43. The final theorem composes these relations by transitivity.

At no stage is equal content used to infer an incidence, order, or metric fact. All geometry is constructed first; content is then transported forward. Consequently I.44 needs neither the positivity principle used at the converse boundary I.39 nor any numerical area theorem.

44.10.3 Representation

The public orientation of the target is

```
IsParallelogram Geo A B M L
Geo.AngleCongruent A B M X Y Z
```

This orientation matters. The prescribed angle is the angle at *B* between the actual adjacent sides *BA* and *BM*. It should not be represented using a diagonal arm.

The I.43 bridge has a second representation issue. I.43 returns complements as explicit sums of two triangle terms, while I.44 wants a whole parallelogram term. The helper

```
i44_i43_content_bridge
```

uses alternate triangulations and triangle permutation identities to prove that these are the same formal scissors objects up to `HilbertScissorsEq`. No new polygon-region semantics is introduced.

The formal figures therefore represent finite triangle sums and parallelogram terms, not geometric interiors with a measure attached to them.

44.10.4 Axiomatic status

Proposition I.44 is Euclidean, not neutral. The explicit type-class context is

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

Hilbert Group IV (the Euclidean parallel axiom) is therefore used. Its formal role is uniqueness/transitivity of parallels and the intersection arguments described above. Parallel existence is provided by the developed construction layer; the specifically Euclidean strength is needed when the proof excludes a second distinct parallel through the same point. This has the same axiomatic strength as the classical use of Postulate 5, although the formal proof uses a different equivalent route.

Group III congruence construction is also essential: segment layoff, angle construction, SAS, SSS, and opposite-side congruence of parallelograms are used in the rigid-placement certificate. No Group V continuity principle is used.

The current proposition introduces no active local axiom. In particular, the provisional `i44_transport_onto_line` axiom has been replaced by the derived theorem of the same mathematical purpose. No new theorem is added to `HilbertInterface`, `HilbertBookZero`, or `HilbertScissors` by I.44 itself.

One source-level comment at the beginning of the production file still describes steps 2–6 as being packaged by a local axiom. That comment is historical and is now stale: the active declaration later in the same file is the derived theorem `i44_transport_onto_line`, while the former axiom declaration survives only inside a block comment. This is a documentary discrepancy, not an active foundational dependency.

44.11 Dependency Summary

The formal dependency chain is

```

Hilbert incidence, order, congruence, and Euclidean parallel axioms
      ↓
segment/angle construction + SameRay + SameSide/OppositeSide
      ↓
parallel uniqueness/transitivity + parallelogram infrastructure
      ↓
Proposition I.42: content-and-angle parallelogram
      ↓
i44_place_parallelogram_shape
      ↓
i44_placed_parallelogram_content
      ↓
ordered construction of  $H, K, L, M$ 
      ↓
large and diagonal parallelogram configuration
      ↓
Proposition I.43 + i44_i43_content_bridge
      ↓
i44_positioned_content
      ↓
I.15 / vertical-angle transfer + content transitivity
      ↓
i44_transport_onto_line
      ↓
euclid_proposition_44.

```

The classical and formal DAGs may be summarized as

```

Euclid: I.42 → placement → I.31/I.29/Post.5
          → I.43 → I.15,
Lean: I.42 → derived rigid placement
          → Euclidean parallel/intersection order
          → I.43 bridge → VerticalAngles.

```

New geometric axiom:	no
New content axiom:	no
Provisional I.44 transport axiom retained as active axiom:	no
New proposition-local helpers:	yes
New reusable high-level transport helper:	yes
New Book Zero theorem:	no
New Interface/Scissors theorem:	no
Numerical area introduced:	no
Hilbert Group IV used:	yes
Continuity used:	no
Public <code>Proposition44.lean</code> compiles:	yes
Full <code>lake build</code> re-run in this documentation pass:	no
GitHub CI status checks on the committed code:	none reported
<code>sorry</code> introduced in <code>Proposition44.lean</code> :	no

Diagrammatic audit. The revised documentation uses six figures, each tied to a genuinely new geometric state. Figure 44.1 is reserved for the classical Euclidean construction. Figure 44.2 isolates the derived rigid placement on the prescribed base, while Figure 44.3 records the distinct two-triangle scissors certificate proving that the rigid copy preserves content. Figures 44.4 and 44.5 expose the two ordered intersection stages that the classical proof compresses into produced-line language. Figure 44.6 is used only after L and M have been constructed, when the complete ordered I.43 configuration first exists. No separate figure is added for the four parallelogram-recognition helpers, the I.43 representation bridge, the vertical-angle theorem, or the final transitivity steps: those operations introduce no new points, lines, intersections, or regions beyond the already displayed configurations. In particular, the figures record geometry rather than theorem-call or normalization state.

44.12 Conclusion

Proposition I.44 is a major architectural checkpoint in the Book I reconstruction. The classical proof combines I.42 and I.43 with a short parallel construction and a vertical-angle observation. The formal proof shows that the apparent simplicity hides two genuine tasks: a rigid placement of the I.42 parallelogram on a prescribed segment, and a large amount of order information needed to instantiate I.43 correctly.

Both tasks are now derived rather than assumed. The rigid placement is built from segment and angle construction and certified by SAS/SSS triangulation. The positioned construction derives H, K, L, M , recovers the strict order relations by plane separation, recognizes the four required parallelograms, and then invokes I.43 through a representation bridge. Vertical angles close the prescribed-angle condition, and equicomplementability transitivity closes the content condition.

The proposition therefore introduces no new foundational debt. It remains fully synthetic: no numerical area, no polygon measure, no continuity axiom, and no application-of-area transport axiom. At the same time it produces a useful high-level theorem, `i44_transport_onto_line`, which packages a substantial geometric construction into a form suitable for later Book I work. This is precisely the kind of compression the Hilbert/Book Zero architecture is intended to provide: the final theorem is short because the hidden geometry has been made explicit and proved once.

Chapter 45

Proposition I.45

45.1 Introduction

Proposition I.45 is the culmination of the application-of-areas sequence I.42–I.45. Proposition I.42 constructs a parallelogram equal in content to one triangle and having a prescribed angle. Proposition I.44 adds the location constraint that such a parallelogram be applied to a prescribed segment. Proposition I.45 passes from one triangle to a rectilinear figure. Classically, the new idea is finite composition: triangulate the figure, apply the preceding construction to its triangular pieces, and combine the resulting parallelograms into one parallelogram with the prescribed angle.

The current Lean reconstruction follows that compositional idea, but makes its data flow explicit. A triangulated figure is represented by `HilbertTriangulatedFigure Geo`, and its executable scissors content is the sum of the corresponding triangle terms. For a single triangle the proof calls I.42 directly. For a longer list, the induction hypothesis produces a parallelogram for the tail, the next triangle is converted into an oriented adjacent parallelogram, and the two parallelograms are pasted into a larger one. The paste must preserve both the synthetic content and the prescribed angle.

The final theorem remains entirely synthetic. It does not attach a number to a figure and does not use a numerical area formula. Equality of content is represented by

```
HilbertScissorsEquicomplementable
```

If L is the input list of triangles, the theorem constructs a parallelogram $STUV$ satisfying

$$\angle STU \cong \angle XYZ$$

and

$$\text{Term}(L) \approx \text{Par}(STUV),$$

where $\approx$ denotes the project's equicomplementability relation.

I.45 also resolves an architectural question left deliberately open earlier in the area block. The existing Hilbert layer already contains `HilbertTriangle`, `HilbertTriangulatedFigure`, and relations between triangulations. For the constructive content theorem of I.45, this list-of-triangles representation is sufficient: the proof needs only finite addition of triangle content and an induction step that adjoins one triangle. No new polygon datatype and no numerical area layer are required.

There is, however, an important representation boundary. In the current library,

```
abbrev HilbertTriangulatedFigure
  (Geo : Geometry.Geo) :=
  List (HilbertTriangle Geo)
```

is only a finite list of triangles. The statement of `euclid_proposition_45` assumes that the list is nonempty and that each triangle is nondegenerate. It does not additionally assert that the triangles have disjoint interiors, fit along common edges, or form the interior of one simple polygon. Thus the formal theorem is a valid and in one sense stronger additive statement about any finite nonempty list of nondegenerate triangles. To read such a list as one geometric rectilinear figure requires a representation invariant supplied outside this theorem. The documentation therefore distinguishes the proved scissors statement from a full internal semantics of polygonal regions.

As in the preceding content propositions, the proof has three distinct layers.

- **Geometry:** extensions, SameRay orientation, SameSide/OppositeSide separation, parallel uniqueness, Pasch, strict betweenness, and recognition of the pasted outer parallelogram.
- **Content:** scissors refinement, equicomplementability of the new triangular piece with an adjacent parallelogram, addition, and transitivity.
- **Representation:** a rectilinear input is represented by a triangle list and a multiset term; the I.44 construction is locally strengthened to retain one strict order fact that its public wrapper intentionally hides.

The most important foundational result is negative: I.45 introduces no new geometric axiom, no new content axiom, no continuity principle, and no positivity or cancellation principle. The temporary `i45_extend_parallelogram` assumption used during development has been completely eliminated and replaced by a theorem.

45.2 The Classical Configuration

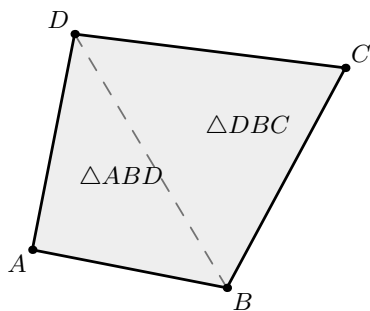
Euclid states the proposition for a given rectilinear figure and a given rectilinear angle. In the first nontrivial case, let the figure be a quadrilateral $ABCD$ divided by the diagonal BD into the two triangles ABD and DBC .

First construct, by I.42, a parallelogram equal to triangle ABD in the given angle. Then apply, by I.44, a second parallelogram equal to triangle DBC to a side of the first parallelogram. The second parallelogram is placed so that the two pieces are adjacent rather than overlapping. Euclid then uses parallel-line angle relations and straight-line criteria to show that the outer boundary is again a parallelogram. Since the two inner parallelograms are respectively equal to the two triangles, their union is equal to the whole rectilinear figure.

For a rectilinear figure with more triangular pieces, the same operation is repeated. One may therefore read the classical construction recursively: start with a parallelogram for one triangle and append one triangle at a time by applying an adjacent parallelogram to the current boundary.

Figure 45.1 separates the historical two-triangle construction from the later formal normalization. The left panel shows a representative quadrilateral divided by a diagonal; the right panel shows the classical I.42/I.44 operation of replacing the two triangular contents by two adjacent parallelograms whose outer boundary is again a parallelogram.

(a) triangulate the given figure



(b) apply the two triangular contents

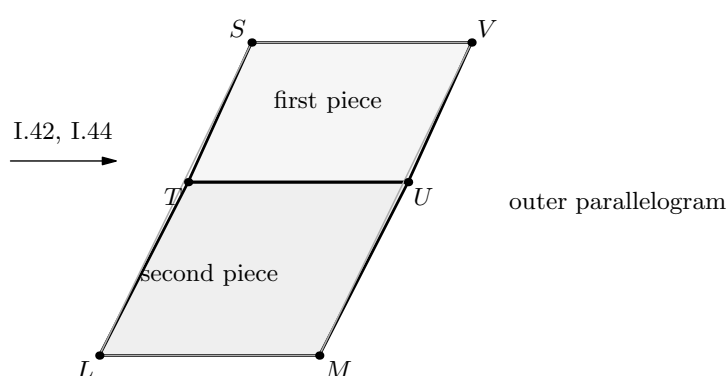


Figure 45.1: The first nontrivial classical case of I.45. A rectilinear quadrilateral is triangulated by a diagonal. I.42 constructs a parallelogram equal to the first triangle in the prescribed angle, and I.44 applies a second parallelogram equal to the second triangle along the current boundary. The two applied pieces form one larger parallelogram.

The current Lean proof adopts exactly this recursive shape, but its local paste configuration is normalized to the notation

$$STUV, \quad TUML, \quad SLMV.$$

Here $STUV$ is the already constructed parallelogram, $TUML$ represents the next triangle, and $SLMV$ is the pasted outer parallelogram. The strict order conditions are

$$S - T - L, \quad V - U - M.$$

These relations, which are proved rather than read from a drawing, are the formal content of saying that the second piece is attached externally along TU . The corresponding formal geometry is displayed only after those order facts have been constructed in Figure 45.2.

45.3 Classical Proof

The classical proof is a finite composition of I.42 and I.44.

For a quadrilateral $ABCD$, draw the diagonal BD . Proposition I.42 gives a parallelogram equal to triangle ABD in the prescribed angle. Proposition I.44 then applies to one side of that parallelogram a second parallelogram equal to triangle DBC , in the supplementary orientation required for the two pieces to form a straight outer boundary.

The angle equalities supplied by the two constructions, together with I.29, show that the relevant sides of the two parallelograms are parallel. Euclid uses I.14 to recognize the straight-line relations created at the common side. The outer opposite sides are then shown parallel using the established parallel theory, including I.30 and the parallelogram criterion behind I.33. Thus the two applied pieces form one larger parallelogram.

The content conclusion is purely additive. One inner parallelogram is equal to triangle ABD , the other to triangle DBC , and the two triangles are the whole rectilinear figure. Therefore the outer parallelogram is equal to the given rectilinear figure. The prescribed outer angle is inherited from the first construction.

The two-triangle historical DAG is therefore

$$\begin{array}{c}
 ABCD \text{ divided by } BD \\
 \Downarrow \\
 \triangle ABD + \triangle DBC \\
 \Downarrow \text{ I.42 and I.44} \\
 \text{two adjacent parallelograms} \\
 \Downarrow \text{ I.29, I.14, I.30, I.33} \\
 \text{one outer parallelogram} \\
 \Downarrow \text{ additivity of equality of figures} \\
 \text{outer parallelogram} = ABCD.
 \end{array}$$

The general finite case repeats the same step for each triangular component. This is the part of the classical argument that the final Lean induction captures most directly.

45.4 Triangulated Figures and the Representation Boundary

The proposition is the first point in Book I where the input is not naturally one triangle or one quadrilateral but an arbitrary rectilineal figure. This forces a distinction between a geometric region and a formal decomposition of that region.

The Hilbert layer already contains

```

structure HilbertTriangle
  (Geo : Geometry.Geo) where
  A : Geo.Point
  B : Geo.Point
  C : Geo.Point

abbrev HilbertTriangulatedFigure
  (Geo : Geometry.Geo) :=
  List (HilbertTriangle Geo)

```

and a relation `HilbertSameFigure` generated by triangle splitting and triangle permutations. This is genuine infrastructure for reasoning about triangulations. It is intentionally more structural than a coordinate polygon representation.

Proposition I.45 introduces the executable term

```

def rectilinealTerm
  (L : HilbertTriangulatedFigure Geo) :
  HilbertScissorsTerm Geo :=
  (L.map
    (fun t =>
      hilbertScissorsTriangle Geo t.A t.B t.C)).sum

```

which forgets all incidence between different list entries and retains exactly what the proof needs for content: a finite formal sum of triangle terms.

This choice has two consequences.

First, it is sufficient for the theorem actually proved. The induction step uses only the identity

$$\text{Term}(T :: L) = \text{Tri}(T) + \text{Term}(L),$$

followed by addition and transitivity of equicomplementability. No operation in the proof needs to inspect a polygon boundary.

Second, the theorem should not be described as proving from first principles that every such list is a simple rectilinear region. A geometric rectilinear figure requires a compatible finite triangulation of one region, in particular the appropriate nonoverlap and coverage conditions. The current Lean theorem does not carry those conditions as hypotheses. When L comes from a genuine triangulation, the theorem formalizes Euclid I.45. Abstractly, it proves a slightly more general finite-additive statement.

This is an important architectural checkpoint. The existing `HilbertTriangulatedFigure` plus scissors terms are sufficient for the constructive content calculus of I.45. A stronger polygon-region type may still be useful later for theorems whose hypotheses or conclusions depend on the boundary, interior, overlap, containment, or equality of geometric regions. I.45 itself does not force that extension.

45.5 Proof Architecture of the Reconstruction

The 2300-line Lean file is best understood as eight conceptual stages rather than as a sequence of low-level incidence manipulations.

45.5.1 Step 1: turn a triangle list into a scissors term

The small definitions

```
rectilinearTerm
rectilinearTerm_cons
rectilinearTerm_nil
```

form the additive interface between the list representation and `HilbertScissorsTerm`. They isolate the only list algebra used by the final induction.

This stage is purely representational. It introduces no new geometric relation and no content axiom.

45.5.2 Step 2: prove the geometry of pasting adjacent parallelograms

Suppose $ABCD$ and $BCML$ are parallelograms and the strict orders are

$$D - C - M, \quad A - B - L.$$

The helpers

```
i45_outer_first_parallel
i45_outer_second_parallel
i45_paste_parallelograms_geometry
```

prove that the outer quadrilateral $ALMD$ is a parallelogram.

The first outer parallel is obtained mostly by transporting parallelism along the two collinear produced sides. The second is subtler: the proof has two lines parallel to the common carrier

and must show that the two candidate outer carriers are distinct before invoking Euclidean transitivity/uniqueness of parallels. This is one of the places where the formal proof uses Group IV in a way hidden by the ordinary picture.

Figure 45.2 records exactly this geometric paste, and nothing from the later scissors refinement.

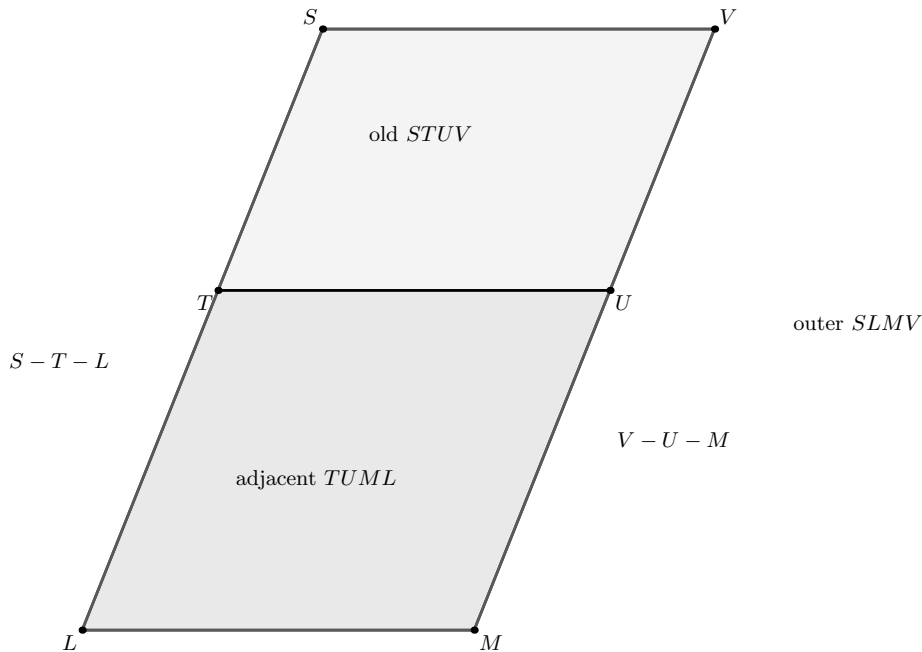


Figure 45.2: The formal paste configuration. The old parallelogram is $STUV$, the adjacent one is $TUML$, and the strict orders $S - T - L$ and $V - U - M$ place the second piece externally. The outer boundary is the parallelogram $SLMV$. No crossing point is shown here: its existence is a separate later theorem used only for the scissors identity.

45.5.3 Step 3: refine the paste into a scissors identity

The geometric union of two adjacent parallelograms is not automatically an identity of the project's multiset terms. The proof therefore constructs an internal crossing point.

The helper

```
i45_paste_crossing
```

returns a point X satisfying

$$A - X - M, \quad B - X - C.$$

Its existence is obtained by side-of-line information and a forced Pasch argument. Parallel/order transport is then used to identify the second betweenness relation.

With this witness,

```
i45_paste_parallelgrams_content_of_crossing
```

splits the relevant triangles and uses the generic diagonal-flip theorem

```
crossing_quadrilateral_two_triangulations
```

Explicit multiset associativity and commutativity then normalize the pieces. The packaged result

```
i45_paste_parallelgrams_content
```

is the formal identity

$$\text{Par}(ALMD) \sim \text{Par}(ABCD) + \text{Par}(BCML).$$

Before packaging the equicomplementability step, it is useful to make the actual scissors geometry visible. Figure 45.3 separates the three states consumed by the proof: the outer triangulation, the crossing witness X , and the diagonal flip inside the central quadrilateral. In the notation of the helper these relations are

$$A - X - M, \quad B - X - C.$$

They permit the replacement

$$\triangle ABM + \triangle AMC \sim \triangle ABC + \triangle BCM,$$

while the two remaining triangles are left unchanged.

(a) split the outer parallelogram

(b) crossing witness

(c) regroup as the two inner pieces

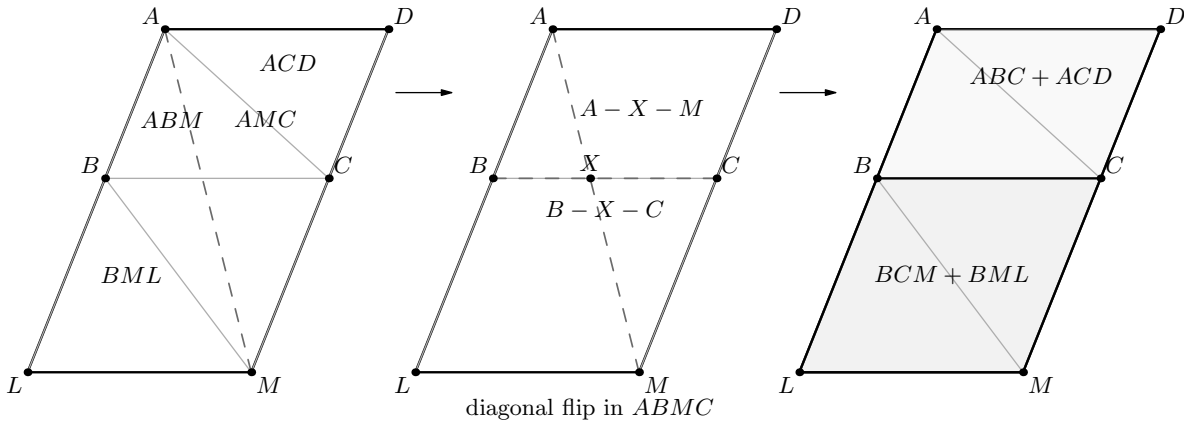


Figure 45.3: The scissors proof of the paste identity. First the outer parallelogram is split into the four triangle terms ABM , BML , AMC , and ACD . The point X with $A - X - M$ and $B - X - C$ certifies the two triangulations of the central quadrilateral $ABMC$. After the diagonal flip, the pieces regroup as $(ABC + ACD) + (BCM + BML)$, exactly the two inner parallelogram terms.

The companion

```
i45_paste_equicomplementable
```

combines that scissors equality with an equicomplementability certificate for the right-hand piece. This is the additive content engine used by the induction step.

45.5.4 Step 4: preserve the prescribed angle

The helper

i45_paste_angle

proves that pasting does not change the prescribed angle carried by the first parallelogram.

For the outer parallelogram $ALMD$, opposite angles give

$$\angle ALM \cong \angle MDA.$$

The order $D - C - M$ identifies the rays DC and DM , so

$$\angle MDA = \angle CDA.$$

The old parallelogram $ABCD$ gives

$$\angle ABC \cong \angle CDA.$$

Transitivity therefore yields

$$\angle ALM \cong \angle ABC.$$

This argument uses only the developed parallelogram-angle theorem and SameRay angle transport; it introduces no new angle principle.

45.5.5 Step 5: reduce the orientation problem to one strict order

The main obstruction to using public I.44 directly is orientation. Public I.44 returns a parallelogram on a prescribed segment, but it does not expose which side of the segment contains the new parallelogram. I.45 needs the new piece to be external to the old one.

The first reduction is

i45_other_side_order

which shows that for parallelograms $STUV$ and $TUML$,

$$V - U - M \implies S - T - L.$$

The proof uses TU as a separator. Parallel endpoints give SameSide data for V, S and for M, L . Since $V - U - M$, points V and M are on opposite sides of the carrier TU . OppositeSide transport therefore puts S and L on opposite sides. Euclidean uniqueness of the parallel identifies the carrier SL through T , and incidence uniqueness identifies the crossing point with T . Thus only one strict order must be controlled by the construction.

The second reduction is

i45_extension_angle

If $STUV$ is a parallelogram and $T - U - E$, then

$$\angle EUV \cong \angle STU.$$

The proof explicitly constructs the side-of-line configuration for the transversal, invokes the developed alternate-angle theorem for parallel lines, and replaces interior points by the corresponding SameRay endpoints. This lemma makes it possible to orient the copied I.44 shape using the old ray UV itself.

45.5.6 Step 6: construct the new triangle piece on the correct side

The central new helper is

```
i45_oriented_triangle_placement
```

Starting from $STUV$ and a nondegenerate triangle PQR , it constructs points E, F, G such that

$$T - U - E, \quad UFG \text{ is a parallelogram}, \quad G \text{ lies on the ray } UV,$$

and

$$\triangle PQR \approx UFG.$$

The proof first uses I.42 to obtain a source parallelogram with the correct triangle content and the angle STU . It then extends TU beyond U and lays off one copied side along that extension. The other copied side is laid off directly on the old ray UV . The lemma `i45_extension_angle` identifies the included angle, and the existing I.44 helper `i44_placed_parallelogram_content` certifies the piecewise scissors content.

Figure 45.4 shows this stage without imposing an order between V and G that Lean does not prove. The first panel fixes the old ray UV ; the second shows the copied parallelogram and places G on that ray, but omits V from the second panel. Thus the diagram records precisely a **SameRay** fact rather than an accidental betweenness relation.

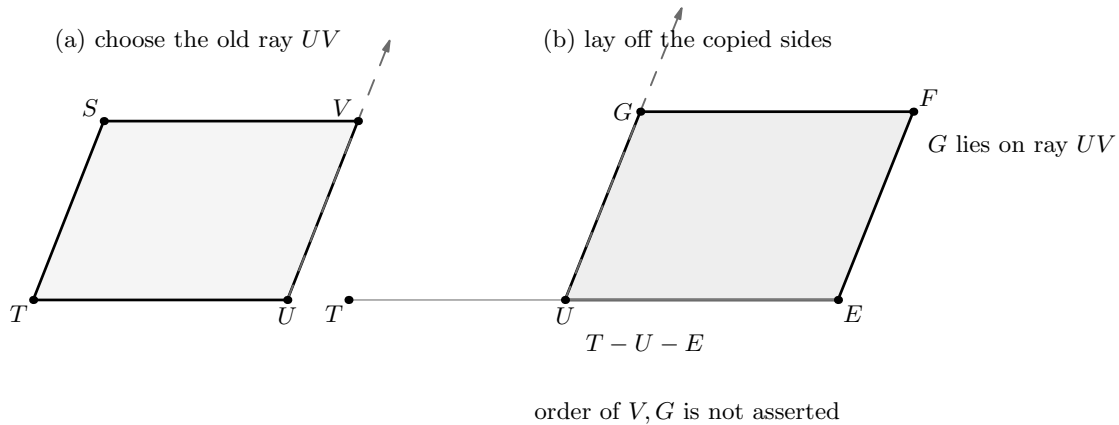


Figure 45.4: The orientation step before the ordered I.44 construction. Panel (a) fixes the selected ray UV of the old parallelogram. Panel (b) extends TU to E , giving $T - U - E$, and lays off G on the same ray from U as V before completing UFG . No order between V and G is asserted; the formal conclusion is only $\text{SameRay}(U; V, G)$.

The next helper

```
i45_positioned_content_ordered
```

is a deliberately stronger local wrapper over the internal I.44 construction. The public helper `i44_positioned_content` returns the target parallelogram and its content equivalence but discards the intermediate order fact $G - B - M$. I.45 needs that fact, so it reuses the same I.44 construction of H, K, L, M and retains

$$G - B - M$$

before hiding the internal witnesses.

This is a representation/API issue, not a new geometric principle. No part of I.44 is assumed again. The local theorem replays already proved I.44 helpers and exposes one conclusion needed by the stronger downstream orientation invariant.

Combining these two helpers gives

```
i45_adjacent_triangle_parallelogram
```

with the clean output

$$TUML \text{ parallelogram, } \quad V - U - M, \quad \triangle PQR \approx TUML.$$

This is the exact object required by the paste lemmas.

Figure 45.5 isolates the order bridge hidden inside this composition. The ordered internal I.44 construction retains $G - U - M$; the previous step supplies $\text{SameRay}(U; V, G)$. Betweenness transport then replaces G by the old boundary point V and yields exactly $V - U - M$.

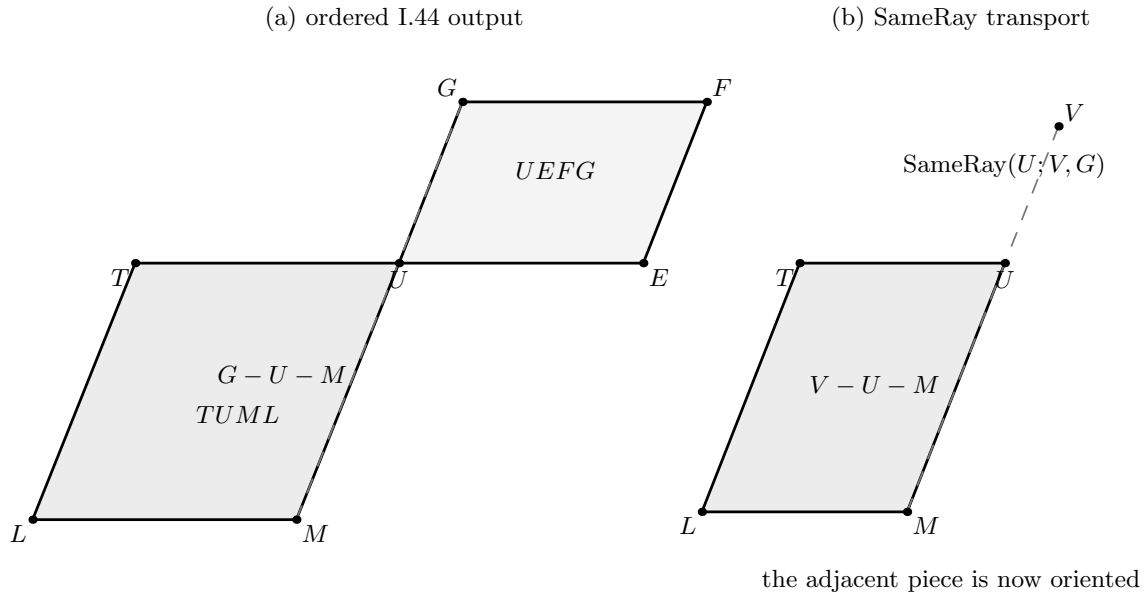


Figure 45.5: Panel (a) shows the extra strict order retained by the local I.44 wrapper: $G - U - M$, together with the source parallelogram $UEFG$ and the target parallelogram $TUML$. Panel (b) uses the previously established SameRay relation between V and G to transport that order to $V - U - M$. The relative order of V and G on their common ray is never needed or asserted.

45.5.7 Step 7: package one induction extension

The theorem

```
i45_extend_parallelogram
```

is the mathematical induction step.

Given an old parallelogram $STUV$ whose angle STU is congruent to the prescribed angle and a new triangle PQR , it first constructs the adjacent parallelogram $TUML$. The order theorem supplies $S - T - L$. The paste theorem produces the outer parallelogram $SLMV$. The angle helper preserves the prescribed angle, while the content helper proves

$$\text{Par}(SLMV) \approx \triangle PQR + \text{Par}(STUV).$$

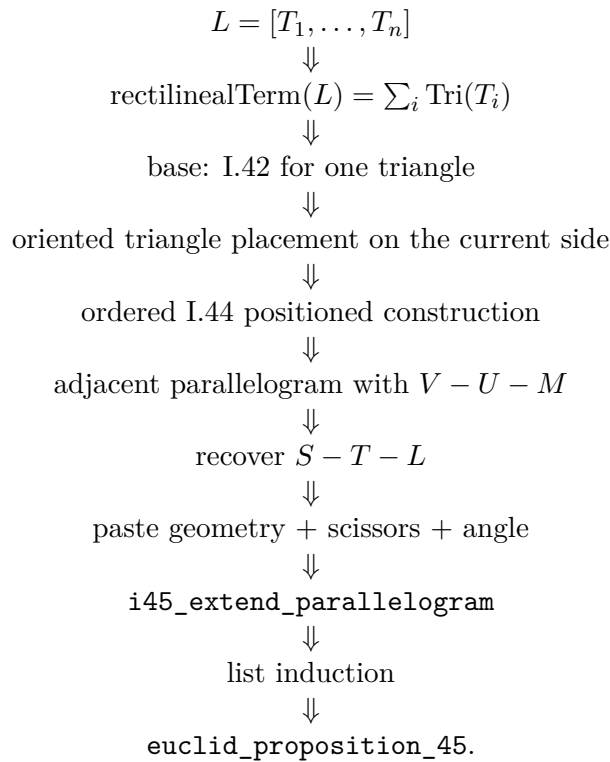
This theorem was initially isolated as a provisional local axiom while the required orientation and paste geometry were being developed. In the final file it is a derived theorem. Eliminating that provisional assumption is the main foundational closure of I.45.

45.5.8 Step 8: induction on the triangle list

The public proof finally becomes short.

If the list contains one triangle, I.42 gives the required parallelogram directly. If the list contains a head triangle and a nonempty tail, the induction hypothesis gives a parallelogram for the tail and `i45_extend_parallelogram` adjoins the head triangle. The only remaining content step is `equicomplementable_add` followed by transitivity.

The actual Lean DAG is therefore



45.6 Lean Formalization

The current file imports only the immediately preceding proposition:

```
import CGJteamLab.Proposition44
```

All earlier dependencies therefore enter through the established module DAG.

45.6.1 The rectilinear term

The executable content of a triangulated input is

```
def rectilinearTerm
  (L : HilbertTriangulatedFigure Geo) :
```



```
HilbertScissorsTerm Geo :=
(L.map
  (fun t =>
    hilbertScissorsTriangle Geo t.A t.B t.C)).sum
```

The two normalization lemmas for cons and nil are used directly by the final induction.

45.6.2 The adjacent-piece theorem

The decisive geometric/content constructor has the following shape, written in the documentation's ASCII convention:

```
theorem i45_adjacent_triangle_parallelogram
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (S T U V P Q R : Geo.Point)
  (hLeft : IsParallelogram Geo S T U V)
  (hPQR : Not (Collinear Geo P Q R)) :
  exists M L : Geo.Point,
    IsParallelogram Geo T U M L /\
    Geo.Between V U M /\
    HilbertScissorsEquicomplementable Geo
      (hilbertScissorsTriangle Geo P Q R)
      (hilbertParallelogramTerm Geo T U M L)
```

The order clause is part of the public output of this helper because the paste step genuinely depends on orientation.

45.6.3 The induction-step theorem

The formerly provisional construction is now a theorem:

```
theorem i45_extend_parallelogram
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (S T U V P Q R X Y Z : Geo.Point)
  (hParallelogram : IsParallelogram Geo S T U V)
  (hAngle : Geo.AngleCongruent S T U X Y Z)
  (hPQR : Not (Collinear Geo P Q R)) :
  exists T' U' V' : Geo.Point,
    IsParallelogram Geo S T' U' V' /\
    Geo.AngleCongruent S T' U' X Y Z /\
    HilbertScissorsEquicomplementable Geo
      (hilbertParallelogramTerm Geo S T' U' V')
      (hilbertScissorsTriangle Geo P Q R +
        hilbertParallelogramTerm Geo S T U V)
```

This is the clean mathematical interface between the large local geometry and the final list induction.

45.6.4 The public proposition

The public theorem is

```

theorem euclid_proposition_45
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (X Y Z : Geo.Point)
  (hXYZ : Not (Collinear Geo X Y Z))
  (L : HilbertTriangulatedFigure Geo)
  (hNonempty : L != [])
  (hTriangles :
    forall t in L,
      Not (Collinear Geo t.A t.B t.C)) :
exists S T U V : Geo.Point,
  IsParallelogram Geo S T U V /\
  Geo.AngleCongruent S T U X Y Z /\
  HilbertScissorsEquicomplementable Geo
    (rectilinearTerm Geo L)
    (hilbertParallelogramTerm Geo S T U V)

```

The displayed code is documentation-friendly ASCII. The compiled source uses the corresponding Lean quantifier and membership notation.

The theorem requires no compatibility condition between distinct triangles in L . This is intentional in the current statement and is the reason the representation discussion above is part of the formal result rather than a purely historical aside.

45.7 Relationship with Earlier Propositions

45.7.1 Proposition I.42

I.42 is the base case of the induction and also the shape/content source inside

```
i45_oriented_triangle_placement
```

It already knows how to turn one nondegenerate triangle into a parallelogram with a prescribed angle. I.45 does not reprove that content theorem.

The new task is to place the I.42 shape in a controlled orientation relative to an already constructed parallelogram.

45.7.2 Proposition I.44

I.44 is the principal direct dependency of the induction step. Its internal rigid-placement certificate

```
i44_placed_parallelogram_content
```

is used to certify the oriented auxiliary copy. Its ordered construction of H, K, L, M is reused by `i45_positioned_content_ordered` so that I.45 can retain the strict order $G - B - M$.

This is a useful API lesson. Public I.44 intentionally exposes only the parallelogram, its prescribed

angle, and its content. I.45 needs a stronger orientation invariant for adjacency. Rather than add an axiom or alter the meaning of public I.44, the local wrapper reuses the proved internal construction and exposes the one extra order fact downstream.

45.7.3 Proposition I.43

I.43 does not appear explicitly in the final induction theorem, but it remains inside the positioned I.44 construction. The content equivalence between the placed auxiliary parallelogram and the target parallelogram on the prescribed side is still the I.43 complement theorem, transported through the I.44 representation bridge.

Thus the content DAG is genuinely cumulative:

$$\text{I.42} \longrightarrow \text{I.44 (using I.43)} \longrightarrow \text{I.45}.$$

45.7.4 The Euclidean parallel block

Classically I.45 invokes I.29, I.30, and I.33 to recognize the final parallelogram. The Lean proof consumes the corresponding Hilbert-level parallel infrastructure rather than replaying those proposition numbers literally.

The important operations are transport of parallelism along collinear carriers, transitivity of parallels, and uniqueness of a parallel through a point. These are exactly the operations used in the paste geometry and in the proof that the second strict order follows from the first.

45.7.5 The content block I.35–I.41

These propositions are present transitively through I.42, I.43, and I.44. I.45 itself does not use a converse-content theorem and never infers geometry from equality of content. It only composes forward content certificates. Consequently it does not require the proper-part/positivity machinery that marks the difficult converse boundary at I.39.

45.8 Hilbert and Book Zero Components Used

45.8.1 Betweenness and SameRay

Strict betweenness is essential to adjacency. The two decisive relations for one paste are

$$S - T - L, \quad V - U - M.$$

They say more than collinearity: they determine which side of the common segment contains the new parallelogram.

SameRay transport is used at two levels. First, segment construction places copied side lengths on chosen rays. Second, angle identities replace an auxiliary point by another point on the same ray without changing the angle. The oriented placement depends critically on the fact that G is not merely collinear with U, V , but lies on the same ray from U as V .

45.8.2 SameSide, OppositeSide, and plane separation

The theorem `i45_other_side_order` is a concentrated use of Hilbert plane separation. Parallel endpoints supply SameSide facts, one strict betweenness supplies an OppositeSide witness, and OppositeSide transport propagates the orientation to the opposite boundary.

The crossing theorem also uses side-of-line data. The common side carrier is used as a separator, and the proof rules out one segment-line intersection before applying a forced Pasch argument. These are exactly the kinds of order facts that are visible in Euclid's diagram but must be named in Lean.

45.8.3 Pasch and crossing

The scissors proof requires an actual point X lying on both the outer diagonal-like segment and the common side. The helper `i45_paste_crossing` obtains it from the order/separation machinery rather than assuming the obvious visual crossing.

The conclusion is stronger than bare line intersection:

$$A - X - M, \quad B - X - C.$$

These strict relations are what allow the triangle-splitting constructors in `HilbertScissorsEq` to be invoked.

45.8.4 Euclidean uniqueness of parallels

The proposition is Euclidean. The type-class context

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

appears throughout the major local helpers.

Group IV enters when the proof identifies carriers by uniqueness/transitivity of parallels. It is used in the outer-parallelogram geometry, in the orientation theorem that recovers $S - T - L$, and inside the I.44 machinery used for positioned content. This is the formal counterpart of the specifically Euclidean parallel reasoning in the classical I.45 construction.

45.8.5 Congruence and angle construction

The oriented copy uses segment construction on prescribed rays and the existing congruence theory of parallelograms. The initial source parallelogram comes from I.42. No new SAS, SSS, or angle-construction theorem is introduced in I.45; those results enter through the established interface and through I.44.

45.8.6 Scissors operations

The content layer uses the existing operations

```
HilbertScissorsEq.split
HilbertScissorsEq.add
```

```
HilbertScissorsEq.trans
equicomplementable_of_scissorsEq
equicomplementable_refl
equicomplementable_symm
equicomplementable_trans
equicomplementable_add
crossing_quadrilateral_two_triangulations
```

No subtraction, cancellation, positivity, or scalar area is required. The entire direction is constructive and additive.

45.9 Formalization Notes

45.9.1 Geometry

The geometric induction step requires substantially more than the phrase “apply the next parallelogram adjacent to the first.” The formal proof must establish:

- a produced point E with $T - U - E$;
- a copied point G on the selected ray UV ;
- noncollinearity needed to complete the auxiliary parallelogram;
- the ordered I.44 output $G - U - M$ and its transport to $V - U - M$;
- the opposite boundary order $S - T - L$;
- both outer pairs of opposite sides parallel;
- an actual crossing point with two strict betweenness relations for the scissors refinement;
- preservation of the prescribed angle after the paste.

None of these facts is inferred from a drawing. They are proved by the existing Hilbert order, separation, incidence, congruence, and Euclidean parallel theory.

45.9.2 Content

The content proof is forward and additive.

The new triangle is first shown equicomplementable with the oriented auxiliary parallelogram. The ordered I.44 construction transfers that content to the adjacent parallelogram on the common side. The paste theorem gives a direct scissors equality between the outer parallelogram and the formal sum of the two inner parallelograms. Finally `equicomplementable_add` and transitivity combine the previous parallelogram with the new triangle.

No equal-content hypothesis is ever used to deduce a line, angle, side, or order relation. Therefore no de Zolt/proper-part principle is needed. I.45 also does not use numerical area, multiplication of lengths, or cancellation of an area factor.

45.9.3 Representation

There are two representation issues.

The first is the triangle-list input. `rectilinearTerm` is a sum of triangle terms, and the public theorem only requires each listed triangle to be nondegenerate. It does not prove that the list is a valid nonoverlapping triangulation of one polygonal region. This should remain explicit in any reuse of I.45.

The second is the orientation information hidden by public I.44. The theorem

```
i45_positioned_content_ordered
```

duplicates no axiom; it simply replays the proved I.44 construction while retaining the intermediate order $G - B - M$. This is a local API strengthening needed to compose constructions. A future refactoring could expose such oriented placement as a shared theorem, but the present I.45 proof does not require changing the public I.44 statement.

45.9.4 Axiomatic status

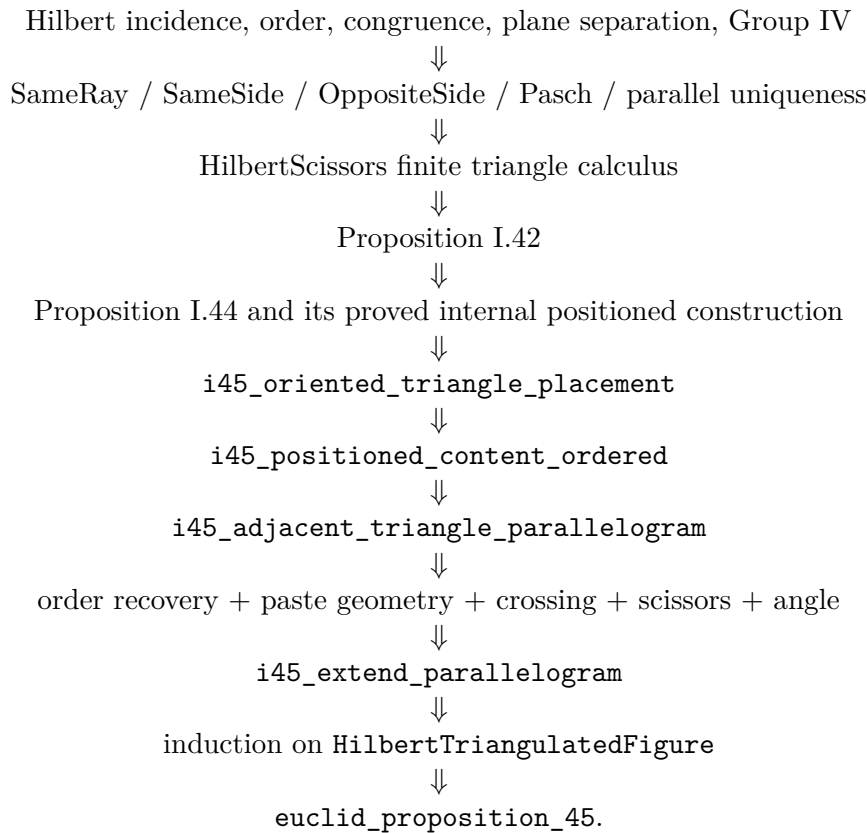
Proposition I.45 is Euclidean. Hilbert Group IV is used through the developed uniqueness/transitivity theory of parallels. The proposition also uses the incidence, order, plane-separation, congruence, and angle-construction layers inherited from the Hilbert interface and from I.44.

No continuity axiom is used by I.45. No Archimedean axiom is used. No new content axiom is introduced. The conclusion is equicomplementability, not a stronger universal equidecomposability claim.

The current source contains no active `axiom`, `sorry`, or `admit` declaration for I.45. The former provisional `i45_extend_parallelogram` assumption has been replaced by the derived theorem documented above.

45.10 Dependency Summary

The formal dependency chain is



The classical and formal DAGs can be compared succinctly as

Euclid: triangulate $\rightarrow$ I.42/I.44 pieces $\rightarrow$ parallel paste $\rightarrow$ add equality,
 Lean: triangle list $\rightarrow$ oriented I.44 piece $\rightarrow$ strict-order recovery
 $\rightarrow$ scissors paste $\rightarrow$ induction.

New geometric axiom:	no
New content axiom:	no
New proposition-local helpers:	yes
New reusable helper promoted to Interface/Scissors:	no
New Book Zero theorem:	no
New Interface/Scissors theorem:	no
New numerical area semantics:	no
New full polygon-region semantics:	no
Continuity used:	no
Archimedean axiom used:	no
Hilbert Group IV used:	yes
Public theorem compiles:	yes
Full lake build:	yes, 1458 jobs
Pre-existing unrelated linter warning:	yes, unused hBC
sorry in Proposition45.lean:	no

Diagrammatic audit. The final documentation uses five figures, each with one geometric role. Figure 45.1 is historical: it records Euclid’s two-triangle I.42/I.44 composition and does not claim

to be the full Lean witness configuration. Figure 45.4 records the formal ray-controlled copy and deliberately avoids asserting an order between the two SameRay points. Figure 45.5 records the additional strict order retained from the internal I.44 construction and its transport to the old boundary ray. Figure 45.2 records only the geometric union of the two adjacent parallelograms. Finally, Figure 45.3 adds the crossing witness and the exact retriangulation required by the scissors equality. No separate diagram is used for list induction, multiset associativity/commutativity, or final angle transitivity, since those steps introduce no new geometric configuration.

45.11 Conclusion

Proposition I.45 closes the application-of-areas construction by showing that the triangle-level machinery can be iterated without introducing a new area axiom. The final Lean proof is not a literal transcription of Euclid’s quadrilateral diagram. Its central operation is a derived extension theorem: construct the next triangular content as an oriented adjacent parallelogram, recover the strict boundary order, paste the geometry, refine the paste in the scissors calculus, and preserve the prescribed angle.

The most delicate issue is orientation. Public I.44 intentionally hides the side on which its target parallelogram lies. I.45 solves this synthetically, not axiomatically: one copied side is placed on the produced base ray, the other on the old parallelogram ray, the ordered internal I.44 construction is reused, and plane separation converts one retained order relation into the second. Once this is done, the rest of the proof becomes a clean additive induction.

The proposition also answers the earlier representation checkpoint in a qualified way. The existing `HilbertTriangulatedFigure` and `HilbertScissorsTerm` layers are sufficient for the constructive content statement of I.45. No stronger polygon datatype is forced by this proof. At the same time, the public theorem treats a triangulated figure as a finite list of nondegenerate triangles and does not internalize nonoverlap, coverage, boundary, or simple-polygon semantics. Those notions remain a separate possible extension of the figure layer rather than a hidden claim of I.45.

With the provisional extension axiom eliminated and the full project build successful, I.45 is therefore closed as a theorem of the existing Hilbert plus scissors architecture.

Chapter 46

Proposition I.46

46.1 Introduction

Proposition I.46 constructs a square on a prescribed segment. It is the last construction immediately before Proposition I.47, and in the present formalization it supplies exactly the square data needed by the Pythagorean theorem.

The proposition is geometrically much cleaner than the equal-content block I.35–I.45. No scissors relation, equicomplementability relation, polygon representation, positivity principle, or numerical area is involved. The proof returns to incidence, order, segment congruence, right angles, parallels, and the developed Euclidean theory of parallelograms.

The final Lean proof has two levels.

- A proposition-local helper, `i46_erect_equal_perpendicular`, constructs the second side of the square at the endpoint of the given segment.
- The main theorem completes the parallelogram, obtains the four equal sides from I.34, and obtains the four right angles from the newly packaged parallelogram-angle interface.

A significant architectural result emerged during the formalization. The initial proof of the adjacent-right-angle step contained a long SameSide/OppositeSide and transversal normalization argument. That argument was not specific to squares. It was factored into reusable parallelogram lemmas in `HilbertInterface`. Proposition I.46 now consumes the general theorem

```
parallelogram_adjacent_right_angle
```

instead of carrying the I.29 normalization locally.

The final source contains no proposition-local geometric axiom and no `sorry`.

46.2 The Classical Configuration

Let AB be the given segment. The final square is written in the Lean vertex order

A, B, C, D .

Thus AB is the given side, AD is the erected perpendicular side, and the constructed quadrilateral is

$$ABCD.$$

The classical construction may be read as follows.

1. At A , erect a perpendicular to AB .
2. On that perpendicular choose D so that

$$AD \cong AB.$$

3. Through D draw the parallel to AB , and through B draw the parallel to AD . Their intersection is C .
4. The quadrilateral $ABCD$ is then a parallelogram.
5. Proposition I.34 makes the opposite sides congruent; since $AD \cong AB$, all four sides are congruent.
6. Proposition I.29 transfers the right-angle information to the adjacent vertex, and I.34 transfers rightness to the opposite angles.

Figure 46.1 records only this classical construction. The formal witnesses needed to implement the endpoint perpendicular are kept out of the classical diagram and are displayed separately in Figure 46.2 below.

46.3 Classical Proof

Euclid's proof uses the standard Book I construction chain.

First apply I.11 to erect a perpendicular at the endpoint A of the given segment AB . Then use I.3 to cut off on that perpendicular a segment AD equal to AB . Through D draw the parallel to AB , and through B draw the parallel to AD ; this is the I.31 parallel construction.

The resulting quadrilateral is a parallelogram. By I.34 its opposite sides are equal:

$$AB \cong CD, \quad BC \cong DA.$$

Since construction gave

$$DA \cong AB,$$

transitivity yields equality of all four sides.

It remains to show that all four angles are right. The given angle at A is right by construction. Since one side of the parallelogram is a transversal of the opposite parallel sides, I.29 gives the appropriate supplementary or alternate-angle relation at the adjacent vertex. Because the angle at A is right, the adjacent interior angle is also right. Proposition I.34 then gives the congruence of opposite angles, so the two remaining angles are right.

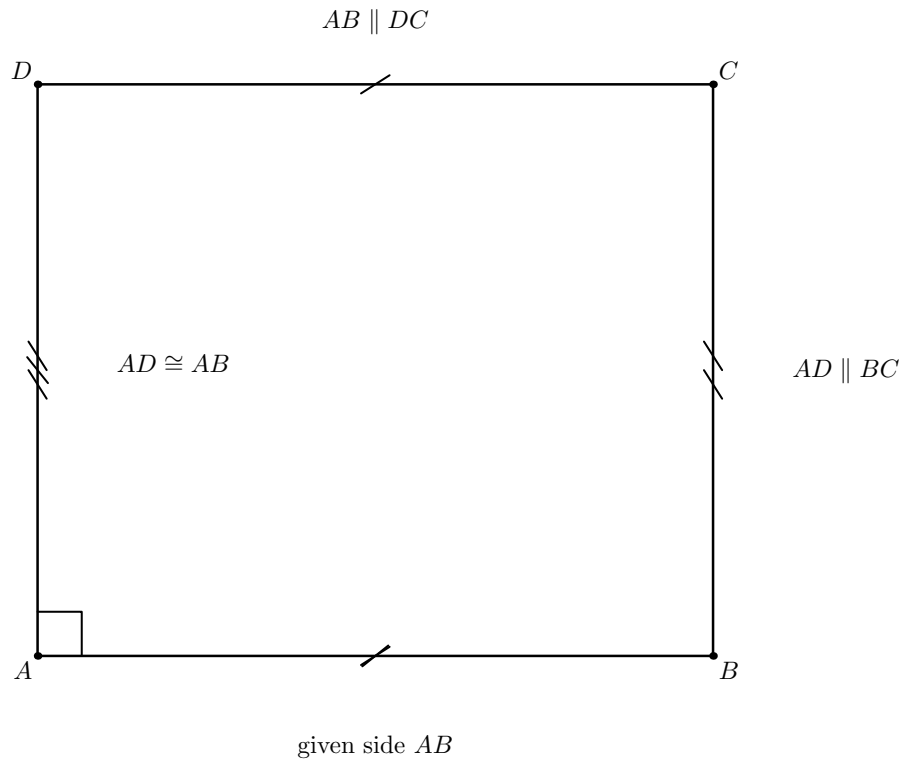


Figure 46.1: The classical I.46 configuration. The prescribed side is AB ; at A one erects the equal perpendicular side AD , and the parallels through D and B meet at C . Thus $ABCD$ is the parallelogram from which I.34 and I.29 complete the proof that all four sides are equal and all four angles are right. No formal endpoint witnesses are shown here.

The historical dependency may therefore be summarized as

$$\begin{aligned}
 &AB \text{ given} \\
 &\Downarrow \text{ I.11, I.3} \\
 &AD \perp AB, \quad AD \cong AB \\
 &\Downarrow \text{ I.31} \\
 &ABCD \text{ parallelogram} \\
 &\Downarrow \text{ I.34} \\
 &AB \cong BC \cong CD \cong DA \\
 &\Downarrow \text{ I.29, I.34} \\
 &\text{all four angles right} \\
 &\Downarrow \\
 &ABCD \text{ square.}
 \end{aligned}$$

46.4 The Square Predicate and the Representation Choice

Euclid defines a square as an equilateral and right-angled quadrilateral. The Lean development records slightly more structure: the four vertices must also form a parallelogram.

In documentation-friendly ASCII notation, the definition is

```
def IsSquare
  (A B C D : Geo.Point) : Prop :=
  IsParallelogram Geo A B C D /\
  Geo.Congruent A B B C /\
```

```

Geo.Congruent B C C D /\
Geo.Congruent C D D A /\
HilbertRightAngle Geo D A B /\
HilbertRightAngle Geo A B C /\
HilbertRightAngle Geo B C D /\
HilbertRightAngle Geo C D A

```

This is a representation strengthening, not an additional geometric claim proved later. The construction already produces a parallelogram, so the extra clause records the incidence structure that distinguishes the intended quadrilateral from a crossed four-point configuration.

For Proposition I.46 this choice is useful in two ways.

- Opposite-side and opposite-angle theorems can be applied directly.
- Downstream propositions can consume a square together with its parallelogram structure without reconstructing the boundary order from the four metric and angular clauses.

The representation remains entirely synthetic. No coordinates, side lengths as numbers, or area values are introduced.

46.5 Proof Architecture of the Reconstruction

The current Lean proof is best read as five conceptual stages.

46.5.1 Step 1: erect an equal perpendicular side

The local theorem

```
i46_erect_equal_perpendicular
```

takes distinct points A, B and constructs a point D such that

$$A, B, D \text{ are noncollinear, } \quad \angle DAB \text{ is right, } \quad AD \cong AB.$$

The available right-angle constructor is formulated at an interior point of a straight line. Therefore the proof first extends BA beyond A :

$$B - A - F.$$

It then applies the nondegenerate right-angle existence theorem to obtain a point X off the carrier BA . Hilbert III.1, represented by `segment_construction`, lays off the length AB on the ray AX , producing D .

The last part of the helper is representational. Since D and X lie on the same ray from A , the angle with arm AX is transported to the angle with arm AD . The unoriented angle calculus then exchanges the arms so that the output has exactly the orientation required by the main theorem:

$$\text{HilbertRightAngle}(D, A, B).$$

Figure 46.2 separates the two formal states of this helper. This separation is important: the proof establishes that D lies on the same ray from A as X , but it does not establish an order relation such as $A - X - D$ or $A - D - X$. The first panel therefore shows the direction point X , while the second panel shows the laid-off point D on the chosen ray without placing X and D simultaneously.

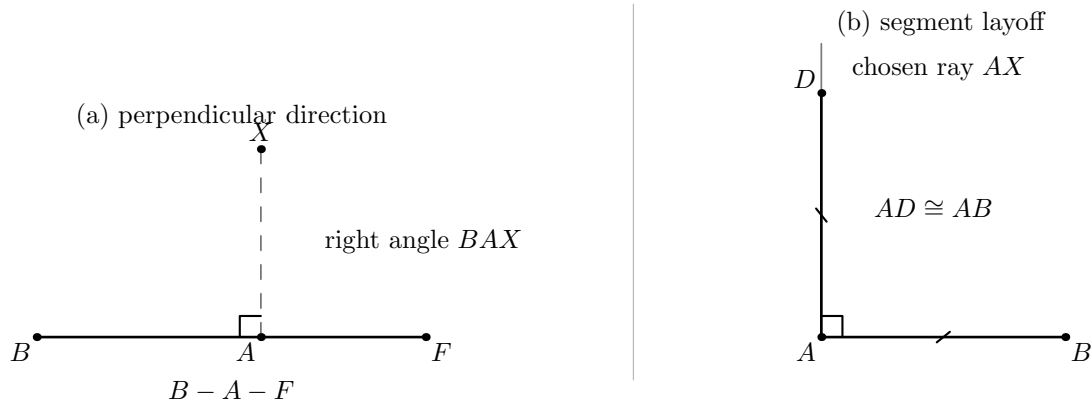


Figure 46.2: The formal implementation of the endpoint construction. In (a) the segment BA is extended to $B - A - F$, so the developed I.11 theorem can construct a nondegenerate right angle BAX . In (b) Hilbert III.1 lays off the length AB on that chosen ray, producing D with $AD \cong AB$. The panels deliberately do not impose any betweenness relation between X and D ; the Lean proof uses only the `SameRay` relation.

46.5.2 Step 2: complete the parallelogram

From noncollinearity of A, B, D , the theorem

```
hilbert_parallelogram_fourth_vertex_exists
```

constructs the fourth point C and returns

$$\text{IsParallelogram}(A, B, C, D).$$

This theorem packages the two parallel constructions and their intersection. The public I.46 proof does not replay carrier uniqueness or parallel transitivity at this point.

46.5.3 Step 3: prove all four sides congruent

The main theorem invokes Proposition I.34:

```
euclid_proposition_34
```

and extracts

```
OppositeSidesCongruent Geo A B C D
```

Together with the constructed equality

$$AD \cong AB,$$

ordinary congruence symmetry and transitivity produce

$$AB \cong BC, \quad BC \cong CD, \quad CD \cong DA.$$

No new segment theorem is introduced here. The proof is a short calculation in the established congruence API.

46.5.4 Step 4: propagate the right angle to the adjacent vertex

This was the main refactoring point of I.46.

The final proposition now calls

```
parallelogram_adjacent_right_angle
```

with the parallelogram $ABCD$ and the right angle DAB . The result is

$$\angle ABC \text{ right.}$$

The theorem is backed by the more geometric interface lemma

```
parallelogram_adjacent_exterior_angle_congruent
```

which extends the side CB through B to a point F :

$$F - B - C,$$

and proves

$$\angle DAB \cong \angle FBA.$$

Its proof is the exact I.29 configuration-normalization that was initially local to Proposition I.46. The line AB is fixed as a transversal. The parallel sides of the parallelogram provide SameSide data for C, D . Extending CB produces F on the opposite side of the transversal. The parallel relation is transported from BC to BF , and Hilbert's Euclidean alternate-angle theorem gives the required exterior-angle congruence.

Figure 46.3 displays this reusable configuration.

46.5.5 Step 5: obtain the two opposite right angles

I.34 also returns

```
OppositeAnglesCongruent Geo A B C D
```

The established right angle DAB is transported to the opposite angle BCD . The newly established right angle ABC is transported to the opposite angle CDA . The helper

```
parallelogram_vertices_noncollinear
```

supplies all four nondegeneracy conditions required by the angle transport theorems.

The eight clauses of `IsSquare` are then assembled directly.

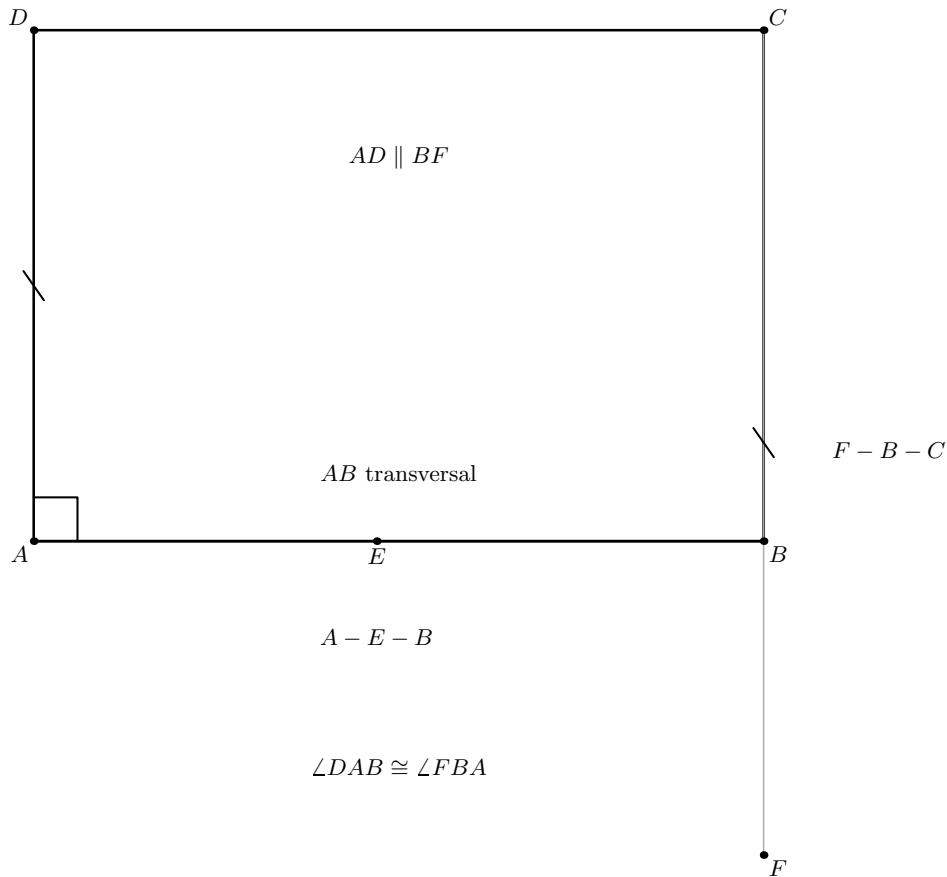


Figure 46.3: The reusable I.29 configuration consumed by Proposition I.46. The side CB is extended through B to F , so $F - B - C$, and AB is the transversal of the parallel carriers AD and BF . The auxiliary point E satisfies $A - E - B$ and is the interior normalization point used by the line-level alternate-angle theorem. After SameRay normalization the theorem gives $\angle DAB \cong \angle FBA$. Rightness then transfers to $\angle FBA$ and across the straight extension $F - B - C$ to $\angle ABC$.

46.6 Lean Formalization

The current proposition imports exactly the two earlier Euclidean propositions used explicitly by the source:

```
import CGJteamLab.Proposition11
import CGJteamLab.Proposition34
```

Earlier parallel and Book Zero infrastructure enters transitively through those modules and through HilbertInterface.

46.6.1 The local endpoint construction

In ASCII notation, the proposition-local helper has the signature

```
theorem i46_erect_equal_perpendicular
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B : Geo.Point)
```

```
(hAB : Ne A B) :
exists D : Geo.Point,
  Not (Collinear Geo A B D) /\
  HilbertRightAngle Geo D A B /\
  Geo.Congruent A D A B
```

This theorem is local because its packaging is tailored to the exact input required by the square construction: an equal perpendicular side based at the first endpoint of the given segment.

46.6.2 The public proposition

The public theorem is

```
theorem euclid_proposition_46
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B : Geo.Point)
  (hAB : Ne A B) :
  exists C D : Geo.Point,
    IsSquare Geo A B C D
```

The proof first constructs D , then constructs C , obtains I.34 once, and uses the resulting side and angle packages twice. After the interface refactor, no SameSide/OppositeSide argument remains in the public proposition.

46.6.3 The downstream side package

For Proposition I.47 the file exposes

```
theorem euclid_proposition_46_side
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D : Geo.Point)
  (hSquare : IsSquare Geo A B C D) :
  Geo.Congruent A B B C /\
  Geo.Congruent A B C D /\
  Geo.Congruent A B D A
```

This theorem does not add geometry. It packages the three side congruences in the orientation used by the Pythagorean construction.

46.6.4 Reusable helpers promoted to the interface

The formalization of I.46 produced four reusable declarations:

```
parallel_first_not_collinear
parallelogram_vertices_noncollinear
parallelogram_adjacent_exterior_angle_congruent
parallelogram_adjacent_right_angle
```

The first two are neutral consequences of strict parallelism and order. The last two are Euclidean:

they use the direction

parallel lines $\implies$ congruent alternate angles,

which is the Hilbert Group IV side of Theorem 30.

This extraction is an architectural improvement rather than a change in the mathematics of I.46. The proposition now consumes a general parallelogram theory in the same way that I.34 and later propositions consume the previously developed opposite-side and opposite-angle API.

46.7 Relationship with Earlier Propositions

46.7.1 Proposition I.11

The first local helper is the endpoint form of the perpendicular construction. The project theorem used internally is

```
hilbert_right_angle_exists_nondegenerate
```

which corresponds to the developed Book Zero/Hilbert form of I.11.

Because its hypotheses use a point strictly between two collinear points, the proof first produces BA beyond A . This is hidden in the ordinary Euclidean instruction “draw a perpendicular at A ” but explicit in the Lean data.

46.7.2 Proposition I.3 and Hilbert III.1

Euclid uses I.3 to cut off

$$AD = AB.$$

The reconstruction performs the same geometric operation through Hilbert’s segment-construction axiom/theorem:

```
HilbertCongruence.segment_construction
```

The point D is not obtained merely as an arbitrary point on the perpendicular line. It is constructed on the chosen ray, which preserves the orientation needed by the subsequent angle transport.

46.7.3 Proposition I.29

I.29 is the unique visibly Euclidean angle step of the proposition. In the interface it appears through the theorem converting parallelism into alternate angle congruence.

The important formal point is that the final I.46 file no longer performs the transversal normalization itself. That work has been generalized to the two interface theorems

```
parallelogram_adjacent_exterior_angle_congruent
parallelogram_adjacent_right_angle
```

46.7.4 Proposition I.31

The classical instruction “draw parallels through D and B ” is consumed through

```
hilbert_parallelogram_fourth_vertex_exists
```

This theorem packages the existence of the fourth vertex and returns the `IsParallelogram` structure directly.

46.7.5 Proposition I.34

I.34 is the principal direct proposition dependency. One call returns both the opposite-side and opposite-angle data. I.46 uses the first component to make the quadrilateral equilateral and the second component to transport the two already known right angles to the opposite vertices.

46.7.6 Proposition I.47

The construction is designed for immediate reuse by I.47. The helper

```
euclid_proposition_46_side
```

exposes the side congruences of a square in a convenient orientation. No additional square theory needs to be reconstructed in the next proposition.

46.8 Hilbert and Book Zero Components Used

46.8.1 Betweenness and extension

The endpoint perpendicular construction requires an explicit produced segment

$$B - A - F.$$

The interface lemma `ExtendSegmentBeyond` supplies this witness. The strict order is not decorative: it is what makes the nondegenerate right-angle constructor applicable at A .

The adjacent-angle theorem uses another produced segment

$$F - B - C.$$

That strict relation is what allows a right exterior angle at B to be transported across the straight line to the interior angle ABC .

46.8.2 SameRay and angle invariance

After the perpendicular direction AX has been constructed, Hilbert III.1 places D on the same ray from A . The equality

$$\angle BAX = \angle BAD$$

is therefore a ray-invariance statement, not a metric calculation.

The proof then uses the project’s unoriented angle calculus to reverse the two arms and obtain the orientation DAB expected by the square predicate.

46.8.3 Parallel carriers and plane separation

The extracted exterior-angle theorem uses the developed carrier-line theory. Its proof fixes the line AB as a transversal, places the relevant endpoints on the required sides of that line, extends the second parallel carrier, and transports the parallel relation to the produced point.

The public I.46 theorem does not expose these witnesses, but they are genuine logical data behind the convenient theorem `parallelogram_adjacent_right_angle`.

46.8.4 Right-angle transport

Rightness is represented synthetically by `HilbertRightAngle`; no numerical angle measure is present.

The theorem

```
hilbert_right_angle_transport
```

is used repeatedly. Once two nondegenerate angles are congruent, rightness of one angle can be transported to the other. This is how I.34's opposite-angle congruences become right-angle statements.

46.9 Formalization Notes

46.9.1 Geometry

The hidden configuration data are concentrated in two places.

For the local endpoint construction Lean must record:

- $B - A - F$ after extending the given carrier;
- noncollinearity of the perpendicular point;
- the `SameRay` relation placing D on the perpendicular ray;
- the noncollinearity permutations required by angle transport.

For the reusable adjacent-angle theorem the interface proof records:

- the carrier of AB ;
- `SameSide` data for the two relevant parallelogram vertices;
- the extension $F - B - C$;
- `OppositeSide` data after the extension;
- transport of the parallel relation from BC to BF ;
- an interior point on the transversal used by the line-level form of Hilbert Theorem 30;
- `SameRay` normalization of the resulting angle arms.

These facts are exactly the kind of information suppressed by the classical diagram but required by a formal synthetic proof.

46.9.2 Content

There is no content layer in Proposition I.46.

This is important because I.46 follows immediately after the long area and equicomplementability development I.35–I.45. The proposition uses none of the following:

```
HilbertScissorsEq
HilbertScissorsEquicomplementable
equicomplementable_add
content comparison
proper-part positivity
```

No cancellation, no de Zolt principle, no multiplication of segment lengths, and no numerical area is involved. The proof is ordinary Euclidean construction geometry.

46.9.3 Representation

The main representation decision is the definition

```
IsSquare
```

whose stored data include an explicit

```
IsParallelogram
```

clause in addition to the four equal-side and four right-angle conditions.

This is stronger as stored data than Euclid’s verbal Definition I.22, but it does not strengthen the constructed conclusion beyond the actual geometry: the proof has already built the figure as a parallelogram. The clause simply retains useful incidence and parallel structure for downstream use.

46.9.4 Axiomatic status

Proposition I.46 is Euclidean.

Hilbert Group IV enters in the developed theory of parallels. In particular, the theorem producing the adjacent-angle congruence uses the Euclidean direction of Hilbert Theorem 30:

parallel $\implies$ equal alternate angles.

The fourth-vertex construction also relies on the developed Euclidean parallel infrastructure needed to identify the appropriate intersection.

The endpoint perpendicular and segment-layoff construction is neutral: it uses incidence, order, and congruence, but not the uniqueness of parallels.

No continuity axiom is used. No Archimedean axiom is used. No new geometric axiom is introduced. No content axiom is used.

46.10 Source Audit

The source audit separates the historical proof from the formal architecture.

Euclid, together with the standard Heath/Joyce and Wilkins presentations, controls the classical dependency chain: erect the perpendicular, cut off the equal side, draw the two parallels, use I.34 for opposite sides, and use I.29 plus I.34 for the right angles.

Hilbert controls the logical placement of the parallel-angle step. The direction from equal alternate angles to parallel lines belongs to the neutral development, while the converse direction from parallel lines to equal alternate angles uses Group IV. Proposition I.46 therefore crosses the Euclidean boundary precisely where the ordinary construction expects it.

Borsuk–Szmielew and Greenberg are control sources for the developed theory of parallel carriers, parallelograms, and plane separation. They support placing the adjacent-angle result in the reusable Euclidean interface rather than leaving it as a proposition-local fact.

The Beeson proof corpus is useful here mainly as a check on hidden configuration conditions: endpoint extension, ray orientation, explicit transversal data, and noncollinearity. The final proof is not organized around Tarski-style tactics.

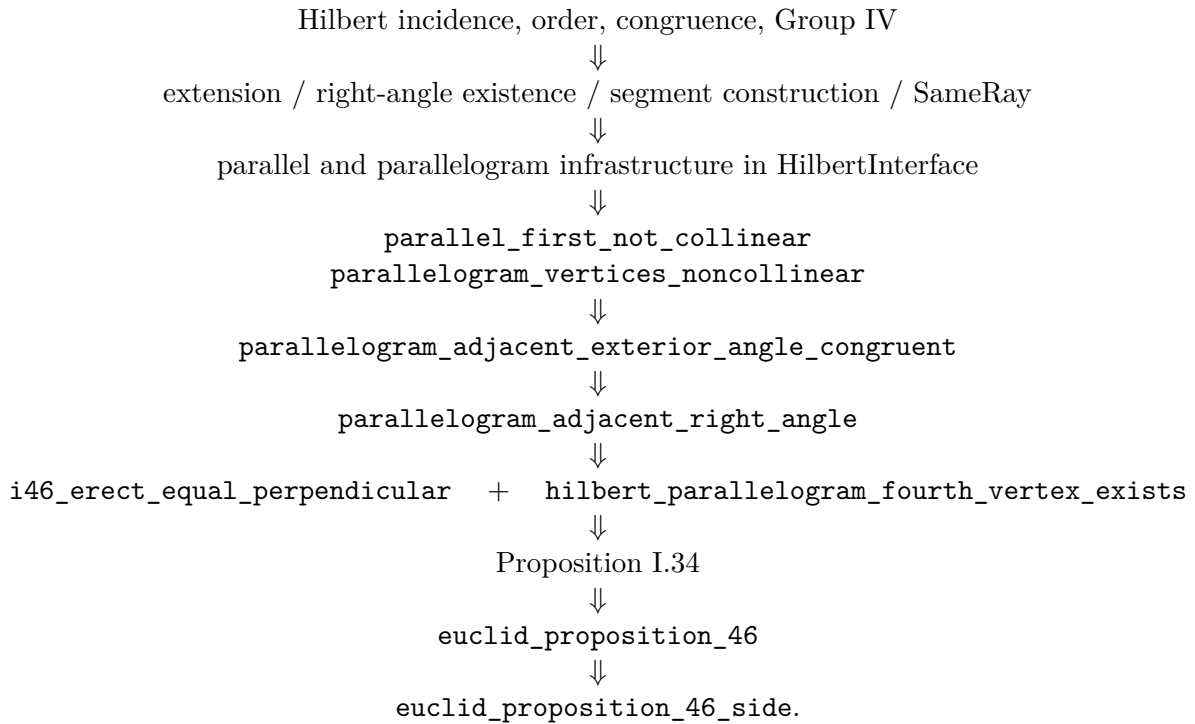
Hartshorne adds no special content issue at I.46. In contrast with I.35–I.45, there is no equal-content semantic boundary to control.

The three-way audit requested for propositions from I.35 onward is therefore particularly simple:

- **Geometry:** nontrivial; perpendicular construction, segment layoff, parallel completion, I.29, and I.34.
- **Content:** absent.
- **Representation:** the explicit `IsParallelogram` clause in `IsSquare`, plus the packaging theorem prepared for I.47.

46.11 Dependency Summary

The formal dependency chain is



The classical and Lean DAGs can be compared succinctly as

Euclid: I.11 + I.3 $\rightarrow$ I.31 $\rightarrow$ I.34 $\rightarrow$ I.29/I.34,
 Lean: endpoint extension + segment construction $\rightarrow$ fourth-vertex theorem
 $\rightarrow$ I.34 $\rightarrow$ interface adjacent-right-angle theorem.

New geometric axiom:	no
New content axiom:	no
New proposition-local helper:	yes
New reusable helper:	yes
New Book Zero theorem:	no
New Interface theorem:	yes, four
Numerical area semantics:	no
Continuity used:	no
Archimedean axiom used:	no
Hilbert Group IV used:	yes
Public theorem compiles:	yes
Full lake build:	clean
Active local axiom:	no
sorry in Proposition46.lean:	no

Diagrammatic audit. The final figure set separates the three geometric roles that were conflated in the previous presentation. Figure 46.1 is the classical Euclidean construction and therefore contains no proposition-local formal witnesses. Figure 46.2 records the two successive states of `i46_erect_equal_perpendicular`; in particular it does not turn the `SameRay` conclusion for X and D into an unsupported betweenness claim. Figure 46.3 records the genuinely geometric configuration hidden behind the reusable adjacent-right-angle interface, including the

normalization point E with $A - E - B$. No additional figure is introduced for the fourth-vertex theorem, the I.34 congruence bookkeeping, or `euclid_proposition_46_side`, since those steps do not create a new geometric state needed by the proposition-level argument.

46.12 Conclusion

Proposition I.46 is a compact construction theorem whose formalization exposes two useful structural facts.

First, the endpoint version of the perpendicular construction is not definitionally trivial in the Hilbert API. It requires an extension witness, a nondegenerate perpendicular direction, a segment layoff on a chosen ray, and angle transport. These details are packaged locally in `i46_erect_equal_perpendicular`.

Second, the ordinary statement that an adjacent angle of a parallelogram is right when one angle is right is not a square-specific fact. Its proof is a general application of I.29 together with plane-separation and ray normalization. Extracting that argument into `HilbertInterface` removes a long low-level configuration proof from I.46 and strengthens the reusable Euclidean parallelogram API.

After that extraction the public proposition has the expected classical shape: construct one equal perpendicular side, complete the parallelogram, use I.34 for opposite sides and angles, and assemble the square. The resulting `IsSquare` object is immediately prepared for Proposition I.47 through `euclid_proposition_46_side`.

Chapter 47

Proposition I.47

47.1 Introduction

Proposition I.47 is Euclid’s theorem of Pythagoras. In the classical statement, if ABC is right-angled at A , then the square on the hypotenuse BC is equal to the sum of the squares on the legs AB and AC .

The word “equal” is crucial. The present formalization does not replace Euclid’s statement by the modern numerical equation

$$a^2 + b^2 = c^2.$$

Instead, the theorem remains inside the synthetic content calculus developed for I.35–I.45. The square on BC is represented by a parallelogram term, the two leg squares are represented by two parallelogram terms, and the conclusion is

`HilbertScissorsEquicomplementable`

between the hypotenuse square and the formal sum of the two leg squares.

This distinction is not cosmetic. Hartshorne explicitly presents I.47 as an equal-content theorem, and Hilbert’s discussion of the Euclidean theory of polygons formulates Pythagoras as equicomplementability of the three squares. The formal theorem therefore stays close to the synthetic meaning of Book I rather than importing a numerical area function.

The current Lean source is substantially longer than the classical proof. The reason is geometric placement. Euclid reads from the diagram that the three squares are constructed outward, that the perpendicular from A to BC lands inside BC , that the cut reaches the opposite side of the square, and that the auxiliary diagonal intersects the cut internally. In Lean all of these are strict order or separation statements that must be proved.

The final source is organized around four large layers:

1. construct the three outward squares;
2. construct and order the internal cut of the hypotenuse square;
3. prove the angle and noncollinearity data needed for two SAS congruences;
4. convert the congruent triangles into equal-content statements and assemble the scissors decomposition.

No proposition-local geometric axiom remains in the file, and the complete project build is clean.

47.2 The Classical Configuration

Let ABC be a nondegenerate triangle with a right angle at A . The Lean labelling is fixed as follows:

$BCED$ square on the hypotenuse BC ,
 $ABFG$ square on the leg AB ,
 $ACKH$ square on the leg AC .

The square on BC is constructed on the side of the carrier BC opposite A . Likewise the square on AB is constructed on the side opposite C , and the square on AC on the side opposite B . These are not merely visual conventions. The formal theorem `i47_outward_squares` returns the corresponding `HilbertOppositeSide` data.

The standard Euclidean auxiliary line through A is parallel to DB . It meets BC at M and the opposite side DE at L . Figure 47.1 shows this classical configuration and deliberately stops there: the additional point N belongs to the later formal scissors construction, not to Euclid's visible diagram.

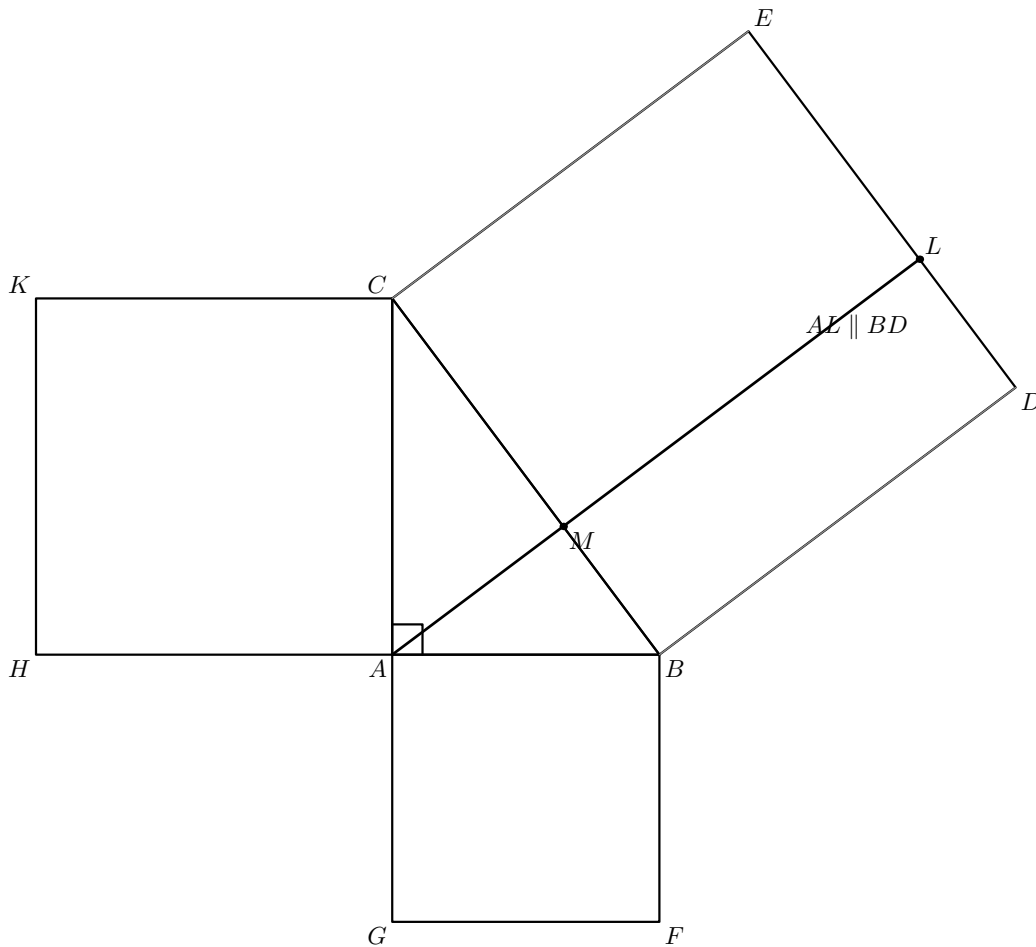


Figure 47.1: The classical configuration of I.47 in the Lean vertex order. The three squares are constructed outward from the right triangle ABC , and the auxiliary line through A is parallel to BD , meeting BC at M and DE at L . No formal crossing point N is shown at this stage.

The formal reconstruction later augments this classical picture with the diagonal DC of the square on the hypotenuse and its intersection N with the cut. The strict order information required by the exact dissection is

$$B - M - C, \quad D - L - E, \quad D - N - C, \quad M - N - L.$$

The two pieces of the square on BC are the parallelograms

$$LDBM \quad \text{and} \quad CELM.$$

They are later rotated in the scissors representation to the vertex orders needed by the final formal sum.

47.3 Classical Proof

Euclid's classical proof has a short dependency graph. In the usual labelling, the essential cited propositions are I.4, I.14, I.31, I.41, and I.46.

First construct the three squares by I.46. Through A draw the line parallel to BD and extend it through the square on BC ; this is the I.31 step. Because the relevant angles at A are right and the two leg squares are constructed externally, I.14 gives the straight-line extensions required to identify the side parallels of the leg squares with the opposite legs of the triangle.

At B , Euclid compares the triangles

$$\triangle BAD \quad \text{and} \quad \triangle BFC.$$

The square data give

$$BA \cong BF, \quad BD \cong BC.$$

The angle step gives

$$\angle ABD \cong \angle FBC.$$

Hence I.4 gives

$$\triangle BAD \cong \triangle BFC.$$

Now I.41 is applied twice. The rectangle $LDBM$ is double a triangle on the base BD between the parallels BD and AL , while the square $ABFG$ is double a congruent triangle on the base BF . Consequently

$$LDBM \approx ABFG,$$

where $\approx$ denotes equal content in the present documentation.

The symmetric argument at C gives

$$MCEL \approx ACKH.$$

Finally the two rectangles make up the whole square $BCED$, and Common Notion 2, expressed in the formal scissors layer by compatibility with addition, yields

$$BCED \approx ABFG + ACKH.$$

The classical dependency graph can be summarized as

$$\begin{aligned} & \text{I.46: three squares} \Downarrow \\ & \text{I.14 + I.31: straight extensions and parallels} \Downarrow \\ & \text{I.4: two congruent-triangle bridges} \Downarrow \\ & \text{I.41: rectangle = double triangle = square} \Downarrow \\ & \text{add the two equal-content relations} \Downarrow \\ & \text{I.47.} \end{aligned}$$

This proof is synthetic throughout. It does not assign numbers to segment lengths or to the contents of the squares.

47.4 Equal Content, Not Numerical Area

Proposition I.47 belongs to the area/content block beginning at I.35. The formal statement must therefore preserve the distinction between three notions:

- geometric congruence of segments and triangles;
- scissors equality, represented by `HilbertScissorsEq`;
- equal content, represented by `HilbertScissorsEqicomplementable`.

The theorem does not assert that the square on the hypotenuse is directly `HilbertScissorsEq` to the sum of the two leg squares. The final relation is equicomplementability. This is exactly the level required by Euclid's use of equality of plane figures and by Hilbert's reconstruction of that equality through cutting and adding pieces.

A stronger scissors equality does occur internally. The theorem `i47_square_split` proves that the square on BC is actually dissected into the two parallelogram terms arising from the cut:

```
theorem i47_square_split
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (B C D E L M N : Geo.Point)
  (hParallelogram : IsParallelogram Geo B C E D)
  (hBMC : Geo.Between B M C)
  (hDLE : Geo.Between D L E)
  (hDNC : Geo.Between D N C)
  (hMNL : Geo.Between M N L) :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo B C E D)
    (hilbertParallelogramTerm Geo M L D B +
     hilbertParallelogramTerm Geo C E L M)
```

Thus the final argument has a clean separation:

exact dissection of the hypotenuse square $\Downarrow$
 equal-content comparison of each rectangle with one leg square $\Downarrow$
 addition and transport $\Downarrow$
 Pythagorean equal-content conclusion.

No numerical area function is introduced anywhere in this chain.

47.5 Proof Architecture of the Reconstruction

The Lean proof is best read as a sequence of geometric packages rather than as a line-by-line translation of Euclid's text.

47.5.1 Step 1: construct the three outward squares

The first substantial theorem is

```
i47_outward_squares
```

It chooses the three side carriers and constructs squares on BC , AB , and AC with explicit side control. In ASCII notation, its essential output is

```
exists lBC lAB lAC : Geo.Line,
exists D E F G H K : Geo.Point,
  IsSquare Geo B C E D /\
  IsSquare Geo A B F G /\
  IsSquare Geo A C K H /\
  HilbertOppositeSide Geo D A lBC /\
  HilbertOppositeSide Geo E A lBC /\
  HilbertOppositeSide Geo G C lAB /\
  HilbertOppositeSide Geo F C lAB /\
  HilbertOppositeSide Geo H B lAC /\
  HilbertOppositeSide Geo K B lAC
```

This is stronger positional information than the public I.46 theorem alone returns. The local reusable constructor

```
hilbert_square_exists_opposite_side
```

rebuilds the I.46 construction with a prescribed choice of side. The key lower-level helper

```
hilbert_erect_equal_perpendicular_opposite_side
```

constructs the equal perpendicular side on the chosen side of the carrier.

These helpers were introduced for I.47, but their mathematical content is not specific to the Pythagorean theorem. They are plausible candidates for later promotion if another proposition requires side-controlled square construction.

47.5.2 Step 2: recover the straight extensions at the right-angle vertex

The outward placement of the two leg squares forces their inner sides to be opposite right-angle rays at A . The local theorem

```
i47_aux_opposite_right_rays_straight
```

packages the general fact that two right-angle rays based at the same point, on opposite sides of the supporting line, are opposite rays.

Applied to the two leg squares it yields

$$G - A - C, \quad H - A - B.$$

These strict betweenness relations are then converted into the parallels

$$AC \parallel BF, \quad AB \parallel CK.$$

The corresponding helpers are

```
i47_aux_square_AB_extension
i47_aux_square_AC_extension
i47_aux_square_AB_parallel_AC
i47_aux_square_AC_parallel_AB
```

Classically this is the short I.14 stage. Formally it is a plane-separation and ray-orientation argument because the straight extensions must be proved rather than read from the figure.

47.5.3 Step 3: prepare the two SAS congruences

The two triangle comparisons need the angle congruences

$$\angle ABD \cong \angle FBC, \quad \angle ACE \cong \angle KCB.$$

They are established by

```
i47_aux_angle_ABD_FBC
i47_aux_angle_ACE_KCB
```

using congruence of all right angles together with angle addition and the unoriented angle representation.

Before SAS can be applied, Lean also needs four explicit noncollinearity conditions. Two are nontrivial:

```
i47_aux_noncollinear_BAD
i47_aux_noncollinear_CAE
```

They rule out degenerate placements by showing that a nondegenerate triangle cannot have two right angles. That contradiction is packaged in

```
i47_aux_two_right_angles_impossible
```

and ultimately uses I.17.

The remaining two noncollinearities follow from the already established strict parallels and are bundled by

```
i47_aux_parallel_noncollinearities
```

This layer produces exactly the geometric data consumed by the final two SAS calls. It does not yet use scissors content.

47.5.4 Step 4: construct the internal cut of the hypotenuse square

This is the longest new geometric component of I.47. Its final package is `i47_cut_core`.

The perpendicular foot

From A drop a perpendicular to the carrier BC . The helper

i47_aux_perpendicular_foot_on_BC

returns a foot M on the carrier and the corresponding right-angle data. At this point it does *not* assume that M lies between B and C .

The strict order

$$B - M - C$$

is proved separately by

i47_aux_perpendicular_foot_between_BC

The proof excludes the two exterior orders with the exterior-angle theorem I.16. The auxiliary angle lemmas

i47_aux_angle_ABC_less_right_BAC

i47_aux_angle_ACB_less_right_BAC

show that both acute angles of the right triangle are strictly smaller than the right angle at A .

This is a good example of hidden diagram data becoming a real theorem. The ordinary picture makes the altitude foot look internal; the formal proof must exclude both exterior collinear possibilities.

Figure 47.2 records exactly the geometric state reached after this order argument.

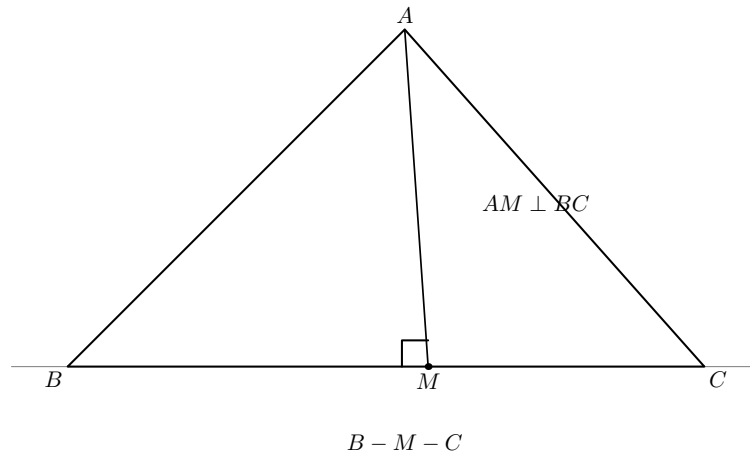


Figure 47.2: The first formal cut state. The perpendicular from A to the carrier BC has foot M , and the nontrivial order theorem proves $B - M - C$. The picture records the proved state; it does not serve as evidence that the foot is internal.

Make the cut parallel to the square side

Once $B - M - C$ is known, the perpendicular through M is shown parallel to the square side CE :

$$MA \parallel CE.$$

The helper

i47_aux_perpendicular_foot_parallel_CE

chooses a point between M and C , observes that the relevant angles are both right, and applies the alternate-angle criterion with the required OppositeSide data.

Construct the upper cut point

The point L is obtained by completing

$$C, E, M$$

to the parallelogram

$$CELM.$$

This is the role of

```
i47_aux_construct_L_parallelogram
```

Since both MA and ML are parallel to CE and pass through M , Euclidean uniqueness of parallels identifies their carriers. Thus

$$A, M, L \text{ are collinear.}$$

This is packaged by

```
i47_aux_cut_line_collinear
```

A second carrier-identification argument proves that L lies on the upper side carrier DE of the square:

```
i47_aux_L_on_DE
```

At this stage the proof has collinearity of D, L, E , but not yet the strict order $D - L - E$.

Recognize the left parallelogram and transfer order

Using the two known carriers and the square parallels, the proof constructs

$$LDBM$$

as a parallelogram. The theorem

```
i47_aux_left_cut_parallelogram
```

contains one delicate nondegeneracy argument: it must prove $L \neq D$ before strict parallelism can be transported. The proof uses

$$CM < CB$$

from $B - M - C$ and transports this proper-part relation through opposite-side congruences of the two parallelograms.

Now the lower-side order can be transferred to the upper side. Opposite-side congruences give

$$BM \cong DL, \quad BC \cong DE, \quad MC \cong LE.$$

Hilbert's three-point transfer theorem then yields

$$D - L - E.$$

This step is isolated as

i47_aux_upper_cut_between

At this point the cut has reached the opposite side and the two pieces are genuine parallelograms, but the diagonal-crossing witness N has not yet been introduced. Figure 47.3 keeps that intermediate state separate from the later Pasch argument.

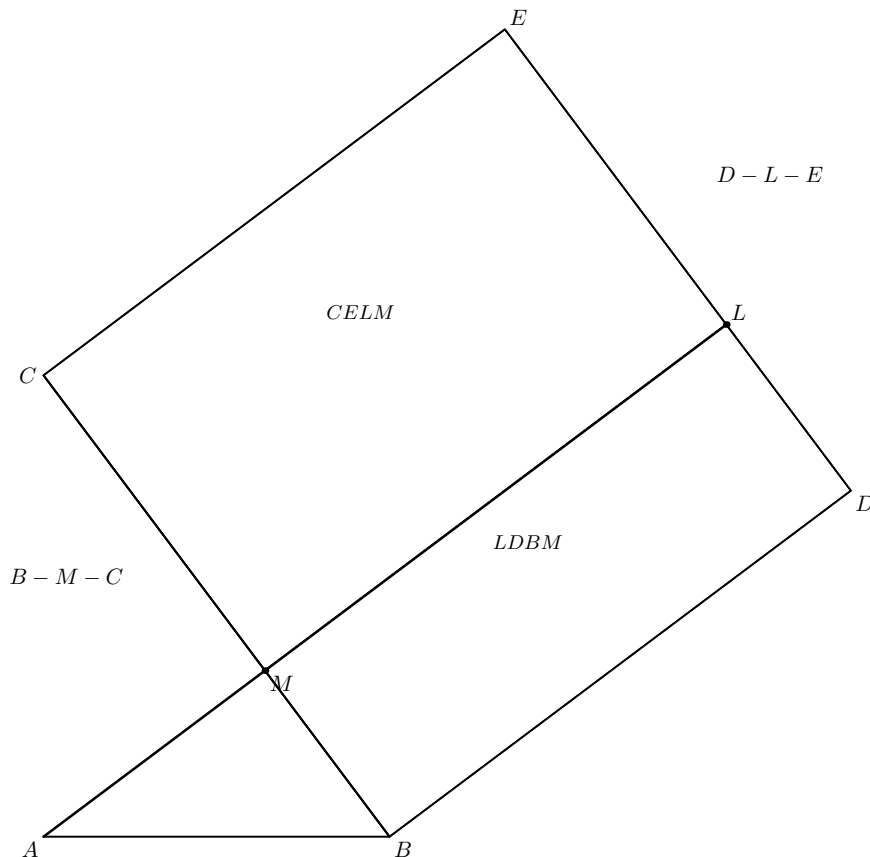


Figure 47.3: The cut after constructing L . The proved orders are $B - M - C$ and $D - L - E$; the two pieces are the parallelograms $LDBM$ and $CELM$, and A, M, L lie on one carrier. The point N and the diagonal DC belong to the next stage.

Construct the crossing point N

The final order witness is the intersection of ML with the diagonal DC . The helper

i47_aux_diagonal_cut_intersection

returns

```
exists N : Geo.Point,
  Geo.Between D N C /\
  Geo.Between M N L
```

The proof applies forced Pasch to triangle CBD . The line ML enters the triangle through the open side CB at M . It cannot meet BD , because $DB \parallel ML$. Pasch therefore forces an intersection with the open side CD , giving $D - N - C$. A parallel-order theorem then gives $M - N - L$.

Figure 47.4 isolates precisely the new Pasch state rather than repeating the whole cut construction.

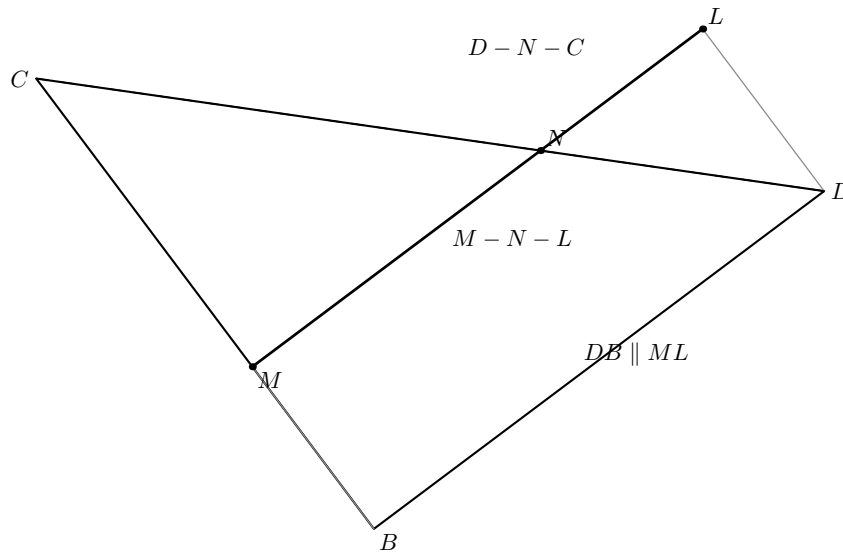


Figure 47.4: The final geometric step inside `i47_cut_core`. In triangle CBD , the line ML enters through CB at M and cannot meet BD because $DB \parallel ML$. Forced Pasch therefore produces N on CD , with $D - N - C$; the parallel-order argument then gives $M - N - L$.

47.5.5 Step 5: package the complete diagram

The theorem

```
i47_diagram
```

is not an axiom. It is now a theorem assembling the three outward squares, the complete cut, the two leg-square parallels, the two angle congruences, and the four noncollinearity conditions.

Its role is architectural. The public theorem can destructure one geometric package and then perform the content argument without interleaving it with further plane-separation proofs.

47.5.6 Step 6: the two SAS bridges

The square-side congruences provide

$$BA \cong BF, \quad BD \cong BC,$$

and the prepared angle theorem gives

$$\angle BAD \cong \angle BFC$$

in the exact orientations required by the project's SAS interface. Hence

$$\triangle BAD \cong \triangle BFC.$$

On the other side,

$$CA \cong CK, \quad CE \cong CB, \quad \angle CAE \cong \angle CKB$$

yield

$$\triangle CAE \cong \triangle CKB.$$

Congruent triangles are immediately converted to `HilbertScissorsEq` by `scissors_congruent`. No area comparison is needed at the SAS stage.

Figure 47.5 shows the two symmetric bridges used in the content proof.

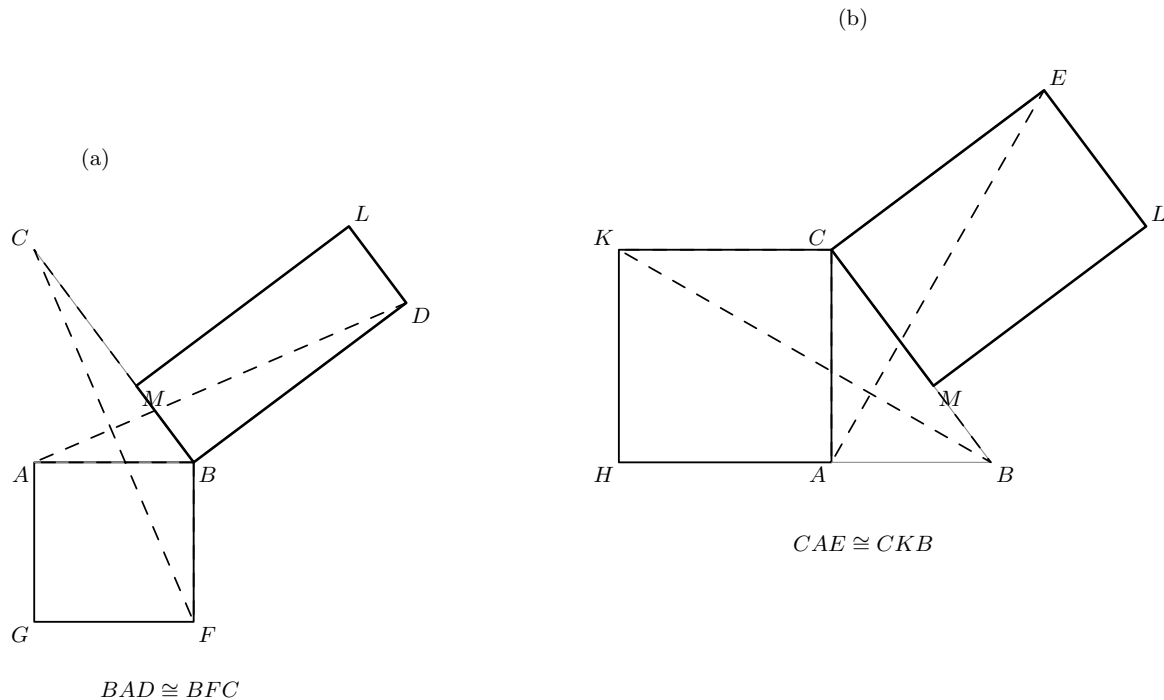


Figure 47.5: The two symmetric content chains. In panel (a), the dashed triangles BAD and BFC are congruent and connect the square $ABFG$ to the cut parallelogram $LDBM$ via I.41. Panel (b) shows the corresponding triangles CAE and CKB , connecting $ACKH$ to $MCEL$.

47.5.7 Step 7: use I.41 to pass from triangles to quadrilaterals

For the left side, I.41 is applied to the parallelogram $LDBM$ with the point A on the parallel cut carrier. It proves that the parallelogram has the content of two copies of triangle ADB .

A second application of I.41 to the square $ABFG$ with the point C proves that the square has the content of two copies of triangle CBF . Since the corresponding triangles are congruent, addition and transitivity yield

$$LDBM \approx ABFG.$$

The same argument gives

$$MCEL \approx ACKH.$$

This is the exact formal counterpart of Euclid's use of I.41. The Lean proof uses no cancellation and no converse-content theorem.

47.5.8 Step 8: dissect the hypotenuse square and add the two relations

The theorem `i47_square_split` cuts the parallelogram term for $BCED$ along the diagonal DC , splits the two resulting triangles at M and L , and then re-triangulates the crossing quadrilateral $DMCL$ using the intersection point N .

The result is the exact scissors equality

$$BCED = (MLDB) + (CELM)$$

at the level of `HilbertScissorsEq`.

Figure 47.6 displays the geometric exchange performed inside `i47_square_split`: first triangulate along DC , split at M and L , and then retriangulate the crossing quadrilateral $DMCL$ along ML before regrouping the pieces into the two cut parallelograms.

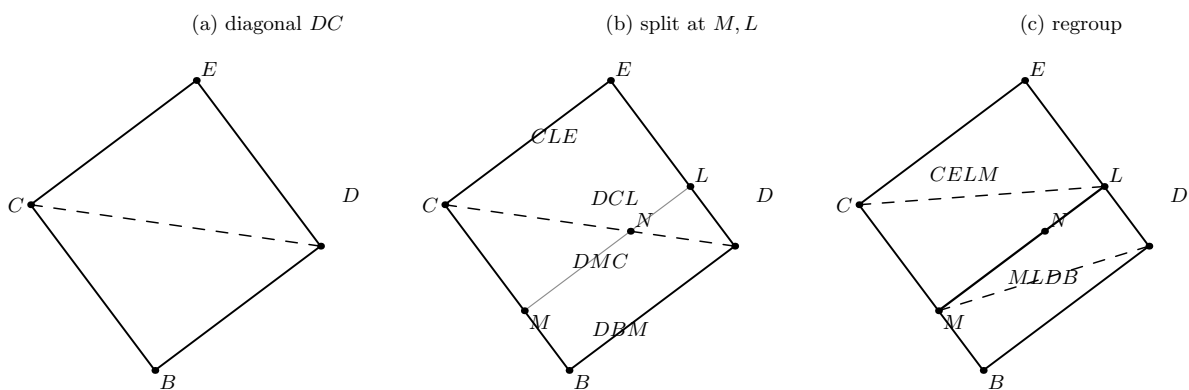


Figure 47.6: The exact dissection used by `i47_square_split`. Panel (a) uses the diagonal DC of the hypotenuse square. Panel (b) splits the two diagonal triangles at M and L ; the central quadrilateral has crossing diagonals DC and ML meeting at N . Panel (c) exchanges the central triangulation to ML and regroupes the four pieces as the parallelograms $MLDB$ and $CELM$.

The local algebraic helper

```
i47_aux_equicomplementable_add
```

combines the two rectangle-to-square equicomplementabilities. The final transport is performed by

```
equicomplementable_transport
```

which carries that content equality across the exact dissection of the hypotenuse square.

This closes the theorem without subtraction, positivity, numerical area, or a new content axiom.

47.6 Lean Formalization

47.6.1 Imports

The current file imports exactly

```
import CGJteamLab.Proposition12
```

```

import CGJteamLab.Proposition14
import CGJteamLab.Proposition17
import CGJteamLab.HilbertRightAngle
import CGJteamLab.Proposition41
import CGJteamLab.Proposition46

```

These imports already show that the formal dependency graph is not identical to Euclid's proposition list. I.12 supplies the perpendicular-from-an-external point interface used in the cut construction; I.17 supplies the two-right-angle contradiction; I.41 and I.46 are the two large direct Euclidean dependencies.

47.6.2 The cut-core theorem

In ASCII notation, the central geometric theorem has the form

```

theorem i47_cut_core
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D E : Geo.Point)
  (base : Geo.Line)
  (hABC : Not (Collinear Geo A B C))
  (hRight : HilbertRightAngle Geo B A C)
  (hBbase : HilbertIncidence.OnLine B base)
  (hCbase : HilbertIncidence.OnLine C base)
  (hSquare : IsSquare Geo B C E D)
  (hOppDA : HilbertOppositeSide Geo D A base)
  (hOppEA : HilbertOppositeSide Geo E A base) :
  exists L M N : Geo.Point,
    Geo.Between B M C /\
    Geo.Between D L E /\
    Geo.Between D N C /\
    Geo.Between M N L /\
    IsParallelogram Geo L D B M /\
    IsParallelogram Geo C E L M /\
    Geo.Parallel L A D B /\
    Geo.Parallel M A C E

```

This theorem is the main formal replacement for the positional information that Euclid leaves in the diagram.

47.6.3 The full diagram theorem

The theorem `i47_diagram` assembles the complete configuration. Its output includes:

- the three `IsSquare` objects;
- the four strict betweenness relations $B - M - C$, $D - L - E$, $D - N - C$, $M - N - L$;
- both cut parallelograms;
- $LA \parallel DB$ and $MA \parallel CE$;
- $AC \parallel BF$ and $AB \parallel CK$;

- the two angle congruences needed by SAS;
- the four noncollinearity facts needed by the angle and SAS interfaces.

The theorem deliberately returns more than the public statement. It is an internal geometry package from which the content proof can proceed without reconstructing positions.

47.6.4 The public proposition

The public theorem is

```
theorem euclid_proposition_47
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C : Geo.Point)
  (hABC : Not (Collinear Geo A B C))
  (hRight : HilbertRightAngle Geo B A C) :
  exists D E F G H K : Geo.Point,
    IsSquare Geo B C E D /\
    IsSquare Geo A B F G /\
    IsSquare Geo A C K H /\
    HilbertScissorsEquicomplementable Geo
    (hilbertParallelogramTerm Geo B C E D)
    (hilbertParallelogramTerm Geo A B F G +
     hilbertParallelogramTerm Geo A C K H)
```

The conclusion is therefore exactly the synthetic Pythagorean statement in the current project language: existence of the three squares plus an equicomplementability relation between the hypotenuse square and the formal sum of the leg squares.

47.7 Relationship with Earlier Propositions

47.7.1 Proposition I.4

The two central triangle comparisons are SAS. In Euclid they are explicit applications of I.4; in the Lean reconstruction they are calls to the established Hilbert SAS interface. The substantial local work lies in producing the correctly oriented angle congruences and noncollinearity data before SAS is called.

47.7.2 Proposition I.12

The internal cut begins by dropping a perpendicular from A to the carrier BC . The project imports I.12 because the general theorem

```
hilbert_perpendicular_from_point_exists
```

is the existence mechanism used by `i47_aux_perpendicular_foot_on_BC`.

This differs from Euclid's visible I.47 dependency list. The historical proof simply draws the line through A parallel to BD . The formal reconstruction finds the same carrier by first constructing a perpendicular to BC and then using right-angle and parallel uniqueness.

47.7.3 Proposition I.14

I.14 is the classical straight-line step at A . The formal proof recovers its geometric effect through the stronger local theorem about opposite right-angle rays. The outward placement of the leg squares supplies the required side information, and the result is the strict orders $G - A - C$ and $H - A - B$.

47.7.4 Propositions I.16 and I.17

These propositions enter the hidden-order proof.

I.16 excludes the two exterior locations of the perpendicular foot M on the carrier BC . The proof makes both acute angles of the right triangle strictly smaller than the right angle at A and derives a contradiction in each exterior case.

I.17 supplies the contradiction used when a degenerate collinear placement would force a second right angle in triangle ABC . It is therefore part of the proof that the SAS triangles BAD and CAE are genuine triangles.

47.7.5 Proposition I.31 and Euclidean parallel uniqueness

Euclid draws the cut through A parallel to BD . In the reconstruction the existence and uniqueness of parallel carriers appear through the developed Hilbert Euclidean interface. Carrier equality is used several times:

- to identify AM with ML ;
- to identify EL with the upper square carrier ED ;
- to transport the final cut parallelism to the exact endpoints required by I.41.

This is one of the main places where Group IV is visible in the formal proof.

47.7.6 Proposition I.41

I.41 is the main content theorem consumed by I.47. It is called four times:

- rectangle $LDBM$ versus two copies of triangle ADB ;
- square $ABFG$ versus two copies of triangle CBF ;
- rectangle $MCEL$ versus two copies of triangle ACE ;
- square $ACKH$ versus two copies of triangle BCK .

The two SAS congruences identify the doubled triangle terms, so each rectangle is equicomplementable with the corresponding leg square.

47.7.7 Proposition I.46

I.46 supplies the square predicate and the square-construction geometry. I.47 needs a stronger positional version: the square must be placed on a specified side of its supporting line. The helper `hilbert_square_exists_opposite_side` adapts the I.46 construction to that requirement.

Thus I.47 does not weaken or bypass I.46. It refines its existence result by adding the side choice that Euclid's diagram assumes.

47.8 Hilbert and Book Zero Components Used

47.8.1 Incidence, carriers, and strict parallelism

The reconstruction repeatedly moves between Hilbert incidence lines and the extensional `Geo.PointLine` representation used by strict parallelism. Carrier uniqueness is central to the cut construction. Two distinct carriers through the same point cannot both be parallel to a third carrier; this is used to identify AM with ML and EL with ED .

The helper family

```
ParallelSymmetry
ParallelSwapFirstLine
ParallelSwapSecondLine
ParallelCollinearLeft
collinear_parallel_trans
hilbert_parallel_transitive_distinct
```

normalizes endpoint order after those carrier identifications.

47.8.2 Betweenness and segment order

The formal proof uses strict betweenness in a way that the classical text does not display. The central orders are

$$B - M - C, \quad D - L - E, \quad D - N - C, \quad M - N - L.$$

The proof of $B - M - C$ uses betweenness trichotomy and angle-order contradictions. The proof of $D - L - E$ uses congruence transport across the two parallelograms and Hilbert's three-point order-transfer theorem. Segment proper-part comparison is also used to prove $L \neq D$ before strict parallel transport can be applied.

47.8.3 SameSide and OppositeSide

The three outward squares are selected by explicit `HilbertOppositeSide` conditions. These side data control which of the two possible perpendicular rays is used and prevent the leg squares from falling into the triangle.

`OppositeSide` is also required by the alternate-angle criterion used to prove $MA \parallel CE$. Thus side-of-line data are not merely construction annotations; they enter subsequent theorems as hypotheses.

47.8.4 Right angles and angle order

The proof uses `HilbertRightAngle` throughout. No numerical angle measure appears. Rightness is transported through angle congruence, and all right angles are compared synthetically.

Strict angle comparison enters only to prove that the perpendicular foot lies inside BC and to rule out degenerate two-right-angle triangles. The relevant infrastructure includes

```
HilbertAngleLess
hilbert_angleLess_trans
hilbert_angleLess_irrefl
hilbert_all_right_angles_congruent
hilbert_right_angle_transport
```

47.8.5 Pasch

The point N is not postulated as the visually obvious intersection of DC and ML . Forced Pasch constructs it as an interior intersection of the side CD of triangle CBD . Parallelism $DB \parallel ML$ excludes the alternative exit through BD .

This is one of the clearest places where formal plane order replaces an unspoken diagrammatic fact.

47.8.6 Scissors operations

The content layer uses the established operations

```
HilbertScissorsEq.split
HilbertScissorsEq.add
HilbertScissorsEq.trans
scissors_congruent
equicomplementtable_of_scissorsEq
equicomplementtable_add
equicomplementtable_symm
equicomplementtable_trans
equicomplementtable_transport
crossing_quadrilateral_two_triangulations
```

The local helper `i47_aux_equicomplementtable_add` is a packaging lemma for componentwise addition of two equicomplementability relations. It does not introduce a new content principle.

47.9 Formalization Notes

47.9.1 Geometry

Geometry is the largest layer of Proposition I.47. The formal proof must establish all of the following data explicitly:

- each square lies on the outward side of its supporting line;
- $G - A - C$ and $H - A - B$ are strict straight extensions;

- the two leg-square sides are parallel to the opposite triangle legs;
- the two SAS angles have the exact argument order required by Lean;
- all four SAS triangle orientations are noncollinear;
- the perpendicular foot M lies in the open segment BC ;
- A, M, L lie on one carrier;
- D, L, E lie on the upper square carrier;
- $LDBM$ and $CELM$ are parallelograms;
- the upper order is $D - L - E$, not one of the two exterior orders;
- DC and ML meet at an interior point N with both strict betweenness relations.

None of these is inferred from coordinates. The proof remains synthetic.

47.9.2 Content

The content direction is forward only. Geometry and congruence produce content equalities; the proof never starts from equal content and tries to recover geometry. Consequently I.47 does not need the proper-part/positivity machinery characteristic of the converse propositions I.39–I.40.

There is also no subtraction. The final operation is addition of two rectangle-to-square equicomplementabilities after an exact scissors split of the hypotenuse square.

47.9.3 Representation

Three representation choices matter.

First, a square is stored as `IsSquare`, which includes an `IsParallelogram` component. I.47 uses that component repeatedly for opposite-side parallels and congruences.

Second, a triangle in the scissors layer is a term insensitive to vertex permutation. The final proof therefore contains several explicit triangle term rotations and swaps. These are representation normalizations, not new geometry.

Third, the cut parallelograms occur in different vertex orders in the geometric and scissors stages. Theorems such as `parallelogram_term_rotateOne` convert between those representations without changing content.

47.9.4 Axiomatic status

Proposition I.47 is Euclidean in the current project API:

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

Hilbert Group IV enters through uniqueness and transitivity of parallels and through the direction from parallel lines to the required angle relations. The long cut-core proof uses this Euclidean parallel infrastructure several times to identify carriers.

The order arguments, Pasch, segment congruence, angle comparison, and SAS belong to the incidence/order/congruence layers. The side-controlled perpendicular construction itself does not require numerical coordinates or an area function.

No new geometric axiom is introduced. No new content axiom is introduced. No continuity axiom is used by the final I.47 source. No Archimedean axiom is used. There is no `sorry` or `admit`.

47.10 Source Audit

The classical and formal sources agree on the mathematical core but organize it differently.

Euclid and the standard Book I dependency tables give the compact historical DAG

$$\text{I.4, rI.14, rI.31, rI.41, rI.46} \longrightarrow \text{I.47.}$$

The classical diagram suppresses most of the order and side-of-line work.

Hartshorne is particularly important for the content semantics. He treats I.35–I.48 as Euclid’s theory of area in the synthetic sense of equal content. For I.47 he describes the same chain used by the formal proof: a leg square is double one triangle, that triangle is congruent to a triangle adjacent to the hypotenuse square, and I.37/I.41-style equal-content transport identifies the corresponding rectangle. Repeating the argument on the other side and adding gives the result. Hartshorne explicitly distinguishes this from the later numerical formula obtained after coordinates and an area function are introduced.

Hilbert gives an even sharper foundational control. In his discussion of the Euclidean theory of polygons he states Pythagoras as equicomplementability of the two leg squares with the hypotenuse square. This supports the project’s choice of `HilbertScissorsEquicomplementable` as the public content relation rather than a real-valued area equality.

Borsuk–Szmielew and Greenberg are useful mainly as control sources for the synthetic geometry underneath the construction: perpendiculars, parallel carriers, parallelograms, congruence, and plane separation. The present proof does not import a coordinate or field model from those sources.

The Beeson proof corpus is again most useful as a warning about hidden configuration data. The formal proof must name the side choice of each square, the internal foot order, the carrier identifications, and the Pasch intersection. These are exactly the kinds of conditions that disappear in a short diagrammatic proof.

The project Book I documentation provides the immediate local context. I.41 already supplies the double-triangle content theorem, and I.46 already supplies the square predicate and construction. I.47 is therefore the point where the construction geometry of I.46 and the forward content machinery of I.41 are combined in a single theorem.

For the required three-way audit:

- **Geometry:** substantial. Outward square placement, strict betweenness, carrier uniqueness, perpendiculars, parallels, Pasch, angle congruence, and SAS all occur.
- **Content:** forward and additive. Four uses of I.41, two congruent-triangle bridges, one exact scissors split, and final equicomplementability addition.

- **Representation:** nontrivial but separate from the mathematics. Square predicates carry parallelogram structure; triangle terms are permutation-invariant; parallelogram terms are rotated to the vertex order required by the dissection.

47.11 Dependency Summary

The formal dependency chain is

```

Hilbert incidence + order + congruence + Group IV ↓
right angles / perpendiculars / SameSide-OppositeSide / Pasch ↓
parallel carrier uniqueness + parallelogram infrastructure ↓
I.12 + I.14 + I.16/I.17 infrastructure + I.41 + I.46 ↓
  hilbert_square_exists_opposite_side ↓
    i47_outward_squares ↓
leg-square extension / parallel / angle / noncollinearity helpers ↓
  i47_aux_perpendicular_foot_between_BC ↓
  i47_aux_perpendicular_foot_parallel_CE ↓
    i47_aux_construct_L_parallelogram ↓
      i47_aux_cut_line_collinear ↓
        i47_aux_L_on_DE ↓
          i47_aux_left_cut_parallelogram ↓
            i47_aux_upper_cut_between ↓
              i47_aux_diagonal_cut_intersection ↓
                i47_cut_core ↓
                  i47_diagram ↓
                    two SAS congruences + four I.41 applications ↓
  i47_square_split + i47_aux_equicomplementable_add ↓
    euclid_proposition_47.

```

The contrast with the classical DAG is therefore

```

Euclid: I.46 + I.14 + I.31 + I.4 + I.41 ↓
Pythagorean dissection,

```

while the Lean proof expands the hidden placement and order obligations into a separate geometric construction layer before entering the content argument.

New geometric axiom:	no
New content axiom:	no
New proposition-local helper:	yes
New reusable helper:	yes, side-controlled square construction
New Book Zero theorem:	no
New Interface theorem:	no in the final cleanup
Numerical area semantics:	no
Exact scissors dissection used:	yes
Equicomplementability used:	yes
Continuity used:	no
Archimedean axiom used:	no
Hilbert Group IV used:	yes
Public theorem compiles:	yes
Full lake build:	clean
Active local axiom:	no
sorry/admit in Proposition47.lean:	no

Diagrammatic audit. The final figure set deliberately separates the classical and formal roles. Figure 47.1 is the classical Euclidean configuration and therefore omits the formal crossing point N . Figures 47.2, 47.3, and 47.4 record three successive formal geometric states of `i47_cut_core`: internal perpendicular foot, completed upper cut, and Pasch crossing. Figure 47.5 records the two symmetric SAS/I.41 bridges, while Figure 47.6 records the genuinely geometric retriangulation inside the exact scissors split. No separate figures are introduced for triangle-term permutations, parallelogram-term rotations, or associativity/commutativity bookkeeping, since those operations do not change the geometric configuration.

47.12 Conclusion

Proposition I.47 is the point where several strands of the Book I reconstruction meet.

The classical proof is famously short because its diagram carries a large amount of geometric information implicitly. The formal proof exposes that information: the three squares must be placed on specified sides, the perpendicular foot must be proved internal, the upper cut point must be placed between the correct endpoints, and the diagonal must be shown to cross the cut internally. The central theorem `i47_cut_core` is the compact package of this hidden geometry.

Once the geometry is complete, the content proof is comparatively small and close to Euclid. Two SAS congruences connect the leg-square triangles with the triangles adjacent to the hypotenuse square. Four applications of I.41 turn those triangle congruences into two rectangle-to-square equal-content relations. The theorem `i47_square_split` gives an exact dissection of the hypotenuse square into those rectangles, and addition completes the argument.

The final theorem therefore preserves the genuinely synthetic form of Pythagoras:

$$\text{square on } BC \approx \text{square on } AB + \text{square on } AC,$$

where $\approx$ is equicomplementability, not numerical area equality.

The resulting source contains no local axiom and no unfinished proof. It is also prepared for Proposition I.48, which can now consume I.47 as an ordinary compiled theorem rather than depending on provisional diagram data.

Chapter 48

Proposition I.48

48.1 Introduction

Proposition I.48 is the converse of the Pythagorean theorem in the synthetic language of Euclid's Book I. Given a nondegenerate triangle ABC , suppose that the square on BC has the same content as the sum of the squares on AB and AC . The conclusion is that the angle BAC is right.

The statement must be read at the same level as Proposition I.47. It is not the numerical assertion

$$BC^2 = AB^2 + AC^2$$

followed by a calculation with real numbers. In the current reconstruction the three squares are explicit geometric objects and the hypothesis is an instance of

`HilbertScissorsEquicomplementable`

relating their formal scissors terms. The conclusion is the synthetic predicate

`HilbertRightAngle Geo B A C`

with no degree measure attached to the angle.

Classically, Euclid's proof is short. Construct a second triangle ADC with $AD \cong AB$ and $\angle DAC$ right. Proposition I.47 gives

$$\text{Sq}(DC) \approx \text{Sq}(DA) + \text{Sq}(AC),$$

where $\approx$ denotes equal content. Since $AD \cong AB$, the square on DA equals the square on BA . The hypothesis then gives

$$\text{Sq}(DC) \approx \text{Sq}(BC).$$

Euclid passes immediately from equality of these squares to equality of their sides, so $DC \cong BC$. SSS applied to DAC and BAC then transfers the right angle at A .

The formal reconstruction exposes that the sentence

equal squares have equal sides

is the difficult direction. The established scissors calculus gives many ways to prove equal content from geometry, but equal content does not by itself contain an order relation that can distinguish a figure from a proper part. Proposition I.48 therefore needs a strictness argument.

The final file proves this strictness rather than postulating an I.48-specific cancellation theorem. A strict inequality between two square sides is converted into an inner square inside the larger square. The larger square is then dissected into the inner square plus a nonempty triangulated remainder. The project-wide scissors positivity principle shows that the remainder has nonzero content, so the two squares cannot be equicomplementable. Segment trichotomy leaves congruence as the only possibility for sides of equicomplementable squares.

Thus the final proof has two quite different scales:

1. a short Euclidean outer proof using a constructed right triangle, I.47, SSS, and right-angle transport;
2. a longer converse-content core proving that equal square content forces congruent sides.

There is no proposition-local axiom in the final source and no `sorry` or `admit`. The complete project build is clean. The proof does, however, depend transitively on the project-wide temporary content assumption `hilbert_scissors_triangle_positive`. This status is recorded explicitly below.

48.2 The Classical Configuration

Let ABC be the given nondegenerate triangle. The formal theorem receives three explicit squares as data:

$$BCPQ, \quad ABRS, \quad ACTU,$$

with the hypothesis

$$BCPQ \approx ABRS + ACTU.$$

Here and throughout this document $\approx$ denotes the relation

`HilbertScissorsEquicomplementable`

not equality of numerical areas.

At A the proof constructs a new point D such that

$$AD \cong AB, \quad \angle DAC \text{ is right.}$$

This is the formal counterpart of Euclid's use of I.11 followed by I.3. The point D need not be placed in the diagram by assuming the conclusion about $\angle BAC$. In particular, the Lean theorem does not require D, A, B to be collinear. The construction only needs the prescribed perpendicular and the prescribed length.

The triangle ADC is therefore a genuine right triangle. Applying I.47 to it constructs three further squares: one on DC , one on AD , and one on AC , together with the equal-content relation

$$\text{Sq}(DC) \approx \text{Sq}(AD) + \text{Sq}(AC).$$

The two leg squares produced by I.47 are not definitionally the same objects as the given squares on AB and AC . A square is not determined in the formal language by its side alone. The theorem `i48_square_transport` therefore transports content between squares erected on congruent segments.

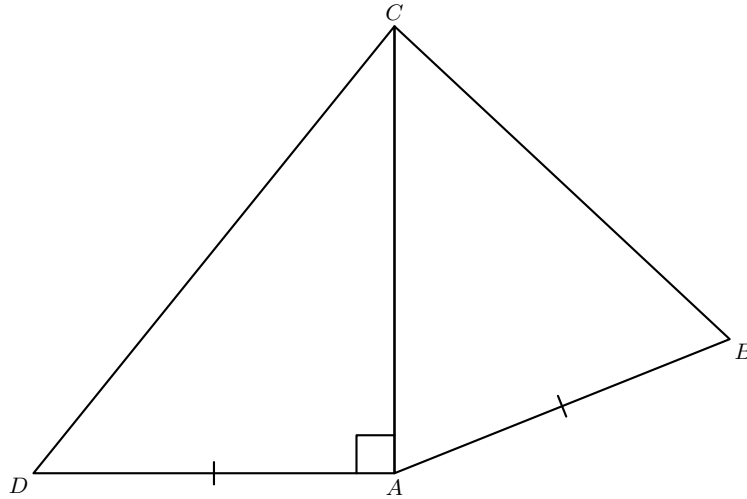


Figure 48.1: Euclid's classical comparison configuration. The auxiliary segment AD is erected at right angles to AC with $AD \cong AB$, and DC is joined. The displayed placement of B and D on opposite sides of AC is the classical diagrammatic arrangement; it is not an additional side hypothesis of the Lean theorem.

After the two transports, transitivity gives

$$\text{Sq}(DC) \approx \text{Sq}(AB) + \text{Sq}(AC) \approx \text{Sq}(BC).$$

The theorem `i48_congruent_of_equal_squares` converts the last content equality into

$$DC \cong BC.$$

Now the triangles DAC and BAC have three corresponding congruent sides:

$$DA \cong BA, \quad AC \cong AC, \quad DC \cong BC.$$

SSS gives the required angle congruence at A , and rightness is transported from $\angle DAC$ to $\angle BAC$.

48.3 Classical Proof

Euclid's proof can be represented by a very small dependency graph.

First erect AD at right angles to AC and choose it so that

$$AD \cong AB.$$

This is the I.11/I.3 construction stage. Join DC . Since $\angle DAC$ is right, I.47 gives

$$\text{Sq}(DC) \approx \text{Sq}(DA) + \text{Sq}(AC).$$

The equality $DA \cong AB$ allows the square on DA to be replaced by the square on BA . Therefore

$$\text{Sq}(DC) \approx \text{Sq}(BA) + \text{Sq}(AC).$$

By the hypothesis of I.48, the right-hand side is also equal to the square on BC . Hence

$$\text{Sq}(DC) \approx \text{Sq}(BC).$$

Euclid then concludes that

$$DC \cong BC.$$

Finally, with

$$DA \cong BA, \quad AC \cong AC, \quad DC \cong BC,$$

Proposition I.8 gives

$$\angle DAC \cong \angle BAC.$$

Since $\angle DAC$ is right, $\angle BAC$ is right.

The classical DAG is therefore

$$\begin{array}{c}
 \text{I.11 + I.3: construct } AD \perp AC, AD \cong AB \\
 \Downarrow \\
 \text{I.47 on } ADC \\
 \Downarrow \\
 \text{Sq}(DC) \approx \text{Sq}(DA) + \text{Sq}(AC) \\
 \Downarrow \\
 AD \cong AB \quad + \quad \text{I.48 hypothesis} \\
 \Downarrow \\
 \text{Sq}(DC) \approx \text{Sq}(BC) \\
 \Downarrow \\
 DC \cong BC \\
 \Downarrow \\
 \text{I.8 / SSS} \\
 \Downarrow \\
 \angle BAC \text{ right.}
 \end{array}$$

The formal reconstruction follows this outer structure closely, but the implication

$$\text{Sq}(DC) \approx \text{Sq}(BC) \implies DC \cong BC$$

expands into the main new proof of Proposition I.48.

48.4 Converse Content, Proper Parts, and Positivity

48.4.1 Why the converse direction is different

The forward content theorems of I.35–I.47 start with geometric information and produce content equality. Congruence, parallelism, common bases, triangulation, and finite addition are enough for this direction.

I.48 reverses the direction at a decisive point. It starts with equality of content of two squares and must recover a metric statement about their sides. For this implication, an equivalence relation and an additive scissors calculus are not enough. One needs a strictness principle saying that adjoining a genuine nondegenerate piece changes content.

The current project isolates that strictness in the global declaration

```
hilbert_scissors_triangle_positive
```

from `HilbertScissorsPositivity`. In schematic form it says

$$P + T \not\approx P$$

whenever T is a nondegenerate triangle term. It is configuration-free: it does not mention squares, betweenness, or Proposition I.48.

The local I.48 proof derives the needed square monotonicity from this one principle. It does not assume a new axiom saying that equal squares have equal sides.

48.4.2 From a shorter side to an inner square

Suppose two squares have sides AB and $A'B'$ and assume

$$AB < A'B'.$$

The relation `HilbertSegmentLess` contains a proper cut on the longer segment. Thus a point X is obtained with

$$A' - X - B', \quad A'X \cong AB.$$

Since adjacent sides of the large square are congruent, the strict comparison is transported to $A'D'$, giving another point Y with

$$A' - Y - D', \quad A'Y \cong AB.$$

Hence

$$A'X \cong A'Y.$$

The rays $A'X$ and $A'B'$ coincide, as do $A'Y$ and $A'D'$. The right angle of the large square is therefore transported to $\angle YA'X$. Completing the equal perpendicular sides constructs a square

$$A'XZY$$

strictly contained at the corner A' of the larger square. The theorem `i48_square_transport` simultaneously shows that the original smaller square has the same content as this inner square.

This stage is packaged by

```
i48_aux_square_side_cuts
i48_aux_complete_square
i48_aux_inner_square_of_less
i48_aux_inner_square_transport
```

48.4.3 The exact remainder decomposition

The next stage is an exact scissors calculation, not merely an intuitive claim that the inner square is a proper subset.

For any parallelogram $ABCD$, the theorem

```
i48_aux_parallelogram_double_corner
```

proves

$$ABCD \sim ABD + ABD,$$

where $\sim$ denotes `HilbertScissorsEq`. The proof uses the two triangulations of a parallelogram and I.34 to identify the two diagonal halves.

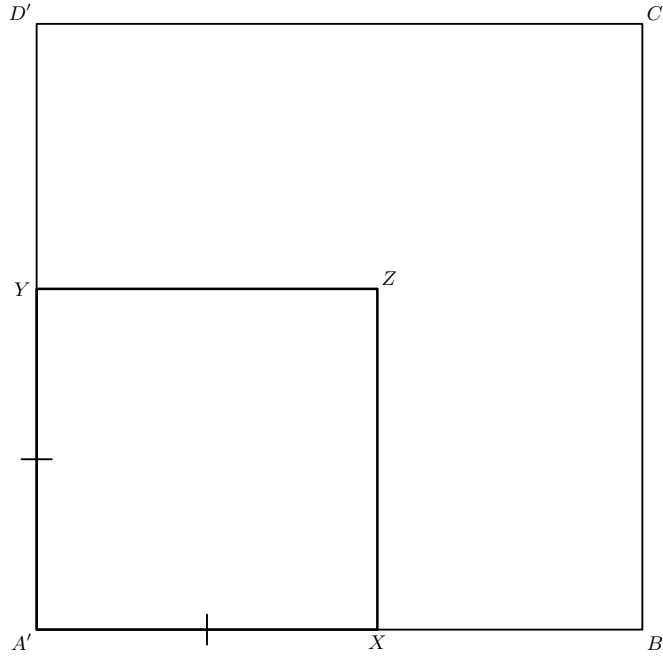


Figure 48.2: The geometric state produced from $AB < A'B'$. The strict cuts $A' - X - B'$ and $A' - Y - D'$ determine equal initial segments on adjacent sides of the larger square, and the construction completes them to the inner square $A'XYZ$. No remainder cuts have yet been introduced.

Now assume

$$A - X - B, \quad A - Y - D.$$

Successive triangle splits give

$$ABD \sim AXY + YDX + XBD.$$

This is the theorem

`i48_aux_corner_triangle_split`

Combining the two doubled-triangle descriptions of the outer and inner squares gives the exact identity

$$\begin{aligned} \text{Sq}(ABCD) &\sim \text{Sq}(AXZY) \\ &\quad + (YDX + XBD) + (YDX + XBD). \end{aligned}$$

This is `i48_aux_square_split_with_remainder`.

The remainder is therefore not an abstract difference. It is an explicit formal sum of four triangles. This is important because the current scissors language has no primitive subtraction of figures and no numerical area.

48.4.4 Why the remainder is genuinely nonzero

The two triangle types occurring in the remainder are

$$YDX, \quad XBD.$$

The theorem

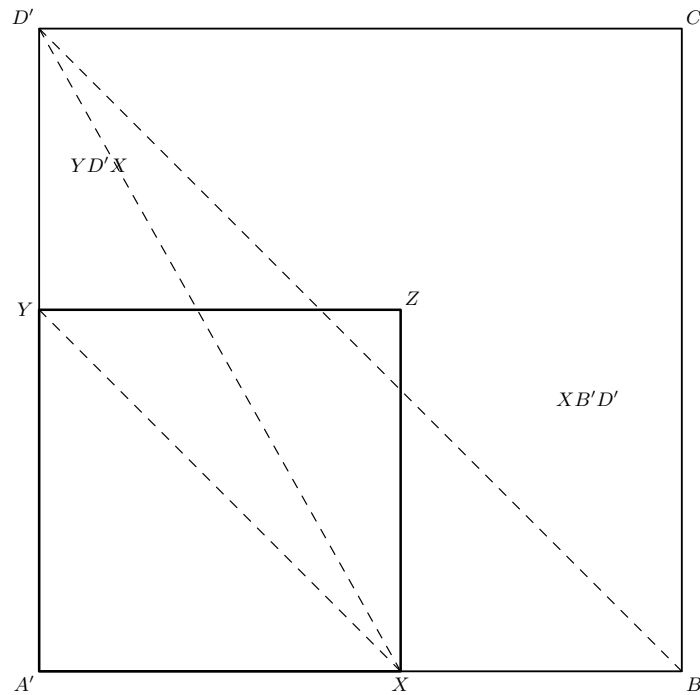


Figure 48.3: The additional cuts used by the exact scissors decomposition. The diagonal $B'D'$, the inner diagonal XY , and the segment $D'X$ split one diagonal half into the inner triangle and the two remainder triangles $YD'X$ and $XB'D'$. Doubling this half produces the four triangle remainder in `i48_aux_square_split_with_remainder`.

```
i48_aux_remainder_triangles_nondegenerate
```

proves that both are noncollinear from the outer square and the strict betweenness relations. The four pieces are then packaged as the nonempty triangulated figure

$$[YDX, XBD, YDX, XBD].$$

The theorem `i48_aux_remainder_figure` verifies both nonemptiness and nondegeneracy of every member of this list and identifies its `rectilinearTerm` with the exact remainder above.

To apply the global triangle positivity axiom to an arbitrary nonempty triangulated remainder, the proof constructs a reduction chain. Proposition I.45 converts the rectilinear figure into an equicomplementable parallelogram. The local theorem

```
i48_aux_parallelogram_to_triangle
```

then converts that parallelogram into one nondegenerate triangle. Hence every such remainder is equicomplementable with a nondegenerate triangle. If the remainder were equicomplementable with zero, transitivity would make that triangle equicomplementable with zero, contradicting `hilbert_scissors_triangle_positive`.

The corresponding chain is

$$\begin{aligned}
 &\text{nonempty triangulated nondegenerate remainder} \\
 &\quad \Downarrow \text{I.45} \\
 &\text{equicomplementtable parallelogram} \\
 &\quad \Downarrow \\
 &\text{equicomplementtable nondegenerate triangle} \\
 &\quad \Downarrow \text{positivity} \\
 &\text{remainder} \not\approx 0.
 \end{aligned}$$

This is implemented by

```
i48_aux_parallelogram_to_triangle
i48_aux_rectilinear_to_triangle
i48_aux_rectilinear_positive
```

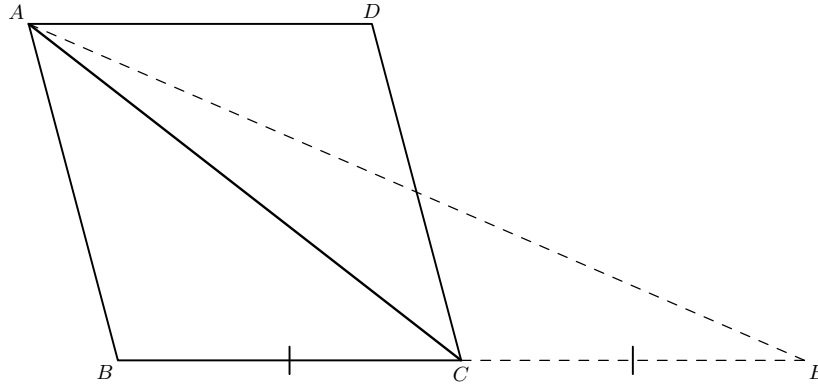


Figure 48.4: The geometric core of `i48_aux_parallelogram_to_triangle`. The side BC is extended to E with $B - C - E$ and $BC \cong CE$, so C is the midpoint of BE . The diagonal AC divides the parallelogram by I.34 and is simultaneously the median used by I.42 in triangle ABE . This converts the parallelogram content to the content of one nondegenerate triangle.

48.4.5 Cancellation and the proper-part theorem

Suppose

$$Q \sim P + R$$

and $R \not\approx 0$. If one also had $Q \approx P$, transitivity would give

$$P + R \approx P.$$

The local algebraic cancellation theorem

```
i48_aux_equicomplementtable_cancel_right
```

removes the common term P after commuting the formal multiset sum, yielding $R \approx 0$, a contradiction.

Therefore

$$Q \sim P + R, \quad R \not\approx 0 \implies Q \not\approx P.$$

The generic packaging is

```
i48_aux_not_equal_of_positive_remainder
i48_aux_not_equal_of_rectilineal_remainder
```

Applied to the inner-square decomposition, it gives

```
i48_aux_inner_square_proper_part
```

and therefore

$$AB < A'B' \implies \text{Sq}(AB) \not\approx \text{Sq}(A'B').$$

The last statement is

```
i48_aux_square_not_equal_of_less
```

48.4.6 Segment trichotomy completes the converse

Now suppose two squares are equicomplementable. The strict inequality $AB < A'B'$ is impossible by the theorem just proved. By symmetry the reverse strict inequality $A'B' < AB$ is also impossible. Both sides are nondegenerate because they are sides of squares. The Book Zero segment trichotomy theorem therefore leaves only

$$AB \cong A'B'.$$

This is the content of

```
i48_congruent_of_equal_squares
```

The theorem is the formal replacement for Euclid's apparently immediate step from equal squares to equal sides.

48.5 Proof Architecture of the Reconstruction

The final Lean source is best read as eight mathematical stages.

48.5.1 Step 1: construct the prescribed perpendicular segment

The theorem

```
i48_erect_perpendicular
```

packages the I.11/I.3 operation needed by Euclid. In ASCII notation its interface is

```
theorem i48_erect_perpendicular
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A C P Q : Geo.Point)
  (hAC : Not (A = C)) :
  exists D : Geo.Point,
    Not (Collinear Geo D A C) /\
```

```
HilbertRightAngle Geo D A C /\
Geo.Congruent A D P Q
```

The proof first extends CA beyond A so that the established right-angle existence theorem can be applied at A . It then lays off the independently prescribed segment PQ on the resulting perpendicular ray. Same-ray angle transport shows that replacing the initially constructed point by the laid-off point preserves the right angle. Finally the angle arms are swapped into the orientation DAC used by I.48.

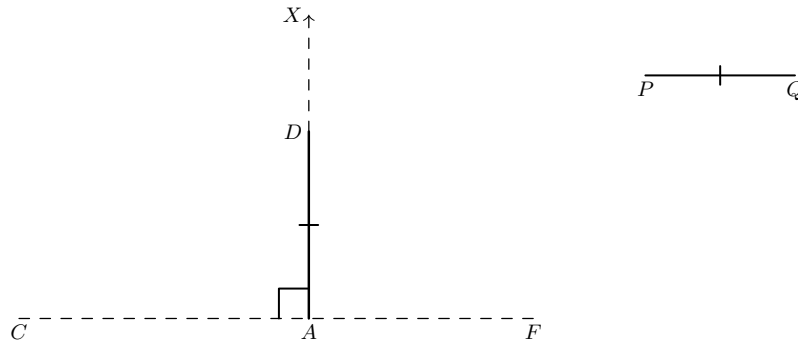


Figure 48.5: The hidden witnesses in `i48_erect_perpendicular`. First CA is extended beyond A to F ; I.11 supplies a perpendicular point X ; then the prescribed segment PQ is laid off as AD on the same ray from A . The drawing chooses one order of D and X only for visibility: the proof uses `SameRay`, not a betweenness relation between those two points.

This helper is mathematically reusable: it erects a perpendicular at a specified endpoint with a length unrelated to the carrier segment.

48.5.2 Step 2: transport content between squares on congruent sides

The theorem

```
i48_square_transport
```

proves that squares erected on congruent segments are equicomplementable. The proof is genuinely geometric rather than definitional.

First the congruence of one pair of sides is propagated around the two squares. The right angles at corresponding vertices are congruent by `hilbert_all_right_angles_congruent`. SAS then identifies the corresponding diagonals. Each of the two diagonal halves is matched by SSS, and the two triangle congruences are added as a `HilbertScissorsEq`. Finally exact scissors equality is weakened to equicomplementability.

This theorem formalizes Euclid's sentence that equal straight lines have equal squares described on them. It is used twice in the public I.48 proof: for the square on AD versus the given square on AB , and for the square on AC produced by I.47 versus the given square on AC .

48.5.3 Step 3: turn strict side inequality into an inner square

The helper

```
i48_aux_square_side_cuts
```

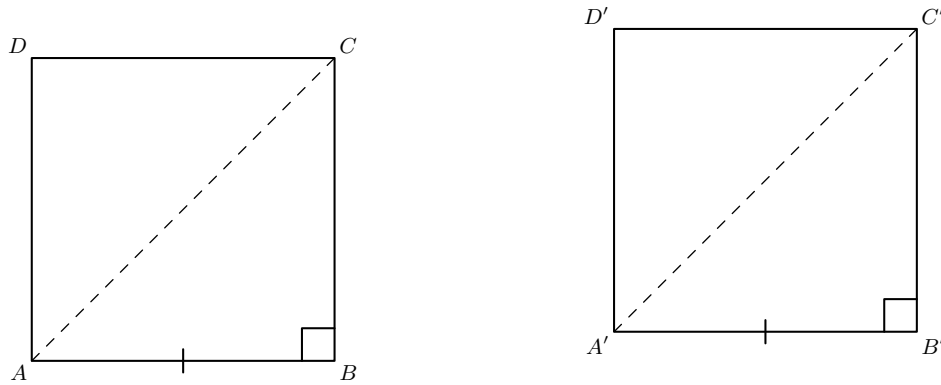


Figure 48.6: The geometric content of `i48_square_transport`. Congruent base sides propagate around the two squares; right angles at B and B' give the congruent diagonals AC and $A'C'$ by SAS, after which the two diagonal halves correspond by SSS. The figure records the triangulation, not a numerical area argument.

uses `HilbertSegmentLess` together with the side congruences of I.46 to cut equal proper initial segments on two adjacent sides of the larger square.

The theorem

```
i48_aux_complete_square
```

completes equal perpendicular segments to a square. It first constructs the fourth vertex of a parallelogram, then uses I.34, opposite-side congruences, and right-angle transport to establish all four square conditions.

The theorem

```
i48_aux_inner_square_of_less
```

combines these steps. It must prove that the cut rays are noncollinear and that their angle is right. The latter is obtained by transporting the right angle of the large square along the same rays determined by the betweenness data.

Finally

```
i48_aux_inner_square_transport
```

uses `i48_square_transport` to identify the content of the original smaller square with the newly constructed inner square.

48.5.4 Step 4: expose the remainder by exact triangle splits

The theorem

```
i48_aux_parallelogram_double_corner
```

retriangulates a parallelogram along the diagonal joining the other two vertices. I.34 identifies the two resulting triangles, so the parallelogram term is scissors-equal to two copies of a corner triangle.

Next,

```
i48_aux_corner_triangle_split
```

uses the two strict betweenness relations to split that corner triangle into one inner-square triangle and two remainder triangles.

The theorem

```
i48_aux_square_split_with_remainder
```

performs the same substitution in both copies and then replaces the two inner triangles by the inner square term. The output is an exact `HilbertScissorsEq`, not merely an equicomplementability statement.

48.5.5 Step 5: certify positivity of the explicit remainder

The theorem

```
i48_aux_remainder_triangles_nondegenerate
```

shows that neither remainder triangle collapses. Both proofs are incidence arguments: a hypothetical collinearity is transported through the betweenness carriers until it contradicts noncollinearity of three vertices of the outer square.

The theorem

```
i48_aux_remainder_figure
```

packages the four pieces as a `HilbertTriangulatedFigure` and proves that the corresponding `rectilinealTerm` is exactly the remainder in the square split.

The two conversion helpers

```
i48_aux_parallellogram_to_triangle  
i48_aux_rectilineal_to_triangle
```

then reduce an arbitrary nonempty triangulated nondegenerate figure to one nondegenerate triangle. The second helper calls I.45. The first uses a segment extension, I.42's median-content theorem, I.34, and an exact triangle split.

The result

```
i48_aux_rectilineal_positive
```

is the point at which the project-wide positivity axiom enters I.48.

48.5.6 Step 6: derive strict monotonicity of square content

The theorem

```
i48_aux_equicomplementable_cancel_right
```


is a purely algebraic consequence of the definition of equicomplementability. It is not a new positivity assumption. It rearranges the complement witnesses so that a common right summand can be cancelled.

Together with positivity this yields

```
i48_aux_not_equal_of_positive_remainder
i48_aux_not_equal_of_rectilineal_remainder
i48_aux_inner_square_proper_part
i48_aux_square_not_equal_of_less
```

The mathematical conclusion is strict monotonicity:

$$AB < A'B' \implies \text{Sq}(AB) \not\approx \text{Sq}(A'B').$$

48.5.7 Step 7: recover side congruence from equal square content

The central converse theorem has the current source signature

```
theorem i48_congruent_of_equal_squares
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D A' B' C' D' : Geo.Point)
  (hSquare : IsSquare Geo A B C D)
  (hSquare' : IsSquare Geo A' B' C' D')
  (hEqual :
    HilbertScissorsEquicomplementable Geo
      (hilbertParallelogramTerm Geo A B C D)
      (hilbertParallelogramTerm Geo A' B' C' D')) :
  Geo.Congruent A B A' B'
```

The proof excludes both strict segment orders using `i48_aux_square_not_equal_of_less`. It then applies `bookZero_31_trichotomy1`. This is the exact point where the strict-content work reconnects to ordinary Hilbert segment geometry.

48.5.8 Step 8: the public I.48 theorem

The public theorem now becomes short. Construct D , apply I.47 to ADC , transport the two leg squares, compose with the given content equality, use `i48_congruent_of_equal_squares` to obtain $DC \cong BC$, apply SSS to DAC and BAC , and transport the right angle.

The final proof therefore has the schematic form

$$\begin{array}{c}
AD \perp AC, AD \cong AB \\
\Downarrow \\
\text{I.47 on } ADC \\
\Downarrow \\
\text{Sq}(DC) \approx \text{Sq}(AD) + \text{Sq}(AC) \\
\Downarrow \text{ square transport} \\
\text{Sq}(DC) \approx \text{Sq}(AB) + \text{Sq}(AC) \\
\Downarrow \text{ I.48 hypothesis} \\
\text{Sq}(DC) \approx \text{Sq}(BC) \\
\Downarrow \text{ strict square monotonicity + trichotomy} \\
DC \cong BC \\
\Downarrow \text{ SSS} \\
\angle DAC \cong \angle BAC \\
\Downarrow \\
\angle BAC \text{ right.}
\end{array}$$

Diagrammatic audit. Figure 48.1 is retained as the classical Euclidean configuration. Figures 48.2, 48.3, 48.4, 48.5, and 48.6 record the distinct formal geometric states introduced by the current Lean proof: the strict inner square, the explicit remainder cuts, the parallelogram-to-triangle positivity bridge, the hidden perpendicular-construction witnesses, and diagonal transport between squares. No separate figures are added for congruence symmetry, transitivity, scissors cancellation, segment trichotomy, or the final SSS/right-angle transport, since those steps do not introduce a new underlying geometric configuration.

48.6 Lean Formalization

48.6.1 Imports

The current source imports exactly

```
import CGJteamLab.Proposition47
import CGJteamLab.Proposition42
import CGJteamLab.Proposition45
import CGJteamLab.HilbertScissorsPositivity
```

These imports reveal the real formal structure. I.47 supplies the forward Pythagorean content equality. I.42 and I.45 are used inside the proof that a nonempty rectilinear remainder has positive content. The positivity module supplies the one project-wide strict-content assumption.

48.6.2 The public proposition

In ASCII-normalized documentation notation, the public theorem is

```
theorem euclid_proposition_48
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C P Q R S T U : Geo.Point)
  (hABC : Not (Collinear Geo A B C))
```

```

(hSqBC : IsSquare Geo B C P Q)
(hSqAB : IsSquare Geo A B R S)
(hSqAC : IsSquare Geo A C T U)
(hEqual :
  HilbertScissorsEquicomplementable Geo
    (hilbertParallelogramTerm Geo B C P Q)
    (hilbertParallelogramTerm Geo A B R S +
      hilbertParallelogramTerm Geo A C T U)) :
  HilbertRightAngle Geo B A C

```

The three squares are hypotheses rather than existentially reconstructed inside I.48. This is deliberate. The content hypothesis refers to the actual square terms, and the current geometry language does not quotient all squares on congruent sides into one definitional object. Equality between different choices is instead proved by `i48_square_transport`.

48.6.3 The strict square inequality theorem

The key monotonicity theorem has the form

```

theorem i48_aux_square_not_equal_of_less
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D A' B' C' D' : Geo.Point)
  (hSquare : IsSquare Geo A B C D)
  (hSquare' : IsSquare Geo A' B' C' D')
  (hLess : HilbertSegmentLess Geo A B A' B') :
  Not
    (HilbertScissorsEquicomplementable Geo
      (hilbertParallelogramTerm Geo A B C D)
      (hilbertParallelogramTerm Geo A' B' C' D'))

```

The statement contains no numerical length and no numerical area. Strict segment comparison is entirely synthetic, and the conclusion is strict inequality at the level of equicomplementability.

48.6.4 The derived square-side converse

The theorem consumed by the public proposition is

```

theorem i48_congruent_of_equal_squares
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D A' B' C' D' : Geo.Point)
  (hSquare : IsSquare Geo A B C D)
  (hSquare' : IsSquare Geo A' B' C' D')
  (hEqual :
    HilbertScissorsEquicomplementable Geo
      (hilbertParallelogramTerm Geo A B C D)
      (hilbertParallelogramTerm Geo A' B' C' D')) :
  Geo.Congruent A B A' B'

```

This theorem was previously the unresolved local assumption in the I.48 sketch. In the current source it is a theorem proved by the full strictness chain described above.

48.6.5 Right-angle congruence

The other former local assumption has also disappeared. The wrapper theorem

```
i48_right_angles_congruent
```

is now an immediate call to

```
hilbert_all_right_angles_congruent
```

Thus Euclid’s Postulate 4 is supplied by the established Hilbert right-angle infrastructure rather than by a proposition-local axiom.

48.7 Relationship with Earlier Propositions

48.7.1 Propositions I.3 and I.11

Euclid first erects a perpendicular at A and then makes its new segment equal to AB . The formal proof packages these operations in `i48_erect_perpendicular`. Segment extension, right-angle existence, and Hilbert segment construction perform the work that the classical proof cites as I.11 and I.3.

The exact Book I theorem names need not appear in the call graph because the corresponding construction operations already live in the developed Hilbert interface. The classical and formal dependency graphs are therefore not identical.

48.7.2 Proposition I.8

The last geometric step is SSS. Once $DC \cong BC$ has been recovered, the triangles DAC and BAC have all three corresponding sides congruent. The Lean proof calls `HilbertSSS` and then uses the returned angle congruence to transport rightness.

Thus the classical role of I.8 is present exactly, although the formal source uses the Hilbert theorem rather than the Book I proposition wrapper.

48.7.3 Proposition I.34

I.34 enters the strictness core rather than the final Euclidean shell. It is used to compare diagonal halves of parallelograms. This supports both the “parallelogram equals two copies of a corner triangle” lemma and the parallelogram-to-triangle reduction used in positivity.

48.7.4 Proposition I.42

The local theorem `i48_aux_parallelogram_to_triangle` uses `i42_median_bisects_area`. A segment is extended beyond a vertex so that a midpoint configuration is created; I.42 then identifies the two half-triangles in content. Together with the diagonal decomposition of the parallelogram this produces one triangle equicomplementable with the whole parallelogram.

I.42 is therefore not part of Euclid's visible I.48 proof. It is a formal support theorem used only to convert a general positive rectilinear remainder to the single-triangle form expected by the global positivity axiom.

48.7.5 Proposition I.45

I.45 is the main bridge from an arbitrary finite rectilinear remainder to a parallelogram. The theorem `i48_aux_rectilinear_to_triangle` first applies `euclid_proposition_45` to the nonempty triangulated figure, then applies the local parallelogram-to-triangle theorem.

This is a particularly useful example of the difference between the classical and formal DAGs. Euclid never needs to replace the square remainder by an arbitrary equivalent parallelogram in the visible I.48 proof. The formal proof does so only because the global strictness principle is stated for a single nondegenerate triangle.

48.7.6 Proposition I.46

The square predicate and its side geometry come from the infrastructure developed for I.46. In particular,

```
euclid_proposition_46_side
```

is used when a strict comparison with one side of the large square must be transported to the adjacent side.

I.46 also provides the conceptual background for treating a square as an explicit geometric object rather than as a numerical square of a segment length.

48.7.7 Proposition I.47

I.47 is the headline dependency of the proposition. It is applied once to the constructed right triangle *ADC*. Its output consists of three explicit squares and an equicomplementability relation between the hypotenuse square and the sum of the two leg squares.

The public I.48 proof then uses

```
i47_aux_equicomplementable_add
```

to replace both I.47 leg squares simultaneously by the squares occurring in the I.48 hypothesis.

The contrast between the propositions is fundamental:

I.47: right angle $\implies$ equal square content,
I.48: equal square content $\implies$ right angle.

The reverse arrow requires the new strictness argument.

48.8 Hilbert and Book Zero Components Used

48.8.1 Segment order and trichotomy

The local definition `HilbertSegmentLess` supplies the witness used to cut the shorter length from the longer square side. Congruence transport on the right endpoint moves that strict comparison to the adjacent side of the square.

At the end of the converse-content argument,

```
bookZero_31_trichotomy1
```

is decisive. After both strict orders are excluded, trichotomy returns segment congruence. No numerical length function is needed.

48.8.2 Betweenness and SameRay

The strict comparison witnesses produce

$$A' - X - B', \quad A' - Y - D'.$$

Betweenness provides nondegeneracy and carrier collinearity. `SameRay` conversion then shows that the rays $A'X$ and $A'B'$ agree and that the rays $A'Y$ and $A'D'$ agree. The right angle of the large square can therefore be transported to the two cut segments.

These order data are central. Without strict betweenness, the constructed inner square could collapse onto the outer boundary and the proper-part argument would fail.

48.8.3 Parallelogram and square infrastructure

The proof repeatedly extracts the `IsParallelogram` component of `IsSquare`. The main tools include opposite-side congruences, noncollinearity of parallelogram vertices, construction of a fourth vertex, and the two diagonal triangulations.

This infrastructure turns the intuitive phrase “a square is twice one of its diagonal triangles” into an exact theorem in the scissors calculus.

48.8.4 Congruence, SAS, SSS, and right-angle transport

The square transport theorem uses both SAS and SSS. SAS first establishes congruence of the diagonals of two squares on congruent sides; SSS then matches the two triangle halves. In the public theorem, SSS appears again in the classical role of I.8.

Right angles are handled synthetically through

```
hilbert_all_right_angles_congruent
hilbert_right_angle_transport
```

No angle measure or arithmetic with 90 degrees is introduced.

48.8.5 Scissors operations

The content proof uses the established constructors and operations

```
HilbertScissorsEq.refl
HilbertScissorsEq.symm
HilbertScissorsEq.trans
HilbertScissorsEq.add
HilbertScissorsEq.split
scissors_congruent
equicomplementable_of_scissorsEq
equicomplementable_add
equicomplementable_symm
equicomplementable_trans
```

The important distinction is that exact dissection is recorded by `HilbertScissorsEq`, while the public equality of contents is expressed by `HilbertScissorsEquicomplementable`. The proof moves from exact splits to equicomplementability only when required.

48.8.6 Scissors positivity

The only strict-content assumption is inherited from `HilbertScissorsPositivity`:

```
hilbert_scissors_triangle_positive
```

The local I.48 theorems derive all square-specific consequences from it. Consequently there is no active axiom in `Proposition48.lean`, but the public theorem is not independent of the project-wide positivity assumption.

48.9 Formalization Notes

48.9.1 Geometry

The geometry layer of I.48 has two distinct parts.

The outer Euclidean proof constructs only one new point D . It proves

- D, A, C are noncollinear;
- $\angle DAC$ is right;
- $AD \cong AB$.

After this, I.47 supplies the squares needed for the comparison.

The strictness core has a different geometry. From a strict side inequality it constructs X and Y as proper points on two adjacent sides of the larger square, proves the cut rays are perpendicular and equal, completes an inner square, and proves the two remainder triangle types are nondegenerate.

No coordinate system is introduced at either level.

48.9.2 Content

The content layer is the central novelty of the proposition.

Forward transport is routine: congruent squares are shown equicomplementable, I.47 gives a sum of two square contents, and transitivity combines the equalities.

The converse is not routine. The proof needs the implication

$$\text{strictly shorter side} \implies \text{strictly smaller square content}.$$

This is derived by an exact proper-part decomposition plus positivity. The proof never introduces a real-valued area, multiplication of segment lengths, or subtraction of area values.

The resulting theorem `i48_congruent_of_equal_squares` is therefore synthetic: it recovers segment congruence from content equality using segment order and a De Zolt-type strictness principle.

48.9.3 Representation

Several representation issues are mathematically harmless but formally visible.

First, a square is an `IsSquare` predicate on four named points. Two squares on congruent sides are not definitionally identical, so explicit content transport is required.

Second, a parallelogram term has a fixed triangulation in its definition, but the proper-part argument is easiest in the other diagonal triangulation. The theorem

```
i48_aux_parallelogram_double_corner
```

bridges those two representations.

Third, the remainder is a multiset-level sum of repeated triangle terms. The proof uses associativity and commutativity of `Multiset` addition to regroup the two copies of the corner decomposition. This bookkeeping does not add geometric content.

Fourth, I.45 consumes a triangulated figure represented by the type

```
HilbertTriangulatedFigure Geo
```

while the remainder of the square split is a `HilbertScissorsTerm`. The theorem

```
i48_aux_remainder_figure
```

explicitly identifies the list's `rectilinealTerm` with the multiset expression produced by the split.

48.9.4 Hidden configuration data

The Lean proof makes the following data explicit even though a classical diagram tends to suppress them:

- $A \neq C$ is extracted from noncollinearity before the perpendicular construction can be called;

- the constructed point D must be proved noncollinear with A, C ;
- replacing the initial perpendicular point by the point obtained from segment layoff requires SameRay angle transport;
- the two squares on AC occurring in I.47 and in the hypothesis are distinct named objects and must be transported;
- a strict side comparison must be transported from $A'B'$ to $A'D'$;
- the two cut points X, Y must be strict interior points of the relevant sides;
- the cut rays must be proved noncollinear before a right angle can be transported to them;
- both remainder triangle types must be proved nondegenerate;
- the four-triangle remainder must be proved nonempty as a list before I.45 can be applied.

The formal proof does *not* need the classical pictorial convention that the newly erected D lies on the opposite side of AC from B . SSS at the end works from side congruences and the established right angle without such a side-of-line hypothesis.

48.9.5 Axiomatic status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

so Proposition I.48 is Euclidean in the current project API. Group IV is available through the imported Euclidean plane infrastructure and is used transitively in the square, parallelogram, I.45, and I.47 machinery.

No continuity typeclass is assumed by `Proposition48.lean`. No Archimedean axiom is introduced in the file. No numerical field of segment lengths and no numerical area function are used.

There is no proposition-local axiom. The former assumptions “all right angles are congruent” and “equal squares have congruent sides” are both theorems in the final source. The second of these, however, depends on the project-wide temporary content axiom

```
hilbert_scissors_triangle_positive
```

imported from `HilbertScissorsPositivity`. Thus the correct status is: I.48 adds no new foundational debt, but it consumes the already isolated scissors-positivity debt.

48.10 Source Audit

48.10.1 Euclid and Wilkins

The classical sources agree on the short proof architecture. A right triangle ADC is constructed with AC common, $AD \cong AB$, and the angle at A right. I.47 then gives equality of the square

on DC with the sum of the squares on DA and AC . The hypothesis gives the same sum as the square on BC , hence the squares on DC and BC are equal. SSS then compares ADC with ABC and transfers the right angle.

Wilkins' study note is particularly useful because it isolates exactly the chain

$$\text{Sq}(DC) = \text{Sq}(DA) + \text{Sq}(AC) = \text{Sq}(BA) + \text{Sq}(AC) = \text{Sq}(BC),$$

followed by the statement that DC and BC are equal and then by I.8. The formal proof preserves this outer chain.

The point at which the reconstruction expands the classical argument is the transition from equal squares to equal sides. The classical tradition itself recognized this as a separate converse statement. Heath, in his commentary on I.46, records Proclus' proof of both directions: squares on equal straight lines are equal, and conversely equal squares stand on equal straight lines. The current term-level scissors semantics therefore does not invent a new problem; it makes an old converse-content step explicit as a theorem.

48.10.2 Hartshorne: equal content and property (Z)

Hartshorne's discussion of Euclid's area theory is the most useful semantic control for the strictness issue. He distinguishes the equal-content relation from numerical area and lists the "whole is greater than the part" property as the condition that a properly contained figure cannot have the same content as the whole. This is property (Z) in his development.

For I.48 Hartshorne makes the dependence completely explicit: Euclid uses the fact that squares with equal content have equal sides, and Hartshorne states that this step depends on (Z); Exercise 22.6 asks precisely for the theorem that equal-content squares stand on congruent segments, assuming (Z). Thus the central local theorem of the Lean file,

```
i48_congruent_of_equal_squares
```

lands exactly on the semantic boundary identified in the modern synthetic analysis of Euclid.

That observation explains the formal architecture accurately. The equality operations of the scissors calculus support the forward Pythagorean theorem, but the converse step needs a strictness principle. The present I.48 proof derives the required square-specific (Z) consequence from the project-wide triangle positivity assumption and an explicit inner-square decomposition. It does not claim that triangle positivity and Hartshorne's full property (Z) are definitionally or logically identical in the current API.

48.10.3 Hilbert: Theorem 48 and De Zolt

Hilbert's Theorem 48 in Chapter IV should not be confused with Euclid's Proposition I.48. Hilbert's theorem is the converse of his equal-base, equal-altitude triangle theorem: equicomplementable triangles with the same base have the same altitude. Hilbert identifies it with Euclid I.39. It is therefore a general converse-content theorem rather than a statement about Pythagoras.

The structural relevance to the present proposition is nevertheless strong. Hilbert's later area theory shows how equal content can recover metric information, and his De Zolt theorem states that after a rectilinear figure is decomposed into triangular pieces, omitting a genuine triangle cannot leave enough content to reconstruct the whole.

The current Lean development does not import Hilbert’s numerical segment-arithmetic proof of these results. Instead, it isolates the needed strictness in

```
hilbert_scissors_triangle_positive
```

and derives the square-specific proper-part theorem synthetically from finite triangle splits.

48.10.4 Borsuk–Szmielw and Greenberg

Borsuk–Szmielw and Greenberg remain control sources for the underlying synthetic geometry: incidence, order, congruence, segment construction, perpendiculars, parallelograms, and separation arguments. Proposition I.48 does not import a coordinate model or an ordered-field representation from those sources.

Their role is therefore geometric rather than semantic. The content strictness is kept explicitly in the Hilbert scissors layer instead of being hidden inside coordinates.

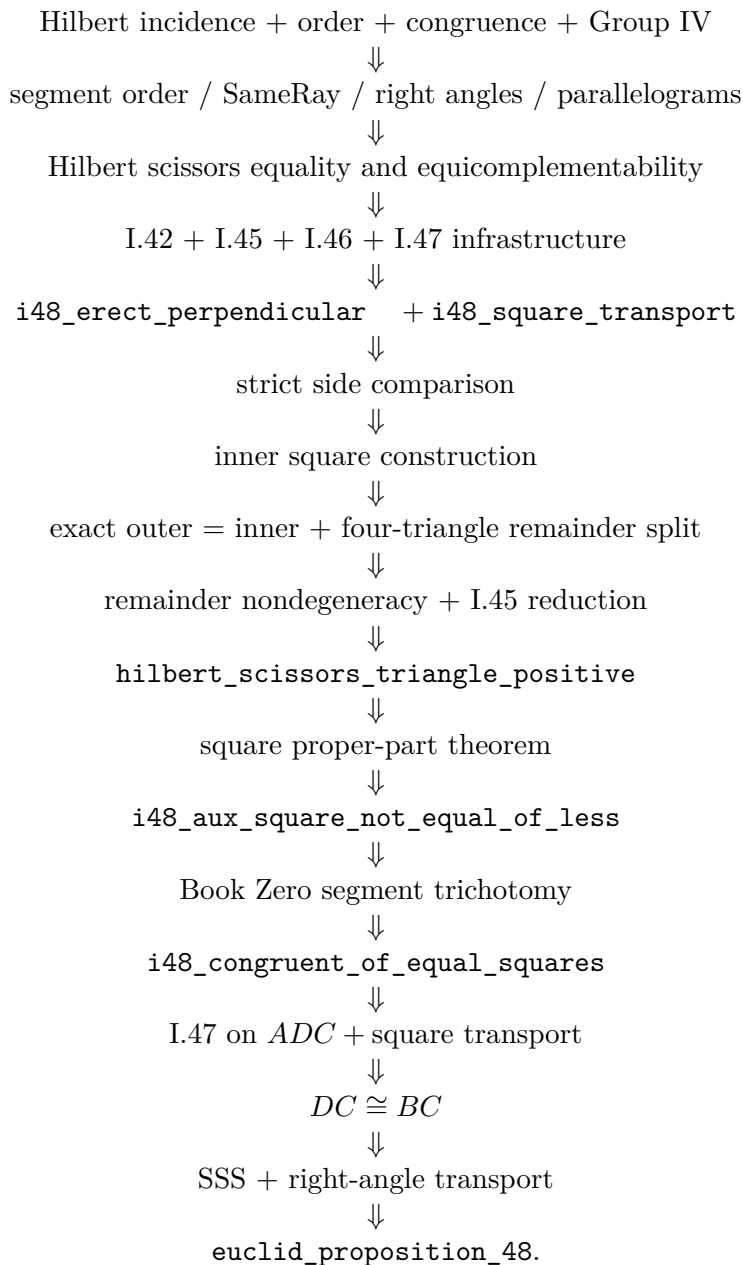
48.10.5 Geometry, content, and representation audit

For the required three-way audit:

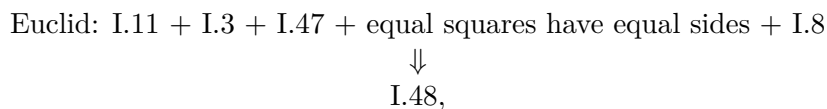
- **Geometry:** moderate in the outer proof and substantial in the strictness core. The proof erects a prescribed perpendicular segment, constructs an inner square from strict segment comparison, establishes strict betweenness, and proves nondegeneracy of the remainder triangles.
- **Content:** central. The proof uses I.47, square transport, exact triangle splits, equicomplementability transitivity, cancellation, and the global positivity principle to derive strict monotonicity of square content.
- **Representation:** substantial but separable. Explicit square choices, alternative parallelogram triangulations, multiset regrouping, and conversion between a triangle list and a scissors term all appear.

48.11 Dependency Summary

The actual formal dependency chain is



The contrast with the classical DAG is



while the Lean proof expands the equal-squares step into a separate proper-part and positivity theory.

New geometric axiom in I.48:	no
New content axiom in I.48:	no
Inherited temporary content assumption:	yes, scissors positivity
New proposition-local helpers:	yes
New reusable local helper candidates:	yes
New Book Zero theorem:	no
New Interface/Scissors theorem:	no
Numerical area semantics:	no
Segment multiplication / field theory:	no
Exact scissors dissection used:	yes
Equicomplementability used:	yes
Hilbert Group IV available/used transitively:	yes
Continuity assumption in the file:	no
Archimedean assumption in the file:	no
Public theorem compiles:	yes
Full lake build:	clean
Active proposition-local axiom:	no
sorry/admit in Proposition48.lean:	no

48.12 Conclusion

Proposition I.48 closes Book I by reversing the Pythagorean theorem at the level at which Euclid actually states it: equality of plane content, not an equation between numerical squared lengths.

The outer proof remains close to Euclid. A prescribed perpendicular segment creates a right triangle ADC with $AD \cong AB$. Proposition I.47 identifies the square on DC with the sum of the two leg squares. Transport replaces those squares by the ones occurring in the hypothesis. Once $DC \cong BC$ is known, SSS transfers the right angle immediately.

The main formal contribution is proving that intermediate side congruence. The source does not assume that equal square content implies congruent sides. Instead it proves strict monotonicity: a strictly shorter side produces a strictly smaller square in the scissors sense. The proof constructs an inner square, exposes the outer square as that inner square plus an explicit four-triangle remainder, proves the remainder nondegenerate, reduces it to one nondegenerate triangle, and invokes the already isolated scissors positivity principle. Segment trichotomy then converts the exclusion of both strict orders into congruence.

This makes the foundational status transparent. No new I.48-specific axiom is introduced, and the two provisional local assumptions from the original sketch have both disappeared. The remaining debt is the project-wide configuration-free positivity principle already isolated in `HilbertScissorsPositivity`. No numerical area, segment multiplication, field construction, continuity axiom, or Archimedean axiom is added to close the proposition.

The final Book I theorem is therefore both synthetic and modular: the classical converse-Pythagoras shell is short, while the nontrivial converse content principle has been factored into a reusable strict-square argument whose exact dependence on scissors positivity is visible in the code.

Appendix A

Greenberg: A Hilbert-Style Reconstruction

.1 Euclid I.1–I.20 in Greenberg’s Hilbert-Style Development

.1.1 Purpose of the comparison

This appendix records the status of Euclid’s first twenty propositions in Marvin J. Greenberg’s Hilbert-style development of elementary geometry. The comparison is deliberately stricter than a list of mathematically related results. For each proposition I.1–I.20, the question is whether Greenberg provides a corresponding result and, if so, in what form: a complete synthetic proof, an axiom, a proposition whose proof is left as an exercise, a proof outline or hint, or a proof that already uses numerical measurement.

This distinction is important for the present formalization. The fact that the mathematical content of a Euclidean proposition occurs in a modern axiomatic treatment does not imply that a complete synthetic proof, suitable for direct reconstruction in Lean, is actually given there.

.1.2 Comparison table

Euclid	Greenberg counterpart	Status in Greenberg	Relation to the present formalization
I.1	Equilateral triangle construction; explicitly discussed as Euclid’s Proposition 1.	Euclid’s construction is discussed together with its continuity gap. Greenberg repairs the existence argument by means of circular continuity.	The proposition is not simply a consequence of incidence, order, and congruence alone in Greenberg’s presentation. It therefore has a different foundational status from many later propositions.
I.2	Segment laying-off; Congruence Axiom C-1.	The essential construction is incorporated into the congruence axioms rather than reproved as Euclid’s proposition.	In the Hilbert layer this operation belongs naturally to the primitive segment-construction infrastructure.
I.3	Cutting off a segment equal to a given shorter segment.	Its content is distributed between segment construction and segment ordering; no separate reconstruction of Euclid I.3 is required in Greenberg’s organization.	The formal proof is assembled from lower-level construction and order lemmas rather than imported as a named Euclidean theorem.
I.4	SAS triangle congruence.	SAS is Congruence Axiom C-6. Greenberg explicitly discusses why Euclid’s superposition argument is not accepted as an axiomatic proof.	In a Hilbert-style formalization I.4 is therefore foundational rather than a theorem derived from the preceding Euclidean propositions.
I.5	Proposition 3.10: base angles of an isosceles triangle.	A complete synthetic proof is given from SAS.	This is a direct and useful Hilbert-style counterpart of Euclid I.5.
I.6	Proposition 3.18: converse of the isosceles base-angle theorem.	The proposition is stated, but its proof is left as an exercise.	Greenberg confirms the theorem and its place in the theory, but does not supply a complete proof for direct transcription into Lean.
I.7	Uniqueness phenomenon for triangles constructed on a fixed base.	No direct proposition-by-proposition reconstruction of Euclid I.7 was identified. Related uniqueness information is built into the congruence and triangle-laying-off machinery.	The formalization must expose the exact uniqueness statement required by the Euclidean proof rather than relying on a named Greenberg counterpart.
I.8	Proposition 3.22: SSS criterion for congruence.	SSS is stated as a proposition. Its proof is assigned as an exercise, with a substantial hint based on the corollary to SAS and a case analysis.	Greenberg supplies the mathematical route but not a complete printed synthetic proof.
I.9	Angle bisector construction; Proposition 4.4.	The result occurs in the neutral-geometry development, but the proof is assigned to the reader.	The existence of the construction is standard in the theory, while the formal reconstruction still requires the missing synthetic details.
I.10	Midpoint construction; Proposition 4.3 and associated exercise.	Greenberg supplies a detailed step-by-step proof scheme in the exercises, with the individual steps to be justified.	This is close to a proof blueprint, but it is not presented as a complete finished proof in the main text.
I.11–I.12	Proposition 3.16: existence of a perpendicular through a point.	A complete synthetic proof is given, treating separately the cases in which the point lies on the line and does not lie on the line.	The single Hilbert-style proposition covers the geometric content split by Euclid into I.11 and I.12.
I.13–I.14	Supplementary angles, rays, angle addition/subtraction, and related order results.	The required content is distributed through the congruence and betweenness theory rather than organized as two direct Euclidean counterparts.	These propositions illustrate why a proposition-number correspondence is insufficient: the modern proof infrastructure reorganizes the underlying elementary geometry.

.1.3 Result of the query

The comparison shows that Greenberg does *not* contain a continuous sequence of twenty complete synthetic proofs corresponding to Euclid I.1–I.20.

Instead, the first twenty propositions are distributed among several different layers of the Hilbert-style theory. Some Euclidean propositions become axioms or immediate consequences of axioms; some receive complete synthetic proofs; some are stated but assigned as exercises; some are represented by more general modern propositions; and some are treated only after numerical measurement has been introduced.

For the present project, this is a significant distinction. Greenberg is an important reference for the mathematical correctness and dependency structure of the Hilbert reconstruction, but it is not a source from which the Lean proofs of I.1–I.20 can simply be transcribed.

.1.4 Consequences for the formal reconstruction

The comparison also clarifies the role of the intermediate Hilbert library developed in the project. A textbook proof can suppress operations that become explicit obligations in Lean: permutations of collinearity, orientations of betweenness, transport along rays, normalization of unoriented angles, transport of segment and angle order under congruence, and repeated proofs of noncollinearity.

Thus the formal reconstruction has two distinct source levels:

1. the classical mathematical argument, for which Greenberg and related literature provide reference results and proof ideas;
2. the explicit synthetic proof infrastructure required to make those arguments executable in Lean.

The second level is represented by the project’s Book Zero and Hilbert interface. It should not be confused with a claim that the underlying Euclidean theorems are new. Its purpose is to expose and organize the elementary proof operations that conventional mathematical prose routinely leaves implicit.

.1.5 Current checkpoint

At the present checkpoint, Euclid Book I, Propositions I.1–I.20 have been reconstructed in the project. The Greenberg comparison should therefore be read as a source audit for this completed initial block, not as a claim about the status of the remaining propositions of Book I. Further sources can be added to the comparison independently as the literature review proceeds.

Appendix A

Hartshorne: Euclid in a Hilbert Plane

A.1 Purpose and scope

This appendix records a source audit of Euclid Book I, Propositions I.1–I.20, against Robin Hartshorne’s *Geometry: Euclid and Beyond* [5].

The audit has a specific purpose. Hartshorne begins with Euclid’s own proofs and constructions, analyzes where their logical structure is incomplete, and then reconstructs the relevant part of Book I inside a Hilbert-plane framework. For each proposition I.1–I.20, we record:

- whether Hartshorne regards Euclid’s proof as valid as it stands;
- whether the proposition is replaced by an axiom or by a more general Hilbert-plane theorem;
- whether Hartshorne supplies a complete proof, a proof sketch, or an exercise;
- which hidden order, betweenness, separation, congruence, or existence issue he explicitly identifies;
- which points are especially relevant to a fully formal synthetic reconstruction.

The purpose is not to claim novelty for the classical mathematics. Rather, the audit identifies the exact level at which a conventional synthetic proof still leaves obligations that must be made explicit in a formal proof assistant.

A.2 Hartshorne’s organization of the problem

Hartshorne separates the discussion into two stages. Chapter 1 studies Euclid’s own geometry and repeatedly asks whether a proof depends on a diagram, an unstated positional assumption, or a construction whose existence has not yet been justified. Chapter 2 then introduces Hilbert’s axioms of incidence, betweenness, and congruence and reconstructs the early propositions inside that framework.

The key structural statement appears in the section on Hilbert planes: apart from I.1 and I.22, the results of Book I up to that point can be recovered in an arbitrary Hilbert plane. Proposition I.1 requires an additional intersection or continuity assumption, and I.22 is another exception. This makes I.1–I.20 a particularly natural block for testing the Hilbert reconstruction.

A.3 Comparison table

Table A.1: Euclid I.1–I.20 in Hartshorne’s Hilbert-plane development

Euclid	Hartshorne counterpart	Status in Hartshorne	Source-audit remarks and relevance to formalization
I.1	Construction of an equilateral triangle on a given segment.	Hartshorne treats I.1 as the first genuine exception to the Hilbert-plane recovery of Book I. The intersection of the two circles used by Euclid cannot be derived from the basic Hilbert-plane axioms alone.	Chapter 1 already isolates the hidden existence problem: the diagram shows two circles meeting, but Euclid’s stated postulates do not guarantee that they do. Hartshorne therefore requires an additional intersection or continuity principle. This is a foundational dependency, not a minor gap in presentation.
I.2	Transporting a given segment to a specified point.	Hartshorne states that I.2 is effectively replaced by the segment congruence construction axiom C1.	The Euclidean proposition becomes primitive infrastructure in the Hilbert system. This is important for later segment arithmetic and for the synthetic interpretation of “adding” segments by extending a ray and laying off a congruent copy.
I.3	Cutting off from a greater segment a segment equal to a given smaller segment.	Hartshorne groups I.3 with I.2 as effectively replaced by C1 together with the order theory of segments.	The proposition is no longer structurally fundamental as a named theorem. Its mathematical content is decomposed into segment construction, uniqueness, betweenness, and segment ordering.
I.4	SAS triangle congruence.	SAS is taken as axiom C6. Hartshorne analyzes Euclid’s superposition argument and concludes that it relies on an unstated theory of rigid motions.	This is one of the sharpest differences between Euclid’s proposition order and a Hilbert-style dependency order. The proof by superposition is not accepted as a derivation from Euclid’s explicit postulates and common notions.
I.5	Base angles of an isosceles triangle.	Hartshorne explicitly reinterprets Euclid’s proof in the Hilbert system and gives the proof in detail using segment construction and SAS.	This is a direct example of the program of recovering Euclid inside a Hilbert plane. Hartshorne also points out the hidden betweenness facts needed to justify some of Euclid’s positional claims.
I.6	Converse of the isosceles base-angle theorem.	Hartshorne says that Euclid’s argument is essentially valid, but one point must be rewritten using the formally defined order relation on angles.	The phrase “less than” cannot be left intuitive. The proof depends on having an explicit angle-order relation and on using its asymmetry and trichotomy correctly. This is precisely the kind of obligation that becomes visible in Lean.

Continued on next page

Table A.1 – continued from previous page

Euclid	Hartshorne counterpart	Status in Hartshorne	Source-audit remarks and relevance to formalization
I.7	Uniqueness configuration for triangles on the same base and on the same side.	Hartshorne identifies a hidden dependence on the relative position of the lines and points. The missing betweenness justification is assigned for completion in the Hilbert-plane exercises.	This proposition is a canonical example of a proof whose diagram suppresses case distinctions. Hartshorne stresses that one must know where the lines and intersection points lie before the angle inequalities used in the proof are legitimate.
I.8	SSS triangle congruence.	Hartshorne replaces Euclid's superposition proof with a new theorem, Proposition 10.1 (SSS), proved from the Hilbert congruence axioms.	A complete Hilbert-style proof is supplied. The construction creates a copy of one triangle on the opposite side of a side of the other and uses angle addition and SAS. The positional cases are explicitly noted.
I.9	Construction of an angle bisector.	Hartshorne reconstructs I.9 using Hilbert tools. Since I.1 is not available in an arbitrary Hilbert plane, he replaces Euclid's equilateral-triangle step by the construction of an isosceles triangle.	He explicitly notes that it is not enough to perform the construction: one must prove that the constructed ray really lies in the interior of the original angle. That requires the order/separation machinery.
I.10	Construction of the midpoint of a segment.	Hartshorne reconstructs the proposition by the same strategy: replace Euclid's use of I.1 by an isosceles triangle. The corresponding Hilbert-tools construction is left as Exercise 10.2.	The proposition is therefore recoverable in a Hilbert plane without using the circle-intersection principle required by I.1. This is an important alternative dependency route.
I.11	Erecting a perpendicular at a point of a line.	Hartshorne explains that the construction can be recovered using Hilbert tools and assigns the explicit construction as Exercise 10.3.	The reconstruction also establishes the existence of right angles, which is not taken for granted beforehand. The proof is therefore part of the basic congruence-and-angle infrastructure rather than a compass construction in Euclid's original sense.
I.12	Dropping a perpendicular from a point to a line.	Hartshorne explicitly says that Euclid's original compass construction does not work in an arbitrary Hilbert plane. A new existence proof is required; the explicit construction is Exercise 10.4.	This is a significant foundational note. The failure is not that the theorem is false, but that Euclid's chosen construction imports an intersection principle unavailable in the bare Hilbert system.
I.13	Angles formed when a straight line stands on another straight line.	Hartshorne replaces the Euclidean proposition by Proposition 9.2 on supplements of congruent angles.	He remarks explicitly that Proposition 9.2 may be viewed as the proper Hilbert-style replacement for I.13. The modern statement is more general and fits directly into the angle-congruence infrastructure.

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Table A.1 – continued from previous page

Euclid	Hartshorne counterpart	Status in Hartshorne	Source-audit remarks and relevance to formalization
I.14	Converse straight-line criterion from adjacent supplementary angles.	Hartshorne treats I.14 as an easy consequence of the preceding supplement theory and asks for a careful Hilbert-plane formulation in Exercise 10.7.	The theorem is mathematically routine only after the modern notions of rays, supplementary angles, and angle congruence have been made precise. The formal version therefore belongs naturally to the low-level angle interface.
I.15	Vertical angles are equal.	Corollary 9.3: vertical angles are congruent. A complete short proof is given from the theory of supplementary angles.	This is a clean example in which Hartshorne replaces Euclid’s common notion argument by an explicitly stated congruence theorem.
I.16	Exterior Angle Theorem.	Proposition 10.3 gives a full Hilbert-plane proof.	Hartshorne returns to the exact point he had criticized in Chapter 1: the constructed ray must really lie inside the relevant angle. He closes this gap using betweenness and plane separation. Thus the proof makes explicit the hidden positional content of Euclid’s diagram.
I.17	Any two angles of a triangle are together less than two right angles.	Hartshorne says that I.17–I.21 are valid in a Hilbert plane after reinterpretation. For I.17 he avoids treating “the sum of two angles” as a primitive object and reformulates the statement using supplements and angle order.	This is especially relevant to formalization: the familiar informal statement must be replaced by a proposition expressible in the available angle language.
I.18	The greater side of a triangle subtends the greater opposite angle.	Hartshorne regards the Euclidean argument as recoverable but assigns a careful Hilbert-plane proof as Exercise 10.6.	The exercise explicitly calls attention to betweenness and inequalities. These are exactly the two classes of hidden obligations that dominate a formal reconstruction of I.18.
I.19	The greater angle of a triangle subtends the greater opposite side.	Included in Hartshorne’s statement that I.17–I.21 are valid in a Hilbert plane after the preceding reconstruction of order and congruence.	Hartshorne does not single out a new main-text proof comparable to Proposition 10.3. Its validity rests on the now-formalized order relations for sides and angles and on the preceding comparison theorem.
I.20	Triangle inequality.	Hartshorne says that I.17–I.21 are valid in a Hilbert plane and assigns a careful proof of I.20 as Exercise 10.8.	The theorem therefore belongs to neutral Hilbert geometry. Hartshorne’s exercise status is important: the mathematics is declared recoverable, but the complete chain of synthetic justifications is left to the reader. A formal proof must expose all those omitted steps explicitly.

A.4 Foundational observations retained from Hartshorne

Several remarks in Hartshorne are important beyond the immediate I.1–I.20 audit and should be preserved for later work.

- Hartshorne repeatedly uses a proof from a diagram as a warning sign. When a step depends on the relative position of points, lines, or rays, that position must be derived from the axioms rather than read from the figure.
- Betweenness and plane separation are not secondary bookkeeping devices. They are the mechanism that repairs several of the early Euclidean proofs, especially I.7, I.9, and I.16.
- Hartshorne’s Hilbert plane is built from incidence, betweenness, and congruence. The parallel postulate is kept separate so that one can identify which propositions belong to neutral geometry.
- The first major exception is I.1. The basic Hilbert-plane axioms do not guarantee the intersection of the two circles used by Euclid. Hence an additional intersection/continuity principle is required.
- Euclid’s constructions and Hilbert’s existence axioms should not be conflated. A Euclidean compass construction may fail as a proof in an arbitrary Hilbert plane even when the corresponding geometric object can still be proved to exist by another route.
- Hartshorne replaces Euclid’s superposition method by axiomatic SAS. He interprets superposition as requiring a theory of rigid motions, which is not supplied by Euclid’s stated postulates.
- Segment addition is formalized synthetically. Hartshorne defines the sum of two segments by laying off a congruent copy on a ray and proves that this construction is well-defined up to congruence.
- Segment order is then defined using betweenness and congruence, and Hartshorne proves its compatibility with congruence and its trichotomy properties.
- The angle theory is developed in parallel: supplementary angles, vertical angles, angle addition, and angle order are all made explicit before the later Euclidean propositions are recovered.
- I.17 illustrates an important language issue. An informal sum of angles need not correspond directly to a primitive formal object, so the theorem is reformulated using inequalities and supplements.
- Hartshorne explicitly identifies I.22 as another exception beyond the current checkpoint. Thus the continuity/intersection issue encountered at I.1 returns immediately after I.20 and is likely to matter in the next phase of Book I.

A.5 Result of the Hartshorne audit

Hartshorne does not present Euclid I.1–I.20 as a sequence of twenty independent modern theorems. Instead, he reorganizes the material according to the logical structure of a Hilbert plane.

Some Euclidean propositions become axioms or primitive constructions (I.2–I.4); some receive complete Hilbert-style proofs (notably I.5, I.8, I.15, and I.16); some are reconstructed by replacing

Euclid's original construction with one valid in a Hilbert plane (I.9–I.12); and several are declared recoverable but are left to the reader as careful exercises (including I.18 and I.20).

The most important feature of the source for formal work is not merely which theorem is available. Hartshorne repeatedly identifies the exact kind of hidden obligation that must be supplied: betweenness, side-of-line information, angle order, segment order, interior-of-angle claims, and existence of intersections.

For a proof assistant, these are not editorial details. They are explicit proof terms and intermediate lemmas. Hartshorne therefore provides an unusually close mathematical guide to the level of granularity required by a fully formal synthetic reconstruction, even where the final derivation is still left as an exercise.

A.6 Current checkpoint

This appendix concerns only Euclid Book I, Propositions I.1–I.20. Observations relevant to later propositions have been retained when Hartshorne explicitly connects them to the present material, but no claim is made here about the complete status of the remainder of Book I.

Appendix B

Wilkins: The Euclidean Dependency Graph

B.1 Purpose and scope

This appendix records a pilot source audit of David R. Wilkins’s *Study Notes on Euclid’s Elements*, Book I, Propositions I.27–I.32. Unlike the Greenberg and Hartshorne appendices, its primary object is the internal Euclidean dependency DAG: earlier propositions and Common Notions used, converse statements, alternative proofs, equivalences, and configuration variants.

B.2 Proposition-by-proposition audit

Prop.	Euclidean dependency DAG	Wilkins additions	Significance for formalization
I.27	I.16 $\rightarrow$ I.27. If the two lines met on either side of the transversal, one of the equal alternate angles would be an exterior angle of a triangle and hence greater than the remote interior angle.	Wilkins explicitly treats both possible sides of intersection and identifies I.27 as the converse of the Alternate Angles Theorem obtained later from I.29.	The proof DAG is exceptionally small. Parallelism is obtained from the neutral exterior-angle inequality; the formal burden is chiefly the two possible intersection orientations.
I.28	First branch: I.15 $\rightarrow$ I.27. Corresponding-angle equality is converted by vertical angles into alternate-angle equality. Second branch: I.13 + Common Notion 3 $\rightarrow$ I.27.	Wilkins treats the two criteria separately. The second argument displays the Common Notion cancellation step rather than hiding it in angle arithmetic.	I.28 naturally splits into two wrappers over I.27 after different angle normalizations. The geometric engine remains I.27.

Prop.	Euclidean dependency DAG	Wilkins additions	Significance for formalization
I.29	Fifth Postulate + I.13 $\rightarrow$ same-side interior angles equal two right angles $\rightarrow$ alternate and corresponding angle equalities.	Wilkins explicitly states that I.29 is the first proposition in Book I requiring the Fifth Postulate. He relates its conclusions to the converses already proved in I.27 and I.28.	This is the exact reversal of the preceding neutral direction: I.27–I.28 infer parallelism from angle relations; I.29 infers angle relations from parallelism.
I.30	Euclid’s route: two uses of I.29 $\rightarrow$ equality of suitable angles $\rightarrow$ I.27 $\rightarrow$ transitivity of parallelism. A detailed variant inserts I.15.	Wilkins gives another route using corresponding angles and I.28. More importantly, he proves that I.30 is logically equivalent to Playfair’s Axiom and shows that Playfair + I.27 implies I.29.	I.30 becomes a bridge between Euclid’s DAG and an axiomatic interface for Euclidean parallelism, rather than merely the next theorem in the linear sequence.
I.31	I.23 $\rightarrow$ construction of an equal alternate angle; I.27 $\rightarrow$ the constructed line is parallel.	Wilkins separately notes that I.29 gives uniqueness of the infinite line through the point parallel to the given line.	The DAG isolates existence from uniqueness: I.23 + I.27 suffice for existence; I.29 enters only for uniqueness. This distinction should remain explicit in a Hilbert-style interface.
I.32	I.31 $\rightarrow$ auxiliary parallel; I.29 twice $\rightarrow$ the two required angle equalities; I.13 $\rightarrow$ the two-right-angles conclusion.	Wilkins also presents the proof attributed by Eudemos to the Pythagoreans and explains how its configuration is related to Euclid’s by extension, reorientation, and relabelling.	The invariant mechanism is clearer than the particular diagram: construct a parallel, transport two angles using I.29, and close the straight angle. Equivalent configurations may have very different formal costs in Lean.

B.3 Dependency structure exposed by the pilot

The six Study Notes reveal the local structure

$$I.16 \longrightarrow I.27,$$

with I.28 obtained from I.27 after angle normalization using I.15 or I.13 together with Common Notion 3. The Euclidean transition is

$$\text{Fifth Postulate} \longrightarrow I.29.$$

Proposition I.29 then becomes the principal parallel-to-angle transport theorem. It feeds I.30, supplies the uniqueness observation attached to I.31, and is used twice in I.32.

A schematic local DAG is

$$\begin{array}{ccccc}
 I.16 & \longrightarrow & I.27 & \longrightarrow & I.28 \\
 & & \downarrow & & \\
 \text{Fifth Postulate} & \longrightarrow & I.29 & \longrightarrow & I.30 \\
 & & \downarrow & & \\
 I.23 & \longrightarrow & I.31 & \longrightarrow & I.32.
 \end{array}$$

This is intentionally schematic: I.31 does not depend on I.29 for existence, while I.32 depends on both I.31 and I.29.

B.4 Converses and equivalences

- I.27 is the converse of the Alternate Angles Theorem contained in I.29.
- The corresponding-angle part of I.28 is the converse of the Corresponding Angles Theorem contained in I.29.
- I.30 is logically equivalent to Playfair’s Axiom.
- Playfair’s Axiom together with I.27 implies I.29.
- I.31 proves existence using I.23 and I.27, whereas I.29 supplies uniqueness.
- The Euclidean and Pythagorean proofs associated with I.32 use different diagrams but share the same parallel-angle transport mechanism.

B.5 Pilot conclusion

The I.27–I.32 sample is sufficient to justify a separate Wilkins appendix. The strongest evidence is I.30: its Study Note gives alternative proof paths, proves equivalence with Playfair’s Axiom, and reconstructs I.29 from Playfair plus I.27. Thus the linear sequence of propositions becomes a network of implications and equivalences.

For the formal project, the role of the Wilkins audit is to record the Euclidean dependency DAG, alternative proof paths, converses, equivalences, and configuration variants, and ultimately to compare these with the dependency graph emerging from the Lean/Hilbert reconstruction.

The next extension should proceed backward through I.1–I.26 without changing the structure established by this pilot. Later propositions can then be appended as the Book I reconstruction advances.

Appendix C

Hilbert: The *Grundlagen* Architecture

C.1 Purpose and scope

This appendix records the Hilbert-side source audit for Euclid Book II at the current checkpoint, Propositions II.1–II.4. Hilbert is relevant in two distinct ways: Chapter IV supplies the synthetic theory of plane content, whereas Section 15 later develops an arithmetic of segments. The documentary purpose is to keep those layers separate and to identify which one is actually used by the current formal reconstruction.

C.2 Hilbert’s organization of the relevant theory

Hilbert’s Chapter IV distinguishes finite equidecomposability from the more general relation of equicomplementability. The theory of plane area begins on the basis of Groups I–IV and does not need to be introduced as a numerical area function [2].

For the present project the useful hierarchy is

exact finite dissection $\implies$ equicomplementability $\implies$ later numerical area semantics,

with no converse inserted merely for convenience.

Section 15 is a different layer. Hilbert there defines segment addition and a geometric product, using a fixed unit segment and parallel construction, and then proves the familiar algebraic laws, including commutativity and distributivity [2]. That later arithmetic is a valuable control source precisely because it shows what would be required if Book II were to be formalized through segment multiplication. The current Lean development does not take that route.

C.3 Comparison table

Table C.1: Euclid II.1–II.4 against Hilbert’s content and segment-arithmetic layers

Euclid	Hilbert counterpart	Status in the Grundlagen	Source-audit remarks and relevance to formalization
II.1	Chapter IV finite decomposition and content theory; later Section 15 distributivity of segment multiplication is a structural analogue.	No one-to-one numbered Hilbert theorem reproducing Euclid II.1 was located in the current audit. The relevant material is organized by content theory and, later, by segment arithmetic rather than by Euclid’s Book II numbering.	The formal proof belongs to the earlier content/dissection layer. Additivity is obtained from an actual rectangle cut and representative transport, not by invoking the later distributive law for a segment product.
II.2	Chapter IV content plus the geometric symmetry of a rectangle; Section 15 commutativity is only a later arithmetic analogue.	Again the source correspondence is structural rather than proposition-level. Hilbert’s later commutative segment multiplication must not be read backward into the scissors proof.	The Lean theorem makes the distinction concrete: the required exchange of the two containing sides is implemented by cyclic rotation of a rectangle representative, preserving exact scissors content.
II.3	Chapter IV content/dissection; later Section 15 distributivity gives the arithmetic pattern only after segment multiplication has been defined.	No separate proposition-level Hilbert counterpart was located. The source supports the distinction between geometric content operations and the later arithmetic of segments.	II.3 is a direct specialization of the binary II.1 cut with the chosen segment used as the fixed side. The square is packaged as a rectangle contained by the same segment twice. No segment product and no side-order transport are needed.
II.4	Chapter IV finite content composition; later Section 15 distributivity and commutativity give an arithmetic analogue of the square expansion.	No one-to-one Hilbert proposition corresponding to Euclid II.4 was located in the current audit. Hilbert’s source role remains architectural.	The formal theorem composes exact II.2 with two exact II.3 instances and a geometric rectangle rotation. Despite the modern shadow $(a + b)^2 = a^2 + b^2 + 2ab$, no segment product, unit segment, or numerical area is used.

C.4 Structural observations retained from Hilbert

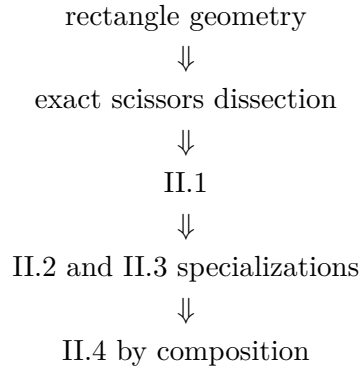
The following distinctions should be preserved as Book II grows.

- Plane content and segment arithmetic are separate theories in the Grundlagen and should remain separate in the documentation.
- Exact finite dissection is stronger than the general content relation; when Lean proves `HilbertScissorsEq`, that fact comes from the local proof architecture rather than from a blanket identification of all equal-content figures with equidecomposable ones.
- Segment multiplication is not primitive. It requires a unit segment, proportion geometry, and parallel constructions before algebraic laws such as commutativity and distributivity can even be stated.
- Consequently, a distributive-looking Euclidean proposition can be formalized below the arithmetic layer when its geometry is literally a finite cut-and-paste argument.

C.5 II.1–II.4: exact dissection below segment arithmetic

The opening four propositions give a useful stress test of the separation. II.1 establishes a reusable exact rectangle split. II.2 shows that the apparent exchange of factors can still be implemented geometrically by rotating a rectangle presentation. II.3 then shows that even this extra symmetry operation is unnecessary when the ordered side data already match the binary II.1 interface. II.4 is the first compositional step: it combines II.2 with two II.3 expansions and uses only exact scissors addition and representation transport.

The resulting formal chain is therefore



not

segment field $\longrightarrow$ multiplication $\longrightarrow$ algebraic simplification.

C.6 Result of the Grundlagen audit

At the II.4 checkpoint Hilbert’s main contribution is architectural. Chapter IV validates the use of a synthetic content layer independently of numerical area, while Section 15 marks a later arithmetic boundary. The current Book II formalization remains on the earlier side of that boundary.

The rectangle API therefore has a clear foundational interpretation: congruent segments and right angles determine representative rectangles, and the scissors layer compares their content. The algebra-like behavior observed in II.1–II.3 is derived from that geometry.

C.7 Current checkpoint

This appendix now concerns Euclid Book II, Propositions II.1–II.4. At this stage the durable Hilbert-side picture is

Groups I–IV geometry $\longrightarrow$ plane-content/dissection layer $\longrightarrow$ rectangle calculus for II.1–II.4,

while Section 15 segment arithmetic remains a later comparison layer rather than a dependency of the proofs.

Appendix D

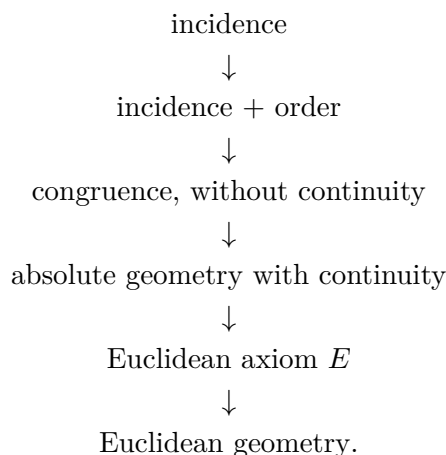
Borsuk–Szmielw: The Axiomatic Layers

D.1 Why this source matters

Karol Borsuk and Wanda Szmielw organize elementary geometry in a way that is unusually close to the architectural questions arising in a formal reconstruction.

Their primitive geometric language consists of points, lines, planes, betweenness, and equidistance. The axioms are divided into five groups: incidence, order, congruence, continuity, and finally either the Euclidean axiom E or the Bolyai–Lobachevskian axiom. The first four groups constitute absolute geometry.

More important for the present purpose, the authors explicitly state that one should study not only which theorems are true, but also which means are necessary to prove them. Their exposition is therefore deliberately layered:



This makes the book especially useful for deciding where a theorem belongs in a formal library. A result corresponding to a Euclidean proposition may turn out to be:

- part of the incidence/order language;
- a consequence of congruence alone;

- available before continuity;
- available only after continuity;
- genuinely Euclidean, requiring E ;
- or absent as a named theorem because Borsuk–Szmielew replace it by a stronger structural result.

The last column of the table below therefore asks a practical question: what does the Borsuk–Szmielew placement suggest for the Hilbert interface and the Book Zero layer?

D.2 Layer notation

For compactness the table uses the following descriptions.

Layer	Meaning
Inc/Ord	Chapter I: incidence and order, including betweenness, half-lines, half-planes, angles as ordered ray configurations, triangles, quadrangles, Pasch-type results, polygons, and triangulation.
Congr-no-Co	Chapter II: congruence added, but the Axiom of Continuity is not used.
Absolute	Chapter III: the Axiom of Continuity is added; the full first four groups of axioms are now available.
Euclidean E	Chapter V: absolute geometry plus the Axiom of Euclid E , equivalent in the book to Euclid’s Fifth Postulate.

D.3 Euclid in the Borsuk–Szmielew development

Table D.1: Euclidean propositions in the axiomatic layers of Borsuk–Szmielew

Euclid	Borsuk–Szmielew counterpart	Layer	Structural significance for the reconstruction
I.1	No one-to-one construction theorem was located. Equilateral triangles are defined in Chapter II, but the audited text does not reproduce Euclid’s two-circle construction as a named theorem.	No direct counterpart	This is a useful absence. Borsuk–Szmielew do not organize the congruence theory by reconstructing Euclid’s first circle construction. The formal project should therefore keep the classical construction DAG separate from the axiomatic layer in which an equilateral-point existence theorem is eventually supplied.
I.2	Axiom C5: on every half-line with origin a , for every segment pq , there exists exactly one point b such that $ab = pq$.	Congr-no-Co	Euclid’s segment transport has become primitive and is strengthened to existence plus uniqueness on a prescribed half-line. This is precisely interface-level segment layoff, not a proposition that later proofs should reconstruct from circles.

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Table D.1 – continued from previous page

Euclid	Borsuk–Szmielw counterpart	Layer	Structural significance for the reconstruction
I.3	Axiom C5 together with segment order and betweenness.	Congr-no-Co	Cutting a prescribed length from a longer segment is absorbed into ray layoff plus order. The natural formal location is a small Book Zero lemma over segment layoff and betweenness.
I.4	Theorem 22(I): two sides and the included angle determine a congruent triangle. The text immediately records this as the SAS congruence criterion.	Congr-no-Co	SAS is developed before continuity. This is strong evidence for keeping it in the neutral congruence interface. Euclid’s superposition argument is not part of the Borsuk–Szmielw proof architecture.
I.5	Theorem 23: in a triangle, congruent sides are opposite congruent angles; in particular, the base angles of an isosceles triangle are congruent.	Congr-no-Co	The isosceles theorem is packaged together with its converse in one biconditional result. The reusable API should reflect this stronger relationship instead of treating I.5 and I.6 as unrelated endpoints.
I.6	The converse direction of Theorem 23: congruent opposite angles imply congruent opposite sides.	Congr-no-Co	Borsuk–Szmielw eliminate Euclid’s separate reductio construction. Once segment and angle congruence are organized properly, I.6 is simply the reverse implication of the same structural theorem as I.5.
I.7	No separate theorem corresponding exactly to Euclid I.7 was located. Its uniqueness content is absorbed into the order theory, unique angle construction (Theorem 13), and the strong congruence axioms.	Congr-no-Co	This is an example of a Euclidean proposition disappearing under a stronger interface. The formal question is not how to imitate I.7, but which general uniqueness and side-of-line lemmas make its role unnecessary.
I.8	Theorem 21: if the three corresponding sides of two triangles are congruent, then the three corresponding angles are congruent.	Congr-no-Co	This is the SSS content in the Borsuk–Szmielw organization, and it is available without continuity and without the Euclidean axiom. For the formal project this strongly supports treating SSS as derived neutral congruence infrastructure rather than Euclidean input.
I.9	Theorems 33–34: a bisector lies between the sides of the angle, and every angle has exactly one bisector.	Congr-no-Co	The source gives a stronger result than Euclid’s bare construction: existence and uniqueness are packaged together. Angle-bisector existence therefore belongs naturally in the reusable interface.
I.10	Theorems 31–32: an equidistant collinear point is between the endpoints, and every segment has exactly one midpoint.	Congr-no-Co	This is especially valuable formally. Borsuk–Szmielw do not take the diagrammatic position of the midpoint for granted; they first prove the betweenness theorem and only then existence and uniqueness. This is exactly the separation needed in Lean.
I.11	Theorem 76: through a point lying on a line there exists exactly one line in the plane perpendicular to the given line.	Congr-no-Co	Perpendicular erection is already a neutral interface theorem. Later propositions should consume it directly rather than replay an auxiliary Euclidean triangle construction.

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Table D.1 – continued from previous page

Euclid	Borsuk–Szmielew counterpart	Layer	Structural significance for the reconstruction
I.12	Theorem 77: through a point outside a given line there exists exactly one perpendicular to that line.	Congr-no-Co	Dropping a perpendicular is likewise available before continuity. The theorem is stronger and cleaner than Euclid’s construction-oriented formulation and belongs naturally in the Hilbert interface.
I.13	Sections 21–24 of Chapter II: addition of free angles, adjacent angles, right angles, and the fact that an angle adjacent to a right angle is right (Theorems 63, 69–71).	Congr-no-Co	Borsuk–Szmielew replace informal angle arithmetic by an explicit calculus of free angles. This is a strong source precedent for using synthetic supplement/addition relations rather than numerical degrees in the formalization.
I.14	Theorem 13 gives unique construction of a congruent angle on a specified side; Theorems 14–15 control adjacent angles and their uniqueness.	Congr-no-Co	The converse straight-line criterion is absorbed into a more general uniqueness theory for angle placement. The formal analogue should be a small collinearity/adjacency consequence of that interface.
I.15	Theorem 16: vertical angles are congruent.	Congr-no-Co	A direct reusable theorem. Its low position in the congruence chapter confirms that vertical-angle congruence should be generic infrastructure for later triangle and parallel arguments.
I.16	Theorem 35: an exterior angle of a triangle is greater than each nonadjacent interior angle.	Congr-no-Co	This is one of the central neutral theorems. The proof explicitly uses midpoint, vertical-angle congruence, SAS, and angle order. It is also the engine for constructing nonintersecting lines before the Euclidean axiom enters.
I.17	A short consequence of Theorem 35 together with the developed theory of adjacent and right angles. It is not necessary to use the later Saccheri–Legendre theorem for this Euclidean proposition.	Congr-no-Co	This distinction prevents an important misclassification. Borsuk–Szmielew later prove stronger angle-sum results after continuity, but Euclid I.17 itself already belongs to the earlier neutral congruence layer.
I.18	Theorem 40, one direction: in a triangle the greater side lies opposite the greater angle.	Congr-no-Co	I.18 and I.19 are unified. Borsuk–Szmielew package strict side and angle comparison into a biconditional theorem, exactly the sort of strong reusable comparison lemma useful in Book Zero.
I.19	Theorem 40, converse direction: the greater angle lies opposite the greater side.	Congr-no-Co	The converse is not a separate construction. It is the reverse direction of the same strict comparison equivalence used for I.18.
I.20	Theorem 50: for distinct points a, b, c , $[ab] + [bc] > [ac]$, with equality exactly when $B(a, b, c)$.	Congr-no-Co	This is substantially stronger than the triangle-only formulation. It combines triangle inequality with the collinear equality case and uses the explicit algebra of free segments. It is an excellent model for a robust segment-order interface.

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Table D.1 – continued from previous page

Euclid	Borsuk–Szmielw counterpart	Layer	Structural significance for the reconstruction
I.21	Theorem 52 generalizes the side-sum part to a convex polygonal line inside a triangle; the angle part is controlled by the Exterior Angle Theorem 35 and the order theory of rays.	Congr-no-Co	The source replaces the one-off proposition by a more general polygonal inequality. For formal work the useful target is therefore the reusable generalized lemma rather than the exact Euclidean diagram.
I.22	No one-to-one prescribed-three-sides construction theorem was located in the audited elementary sequence. The book has the triangle inequality and a strong congruence construction axiom, but does not replay Euclid’s circle-intersection construction here.	No direct counterpart	This should be kept as an explicit gap in the source map. Existence of a triangle with three prescribed side lengths is logically different from SSS uniqueness/congruence and should not be silently identified with Theorem 21.
I.23	Theorem 13: for a given ray, chosen half-plane, and given angle, there exists exactly one ray on that side making a congruent angle.	Congr-no-Co	Euclid’s angle-copy construction becomes a primitive-style transport theorem with uniqueness. This is one of the strongest direct precedents for the angle-construction API already used in the Hilbert route.
I.24	Theorem 41: with two corresponding sides fixed, comparison of the included angles is equivalent to comparison of the third sides.	Congr-no-Co	Borsuk–Szmielw combine the Euclidean comparison theorem and its converse into one biconditional. This is a natural interface theorem for replacing configuration-heavy local proofs.
I.25	The converse direction of Theorem 41.	Congr-no-Co	As with I.18/I.19, the source suggests exposing one symmetric comparison theorem rather than maintaining two large proposition-sized proofs.
I.26	Theorem 22(II): one side and the two angles adjacent to it determine a congruent triangle (ASA).	Congr-no-Co	The ASA part of I.26 is explicit and neutral. The audited text does not package Euclid’s second AAS configuration under a separate theorem number here, so the appendix does not silently identify the two formulations.
I.27	Theorems 37–39. Equal appropriate transversal angles give nonintersection; Theorem 39 concludes that through an external point there is at least one line not intersecting the given line.	Congr-no-Co	This is one of the most important source placements in the book. The angle-to-parallel direction and existence of a nonintersecting line are established before continuity and before the Euclidean axiom.
I.28	A consequence of the same Theorems 37–38 together with the earlier adjacent/vertical-angle normalization theorems.	Congr-no-Co	The two Euclidean angle criteria are wrappers over the neutral nonintersection theorem. The conclusion “parallel” does not by itself make the result Euclidean.

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Table D.1 – continued from previous page

Euclid	Borsuk–Szmielew counterpart	Layer	Structural significance for the reconstruction
I.29	After E , Chapter V develops the Euclidean characterization of parallel lines and Theorem 10: angles whose corresponding sides are parallel are congruent.	Euclidean E	This is the reversal of I.27. Before E , appropriate angle data produce nonintersection; after E , parallelism can be converted back into angle congruence. The source therefore marks the same neutral-to-Euclidean boundary that the formal reconstruction uses.
I.30	Chapter V, Theorem 7: parallelism of lines is transitive.	Euclidean E	Parallel transitivity is explicitly downstream of the Euclidean axiom. It should therefore be classified as Euclidean interface infrastructure, not as a neutral convenience lemma.
I.31	Chapter II, Theorem 39 gives existence of a non-intersecting line. Axiom E plus that theorem gives Chapter V, Theorem 1, uniqueness; Theorem 6 states the corresponding unique parallel through a point.	Existence: Congr-no-Co; uniqueness: Euclidean E	Borsuk–Szmielew make the existence/uniqueness split exceptionally clean. This is a direct architectural model for keeping the two layers separate in Lean.
I.32	Chapter V, Theorem 2: the sum of the angles of every triangle is π .	Euclidean E	The equality form of the triangle angle-sum theorem is placed immediately after the Euclidean axiom. A synthetic Lean statement need not use numerical measure, but its dependency must remain Euclidean.
I.33	Chapter V, Theorem 12, sufficiency direction: a plane quadrangle is a parallelogram when one pair of corresponding directed sides is parallel and the corresponding segments are congruent.	Euclidean E	This theorem is stronger and more useful than a literal copy of I.33. It characterizes a parallelogram from exactly the sort of parallel-plus-congruence data that arise in I.33. It is a strong candidate for reusable interface infrastructure.
I.34	Chapter V, Theorem 12, necessity direction and proof: in a parallelogram the diagonal triangles are congruent, yielding equality of opposite sides and the corresponding angle data.	Euclidean E	The proof architecture is almost tailor-made for the formal I.34 interface: parallel half-lines give angle congruences, then triangle congruence supplies opposite-side data. I.34 should expose this package for immediate downstream use.
I.35	No Euclidean equal-content or equicomplementability theorem corresponding directly to I.35 was located. Chapter I already has a substantial theory of polygons and triangulation, while Chapter V has the parallelogram theory needed immediately before I.35.	No direct counterpart	The absence is informative. Borsuk–Szmielew supply rigorous figure/triangulation infrastructure and Euclidean parallelogram infrastructure, but they do not insert a Hilbert/Hartshorne-style equal-content layer at this point. Thus they are highly useful for the components feeding I.35, but not the primary source for the semantics of its conclusion.

D.4 The strongest architectural correspondences

The most useful feature of the Borsuk–Szmielw source is that several Euclidean propositions are replaced by stronger interface theorems.

D.4.1 Transport is primitive-strength infrastructure

Axiom C5 is stronger than the bare construction in I.2: it gives exactly one point on a chosen half-line at a prescribed distance. Theorem 13 is the angular analogue: on a chosen side of a ray, an angle congruent to a given angle can be constructed uniquely.

Thus two operations that appear relatively late or require elaborate construction in Euclid become basic transport mechanisms:

segment layoff and angle layoff.

This is directly consonant with the Hilbert-style formal interface.

D.4.2 Pairs of Euclidean propositions become biconditionals

Theorem 23 unifies I.5 and I.6:

equal sides $\iff$ equal opposite angles.

Theorem 40 unifies I.18 and I.19:

greater side $\iff$ greater opposite angle.

Theorem 41 similarly unifies I.24 and I.25 when two corresponding sides are fixed:

greater included angle $\iff$ greater third side.

For library design, these are more informative than the proposition numbers. They identify the natural reusable theorem boundaries.

D.4.3 Midpoints and bisectors include positional information

The midpoint discussion is especially close to the needs of formal geometry. Borsuk–Szmielw first prove that a collinear point equidistant from the endpoints of a segment must lie between them. Only then do they prove existence and uniqueness of the midpoint.

The angle-bisector discussion has the same pattern: first the bisector is shown to lie between the sides of the angle, then existence and uniqueness are established.

The lesson for formalization is precise. “Midpoint” and “bisector” should not be treated merely as metric equalities; their order/ray placement is part of the reusable theorem package.

D.5 The neutral parallel theory is already large

The source is particularly valuable around Euclid I.27–I.31.

From the Exterior Angle Theorem and the congruence/order theory, Borsuk–Szmielw prove that an appropriate equality of transversal angles forces two lines to be disjoint. They then obtain Theorem 39:

through an external point there exists at least one nonintersecting line.

This theorem belongs to Chapter II, before continuity and before the Euclidean axiom.

Only much later, in Chapter V, is Axiom *E* added:

through an external point there is at most one nonintersecting line.

Together the two statements give uniqueness. Chapter V then derives the Euclidean parallel theory: characterization of disjoint coplanar lines, uniqueness of the parallel, transitivity, angles with parallel sides, and parallelograms.

This produces a clean formal boundary:

angle criterion $\Rightarrow$ nonintersection	before <i>E</i> ,
existence of a nonintersecting line	

while

uniqueness of the parallel	after <i>E</i> .
parallelism $\Rightarrow$ angle congruence	
parallel transitivity	
triangle angle sum	

For the present reconstruction this is one of the strongest independent checks on the placement of I.27, I.29, I.30, I.31, and I.32.

D.6 Theorem 12 of Chapter V as an interface theorem

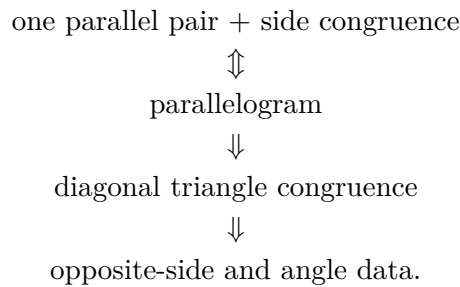
The parallelogram theorem deserves special attention.

Borsuk–Szmielw define a parallelogram in the expected way, by two pairs of opposite sides lying on parallel lines. Their Theorem 12 then proves a necessary and sufficient condition using only one pair of directed parallel sides and equality of the corresponding segments.

In the necessity direction, the proof starts from a parallelogram, transports angles using parallel half-lines, and obtains congruence of the two triangles cut by a diagonal. Opposite-side congruence follows.

In the sufficiency direction, one parallel pair together with one segment congruence forces the remaining pair to be parallel.

This theorem simultaneously explains why I.33 and I.34 should sit close to the reusable Euclidean parallelogram interface:



It is therefore not merely a historical counterpart. It is a plausible API boundary for subsequent formal work.

D.7 What the book contributes at I.35

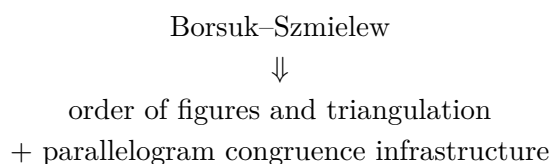
The transition at I.35 is revealing because the source provides two sides of the problem but not the final semantic relation.

Chapter I develops polygons, intersections of polygons with half-planes, and triangulation. This is a rigorous combinatorial theory of plane figures built very low in the axiomatic hierarchy.

Chapter V develops the Euclidean parallelogram theory and supplies the congruence infrastructure immediately preceding I.35.

What is not developed there is a Euclidean theory of equal content or equicomplementability comparable to Hilbert Chapter IV or Hartshorne Section 22. Later references to area in the Bolyai–Lobachevskian part do not supply such a replacement; the authors explicitly say that they will not develop a general theory of area there.

For the formal reconstruction the source therefore has a sharply defined role:



but

Hilbert/Hartshorne remain the control sources for equicomplementability/equal content.

This is not a weakness of the audit. It tells us exactly where one source stops and another must take over.

D.8 Consequences for the formal library

The Borsuk–Szmielw audit suggests a practical classification of reusable results.

D.8.1 Hilbert interface candidates

The following belong naturally to a general synthetic interface rather than to individual Euclidean proposition files:

- unique segment layoff on a ray;
- unique angle layoff on a chosen side;
- SAS and SSS-style triangle congruence;
- isosceles theorem and converse;
- midpoint existence, uniqueness, and betweenness;
- angle-bisector existence, uniqueness, and ray order;
- vertical angles;
- Exterior Angle Theorem;
- strict side/angle comparison;
- perpendicular existence and uniqueness;
- neutral angle criteria for nonintersection.

D.8.2 Book Zero candidates

The developed calculus of free segments and free angles also points to a natural macro-language:

- segment less-than and trichotomy;
- transport of order under congruence;
- segment addition and subtraction;
- triangle inequality with the collinear equality case;
- angle less-than and trichotomy;
- angle addition and adjacency;
- supplement/right-angle normalization.

These are exactly the sorts of operations that prevent later proposition files from expanding into repeated low-level incidence and order proofs.

D.8.3 Euclidean interface candidates

After Axiom *E*, the natural reusable layer contains:

- unique parallel through a point;
- characterization of Euclidean parallels;
- parallel transitivity;
- congruence of angles with respectively parallel sides;
- Euclidean triangle angle sum;
- the Chapter V parallelogram characterization.

The source thus gives not just another list of theorem counterparts, but a plausible decomposition of the formal library itself.

D.9 Conclusion

Borsuk and Szmielew are particularly valuable for the reconstruction because their book asks almost the same structural question as the formal project: not only whether a theorem is true, but which axiomatic resources are actually needed to prove it.

Their treatment makes several boundaries unusually visible. Much of the elementary order and congruence theory, including the Exterior Angle Theorem, midpoints, bisectors, perpendiculars, triangle comparison, and existence of nonintersecting lines, is developed before continuity. The Axiom of Euclid then contributes the uniqueness needed to obtain the full Euclidean theory of parallels. Their parallelogram theorem packages the material around I.33–I.34 in a form that is especially attractive as formal interface infrastructure.

At I.35 the source becomes useful in a different way. Its early theory of polygons and triangulation and its later parallelogram theory provide important components of the proof, while the semantic theory of equal content is left to other sources.

The resulting appendix should therefore be read as an axiomatic-layer map and an API-design aid, not merely as another historical comparison with Euclid.

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